HL7 FHIR R4 — Chapter 12: Workflow Module (sections 12 – 12.23.9)

Source: HL7 FHIR R4 (v4.0.1) specification, https://hl7.org/fhir/R4/ — text-only rendition for AI ingestion; page order follows toc.html.

12 Workflow Module

URL: https://hl7.org/fhir/R4/workflow-module.html
Work Group FHIR Infrastructure | Standards Status: Informative

12.0 Workflow Module

The workflow module focuses on the coordination of activities within and across systems. This includes three primary aspects:

• How do we ask for another person, device or system to do something?

• How do we track the linkages and dependencies between activities - actions to their authorizations, complex activities to individual steps, protocols to plans to orders, etc.?

• How do we define what activities are possible and the expected order and dependencies of the steps within those activities? I.e. process/orchestration definition

12.0 Index

Infrastructure | Start here: Overview Resource: Task Patterns: Definition, Request, Event Documentation: Communication Patterns, Ad-Hoc Patterns, Management Patterns & Examples
Scheduling | Appointments: Appointment / AppointmentResponse Availability: Schedule / Slot
Clinical Process | Referrals: ServiceRequest Orders: NutritionOrder, VisionPrescription Definitions: ActivityDefinition, PlanDefinition
Miscellaneous: DeviceRequest & DeviceUseStatement, SupplyRequest & SupplyDelivery

12.0.1 Introduction

Workflows can be performed through direct posting of resources to a target server (combined with a specific tag), by using the Task resource, through the use of messaging or via FHIR services. This specification includes a workflow page that describes the concepts underlying the discussion of workflows, and points to a number of different communication and architectural workflow patterns.

In addition to the Task resource, this specification defines three logical models - Definition, Request and Event that define the patterns for resources that are typically involved in workflow. These patterns include elements defining common attributes of each type of resource as well as relationships between them. These relationships are summarized on the workflow page, along with a complete list of resources that follow (or are hoped to soon follow) the request and event patterns.

Finally the PlanDefinition and ActivityDefinition resources combine to support the creation of protocols, orders sets, guidelines and other workflow definitions by describing the types of activities that can occur and setting rules about their composition, sequencing, interdependencies and flow.

12.0.2 Common use Cases

Workflow manifests in many places in the healthcare environment:

• Creating a lab order, drug prescription, or other clinical order or an insurance claim, enrollment request, Appointment or similar administrative request and asking for it to be actioned by a specific organization or practitioner

• Negotiating a fulfillment process, such as requesting further information before accepting a claim or referral or proposing an alternative therapy when processing an order

• Letting an ordering physician know what the current progress is in fulfilling an order (e.g. blood has been drawn, sample is being processed, preliminary results are in, etc.)

• Defining a plan or recommendation for a set of clinical and/or administrative activities to manage a patient's care and then tracking how those plans and recommendations are (or are not) acted upon over time.

• Communicating a state change to a request or order (e.g. suspension, update, cancellation, etc.) to a fulfilling system so that they can take appropriate action

• Asking for a state change, requesting the merge of a couple of patients or the invoking of some operation or decision support in an asynchronous manner - for example, one where human intervention is required

• Designing or adhering to a study protocol, chemotherapy protocol, instantiating an order set or other plan definition

FHIR provides multiple ways to enable these scenarios (and many others). Common mechanisms, along with their pros and cons can be found in the workflow sections on patterns.

12.0.3 Security and Privacy

Resources related to workflow need to adhere to the same security and privacy guidelines that apply to all FHIR resources, including specific considerations for those that may contain personally-identifying information. There are a couple of additional security and privacy considerations specific to workflow:

1. Some workflows are ad-hoc without pre-defined participants or flows. These can be challenging for security and privacy processes to manage appropriately

2. Workflow can drive automated behavior. I.e. The mere existence of an electronic record can cause information to flow, procedures to be performed, records to be changed and money to be transferred, potentially without any intervention, oversight or sanity checking by a human being. As such, even greater care must be taken to ensure that:

• constraints are placed on what systems (and users) can initiate workflow processes

• requests for action are appropriately authenticated before action is taken

• patient consents and other relevant policies are enforced either by the system storing the request or the system acting upon it (and that if enforcement is not performed by the actor, that they are confident that relevant policies have been enforced on the request prior to action)

For more general considerations, see the Security and Privacy module.

12.0.4 Developmental Roadmap

Initial work has taken place on aligning most (though not yet all) resources with the Definition, Request and Event patterns. In the lead-up to R5, we'll be moving the alignment checks into the build process and more formally documenting (and potentially reporting) on variations along with their justifications. Further alignment is also possible (where beneficial to implementers). We'll also be examining the potential for exposing alignment with the patterns in a computably useful manner (e.g. as interfaces).

Work will continue on the workflow patterns, including vetting the patterns against various clinical scenarios and enhancing pattern documentation. We also hope to examine both messaging and services in more detail with further guidance about when and how such mechanisms should be used for workflow and how they relate to the Task resource. As well, we'll examine the possibility for developing "standardized" workflows for certain domains and how such patterns might be documented, particularly through the use of the ExampleScenario resource. We will look for implementer feedback to guide this work.

The PlanDefinition and ActivityDefinition resources will continue to evolve based on feedback from the implementer community. We'll explore using them in a variety of ways, including clinical order sets, medication protocols, workflow protocols, clinical pathways, administrative protocols, etc. We hope to develop several example workflow protocols.

Additional topics for future work include:

• The initial effort to align with workflow patterns has been a bit over-zealous for some resources, resulting in the loss of domain-specific context or occasionally the introduction of elements that might be more properly represented as extensions. In R5, we'll continue to work on improving the balance, ensuring that consistency with patterns does not overshadow the essential requirements for implementer intuitiveness and simplicity

• Resolving the overlap between the SupplyRequest, DeviceRequest and VisionPrescription resources

• Improving mapping and alignment of the elements and status codes of the Task resource with the WS-HumanTask specification

• Creating "best practice" guides for how to implement workflow for different business patterns

• Examining how workflow is used for compensating actions E.g. account transactions and reversals

12.1 Resource Task - Content

URL: https://hl7.org/fhir/R4/task.html
Orders and Observations Work Group | Maturity Level: 2 | Trial Use | Security Category: Not Classified | Compartments: Not linked to any defined compartments

A task to be performed.

12.1.1 Scope and Usage

A task resource describes an activity that can be performed and tracks the state of completion of that activity. It is a representation that an activity should be or has been initiated, and eventually, represents the successful or unsuccessful completion of that activity.

Note that there are a variety of processes associated with making and processing orders. Some orders may be handled immediately by automated systems but most require real world actions by one or more humans. Some orders can only be processed when other real world actions happen, such as a patient presenting themselves so that the action to be performed can actually be performed. Often these real world dependencies are only implicit in the order details.

12.1.2 Background and Context

12.1.2.1 Using Tasks in a RESTful context

In a RESTful context, a server functions as a repository of tasks. The server itself, or other agents are expected to monitor task activity and initiate appropriate actions to ensure task completion, updating the status of the task as it proceeds through its various stages of completion. These agents can be coordinated, following well choreographed business logic to ensure that tasks are completed. Task management may also be centrally controlled using some form of a workflow engine, in which case, the workflow engine itself may update and maintain the task resources in the server, and through its orchestration activities ensure that tasks are completed. The task resource enables either model of task management yet provides a consistent view of the status of tasks being executed in support of healthcare workflows.

The assignment of tasks into categories by type of task, type of performer and task status enable the server to function as a queue of work items. This queue can be polled or subscribed to by various agents, enabling automation of workflows in FHIR using existing search and subscription mechanisms. Owners, requesters, other agents (e.g. workflow managers) can thus be ready to initiate the next steps in a complex workflow.

12.1.2.2 Tasks State Machine

Tasks start in a Created state. Once they have been assigned to an owner they transition to the Ready state, indicating that they are ready to be performed. Once the owner initiates activity on the task, the task transitions to the In Progress state, indicating that work is being performed. Upon normal completion, the task enters the Completed state. If there is a failure during the task execution that prevents the task from being completed, it can also enter a Failed state, indicating an abnormal termination of the task. A task in any non-terminal state may also be Cancelled, representing an abnormal termination of the task due to external forces, rather than an error condition.

Tasks in any non-terminal state (Created, Ready, In Progress) can be suspended and resumed. When a task is suspended, it is typically resumed in the state it was in when it was originally suspended. Suspending a task suspends all of its children as well. Resuming a task resumes all of its children.

An In-progress task can also be stopped, returning it to a Ready state. This may be in preparation for delegation or reassignment (e.g., because it cannot be completed by the current owner), to restart a task due to a temporary failure (e.g., to reattempt completion of the activity), or in preparation to allow others to claim the task.

The task history allows applications monitoring the state of a workflow to identify tasks that are long running, perhaps stuck in some queue, to enable management activities that could ensure completion. It also enables tracking of task statistics such as wait time, or time in progress, or time to completion, enabling capture of important task metrics in support of optimization and quality improvement.

Note: Currently, task pre-requisites can be represented using Task's `description` element or the RequestGroup resource. We are seeking input from the implementer community in evaluating whether there is need for adding another mechanism to this resource for this purpose.

Feedback here .

12.1.3 Boundaries and Relationships

Task spans both intent and event and tracks the execution through to completion. A Task is a workflow step such as cancelling an order, fulfilling an order, signing an order, merging a set of records, admitting a patient. In contrast, Procedures are actions that are intended to result in a physical or mental change to or for the subject (for example, surgery, physiotherapy, training, counseling). A Task resource often exists in parallel with clinical resources. For example, a Task could request fulfillment of a ServiceRequest ordering a Procedure that would result in a Procedure, Observation, DiagnosticReport, ImagingStudy or similar resource. Another example would be a Task that requests fulfillment of a CommunicationRequest to be performed between various actors.

As stated above, the task resource tracks the state of a task, enabling systems to ensure that tasks are completed. This information is kept separate from the operational details necessary to complete the task, as those details vary across and even within workflows. That detail is expected to be carried in the referenced `focus` of the task.

Tasks may have named inputs and outputs. Inputs represent information that may or must be present in order for a task to complete. Outputs represent intermediate or final results produced by a task. For example, in a task supporting reading of radiology image, the inputs might include both the imaging study to be read, as well as relevant prior images. Outputs could represent radiology measurements as well as the Radiologist's diagnostic report.

Inputs and outputs are tracked by the task because workflow management activity may automate the transfer of outputs from one task to inputs to a subsequent task.

To facilitate the integration of off the shelf workflow applications with FHIR, the task resource may reference a definition. This definition can represent a description of the workflow activity to be performed, using a standard workflow description language such as BPEL, BPMN, or XPDL, a workflow definition such as those defined in IHE profiles, or even simple written instructions explaining a process to be performed.

This resource is referenced by CarePlan, ImagingStudy, PaymentReconciliation and itself

12.1.4 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
Task | ITU | | DomainResource | A task to be performed + Rule: Last modified date must be greater than or equal to authored-on date. Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | | 0..* | Identifier | Task Instance Identifier
instantiatesCanonical | Σ | 0..1 | canonical(ActivityDefinition) | Formal definition of task
instantiatesUri | Σ | 0..1 | uri | Formal definition of task
basedOn | Σ | 0..* | Reference(Any) | Request fulfilled by this task
groupIdentifier | Σ | 0..1 | Identifier | Requisition or grouper id
partOf | Σ | 0..* | Reference(Task) | Composite task
status | ?!Σ | 1..1 | code | draft | requested | received | accepted | + TaskStatus (Required)
statusReason | Σ | 0..1 | CodeableConcept | Reason for current status
businessStatus | Σ | 0..1 | CodeableConcept | E.g. "Specimen collected", "IV prepped"
intent | Σ | 1..1 | code | unknown | proposal | plan | order | original-order | reflex-order | filler-order | instance-order | option TaskIntent (Required)
priority | | 0..1 | code | routine | urgent | asap | stat Request priority (Required)
code | Σ | 0..1 | CodeableConcept | Task Type Task Codes (Example)
description | Σ | 0..1 | string | Human-readable explanation of task
focus | Σ | 0..1 | Reference(Any) | What task is acting on
for | Σ | 0..1 | Reference(Any) | Beneficiary of the Task
encounter | Σ | 0..1 | Reference(Encounter) | Healthcare event during which this task originated
executionPeriod | Σ | 0..1 | Period | Start and end time of execution
authoredOn | I | 0..1 | dateTime | Task Creation Date
lastModified | ΣI | 0..1 | dateTime | Task Last Modified Date
requester | Σ | 0..1 | Reference(Device | Organization | Patient | Practitioner | PractitionerRole | RelatedPerson) | Who is asking for task to be done
performerType | | 0..* | CodeableConcept | Requested performer Procedure Performer Role Codes (Preferred)
owner | Σ | 0..1 | Reference(Practitioner | PractitionerRole | Organization | CareTeam | HealthcareService | Patient | Device | RelatedPerson) | Responsible individual
location | Σ | 0..1 | Reference(Location) | Where task occurs
reasonCode | | 0..1 | CodeableConcept | Why task is needed
reasonReference | | 0..1 | Reference(Any) | Why task is needed
insurance | | 0..* | Reference(Coverage | ClaimResponse) | Associated insurance coverage
note | | 0..* | Annotation | Comments made about the task
relevantHistory | | 0..* | Reference(Provenance) | Key events in history of the Task
restriction | | 0..1 | BackboneElement | Constraints on fulfillment tasks
repetitions | | 0..1 | positiveInt | How many times to repeat
period | | 0..1 | Period | When fulfillment sought
recipient | | 0..* | Reference(Patient | Practitioner | PractitionerRole | RelatedPerson | Group | Organization) | For whom is fulfillment sought?
input | | 0..* | BackboneElement | Information used to perform task
type | | 1..1 | CodeableConcept | Label for the input
value[x] | | 1..1 | * | Content to use in performing the task
output | | 0..* | BackboneElement | Information produced as part of task
type | | 1..1 | CodeableConcept | Label for output
value[x] | | 1..1 | * | Result of output
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Task xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Task Instance Identifier --></identifier>
 <instantiatesCanonical><!-- 0..1 canonical(ActivityDefinition) Formal definition of task --></instantiatesCanonical>
 <instantiatesUri value="[uri]"/><!-- 0..1 Formal definition of task -->
 <basedOn><!-- 0..* Reference(Any) Request fulfilled by this task --></basedOn>
 <groupIdentifier><!-- 0..1 Identifier Requisition or grouper id --></groupIdentifier>
 <partOf><!-- 0..* Reference(Task) Composite task --></partOf>
 <status value="[code]"/><!-- 1..1 draft | requested | received | accepted | + -->
 <statusReason><!-- 0..1 CodeableConcept Reason for current status --></statusReason>
```

```
 <businessStatus><!-- 0..1 CodeableConcept E.g. "Specimen collected", "IV prepped" --></businessStatus>
 <intent value="[code]"/><!-- 1..1 unknown | proposal | plan | order | original-order | reflex-order | filler-order |
    instance-order | option -->
 <priority value="[code]"/><!-- 0..1 routine | urgent | asap | stat -->
 <code><!-- 0..1 CodeableConcept Task Type --></code>
 <description value="[string]"/><!-- 0..1 Human-readable explanation of task -->
 <focus><!-- 0..1 Reference(Any) What task is acting on --></focus>
 <for><!-- 0..1 Reference(Any) Beneficiary of the Task --></for>
 <encounter><!-- 0..1 Reference(Encounter) Healthcare event during which this task originated --></encounter>
 <executionPeriod><!-- 0..1 Period Start and end time of execution --></executionPeriod>
 <authoredOn value="[dateTime]"/><!--  0..1 Task Creation Date -->
 <lastModified value="[dateTime]"/><!--  0..1 Task Last Modified Date -->
 <requester><!-- 0..1 Reference(Device|Organization|Patient|Practitioner|
   PractitionerRole|RelatedPerson) Who is asking for task to be done --></requester>
 <performerType><!-- 0..* CodeableConcept Requested performer --></performerType>
 <owner><!-- 0..1 Reference(Practitioner|PractitionerRole|Organization|CareTeam|
   HealthcareService|Patient|Device|RelatedPerson) Responsible individual --></owner>
 <location><!-- 0..1 Reference(Location) Where task occurs --></location>
 <reasonCode><!-- 0..1 CodeableConcept Why task is needed --></reasonCode>
 <reasonReference><!-- 0..1 Reference(Any) Why task is needed --></reasonReference>
 <insurance><!-- 0..* Reference(Coverage|ClaimResponse) Associated insurance coverage --></insurance>
 <note><!-- 0..* Annotation Comments made about the task --></note>
 <relevantHistory><!-- 0..* Reference(Provenance) Key events in history of the Task --></relevantHistory>
 <restriction>  <!-- 0..1 Constraints on fulfillment tasks -->
  <repetitions value="[positiveInt]"/><!-- 0..1 How many times to repeat -->
  <period><!-- 0..1 Period When fulfillment sought --></period>
  <recipient><!-- 0..* Reference(Patient|Practitioner|PractitionerRole|
    RelatedPerson|Group|Organization) For whom is fulfillment sought? --></recipient>
 </restriction>
 <input>  <!-- 0..* Information used to perform task -->
  <type><!-- 1..1 CodeableConcept Label for the input --></type>
  <value[x]><!-- 1..1 * Content to use in performing the task --></value[x]>
 </input>
 <output>  <!-- 0..* Information produced as part of task -->
  <type><!-- 1..1 CodeableConcept Label for output --></type>
  <value[x]><!-- 1..1 * Result of output --></value[x]>
 </output>
</Task>
```

JSON Template

```
{
  "resourceType" : "Task",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Task Instance Identifier
  "instantiatesCanonical" : { canonical(ActivityDefinition) }, // Formal definition of task
  "instantiatesUri" : "<uri>", // Formal definition of task
  "basedOn" : [{ Reference(Any) }], // Request fulfilled by this task
  "groupIdentifier" : { Identifier }, // Requisition or grouper id
  "partOf" : [{ Reference(Task) }], // Composite task
  "status" : "<code>", // R!  draft | requested | received | accepted | +
  "statusReason" : { CodeableConcept }, // Reason for current status
  "businessStatus" : { CodeableConcept }, // E.g. "Specimen collected", "IV prepped"
  "intent" : "<code>", // R!  unknown | proposal | plan | order | original-order | reflex-order | filler-order |
    instance-order | option
  "priority" : "<code>", // routine | urgent | asap | stat
  "code" : { CodeableConcept }, // Task Type
  "description" : "<string>", // Human-readable explanation of task
  "focus" : { Reference(Any) }, // What task is acting on
  "for" : { Reference(Any) }, // Beneficiary of the Task
  "encounter" : { Reference(Encounter) }, // Healthcare event during which this task originated
  "executionPeriod" : { Period }, // Start and end time of execution
  "authoredOn" : "<dateTime>", // C? Task Creation Date
  "lastModified" : "<dateTime>", // C? Task Last Modified Date
  "requester" : { Reference(Device|Organization|Patient|Practitioner|
   PractitionerRole|RelatedPerson) }, // Who is asking for task to be done
  "performerType" : [{ CodeableConcept }], // Requested performer
  "owner" : { Reference(Practitioner|PractitionerRole|Organization|CareTeam|
   HealthcareService|Patient|Device|RelatedPerson) }, // Responsible individual
  "location" : { Reference(Location) }, // Where task occurs
  "reasonCode" : { CodeableConcept }, // Why task is needed
  "reasonReference" : { Reference(Any) }, // Why task is needed
  "insurance" : [{ Reference(Coverage|ClaimResponse) }], // Associated insurance coverage
  "note" : [{ Annotation }], // Comments made about the task
  "relevantHistory" : [{ Reference(Provenance) }], // Key events in history of the Task
  "restriction" : { // Constraints on fulfillment tasks
    "repetitions" : "<positiveInt>", // How many times to repeat
    "period" : { Period }, // When fulfillment sought
    "recipient" : [{ Reference(Patient|Practitioner|PractitionerRole|
    RelatedPerson|Group|Organization) }] // For whom is fulfillment sought?
  },
  "input" : [{ // Information used to perform task
    "type" : { CodeableConcept }, // R!  Label for the input
    // value[x]: Content to use in performing the task. One of these 50:
    "valueBase64Binary" : "<base64Binary>"
    "valueBoolean" : <boolean>
    "valueCanonical" : "<canonical>"
    "valueCode" : "<code>"
    "valueDate" : "<date>"
```

```
    "valueDateTime" : "<dateTime>"
    "valueDecimal" : <decimal>
    "valueId" : "<id>"
    "valueInstant" : "<instant>"
    "valueInteger" : <integer>
    "valueMarkdown" : "<markdown>"
    "valueOid" : "<oid>"
    "valuePositiveInt" : "<positiveInt>"
    "valueString" : "<string>"
    "valueTime" : "<time>"
    "valueUnsignedInt" : "<unsignedInt>"
    "valueUri" : "<uri>"
    "valueUrl" : "<url>"
    "valueUuid" : "<uuid>"
    "valueAddress" : { Address }
    "valueAge" : { Age }
    "valueAnnotation" : { Annotation }
    "valueAttachment" : { Attachment }
    "valueCodeableConcept" : { CodeableConcept }
    "valueCoding" : { Coding }
    "valueContactPoint" : { ContactPoint }
    "valueCount" : { Count }
    "valueDistance" : { Distance }
    "valueDuration" : { Duration }
    "valueHumanName" : { HumanName }
    "valueIdentifier" : { Identifier }
    "valueMoney" : { Money }
    "valuePeriod" : { Period }
    "valueQuantity" : { Quantity }
    "valueRange" : { Range }
    "valueRatio" : { Ratio }
    "valueReference" : { Reference }
    "valueSampledData" : { SampledData }
    "valueSignature" : { Signature }
    "valueTiming" : { Timing }
    "valueContactDetail" : { ContactDetail }
    "valueContributor" : { Contributor }
    "valueDataRequirement" : { DataRequirement }
    "valueExpression" : { Expression }
    "valueParameterDefinition" : { ParameterDefinition }
    "valueRelatedArtifact" : { RelatedArtifact }
    "valueTriggerDefinition" : { TriggerDefinition }
    "valueUsageContext" : { UsageContext }
    "valueDosage" : { Dosage }
    "valueMeta" : { Meta }
  }],
  "output" : [{ // Information produced as part of task
    "type" : { CodeableConcept }, // R!  Label for output
    // value[x]: Result of output. One of these 50:
    "valueBase64Binary" : "<base64Binary>"
    "valueBoolean" : <boolean>
    "valueCanonical" : "<canonical>"
    "valueCode" : "<code>"
    "valueDate" : "<date>"
    "valueDateTime" : "<dateTime>"
    "valueDecimal" : <decimal>
    "valueId" : "<id>"
    "valueInstant" : "<instant>"
    "valueInteger" : <integer>
    "valueMarkdown" : "<markdown>"
    "valueOid" : "<oid>"
    "valuePositiveInt" : "<positiveInt>"
    "valueString" : "<string>"
    "valueTime" : "<time>"
    "valueUnsignedInt" : "<unsignedInt>"
    "valueUri" : "<uri>"
    "valueUrl" : "<url>"
    "valueUuid" : "<uuid>"
    "valueAddress" : { Address }
    "valueAge" : { Age }
    "valueAnnotation" : { Annotation }
    "valueAttachment" : { Attachment }
    "valueCodeableConcept" : { CodeableConcept }
    "valueCoding" : { Coding }
    "valueContactPoint" : { ContactPoint }
    "valueCount" : { Count }
    "valueDistance" : { Distance }
    "valueDuration" : { Duration }
    "valueHumanName" : { HumanName }
    "valueIdentifier" : { Identifier }
    "valueMoney" : { Money }
    "valuePeriod" : { Period }
    "valueQuantity" : { Quantity }
    "valueRange" : { Range }
    "valueRatio" : { Ratio }
    "valueReference" : { Reference }
    "valueSampledData" : { SampledData }
    "valueSignature" : { Signature }
    "valueTiming" : { Timing }
```

```
    "valueContactDetail" : { ContactDetail }
    "valueContributor" : { Contributor }
    "valueDataRequirement" : { DataRequirement }
    "valueExpression" : { Expression }
    "valueParameterDefinition" : { ParameterDefinition }
    "valueRelatedArtifact" : { RelatedArtifact }
    "valueTriggerDefinition" : { TriggerDefinition }
    "valueUsageContext" : { UsageContext }
    "valueDosage" : { Dosage }
    "valueMeta" : { Meta }
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Task;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Task.identifier [ Identifier ], ... ; # 0..* Task Instance Identifier
  fhir:Task.instantiatesCanonical [ canonical(ActivityDefinition) ]; # 0..1 Formal definition of task
  fhir:Task.instantiatesUri [ uri ]; # 0..1 Formal definition of task
  fhir:Task.basedOn [ Reference(Any) ], ... ; # 0..* Request fulfilled by this task
  fhir:Task.groupIdentifier [ Identifier ]; # 0..1 Requisition or grouper id
  fhir:Task.partOf [ Reference(Task) ], ... ; # 0..* Composite task
  fhir:Task.status [ code ]; # 1..1 draft | requested | received | accepted | +
  fhir:Task.statusReason [ CodeableConcept ]; # 0..1 Reason for current status
  fhir:Task.businessStatus [ CodeableConcept ]; # 0..1 E.g. "Specimen collected", "IV prepped"
  fhir:Task.intent [ code ]; # 1..1 unknown | proposal | plan | order | original-order | reflex-order | filler-order |
    instance-order | option
  fhir:Task.priority [ code ]; # 0..1 routine | urgent | asap | stat
  fhir:Task.code [ CodeableConcept ]; # 0..1 Task Type
  fhir:Task.description [ string ]; # 0..1 Human-readable explanation of task
  fhir:Task.focus [ Reference(Any) ]; # 0..1 What task is acting on
  fhir:Task.for [ Reference(Any) ]; # 0..1 Beneficiary of the Task
  fhir:Task.encounter [ Reference(Encounter) ]; # 0..1 Healthcare event during which this task originated
  fhir:Task.executionPeriod [ Period ]; # 0..1 Start and end time of execution
  fhir:Task.authoredOn [ dateTime ]; # 0..1 Task Creation Date
  fhir:Task.lastModified [ dateTime ]; # 0..1 Task Last Modified Date
  fhir:Task.requester [ Reference(Device|Organization|Patient|Practitioner|PractitionerRole|RelatedPerson) ]; # 0..1 Who
     is asking for task to be done
  fhir:Task.performerType [ CodeableConcept ], ... ; # 0..* Requested performer
  fhir:Task.owner [ Reference(Practitioner|PractitionerRole|Organization|CareTeam|HealthcareService|Patient|
  Device|RelatedPerson) ]; # 0..1 Responsible individual
  fhir:Task.location [ Reference(Location) ]; # 0..1 Where task occurs
  fhir:Task.reasonCode [ CodeableConcept ]; # 0..1 Why task is needed
  fhir:Task.reasonReference [ Reference(Any) ]; # 0..1 Why task is needed
  fhir:Task.insurance [ Reference(Coverage|ClaimResponse) ], ... ; # 0..* Associated insurance coverage
  fhir:Task.note [ Annotation ], ... ; # 0..* Comments made about the task
  fhir:Task.relevantHistory [ Reference(Provenance) ], ... ; # 0..* Key events in history of the Task
  fhir:Task.restriction [ # 0..1 Constraints on fulfillment tasks
    fhir:Task.restriction.repetitions [ positiveInt ]; # 0..1 How many times to repeat
    fhir:Task.restriction.period [ Period ]; # 0..1 When fulfillment sought
    fhir:Task.restriction.recipient [ Reference(Patient|Practitioner|PractitionerRole|RelatedPerson|Group|Organization)
    ], ... ; # 0..* For whom is fulfillment sought?
  ];
  fhir:Task.input [ # 0..* Information used to perform task
    fhir:Task.input.type [ CodeableConcept ]; # 1..1 Label for the input
    # Task.input.value[x] : 1..1 Content to use in performing the task. One of these 50
      fhir:Task.input.valueBase64Binary [ base64Binary ]
      fhir:Task.input.valueBoolean [ boolean ]
      fhir:Task.input.valueCanonical [ canonical ]
      fhir:Task.input.valueCode [ code ]
      fhir:Task.input.valueDate [ date ]
      fhir:Task.input.valueDateTime [ dateTime ]
      fhir:Task.input.valueDecimal [ decimal ]
      fhir:Task.input.valueId [ id ]
      fhir:Task.input.valueInstant [ instant ]
      fhir:Task.input.valueInteger [ integer ]
      fhir:Task.input.valueMarkdown [ markdown ]
      fhir:Task.input.valueOid [ oid ]
      fhir:Task.input.valuePositiveInt [ positiveInt ]
      fhir:Task.input.valueString [ string ]
      fhir:Task.input.valueTime [ time ]
      fhir:Task.input.valueUnsignedInt [ unsignedInt ]
      fhir:Task.input.valueUri [ uri ]
      fhir:Task.input.valueUrl [ url ]
      fhir:Task.input.valueUuid [ uuid ]
      fhir:Task.input.valueAddress [ Address ]
      fhir:Task.input.valueAge [ Age ]
      fhir:Task.input.valueAnnotation [ Annotation ]
      fhir:Task.input.valueAttachment [ Attachment ]
      fhir:Task.input.valueCodeableConcept [ CodeableConcept ]
      fhir:Task.input.valueCoding [ Coding ]
      fhir:Task.input.valueContactPoint [ ContactPoint ]
      fhir:Task.input.valueCount [ Count ]
```

```
      fhir:Task.input.valueDistance [ Distance ]
      fhir:Task.input.valueDuration [ Duration ]
      fhir:Task.input.valueHumanName [ HumanName ]
      fhir:Task.input.valueIdentifier [ Identifier ]
      fhir:Task.input.valueMoney [ Money ]
      fhir:Task.input.valuePeriod [ Period ]
      fhir:Task.input.valueQuantity [ Quantity ]
      fhir:Task.input.valueRange [ Range ]
      fhir:Task.input.valueRatio [ Ratio ]
      fhir:Task.input.valueReference [ Reference ]
      fhir:Task.input.valueSampledData [ SampledData ]
      fhir:Task.input.valueSignature [ Signature ]
      fhir:Task.input.valueTiming [ Timing ]
      fhir:Task.input.valueContactDetail [ ContactDetail ]
      fhir:Task.input.valueContributor [ Contributor ]
      fhir:Task.input.valueDataRequirement [ DataRequirement ]
      fhir:Task.input.valueExpression [ Expression ]
      fhir:Task.input.valueParameterDefinition [ ParameterDefinition ]
      fhir:Task.input.valueRelatedArtifact [ RelatedArtifact ]
      fhir:Task.input.valueTriggerDefinition [ TriggerDefinition ]
      fhir:Task.input.valueUsageContext [ UsageContext ]
      fhir:Task.input.valueDosage [ Dosage ]
      fhir:Task.input.valueMeta [ Meta ]
  ], ...;
  fhir:Task.output [ # 0..* Information produced as part of task
    fhir:Task.output.type [ CodeableConcept ]; # 1..1 Label for output
    # Task.output.value[x] : 1..1 Result of output. One of these 50
      fhir:Task.output.valueBase64Binary [ base64Binary ]
      fhir:Task.output.valueBoolean [ boolean ]
      fhir:Task.output.valueCanonical [ canonical ]
      fhir:Task.output.valueCode [ code ]
      fhir:Task.output.valueDate [ date ]
      fhir:Task.output.valueDateTime [ dateTime ]
      fhir:Task.output.valueDecimal [ decimal ]
      fhir:Task.output.valueId [ id ]
      fhir:Task.output.valueInstant [ instant ]
      fhir:Task.output.valueInteger [ integer ]
      fhir:Task.output.valueMarkdown [ markdown ]
      fhir:Task.output.valueOid [ oid ]
      fhir:Task.output.valuePositiveInt [ positiveInt ]
      fhir:Task.output.valueString [ string ]
      fhir:Task.output.valueTime [ time ]
      fhir:Task.output.valueUnsignedInt [ unsignedInt ]
      fhir:Task.output.valueUri [ uri ]
      fhir:Task.output.valueUrl [ url ]
      fhir:Task.output.valueUuid [ uuid ]
      fhir:Task.output.valueAddress [ Address ]
      fhir:Task.output.valueAge [ Age ]
      fhir:Task.output.valueAnnotation [ Annotation ]
      fhir:Task.output.valueAttachment [ Attachment ]
      fhir:Task.output.valueCodeableConcept [ CodeableConcept ]
      fhir:Task.output.valueCoding [ Coding ]
      fhir:Task.output.valueContactPoint [ ContactPoint ]
      fhir:Task.output.valueCount [ Count ]
      fhir:Task.output.valueDistance [ Distance ]
      fhir:Task.output.valueDuration [ Duration ]
      fhir:Task.output.valueHumanName [ HumanName ]
      fhir:Task.output.valueIdentifier [ Identifier ]
      fhir:Task.output.valueMoney [ Money ]
      fhir:Task.output.valuePeriod [ Period ]
      fhir:Task.output.valueQuantity [ Quantity ]
      fhir:Task.output.valueRange [ Range ]
      fhir:Task.output.valueRatio [ Ratio ]
      fhir:Task.output.valueReference [ Reference ]
      fhir:Task.output.valueSampledData [ SampledData ]
      fhir:Task.output.valueSignature [ Signature ]
      fhir:Task.output.valueTiming [ Timing ]
      fhir:Task.output.valueContactDetail [ ContactDetail ]
      fhir:Task.output.valueContributor [ Contributor ]
      fhir:Task.output.valueDataRequirement [ DataRequirement ]
      fhir:Task.output.valueExpression [ Expression ]
      fhir:Task.output.valueParameterDefinition [ ParameterDefinition ]
      fhir:Task.output.valueRelatedArtifact [ RelatedArtifact ]
      fhir:Task.output.valueTriggerDefinition [ TriggerDefinition ]
      fhir:Task.output.valueUsageContext [ UsageContext ]
      fhir:Task.output.valueDosage [ Dosage ]
      fhir:Task.output.valueMeta [ Meta ]
  ], ...;
]
```

Changes since R3

Task

Task.instantiatesCanonical | Added Element

Task.instantiatesUri | Added Element

Task.status | Change value set from http://hl7.org/fhir/ValueSet/task-status to http://hl7.org/fhir/ValueSet/task-status|4.0.1 Now marked as Modifier

Task.intent | Change value set from http://hl7.org/fhir/ValueSet/request-intent to http://hl7.org/fhir/ValueSet/task-intent|4.0.1

Task.priority | Change value set from http://hl7.org/fhir/ValueSet/request-priority to http://hl7.org/fhir/ValueSet/request-priority|4.0.1
Task.encounter | Added Element
Task.requester | Type changed from BackboneElement to Reference(Device | Organization | Patient | Practitioner | PractitionerRole | RelatedPerson)
Task.owner | Type Reference: Added Target Types PractitionerRole, CareTeam, HealthcareService
Task.location | Added Element
Task.reasonCode | Added Element
Task.reasonReference | Added Element
Task.insurance | Added Element
Task.restriction.recipient | Type Reference: Added Target Type PractitionerRole
Task.input.value[x] | Add Types canonical, url, uuid, ContactDetail, Contributor, DataRequirement, Expression, ParameterDefinition, RelatedArtifact, TriggerDefinition, UsageContext, Dosage
Task.output.value[x] | Add Types canonical, url, uuid, ContactDetail, Contributor, DataRequirement, Expression, ParameterDefinition, RelatedArtifact, TriggerDefinition, UsageContext, Dosage
Task.definition[x] | deleted
Task.context | deleted
Task.requester.agent | deleted
Task.requester.onBehalfOf | deleted
Task.reason | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 6 tests that all execute ok. 3 fail round-trip testing and 1 r3 resources are invalid (0 errors).)

Structure

Name | Flags | Card. | Type | Description & Constraints
Task | ITU | | DomainResource | A task to be performed + Rule: Last modified date must be greater than or equal to authored-on date. Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | | 0..* | Identifier | Task Instance Identifier
instantiatesCanonical | Σ | 0..1 | canonical(ActivityDefinition) | Formal definition of task
instantiatesUri | Σ | 0..1 | uri | Formal definition of task
basedOn | Σ | 0..* | Reference(Any) | Request fulfilled by this task
groupIdentifier | Σ | 0..1 | Identifier | Requisition or grouper id
partOf | Σ | 0..* | Reference(Task) | Composite task
status | ?!Σ | 1..1 | code | draft | requested | received | accepted | + TaskStatus (Required)
statusReason | Σ | 0..1 | CodeableConcept | Reason for current status
businessStatus | Σ | 0..1 | CodeableConcept | E.g. "Specimen collected", "IV prepped"
intent | Σ | 1..1 | code | unknown | proposal | plan | order | original-order | reflex-order | filler-order | instance-order | option TaskIntent (Required)
priority | | 0..1 | code | routine | urgent | asap | stat Request priority (Required)
code | Σ | 0..1 | CodeableConcept | Task Type Task Codes (Example)
description | Σ | 0..1 | string | Human-readable explanation of task
focus | Σ | 0..1 | Reference(Any) | What task is acting on
for | Σ | 0..1 | Reference(Any) | Beneficiary of the Task
encounter | Σ | 0..1 | Reference(Encounter) | Healthcare event during which this task originated
executionPeriod | Σ | 0..1 | Period | Start and end time of execution
authoredOn | I | 0..1 | dateTime | Task Creation Date
lastModified | ΣI | 0..1 | dateTime | Task Last Modified Date
requester | Σ | 0..1 | Reference(Device | Organization | Patient | Practitioner | PractitionerRole | RelatedPerson) | Who is asking for task to be done
performerType | | 0..* | CodeableConcept | Requested performer Procedure Performer Role Codes (Preferred)
owner | Σ | 0..1 | Reference(Practitioner | PractitionerRole | Organization | CareTeam | HealthcareService | Patient | Device | RelatedPerson) | Responsible individual
location | Σ | 0..1 | Reference(Location) | Where task occurs
reasonCode | | 0..1 | CodeableConcept | Why task is needed
reasonReference | | 0..1 | Reference(Any) | Why task is needed
insurance | | 0..* | Reference(Coverage | ClaimResponse) | Associated insurance coverage
note | | 0..* | Annotation | Comments made about the task
relevantHistory | | 0..* | Reference(Provenance) | Key events in history of the Task
restriction | | 0..1 | BackboneElement | Constraints on fulfillment tasks
repetitions | | 0..1 | positiveInt | How many times to repeat
period | | 0..1 | Period | When fulfillment sought
recipient | | 0..* | Reference(Patient | Practitioner | PractitionerRole | RelatedPerson | Group | Organization) | For whom is fulfillment sought?
input | | 0..* | BackboneElement | Information used to perform task
type | | 1..1 | CodeableConcept | Label for the input
value[x] | | 1..1 | * | Content to use in performing the task
output | | 0..* | BackboneElement | Information produced as part of task
type | | 1..1 | CodeableConcept | Label for output
value[x] | | 1..1 | * | Result of output
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Task xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Task Instance Identifier --></identifier>
 <instantiatesCanonical><!-- 0..1 canonical(ActivityDefinition) Formal definition of task --></instantiatesCanonical>
 <instantiatesUri value="[uri]"/><!-- 0..1 Formal definition of task -->
 <basedOn><!-- 0..* Reference(Any) Request fulfilled by this task --></basedOn>
 <groupIdentifier><!-- 0..1 Identifier Requisition or grouper id --></groupIdentifier>
 <partOf><!-- 0..* Reference(Task) Composite task --></partOf>
 <status value="[code]"/><!-- 1..1 draft | requested | received | accepted | + -->
 <statusReason><!-- 0..1 CodeableConcept Reason for current status --></statusReason>
 <businessStatus><!-- 0..1 CodeableConcept E.g. "Specimen collected", "IV prepped" --></businessStatus>
 <intent value="[code]"/><!-- 1..1 unknown | proposal | plan | order | original-order | reflex-order | filler-order |
    instance-order | option -->
 <priority value="[code]"/><!-- 0..1 routine | urgent | asap | stat -->
 <code><!-- 0..1 CodeableConcept Task Type --></code>
 <description value="[string]"/><!-- 0..1 Human-readable explanation of task -->
 <focus><!-- 0..1 Reference(Any) What task is acting on --></focus>
 <for><!-- 0..1 Reference(Any) Beneficiary of the Task --></for>
 <encounter><!-- 0..1 Reference(Encounter) Healthcare event during which this task originated --></encounter>
 <executionPeriod><!-- 0..1 Period Start and end time of execution --></executionPeriod>
 <authoredOn value="[dateTime]"/><!--  0..1 Task Creation Date -->
 <lastModified value="[dateTime]"/><!--  0..1 Task Last Modified Date -->
 <requester><!-- 0..1 Reference(Device|Organization|Patient|Practitioner|
   PractitionerRole|RelatedPerson) Who is asking for task to be done --></requester>
 <performerType><!-- 0..* CodeableConcept Requested performer --></performerType>
 <owner><!-- 0..1 Reference(Practitioner|PractitionerRole|Organization|CareTeam|
   HealthcareService|Patient|Device|RelatedPerson) Responsible individual --></owner>
 <location><!-- 0..1 Reference(Location) Where task occurs --></location>
 <reasonCode><!-- 0..1 CodeableConcept Why task is needed --></reasonCode>
 <reasonReference><!-- 0..1 Reference(Any) Why task is needed --></reasonReference>
 <insurance><!-- 0..* Reference(Coverage|ClaimResponse) Associated insurance coverage --></insurance>
 <note><!-- 0..* Annotation Comments made about the task --></note>
 <relevantHistory><!-- 0..* Reference(Provenance) Key events in history of the Task --></relevantHistory>
 <restriction>  <!-- 0..1 Constraints on fulfillment tasks -->
  <repetitions value="[positiveInt]"/><!-- 0..1 How many times to repeat -->
  <period><!-- 0..1 Period When fulfillment sought --></period>
  <recipient><!-- 0..* Reference(Patient|Practitioner|PractitionerRole|
    RelatedPerson|Group|Organization) For whom is fulfillment sought? --></recipient>
 </restriction>
 <input>  <!-- 0..* Information used to perform task -->
  <type><!-- 1..1 CodeableConcept Label for the input --></type>
  <value[x]><!-- 1..1 * Content to use in performing the task --></value[x]>
 </input>
 <output>  <!-- 0..* Information produced as part of task -->
  <type><!-- 1..1 CodeableConcept Label for output --></type>
  <value[x]><!-- 1..1 * Result of output --></value[x]>
 </output>
</Task>
```

JSON Template

```
{
  "resourceType" : "Task",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Task Instance Identifier
  "instantiatesCanonical" : { canonical(ActivityDefinition) }, // Formal definition of task
  "instantiatesUri" : "<uri>", // Formal definition of task
  "basedOn" : [{ Reference(Any) }], // Request fulfilled by this task
  "groupIdentifier" : { Identifier }, // Requisition or grouper id
  "partOf" : [{ Reference(Task) }], // Composite task
  "status" : "<code>", // R!  draft | requested | received | accepted | +
  "statusReason" : { CodeableConcept }, // Reason for current status
  "businessStatus" : { CodeableConcept }, // E.g. "Specimen collected", "IV prepped"
  "intent" : "<code>", // R!  unknown | proposal | plan | order | original-order | reflex-order | filler-order |
    instance-order | option
  "priority" : "<code>", // routine | urgent | asap | stat
  "code" : { CodeableConcept }, // Task Type
  "description" : "<string>", // Human-readable explanation of task
  "focus" : { Reference(Any) }, // What task is acting on
  "for" : { Reference(Any) }, // Beneficiary of the Task
  "encounter" : { Reference(Encounter) }, // Healthcare event during which this task originated
  "executionPeriod" : { Period }, // Start and end time of execution
  "authoredOn" : "<dateTime>", // C? Task Creation Date
  "lastModified" : "<dateTime>", // C? Task Last Modified Date
  "requester" : { Reference(Device|Organization|Patient|Practitioner|
   PractitionerRole|RelatedPerson) }, // Who is asking for task to be done
  "performerType" : [{ CodeableConcept }], // Requested performer
  "owner" : { Reference(Practitioner|PractitionerRole|Organization|CareTeam|
   HealthcareService|Patient|Device|RelatedPerson) }, // Responsible individual
  "location" : { Reference(Location) }, // Where task occurs
  "reasonCode" : { CodeableConcept }, // Why task is needed
  "reasonReference" : { Reference(Any) }, // Why task is needed
  "insurance" : [{ Reference(Coverage|ClaimResponse) }], // Associated insurance coverage
  "note" : [{ Annotation }], // Comments made about the task
  "relevantHistory" : [{ Reference(Provenance) }], // Key events in history of the Task
```

```
"restriction" : { // Constraints on fulfillment tasks
  "repetitions" : "<positiveInt>", // How many times to repeat
  "period" : { Period }, // When fulfillment sought
  "recipient" : [{ Reference(Patient|Practitioner|PractitionerRole|
  RelatedPerson|Group|Organization) }] // For whom is fulfillment sought?
},
"input" : [{ // Information used to perform task
  "type" : { CodeableConcept }, // R!  Label for the input
  // value[x]: Content to use in performing the task. One of these 50:
  "valueBase64Binary" : "<base64Binary>"
  "valueBoolean" : <boolean>
  "valueCanonical" : "<canonical>"
  "valueCode" : "<code>"
  "valueDate" : "<date>"
  "valueDateTime" : "<dateTime>"
  "valueDecimal" : <decimal>
  "valueId" : "<id>"
  "valueInstant" : "<instant>"
  "valueInteger" : <integer>
  "valueMarkdown" : "<markdown>"
  "valueOid" : "<oid>"
  "valuePositiveInt" : "<positiveInt>"
  "valueString" : "<string>"
  "valueTime" : "<time>"
  "valueUnsignedInt" : "<unsignedInt>"
  "valueUri" : "<uri>"
  "valueUrl" : "<url>"
  "valueUuid" : "<uuid>"
  "valueAddress" : { Address }
  "valueAge" : { Age }
  "valueAnnotation" : { Annotation }
  "valueAttachment" : { Attachment }
  "valueCodeableConcept" : { CodeableConcept }
  "valueCoding" : { Coding }
  "valueContactPoint" : { ContactPoint }
  "valueCount" : { Count }
  "valueDistance" : { Distance }
  "valueDuration" : { Duration }
  "valueHumanName" : { HumanName }
  "valueIdentifier" : { Identifier }
  "valueMoney" : { Money }
  "valuePeriod" : { Period }
  "valueQuantity" : { Quantity }
  "valueRange" : { Range }
  "valueRatio" : { Ratio }
  "valueReference" : { Reference }
  "valueSampledData" : { SampledData }
  "valueSignature" : { Signature }
  "valueTiming" : { Timing }
  "valueContactDetail" : { ContactDetail }
  "valueContributor" : { Contributor }
  "valueDataRequirement" : { DataRequirement }
  "valueExpression" : { Expression }
  "valueParameterDefinition" : { ParameterDefinition }
  "valueRelatedArtifact" : { RelatedArtifact }
  "valueTriggerDefinition" : { TriggerDefinition }
  "valueUsageContext" : { UsageContext }
  "valueDosage" : { Dosage }
  "valueMeta" : { Meta }
}],
"output" : [{ // Information produced as part of task
  "type" : { CodeableConcept }, // R!  Label for output
  // value[x]: Result of output. One of these 50:
  "valueBase64Binary" : "<base64Binary>"
  "valueBoolean" : <boolean>
  "valueCanonical" : "<canonical>"
  "valueCode" : "<code>"
  "valueDate" : "<date>"
  "valueDateTime" : "<dateTime>"
  "valueDecimal" : <decimal>
  "valueId" : "<id>"
  "valueInstant" : "<instant>"
  "valueInteger" : <integer>
  "valueMarkdown" : "<markdown>"
  "valueOid" : "<oid>"
  "valuePositiveInt" : "<positiveInt>"
  "valueString" : "<string>"
  "valueTime" : "<time>"
  "valueUnsignedInt" : "<unsignedInt>"
  "valueUri" : "<uri>"
  "valueUrl" : "<url>"
  "valueUuid" : "<uuid>"
  "valueAddress" : { Address }
  "valueAge" : { Age }
  "valueAnnotation" : { Annotation }
  "valueAttachment" : { Attachment }
  "valueCodeableConcept" : { CodeableConcept }
  "valueCoding" : { Coding }
  "valueContactPoint" : { ContactPoint }
```

```
    "valueCount" : { Count }
    "valueDistance" : { Distance }
    "valueDuration" : { Duration }
    "valueHumanName" : { HumanName }
    "valueIdentifier" : { Identifier }
    "valueMoney" : { Money }
    "valuePeriod" : { Period }
    "valueQuantity" : { Quantity }
    "valueRange" : { Range }
    "valueRatio" : { Ratio }
    "valueReference" : { Reference }
    "valueSampledData" : { SampledData }
    "valueSignature" : { Signature }
    "valueTiming" : { Timing }
    "valueContactDetail" : { ContactDetail }
    "valueContributor" : { Contributor }
    "valueDataRequirement" : { DataRequirement }
    "valueExpression" : { Expression }
    "valueParameterDefinition" : { ParameterDefinition }
    "valueRelatedArtifact" : { RelatedArtifact }
    "valueTriggerDefinition" : { TriggerDefinition }
    "valueUsageContext" : { UsageContext }
    "valueDosage" : { Dosage }
    "valueMeta" : { Meta }
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Task;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Task.identifier [ Identifier ], ... ; # 0..* Task Instance Identifier
  fhir:Task.instantiatesCanonical [ canonical(ActivityDefinition) ]; # 0..1 Formal definition of task
  fhir:Task.instantiatesUri [ uri ]; # 0..1 Formal definition of task
  fhir:Task.basedOn [ Reference(Any) ], ... ; # 0..* Request fulfilled by this task
  fhir:Task.groupIdentifier [ Identifier ]; # 0..1 Requisition or grouper id
  fhir:Task.partOf [ Reference(Task) ], ... ; # 0..* Composite task
  fhir:Task.status [ code ]; # 1..1 draft | requested | received | accepted | +
  fhir:Task.statusReason [ CodeableConcept ]; # 0..1 Reason for current status
  fhir:Task.businessStatus [ CodeableConcept ]; # 0..1 E.g. "Specimen collected", "IV prepped"
  fhir:Task.intent [ code ]; # 1..1 unknown | proposal | plan | order | original-order | reflex-order | filler-order |
    instance-order | option
  fhir:Task.priority [ code ]; # 0..1 routine | urgent | asap | stat
  fhir:Task.code [ CodeableConcept ]; # 0..1 Task Type
  fhir:Task.description [ string ]; # 0..1 Human-readable explanation of task
  fhir:Task.focus [ Reference(Any) ]; # 0..1 What task is acting on
  fhir:Task.for [ Reference(Any) ]; # 0..1 Beneficiary of the Task
  fhir:Task.encounter [ Reference(Encounter) ]; # 0..1 Healthcare event during which this task originated
  fhir:Task.executionPeriod [ Period ]; # 0..1 Start and end time of execution
  fhir:Task.authoredOn [ dateTime ]; # 0..1 Task Creation Date
  fhir:Task.lastModified [ dateTime ]; # 0..1 Task Last Modified Date
  fhir:Task.requester [ Reference(Device|Organization|Patient|Practitioner|PractitionerRole|RelatedPerson) ]; # 0..1 Who
     is asking for task to be done
  fhir:Task.performerType [ CodeableConcept ], ... ; # 0..* Requested performer
  fhir:Task.owner [ Reference(Practitioner|PractitionerRole|Organization|CareTeam|HealthcareService|Patient|
  Device|RelatedPerson) ]; # 0..1 Responsible individual
  fhir:Task.location [ Reference(Location) ]; # 0..1 Where task occurs
  fhir:Task.reasonCode [ CodeableConcept ]; # 0..1 Why task is needed
  fhir:Task.reasonReference [ Reference(Any) ]; # 0..1 Why task is needed
  fhir:Task.insurance [ Reference(Coverage|ClaimResponse) ], ... ; # 0..* Associated insurance coverage
  fhir:Task.note [ Annotation ], ... ; # 0..* Comments made about the task
  fhir:Task.relevantHistory [ Reference(Provenance) ], ... ; # 0..* Key events in history of the Task
  fhir:Task.restriction [ # 0..1 Constraints on fulfillment tasks
    fhir:Task.restriction.repetitions [ positiveInt ]; # 0..1 How many times to repeat
    fhir:Task.restriction.period [ Period ]; # 0..1 When fulfillment sought
    fhir:Task.restriction.recipient [ Reference(Patient|Practitioner|PractitionerRole|RelatedPerson|Group|Organization)
    ], ... ; # 0..* For whom is fulfillment sought?
  ];
  fhir:Task.input [ # 0..* Information used to perform task
    fhir:Task.input.type [ CodeableConcept ]; # 1..1 Label for the input
    # Task.input.value[x] : 1..1 Content to use in performing the task. One of these 50
      fhir:Task.input.valueBase64Binary [ base64Binary ]
      fhir:Task.input.valueBoolean [ boolean ]
      fhir:Task.input.valueCanonical [ canonical ]
      fhir:Task.input.valueCode [ code ]
      fhir:Task.input.valueDate [ date ]
      fhir:Task.input.valueDateTime [ dateTime ]
      fhir:Task.input.valueDecimal [ decimal ]
      fhir:Task.input.valueId [ id ]
      fhir:Task.input.valueInstant [ instant ]
      fhir:Task.input.valueInteger [ integer ]
      fhir:Task.input.valueMarkdown [ markdown ]
      fhir:Task.input.valueOid [ oid ]
      fhir:Task.input.valuePositiveInt [ positiveInt ]
```

```
    fhir:Task.input.valueString [ string ]
    fhir:Task.input.valueTime [ time ]
    fhir:Task.input.valueUnsignedInt [ unsignedInt ]
    fhir:Task.input.valueUri [ uri ]
    fhir:Task.input.valueUrl [ url ]
    fhir:Task.input.valueUuid [ uuid ]
    fhir:Task.input.valueAddress [ Address ]
    fhir:Task.input.valueAge [ Age ]
    fhir:Task.input.valueAnnotation [ Annotation ]
    fhir:Task.input.valueAttachment [ Attachment ]
    fhir:Task.input.valueCodeableConcept [ CodeableConcept ]
    fhir:Task.input.valueCoding [ Coding ]
    fhir:Task.input.valueContactPoint [ ContactPoint ]
    fhir:Task.input.valueCount [ Count ]
    fhir:Task.input.valueDistance [ Distance ]
    fhir:Task.input.valueDuration [ Duration ]
    fhir:Task.input.valueHumanName [ HumanName ]
    fhir:Task.input.valueIdentifier [ Identifier ]
    fhir:Task.input.valueMoney [ Money ]
    fhir:Task.input.valuePeriod [ Period ]
    fhir:Task.input.valueQuantity [ Quantity ]
    fhir:Task.input.valueRange [ Range ]
    fhir:Task.input.valueRatio [ Ratio ]
    fhir:Task.input.valueReference [ Reference ]
    fhir:Task.input.valueSampledData [ SampledData ]
    fhir:Task.input.valueSignature [ Signature ]
    fhir:Task.input.valueTiming [ Timing ]
    fhir:Task.input.valueContactDetail [ ContactDetail ]
    fhir:Task.input.valueContributor [ Contributor ]
    fhir:Task.input.valueDataRequirement [ DataRequirement ]
    fhir:Task.input.valueExpression [ Expression ]
    fhir:Task.input.valueParameterDefinition [ ParameterDefinition ]
    fhir:Task.input.valueRelatedArtifact [ RelatedArtifact ]
    fhir:Task.input.valueTriggerDefinition [ TriggerDefinition ]
    fhir:Task.input.valueUsageContext [ UsageContext ]
    fhir:Task.input.valueDosage [ Dosage ]
    fhir:Task.input.valueMeta [ Meta ]
], ...;
fhir:Task.output [ # 0..* Information produced as part of task
  fhir:Task.output.type [ CodeableConcept ]; # 1..1 Label for output
  # Task.output.value[x] : 1..1 Result of output. One of these 50
    fhir:Task.output.valueBase64Binary [ base64Binary ]
    fhir:Task.output.valueBoolean [ boolean ]
    fhir:Task.output.valueCanonical [ canonical ]
    fhir:Task.output.valueCode [ code ]
    fhir:Task.output.valueDate [ date ]
    fhir:Task.output.valueDateTime [ dateTime ]
    fhir:Task.output.valueDecimal [ decimal ]
    fhir:Task.output.valueId [ id ]
    fhir:Task.output.valueInstant [ instant ]
    fhir:Task.output.valueInteger [ integer ]
    fhir:Task.output.valueMarkdown [ markdown ]
    fhir:Task.output.valueOid [ oid ]
    fhir:Task.output.valuePositiveInt [ positiveInt ]
    fhir:Task.output.valueString [ string ]
    fhir:Task.output.valueTime [ time ]
    fhir:Task.output.valueUnsignedInt [ unsignedInt ]
    fhir:Task.output.valueUri [ uri ]
    fhir:Task.output.valueUrl [ url ]
    fhir:Task.output.valueUuid [ uuid ]
    fhir:Task.output.valueAddress [ Address ]
    fhir:Task.output.valueAge [ Age ]
    fhir:Task.output.valueAnnotation [ Annotation ]
    fhir:Task.output.valueAttachment [ Attachment ]
    fhir:Task.output.valueCodeableConcept [ CodeableConcept ]
    fhir:Task.output.valueCoding [ Coding ]
    fhir:Task.output.valueContactPoint [ ContactPoint ]
    fhir:Task.output.valueCount [ Count ]
    fhir:Task.output.valueDistance [ Distance ]
    fhir:Task.output.valueDuration [ Duration ]
    fhir:Task.output.valueHumanName [ HumanName ]
    fhir:Task.output.valueIdentifier [ Identifier ]
    fhir:Task.output.valueMoney [ Money ]
    fhir:Task.output.valuePeriod [ Period ]
    fhir:Task.output.valueQuantity [ Quantity ]
    fhir:Task.output.valueRange [ Range ]
    fhir:Task.output.valueRatio [ Ratio ]
    fhir:Task.output.valueReference [ Reference ]
    fhir:Task.output.valueSampledData [ SampledData ]
    fhir:Task.output.valueSignature [ Signature ]
    fhir:Task.output.valueTiming [ Timing ]
    fhir:Task.output.valueContactDetail [ ContactDetail ]
    fhir:Task.output.valueContributor [ Contributor ]
    fhir:Task.output.valueDataRequirement [ DataRequirement ]
    fhir:Task.output.valueExpression [ Expression ]
    fhir:Task.output.valueParameterDefinition [ ParameterDefinition ]
    fhir:Task.output.valueRelatedArtifact [ RelatedArtifact ]
    fhir:Task.output.valueTriggerDefinition [ TriggerDefinition ]
    fhir:Task.output.valueUsageContext [ UsageContext ]
```

```
      fhir:Task.output.valueDosage [ Dosage ]
      fhir:Task.output.valueMeta [ Meta ]
  ], ...;
]
```

Changes since Release 3

Task
Task.instantiatesCanonical | Added Element
Task.instantiatesUri | Added Element
Task.status | Change value set from http://hl7.org/fhir/ValueSet/task-status to http://hl7.org/fhir/ValueSet/task-status|4.0.1 Now marked as Modifier
Task.intent | Change value set from http://hl7.org/fhir/ValueSet/request-intent to http://hl7.org/fhir/ValueSet/task-intent|4.0.1
Task.priority | Change value set from http://hl7.org/fhir/ValueSet/request-priority to http://hl7.org/fhir/ValueSet/request-priority|4.0.1
Task.encounter | Added Element
Task.requester | Type changed from BackboneElement to Reference(Device | Organization | Patient | Practitioner | PractitionerRole | RelatedPerson)
Task.owner | Type Reference: Added Target Types PractitionerRole, CareTeam, HealthcareService
Task.location | Added Element
Task.reasonCode | Added Element
Task.reasonReference | Added Element
Task.insurance | Added Element
Task.restriction.recipient | Type Reference: Added Target Type PractitionerRole
Task.input.value[x] | Add Types canonical, url, uuid, ContactDetail, Contributor, DataRequirement, Expression, ParameterDefinition, RelatedArtifact, TriggerDefinition, UsageContext, Dosage
Task.output.value[x] | Add Types canonical, url, uuid, ContactDetail, Contributor, DataRequirement, Expression, ParameterDefinition, RelatedArtifact, TriggerDefinition, UsageContext, Dosage
Task.definition[x] | deleted
Task.context | deleted
Task.requester.agent | deleted
Task.requester.onBehalfOf | deleted
Task.reason | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 6 tests that all execute ok. 3 fail round-trip testing and 1 r3 resources are invalid (0 errors).)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

12.1.4.1 Terminology Bindings

Path | Definition | Type | Reference
Task.status | The current status of the task. | Required | TaskStatus
Task.statusReason | Codes to identify the reason for current status. These will typically be specific to a particular workflow. | Unknown | No details provided yet
Task.businessStatus | The domain-specific business-contextual sub-state of the task. For example: "Blood drawn", "IV inserted", "Awaiting physician signature", etc. | Unknown | No details provided yet
Task.intent | Distinguishes whether the task is a proposal, plan or full order. | Required | TaskIntent
Task.priority | The task's priority. | Required | RequestPriority
Task.code | Codes to identify what the task involves. These will typically be specific to a particular workflow. | Example | TaskCode
Task.performerType | The type(s) of task performers allowed. | Preferred | ProcedurePerformerRoleCodes
Task.reasonCode | Indicates why the task is needed. E.g. Suspended because patient admitted to hospital. | Unknown | No details provided yet
Task.input.type | Codes to identify types of input parameters. These will typically be specific to a particular workflow. E.g. "Comparison source", "Applicable consent", "Concomitent Medications", etc. | Unknown | No details provided yet
Task.output.type | Codes to identify types of input parameters. These will typically be specific to a particular workflow. E.g. "Identified issues", "Preliminary results", "Filler order", "Final results", etc. | Unknown | No details provided yet

12.1.4.2 Constraints

id | Level | Location | Description | Expression
inv-1 | Rule | (base) | Last modified date must be greater than or equal to authored-on date. | lastModified.exists().not() or authoredOn.exists().not() or lastModified >= authoredOn

12.1.5 Task Titles

Tasks often have titles (eg "My Tasks", "Outstanding Tasks for Patient X") which can be presented in a list. The task title should go into the Task.code as a coded concept and/or text.

12.1.6 Task state machine

The following diagram reflects the "typical" state machine for Task. Note that not all states will be supported by all workflows and that some workflows may support additional transitions, including transitions from terminal states (e.g. back to "in-progress" from "failed" or "completed").

12.1.7 The Cancelled state

While the intention of a "cancelled" task is that all work authorized by the task should cease, this might not always be possible practice. It is possible that the originally requested action could still be completed and still attached to the Task but this would not change the status of the task. If the placer cancels a task, it signals they no longer care about the outcome of the task.

12.1.8 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
authored-on | date | Search by creation date | Task.authoredOn
based-on | reference | Search by requests this task is based on | Task.basedOn (Any)
business-status | token | Search by business status | Task.businessStatus
code | token | Search by task code | Task.code
encounter | reference | Search by encounter | Task.encounter (Encounter)
focus | reference | Search by task focus | Task.focus (Any)
group-identifier | token | Search by group identifier | Task.groupIdentifier
identifier | token | Search for a task instance by its business identifier | Task.identifier
intent | token | Search by task intent | Task.intent
modified | date | Search by last modification date | Task.lastModified
owner | reference | Search by task owner | Task.owner (Practitioner, Organization, CareTeam, Device, Patient, HealthcareService, PractitionerRole, RelatedPerson)
part-of | reference | Search by task this task is part of | Task.partOf (Task)
patient | reference | Search by patient | Task.for.where(resolve() is Patient) (Patient)
performer | token | Search by recommended type of performer (e.g., Requester, Performer, Scheduler). | Task.performerType
period | date | Search by period Task is/was underway | Task.executionPeriod
priority | token | Search by task priority | Task.priority
requester | reference | Search by task requester | Task.requester (Practitioner, Organization, Device, Patient, PractitionerRole, RelatedPerson)
status | token | Search by task status | Task.status
subject | reference | Search by subject | Task.for (Any)

12.1.9 Resource Task - Examples

URL: https://hl7.org/fhir/R4/task-examples.html

12.1.9 Resource Task - Examples

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Not Classified | Compartments: Not linked to any defined compartments

Example Name | id | Format
Example of in-progress/awaiting specimen task for a lipid panel order | example1 | XML | JSON | Turtle
Example of an accepted, filler generated specimen collection subtask | example2 | XML | JSON | Turtle
Example of a completed filler generated specimen collection subtask | example4 | XML | JSON | Turtle
Example of in-progress/test results pending task for a lipid panel order | example5 | XML | JSON | Turtle
Example of completed task for a lipid panel order | example6 | XML | JSON | Turtle
Example of simple prescription task | example3 | XML | JSON | Turtle
Example of the cancellation of a preauthorization | fm-example1 | XML | JSON | Turtle
Example of polling for queued resources | fm-example2 | XML | JSON | Turtle
Example of the release of the balance of reserved funds for a preauthorization | fm-example3 | XML | JSON | Turtle
Example of the cancellation of a preauthorization | fm-example4 | XML | JSON | Turtle
Example of checking the processing status of a preauthorization | fm-example5 | XML | JSON | Turtle
Example of the response to checking the processing status of a preauthorization | fm-example6 | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

12.1.10 Resource Task - Detailed Descriptions

URL: https://hl7.org/fhir/R4/task-definitions.html

12.1.10 Resource Task - Detailed Descriptions

Orders and Observations Work Group | Maturity Level: 2 | Trial Use | Security Category: Not Classified | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the Task resource.

Task
Element Id | Task
Definition | A task to be performed.
Cardinality | 0..*
Type | DomainResource

Invariants | Defined on this element inv-1 Rule Last modified date must be greater than or equal to authored-on date. lastModified.exists().not() or authoredOn.exists().not() or lastModified >= authoredOn
Defined on this element
inv-1 | Rule | Last modified date must be greater than or equal to authored-on date. | lastModified.exists().not() or authoredOn.exists().not() or lastModified >= authoredOn
Task.identifier
Element Id | Task.identifier
Definition | The business identifier for this task.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Task.instantiatesCanonical
Element Id | Task.instantiatesCanonical
Definition | The URL pointing to a FHIR-defined protocol, guideline, orderset or other definition that is adhered to in whole or in part by this Task.
Cardinality | 0..1
Type | canonical(ActivityDefinition)
Requirements | Enables a formal definition of how he task is to be performed, enabling automation.
Summary | true
Task.instantiatesUri
Element Id | Task.instantiatesUri
Definition | The URL pointing to an externally maintained protocol, guideline, orderset or other definition that is adhered to in whole or in part by this Task.
Cardinality | 0..1
Type | uri
Requirements | Enables a formal definition of how he task is to be performed (e.g. using BPMN, BPEL, XPDL or other formal notation to be associated with a task), enabling automation.
Summary | true
Task.basedOn
Element Id | Task.basedOn
Definition | BasedOn refers to a higher-level authorization that triggered the creation of the task. It references a "request" resource such as a ServiceRequest, MedicationRequest, ServiceRequest, CarePlan, etc. which is distinct from the "request" resource the task is seeking to fulfill. This latter resource is referenced by FocusOn. For example, based on a ServiceRequest (= BasedOn), a task is created to fulfill a procedureRequest (= FocusOn) to collect a specimen from a patient.
Cardinality | 0..*
Type | Reference(Any)
Summary | true
Task.groupIdentifier
Element Id | Task.groupIdentifier
Definition | An identifier that links together multiple tasks and other requests that were created in the same context.
Cardinality | 0..1
Type | Identifier
Requirements | Billing and/or reporting can be linked to whether multiple requests were created as a single unit.
Summary | true
Task.partOf
Element Id | Task.partOf
Definition | Task that this particular task is part of.
Cardinality | 0..*
Type | Reference(Task)
Requirements | Allows tasks to be broken down into sub-steps (and this division can occur independent of the original task).
Summary | true
Comments | This should usually be 0..1.
Task.status
Element Id | Task.status
Definition | The current status of the task.
Cardinality | 1..1
Terminology Binding | TaskStatus (Required)
Type | code
Is Modifier | true (Reason: This element is labeled as a modifier because it is a status element that contains status entered-in-error which means that the resource should not be treated as valid)
Requirements | These states enable coordination of task status with off-the-shelf workflow solutions that support automation of tasks.
Summary | true
Task.statusReason
Element Id | Task.statusReason
Definition | An explanation as to why this task is held, failed, was refused, etc.
Cardinality | 0..1
Terminology Binding | TaskStatusReason:
Type | CodeableConcept
Summary | true
Comments | This applies to the current status. Look at the history of the task to see reasons for past statuses.
Task.businessStatus
Element Id | Task.businessStatus

Definition | Contains business-specific nuances of the business state.
Cardinality | 0..1
Terminology Binding | TaskBusinessStatus:
Type | CodeableConcept
Requirements | There's often a need to track substates of a task - this is often variable by specific workflow implementation.
Summary | true
Task.intent
Element Id | Task.intent
Definition | Indicates the "level" of actionability associated with the Task, i.e. i+R[9]Cs this a proposed task, a planned task, an actionable task, etc.
Cardinality | 1..1
Terminology Binding | TaskIntent (Required)
Type | code
Summary | true
Comments | This element is immutable. Proposed tasks, planned tasks, etc. must be distinct instances. In most cases, Tasks will have an intent of "order".
Task.priority
Element Id | Task.priority
Definition | Indicates how quickly the Task should be addressed with respect to other requests.
Cardinality | 0..1
Terminology Binding | Request priority (Required)
Type | code
Meaning if Missing | If missing, this task should be performed with normal priority
Requirements | Used to identify the service level expected while performing a task.
Task.code
Element Id | Task.code
Definition | A name or code (or both) briefly describing what the task involves.
Cardinality | 0..1
Terminology Binding | Task Codes (Example)
Type | CodeableConcept
Summary | true
Comments | The title (eg "My Tasks", "Outstanding Tasks for Patient X") should go into the code.
Task.description
Element Id | Task.description
Definition | A free-text description of what is to be performed.
Cardinality | 0..1
Type | string
Summary | true
Task.focus
Element Id | Task.focus
Definition | The request being actioned or the resource being manipulated by this task.
Cardinality | 0..1
Type | Reference(Any)
Requirements | Used to identify the thing to be done.
Summary | true
Comments | If multiple resources need to be manipulated, use sub-tasks. (This ensures that status can be tracked independently for each referenced resource.).
Task.for
Element Id | Task.for
Definition | The entity who benefits from the performance of the service specified in the task (e.g., the patient).
Cardinality | 0..1
Type | Reference(Any)
Requirements | Used to track tasks outstanding for a beneficiary. Do not use to track the task owner or creator (see owner and creator respectively). This can also affect access control.
Alternate Names | Patient
Summary | true
Task.encounter
Element Id | Task.encounter
Definition | The healthcare event (e.g. a patient and healthcare provider interaction) during which this task was created.
Cardinality | 0..1
Type | Reference(Encounter)
Requirements | For some tasks it may be important to know the link between the encounter the task originated within.
Summary | true
Task.executionPeriod
Element Id | Task.executionPeriod
Definition | Identifies the time action was first taken against the task (start) and/or the time final action was taken against the task prior to marking it as completed (end).
Cardinality | 0..1
Type | Period
Summary | true
Task.authoredOn
Element Id | Task.authoredOn
Definition | The date and time this task was created.

Cardinality | 0..1
Type | dateTime
Requirements | Most often used along with lastUpdated to track duration of task to supporting monitoring and management.
Alternate Names | Created Date
Invariants | Affect this element inv-1 Rule Last modified date must be greater than or equal to authored-on date. lastModified.exists().not() or authoredOn.exists().not() or lastModified >= authoredOn
Affect this element
inv-1 | Rule | Last modified date must be greater than or equal to authored-on date. | lastModified.exists().not() or authoredOn.exists().not() or lastModified >= authoredOn
Task.lastModified
Element Id | Task.lastModified
Definition | The date and time of last modification to this task.
Cardinality | 0..1
Type | dateTime
Requirements | Used along with history to track task activity and time in a particular task state. This enables monitoring and management.
Alternate Names | Update Date
Summary | true
Invariants | Affect this element inv-1 Rule Last modified date must be greater than or equal to authored-on date. lastModified.exists().not() or authoredOn.exists().not() or lastModified >= authoredOn
Affect this element
inv-1 | Rule | Last modified date must be greater than or equal to authored-on date. | lastModified.exists().not() or authoredOn.exists().not() or lastModified >= authoredOn
Task.requester
Element Id | Task.requester
Definition | The creator of the task.
Cardinality | 0..1
Type | Reference(Device | Organization | Patient | Practitioner | PractitionerRole | RelatedPerson)
Requirements | Identifies who created this task. May be used by access control mechanisms (e.g., to ensure that only the creator can cancel a task).
Summary | true
Task.performerType
Element Id | Task.performerType
Definition | The kind of participant that should perform the task.
Cardinality | 0..*
Terminology Binding | Procedure Performer Role Codes (Preferred)
Type | CodeableConcept
Requirements | Use to distinguish tasks on different activity queues.
Task.owner
Element Id | Task.owner
Definition | Individual organization or Device currently responsible for task execution.
Cardinality | 0..1
Type | Reference(Practitioner | PractitionerRole | Organization | CareTeam | HealthcareService | Patient | Device | RelatedPerson)
Requirements | Identifies who is expected to perform this task.
Alternate Names | Performer; Executer
Summary | true
Comments | Tasks may be created with an owner not yet identified.
Task.location
Element Id | Task.location
Definition | Principal physical location where the this task is performed.
Cardinality | 0..1
Type | Reference(Location)
Requirements | Ties the event to where the records are likely kept and provides context around the event occurrence (e.g. if it occurred inside or outside a dedicated healthcare setting).
Summary | true
Task.reasonCode
Element Id | Task.reasonCode
Definition | A description or code indicating why this task needs to be performed.
Cardinality | 0..1
Terminology Binding | TaskReason:
Type | CodeableConcept
Comments | This should only be included if there is no focus or if it differs from the reason indicated on the focus.
Task.reasonReference
Element Id | Task.reasonReference
Definition | A resource reference indicating why this task needs to be performed.
Cardinality | 0..1
Type | Reference(Any)
Comments | Tasks might be justified based on an Observation, a Condition, a past or planned procedure, etc. This should only be included if there is no focus or if it differs from the reason indicated on the focus. Use the CodeableConcept text element in Task.reasonCode if the data is free (uncoded) text.
Task.insurance
Element Id | Task.insurance
Definition | Insurance plans, coverage extensions, pre-authorizations and/or pre-determinations that may be relevant to the Task.

Cardinality | 0..*
Type | Reference(Coverage | ClaimResponse)
Task.note
Element Id | Task.note
Definition | Free-text information captured about the task as it progresses.
Cardinality | 0..*
Type | Annotation
Task.relevantHistory
Element Id | Task.relevantHistory
Definition | Links to Provenance records for past versions of this Task that identify key state transitions or updates that are likely to be relevant to a user looking at the current version of the task.
Cardinality | 0..*
Type | Reference(Provenance)
Alternate Names | Status History
Comments | This element does not point to the Provenance associated with the current version of the resource - as it would be created after this version existed. The Provenance for the current version can be retrieved with a _revinclude.
Task.restriction
Element Id | Task.restriction
Definition | If the Task.focus is a request resource and the task is seeking fulfillment (i.e. is asking for the request to be actioned), this element identifies any limitations on what parts of the referenced request should be actioned.
Cardinality | 0..1
Requirements | Sometimes when fulfillment is sought, you don't want full fulfillment.
Task.restriction.repetitions
Element Id | Task.restriction.repetitions
Definition | Indicates the number of times the requested action should occur.
Cardinality | 0..1
Type | positiveInt
Requirements | E.g. order that requests monthly lab tests, fulfillment is sought for 1.
Task.restriction.period
Element Id | Task.restriction.period
Definition | Over what time-period is fulfillment sought.
Cardinality | 0..1
Type | Period
Requirements | E.g. order that authorizes 1 year's services. Fulfillment is sought for next 3 months.
Comments | Note that period.high is the due date representing the time by which the task should be completed.
Task.restriction.recipient
Element Id | Task.restriction.recipient
Definition | For requests that are targeted to more than on potential recipient/target, for whom is fulfillment sought?
Cardinality | 0..*
Type | Reference(Patient | Practitioner | PractitionerRole | RelatedPerson | Group | Organization)
Task.input
Element Id | Task.input
Definition | Additional information that may be needed in the execution of the task.
Cardinality | 0..*
Requirements | Resources and data used to perform the task. This data is used in the business logic of task execution, and is stored separately because it varies between workflows.
Alternate Names | Supporting Information
Task.input.type
Element Id | Task.input.type
Definition | A code or description indicating how the input is intended to be used as part of the task execution.
Cardinality | 1..1
Terminology Binding | TaskInputParameterType:
Type | CodeableConcept
Requirements | Inputs are named to enable task automation to bind data and pass it from one task to the next.
Alternate Names | Name
Comments | If referencing a BPMN workflow or Protocol, the "system" is the URL for the workflow definition and the code is the "name" of the required input.
Task.input.value[x]
Element Id | Task.input.value[x]
Definition | The value of the input parameter as a basic type.
Cardinality | 1..1
Type | *
[x] Note | See Choice of Data Types for further information about how to use [x]
Task.output
Element Id | Task.output
Definition | Outputs produced by the Task.
Cardinality | 0..*
Requirements | Resources and data produced during the execution the task. This data is generated by the business logic of task execution, and is stored separately because it varies between workflows.
Task.output.type
Element Id | Task.output.type
Definition | The name of the Output parameter.
Cardinality | 1..1

Terminology Binding | TaskOutputParameterType:
Type | CodeableConcept
Requirements | Outputs are named to enable task automation to bind data and pass it from one task to the next.
Alternate Names | Name
Task.output.value[x]
Element Id | Task.output.value[x]
Definition | The value of the Output parameter as a basic type.
Cardinality | 1..1
Type | *
[x] Note | See Choice of Data Types for further information about how to use [x]
Requirements | Task outputs can take any form.

12.1.11 Resource Task - Mappings

URL: https://hl7.org/fhir/R4/task-mappings.html

12.1.11 Resource Task - Mappings

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Not Classified | Compartments: Not linked to any defined compartments

Mappings:

FiveWs Pattern Mapping

Workflow Pattern

HL7 v2 Mapping

RIM Mapping

Mappings for the task resource (see Mappings to Other Standards for further information & status).

12.1.11.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

Task
identifier | FiveWs.identifier
status | FiveWs.status
intent | FiveWs.class
priority | FiveWs.grade
code | FiveWs.what[x]
focus | FiveWs.what[x]
for | FiveWs.subject[x]
encounter | FiveWs.context
executionPeriod | FiveWs.done[x]
authoredOn | FiveWs.recorded
requester | FiveWs.author
performerType | FiveWs.actor
owner | FiveWs.actor
location | FiveWs.where[x]
reasonCode | FiveWs.why[x]
reasonReference | FiveWs.why[x]

12.1.11.2 Workflow Pattern (http://hl7.org/fhir/workflow)

Task | Request, Event
identifier | Request.identifier, Event.identifier
instantiatesCanonical | Request.instantiatesCanonical, Event.instantiatesCanonical
instantiatesUri | Event.instantiatesUrl
basedOn | Request.basedOn, Event.basedOn
groupIdentifier | Request.groupIdentifier
partOf | Event.partOf
status | Request.status, Event.status
intent | Request.intent
priority | Request.priority
code | Request.code, Event.code
for | Request.subject, Event.subject
encounter | Request.context, Event.context
executionPeriod | Event.occurrence[x]
authoredOn | Request.authoredOn
requester | Request.requester
performerType | Event.performer.role, Request.performerType
owner | Event.performer.actor, Request.performer
location | Request.reasonCode, Event.reasonCode
reasonCode | Event.location
reasonReference | Request.reasonReference, Event.reasonReference
insurance | Request.insurance
note | Request.note, Event.note

relevantHistory | Request.relevantHistory

12.1.11.3 HL7 v2 Mapping (http://hl7.org/v2)

Task
identifier
instantiatesCanonical
instantiatesUri
basedOn
groupIdentifier
partOf
status
statusReason
businessStatus
intent
priority
code
description
focus
for
encounter
executionPeriod
authoredOn
lastModified
requester
performerType
owner
location
reasonCode | EVN.7
reasonReference
insurance | IN1/IN2
note
relevantHistory
restriction
repetitions
period
recipient
input
type
value[x]
output
type
value[x]

12.1.11.4 RIM Mapping (http://hl7.org/v3)

Task | ControlAct[moodCode=INT]
identifier | .id
instantiatesCanonical | .outboundRelationship[typeCode=DEFN].target
instantiatesUri | .outboundRelationship[typeCode=DEFN].target
basedOn | .outboundRelationship[typeCode=FLFS].target[moodCode=INT]
groupIdentifier | .inboundRelationship[typeCode=COMP].source[moodCode=INT].id
partOf | .inboundRelationship[typeCode=COMP].source[moodCode=INT]
status | .statusCode
statusReason | .inboundRelationship[typeCode=SUBJ].source[classCode=CACT, moodCode=EVN, code="status change"].reasonCode
businessStatus | .inboundRelationship[typeCode=SUBJ].source[classCode=OBS, moodCode=EVN, code="business status"]
intent | .moodCode
priority | .priorityCode
code | .code
description | .text
focus | .outboundRelationship[typeCode=SUBJ].target
for | .participation[typeCode=RCT].role
encounter | .inboundRelationship[typeCode=COMP].source[classCode=PCPR, moodCode=EVN]
executionPeriod | .effectiveTime
authoredOn | .participation[typeCode=AUT].time
lastModified | .inboundRelationship[typeCode=SUBJ,].source[classCode=CACT, moodCode=EVN].effectiveTime
requester | .participation[typeCode=AUT].role
performerType | .participation[typeCode=PRF].role.code
owner | .participation[typeCode=PRF].role
location | .participation[typeCode=LOC].role
reasonCode | .reasonCode
reasonReference | .outboundRelationship[typeCode=RSON].target
insurance | .outboundRelationship[typeCode=COVBY].target

note | .inboundRelationship[typeCode=SUBJ,].source[classCode=OBS, moodCode=EVN, code="annotation"].value(string)
relevantHistory | .inboundRelationship(typeCode=SUBJ].source[classCode=CACT, moodCode=EVN]
restriction | Instead of pointing to request, would point to component of request, having these characteristics
repetitions | .repeatNumber
period | .effectiveTime(IVL<TS>)
recipient | .participation[typeCode=SBJ].role
input | ???
type | ???
value[x] | ???
output | ???
type | ???
value[x] | ???

12.1.12 Resource Task - Extensions & Profiles

URL: https://hl7.org/fhir/R4/task-profiles.html

12.1.12 Resource Task - Extensions & Profiles

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Not Classified | Compartments: Not linked to any defined compartments

This table lists profiles and extensions for the Task resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

12.1.12.1 Profiles

No Profiles defined for this resource

12.1.12.2 Extensions

event-eventHistory | Task | eventHistory | for Event Pattern HL7 Extensions
task-candidateList | Task | candidateList | for Task HL7 Extensions
task-replaces | Task | replaces | for Task HL7 Extensions
workflow-researchStudy | Task | researchStudy | for Workflow Pattern HL7 Extensions
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

task-replaces | replaces | for Task HL7 Extensions
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

12.1.12.3 Search Extensions

No Search Extensions defined for this resource

12.1.13 Resource Task - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/task-version-maps.html

No validation errors - all conversions are clean

12.1.13 Resource Task - R3/R4 Conversions

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Not Classified | Compartments: Not linked to any defined compartments

R3 : R4 Conversion maps for Task.

Functional status for this map: 6 tests that all execute ok. 3 fail round-trip testing and 1 r3 resources are invalid (0 errors). (see documentation)

12.1.13.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/Task3to4" = "R3 to R4 Conversions for Task"

uses "http://hl7.org/fhir/3.0/StructureDefinition/Task" alias TaskR3 as source
uses "http://hl7.org/fhir/StructureDefinition/Task" alias Task as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group Task(source src : TaskR3, target tgt : Task) extends DomainResource <<type+>> {
```

```
  src.identifier -> tgt.identifier;
  src.definition : uri as vs -> tgt.instantiatesUri = vs;
  src.definition : Reference as vs -> tgt.instantiatesCanonical = vs;
  src.basedOn -> tgt.basedOn;
  src.groupIdentifier -> tgt.groupIdentifier;
  src.partOf -> tgt.partOf;
  src.status -> tgt.status;
  src.statusReason -> tgt.statusReason;
  src.businessStatus -> tgt.businessStatus;
  src.intent -> tgt.intent;
  src.priority -> tgt.priority;
  src.code -> tgt.code;
  src.description -> tgt.description;
  src.focus -> tgt.focus;
  src.for -> tgt.for;
  src.context -> tgt.encounter;
  src.executionPeriod -> tgt.executionPeriod;
  src.authoredOn -> tgt.authoredOn;
  src.lastModified -> tgt.lastModified;
  src.requester as vs then {
    vs.agent -> tgt.requester;
  };
  src.performerType -> tgt.performerType;
  src.owner -> tgt.owner;
  src.location -> tgt.location;
  src.reason -> tgt.reasonCode;
  src.note -> tgt.note;
  src.relevantHistory -> tgt.relevantHistory;
  src.restriction as s -> tgt.restriction as t then TaskRestriction(s, t);
  src.input as s -> tgt.input as t then TaskInput(s, t);
  src.output as s -> tgt.output as t then TaskOutput(s, t);
}

group TaskRestriction(source src, target tgt) extends BackboneElement {
  src.repetitions -> tgt.repetitions;
  src.period -> tgt.period;
  src.recipient -> tgt.recipient;
}

group TaskInput(source src, target tgt) extends BackboneElement {
  src.type -> tgt.type;
  src.value -> tgt.value;
}

group TaskOutput(source src, target tgt) extends BackboneElement {
  src.type -> tgt.type;
  src.value -> tgt.value;
}
```

12.1.13.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/Task4to3" = "R4 to R3 Conversion for Task"

uses "http://hl7.org/fhir/StructureDefinition/Task" alias Task as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/Task" alias TaskR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group Task(source src : TaskR3, target tgt : Task) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.instantiatesUri -> tgt.definition;
  src.instantiatesCanonical -> tgt.definition;
  src.basedOn -> tgt.basedOn;
  src.groupIdentifier -> tgt.groupIdentifier;
  src.partOf -> tgt.partOf;
  src.status -> tgt.status;
  src.statusReason -> tgt.statusReason;
  src.businessStatus -> tgt.businessStatus;
  src.intent -> tgt.intent;
  src.priority -> tgt.priority;
  src.code -> tgt.code;
  src.description -> tgt.description;
  src.focus -> tgt.focus;
  src.for -> tgt.for;
  src.encounter -> tgt.context;
  src.executionPeriod -> tgt.executionPeriod;
  src.authoredOn -> tgt.authoredOn;
  src.lastModified -> tgt.lastModified;
  src.requester as vs ->  tgt.requester as vt,  vt.agent = vs;
  src.performerType -> tgt.performerType;
  src.owner -> tgt.owner;
  src.location -> tgt.location;
  src.reasonCode -> tgt.reason;
  src.note -> tgt.note;
  src.relevantHistory -> tgt.relevantHistory;
  src.restriction as s -> tgt.restriction as t then TaskRestriction(s, t);
  src.input as s -> tgt.input as t then TaskInput(s, t);
  src.output as s -> tgt.output as t then TaskOutput(s, t);
}
```

```
group TaskRestriction(source src, target tgt) extends BackboneElement {
  src.repetitions -> tgt.repetitions;
  src.period -> tgt.period;
  src.recipient -> tgt.recipient;
}

group TaskInput(source src, target tgt) extends BackboneElement {
  src.type -> tgt.typeendgroup;
  src.value -> tgt.value;
}

group TaskOutput(source src, target tgt) extends BackboneElement {
  src.type -> tgt.type;
  src.value -> tgt.value;
}
```

12.1.13.3 R4 Validation Errors

12.1.14 Task HL7 Extensions

URL: https://hl7.org/fhir/R4/task-extensions.html

Defines common extensions used with or related to the Task resource

12.1.14 Task HL7 Extensions

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative

12.1.14.1 Content

Extensions:

task-candidateList | candidateList : Identifies the individuals who are candidates for being the owner of the task.

task-replaces | replaces : Completed or terminated task(s) whose function is taken by this new task.

12.1.15 JSON Schema for Task

URL: https://hl7.org/fhir/R4/task.schema.json.html

12.1.15 JSON Schema for Task

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Not linked to any defined compartments

This is a representation of the json schema for Task, which is just a part of the full JSON Schema.

```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/Task",
  "$ref": "#/definitions/Task",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "Task": {
      "description": "A task to be performed.",
      "properties": {
        "resourceType": {
          "description": "This is a Task resource",
          "const": "Task"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
          "$ref": "#/definitions/code"
        },
        "_language": {
          "description": "Extensions for language",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "text": {
```

```
      "description": "A human-readable narrative that contains a summary of the resource and can be used to
represent the content of the resource to a human. The narrative need not encode all the structured data, but is
required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
      "$ref": "Narrative.schema.json#/definitions/Narrative"
    },
    "contained": {
      "description": "These resources do not have an independent existence apart from the resource that contains
them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
      "items": {
        "$ref": "ResourceList.schema.json#/definitions/ResourceList"
      },
      "type": "array"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
 definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource and that modifies the understanding of the element that contains it and/or the understanding of the
containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
met as part of the definition of the extension. Applications processing a resource are required to check for
modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "identifier": {
      "description": "The business identifier for this task.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "instantiatesCanonical": {
      "description": "The URL pointing to a *FHIR*-defined protocol, guideline, orderset or other definition that is
 adhered to in whole or in part by this Task.",
      "$ref": "canonical.schema.json#/definitions/canonical"
    },
    "instantiatesUri": {
      "description": "The URL pointing to an *externally* maintained  protocol, guideline, orderset or other
definition that is adhered to in whole or in part by this Task.",
      "$ref": "#/definitions/uri"
    },
    "_instantiatesUri": {
      "description": "Extensions for instantiatesUri",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "basedOn": {
      "description": "BasedOn refers to a higher-level authorization that triggered the creation of the task.  It
references a \"request\" resource such as a ServiceRequest, MedicationRequest, ServiceRequest, CarePlan, etc. which
is distinct from the \"request\" resource the task is seeking to fulfill.  This latter resource is referenced by
FocusOn.  For example, based on a ServiceRequest (\u003d BasedOn), a task is created to fulfill a procedureRequest (
 \u003d FocusOn ) to collect a specimen from a patient.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "groupIdentifier": {
      "description": "An identifier that links together multiple tasks and other requests that were created in the
same context.",
      "$ref": "Identifier.schema.json#/definitions/Identifier"
    },
    "partOf": {
      "description": "Task that this particular task is part of.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "status": {
      "description": "The current status of the task.",
      "enum": [
        "draft",
        "requested",
        "received",
        "accepted",
```

```
        "rejected",
        "ready",
        "cancelled",
        "in-progress",
        "on-hold",
        "failed",
        "completed",
        "entered-in-error"
      ]
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "statusReason": {
      "description": "An explanation as to why this task is held, failed, was refused, etc.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "businessStatus": {
      "description": "Contains business-specific nuances of the business state.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "intent": {
      "description": "Indicates the \"level\" of actionability associated with the Task, i.e. i+R[9]Cs this a
proposed task, a planned task, an actionable task, etc.",
      "enum": [
        "unknown",
        "proposal",
        "plan",
        "order",
        "original-order",
        "reflex-order",
        "filler-order",
        "instance-order",
        "option"
      ]
    },
    "_intent": {
      "description": "Extensions for intent",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "priority": {
      "description": "Indicates how quickly the Task should be addressed with respect to other requests.",
      "$ref": "#/definitions/code"
    },
    "_priority": {
      "description": "Extensions for priority",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "code": {
      "description": "A name or code (or both) briefly describing what the task involves.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "description": {
      "description": "A free-text description of what is to be performed.",
      "$ref": "#/definitions/string"
    },
    "_description": {
      "description": "Extensions for description",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "focus": {
      "description": "The request being actioned or the resource being manipulated by this task.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "for": {
      "description": "The entity who benefits from the performance of the service specified in the task (e.g., the
patient).",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "encounter": {
      "description": "The healthcare event  (e.g. a patient and healthcare provider interaction) during which this
task was created.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "executionPeriod": {
      "description": "Identifies the time action was first taken against the task (start) and/or the time final
action was taken against the task prior to marking it as completed (end).",
      "$ref": "Period.schema.json#/definitions/Period"
    },
    "authoredOn": {
      "description": "The date and time this task was created.",
      "$ref": "#/definitions/dateTime"
    },
    "_authoredOn": {
      "description": "Extensions for authoredOn",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "lastModified": {
```

```
      "description": "The date and time of last modification to this task.",
      "$ref": "#/definitions/dateTime"
    },
    "_lastModified": {
      "description": "Extensions for lastModified",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "requester": {
      "description": "The creator of the task.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "performerType": {
      "description": "The kind of participant that should perform the task.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "owner": {
      "description": "Individual organization or Device currently responsible for task execution.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "location": {
      "description": "Principal physical location where the this task is performed.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "reasonCode": {
      "description": "A description or code indicating why this task needs to be performed.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "reasonReference": {
      "description": "A resource reference indicating why this task needs to be performed.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "insurance": {
      "description": "Insurance plans, coverage extensions, pre-authorizations and/or pre-determinations that may be
 relevant to the Task.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "note": {
      "description": "Free-text information captured about the task as it progresses.",
      "items": {
        "$ref": "Annotation.schema.json#/definitions/Annotation"
      },
      "type": "array"
    },
    "relevantHistory": {
      "description": "Links to Provenance records for past versions of this Task that identify key state transitions
 or updates that are likely to be relevant to a user looking at the current version of the task.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "restriction": {
      "description": "If the Task.focus is a request resource and the task is seeking fulfillment (i.e. is asking
for the request to be actioned), this element identifies any limitations on what parts of the referenced request
should be actioned.",
      "$ref": "#/definitions/Task_Restriction"
    },
    "input": {
      "description": "Additional information that may be needed in the execution of the task.",
      "items": {
        "$ref": "#/definitions/Task_Input"
      },
      "type": "array"
    },
    "output": {
      "description": "Outputs produced by the Task.",
      "items": {
        "$ref": "#/definitions/Task_Output"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "resourceType"
  ]
},
"Task_Restriction": {
  "description": "A task to be performed.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
```

```
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "repetitions": {
      "description": "Indicates the number of times the requested action should occur.",
      "$ref": "#/definitions/positiveInt"
    },
    "_repetitions": {
      "description": "Extensions for repetitions",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "period": {
      "description": "Over what time-period is fulfillment sought.",
      "$ref": "Period.schema.json#/definitions/Period"
    },
    "recipient": {
      "description": "For requests that are targeted to more than on potential recipient/target, for whom is
fulfillment sought?",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    }
  },
  "additionalProperties": false
},
"Task_Input": {
  "description": "A task to be performed.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "type": {
      "description": "A code or description indicating how the input is intended to be used as part of the task
execution.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "valueBase64Binary": {
      "description": "The value of the input parameter as a basic type.",
      "pattern": "^(\\s*([0-9a-zA-Z\\+/\u003d]){4}\\s*)+$",
```

```
      "type": "string"
    },
    "_valueBase64Binary": {
      "description": "Extensions for valueBase64Binary",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "valueBoolean": {
      "description": "The value of the input parameter as a basic type.",
      "pattern": "^true|false$",
      "type": "boolean"
    },
    "_valueBoolean": {
      "description": "Extensions for valueBoolean",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "valueCanonical": {
      "description": "The value of the input parameter as a basic type.",
      "pattern": "^\\S*$",
      "type": "string"
    },
    "_valueCanonical": {
      "description": "Extensions for valueCanonical",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "valueCode": {
      "description": "The value of the input parameter as a basic type.",
      "pattern": "^[^\\s]+(\\s[^\\s]+)*$",
      "type": "string"
    },
    "_valueCode": {
      "description": "Extensions for valueCode",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "valueDate": {
      "description": "The value of the input parameter as a basic type.",
      "pattern":
"^([0-9]([0-9]([0-9][1-9]|[1-9]0)|[1-9]00)|[1-9]000)(-(0[1-9]|1[0-2])(-(0[1-9]|[1-2][0-9]|3[0-1]))?)?$",
      "type": "string"
    },
    "_valueDate": {
      "description": "Extensions for valueDate",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "valueDateTime": {
      "description": "The value of the input parameter as a basic type.",
      "pattern": "^([0-9]([0-9]([0-9][1-9]|[1-9]0)|[1-9]00)|[1-9]000)(-(0[1-9]|1[0-2])(-(0[1-9]|[1-2][0-9]|3[0-
1])(T([01][0-9]|2[0-3]):[0-5][0-9]:([0-5][0-9]|60)(\\.[0-9]+)?(Z|(\\+|-)((0[0-9]|1[0-3]):[0-5][0-9]|14:00)))?)?)?$",
      "type": "string"
    },
    "_valueDateTime": {
      "description": "Extensions for valueDateTime",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "valueDecimal": {
      "description": "The value of the input parameter as a basic type.",
      "pattern": "^-?(0|[1-9][0-9]*)(\\.[0-9]+)?([eE][+-]?[0-9]+)?$",
      "type": "number"
    },
    "_valueDecimal": {
      "description": "Extensions for valueDecimal",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "valueId": {
      "description": "The value of the input parameter as a basic type.",
      "pattern": "^[A-Za-z0-9\\-\\.]{1,64}$",
      "type": "string"
    },
    "_valueId": {
      "description": "Extensions for valueId",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "valueInstant": {
      "description": "The value of the input parameter as a basic type.",
      "pattern": "^([0-9]([0-9]([0-9][1-9]|[1-9]0)|[1-9]00)|[1-9]000)-(0[1-9]|1[0-2])-(0[1-9]|[1-2][0-9]|3[0-
1])T([01][0-9]|2[0-3]):[0-5][0-9]:([0-5][0-9]|60)(\\.[0-9]+)?(Z|(\\+|-)((0[0-9]|1[0-3]):[0-5][0-9]|14:00))$",
      "type": "string"
    },
    "_valueInstant": {
      "description": "Extensions for valueInstant",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "valueInteger": {
      "description": "The value of the input parameter as a basic type.",
      "pattern": "^-?([0]|([1-9][0-9]*))$",
      "type": "number"
    },
    "_valueInteger": {
      "description": "Extensions for valueInteger",
      "$ref": "Element.schema.json#/definitions/Element"
```

```
},
"valueMarkdown": {
  "description": "The value of the input parameter as a basic type.",
  "pattern": "^[ \\r\\n\\t\\S]+$",
  "type": "string"
},
"_valueMarkdown": {
  "description": "Extensions for valueMarkdown",
  "$ref": "Element.schema.json#/definitions/Element"
},
"valueOid": {
  "description": "The value of the input parameter as a basic type.",
  "pattern": "^urn:oid:[0-2](\\.(0|[1-9][0-9]*))+$",
  "type": "string"
},
"_valueOid": {
  "description": "Extensions for valueOid",
  "$ref": "Element.schema.json#/definitions/Element"
},
"valuePositiveInt": {
  "description": "The value of the input parameter as a basic type.",
  "pattern": "^[1-9][0-9]*$",
  "type": "number"
},
"_valuePositiveInt": {
  "description": "Extensions for valuePositiveInt",
  "$ref": "Element.schema.json#/definitions/Element"
},
"valueString": {
  "description": "The value of the input parameter as a basic type.",
  "pattern": "^[ \\r\\n\\t\\S]+$",
  "type": "string"
},
"_valueString": {
  "description": "Extensions for valueString",
  "$ref": "Element.schema.json#/definitions/Element"
},
"valueTime": {
  "description": "The value of the input parameter as a basic type.",
  "pattern": "^([01][0-9]|2[0-3]):[0-5][0-9]:([0-5][0-9]|60)(\\.[0-9]+)?$",
  "type": "string"
},
"_valueTime": {
  "description": "Extensions for valueTime",
  "$ref": "Element.schema.json#/definitions/Element"
},
"valueUnsignedInt": {
  "description": "The value of the input parameter as a basic type.",
  "pattern": "^[0]|([1-9][0-9]*)$",
  "type": "number"
},
"_valueUnsignedInt": {
  "description": "Extensions for valueUnsignedInt",
  "$ref": "Element.schema.json#/definitions/Element"
},
"valueUri": {
  "description": "The value of the input parameter as a basic type.",
  "pattern": "^\\S*$",
  "type": "string"
},
"_valueUri": {
  "description": "Extensions for valueUri",
  "$ref": "Element.schema.json#/definitions/Element"
},
"valueUrl": {
  "description": "The value of the input parameter as a basic type.",
  "pattern": "^\\S*$",
  "type": "string"
},
"_valueUrl": {
  "description": "Extensions for valueUrl",
  "$ref": "Element.schema.json#/definitions/Element"
},
"valueUuid": {
  "description": "The value of the input parameter as a basic type.",
  "pattern": "^urn:uuid:[0-9a-f]{8}-[0-9a-f]{4}-[0-9a-f]{4}-[0-9a-f]{4}-[0-9a-f]{12}$",
  "type": "string"
},
"_valueUuid": {
  "description": "Extensions for valueUuid",
  "$ref": "Element.schema.json#/definitions/Element"
},
"valueAddress": {
  "description": "The value of the input parameter as a basic type.",
  "$ref": "Address.schema.json#/definitions/Address"
},
"valueAge": {
  "description": "The value of the input parameter as a basic type.",
  "$ref": "Age.schema.json#/definitions/Age"
```

```
},
"valueAnnotation": {
  "description": "The value of the input parameter as a basic type.",
  "$ref": "Annotation.schema.json#/definitions/Annotation"
},
"valueAttachment": {
  "description": "The value of the input parameter as a basic type.",
  "$ref": "Attachment.schema.json#/definitions/Attachment"
},
"valueCodeableConcept": {
  "description": "The value of the input parameter as a basic type.",
  "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
},
"valueCoding": {
  "description": "The value of the input parameter as a basic type.",
  "$ref": "Coding.schema.json#/definitions/Coding"
},
"valueContactPoint": {
  "description": "The value of the input parameter as a basic type.",
  "$ref": "ContactPoint.schema.json#/definitions/ContactPoint"
},
"valueCount": {
  "description": "The value of the input parameter as a basic type.",
  "$ref": "Count.schema.json#/definitions/Count"
},
"valueDistance": {
  "description": "The value of the input parameter as a basic type.",
  "$ref": "Distance.schema.json#/definitions/Distance"
},
"valueDuration": {
  "description": "The value of the input parameter as a basic type.",
  "$ref": "Duration.schema.json#/definitions/Duration"
},
"valueHumanName": {
  "description": "The value of the input parameter as a basic type.",
  "$ref": "HumanName.schema.json#/definitions/HumanName"
},
"valueIdentifier": {
  "description": "The value of the input parameter as a basic type.",
  "$ref": "Identifier.schema.json#/definitions/Identifier"
},
"valueMoney": {
  "description": "The value of the input parameter as a basic type.",
  "$ref": "Money.schema.json#/definitions/Money"
},
"valuePeriod": {
  "description": "The value of the input parameter as a basic type.",
  "$ref": "Period.schema.json#/definitions/Period"
},
"valueQuantity": {
  "description": "The value of the input parameter as a basic type.",
  "$ref": "Quantity.schema.json#/definitions/Quantity"
},
"valueRange": {
  "description": "The value of the input parameter as a basic type.",
  "$ref": "Range.schema.json#/definitions/Range"
},
"valueRatio": {
  "description": "The value of the input parameter as a basic type.",
  "$ref": "Ratio.schema.json#/definitions/Ratio"
},
"valueReference": {
  "description": "The value of the input parameter as a basic type.",
  "$ref": "Reference.schema.json#/definitions/Reference"
},
"valueSampledData": {
  "description": "The value of the input parameter as a basic type.",
  "$ref": "SampledData.schema.json#/definitions/SampledData"
},
"valueSignature": {
  "description": "The value of the input parameter as a basic type.",
  "$ref": "Signature.schema.json#/definitions/Signature"
},
"valueTiming": {
  "description": "The value of the input parameter as a basic type.",
  "$ref": "Timing.schema.json#/definitions/Timing"
},
"valueContactDetail": {
  "description": "The value of the input parameter as a basic type.",
  "$ref": "ContactDetail.schema.json#/definitions/ContactDetail"
},
"valueContributor": {
  "description": "The value of the input parameter as a basic type.",
  "$ref": "Contributor.schema.json#/definitions/Contributor"
},
"valueDataRequirement": {
  "description": "The value of the input parameter as a basic type.",
  "$ref": "DataRequirement.schema.json#/definitions/DataRequirement"
},
```

```
    "valueExpression": {
      "description": "The value of the input parameter as a basic type.",
      "$ref": "Expression.schema.json#/definitions/Expression"
    },
    "valueParameterDefinition": {
      "description": "The value of the input parameter as a basic type.",
      "$ref": "ParameterDefinition.schema.json#/definitions/ParameterDefinition"
    },
    "valueRelatedArtifact": {
      "description": "The value of the input parameter as a basic type.",
      "$ref": "RelatedArtifact.schema.json#/definitions/RelatedArtifact"
    },
    "valueTriggerDefinition": {
      "description": "The value of the input parameter as a basic type.",
      "$ref": "TriggerDefinition.schema.json#/definitions/TriggerDefinition"
    },
    "valueUsageContext": {
      "description": "The value of the input parameter as a basic type.",
      "$ref": "UsageContext.schema.json#/definitions/UsageContext"
    },
    "valueDosage": {
      "description": "The value of the input parameter as a basic type.",
      "$ref": "Dosage.schema.json#/definitions/Dosage"
    },
    "valueMeta": {
      "description": "The value of the input parameter as a basic type.",
      "$ref": "Meta.schema.json#/definitions/Meta"
    }
  },
  "additionalProperties": false,
  "required": [
    "type"
  ]
},
"Task_Output": {
  "description": "A task to be performed.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "type": {
      "description": "The name of the Output parameter.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "valueBase64Binary": {
      "description": "The value of the Output parameter as a basic type.",
      "pattern": "^(\\s*([0-9a-zA-Z\\+/\u003d]){4}\\s*)+$",
      "type": "string"
    },
    "_valueBase64Binary": {
      "description": "Extensions for valueBase64Binary",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "valueBoolean": {
      "description": "The value of the Output parameter as a basic type.",
      "pattern": "^true|false$",
      "type": "boolean"
    },
    "_valueBoolean": {
      "description": "Extensions for valueBoolean",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "valueCanonical": {
```

```
      "description": "The value of the Output parameter as a basic type.",
      "pattern": "^\\S*$",
      "type": "string"
    },
    "_valueCanonical": {
      "description": "Extensions for valueCanonical",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "valueCode": {
      "description": "The value of the Output parameter as a basic type.",
      "pattern": "^[^\\s]+(\\s[^\\s]+)*$",
      "type": "string"
    },
    "_valueCode": {
      "description": "Extensions for valueCode",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "valueDate": {
      "description": "The value of the Output parameter as a basic type.",
      "pattern":
"^([0-9]([0-9]([0-9][1-9]|[1-9]0)|[1-9]00)|[1-9]000)(-(0[1-9]|1[0-2])(-(0[1-9]|[1-2][0-9]|3[0-1]))?)?$",
      "type": "string"
    },
    "_valueDate": {
      "description": "Extensions for valueDate",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "valueDateTime": {
      "description": "The value of the Output parameter as a basic type.",
      "pattern": "^([0-9]([0-9]([0-9][1-9]|[1-9]0)|[1-9]00)|[1-9]000)(-(0[1-9]|1[0-2])(-(0[1-9]|[1-2][0-9]|3[0-
1])(T([01][0-9]|2[0-3]):[0-5][0-9]:([0-5][0-9]|60)(\\.[0-9]+)?(Z|(\\+|-)((0[0-9]|1[0-3]):[0-5][0-9]|14:00)))?)?)?$",
      "type": "string"
    },
    "_valueDateTime": {
      "description": "Extensions for valueDateTime",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "valueDecimal": {
      "description": "The value of the Output parameter as a basic type.",
      "pattern": "^-?(0|[1-9][0-9]*)(\\.[0-9]+)?([eE][+-]?[0-9]+)?$",
      "type": "number"
    },
    "_valueDecimal": {
      "description": "Extensions for valueDecimal",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "valueId": {
      "description": "The value of the Output parameter as a basic type.",
      "pattern": "^[A-Za-z0-9\\-\\.]{1,64}$",
      "type": "string"
    },
    "_valueId": {
      "description": "Extensions for valueId",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "valueInstant": {
      "description": "The value of the Output parameter as a basic type.",
      "pattern": "^([0-9]([0-9]([0-9][1-9]|[1-9]0)|[1-9]00)|[1-9]000)-(0[1-9]|1[0-2])-(0[1-9]|[1-2][0-9]|3[0-
1])T([01][0-9]|2[0-3]):[0-5][0-9]:([0-5][0-9]|60)(\\.[0-9]+)?(Z|(\\+|-)((0[0-9]|1[0-3]):[0-5][0-9]|14:00))$",
      "type": "string"
    },
    "_valueInstant": {
      "description": "Extensions for valueInstant",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "valueInteger": {
      "description": "The value of the Output parameter as a basic type.",
      "pattern": "^-?([0]|([1-9][0-9]*))$",
      "type": "number"
    },
    "_valueInteger": {
      "description": "Extensions for valueInteger",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "valueMarkdown": {
      "description": "The value of the Output parameter as a basic type.",
      "pattern": "^[ \\r\\n\\t\\S]+$",
      "type": "string"
    },
    "_valueMarkdown": {
      "description": "Extensions for valueMarkdown",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "valueOid": {
      "description": "The value of the Output parameter as a basic type.",
      "pattern": "^urn:oid:[0-2](\\.(0|[1-9][0-9]*))+$",
      "type": "string"
    },
    "_valueOid": {
```

```
  "description": "Extensions for valueOid",
  "$ref": "Element.schema.json#/definitions/Element"
},
"valuePositiveInt": {
  "description": "The value of the Output parameter as a basic type.",
  "pattern": "^[1-9][0-9]*$",
  "type": "number"
},
"_valuePositiveInt": {
  "description": "Extensions for valuePositiveInt",
  "$ref": "Element.schema.json#/definitions/Element"
},
"valueString": {
  "description": "The value of the Output parameter as a basic type.",
  "pattern": "^[ \\r\\n\\t\\S]+$",
  "type": "string"
},
"_valueString": {
  "description": "Extensions for valueString",
  "$ref": "Element.schema.json#/definitions/Element"
},
"valueTime": {
  "description": "The value of the Output parameter as a basic type.",
  "pattern": "^([01][0-9]|2[0-3]):[0-5][0-9]:([0-5][0-9]|60)(\\.[0-9]+)?$",
  "type": "string"
},
"_valueTime": {
  "description": "Extensions for valueTime",
  "$ref": "Element.schema.json#/definitions/Element"
},
"valueUnsignedInt": {
  "description": "The value of the Output parameter as a basic type.",
  "pattern": "^[0]|([1-9][0-9]*)$",
  "type": "number"
},
"_valueUnsignedInt": {
  "description": "Extensions for valueUnsignedInt",
  "$ref": "Element.schema.json#/definitions/Element"
},
"valueUri": {
  "description": "The value of the Output parameter as a basic type.",
  "pattern": "^\\S*$",
  "type": "string"
},
"_valueUri": {
  "description": "Extensions for valueUri",
  "$ref": "Element.schema.json#/definitions/Element"
},
"valueUrl": {
  "description": "The value of the Output parameter as a basic type.",
  "pattern": "^\\S*$",
  "type": "string"
},
"_valueUrl": {
  "description": "Extensions for valueUrl",
  "$ref": "Element.schema.json#/definitions/Element"
},
"valueUuid": {
  "description": "The value of the Output parameter as a basic type.",
  "pattern": "^urn:uuid:[0-9a-f]{8}-[0-9a-f]{4}-[0-9a-f]{4}-[0-9a-f]{4}-[0-9a-f]{12}$",
  "type": "string"
},
"_valueUuid": {
  "description": "Extensions for valueUuid",
  "$ref": "Element.schema.json#/definitions/Element"
},
"valueAddress": {
  "description": "The value of the Output parameter as a basic type.",
  "$ref": "Address.schema.json#/definitions/Address"
},
"valueAge": {
  "description": "The value of the Output parameter as a basic type.",
  "$ref": "Age.schema.json#/definitions/Age"
},
"valueAnnotation": {
  "description": "The value of the Output parameter as a basic type.",
  "$ref": "Annotation.schema.json#/definitions/Annotation"
},
"valueAttachment": {
  "description": "The value of the Output parameter as a basic type.",
  "$ref": "Attachment.schema.json#/definitions/Attachment"
},
"valueCodeableConcept": {
  "description": "The value of the Output parameter as a basic type.",
  "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
},
"valueCoding": {
  "description": "The value of the Output parameter as a basic type.",
  "$ref": "Coding.schema.json#/definitions/Coding"
```

```
},
"valueContactPoint": {
  "description": "The value of the Output parameter as a basic type.",
  "$ref": "ContactPoint.schema.json#/definitions/ContactPoint"
},
"valueCount": {
  "description": "The value of the Output parameter as a basic type.",
  "$ref": "Count.schema.json#/definitions/Count"
},
"valueDistance": {
  "description": "The value of the Output parameter as a basic type.",
  "$ref": "Distance.schema.json#/definitions/Distance"
},
"valueDuration": {
  "description": "The value of the Output parameter as a basic type.",
  "$ref": "Duration.schema.json#/definitions/Duration"
},
"valueHumanName": {
  "description": "The value of the Output parameter as a basic type.",
  "$ref": "HumanName.schema.json#/definitions/HumanName"
},
"valueIdentifier": {
  "description": "The value of the Output parameter as a basic type.",
  "$ref": "Identifier.schema.json#/definitions/Identifier"
},
"valueMoney": {
  "description": "The value of the Output parameter as a basic type.",
  "$ref": "Money.schema.json#/definitions/Money"
},
"valuePeriod": {
  "description": "The value of the Output parameter as a basic type.",
  "$ref": "Period.schema.json#/definitions/Period"
},
"valueQuantity": {
  "description": "The value of the Output parameter as a basic type.",
  "$ref": "Quantity.schema.json#/definitions/Quantity"
},
"valueRange": {
  "description": "The value of the Output parameter as a basic type.",
  "$ref": "Range.schema.json#/definitions/Range"
},
"valueRatio": {
  "description": "The value of the Output parameter as a basic type.",
  "$ref": "Ratio.schema.json#/definitions/Ratio"
},
"valueReference": {
  "description": "The value of the Output parameter as a basic type.",
  "$ref": "Reference.schema.json#/definitions/Reference"
},
"valueSampledData": {
  "description": "The value of the Output parameter as a basic type.",
  "$ref": "SampledData.schema.json#/definitions/SampledData"
},
"valueSignature": {
  "description": "The value of the Output parameter as a basic type.",
  "$ref": "Signature.schema.json#/definitions/Signature"
},
"valueTiming": {
  "description": "The value of the Output parameter as a basic type.",
  "$ref": "Timing.schema.json#/definitions/Timing"
},
"valueContactDetail": {
  "description": "The value of the Output parameter as a basic type.",
  "$ref": "ContactDetail.schema.json#/definitions/ContactDetail"
},
"valueContributor": {
  "description": "The value of the Output parameter as a basic type.",
  "$ref": "Contributor.schema.json#/definitions/Contributor"
},
"valueDataRequirement": {
  "description": "The value of the Output parameter as a basic type.",
  "$ref": "DataRequirement.schema.json#/definitions/DataRequirement"
},
"valueExpression": {
  "description": "The value of the Output parameter as a basic type.",
  "$ref": "Expression.schema.json#/definitions/Expression"
},
"valueParameterDefinition": {
  "description": "The value of the Output parameter as a basic type.",
  "$ref": "ParameterDefinition.schema.json#/definitions/ParameterDefinition"
},
"valueRelatedArtifact": {
  "description": "The value of the Output parameter as a basic type.",
  "$ref": "RelatedArtifact.schema.json#/definitions/RelatedArtifact"
},
"valueTriggerDefinition": {
  "description": "The value of the Output parameter as a basic type.",
  "$ref": "TriggerDefinition.schema.json#/definitions/TriggerDefinition"
},
```

```
        "valueUsageContext": {
          "description": "The value of the Output parameter as a basic type.",
          "$ref": "UsageContext.schema.json#/definitions/UsageContext"
        },
        "valueDosage": {
          "description": "The value of the Output parameter as a basic type.",
          "$ref": "Dosage.schema.json#/definitions/Dosage"
        },
        "valueMeta": {
          "description": "The value of the Output parameter as a basic type.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        }
      },
      "additionalProperties": false,
      "required": [
        "type"
      ]
    }
  }
}
```

12.1.16 Resource Task - Maturity Dependency View

URL: https://hl7.org/fhir/R4/task-dependencies.html

12.1.16 Resource Task - Maturity Dependency View

Orders and Observations Work Group | Maturity Level: 2 | Trial Use | Compartments: Not linked to any defined compartments

Dependency Graph for Task FMM level 2

Name | Card. | Type | Dependency Analysis
Task | | DomainResource
identifier | 0..* | Identifier
instantiatesCanonical | 0..1 | canonical(ActivityDefinition)
instantiatesUri | 0..1 | uri
basedOn | 0..* | Reference(Any) | Type Warning: (Any = FMM1-Trial Use vs. Element = FMM2-Trial Use)
groupIdentifier | 0..1 | Identifier
partOf | 0..* | Reference(Task)
status | 1..1 | code
statusReason | 0..1 | CodeableConcept
businessStatus | 0..1 | CodeableConcept
intent | 1..1 | code
priority | 0..1 | code
code | 0..1 | CodeableConcept
description | 0..1 | string
focus | 0..1 | Reference(Any) | Type Warning: (Any = FMM1-Trial Use vs. Element = FMM2-Trial Use)
for | 0..1 | Reference(Any) | Type Warning: (Any = FMM1-Trial Use vs. Element = FMM2-Trial Use)
encounter | 0..1 | Reference(Encounter)
executionPeriod | 0..1 | Period
authoredOn | 0..1 | dateTime
lastModified | 0..1 | dateTime
requester | 0..1 | Reference(Device | Organization | Patient | Practitioner | PractitionerRole | RelatedPerson)
performerType | 0..* | CodeableConcept
owner | 0..1 | Reference(Practitioner | PractitionerRole | Organization | CareTeam | HealthcareService | Patient | Device | RelatedPerson)
location | 0..1 | Reference(Location)
reasonCode | 0..1 | CodeableConcept
reasonReference | 0..1 | Reference(Any) | Type Warning: (Any = FMM1-Trial Use vs. Element = FMM2-Trial Use)
insurance | 0..* | Reference(Coverage | ClaimResponse)
note | 0..* | Annotation
relevantHistory | 0..* | Reference(Provenance)
restriction | 0..1 | BackboneElement
repetitions | 0..1 | positiveInt
period | 0..1 | Period
recipient | 0..* | Reference(Patient | Practitioner | PractitionerRole | RelatedPerson | Group | Organization) | Type Warning: (Group = FMM1-Trial Use vs. Element = FMM2-Trial Use)
input | 0..* | BackboneElement
type | 1..1 | CodeableConcept
value[x] | 1..1 | *
output | 0..* | BackboneElement
type | 1..1 | CodeableConcept
value[x] | 1..1 | *

12.1.17 Value Set Sc-valueset-task-status - Status Code Mapping

URL: https://hl7.org/fhir/R4/sc-valueset-task-status.html

12.1.17 Value Set Sc-valueset-task-status - Status Code Mapping

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Not linked to any defined compartments

This is the narrative for the resource. See also the XML, JSON or Turtle format.

12.1.18 TaskStatusCanonicalMap (http://hl7.org/fhir/ConceptMap/sc-task-status)

Mapping from http://hl7.org/fhir/ValueSet/task-status to http://hl7.org/fhir/ValueSet/resource-status

DRAFT. Published on 01/11/2019 9:29:23 AM by HL7 (FHIR Project) (http://hl7.org/fhir , fhir@lists.hl7.org).

Canonical Mapping for "The current status of the task."

Source Code | Equivalence | Destination Code
entered-in-error | equivalent | error
draft | equivalent | draft
requested | equivalent | requested
received | equivalent | received
rejected | equivalent | declined
accepted | equivalent | accepted
in-progress | equivalent | active
on-hold | equivalent | suspended
failed | equivalent | failed
completed | equivalent | complete
cancelled | equivalent | abandoned
ready | equivalent | on-target

12.5 Workflow Description

URL: https://hl7.org/fhir/R4/workflow.html
FHIR Infrastructure Work Group | Maturity Level: 2 | Standards Status: Trial Use

12.5 Workflow Description

Workflow is an essential part of healthcare - orders, care protocols, referrals are the drivers of most activity within in-patient settings and a great deal of activity in community care as well. FHIR is concerned with workflow when there's a need to share information about workflow state or relationships, when there's a need to coordinate or drive the execution of workflow across systems and when there's a need to define allowed actions, dependencies and conditions on behavior.

Workflow state & relationships

FHIR does not need to be used for the execution of workflow. Orders, care plans, lab results, hospital admissions, claim payments and other records can all be shared using FHIR resources without the process to solicit fulfillment of those orders or requesting payment of those claims being driven by a FHIR transaction. Interoperable support for workflow execution is a more advanced FHIR activity because it requires a higher degree of standardization. Rather than merely standardizing the data to exchange, interoperable workflow execution requires standardization of the processes, roles and activities across the different systems. However, even without using FHIR for workflow execution, there's still a need to standardize the data elements related to workflow: how does an event or a result point to the order that authorized it? How do parent steps and child steps get linked together? How does a care plan identify what protocol it's adhering to?

FHIR defines three categories of resources that are involved in activities - requests, events and definitions. Each of these categories has a "pattern" associated with it. Resources that fall into that category are encouraged to adhere to their respective pattern. These patterns provide standard elements that are typical for most resources of each category. Strict adherence is not required as work groups are expected to align with typical domain behavior and requirements as more authoritative than "desired" architectural patterns. In some cases, capabilities might be supported with extensions rather than core elements where a pattern capability is deemed to be "not common, but still relevant" for a given resource.

A full description of the patterns and their interrelationships can be found in the Workflow Resource Patterns section of this page.

Workflow execution

In addition to defining patterns for resources used in workflow processes, FHIR supports the execution of those processes as well. However, FHIR does not define a "one size fits all" solution for workflow architecture. FHIR supports a variety of interoperability paradigms and most of them (REST, Messaging and Services) provide support for driving workflow execution. (The Document paradigm does not directly support driving behavior, though it can be combined with one of the other patterns to do so.) In addition, several of these paradigms allow multiple approaches to supporting workflow, depending on the context and needs of the

workflow process.

The Workflow Execution and Communication Patterns section describes many options for workflow execution, summarizes their respective pros and cons and makes recommendations for the circumstances in which they might best be used.

Workflow definition

The definition of protocols, order sets, guidelines and other structures that define what sorts of activities should occur, what order they should occur on, what dependencies they have, in what circumstances they should start or end, etc. is handled by a pair of resources:

• PlanDefinition defines the interrelationships of steps and the rules around their execution

• ActivityDefinition defines an activity to be performed as a single step

The use of these two artifacts is discussed here.

12.5.1 Workflow Resource Patterns

Not all resources in FHIR are related to workflow - many are used to describe entities and roles (patients, medications, etc.) or infrastructure (structure definitions, value sets, etc.). However, a large proportion of the FHIR resources are devoted to the description of activities in one fashion or another and almost all of these fall into the realm of workflow - they describe things that can be done (definitions), are desired to be done (requests) or that have been done (events). The table below summarizes the list of workflow-relevant resources:

12.5.1.1 Workflow resources

Definitions | Resources that define something that can potentially happen in a patient and time-independent manner
ActivityDefinition Measure | OperationDefinition | PlanDefinition | Questionnaire
Requests | Resources that ask for or express a desire/intention for something to be done
Appointment* AppointmentResponse* CarePlan Claim CommunicationRequest | Contract CoverageEligibilityRequest DeviceRequest EnrollmentRequest | ImmunizationRecommendation MedicationRequest NutritionOrder ServiceRequest | Task‡ SupplyRequest VisionPrescription
Events | Resources that express that something has been done and which can potentially be done because of a request
ChargeItem ClaimResponse ClinicalImpression Communication Composition Condition (aka Problem) Consent Coverage CoverageEligibilityResponse DeviceUseStatement | DiagnosticReport DocumentManifest DocumentReference Encounter EnrollmentResponse EpisodeOfCare ExplanationOfBenefit FamilyMemberHistory | GuidanceResponse ImagingStudy Immunization MeasureReport Media MedicationAdministration MedicationDispense MedicationStatement | Observation PaymentNotice PaymentReconciliation Procedure QuestionnaireResponse RiskAssessment SupplyDelivery Task‡

* | The Appointment and AppointmentResponse resources do not follow the same sort of request/response pattern as the other resources. Their design is based on iCal conventions, so their model won't align with the pattern as strictly as most other resources. They are included here for completeness.
‡ | The Task resource takes on characteristics of both "requests" and "events" and thus shares characteristics from both patterns

Note that requests, events and definitions don't exist in a 1:1:1 relationship. Some requests and events have obvious pairings. For example, a SupplyRequest will generally always pair with a SupplyDelivery. The same goes for EnrollmentRequest/EnrollmentResponse, etc. On the other hand, for other resources there isn't a strict pairing. A ServiceRequest might be responded to by an Encounter, DiagnosticReport, Procedure, RiskAssessment, etc. Similarly, a Procedure might be triggered by a ServiceRequest. The set of common linkages should be asserted in their respective resources. The specific types of responses for a given request will be governed by the Request.code, any workflow definitions/protocols referenced and local convention.

12.5.1.2 Workflow Resource Relationships

These three patterns of resources have a standard set of relationships, both with themselves, as well as with each other.

Specifically:

• requests, events and definitions can point to their respective definitions

• events and requests can point to the proposals, plans or orders they are based on

• events and definitions can be organized into parent-child relationships of parents and components

• definitions and requests can both replace prior versions of the same type of artifact

This list of relationships is not exhaustive, but covers those that are "standardized" as part of the patterns. Further description and guidance on these relationships can be found in the Request, Event and Definition logical patterns.

12.5.1.3 Request Resource Pattern

Requests are resources that represent the proposal, plan or order for an activity to occur. A Request pattern defines the common elements typically present on all request resources.

The amount of information needed for a Request to be actionable can vary by circumstance. Some request instances might not be "fully specified" - additional information from protocol, patient preference and/or professional decision-making may be necessary before the authorized action can occur. For example, a MedicationRequest might be specified without indicating a strength or route in situations where the pharmacy

(or even nursing station) has the authority to determine those parameters. A VisionPrescription might not be actionable until frames have been chosen and the necessary measurements of the patient's face have been taken to allow the lenses to be positioned appropriately within the frames.

All requests with an intent of "order" authorize something. Whether what is authorized is sufficient to be immediately actionable depends on who is fulfilling the order and the context in which the fulfillment request is made. The determination of whether a given "request" is actionable may be made by the systems involved or the humans being asked to act.

As well, the existence of a "Request" instance doesn't necessarily imply that fulfillment will be requested immediately - or even ever. The decision to request fulfillment may be delegated to the patient or to down-stream practitioners. Such fulfilling practitioners may need to capture additional information prior to executing the fulfillment.

12.5.1.4 Event Resource Pattern

Events are resources that represent the ongoing or completed execution of some activity or observation. For example, a clinical procedure, a financial transaction, the recording of a diagnosis, etc. An Event pattern defines the common elements typically present on all event resources.

12.5.1.5 Definition Resource Pattern

Definitions are resources that represent activities that could be performed in a time and subject-independent manner such as a protocol, order set, clinical guideline, etc. A Definition pattern defines the common elements typically present on all definition resources.

12.5.2 Open Issues

STU Notes:

• It is possible to replace some portions of the MessageHeader with a reference to the Task resource. Doing so would mean consistency in how asynchronous requests are represented using REST and messaging. However, it introduces an additional layer of complexity and formality into the messaging paradigm that may be unwelcome, particularly for those systems that do not currently foresee a need to support both RESTful and messaging invocations of workflow

• The OperationDefinition resource could be used to define types of tasks and the sets of parameters that can go with them. Is this an appropriate use of the OperationDefinition resource?

• The SupplyRequest, DeviceRequest and VisionPrescription resources have a significant degree of overlap. Should they remain distinct resources?

Feedback is welcome here .

12.6 Workflow Execution and Communication Patterns

URL: https://hl7.org/fhir/R4/workflow-communications.html
FHIR Infrastructure Work Group | Maturity Level: 2 | Standards Status: Trial Use

12.6 Workflow Execution and Communication Patterns

As described in previous sections, many FHIR resources can be used within workflows without using FHIR to manage the execution of the workflows. When we want to manage workflow execution with FHIR, there are several mechanisms available. In addition to managing workflow with FHIR, there are also cases where management of workflow execution is not necessary, and supporting workflow becomes simply the use of the resource patterns in an ad-hoc fashion. In deciding how best to interoperate around workflow with FHIR, there are several considerations:

• Is sharing of the state of the workflow necessary among the participants?

• Which paradigm do you want to use (REST, messaging, services, a mix)?

• Who owns/manages the various resources involved in the workflow (placer, filler, another participant)?

• Is there infrastructure in place to support polling, push notifications via subscriptions or both?

• Is there a need for confirmation that the desired performer agrees to act, or can that be presumed?

• Is there a need to negotiate whether/how the requested action will be performed?

• Can the requesting and performing system communicate directly? Are they able to post to each other's servers (if using REST)?

• Is there an ability/need to have a queue server to facilitate workflow execution?

• How many potential actors are involved?

• Will the workflow always be directed or is there a pool of potential performers who could choose to perform the requested action?

The answers to these (and other) questions will guide the selection of communication patterns to be used for a specific interoperability use case. It is recommended that domain workgroups analyze the support for

workflow execution within their domains and provide recommendations for an appropriate subset of patterns, with a discussion on where and how to use them. Implementation guides, by their nature, should prescribe which patterns to use for workflow execution management.

12.6.1 Communication Patterns Conventions

This section highlights some of the more common patterns and identifies their characteristics and limitations and provides recommendations on when each approach may be most useful or relevant. Please note that this list of patterns is not exhaustive. Patterns can be combined in various ways and there are likely some possibilities we haven't thought about yet (feel free to submit additional patterns using the 'submit a change' link at the bottom of the page). As well, the recommendations given here are just that - recommendations. Implementers are free to choose which patterns they wish to support. Because of this, tight interoperability around workflow execution (as with any other tight interoperability using FHIR) will depend on communicating participants doing some up-front negotiation around how they plan to support workflow execution or all communicating partners will need to adhere to an implementation guide that sets out clear interoperability expectations.

Prior to reviewing this list of options, readers are encouraged to be familiar with the following pages and resources: REST, messaging, operations, services and the Subscription resource.

The scenarios used to illustrate the patterns below make use of a few conventions:

• The focus here is on a "request" and the actioning of that request. Almost all workflows can be broken down to a sequence of these steps, though the responsibilities of the different parties may shift for each interaction and there can be more than two parties involved in the overall workflow

• The request could be as simple as "please look at this information" and the response could be as simple as an implicit "it's been looked at" or the request could be for some more involved action that may include reporting back multiple interim and final steps

• The requester is referred to as the "placer" and the performer is referred to as the "filler", which are often seen as order-specific terms. However, in this context, the terms hold whether the request is expressed as a proposal, plan or full-blown order

• Each of the patterns defines the set of steps involved in processing the request, lists some of the benefits and limitations associated with the approach, provides a scenario to illustrate the use of the pattern, and then makes recommendations about when the pattern is most appropriate

• The descriptions of these patterns focus on the notion of requesting fulfillment of a request. However most of these patterns are also applicable to requests for status change, requests for information, etc. If a pattern is limited in the types of execution it can trigger, this will be noted in the "limitations" section.

12.6.2 Workflow State

One of the key distinguishing characteristics in the patterns below is whether the pattern supports the tracking of the workflow's state by both the placer and the filler. The workflow state is represented by the Task resource. The Task state machine, which is also presented here, shows the use of Task.status to represent the general infrastructure state of the Task resource.

The combination of Task.status, Task.statusReason and Task.businessStatus provide the mechanism for tracking the workflow state. The following two examples illustrate how that could be accomplished with regards of specific workflows. Note that these examples avoid any specificity on where the described resources exist, or any reference to a specific communication pattern. The goal of the examples is to demonstrate the use of the Task attributes to represent the workflow state.

12.6.2.1 Example: Workflow states for a lab order for a blood test

The following table shows the steps for a simple blood test order, and the corresponding values of the Task attributes. A lot of details on other attributes have been omitted in order to focus on the state changes.

Workflow step | Task.status | Task.statusReason | Task.businessStatus
Provider orders a blood test for a patient. A ServiceRequest resource is created to represent the order, and a Task resource is created to represent the request to the lab. Task.basedOn points to the ServiceRequest instance. | Requested | New order | Ordered
The lab accepts the order. | Accepted | Able to perform the test. | Accepted
The patient is at the lab, and blood is drawn. | In progress | Obtained the specimen | Specimen available
Work on the blood sample is in progress, preliminary results are available, a DiagnosticReport resource is created, and Task.output.value[1] references the DiagnosticReport resource. | In progress | Preliminary results available | Preliminary results
Final results are available | Complete | Final results are available | Final results

12.6.2.2 Example: Workflow states for a referral from a PCP to a specialist

The following table shows the steps for a referral workflow, and the corresponding values of the Task attributes. A lot of details on other attributes have been omitted in order to focus on the state changes.

Workflow step | Task.status | Task.statusReason | Task.businessStatus
Provider A is seeing a patient with a specific complaint. Based on the exam, Provider A refers the patient to a specialist at clinic B. It is an urgent referral, requesting to see the specialist within 4 days. A ServiceRequest resource is created to represent the referral order, and a Task resource is created with Task.priority set to Urgent, and Task.restriction.period.end set to 4 days in the

future. | Requested | New referral | Referred
Clinic B receives the referral and puts it on the urgent queue to be reviewed. | Received | Received by organization | Pending
Upon reviewing the referral, the clinic determines they have no capacity to see the patient within the specified time period, and they decline the referral. | Rejected | Provider unavailable | Declined
Provider A changes the referral request to clinic C. This creates a new Task resource, with the appropriate information | Requested | Changed the referred to provider | Referred
Clinic C accepts the referral and it is placed on a scheduler's queue. | Accepted | Able to provide the service. | Accepted
An appointment is scheduled for the patient at Clinic C. | In progress | Appointment scheduled | Scheduled
Soon after the appointment is scheduled, the patient calls provider A to ask for a different location for the referral appointment, as he lost his transportation, and he cannot reach Clinic C. The provider requests a cancel for the referral. | Suspended | Patient unable to make appointment. | Cancel requested
Clinic C confirms the cancellation of the existing task. | In progress Failed | Referral cancelled | Cancelled as requested
Provider A changes the referral to Clinic D, which is located close to where the patient lives, and can be conveniently reached using public transportation. This creates a new Task resource, with the appropriate information. | Requested | Changed the referred to provider | Referred
Clinic D accepts the referral and it is placed on a scheduler's queue. | Accepted | Able to provide the service. | Accepted
An appointment is scheduled for the patient at Clinic D. | In progress | Appointment scheduled | Scheduled
Patient is seen at Clinic D. A diagnosis is made by the specialist, a prescription is given, and a follow-up appointment is made at clinic D. Task.output.value[1] references a Compositions resource, which contains the encounter summary. | In progress | Intermediary consultation note available | Preliminary notes
Patient comes for follow-up appointment at clinic D. The specialist is satisfied by the progress the patient has made and creates the final consultation note. Task.output.value[2] references a Compositions resource, which contains consultation note. | Completed | Final consultation note available | Final notes

12.6.3 List of patterns

The patterns that facilitate the execution of workflow using the Task resource are grouped in the Workflow Management Patterns section. The patterns where no Task resource is used are grouped in the Ad-hoc Workflow Patterns section. The list of patterns is as follows:

Ad-Hoc Workflow Patterns
Option A: Simple RESTful POST or PUT
Option B: Direct POST of request to fulfiller's system
Option C: POST of request to placer/queue server system, receiver uses polling or subscription
Option D: Messaging request from placer to filler & acknowledgment
Option E: Services request from placer to filler & acknowledgment
Workflow Management Patterns
Option F: Creation of Task on placer's system
Option G: POST of Task to fulfiller's system
Option H: POST of Task to a workflow broker
Option I: POST of Task to fulfiller's system, POST of sub-Task on placer's system
Option J: Messaging Task from placer to fulfiller
Option K: Service request referencing Task from placer to fullfiller
Additional Scenarios and Variations

TODO: Insert Jose's decision tree here?

12.7 Ad-Hoc Workflow Communication Patterns

URL: https://hl7.org/fhir/R4/workflow-ad-hoc.html
FHIR Infrastructure Work Group | Maturity Level: 2 | Standards Status: Trial Use

12.7 Ad-Hoc Workflow Communication Patterns

TODO: Discussion (or reference to one) on Polling and Subscription

12.7.1 Option A: Simple RESTful POST or PUT

12.7.1.1 Steps

• The placer makes a RESTful call to create or update a record or a POST to invoke an operation over HTTP

• The receiver responds with a 2xx HTTP response indicating whether the request was successfully processed or not and, if appropriate, provides the response to the request in the payload of the HTTP response

12.7.1.2 Benefits

• Simplest of all the possible workflow architectures

• Placer knows whether the request was accepted or not and knows when the task has been done

12.7.1.3 Limitations

• Only works for automated execution where the decision to perform the request and the execution of the request can be done synchronously within the HTTP timeout period (generally on the order of 10s of seconds).

• Requires that the placer have authority to post directly to the placer's system

• Requires that the "request" be expressible as a simple creation, update or operation invocation

• Only works for "fulfillment" requests for Request resources - can't handle request for state changes or information

• Only way to cancel the request is with a separate custom operation

12.7.1.4 Usage Recommendations

This is by far the most common pattern in FHIR for simple changes as it requires the least overhead. However, it covers only automated responses, and does not support any actual workflows of more than one step. If human processing is involved in the request execution, then this approach won't suffice. This approach is listed here to make sure that implementers consider whether they can make this one work first before falling back to one of the more sophisticated patterns.

12.7.1.5 Usage Examples

???

12.7.2 Option B: Direct POST of request to fulfiller's system

12.7.2.1 Steps

• Placer system invokes a create by POSTing a 'request' resource (e.g. MedicationRequest, ServiceRequest, etc.) to the appropriate RESTful resource endpoint (e.g. [base]/MedicationRequest) on the filler system and places an actionable tag on the resource that indicates the request is intended to be acted upon, not merely stored.

• The filler synchronously responds with a "201" indicating that they have received and stored (created) the resource on their system

• At some later point, the filler POSTs an 'event' resource (e.g. MedicationDispense, DiagosticReport, Encounter, etc.) to the appropriate resource endpoint on the placer system, including a basedOn link to the 'request' resource that the action was performed in fulfillment of.

• The placer system synchronously responds with a "201" indicating they've received and store (created) the resource on their system

12.7.2.2 Benefits

• Lowest amount of overhead. No need for Task. No need for polling or subscriptions

• Explicit acknowledgement that filler has received the request

12.7.2.3 Limitations

• Can only use when requesting fulfillment (can't use to request status change or other updates)

• Placer and filler must be able to communicate directly (i.e. know each other's respective endpoints), must each have a FHIR server, and must have "write" permissions to each other's servers. This could become unmanageable if there are a large (or dynamic) number of placers and fillers that need to communicate

• No indication of agreement to act on the request

• There's no ability to negotiate fulfillment - no ability to say "no"

• This completely reverses the usual provenance of resources, as the request only exists on the fulfiller's system, and the event only exists on the placer's system. This greatly limits the use of these resources for any other purpose.

• The only way to stop fulfillment is to update the Request to have a 'cancelled' status

12.7.2.4 Usage Recommendations

Use this approach when there's no ability to have queue servers and no support/need for complexity of Task, polling or pub/sub (and no need for negotiation or the ability for the filler to say "no"). This is a pseudo-messaging architecture that doesn't actually use messaging architecture.

12.7.2.5 Usage Examples

12.7.3 Option C: POST of request to placer/queue server system, receiver uses polling or subscription

12.7.3.1 Steps

• Placer system creates a 'request' resource (e.g. ServiceRequest etc.) by a system action or a POST on either its own system or a third-party queue server system and places an "actionable" tag on the resource that indicates the request is intended to be acted upon. The request explicitly identifies the intended fullfiller

• The filler system uses polling (or a subscription) at the placer or queue server system to see if there are any "new" requests that: are tagged as "actionable", have the filler identified as the intended performer, and are a type of request "of interest" to the filler.

• At some later point, the filler creates an 'event' resource (e.g. MedicationDispense, DiagosticReport, Encounter, etc.) via a system action, or a POST on either its own system, the same queue server as the request was placed on, or some alternate queue server, including a link to the 'request' resource that the

action was performed in fulfillment of

• The placer system uses polling (or a subscription) to the filler or queue server system to see if there are any "new" events that are tied to any outstanding requests the placer has initiated

12.7.3.2 Benefits

• Placer and fulfiller don't have to communicate directly (can act through queue server). This can reduce the number of point-to-point interfaces that need to be supported

• Preserves the conventional management and ownership of the request and event at the placer and fulfiller's systems respectively

• No need for Task

12.7.3.3 Limitations

• Can only use when requesting fulfillment (can't use to request status change or other updates)

• Additional complexity of using subscription or polling

• Polling by the placer for "anything related to these 500 open orders" could be onerous, especially if some orders never get closed.

• Placer and fulfiller must know where to poll or subscribe for content - this could be a large number of systems

• No indication of agreement to act on the request

• There's no ability to negotiate fulfillment - no ability to say "no"

• Placer might not know when (or if) filler system has retrieved the request

• If the Request is cancelled, there's no guarantee the filler will notice or act on the change

12.7.3.4 Usage Recommendations

This pattern could be used when there's no support/need for complexity of Task, usually for simple two step workflows. This is a more typically RESTful approach where data resides on the server "owned" by the data creator and is accessed by other systems.

12.7.3.5 Usage Examples

12.7.4 Option D: Messaging request from placer to filler & acknowledgment

TODO: needs review and update

12.7.4.1 Steps

• Placer sends message to filler system including Request resource (and other relevant resources) along with a MessageHeader with an "event" code saying "please fulfill" and "data" element pointing to the Request resource as the item to fulfill. Message could potentially use Task instead of MessageHeader.event to convey desired action (ongoing discussion)

• Filler system sends a response indicating receipt of the message and, optionally an indication of their intention to fulfill the request

• Filler system may send incremental messages to the placer showing progress (e.g. specimen collected, preliminary results, final results)

12.7.4.2 Benefits

• Reduced number of communications

• All relevant data sent in one package

• Responses can be asynchronous, and content may be routed

• There's an ability to negotiate fulfillment - i.e. the ability to say "no"

• Can request things other than just fulfillment (e.g. please suspend)

• Explicit acknowledgement that filler has received and agreed to act on the request (though no need for the placer to check)

12.7.4.3 Limitations

• Messaging is "heavy"

• Need to negotiate what allowed responses are and what data can be present in request and response messages

• Additional complexity of setting up and maintaining a subscription or polling infrastructure

• Additional complexity of using Task

• Need message delivery infrastructure in place

• Cancellation requires a separate purpose-specific message

12.7.4.4 Usage Recommendations

Existing messaging infrastructure (e.g. v2 LTP, MLTP, WSI Web Services, Direct, VISA, REST, etc.) and a need to stay consistent with that architecture

12.7.5 Option E: Services request from placer to filler & acknowledgment

TODO: Needs review, update and more work - there's not a lot of experience using FHIR services to manage the fulfillment process

12.7.5.1 Steps

• Placer may create and store a Request resource on their own system or a queue server.

• Placer invokes a service on the filler system saying "please fulfill this order", including the content or a reference to the request resource and any other relevant data

• Filler system responds (synchronously if using HTTP, but may be asynchronous if using SOAP or other transport mechanisms) with conformation of receipt and, optionally indication of intention to fulfill and/or results

12.7.5.2 Benefits

• ???

12.7.5.3 Limitations

• ???

12.7.5.4 Usage Recommendations

TBD

12.7.6 Additional Scenarios

TODO: needs review and update. Possibly add options about using messaging and/or services instead of polling/subscription in above scenarios

12.7.6.1 Querying the status of a workflow using REST

• Placer sends query for Task(s) that have a focus of the request of interest to a system (placer system, queue server or filler) that holds tasks related to their request.

• System returns a query response showing all related tasks (typically just one). Task shows current status.

12.7.6.2 Querying the status of a workflow using services

• Placer invokes a "what's the status of this order" service, passing the request business identifier or URL of the request

• Services responds with a Task showing the current state of the fulfillment of the request

12.7.6.3 Cancellation of a Task using REST - placer owns

• Placer sends an update to the Task setting the status to "cancelled" signaling they are no longer concerned about the outcome of the task

• Filler receives notification of the update (because the task is on their system, or because they poll it, or are subscribed to it) and ceases work if they are able

12.7.6.4 Cancellation of a Task using REST - filler owns

• Placer creates a new task requesting cancellation of the original fulfillment task Fulfillment of the "cancellation task" can be requested using any of the mechanisms above

• Filler decides whether they can cancel the task and update the "cancellation" task to indicate either cancellation is complete or has been refused

12.8 Workflow Management Communication Patterns

URL: https://hl7.org/fhir/R4/workflow-management.html
FHIR Infrastructure Work Group | Maturity Level: 2 | Standards Status: Trial Use

12.8 Workflow Management Communication Patterns

TODO: Discussion (or reference to one) on Polling and Subscription

12.8.1 Option F: Creation of Task on placer's system

12.8.1.1 Steps

• Placer creates the request in its own system via POST, an internal action, or POSTs it to a queue server system

• Placer creates a Task resource in its own system via POST, an internal action, or POSTs it to a queue server system, pointing to the request resource and seeking fulfillment. The Task may have a specified performer, in which case step 3 is expected to be done by that performer. If the Task does not have a specified "performer"

(although may have performer type), then this is a case of an "open" task, where any number of fulfillers may attempt to "claim" the task. Who succeeds is determined by local policies and procedures.

• Fulfiller system uses either polling or pub/sub to become aware of the existence of the task A common case may be the conveyance of the Task id to a fulfiller by other means. For example, a lab test is ordered, and the patient takes a requisition to the lab of his choice. The requisition contains the Task id (as a bar code, or stored in the patient's healthcare smart card), and the lab system can execute a direct GET for the Task, thus eliminating the need for subscription or polling.

• Fulfiller system queries to retrieve the referenced request and updates the Task to indicate "acceptance" and agreement to fulfill

• Fulfiller may update the Task to indicate interim progress notes

• Placer is made aware of the acceptance of the Task and any changes to the Task either through its ownership of the resource or using polling or a subscription to a queue server system to determine the same

• Fulfiller creates an event resource in its own system via POST or internal action, or POSTs it to a queue server system

• Fulfiller PUTs an update to the Task resource to change its status to completed and to point to the event resource

• Placer is aware of the completion of the Task either through the ownership of the resource, or via polling or subscription to a queue server system, and retrieves the referenced event resource

• Placer updates the request resource to indicate completion via PUT or an internal action, or PUTs the update to a queue server system

12.8.1.2 Benefits

• Uses the FHIR REST API for managing the workflow

• Both placer and fulfiller can track the state of the workflow

• Can use this approach for request other than just fulfillment (e.g. to request status change or other updates)

• There's an ability to negotiate fulfillment - i.e. the ability to say "no"

• Explicit acknowledgement that filler has received and agreed to act on the request

• Supports cancellation through ability of placer to change the status of the Task to cancelled.

12.8.1.3 Limitations

• Additional complexity of using Task

• Additional complexity of setting up and maintaining a subscription or polling infrastructure

• Placer and fulfiller must have a FHIR server, and placer's FHIR server must allow "write" updates to the Task resource

Placer and filler may need to be able to communicate directly (i.e. know each other's respective endpoints)

• This could become unmanageable if there are a large (or dynamic) number of placers and fillers that need to communicate

• Placer and fulfiller must know where to subscribe for content - this could be a large number of systems

• Might not apply if there's a queue server

• Does not give fulfiller control over whether the task can be cancelled

12.8.1.4 Usage Recommendations

• When following the state of the workflow is important When there is a need to know that a fulfiller has accepted to perform the request When there is a need to be able to change the performance of the request When there are multiple steps in fulfilling the request

• When requests are not (always) directed to a specific filler

• When the placer needs to manage the Task resource

12.8.1.5 Usage examples

Lab order to known performer

A provider orders a bilirubin test for a patient, to be done at the provider's facility lab....

"Open" lab order

A patient is scheduled for their physical, and the provider sends her a secure message to have a fasting blood test performed before the office visit. The message has a link to a requisition, and a list of possible lab locations where the test can be done. The patient chooses a lab location near her place of employment and a test is scheduled. The lab gets the requisition from the patient and, using the information on how to retrieve the Task from the provider's system, claims the test to be performed...

12.8.2 Option G: POST of Task to fulfiller's system

12.8.2.1 Steps

• Placer creates the request in its own system via POST or an internal action, or POSTs it to a queue server system

• Placer POSTs a Task resource to the filler system, pointing to the request resource and seeking fulfillment

• Fulfiller system GETs the referenced request

• Fulfiller updates the Task to indicate acceptance of the task

• Placer either polls the Task to note acceptance or uses a subscription to determine the same

• Fulfiller may further update the Task to reflect the progress made. Using the same method as in step 5, the placer becomes aware of these updates

• Fulfiller creates an event resource in its own system via POST or an internal action, or POSTs it to a queue server system

• Fulfiller Updates the Task resource to change its status to completed and to point to the event resource

• Placer either polls the Task to note completion and changes or uses a subscription to determine the same

• Placer system queries to retrieve the referenced event resource

• Placer updates the request resource to indicate completion via PUT or an internal action, or PUTs the update to a queue server system

12.8.2.2 Benefits

• Uses the FHIR REST API for managing the workflow

• Subscription/polling is targeted to a specific resource instance - much simpler than general subscription/polling infrastructure

• Both placer and fulfiller can track the state of the workflow

• Can use this approach for request other than just when requesting fulfillment (e.g. to request status change or other updates)

• There's an ability to negotiate fulfillment - i.e. the ability to say "no"

• Explicit acknowledgement that filler has received and agreed to act on the request

• Placer can attempt to cancel task in the event of order cancellation, but no guarantee

• Filler has control over whether task can be cancelled

12.8.2.3 Limitations

• Additional complexity of using Task

• Additional complexity of using subscription or polling

• Placer and fulfiller must have a FHIR server and fulfiller's FHIR server must allow "write" creation of the Task resource by the placer

• Placer and filler may need to be able to communicate directly (i.e. know each other's respective endpoints) This could become unmanageable if there are a large (or dynamic) number of placers and fillers that need to communicate Might not apply if there's a queue server

• Placer might not know immediately when filler system has retrieved the request

12.8.2.4 Usage Recommendations

• When following the state of the workflow is important When there is a need to know that a fulfiller has accepted to perform the request When there is a need to be able to change the status of the workflow by either the placer or the fulfiller When there are multiple steps in fulfilling the request

• When the fulfiller needs to manage the Task resource

12.8.2.5 Usage Examples

Outpatient prescription

A patient is suffering from a poison ivy rash and is prescribed a steroid medication to manage the reaction. The provider asks the patient for his preferred pharmacy and creates the Task at the specified pharmacy's system. The patient is running late and by the time he arrives at the pharmacy, it is closed. He calls the clinic where the provider practices and requests that the pharmacy where the order is to be fulfilled is changed to one that is open around the clock. The staff at the clinic cancels the Task at the original pharmacy and creates a new Task at the requested new one. By the time the patient arrives at the new pharmacy, the medication is ready for pickup. Once the request is fulfilled, the clinic receives the notification that the Task is complete, and the prescription is also marked as complete.

12.8.3 Option H: POST of Task to a workflow broker

TODO: Still needs review and update

12.8.3.1 Steps

• Placer POSTs the request to its own system or to a queue server system

• Broker detects that new un-assigned request (without a Task yet created and falling within the scope of the Broker to ensure fulfillment) via polling or subscription

• Broker POSTs a Task resource to its own system or a queue server system, pointing to the request resource and seeking fulfillment from a specific filler Task does not have a specified "performer" (but may have performer type)

• If the Task is rejected by one potential recipient, the broker may create a new task to seek fulfillment from others

• Continue as per Option G

12.8.3.2 Benefits

• Offloads responsibility for seeking fulfillment from the placer system, but more actively solicits fulfillment than a simple "post the task and see who takes it". Also, allows prioritized assignment of tasks (i.e. some fillers may be preferred over others)

12.8.3.3 Limitations

• Requires a broker to exist

• Broker must know all available fillers and their capabilities to allow appropriate assignment

• Broker must understand business rules for permitting changes to tasks such as cancelation

12.8.3.4 Usage Recommendations

Appropriate in environments that have a workflow engine that takes on responsibility for ensuring fulfillment

12.8.3.5 Usage Examples

12.8.4 Option I: POST of Task to fulfiller's system, followed by POST of sub-Task on placer's system

12.8.4.1 Steps

• Placer creates the request in its own system via POST or an internal action

• Placer POSTs a Task resource to the filler system, pointing to the request resource and seeking fulfillment

• Fulfiller system GETs the referenced request

• Fulfiller updates the Task to indicate acceptance of the task

• Fulfiller POSTs a Task, which also points to the request resource and uses the Task.partOf attribute to point to the Task from step 2 (indicating it is a sub-Task)

• Fulfiller may further update both Tasks to reflect the progress made. Since the sub-Task is on the placer's system, the placer is aware of these updates

• Fulfiller creates an event resource in its own system via POST or an internal action

• Fulfiller Updates the Task resource to change its status to completed and to point to the event resource

• Fulfiller updates the sub-Task resource as completed and points to the event resource. Since the sub-Task is on the placer's system, the placer is aware of the completion and changes.

• Placer system queries to retrieve the referenced event resource

• Placer updates the request resource to indicate completion via PUT or an internal action

12.8.4.2 Benefits

• Uses the FHIR REST API for managing the workflow

• No need to use subscription or polling

• Both placer and fulfiller can track and change the state of the workflow

• Can use this approach for request other than just when requesting fulfillment (e.g. to request status change or other updates)

• There's an ability to negotiate fulfillment - i.e. the ability to say "no"

• Explicit acknowledgement that filler has received and agreed to act on the request

• Placer can initiate task cancellation, but filler decides whether it's possible to cancel the task or not

12.8.4.3 Limitations

• Additional complexity of using Task

• Need to keep the two Task states in sync

• Placer and fulfiller must each have a FHIR server and give the other side "write" access to create and update Task resources.

• Placer and filler may need to be able to communicate directly (i.e. know each other's respective endpoints) This could become unmanageable if there are a large (or dynamic) number of placers and fillers that need to communicate Might not apply if there's a queue server

12.8.4.4 Usage Recommendations

• When following the state of the workflow is important When there is a need to know that a fulfiller has accepted to perform the request When there is a need to be able to change the status of the workflow by either the placer or the fulfiller When there are multiple steps in fulfilling the request

• When there is no need for polling or subscription infrastructure

• When us of the FHIR REST API is preferred to any other method of communications

• When the intended fulfiller is known in advance

• When messaging-like functionality is needed, but a messaging infrastructure is not available

12.8.4.5 Usage Examples

Outpatient referral

A patient is seen by her Primary Care Provider (PCP) for shortness of breath and the provider decides to refer her to a cardiologist. Because of the referral, the provider's system creates ServiceRequest resource and POSTs a Task resource to a cardiologist's practice system, requesting a referral. This places the referral request on a referral coordinator's work list to review. Upon accepting the referral, the Task is updated as accepted and a new sub-Task, referencing both the ServiceRequest and the Task created by the provider's system, is created on the provider's system. The patient gets an appointment with the cardiologist and the sub-Task is updated with that information, allowing the provider's system to inform the provider of the change.

The patient misplaces the information about the appointment and misses the visit with the cardiologist. The cardiologist's system updates the sub-Task to indicate the no-show and the PCP is alerted of the missed appointment. The provider's staff gets hold of the patient and schedules another appointment for her with the cardiologist. Because of the cardiologist's system scheduling the appointment, the sub-Task is updated with that information. The patient is seen and the sub-Task is updated accordingly. The cardiologist document the findings about the patient and updates the sub-Task as complete, adding references to the resources representing the findings. The PCP is alerted of the updates and marks the referral as complete, which updates the original Task and the ServiceRequest as complete as well.

12.8.5 Option J: Messaging Task from placer to fulfiller

TODO: needs more details

12.8.5.1 Steps

• Placer sends message to filler with a MessageHeader, where the "data" element points to the Task resource, also contained in the message. The message might or might not contain any other relevant resources (e.g. the actual request resource), or an "event" code saying "please fulfill"

• Filler system sends a response containing the same Task resource, indicating receipt of the message and, optionally, an indication of their intention to fulfill the request

• Filler system may send incremental messages to the placer showing progress (e.g. specimen collected, preliminary results, final results) by including an updated Task resource

• Placer system may also send messages to the fulfiller containing the Task resource and updating the state of the workflow, for example cancelling the task

12.8.5.2 Benefits

• Reduced number of communications

• All relevant data can be sent in one package

• Responses can be asynchronous, and content may be routed

• There's an ability to negotiate fulfillment - i.e. the ability to say "no"

• Can request things other than just fulfillment (e.g. please suspend)

• Explicit acknowledgement that filler has received and agreed to act on the request (though no need for the placer to check)

12.8.5.3 Limitations

• Messaging is "heavy"

• Need to negotiate what allowed responses are and what data can be present in request and response messages

• Additional complexity of using Task

• Need message delivery infrastructure in place

12.8.5.4 Usage Recommendations

Appropriate when existing messaging infrastructure can be used (e.g. HL7 over HTTP, v2 LTP, MLTP, WSI Web Services, Direct, VISA, REST, etc.), and a need to stay consistent with that architecture.

12.8.5.5 Usage examples

12.8.6 Option K: Service request referencing Task from placer to fulfiller

TODO: This scenario needs work - there's not a lot of experience using FHIR services to manage the fulfillment process

12.8.6.1 Steps

• Placer creates a request resource on their own system or a queue server

• Placer may create a Task resource on their own system or a queue server

• Placer invokes a service on the filler system saying "please fulfill this request", including the content or a reference to the request resource and any other relevant data

• Filler system responds (synchronously if using HTTP, but may be asynchronous if using SOAP or other transport mechanisms) with conformation of receipt and, optionally indication of intention to fulfill and/or results

12.8.6.2 Benefits

• No need to expose RESTful endpoints for either requests or tasks - acts like a remote procedure call

12.8.6.3 Limitations

• Requires support for a custom operation by both parties (typically at least a different operation for each type of request)

• No way to get insight into progress without invoking another custom operation

12.8.6.4 Usage Recommendations

TBD

12.8.6.5 Usage examples

12.8.7 Additional Scenarios

TODO: needs review and update. Possibly add options about using messaging and/or services instead of polling/subscription in above scenarios

12.8.7.1 POST of "request" resource for filler system, response via Task

This is a variation ofOption H, where the Workflow broker is essentially merged with the fulfiller. It still allows the placer to only use a POST of the request and be made aware of the changes to the other resources via subscription or polling.

• Placer system invokes a "create" action by POSTing a 'request' resource (e.g. ServiceRequest etc.) to the appropriate RESTful resource endpoint (e.g. [base]/MedicationRequest) on the filler, placer or queue server system and sets a "tag" on the resource that indicates the request is "actionable"

• Filler POSTs a Task resource to its own system or a queue server system, pointing to the request resource and indicating intent to fulfill or refusal to fulfill

• Placer system uses either polling or pub/sub to become aware of the existence of the task and fulfillment intent

• Fulfiller may update the Task to indicate interim progress notes

• Placer either polls the Task to note acceptance and changes or uses a subscription to determine the same

• Fulfiller POSTs an event resource to its own system or to a queue server system

• Fulfiller Updates the Task resource to change its status to completed and to point to the event resource

• Placer system becomes aware of the update via polling or subscription

• Placer system retrieves the event

• Placer system marks the request as "complete"

12.9 Usage Examples for Workflow Patterns

URL: https://hl7.org/fhir/R4/workflow-examples.html
FHIR Infrastructure Work Group | Maturity Level: 2 | Standards Status: Trial Use

12.9 Usage Examples for Workflow Patterns

Future home for examples

12.10 Resource Appointment - Content

URL: https://hl7.org/fhir/R4/appointment.html
Patient Administration Work Group | Maturity Level: 3 | Trial Use | Security Category: Patient | Compartments: Device, Patient, Practitioner, RelatedPerson

A booking of a healthcare event among patient(s), practitioner(s), related person(s) and/or device(s) for a specific date/time. This may result in one or more Encounter(s).

12.10.1 Scope and Usage

Appointment resources are used to provide information about a planned meeting that may be in the future or past. The resource only describes a single meeting, a series of repeating visits would require multiple appointment resources to be created for each instance. Examples include a scheduled surgery, a follow-up for a clinical visit, a scheduled conference call between clinicians to discuss a case, the reservation of a piece of diagnostic equipment for a particular use, etc. The visit scheduled by an appointment may be in person or remote (by phone, video conference, etc.) All that matters is that the time and usage of one or more individuals, locations and/or pieces of equipment is being fully or partially reserved for a designated period of time.

This definition takes the concepts of appointments in a clinical setting and also extends them to be relevant in the community healthcare space, and to ease exposure to other appointment / calendar standards widely used outside of healthcare.

12.10.1.1 The basic workflow to create an appointment

Discovery/Addressing

Before an appointment can be made, the address/endpoint details of the resource that we want to schedule an appointment with must be determined. This is often based on the healthcare service type and any formatting information which indicates how to make the request. This is typically handled via the Schedule resource.

Checking Availability on the Schedule (optional)

This optional step permits the checking of any existing available times (Slot resources associated with a selected Schedule) that can be booked against. Just because a time is indicated as available doesn't guarantee that an appointment can be made. The booking system that is going to process the request may make other qualifying decisions to determine if the appointment can be made, such as permissions, assessments, availability of other resources, etc.

This step is optional, as the creation of the appointment is never a guaranteed action. But by performing this availability check, you can increase the chances of making a successful booking.

Making the Appointment Request

When an appointment is required, a requester creates new Appointment resource with the Appointment.status="proposed". All included participants (optional or mandatory) should have the status="needs-action" to allow filtering and displaying appointments to user-participants for accepting or rejecting new and updated requests. Based on internal system business rules, certain statuses may be automatically updated, for example: "reject because the requested participant is on vacation" or "this type of user is not allowed to request those specific appointments".

Replying to the request

The reply process is simply performed by the person/system handing the requests, updating the participant statuses on the appointment as needed. If there are multiple systems involved, then these will create AppointmentResponse entries with the desired statuses.

Once all participants have their participation status created/updated (and the main system marking the appointment participant records with the AppointmentResponse statuses) then the overall status of the Appointment is updated.

Checking the overall status (Requester)

The requester (organizer) of the appointment checks for the overall status of the appointment (and appointment responses, where applicable) using FHIR pub-sub techniques.

Where the participant statuses indicate that a re-scheduling is required, then the process may start again, with other systems replying to a new set of times.

Waitlisting the Appointment (optional)

This optional step permits creating a waitlisted appointment. This could occur if an appointment needs to be booked into a time that is not ideal for the patient due to lack of available time slots. In this workflow, there would be two appointments, the booked appointment in the time slot that is currently available, and the waitlisted appointment with a requestedPeriod spanning the time that the patient would prefer if new slots become available.

If new time slots become available during the requestedPeriod, the scheduling system, or staff at the scheduling organization, can notify the patient that a new time slot is available. If the patient chooses, the waitlisted appointment would then be booked into that specific slot, and the previously booked appointment would be canceled. The specific business process for notifying patients of new availability is not specified, and is up to the implementing system to determine.

12.10.1.2 There are 2 typical workflows that occur with appointments

Outlook Style - Community

These types of requests are typically handled by selecting a specific time from a list of available slots, then making the request for that timeslot.

Hospital Scheduling - Clinical

Clinical scheduling is often far more complex in its requirements and processing. Often this involves checking multiple availabilities across multiple systems and timing with other internal systems, not just those exposed by the Slot resources.

Consideration should be given to situations where scheduling needs to be handled in more of a queue-like process.

Implementation Note: Note: This type of clinical appointment scheduling has not been specifically covered with this definition of the Appointment resource (and other related resources), however if you would like to contribute to the modification of this resource to cover these use cases, please contact the HL7 Patient Administration work-group.

12.10.2 Boundaries and Relationships

12.10.2.1 Appointment Request/Response Pattern

When using a request-response style of appointment this is done using Appointment and AppointmentResponse resources. The request is made in the form of an Appointment with a proposed or pending status, and the list of actors with a participation status of "needs-action".

Participants in the appointment respond with their acceptance (or not) to the appointment by creating AppointmentResponse resources. Once all the participants have replied, then the Appointment resource is able to be updated with an overall status which collates the results of all the participants and presents the approved details of the appointment.

The participant type property can be used to represent a specific role that a practitioner is required to perform for the appointment. This could be specified without an actor when the actual practitioner is not known, and will be filled in closer to the scheduled time. This property must be the same between the Appointment-participant and the AppointmentResponse so that the appropriate values can be allocated. If you need multiple actors of a specific type, then multiple participants with that type value are included on the appointment.

12.10.2.2 Appointment Statuses and Encounters

Appointments can be considered as Administrative only, and the Encounter is expected to have clinical implications.

In general, it is expected that appointments will result in the creation of an Encounter. The encounter is typically created when the service starts, not when the patient arrives. When the patient arrives, an appointment can be marked with a status of Arrived.

In an Emergency Room context, the appointment Resource is probably not appropriate to be used. In these cases, an Encounter should be created.

The Appointment request pattern used is different from the order-response pattern used elsewhere in FHIR. This is due to the close relationship to the iCal standard. Many non-clinical systems use generic non-health appointment systems which implement this standard, and the desire to integrate with the consumer who has no access to health based software is highly desirable. The mappings to the iCal standard have been provided to guide implementation of gateways between FHIR servers and iCal systems.

12.10.2.3 Appointment Locations and Participation

The location of the appointment is to be defined by using a participant that references a Location or HealthcareService resource. This permits the location to also have its availability checked via a schedule and any conflicts more easily managed.

This resource is referenced by AppointmentResponse, CarePlan, Encounter and ImagingStudy

12.10.3 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints

Appointment | ITU | | DomainResource | A booking of a healthcare event among patient(s), practitioner(s), related person(s) and/or device(s) for a specific date/time. This may result in one or more Encounter(s) + Rule: Either start and end are specified, or neither + Rule: Only proposed or cancelled appointments can be missing start/end dates + Rule: Cancelation reason is only used for appointments that have been cancelled, or no-show Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | External Ids for this item
status | ?!Σ | 1..1 | code | proposed | pending | booked | arrived | fulfilled | cancelled | noshow | entered-in-error | checked-in | waitlist AppointmentStatus (Required)
cancelationReason | Σ | 0..1 | CodeableConcept | The coded reason for the appointment being cancelled Appointment cancellation reason (Example)
serviceCategory | Σ | 0..* | CodeableConcept | A broad categorization of the service that is to be performed during this appointment Service category (Example)
serviceType | Σ | 0..* | CodeableConcept | The specific service that is to be performed during this appointment Service type (Example)
specialty | Σ | 0..* | CodeableConcept | The specialty of a practitioner that would be required to perform the service requested in this appointment Practice Setting Code Value Set (Preferred)
appointmentType | Σ | 0..1 | CodeableConcept | The style of appointment or patient that has been booked in the slot (not service type) v2 Appointment Reason Codes (Preferred)
reasonCode | Σ | 0..* | CodeableConcept | Coded reason this appointment is scheduled Encounter Reason Codes (Preferred)
reasonReference | | 0..* | Reference(Condition | Procedure | Observation | ImmunizationRecommendation) | Reason the appointment is to take place (resource)
priority | | 0..1 | unsignedInt | Used to make informed decisions if needing to re-prioritize
description | | 0..1 | string | Shown on a subject line in a meeting request, or appointment list
supportingInformation | | 0..* | Reference(Any) | Additional information to support the appointment
start | Σ | 0..1 | instant | When appointment is to take place
end | Σ | 0..1 | instant | When appointment is to conclude
minutesDuration | | 0..1 | positiveInt | Can be less than start/end (e.g. estimate)
slot | | 0..* | Reference(Slot) | The slots that this appointment is filling
created | | 0..1 | dateTime | The date that this appointment was initially created
comment | | 0..1 | string | Additional comments
patientInstruction | | 0..1 | string | Detailed information and instructions for the patient
basedOn | | 0..* | Reference(ServiceRequest) | The service request this appointment is allocated to assess
participant | I | 1..* | BackboneElement | Participants involved in appointment + Rule: Either the type or actor on the participant SHALL be specified
type | Σ | 0..* | CodeableConcept | Role of participant in the appointment Participant type (Extensible)
actor | Σ | 0..1 | Reference(Patient | Practitioner | PractitionerRole | RelatedPerson | Device | HealthcareService | Location) | Person, Location/HealthcareService or Device
required | Σ | 0..1 | code | required | optional | information-only ParticipantRequired (Required)
status | Σ | 1..1 | code | accepted | declined | tentative | needs-action ParticipationStatus (Required)
period | | 0..1 | Period | Participation period of the actor
requestedPeriod | | 0..* | Period | Potential date/time interval(s) requested to allocate the appointment within
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Appointment xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier External Ids for this item --></identifier>
 <status value="[code]"/><!-- 1..1 proposed | pending | booked | arrived | fulfilled | cancelled | noshow | entered-in-
    error | checked-in | waitlist -->
 <cancelationReason><!-- 0..1 CodeableConcept The coded reason for the appointment being cancelled
    --></cancelationReason>
 <serviceCategory><!-- 0..* CodeableConcept A broad categorization of the service that is to be performed during this
    appointment --></serviceCategory>
 <serviceType><!-- 0..* CodeableConcept The specific service that is to be performed during this appointment
    --></serviceType>
 <specialty><!-- 0..* CodeableConcept The specialty of a practitioner that would be required to perform the service
    requested in this appointment --></specialty>
 <appointmentType><!-- 0..1 CodeableConcept The style of appointment or patient that has been booked in the slot (not
    service type) --></appointmentType>
 <reasonCode><!-- 0..* CodeableConcept Coded reason this appointment is scheduled --></reasonCode>
 <reasonReference><!-- 0..* Reference(Condition|Procedure|Observation|
   ImmunizationRecommendation) Reason the appointment is to take place (resource) --></reasonReference>
 <priority value="[unsignedInt]"/><!-- 0..1 Used to make informed decisions if needing to re-prioritize -->
 <description value="[string]"/><!-- 0..1 Shown on a subject line in a meeting request, or appointment list -->
 <supportingInformation><!-- 0..* Reference(Any) Additional information to support the appointment
    --></supportingInformation>
 <start value="[instant]"/><!-- 0..1 When appointment is to take place -->
 <end value="[instant]"/><!-- 0..1 When appointment is to conclude -->
 <minutesDuration value="[positiveInt]"/><!-- 0..1 Can be less than start/end (e.g. estimate) -->
 <slot><!-- 0..* Reference(Slot) The slots that this appointment is filling --></slot>
 <created value="[dateTime]"/><!-- 0..1 The date that this appointment was initially created -->
 <comment value="[string]"/><!-- 0..1 Additional comments -->
 <patientInstruction value="[string]"/><!-- 0..1 Detailed information and instructions for the patient -->
 <basedOn><!-- 0..* Reference(ServiceRequest) The service request this appointment is allocated to assess --></basedOn>
 <participant>  <!-- 1..* Participants involved in appointment -->
  <type><!-- 0..* CodeableConcept Role of participant in the appointment --></type>
  <actor><!-- 0..1 Reference(Patient|Practitioner|PractitionerRole|RelatedPerson|
    Device|HealthcareService|Location) Person, Location/HealthcareService or Device --></actor>
```

```
  <required value="[code]"/><!-- 0..1 required | optional | information-only -->
  <status value="[code]"/><!-- 1..1 accepted | declined | tentative | needs-action -->
  <period><!-- 0..1 Period Participation period of the actor --></period>
 </participant>
 <requestedPeriod><!-- 0..* Period Potential date/time interval(s) requested to allocate the appointment within
    --></requestedPeriod>
</Appointment>
```

JSON Template

```
{
  "resourceType" : "Appointment",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // External Ids for this item
  "status" : "<code>", // R!  proposed | pending | booked | arrived | fulfilled | cancelled | noshow | entered-in-error
    | checked-in | waitlist
  "cancelationReason" : { CodeableConcept }, // The coded reason for the appointment being cancelled
  "serviceCategory" : [{ CodeableConcept }], // A broad categorization of the service that is to be performed during
    this appointment
  "serviceType" : [{ CodeableConcept }], // The specific service that is to be performed during this appointment
  "specialty" : [{ CodeableConcept }], // The specialty of a practitioner that would be required to perform the service
    requested in this appointment
  "appointmentType" : { CodeableConcept }, // The style of appointment or patient that has been booked in the slot (not
    service type)
  "reasonCode" : [{ CodeableConcept }], // Coded reason this appointment is scheduled
  "reasonReference" : [{ Reference(Condition|Procedure|Observation|
   ImmunizationRecommendation) }], // Reason the appointment is to take place (resource)
  "priority" : "<unsignedInt>", // Used to make informed decisions if needing to re-prioritize
  "description" : "<string>", // Shown on a subject line in a meeting request, or appointment list
  "supportingInformation" : [{ Reference(Any) }], // Additional information to support the appointment
  "start" : "<instant>", // When appointment is to take place
  "end" : "<instant>", // When appointment is to conclude
  "minutesDuration" : "<positiveInt>", // Can be less than start/end (e.g. estimate)
  "slot" : [{ Reference(Slot) }], // The slots that this appointment is filling
  "created" : "<dateTime>", // The date that this appointment was initially created
  "comment" : "<string>", // Additional comments
  "patientInstruction" : "<string>", // Detailed information and instructions for the patient
  "basedOn" : [{ Reference(ServiceRequest) }], // The service request this appointment is allocated to assess
  "participant" : [{ // R!  Participants involved in appointment
    "type" : [{ CodeableConcept }], // Role of participant in the appointment
    "actor" : { Reference(Patient|Practitioner|PractitionerRole|RelatedPerson|
    Device|HealthcareService|Location) }, // Person, Location/HealthcareService or Device
    "required" : "<code>", // required | optional | information-only
    "status" : "<code>", // R!  accepted | declined | tentative | needs-action
    "period" : { Period } // Participation period of the actor
  }],
  "requestedPeriod" : [{ Period }] // Potential date/time interval(s) requested to allocate the appointment within
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Appointment;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Appointment.identifier [ Identifier ], ... ; # 0..* External Ids for this item
  fhir:Appointment.status [ code ]; # 1..1 proposed | pending | booked | arrived | fulfilled | cancelled | noshow |
    entered-in-error | checked-in | waitlist
  fhir:Appointment.cancelationReason [ CodeableConcept ]; # 0..1 The coded reason for the appointment being cancelled
  fhir:Appointment.serviceCategory [ CodeableConcept ], ... ; # 0..* A broad categorization of the service that is to be
     performed during this appointment
  fhir:Appointment.serviceType [ CodeableConcept ], ... ; # 0..* The specific service that is to be performed during
    this appointment
  fhir:Appointment.specialty [ CodeableConcept ], ... ; # 0..* The specialty of a practitioner that would be required to
     perform the service requested in this appointment
  fhir:Appointment.appointmentType [ CodeableConcept ]; # 0..1 The style of appointment or patient that has been booked
    in the slot (not service type)
  fhir:Appointment.reasonCode [ CodeableConcept ], ... ; # 0..* Coded reason this appointment is scheduled
  fhir:Appointment.reasonReference [ Reference(Condition|Procedure|Observation|ImmunizationRecommendation) ], ... ; #
    0..* Reason the appointment is to take place (resource)
  fhir:Appointment.priority [ unsignedInt ]; # 0..1 Used to make informed decisions if needing to re-prioritize
  fhir:Appointment.description [ string ]; # 0..1 Shown on a subject line in a meeting request, or appointment list
  fhir:Appointment.supportingInformation [ Reference(Any) ], ... ; # 0..* Additional information to support the
    appointment
  fhir:Appointment.start [ instant ]; # 0..1 When appointment is to take place
  fhir:Appointment.end [ instant ]; # 0..1 When appointment is to conclude
  fhir:Appointment.minutesDuration [ positiveInt ]; # 0..1 Can be less than start/end (e.g. estimate)
  fhir:Appointment.slot [ Reference(Slot) ], ... ; # 0..* The slots that this appointment is filling
  fhir:Appointment.created [ dateTime ]; # 0..1 The date that this appointment was initially created
  fhir:Appointment.comment [ string ]; # 0..1 Additional comments
  fhir:Appointment.patientInstruction [ string ]; # 0..1 Detailed information and instructions for the patient
  fhir:Appointment.basedOn [ Reference(ServiceRequest) ], ... ; # 0..* The service request this appointment is allocated
     to assess
  fhir:Appointment.participant [ # 1..* Participants involved in appointment
    fhir:Appointment.participant.type [ CodeableConcept ], ... ; # 0..* Role of participant in the appointment
    fhir:Appointment.participant.actor [
```

```
    Reference(Patient|Practitioner|PractitionerRole|RelatedPerson|Device|HealthcareService|
  Location) ]; # 0..1 Person, Location/HealthcareService or Device
    fhir:Appointment.participant.required [ code ]; # 0..1 required | optional | information-only
    fhir:Appointment.participant.status [ code ]; # 1..1 accepted | declined | tentative | needs-action
    fhir:Appointment.participant.period [ Period ]; # 0..1 Participation period of the actor
  ], ...;
  fhir:Appointment.requestedPeriod [ Period ], ... ; # 0..* Potential date/time interval(s) requested to allocate the
    appointment within
]
```

Changes since R3

Appointment

Appointment.status | Change value set from http://hl7.org/fhir/ValueSet/appointmentstatus to http://hl7.org/fhir/ValueSet/appointmentstatus|4.0.1

Appointment.cancelationReason | Added Element

Appointment.serviceCategory | Max Cardinality changed from 1 to *

Appointment.reasonCode | Added Element

Appointment.reasonReference | Added Element

Appointment.patientInstruction | Added Element

Appointment.basedOn | Added Element

Appointment.participant.actor | Type Reference: Added Target Type PractitionerRole

Appointment.participant.required | Change value set from http://hl7.org/fhir/ValueSet/participantrequired to http://hl7.org/fhir/ValueSet/participantrequired|4.0.1

Appointment.participant.status | Change value set from http://hl7.org/fhir/ValueSet/participationstatus to http://hl7.org/fhir/ValueSet/participationstatus|4.0.1

Appointment.participant.period | Added Element

Appointment.reason | deleted

Appointment.indication | deleted

Appointment.incomingReferral | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 3 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid.)

Structure

Name | Flags | Card. | Type | Description & Constraints

Appointment | ITU | | DomainResource | A booking of a healthcare event among patient(s), practitioner(s), related person(s) and/or device(s) for a specific date/time. This may result in one or more Encounter(s) + Rule: Either start and end are specified, or neither + Rule: Only proposed or cancelled appointments can be missing start/end dates + Rule: Cancelation reason is only used for appointments that have been cancelled, or no-show Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

identifier | Σ | 0..* | Identifier | External Ids for this item

status | ?!Σ | 1..1 | code | proposed | pending | booked | arrived | fulfilled | cancelled | noshow | entered-in-error | checked-in | waitlist AppointmentStatus (Required)

cancelationReason | Σ | 0..1 | CodeableConcept | The coded reason for the appointment being cancelled Appointment cancellation reason (Example)

serviceCategory | Σ | 0..* | CodeableConcept | A broad categorization of the service that is to be performed during this appointment Service category (Example)

serviceType | Σ | 0..* | CodeableConcept | The specific service that is to be performed during this appointment Service type (Example)

specialty | Σ | 0..* | CodeableConcept | The specialty of a practitioner that would be required to perform the service requested in this appointment Practice Setting Code Value Set (Preferred)

appointmentType | Σ | 0..1 | CodeableConcept | The style of appointment or patient that has been booked in the slot (not service type) v2 Appointment Reason Codes (Preferred)

reasonCode | Σ | 0..* | CodeableConcept | Coded reason this appointment is scheduled Encounter Reason Codes (Preferred)

reasonReference | | 0..* | Reference(Condition | Procedure | Observation | ImmunizationRecommendation) | Reason the appointment is to take place (resource)

priority | | 0..1 | unsignedInt | Used to make informed decisions if needing to re-prioritize

description | | 0..1 | string | Shown on a subject line in a meeting request, or appointment list

supportingInformation | | 0..* | Reference(Any) | Additional information to support the appointment

start | Σ | 0..1 | instant | When appointment is to take place

end | Σ | 0..1 | instant | When appointment is to conclude

minutesDuration | | 0..1 | positiveInt | Can be less than start/end (e.g. estimate)

slot | | 0..* | Reference(Slot) | The slots that this appointment is filling

created | | 0..1 | dateTime | The date that this appointment was initially created

comment | | 0..1 | string | Additional comments

patientInstruction | | 0..1 | string | Detailed information and instructions for the patient

basedOn | | 0..* | Reference(ServiceRequest) | The service request this appointment is allocated to assess

participant | I | 1..* | BackboneElement | Participants involved in appointment + Rule: Either the type or actor on the participant SHALL be specified

type | Σ | 0..* | CodeableConcept | Role of participant in the appointment Participant type (Extensible)

actor | Σ | 0..1 | Reference(Patient | Practitioner | PractitionerRole | RelatedPerson | Device | HealthcareService | Location) | Person, Location/HealthcareService or Device

required | Σ | 0..1 | code | required | optional | information-only ParticipantRequired (Required)

status | Σ | 1..1 | code | accepted | declined | tentative | needs-action ParticipationStatus (Required)

period | | 0..1 | Period | Participation period of the actor
requestedPeriod | | 0..* | Period | Potential date/time interval(s) requested to allocate the appointment within
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Appointment xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier External Ids for this item --></identifier>
 <status value="[code]"/><!-- 1..1 proposed | pending | booked | arrived | fulfilled | cancelled | noshow | entered-in-
    error | checked-in | waitlist -->
 <cancelationReason><!-- 0..1 CodeableConcept The coded reason for the appointment being cancelled
    --></cancelationReason>
 <serviceCategory><!-- 0..* CodeableConcept A broad categorization of the service that is to be performed during this
    appointment --></serviceCategory>
 <serviceType><!-- 0..* CodeableConcept The specific service that is to be performed during this appointment
    --></serviceType>
 <specialty><!-- 0..* CodeableConcept The specialty of a practitioner that would be required to perform the service
    requested in this appointment --></specialty>
 <appointmentType><!-- 0..1 CodeableConcept The style of appointment or patient that has been booked in the slot (not
    service type) --></appointmentType>
 <reasonCode><!-- 0..* CodeableConcept Coded reason this appointment is scheduled --></reasonCode>
 <reasonReference><!-- 0..* Reference(Condition|Procedure|Observation|
   ImmunizationRecommendation) Reason the appointment is to take place (resource) --></reasonReference>
 <priority value="[unsignedInt]"/><!-- 0..1 Used to make informed decisions if needing to re-prioritize -->
 <description value="[string]"/><!-- 0..1 Shown on a subject line in a meeting request, or appointment list -->
 <supportingInformation><!-- 0..* Reference(Any) Additional information to support the appointment
    --></supportingInformation>
 <start value="[instant]"/><!-- 0..1 When appointment is to take place -->
 <end value="[instant]"/><!-- 0..1 When appointment is to conclude -->
 <minutesDuration value="[positiveInt]"/><!-- 0..1 Can be less than start/end (e.g. estimate) -->
 <slot><!-- 0..* Reference(Slot) The slots that this appointment is filling --></slot>
 <created value="[dateTime]"/><!-- 0..1 The date that this appointment was initially created -->
 <comment value="[string]"/><!-- 0..1 Additional comments -->
 <patientInstruction value="[string]"/><!-- 0..1 Detailed information and instructions for the patient -->
 <basedOn><!-- 0..* Reference(ServiceRequest) The service request this appointment is allocated to assess --></basedOn>
 <participant>  <!-- 1..* Participants involved in appointment -->
  <type><!-- 0..* CodeableConcept Role of participant in the appointment --></type>
  <actor><!-- 0..1 Reference(Patient|Practitioner|PractitionerRole|RelatedPerson|
    Device|HealthcareService|Location) Person, Location/HealthcareService or Device --></actor>
  <required value="[code]"/><!-- 0..1 required | optional | information-only -->
  <status value="[code]"/><!-- 1..1 accepted | declined | tentative | needs-action -->
  <period><!-- 0..1 Period Participation period of the actor --></period>
 </participant>
 <requestedPeriod><!-- 0..* Period Potential date/time interval(s) requested to allocate the appointment within
    --></requestedPeriod>
</Appointment>
```

JSON Template

```
{
  "resourceType" : "Appointment",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // External Ids for this item
  "status" : "<code>", // R!  proposed | pending | booked | arrived | fulfilled | cancelled | noshow | entered-in-error
    | checked-in | waitlist
  "cancelationReason" : { CodeableConcept }, // The coded reason for the appointment being cancelled
  "serviceCategory" : [{ CodeableConcept }], // A broad categorization of the service that is to be performed during
    this appointment
  "serviceType" : [{ CodeableConcept }], // The specific service that is to be performed during this appointment
  "specialty" : [{ CodeableConcept }], // The specialty of a practitioner that would be required to perform the service
    requested in this appointment
  "appointmentType" : { CodeableConcept }, // The style of appointment or patient that has been booked in the slot (not
    service type)
  "reasonCode" : [{ CodeableConcept }], // Coded reason this appointment is scheduled
  "reasonReference" : [{ Reference(Condition|Procedure|Observation|
   ImmunizationRecommendation) }], // Reason the appointment is to take place (resource)
  "priority" : "<unsignedInt>", // Used to make informed decisions if needing to re-prioritize
  "description" : "<string>", // Shown on a subject line in a meeting request, or appointment list
  "supportingInformation" : [{ Reference(Any) }], // Additional information to support the appointment
  "start" : "<instant>", // When appointment is to take place
  "end" : "<instant>", // When appointment is to conclude
  "minutesDuration" : "<positiveInt>", // Can be less than start/end (e.g. estimate)
  "slot" : [{ Reference(Slot) }], // The slots that this appointment is filling
  "created" : "<dateTime>", // The date that this appointment was initially created
  "comment" : "<string>", // Additional comments
  "patientInstruction" : "<string>", // Detailed information and instructions for the patient
  "basedOn" : [{ Reference(ServiceRequest) }], // The service request this appointment is allocated to assess
  "participant" : [{ // R!  Participants involved in appointment
    "type" : [{ CodeableConcept }], // Role of participant in the appointment
    "actor" : { Reference(Patient|Practitioner|PractitionerRole|RelatedPerson|
    Device|HealthcareService|Location) }, // Person, Location/HealthcareService or Device
    "required" : "<code>", // required | optional | information-only
    "status" : "<code>", // R!  accepted | declined | tentative | needs-action
    "period" : { Period } // Participation period of the actor
  }],
```

```
  "requestedPeriod" : [{ Period }] // Potential date/time interval(s) requested to allocate the appointment within
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Appointment;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Appointment.identifier [ Identifier ], ... ; # 0..* External Ids for this item
  fhir:Appointment.status [ code ]; # 1..1 proposed | pending | booked | arrived | fulfilled | cancelled | noshow |
    entered-in-error | checked-in | waitlist
  fhir:Appointment.cancelationReason [ CodeableConcept ]; # 0..1 The coded reason for the appointment being cancelled
  fhir:Appointment.serviceCategory [ CodeableConcept ], ... ; # 0..* A broad categorization of the service that is to be
     performed during this appointment
  fhir:Appointment.serviceType [ CodeableConcept ], ... ; # 0..* The specific service that is to be performed during
    this appointment
  fhir:Appointment.specialty [ CodeableConcept ], ... ; # 0..* The specialty of a practitioner that would be required to
     perform the service requested in this appointment
  fhir:Appointment.appointmentType [ CodeableConcept ]; # 0..1 The style of appointment or patient that has been booked
    in the slot (not service type)
  fhir:Appointment.reasonCode [ CodeableConcept ], ... ; # 0..* Coded reason this appointment is scheduled
  fhir:Appointment.reasonReference [ Reference(Condition|Procedure|Observation|ImmunizationRecommendation) ], ... ; #
    0..* Reason the appointment is to take place (resource)
  fhir:Appointment.priority [ unsignedInt ]; # 0..1 Used to make informed decisions if needing to re-prioritize
  fhir:Appointment.description [ string ]; # 0..1 Shown on a subject line in a meeting request, or appointment list
  fhir:Appointment.supportingInformation [ Reference(Any) ], ... ; # 0..* Additional information to support the
    appointment
  fhir:Appointment.start [ instant ]; # 0..1 When appointment is to take place
  fhir:Appointment.end [ instant ]; # 0..1 When appointment is to conclude
  fhir:Appointment.minutesDuration [ positiveInt ]; # 0..1 Can be less than start/end (e.g. estimate)
  fhir:Appointment.slot [ Reference(Slot) ], ... ; # 0..* The slots that this appointment is filling
  fhir:Appointment.created [ dateTime ]; # 0..1 The date that this appointment was initially created
  fhir:Appointment.comment [ string ]; # 0..1 Additional comments
  fhir:Appointment.patientInstruction [ string ]; # 0..1 Detailed information and instructions for the patient
  fhir:Appointment.basedOn [ Reference(ServiceRequest) ], ... ; # 0..* The service request this appointment is allocated
     to assess
  fhir:Appointment.participant [ # 1..* Participants involved in appointment
    fhir:Appointment.participant.type [ CodeableConcept ], ... ; # 0..* Role of participant in the appointment
    fhir:Appointment.participant.actor [
    Reference(Patient|Practitioner|PractitionerRole|RelatedPerson|Device|HealthcareService|
  Location) ]; # 0..1 Person, Location/HealthcareService or Device
    fhir:Appointment.participant.required [ code ]; # 0..1 required | optional | information-only
    fhir:Appointment.participant.status [ code ]; # 1..1 accepted | declined | tentative | needs-action
    fhir:Appointment.participant.period [ Period ]; # 0..1 Participation period of the actor
  ], ...;
  fhir:Appointment.requestedPeriod [ Period ], ... ; # 0..* Potential date/time interval(s) requested to allocate the
    appointment within
]
```

Changes since Release 3

Appointment
Appointment.status | Change value set from http://hl7.org/fhir/ValueSet/appointmentstatus to http://hl7.org/fhir/ValueSet/appointmentstatus|4.0.1
Appointment.cancelationReason | Added Element
Appointment.serviceCategory | Max Cardinality changed from 1 to *
Appointment.reasonCode | Added Element
Appointment.reasonReference | Added Element
Appointment.patientInstruction | Added Element
Appointment.basedOn | Added Element
Appointment.participant.actor | Type Reference: Added Target Type PractitionerRole
Appointment.participant.required | Change value set from http://hl7.org/fhir/ValueSet/participantrequired to http://hl7.org/fhir/ValueSet/participantrequired|4.0.1
Appointment.participant.status | Change value set from http://hl7.org/fhir/ValueSet/participationstatus to http://hl7.org/fhir/ValueSet/participationstatus|4.0.1
Appointment.participant.period | Added Element
Appointment.reason | deleted
Appointment.indication | deleted
Appointment.incomingReferral | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 3 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid.)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

12.10.3.1 Terminology Bindings

Path | Definition | Type | Reference

Appointment.status | The free/busy status of an appointment. | Required | AppointmentStatus
Appointment.cancelationReason | | Example | AppointmentCancellationReason
Appointment.serviceCategory | | Example | ServiceCategory
Appointment.serviceType | | Example | ServiceType
Appointment.specialty | | Preferred | PracticeSettingCodeValueSet
Appointment.appointmentType | | Preferred | v2.0276
Appointment.reasonCode | The Reason for the appointment to take place. | Preferred | EncounterReasonCodes
Appointment.participant.type | Role of participant in encounter. | Extensible | ParticipantType
Appointment.participant.required | Is the Participant required to attend the appointment. | Required | ParticipantRequired
Appointment.participant.status | The Participation status of an appointment. | Required | ParticipationStatus

12.10.3.2 Constraints

id | Level | Location | Description | Expression
app-1 | Rule | Appointment.participant | Either the type or actor on the participant SHALL be specified | type.exists() or actor.exists()
app-2 | Rule | (base) | Either start and end are specified, or neither | start.exists() = end.exists()
app-3 | Rule | (base) | Only proposed or cancelled appointments can be missing start/end dates | (start.exists() and end.exists()) or (status in ('proposed' | 'cancelled' | 'waitlist'))
app-4 | Rule | (base) | Cancelation reason is only used for appointments that have been cancelled, or no-show | Appointment.cancelationReason.exists() implies (Appointment.status='no-show' or Appointment.status='cancelled')

12.10.4 Typical Status Transition Examples:

12.10.4.1 Typical flow of statuses for an appointment:

Activity Description | Slot | Appointment | Appointment Response | Encounter
The schedule is created/published (Role: Scheduler) | freeBusyType = FREE
An appointment request is created after locating an available slot (Role: Requester) | | status = pending participant.status = needs-action
The appointment request is processed and the slot status updated (Role: Scheduler) | freeBusyType = BUSY-TENTATIVE
The appointment is accepted as described – by all participants (Role: Participant(s)) | | | participantStatus = accepted
The appointment is confirmed as accepted by all participants (Role: Scheduler) | freeBusyType = BUSY | status = booked participant.status = accepted
Optional: Preparation for the appointment begins – could be preparing a room for the appointment, etc. (Role: Participants/Admin) | | | | status = planned (optional) location.status = planned
The patient arrives for the appointment, often sitting in a waiting room (Role: Admin) | | status = arrived | | status = arrived location.status = present
The practitioner and the patient meet and the provision of the service begins (Role: Scheduler/Participant(s)/Admin) | | status = fulfilled | | status = in-progress
The encounter concludes (Role: Scheduler/Participant(s)/Admin) | | | | status = finished

12.10.4.2 Flow for the rejection/cancellation of an appointment

Activity Description | Slot | Appointment | Appointment Response
The schedule is created/published (Role: Scheduler) | freeBusyType = FREE
An appointment request is created (Role: Requester) | | status = pending participant.status = needs-action
The appointment request is processed and the slot status updated (Role: Scheduler) | freeBusyType = BUSY-TENTATIVE
Participant declines the Appointment (Role: Participant) | | | participantStatus = declined
The appointment is cancelled (Role: Scheduler) | freeBusyType = FREE | status = cancelled participant.status = declined

12.10.4.3 Flow for re-negotiation:

Activity Description | Slot | Appointment | Appointment Response
The schedule is created/published (Role: Scheduler) | freeBusyType = FREE
An appointment is requested (e.g. with Brian and Peter) (Role: Requester) | | status = proposed participant(Brian).status = needs-action participant(Peter).status = needs-action
The schedule is updated to inform others of interest in the slot (Role: Scheduler) | freeBusyType = BUSY-TENTATIVE
Brian accepts the appointment (Role: Participant-Brian) | | | (Brian).participantStatus = accepted
Appointment is updated with Brian's status (Role: Scheduler) | | status = pending participant(Brian).status = accepted
Peter suggests a new time (Role: Participant-Peter) | | | (Peter).participantStatus = tentative (with new time)
Appointment is updated with new time, and indicates that action is needed by both participants (Role: Scheduler) | | (new time details updated) participant(Brian).status = needs-action participant(Peter).status = needs-action
Brian accepts the appointment (Role: Participant-Brian) | | | (Brian).participantStatus = accepted
Appointment updated (Role: Scheduler) | | participant(Brian).status = accepted
(Role: Participant-Peter) | | | (Peter).participantStatus = accepted
Appointment updated (Role: Scheduler) | freeBusyType = BUSY | status = booked participant(Peter).status = accepted

12.10.4.4 Flow for a patient no-show:

Activity Description | Slot | Appointment | Appointment Response | Encounter
(from typical status flow) | freeBusyType = BUSY | status = booked participant.status = accepted
Appointment is updated as a noshow (Role: Scheduler/Admin) | | status = noshow | | (no encounter created)

12.10.4.5 Flow for a patient waitlist:

Activity Description | Appointment (inconvenient) | Appointment (preferred)

An appointment is booked for an inconvenient time using a typical status flow | status = booked participant.status = accepted
Waitlist appointment created | status = booked participant.status = accepted | status = waitlist requestedPeriod = (more convenient time period)
Patient notified of availability of a better slot | status = booked | status = proposed participant.status = needs-action
Patient accepts better slot | status = cancelled | status = booked participant.status = accepted

12.10.5 Notes:

• Placer/Filler (HL7 v2) The appointment information is effectively the same between the filler and placer, and given the nature of the FHIR resource, there is only a single resource for both purposes. The placer is the actor that performs the PUT or POST operation on the resource, and the filler is the actor that receives these resource messages and processes the information and makes a decision if the appointment can be used.

• Terminology - ServiceCategory, ServiceType and Specialty The 3 core clinical terminologies associated with the appointment may have relationships with each other through the terminology server, but not these might not be visible in the actual appointment resource or profile.

• Interaction with other Standards The strong desire is that implementers of this resource should consider providing this resource in the iCalendar format as an alternative representation. Many 3rd party applications and component providers have parsers and user interface controls to display this information. This may lower the entry point to integrate outside the health-care specific applications, and into the consumer space. This would permit the easier creation of a mobile application that creates appointments in the devices native calendar. The iCalendar specification can be found at http://www.ietf.org/rfc/rfc2445.txt .

Trial-Use Note: Implementer feedback is sought on the values for Appointment.priority and how interoperable they are. Using an extension to record a CodeableConcept for named values may be tested at a future Connectathon. Implementer feedback is also sought to clarify desired relationship linkages between ServiceCategory, ServiceType and Specialty, along with how they have approached the definition.

Feedback here .

12.10.6 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
actor | reference | Any one of the individuals participating in the appointment | Appointment.participant.actor (Practitioner, Device, Patient, HealthcareService, PractitionerRole, RelatedPerson, Location)
appointment-type | token | The style of appointment or patient that has been booked in the slot (not service type) | Appointment.appointmentType
based-on | reference | The service request this appointment is allocated to assess | Appointment.basedOn (ServiceRequest)
date | date | Appointment date/time. | Appointment.start
identifier | token | An Identifier of the Appointment | Appointment.identifier
location | reference | This location is listed in the participants of the appointment | Appointment.participant.actor.where(resolve() is Location) (Location)
part-status | token | The Participation status of the subject, or other participant on the appointment. Can be used to locate participants that have not responded to meeting requests. | Appointment.participant.status
patient | reference | One of the individuals of the appointment is this patient | Appointment.participant.actor.where(resolve() is Patient) (Patient)
practitioner | reference | One of the individuals of the appointment is this practitioner | Appointment.participant.actor.where(resolve() is Practitioner) (Practitioner)
reason-code | token | Coded reason this appointment is scheduled | Appointment.reasonCode
reason-reference | reference | Reason the appointment is to take place (resource) | Appointment.reasonReference (Condition, Observation, Procedure, ImmunizationRecommendation)
service-category | token | A broad categorization of the service that is to be performed during this appointment | Appointment.serviceCategory
service-type | token | The specific service that is to be performed during this appointment | Appointment.serviceType
slot | reference | The slots that this appointment is filling | Appointment.slot (Slot)
specialty | token | The specialty of a practitioner that would be required to perform the service requested in this appointment | Appointment.specialty
status | token | The overall status of the appointment | Appointment.status
supporting-info | reference | Additional information to support the appointment | Appointment.supportingInformation (Any)

12.10.7 Resource Appointment - Examples

URL: https://hl7.org/fhir/R4/appointment-examples.html

12.10.7 Resource Appointment - Examples

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Device, Patient, Practitioner, RelatedPerson

Example Name | id | Format
General Person Example | example | XML | JSON | Turtle
Appointment between 2 doctors to discuss MRI results | 2docs | XML | JSON | Turtle
Appointment request | examplereq | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

12.10.8 Resource Appointment - Detailed Descriptions

URL: https://hl7.org/fhir/R4/appointment-definitions.html

12.10.8 Resource Appointment - Detailed Descriptions

Patient Administration Work Group | Maturity Level: 3 | Trial Use | Security Category: Patient | Compartments: Device, Patient, Practitioner, RelatedPerson

Detailed Descriptions for the elements in the Appointment resource.

Appointment
Element Id | Appointment
Definition | A booking of a healthcare event among patient(s), practitioner(s), related person(s) and/or device(s) for a specific date/time. This may result in one or more Encounter(s).
Cardinality | 0..*
Type | DomainResource
Invariants | Defined on this element app-2 Rule Either start and end are specified, or neither start.exists() = end.exists() app-3 Rule Only proposed or cancelled appointments can be missing start/end dates (start.exists() and end.exists()) or (status in ('proposed' | 'cancelled' | 'waitlist')) app-4 Rule Cancelation reason is only used for appointments that have been cancelled, or no-show Appointment.cancelationReason.exists() implies (Appointment.status='no-show' or Appointment.status='cancelled')
Defined on this element
app-2 | Rule | Either start and end are specified, or neither | start.exists() = end.exists()
app-3 | Rule | Only proposed or cancelled appointments can be missing start/end dates | (start.exists() and end.exists()) or (status in ('proposed' | 'cancelled' | 'waitlist'))
app-4 | Rule | Cancelation reason is only used for appointments that have been cancelled, or no-show | Appointment.cancelationReason.exists() implies (Appointment.status='no-show' or Appointment.status='cancelled')
Appointment.identifier
Element Id | Appointment.identifier
Definition | This records identifiers associated with this appointment concern that are defined by business processes and/or used to refer to it when a direct URL reference to the resource itself is not appropriate (e.g. in CDA documents, or in written / printed documentation).
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Summary | true
Appointment.status
Element Id | Appointment.status
Definition | The overall status of the Appointment. Each of the participants has their own participation status which indicates their involvement in the process, however this status indicates the shared status.
Cardinality | 1..1
Terminology Binding | AppointmentStatus (Required)
Type | code
Is Modifier | true (Reason: This element is labelled as a modifier because it is a status element that contains status entered-in-error which means that the resource should not be treated as valid)
Summary | true
Comments | If the Appointment's status is "cancelled" then all participants are expected to have their calendars released for the appointment period, and as such any Slots that were marked as BUSY can be re-set to FREE. This element is labeled as a modifier because the status contains the code entered-in-error that mark the Appointment as not currently valid.
Appointment.cancelationReason
Element Id | Appointment.cancelationReason
Definition | The coded reason for the appointment being cancelled. This is often used in reporting/billing/futher processing to determine if further actions are required, or specific fees apply.
Cardinality | 0..1
Terminology Binding | Appointment cancellation reason (Example)
Type | CodeableConcept
Summary | true
Appointment.serviceCategory
Element Id | Appointment.serviceCategory
Definition | A broad categorization of the service that is to be performed during this appointment.
Cardinality | 0..*
Terminology Binding | Service category (Example)
Type | CodeableConcept
Summary | true
Appointment.serviceType
Element Id | Appointment.serviceType
Definition | The specific service that is to be performed during this appointment.
Cardinality | 0..*
Terminology Binding | Service type (Example)
Type | CodeableConcept
Summary | true

Comments | For a provider to provider appointment the code "FOLLOWUP" may be appropriate, as this is expected to be discussing some patient that was seen in the past.
Appointment.specialty
Element Id | Appointment.specialty
Definition | The specialty of a practitioner that would be required to perform the service requested in this appointment.
Cardinality | 0..*
Terminology Binding | Practice Setting Code Value Set (Preferred)
Type | CodeableConcept
Summary | true
Appointment.appointmentType
Element Id | Appointment.appointmentType
Definition | The style of appointment or patient that has been booked in the slot (not service type).
Cardinality | 0..1
Terminology Binding | v2 Appointment Reason Codes (Preferred)
Type | CodeableConcept
Summary | true
Appointment.reasonCode
Element Id | Appointment.reasonCode
Definition | The coded reason that this appointment is being scheduled. This is more clinical than administrative.
Cardinality | 0..*
Terminology Binding | Encounter Reason Codes (Preferred)
Type | CodeableConcept
Summary | true
Appointment.reasonReference
Element Id | Appointment.reasonReference
Definition | Reason the appointment has been scheduled to take place, as specified using information from another resource. When the patient arrives and the encounter begins it may be used as the admission diagnosis. The indication will typically be a Condition (with other resources referenced in the evidence.detail), or a Procedure.
Cardinality | 0..*
Type | Reference(Condition | Procedure | Observation | ImmunizationRecommendation)
Appointment.priority
Element Id | Appointment.priority
Definition | The priority of the appointment. Can be used to make informed decisions if needing to re-prioritize appointments. (The iCal Standard specifies 0 as undefined, 1 as highest, 9 as lowest priority).
Cardinality | 0..1
Type | unsignedInt
Comments | Seeking implementer feedback on this property and how interoperable it is. Using an extension to record a CodeableConcept for named values may be tested at a future connectathon.
Appointment.description
Element Id | Appointment.description
Definition | The brief description of the appointment as would be shown on a subject line in a meeting request, or appointment list. Detailed or expanded information should be put in the comment field.
Cardinality | 0..1
Type | string
Appointment.supportingInformation
Element Id | Appointment.supportingInformation
Definition | Additional information to support the appointment provided when making the appointment.
Cardinality | 0..*
Type | Reference(Any)
Appointment.start
Element Id | Appointment.start
Definition | Date/Time that the appointment is to take place.
Cardinality | 0..1
Type | instant
Summary | true
Appointment.end
Element Id | Appointment.end
Definition | Date/Time that the appointment is to conclude.
Cardinality | 0..1
Type | instant
Summary | true
Appointment.minutesDuration
Element Id | Appointment.minutesDuration
Definition | Number of minutes that the appointment is to take. This can be less than the duration between the start and end times. For example, where the actual time of appointment is only an estimate or if a 30 minute appointment is being requested, but any time would work. Also, if there is, for example, a planned 15 minute break in the middle of a long appointment, the duration may be 15 minutes less than the difference between the start and end.
Cardinality | 0..1
Type | positiveInt
Appointment.slot
Element Id | Appointment.slot
Definition | The slots from the participants' schedules that will be filled by the appointment.
Cardinality | 0..*

Type | Reference(Slot)
Appointment.created
Element Id | Appointment.created
Definition | The date that this appointment was initially created. This could be different to the meta.lastModified value on the initial entry, as this could have been before the resource was created on the FHIR server, and should remain unchanged over the lifespan of the appointment.
Cardinality | 0..1
Type | dateTime
Comments | This property is required for many use cases where the age of an appointment is considered in processing workflows for scheduling and billing of appointments.
Appointment.comment
Element Id | Appointment.comment
Definition | Additional comments about the appointment.
Cardinality | 0..1
Type | string
Comments | Additional text to aid in facilitating the appointment. For instance, a comment might be, "patient should proceed immediately to infusion room upon arrival" Where this is a planned appointment and the start/end dates are not set then this field can be used to provide additional guidance on the details of the appointment request, including any restrictions on when to book it.
Appointment.patientInstruction
Element Id | Appointment.patientInstruction
Definition | While Appointment.comment contains information for internal use, Appointment.patientInstructions is used to capture patient facing information about the Appointment (e.g. please bring your referral or fast from 8pm night before).
Cardinality | 0..1
Type | string
Appointment.basedOn
Element Id | Appointment.basedOn
Definition | The service request this appointment is allocated to assess (e.g. incoming referral or procedure request).
Cardinality | 0..*
Type | Reference(ServiceRequest)
Alternate Names | incomingReferral
Appointment.participant
Element Id | Appointment.participant
Definition | List of participants involved in the appointment.
Cardinality | 1..*
Invariants | Defined on this element app-1 Rule Either the type or actor on the participant SHALL be specified type.exists() or actor.exists()
Defined on this element
app-1 | Rule | Either the type or actor on the participant SHALL be specified | type.exists() or actor.exists()
Appointment.participant.type
Element Id | Appointment.participant.type
Definition | Role of participant in the appointment.
Cardinality | 0..*
Terminology Binding | Participant type (Extensible)
Type | CodeableConcept
Summary | true
Comments | The role of the participant can be used to declare what the actor will be doing in the scope of this appointment. If the actor is not specified, then it is expected that the actor will be filled in at a later stage of planning. This value SHALL be the same when creating an AppointmentResponse so that they can be matched, and subsequently update the Appointment.
Appointment.participant.actor
Element Id | Appointment.participant.actor
Definition | A Person, Location/HealthcareService or Device that is participating in the appointment.
Cardinality | 0..1
Type | Reference(Patient | Practitioner | PractitionerRole | RelatedPerson | Device | HealthcareService | Location)
Summary | true
Appointment.participant.required
Element Id | Appointment.participant.required
Definition | Whether this participant is required to be present at the meeting. This covers a use-case where two doctors need to meet to discuss the results for a specific patient, and the patient is not required to be present.
Cardinality | 0..1
Terminology Binding | ParticipantRequired (Required)
Type | code
Summary | true
Appointment.participant.status
Element Id | Appointment.participant.status
Definition | Participation status of the actor.
Cardinality | 1..1
Terminology Binding | ParticipationStatus (Required)
Type | code
Summary | true
Appointment.participant.period
Element Id | Appointment.participant.period
Definition | Participation period of the actor.

Cardinality | 0..1
Type | Period
Appointment.requestedPeriod
Element Id | Appointment.requestedPeriod
Definition | A set of date ranges (potentially including times) that the appointment is preferred to be scheduled within. The duration (usually in minutes) could also be provided to indicate the length of the appointment to fill and populate the start/end times for the actual allocated time. However, in other situations the duration may be calculated by the scheduling system.
Cardinality | 0..*
Type | Period
Comments | This does not introduce a capacity for recurring appointments.

12.10.9 Resource Appointment - Mappings

URL: https://hl7.org/fhir/R4/appointment-mappings.html

12.10.9 Resource Appointment - Mappings

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Device, Patient, Practitioner, RelatedPerson

Mappings:

FiveWs Pattern Mapping

Workflow Pattern

HL7 v2 Mapping

RIM Mapping

iCalendar

Mappings for the appointment resource (see Mappings to Other Standards for further information & status).

12.10.9.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

Appointment
identifier | FiveWs.identifier
status | FiveWs.status
specialty | FiveWs.class
priority | FiveWs.class
supportingInformation | FiveWs.context
start | FiveWs.init
end | FiveWs.done[x]
actor | FiveWs.who

12.10.9.2 Workflow Pattern (http://hl7.org/fhir/workflow)

Appointment | Request
identifier | Request.identifier
status | Request.status
serviceType | Request.code
reasonCode | Request.reasonCode
reasonReference | Request.reasonReference
priority | Request.priority
supportingInformation | Request.supportingInfo
start | Request.occurrence[x]
end | Request.occurrence[x]
minutesDuration | Request.occurrence[x]
created | Request.authoredOn
comment | Request.note
basedOn | Request.basedOn
participant | Request.performer, Request.subject
requestedPeriod | Request.occurrence[x]

12.10.9.3 HL7 v2 Mapping (http://hl7.org/v2)

Appointment
identifier | SCH-1, SCH-2, ARQ-1, ARQ-2
status | SCH-25
cancelationReason
serviceCategory
serviceType
specialty
appointmentType | ARQ-7
reasonCode | AIS-3, SCH-7
reasonReference
priority | ARQ-12, SCH-11.6 / TQ1-9
description | NTE-3

supportingInformation
start | ARQ-11.1, SCH-11.4 / TQ1-7
end | SCH-11.5 / TQ1-8/ calculated based on the start and the duration which is in ARQ-9
minutesDuration
slot
created
comment | NTE-3
patientInstruction | NTE-3
basedOn
participant | PID, AIP, AIG, AIL
type | AIP-4, AIG-4
actor | PID-3-Patient ID List | AIL-3 | AIG-3 | AIP-3
required
status | AIP-12, AIG-14
period
requestedPeriod | ARQ-11

12.10.9.4 RIM Mapping (http://hl7.org/v3)

Appointment | Act[@moodCode <= 'INT']
identifier | .id
status | .statusCode
cancelationReason | n/a
serviceCategory | n/a, might be inferred from the ServiceDeliveryLocation
serviceType | n/a, might be inferred from the ServiceDeliveryLocation
specialty | .performer.AssignedPerson.code
appointmentType | .code
reasonCode | .reasonCode
reasonReference | .inboundRelationship[@typeCode = 'RSON'].observation
priority | .priorityCode
description | .text
supportingInformation | .inboundRelationship[@typeCode = 'SPRT'].observation
start | .effectiveTime.low
end | .effectiveTime.high
minutesDuration | .activityTime[@xsi:type = ('SXPR_TS', 'PIVL_TS')].width
slot | .outboundRelationship[@typeCode = 'REFR'].act[@classCode = 'ACT'][@moodCode = 'SLOT']
created | .author.time
comment | .subjectOf.observation[@moodCode = 'EVN'][code = 'annotation'].value
patientInstruction | .subjectOf.observation[@moodCode = 'EVN'][code = 'annotation'].value
basedOn | .outboundRelationship[@typeCode = 'FLFS'].act[@classCode < 'ActCareProvisionRequestType'][@moodCode = 'RQO']
participant | performer | reusableDevice | subject | location
type | (performer | reusableDevice | subject | location).@typeCode
actor | performer.person | reusableDevice.manufacturedDevice | subject.patient | location.serviceDeliveryLocation
required | (performer | reusableDevice | subject | location).@performInd
status | (performer | reusableDevice | subject | location).statusCode
period | n/a
requestedPeriod | .activityTime[@xsi:type = ('SXPR_TS', 'PIVL_TS')]

12.10.9.5 iCalendar (http://ietf.org/rfc/2445)

Appointment | VEVENT
identifier | UID
status | STATUS
cancelationReason
serviceCategory | CATEGORIES
serviceType
specialty
appointmentType | CLASSIFICATION
reasonCode
reasonReference
priority | PRIORITY
description | SUMMARY
supportingInformation | ATTACH
start | DTSTART
end | DTEND
minutesDuration | DURATION (e.g. PT15M)
slot
created | CREATED
comment | DESCRIPTION (for internal use)
patientInstruction | DESCRIPTION (for patient use)
basedOn
participant | ATTENDEE component to be created
type | n/a

actor | ATTENDEE: (Need to provide the common name CN and also the MAILTO properties from the practitioner resource) where the participant type is not "information-only"
required | ROLE=REQ-PARTICIPANT (roleparam)
status | ATTENDEE;CN="John Doe";RSVP=TRUE:mailto:john@doe.com (rsvpparam | partstatparam)
period
requestedPeriod

12.10.10 Resource Appointment - Extensions & Profiles

URL: https://hl7.org/fhir/R4/appointment-profiles.html

12.10.10 Resource Appointment - Extensions & Profiles

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Device, Patient, Practitioner, RelatedPerson

This table lists profiles and extensions for the Appointment resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

12.10.10.1 Profiles

No Profiles defined for this resource

12.10.10.2 Extensions

No Extensions defined for this resource
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

12.10.10.3 Search Extensions

No Search Extensions defined for this resource

12.10.11 Resource Appointment - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/appointment-version-maps.html

No validation errors - all conversions are clean

12.10.11 Resource Appointment - R3/R4 Conversions

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Device, Patient, Practitioner, RelatedPerson

R3 : R4 Conversion maps for Appointment.

Functional status for this map: 3 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid. (see documentation)

12.10.11.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/Appointment3to4" = "R3 to R4 Conversions for Appointment"

uses "http://hl7.org/fhir/3.0/StructureDefinition/Appointment" alias AppointmentR3 as source
uses "http://hl7.org/fhir/StructureDefinition/Appointment" alias Appointment as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group Appointment(source src : AppointmentR3, target tgt : Appointment) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.status -> tgt.status;
  src.serviceCategory -> tgt.serviceCategory;
  src.serviceType -> tgt.serviceType;
  src.specialty -> tgt.specialty;
  src.appointmentType -> tgt.appointmentType;
  src.reason -> tgt.reasonCode;
  src.indication -> tgt.reasonReference;
  src.priority -> tgt.priority;
  src.description -> tgt.description;
```

```
  src.supportingInformation -> tgt.supportingInformation;
  src.start -> tgt.start;
  src.end -> tgt.end;
  src.minutesDuration -> tgt.minutesDuration;
  src.slot -> tgt.slot;
  src.created -> tgt.created;
  src.comment -> tgt.comment;
  src.incomingReferral -> tgt.basedOn "basedOn";
  src.participant as vs0 -> tgt.participant as vt0 then {
    vs0.type -> vt0.type;
    vs0.actor -> vt0.actor;
    vs0.required -> vt0.required;
    vs0.status -> vt0.status;
  };
  src.requestedPeriod -> tgt.requestedPeriod;
}
```

12.10.11.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/Appointment4to3" = "R4 to R3 Conversion for Appointment"

uses "http://hl7.org/fhir/StructureDefinition/Appointment" alias Appointment as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/Appointment" alias AppointmentR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group Appointment(source src : Appointment, target tgt : AppointmentR3) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.status -> tgt.status;
  src.serviceCategory -> tgt.serviceCategory;
  src.serviceType -> tgt.serviceType;
  src.specialty -> tgt.specialty;
  src.appointmentType -> tgt.appointmentType;
  src.reasonCode -> tgt.reason;
  src.reasonReference -> tgt.indication;
  src.priority -> tgt.priority;
  src.description -> tgt.description;
  src.supportingInformation -> tgt.supportingInformation;
  src.start -> tgt.start;
  src.end -> tgt.end;
  src.minutesDuration -> tgt.minutesDuration;
  src.slot -> tgt.slot;
  src.created -> tgt.created;
  src.comment -> tgt.comment;
  src.basedOn -> tgt.incomingReferral "incomingReferral";
  src.participant as vs0 -> tgt.participant as vt0 then {
    vs0.type -> vt0.type;
    vs0.actor -> vt0.actor;
    vs0.required -> vt0.required;
    vs0.status -> vt0.status;
  };
  src.requestedPeriod -> tgt.requestedPeriod;
}
```

12.10.11.3 R4 Validation Errors

12.10.12 JSON Schema for Appointment

URL: https://hl7.org/fhir/R4/appointment.schema.json.html

12.10.12 JSON Schema for Appointment

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Device, Patient, Practitioner, RelatedPerson

This is a representation of the json schema for Appointment, which is just a part of the full JSON Schema.

```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/Appointment",
  "$ref": "#/definitions/Appointment",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "Appointment": {
      "description": "A booking of a healthcare event among patient(s), practitioner(s), related person(s) and/or
    device(s) for a specific date/time. This may result in one or more Encounter(s).",
      "properties": {
        "resourceType": {
          "description": "This is a Appointment resource",
          "const": "Appointment"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
```

```
      "$ref": "Meta.schema.json#/definitions/Meta"
    },
    "implicitRules": {
      "description": "A reference to a set of rules that were followed when the resource was constructed, and which must be understood when processing the content. Often, this is a reference to an implementation guide that defines the special rules along with other profiles etc.",
      "$ref": "#/definitions/uri"
    },
    "_implicitRules": {
      "description": "Extensions for implicitRules",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "language": {
      "description": "The base language in which the resource is written.",
      "$ref": "#/definitions/code"
    },
    "_language": {
      "description": "Extensions for language",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "text": {
      "description": "A human-readable narrative that contains a summary of the resource and can be used to represent the content of the resource to a human. The narrative need not encode all the structured data, but is required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative. Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
      "$ref": "Narrative.schema.json#/definitions/Narrative"
    },
    "contained": {
      "description": "These resources do not have an independent existence apart from the resource that contains them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
      "items": {
        "$ref": "ResourceList.schema.json#/definitions/ResourceList"
      },
      "type": "array"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the resource and that modifies the understanding of the element that contains it and/or the understanding of the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension. Applications processing a resource are required to check for modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "identifier": {
      "description": "This records identifiers associated with this appointment concern that are defined by business processes and/or used to refer to it when a direct URL reference to the resource itself is not appropriate (e.g. in CDA documents, or in written / printed documentation).",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "status": {
      "description": "The overall status of the Appointment. Each of the participants has their own participation status which indicates their involvement in the process, however this status indicates the shared status.",
      "enum": [
        "proposed",
        "pending",
        "booked",
        "arrived",
        "fulfilled",
        "cancelled",
        "noshow",
        "entered-in-error",
        "checked-in",
        "waitlist"
      ]
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "cancelationReason": {
```

```
      "description": "The coded reason for the appointment being cancelled. This is often used in
reporting/billing/futher processing to determine if further actions are required, or specific fees apply.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "serviceCategory": {
      "description": "A broad categorization of the service that is to be performed during this appointment.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "serviceType": {
      "description": "The specific service that is to be performed during this appointment.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "specialty": {
      "description": "The specialty of a practitioner that would be required to perform the service requested in
this appointment.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "appointmentType": {
      "description": "The style of appointment or patient that has been booked in the slot (not service type).",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "reasonCode": {
      "description": "The coded reason that this appointment is being scheduled. This is more clinical than
administrative.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "reasonReference": {
      "description": "Reason the appointment has been scheduled to take place, as specified using information from
another resource. When the patient arrives and the encounter begins it may be used as the admission diagnosis. The
indication will typically be a Condition (with other resources referenced in the evidence.detail), or a Procedure.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "priority": {
      "description": "The priority of the appointment. Can be used to make informed decisions if needing to re-
prioritize appointments. (The iCal Standard specifies 0 as undefined, 1 as highest, 9 as lowest priority).",
      "$ref": "#/definitions/unsignedInt"
    },
    "_priority": {
      "description": "Extensions for priority",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "description": {
      "description": "The brief description of the appointment as would be shown on a subject line in a meeting
request, or appointment list. Detailed or expanded information should be put in the comment field.",
      "$ref": "#/definitions/string"
    },
    "_description": {
      "description": "Extensions for description",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "supportingInformation": {
      "description": "Additional information to support the appointment provided when making the appointment.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "start": {
      "description": "Date/Time that the appointment is to take place.",
      "$ref": "#/definitions/instant"
    },
    "_start": {
      "description": "Extensions for start",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "end": {
      "description": "Date/Time that the appointment is to conclude.",
      "$ref": "#/definitions/instant"
    },
    "_end": {
      "description": "Extensions for end",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "minutesDuration": {
```

```
      "description": "Number of minutes that the appointment is to take. This can be less than the duration between the start and end times.  For example, where the actual time of appointment is only an estimate or if a 30 minute appointment is being requested, but any time would work.  Also, if there is, for example, a planned 15 minute break in the middle of a long appointment, the duration may be 15 minutes less than the difference between the start and end.",
      "$ref": "#/definitions/positiveInt"
    },
    "_minutesDuration": {
      "description": "Extensions for minutesDuration",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "slot": {
      "description": "The slots from the participants\u0027 schedules that will be filled by the appointment.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "created": {
      "description": "The date that this appointment was initially created. This could be different to the meta.lastModified value on the initial entry, as this could have been before the resource was created on the FHIR server, and should remain unchanged over the lifespan of the appointment.",
      "$ref": "#/definitions/dateTime"
    },
    "_created": {
      "description": "Extensions for created",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "comment": {
      "description": "Additional comments about the appointment.",
      "$ref": "#/definitions/string"
    },
    "_comment": {
      "description": "Extensions for comment",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "patientInstruction": {
      "description": "While Appointment.comment contains information for internal use, Appointment.patientInstructions is used to capture patient facing information about the Appointment (e.g. please bring your referral or fast from 8pm night before).",
      "$ref": "#/definitions/string"
    },
    "_patientInstruction": {
      "description": "Extensions for patientInstruction",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "basedOn": {
      "description": "The service request this appointment is allocated to assess (e.g. incoming referral or procedure request).",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "participant": {
      "description": "List of participants involved in the appointment.",
      "items": {
        "$ref": "#/definitions/Appointment_Participant"
      },
      "type": "array"
    },
    "requestedPeriod": {
      "description": "A set of date ranges (potentially including times) that the appointment is preferred to be scheduled within.\n\nThe duration (usually in minutes) could also be provided to indicate the length of the appointment to fill and populate the start/end times for the actual allocated time. However, in other situations the duration may be calculated by the scheduling system.",
      "items": {
        "$ref": "Period.schema.json#/definitions/Period"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "participant",
    "resourceType"
  ]
},
"Appointment_Participant": {
  "description": "A booking of a healthcare event among patient(s), practitioner(s), related person(s) and/or device(s) for a specific date/time. This may result in one or more Encounter(s).",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
```

```
    the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
    definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
    that SHALL be met as part of the definition of the extension.",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "modifierExtension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the element and that modifies the understanding of the element in which it is contained and/or the understanding of
    the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
    use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
    extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
     of the definition of the extension. Applications processing a resource are required to check for modifier
    extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
    (including cannot change the meaning of modifierExtension itself).",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "type": {
          "description": "Role of participant in the appointment.",
          "items": {
            "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
          },
          "type": "array"
        },
        "actor": {
          "description": "A Person, Location/HealthcareService or Device that is participating in the appointment.",
          "$ref": "Reference.schema.json#/definitions/Reference"
        },
        "required": {
          "description": "Whether this participant is required to be present at the meeting. This covers a use-case
    where two doctors need to meet to discuss the results for a specific patient, and the patient is not required to be
    present.",
          "enum": [
            "required",
            "optional",
            "information-only"
          ]
        },
        "_required": {
          "description": "Extensions for required",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "status": {
          "description": "Participation status of the actor.",
          "enum": [
            "accepted",
            "declined",
            "tentative",
            "needs-action"
          ]
        },
        "_status": {
          "description": "Extensions for status",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "period": {
          "description": "Participation period of the actor.",
          "$ref": "Period.schema.json#/definitions/Period"
        }
      },
      "additionalProperties": false
    }
  }
}
```


12.10.13 Resource Appointment - Maturity Dependency View

URL: https://hl7.org/fhir/R4/appointment-dependencies.html

12.10.13 Resource Appointment - Maturity Dependency View

Patient Administration Work Group | Maturity Level: 3 | Trial Use | Compartments: Device, Patient, Practitioner, RelatedPerson

Dependency Graph for Appointment FMM level 3

Name | Card. | Type | Dependency Analysis
Appointment | | DomainResource
identifier | 0..* | Identifier
status | 1..1 | code
cancelationReason | 0..1 | CodeableConcept
serviceCategory | 0..* | CodeableConcept
serviceType | 0..* | CodeableConcept

specialty | 0..* | CodeableConcept
appointmentType | 0..1 | CodeableConcept
reasonCode | 0..* | CodeableConcept
reasonReference | 0..* | Reference(Condition | Procedure | Observation | ImmunizationRecommendation) | Type Warning: (ImmunizationRecommendation = FMM1-Trial Use vs. Element = FMM3-Trial Use)
priority | 0..1 | unsignedInt
description | 0..1 | string
supportingInformation | 0..* | Reference(Any) | Type Warning: (Any = FMM1-Trial Use vs. Element = FMM3-Trial Use)
start | 0..1 | instant
end | 0..1 | instant
minutesDuration | 0..1 | positiveInt
slot | 0..* | Reference(Slot)
created | 0..1 | dateTime
comment | 0..1 | string
patientInstruction | 0..1 | string
basedOn | 0..* | Reference(ServiceRequest) | Type Warning: (ServiceRequest = FMM2-Trial Use vs. Element = FMM3-Trial Use)
participant | 1..* | BackboneElement
type | 0..* | CodeableConcept
actor | 0..1 | Reference(Patient | Practitioner | PractitionerRole | RelatedPerson | Device | HealthcareService | Location) | Type Warning: (PractitionerRole = FMM2-Trial Use vs. Element = FMM3-Trial Use) Type Warning: (RelatedPerson = FMM2-Trial Use vs. Element = FMM3-Trial Use) Type Warning: (Device = FMM2-Trial Use vs. Element = FMM3-Trial Use) Type Warning: (HealthcareService = FMM2-Trial Use vs. Element = FMM3-Trial Use)
required | 0..1 | code
status | 1..1 | code
period | 0..1 | Period
requestedPeriod | 0..* | Period

12.10.14 Value Set Sc-valueset-appointmentstatus - Status Code Mapping

URL: https://hl7.org/fhir/R4/sc-valueset-appointmentstatus.html

12.10.14 Value Set Sc-valueset-appointmentstatus - Status Code Mapping

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Device, Patient, Practitioner, RelatedPerson

This is the narrative for the resource. See also the XML, JSON or Turtle format.

12.10.15 AppointmentStatusCanonicalMap (http://hl7.org/fhir/ConceptMap/sc-appointmentstatus)

Mapping from http://hl7.org/fhir/ValueSet/appointmentstatus to http://hl7.org/fhir/ValueSet/resource-status

DRAFT. Published on 01/11/2019 9:29:23 AM by HL7 (FHIR Project) (http://hl7.org/fhir , fhir@lists.hl7.org).

Canonical Mapping for "The free/busy status of an appointment."

Source Code | Equivalence | Destination Code
entered-in-error | equivalent | error
proposed,waitlist | equivalent | proposed
pending | equivalent | draft
booked | equivalent | accepted
arrived | equivalent | arrived
noshow | equivalent | suspended
fulfilled | equivalent | complete
cancelled | equivalent | abandoned
checked-in | equivalent | partial

12.10.16 Value Set Sc-valueset-participationstatus - Status Code Mapping

URL: https://hl7.org/fhir/R4/sc-valueset-participationstatus.html

12.10.16 Value Set Sc-valueset-participationstatus - Status Code Mapping

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Device, Patient, Practitioner, RelatedPerson

This is the narrative for the resource. See also the XML, JSON or Turtle format.

12.10.17 ParticipationStatusCanonicalMap (http://hl7.org/fhir/ConceptMap/sc-participationstatus)

Mapping from http://hl7.org/fhir/ValueSet/participationstatus to http://hl7.org/fhir/ValueSet/resource-status

Canonical Mapping for "The Participation status of an appointment."

Source Code	Equivalence	Destination Code
entered-in-error	equivalent	error
tentative	equivalent	draft
declined	equivalent	declined
accepted	equivalent	accepted
needs-action	equivalent	failed

12.11 Resource AppointmentResponse - Content

URL: https://hl7.org/fhir/R4/appointmentresponse.html
Patient Administration Work Group | Maturity Level: 3 | Trial Use | Security Category: Patient | Compartments: Device, Patient, Practitioner, RelatedPerson

A reply to an appointment request for a patient and/or practitioner(s), such as a confirmation or rejection.

12.11.1 Scope and Usage

Appointment resources are used to provide information about a planned meeting that may be in the future or past. They may be for a single meeting or for a series of repeating visits. Examples include a scheduled surgery, a follow-up for a clinical visit, a scheduled conference call between clinicians to discuss a case, the reservation of a piece of diagnostic equipment for a particular use, etc. The visit scheduled by an appointment may be in person or remote (by phone, video conference, etc.) All that matters is that the time and usage of one or more individuals, locations and/or pieces of equipment is being fully or partially reserved for a designated period of time.

This definition takes the concepts of appointments in a clinical setting and also extends them to be relevant in the community healthcare space, and also ease exposure to other appointment / calendar standards widely used outside of Healthcare.

12.11.1.1 The basic workflow to create an appointment

Making the Appointment Request

When an appointment is required, a requester creates new Appointment resource with the Appointment.status="proposed". All included participants (optional or mandatory) should have the status="needs-action" to allow filtering and displaying appointments to user-participants for accepting or rejecting new and updated requests. Based on internal system business rules, certain statuses may be automatically updated, for example: "reject because the requested participant is on vacation" or "this type of user is not allowed to request those specific appointments".

Replying to the request

The reply process is simply performed by the person/system handing the requests updating the participant statuses as needed. If there are multiple systems involved, then these will create AppointmentResponse entries with the desired statuses.

Once all participants have their participation status created/updated (and the main system marking the appointment participant records with the AppointmentResponse statuses) then the overall status of the Appointment is updated.

This resource is referenced by ImagingStudy

12.11.2 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name	Flags	Card.	Type	Description & Constraints
AppointmentResponse	ITU		DomainResource	A reply to an appointment request for a patient and/or practitioner(s), such as a confirmation or rejection + Rule: Either the participantType or actor must be specified Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier	Σ	0..*	Identifier	External Ids for this item
appointment	Σ	1..1	Reference(Appointment)	Appointment this response relates to
start		0..1	instant	Time from appointment, or requested new start time
end		0..1	instant	Time from appointment, or requested new end time
participantType	Σ	0..*	CodeableConcept	Role of participant in the appointment Participant type (Extensible)

actor | Σ | 0..1 | Reference(Patient | Practitioner | PractitionerRole | RelatedPerson | Device | HealthcareService | Location) | Person, Location, HealthcareService, or Device

participantStatus | ?!Σ | 1..1 | code | accepted | declined | tentative | needs-action ParticipationStatus (Required)

comment | | 0..1 | string | Additional comments

Documentation for this format

UML Diagram (Legend)

XML Template

```
<AppointmentResponse xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier External Ids for this item --></identifier>
 <appointment><!-- 1..1 Reference(Appointment) Appointment this response relates to --></appointment>
 <start value="[instant]"/><!-- 0..1 Time from appointment, or requested new start time -->
 <end value="[instant]"/><!-- 0..1 Time from appointment, or requested new end time -->
 <participantType><!-- 0..* CodeableConcept Role of participant in the appointment --></participantType>
 <actor><!-- 0..1 Reference(Patient|Practitioner|PractitionerRole|RelatedPerson|
   Device|HealthcareService|Location) Person, Location, HealthcareService, or Device --></actor>
 <participantStatus value="[code]"/><!-- 1..1 accepted | declined | tentative | needs-action -->
 <comment value="[string]"/><!-- 0..1 Additional comments -->
</AppointmentResponse>
```

JSON Template

```
{
  "resourceType" : "AppointmentResponse",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // External Ids for this item
  "appointment" : { Reference(Appointment) }, // R!  Appointment this response relates to
  "start" : "<instant>", // Time from appointment, or requested new start time
  "end" : "<instant>", // Time from appointment, or requested new end time
  "participantType" : [{ CodeableConcept }], // Role of participant in the appointment
  "actor" : { Reference(Patient|Practitioner|PractitionerRole|RelatedPerson|
   Device|HealthcareService|Location) }, // Person, Location, HealthcareService, or Device
  "participantStatus" : "<code>", // R!  accepted | declined | tentative | needs-action
  "comment" : "<string>" // Additional comments
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:AppointmentResponse;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:AppointmentResponse.identifier [ Identifier ], ... ; # 0..* External Ids for this item
  fhir:AppointmentResponse.appointment [ Reference(Appointment) ]; # 1..1 Appointment this response relates to
  fhir:AppointmentResponse.start [ instant ]; # 0..1 Time from appointment, or requested new start time
  fhir:AppointmentResponse.end [ instant ]; # 0..1 Time from appointment, or requested new end time
  fhir:AppointmentResponse.participantType [ CodeableConcept ], ... ; # 0..* Role of participant in the appointment
  fhir:AppointmentResponse.actor [
    Reference(Patient|Practitioner|PractitionerRole|RelatedPerson|Device|HealthcareService|
  Location) ]; # 0..1 Person, Location, HealthcareService, or Device
  fhir:AppointmentResponse.participantStatus [ code ]; # 1..1 accepted | declined | tentative | needs-action
  fhir:AppointmentResponse.comment [ string ]; # 0..1 Additional comments
]
```

Changes since R3

AppointmentResponse

AppointmentResponse.actor | Type Reference: Added Target Type PractitionerRole

AppointmentResponse.participantStatus | Change value set from http://hl7.org/fhir/ValueSet/participationstatus to http://hl7.org/fhir/ValueSet/participationstatus|4.0.1

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 2 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid.)

Structure

Name | Flags | Card. | Type | Description & Constraints

AppointmentResponse | ITU | | DomainResource | A reply to an appointment request for a patient and/or practitioner(s), such as a confirmation or rejection + Rule: Either the participantType or actor must be specified Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

identifier | Σ | 0..* | Identifier | External Ids for this item

appointment | Σ | 1..1 | Reference(Appointment) | Appointment this response relates to

start | | 0..1 | instant | Time from appointment, or requested new start time

end | | 0..1 | instant | Time from appointment, or requested new end time

participantType | Σ | 0..* | CodeableConcept | Role of participant in the appointment Participant type (Extensible)

actor | Σ | 0..1 | Reference(Patient | Practitioner | PractitionerRole | RelatedPerson | Device | HealthcareService | Location) | Person, Location, HealthcareService, or Device

participantStatus | ?!Σ | 1..1 | code | accepted | declined | tentative | needs-action ParticipationStatus (Required)
comment | | 0..1 | string | Additional comments
Documentation for this format

UML Diagram (Legend)

XML Template

```
<AppointmentResponse xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier External Ids for this item --></identifier>
 <appointment><!-- 1..1 Reference(Appointment) Appointment this response relates to --></appointment>
 <start value="[instant]"/><!-- 0..1 Time from appointment, or requested new start time -->
 <end value="[instant]"/><!-- 0..1 Time from appointment, or requested new end time -->
 <participantType><!-- 0..* CodeableConcept Role of participant in the appointment --></participantType>
 <actor><!-- 0..1 Reference(Patient|Practitioner|PractitionerRole|RelatedPerson|
   Device|HealthcareService|Location) Person, Location, HealthcareService, or Device --></actor>
 <participantStatus value="[code]"/><!-- 1..1 accepted | declined | tentative | needs-action -->
 <comment value="[string]"/><!-- 0..1 Additional comments -->
</AppointmentResponse>
```

JSON Template

```
{
  "resourceType" : "AppointmentResponse",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // External Ids for this item
  "appointment" : { Reference(Appointment) }, // R!  Appointment this response relates to
  "start" : "<instant>", // Time from appointment, or requested new start time
  "end" : "<instant>", // Time from appointment, or requested new end time
  "participantType" : [{ CodeableConcept }], // Role of participant in the appointment
  "actor" : { Reference(Patient|Practitioner|PractitionerRole|RelatedPerson|
   Device|HealthcareService|Location) }, // Person, Location, HealthcareService, or Device
  "participantStatus" : "<code>", // R!  accepted | declined | tentative | needs-action
  "comment" : "<string>" // Additional comments
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:AppointmentResponse;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:AppointmentResponse.identifier [ Identifier ], ... ; # 0..* External Ids for this item
  fhir:AppointmentResponse.appointment [ Reference(Appointment) ]; # 1..1 Appointment this response relates to
  fhir:AppointmentResponse.start [ instant ]; # 0..1 Time from appointment, or requested new start time
  fhir:AppointmentResponse.end [ instant ]; # 0..1 Time from appointment, or requested new end time
  fhir:AppointmentResponse.participantType [ CodeableConcept ], ... ; # 0..* Role of participant in the appointment
  fhir:AppointmentResponse.actor [
    Reference(Patient|Practitioner|PractitionerRole|RelatedPerson|Device|HealthcareService|
  Location) ]; # 0..1 Person, Location, HealthcareService, or Device
  fhir:AppointmentResponse.participantStatus [ code ]; # 1..1 accepted | declined | tentative | needs-action
  fhir:AppointmentResponse.comment [ string ]; # 0..1 Additional comments
]
```

Changes since Release 3

AppointmentResponse
AppointmentResponse.actor | Type Reference: Added Target Type PractitionerRole
AppointmentResponse.participantStatus | Change value set from http://hl7.org/fhir/ValueSet/participationstatus to http://hl7.org/fhir/ValueSet/participationstatus|4.0.1

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 2 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid.)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

12.11.2.1 Terminology Bindings

Path | Definition | Type | Reference
AppointmentResponse.participantType | Role of participant in encounter. | Extensible | ParticipantType
AppointmentResponse.participantStatus | The Participation status of an appointment. | Required | ParticipationStatus

12.11.2.2 Constraints

id | Level | Location | Description | Expression
apr-1 | Rule | (base) | Either the participantType or actor must be specified | participantType.exists() or actor.exists()

12.11.3 Notes:

12.11.3.1 Time zones and recurring appointments

Recurring appointments need to have the time zone defined in which the values were entered. Knowing that the start time was at 9:00:00Z+10 does not mean that the same time in 2 weeks is actually the same.

For example, if this was a time in Brisbane Australia, this time would be the same (in respect to its offset from UTC), however if this was for Melbourne Australia, during the daylight savings period Melbourne time zone becomes +11. So, without the additional information as to which time zone it was created in, scheduling a 9am appointment every Wednesday would not be possible.

12.11.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
actor | reference | The Person, Location/HealthcareService or Device that this appointment response replies for | AppointmentResponse.actor (Practitioner, Device, Patient, HealthcareService, PractitionerRole, RelatedPerson, Location)
appointment | reference | The appointment that the response is attached to | AppointmentResponse.appointment (Appointment)
identifier | token | An Identifier in this appointment response | AppointmentResponse.identifier
location | reference | This Response is for this Location | AppointmentResponse.actor.where(resolve() is Location) (Location)
part-status | token | The participants acceptance status for this appointment | AppointmentResponse.participantStatus
patient | reference | This Response is for this Patient | AppointmentResponse.actor.where(resolve() is Patient) (Patient)
practitioner | reference | This Response is for this Practitioner | AppointmentResponse.actor.where(resolve() is Practitioner) (Practitioner)

12.11.5 Resource AppointmentResponse - Examples

URL: https://hl7.org/fhir/R4/appointmentresponse-examples.html

12.11.5 Resource AppointmentResponse - Examples

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Device, Patient, Practitioner, RelatedPerson

Example Name | id | Format
General Person Example | example | XML | JSON | Turtle
Response to renegotiate an appointmet request | exampleresp | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

12.11.6 Resource AppointmentResponse - Detailed Descriptions

URL: https://hl7.org/fhir/R4/appointmentresponse-definitions.html

12.11.6 Resource AppointmentResponse - Detailed Descriptions

Patient Administration Work Group | Maturity Level: 3 | Trial Use | Security Category: Patient | Compartments: Device, Patient, Practitioner, RelatedPerson

Detailed Descriptions for the elements in the AppointmentResponse resource.

AppointmentResponse
Element Id | AppointmentResponse
Definition | A reply to an appointment request for a patient and/or practitioner(s), such as a confirmation or rejection.
Cardinality | 0..*
Type | DomainResource
Invariants | Defined on this element apr-1 Rule Either the participantType or actor must be specified participantType.exists() or actor.exists()
Defined on this element
apr-1 | Rule | Either the participantType or actor must be specified | participantType.exists() or actor.exists()
AppointmentResponse.identifier
Element Id | AppointmentResponse.identifier
Definition | This records identifiers associated with this appointment response concern that are defined by business processes and/ or used to refer to it when a direct URL reference to the resource itself is not appropriate.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Summary | true
AppointmentResponse.appointment
Element Id | AppointmentResponse.appointment
Definition | Appointment that this response is replying to.
Cardinality | 1..1
Type | Reference(Appointment)
Summary | true

AppointmentResponse.start
Element Id | AppointmentResponse.start
Definition | Date/Time that the appointment is to take place, or requested new start time.
Cardinality | 0..1
Type | instant
Comments | This may be either the same as the appointment request to confirm the details of the appointment, or alternately a new time to request a re-negotiation of the start time.
AppointmentResponse.end
Element Id | AppointmentResponse.end
Definition | This may be either the same as the appointment request to confirm the details of the appointment, or alternately a new time to request a re-negotiation of the end time.
Cardinality | 0..1
Type | instant
AppointmentResponse.participantType
Element Id | AppointmentResponse.participantType
Definition | Role of participant in the appointment.
Cardinality | 0..*
Terminology Binding | Participant type (Extensible)
Type | CodeableConcept
Summary | true
Comments | The role of the participant can be used to declare what the actor will be doing in the scope of the referenced appointment. If the actor is not specified, then it is expected that the actor will be filled in at a later stage of planning. This value SHALL be the same as specified on the referenced Appointment so that they can be matched, and subsequently updated.
AppointmentResponse.actor
Element Id | AppointmentResponse.actor
Definition | A Person, Location, HealthcareService, or Device that is participating in the appointment.
Cardinality | 0..1
Type | Reference(Patient | Practitioner | PractitionerRole | RelatedPerson | Device | HealthcareService | Location)
Summary | true
AppointmentResponse.participantStatus
Element Id | AppointmentResponse.participantStatus
Definition | Participation status of the participant. When the status is declined or tentative if the start/end times are different to the appointment, then these times should be interpreted as a requested time change. When the status is accepted, the times can either be the time of the appointment (as a confirmation of the time) or can be empty.
Cardinality | 1..1
Terminology Binding | ParticipationStatus (Required)
Type | code
Is Modifier | true (Reason: Not known why this is labelled a modifier)
Summary | true
Comments | This element is labeled as a modifier because the status contains the code entered-in-error that marks the participant as not currently valid.
AppointmentResponse.comment
Element Id | AppointmentResponse.comment
Definition | Additional comments about the appointment.
Cardinality | 0..1
Type | string
Comments | This comment is particularly important when the responder is declining, tentatively accepting or requesting another time to indicate the reasons why.

12.11.7 Resource AppointmentResponse - Mappings

URL: https://hl7.org/fhir/R4/appointmentresponse-mappings.html

12.11.7 Resource AppointmentResponse - Mappings

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Device, Patient, Practitioner, RelatedPerson

Mappings:

FiveWs Pattern Mapping

Workflow Pattern

HL7 v2 Mapping

RIM Mapping

iCalendar

Mappings for the appointmentresponse resource (see Mappings to Other Standards for further information & status).

12.11.7.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

AppointmentResponse
identifier | FiveWs.identifier
start | FiveWs.init

end | FiveWs.done[x]
actor | FiveWs.who

12.11.7.2 Workflow Pattern (http://hl7.org/fhir/workflow)

AppointmentResponse | Request
identifier | Request.identifier
appointment | Request.basedOn
start | Request.occurrence[x]
end | Request.occurrence[x]
participantType | Request.performerType
actor | Request.performer
participantStatus | Request.status
comment | Request.note

12.11.7.3 HL7 v2 Mapping (http://hl7.org/v2)

AppointmentResponse
identifier
appointment
start
end
participantType
actor | PID-3-Patient ID List
participantStatus
comment

12.11.7.4 RIM Mapping (http://hl7.org/v3)

AppointmentResponse | Appointment[@moodCode <= 'PRMS']
identifier | .id
appointment | scheduleRequest.ActAppointmentRequest[@moodCode <= 'ARQ']
start | .effectiveTime.low
end | .effectiveTime.high
participantType | (performer | reusableDevice | subject | location).@typeCode
actor | performer.person | reusableDevice.manufacturedDevice | subject.patient | location.serviceDeliveryLocation
participantStatus | (performer | reusableDevice | subject | location).statusCode
comment | .subjectOf.observation[@moodCode = 'EVN'][code = 'annotation'].value

12.11.7.5 iCalendar (http://ietf.org/rfc/2445)

AppointmentResponse | VEVENT
identifier | UID
appointment
start
end
participantType | n/a
actor | ATTENDEE: (Need to provide the common name CN and also the MAILTO properties from the practitioner resource)
participantStatus | ATTENDEE;CN="Brian Postlethwaite";RSVP=TRUE:mailto:BPostlethwaite@data.com (rsvpparam | partstatparam)
comment | COMMENT

12.11.8 Resource AppointmentResponse - Extensions & Profiles

URL: https://hl7.org/fhir/R4/appointmentresponse-profiles.html

12.11.8 Resource AppointmentResponse - Extensions & Profiles

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Device, Patient, Practitioner, RelatedPerson

This table lists profiles and extensions for the AppointmentResponse resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

12.11.8.1 Profiles

No Profiles defined for this resource

12.11.8.2 Extensions

No Extensions defined for this resource
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

12.11.8.3 Search Extensions

No Search Extensions defined for this resource

12.11.9 Resource AppointmentResponse - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/appointmentresponse-version-maps.html

No validation errors - all conversions are clean

12.11.9 Resource AppointmentResponse - R3/R4 Conversions

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Device, Patient, Practitioner, RelatedPerson

R3 : R4 Conversion maps for AppointmentResponse.

Functional status for this map: 2 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid. (see documentation)

12.11.9.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/AppointmentResponse3to4" = "R3 to R4 Conversions for AppointmentResponse"

uses "http://hl7.org/fhir/3.0/StructureDefinition/AppointmentResponse" alias AppointmentResponseR3 as source
uses "http://hl7.org/fhir/StructureDefinition/AppointmentResponse" alias AppointmentResponse as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group AppointmentResponse(source src : AppointmentResponseR3, target tgt : AppointmentResponse) extends DomainResource
    <<type+>> {
  src.identifier -> tgt.identifier;
  src.appointment -> tgt.appointment;
  src.start -> tgt.start;
  src.end -> tgt.end;
  src.participantType -> tgt.participantType;
  src.actor -> tgt.actor;
  src.participantStatus -> tgt.participantStatus;
  src.comment -> tgt.comment;
}
```

12.11.9.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/AppointmentResponse4to3" = "R4 to R3 Conversion for AppointmentResponse"

uses "http://hl7.org/fhir/StructureDefinition/AppointmentResponse" alias AppointmentResponse as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/AppointmentResponse" alias AppointmentResponseR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group AppointmentResponse(source src : AppointmentResponse, target tgt : AppointmentResponseR3) extends DomainResource
    <<type+>> {
  src.identifier -> tgt.identifier;
  src.appointment -> tgt.appointment;
  src.start -> tgt.start;
  src.end -> tgt.end;
  src.participantType -> tgt.participantType;
  src.actor -> tgt.actor;
  src.participantStatus -> tgt.participantStatus;
  src.comment -> tgt.comment;
}
```

12.11.9.3 R4 Validation Errors

12.11.10 JSON Schema for AppointmentResponse

URL: https://hl7.org/fhir/R4/appointmentresponse.schema.json.html

12.11.10 JSON Schema for AppointmentResponse

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Device, Patient, Practitioner, RelatedPerson

This is a representation of the json schema for AppointmentResponse, which is just a part of the full JSON Schema.


```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/AppointmentResponse",
  "$ref": "#/definitions/AppointmentResponse",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "AppointmentResponse": {
      "description": "A reply to an appointment request for a patient and/or practitioner(s), such as a confirmation or
    rejection.",
      "properties": {
        "resourceType": {
          "description": "This is a AppointmentResponse resource",
          "const": "AppointmentResponse"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
          "$ref": "#/definitions/code"
        },
        "_language": {
          "description": "Extensions for language",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "text": {
          "description": "A human-readable narrative that contains a summary of the resource and can be used to
    represent the content of the resource to a human. The narrative need not encode all the structured data, but is
    required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
    Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
          "$ref": "Narrative.schema.json#/definitions/Narrative"
        },
        "contained": {
          "description": "These resources do not have an independent existence apart from the resource that contains
    them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
          "items": {
            "$ref": "ResourceList.schema.json#/definitions/ResourceList"
          },
          "type": "array"
        },
        "extension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
     definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
    that SHALL be met as part of the definition of the extension.",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "modifierExtension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the resource and that modifies the understanding of the element that contains it and/or the understanding of the
    containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
    of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
    extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
    met as part of the definition of the extension. Applications processing a resource are required to check for
    modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
    DomainResource (including cannot change the meaning of modifierExtension itself).",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "identifier": {
          "description": "This records identifiers associated with this appointment response concern that are defined by
     business processes and/ or used to refer to it when a direct URL reference to the resource itself is not
    appropriate.",
          "items": {
```

```
          "$ref": "Identifier.schema.json#/definitions/Identifier"
        },
        "type": "array"
      },
      "appointment": {
        "description": "Appointment that this response is replying to.",
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "start": {
        "description": "Date/Time that the appointment is to take place, or requested new start time.",
        "$ref": "#/definitions/instant"
      },
      "_start": {
        "description": "Extensions for start",
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "end": {
        "description": "This may be either the same as the appointment request to confirm the details of the
  appointment, or alternately a new time to request a re-negotiation of the end time.",
        "$ref": "#/definitions/instant"
      },
      "_end": {
        "description": "Extensions for end",
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "participantType": {
        "description": "Role of participant in the appointment.",
        "items": {
          "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
        },
        "type": "array"
      },
      "actor": {
        "description": "A Person, Location, HealthcareService, or Device that is participating in the appointment.",
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "participantStatus": {
        "description": "Participation status of the participant. When the status is declined or tentative if the
  start/end times are different to the appointment, then these times should be interpreted as a requested time change.
   When the status is accepted, the times can either be the time of the appointment (as a confirmation of the time) or
   can be empty.",
        "$ref": "#/definitions/code"
      },
      "_participantStatus": {
        "description": "Extensions for participantStatus",
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "comment": {
        "description": "Additional comments about the appointment.",
        "$ref": "#/definitions/string"
      },
      "_comment": {
        "description": "Extensions for comment",
        "$ref": "Element.schema.json#/definitions/Element"
      }
    },
    "additionalProperties": false,
    "required": [
      "appointment",
      "resourceType"
    ]
  }
 }
}
```


12.11.11 Resource AppointmentResponse - Maturity Dependency View

URL: https://hl7.org/fhir/R4/appointmentresponse-dependencies.html

12.11.11 Resource AppointmentResponse - Maturity Dependency View

Patient Administration Work Group | Maturity Level: 3 | Trial Use | Compartments: Device, Patient, Practitioner, RelatedPerson

Dependency Graph for AppointmentResponse FMM level 3

Name | Card. | Type | Dependency Analysis
AppointmentResponse | | DomainResource
identifier | 0..* | Identifier
appointment | 1..1 | Reference(Appointment)
start | 0..1 | instant
end | 0..1 | instant
participantType | 0..* | CodeableConcept
actor | 0..1 | Reference(Patient | Practitioner | PractitionerRole | RelatedPerson | Device | HealthcareService | Location) | Type Warning: (PractitionerRole = FMM2-Trial Use vs. Element = FMM3-Trial Use) Type Warning: (RelatedPerson = FMM2-Trial Use vs.

Element = FMM3-Trial Use) Type Warning: (Device = FMM2-Trial Use vs. Element = FMM3-Trial Use) Type Warning: (HealthcareService = FMM2-Trial Use vs. Element = FMM3-Trial Use)
participantStatus | 1..1 | code
comment | 0..1 | string

12.12 Resource Schedule - Content

URL: https://hl7.org/fhir/R4/schedule.html
Patient Administration Work Group | Maturity Level: 3 | Trial Use | Security Category: Business | Compartments: Device, Patient, Practitioner, RelatedPerson

A container for slots of time that may be available for booking appointments.

12.12.1 Scope and Usage

Schedule resources provide a container for time-slots that can be booked using an appointment. It provides the window of time (period) that slots are defined for and what type of appointments can be booked. The schedule does not provide any information about actual appointments. This separation greatly assists where access to the appointments would not be permitted for security or privacy reasons, while still being able to determine if an appointment might be available.

Note: A schedule is not used for the delivery of medication, the Timing data type should be used for that purpose.

12.12.1.1 Context

A schedule controls the dates and times available for the performance of a service and/or the use of a resource. One schedule applies to one service or resource, since each service or resource can be reserved independently of the others. If two or more services, people, locations, or things cannot be reserved independently of one another, they are considered to be one activity or resource.

A schedule consists of slots of time during which the controlled service or resource is potentially available for provision or use. Slots are categorized as open, booked, or blocked. An open slot on a schedule indicates that the service or resource is available for provision or use during that period of time. A booked slot indicates that the service or resource is not available during the time period, because an appointment has been scheduled. A blocked slot indicates that a service or resource is unavailable for reasons other than a scheduled appointment.

The real-world, non-automated analogue of the schedule described above is a standard appointment book. These books are generally organized with rows of time slots, during which a service or resource is available.

A slot is one unit on a schedule. A slot represents the smallest unit of time or quantity that a service or resource may be booked. Depending on the nature of the service or resource, there may be more than one defined slot at a given instant of time. For example, if a service is an open group therapy session with twelve available seats, then there are twelve slots for the given block of time.

12.12.1.2 Actor - What the schedule applies to

The schedule belongs to a single instance of a service or resource. This is normally a HealthcareService, Practitioner, Location or Device. In the case where a single resource can provide different services, potentially at different location, then the schedulable resource is considered the composite of the actors. For example, if a practitioner can provide services at multiple locations, they might have one schedule per location, where each schedule includes both the practitioner and location actors. When booking an appointment with multiple schedulable resources, multiple schedules may need to be checked depending on the configuration of the system.

If an appointment has two practitioners, a specific medical device and a room then there could be a schedule for each of these resources that may need to be consulted to ensure that no collisions occur. If the schedule needed to be consulted, then there would be one created covering the planning horizon for the time of the appointment.

12.12.1.3 Checking availability - Searching

When checking availability for an appointment, the creator of the appointment should determine which schedules are applicable, then check for available slots within each schedule.

Determining which schedules should be consulted often will involve searching via the properties of the referenced actors, such as the ServiceCategory on the HealthcareService, or the Address on a Location.

The type parameter can be used to filter the type of services that can be booked within the associated slots.

If all slots are busy, the caller may attempt to book an appointment into an already-booked slot, but the server business rules will dictate whether overbooking is allowed, or whether the appointment may be given a higher precedence and allocated the overbooked slot.

This resource is referenced by TriggerDefinition and Slot

12.12.2 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
Schedule | TU | | DomainResource | A container for slots of time that may be available for booking appointments Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | External Ids for this item
active | ?!Σ | 0..1 | boolean | Whether this schedule is in active use
serviceCategory | Σ | 0..* | CodeableConcept | High-level category Service category (Example)
serviceType | Σ | 0..* | CodeableConcept | Specific service Service type (Example)
specialty | Σ | 0..* | CodeableConcept | Type of specialty needed Practice Setting Code Value Set (Preferred)
actor | Σ | 1..* | Reference(Patient | Practitioner | PractitionerRole | RelatedPerson | Device | HealthcareService | Location) | Resource(s) that availability information is being provided for
planningHorizon | Σ | 0..1 | Period | Period of time covered by schedule
comment | | 0..1 | string | Comments on availability
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Schedule xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier External Ids for this item --></identifier>
 <active value="[boolean]"/><!-- 0..1 Whether this schedule is in active use -->
 <serviceCategory><!-- 0..* CodeableConcept High-level category --></serviceCategory>
 <serviceType><!-- 0..* CodeableConcept Specific service --></serviceType>
 <specialty><!-- 0..* CodeableConcept Type of specialty needed --></specialty>
 <actor><!-- 1..* Reference(Patient|Practitioner|PractitionerRole|RelatedPerson|
   Device|HealthcareService|Location) Resource(s) that availability information is being provided for --></actor>
 <planningHorizon><!-- 0..1 Period Period of time covered by schedule --></planningHorizon>
 <comment value="[string]"/><!-- 0..1 Comments on availability -->
</Schedule>
```

JSON Template

```
{
  "resourceType" : "Schedule",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // External Ids for this item
  "active" : <boolean>, // Whether this schedule is in active use
  "serviceCategory" : [{ CodeableConcept }], // High-level category
  "serviceType" : [{ CodeableConcept }], // Specific service
  "specialty" : [{ CodeableConcept }], // Type of specialty needed
  "actor" : [{ Reference(Patient|Practitioner|PractitionerRole|RelatedPerson|
   Device|HealthcareService|Location) }], // R!  Resource(s) that availability information is being provided for
  "planningHorizon" : { Period }, // Period of time covered by schedule
  "comment" : "<string>" // Comments on availability
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Schedule;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Schedule.identifier [ Identifier ], ... ; # 0..* External Ids for this item
  fhir:Schedule.active [ boolean ]; # 0..1 Whether this schedule is in active use
  fhir:Schedule.serviceCategory [ CodeableConcept ], ... ; # 0..* High-level category
  fhir:Schedule.serviceType [ CodeableConcept ], ... ; # 0..* Specific service
  fhir:Schedule.specialty [ CodeableConcept ], ... ; # 0..* Type of specialty needed
  fhir:Schedule.actor [ Reference(Patient|Practitioner|PractitionerRole|RelatedPerson|Device|HealthcareService|
  Location) ], ... ; # 1..* Resource(s) that availability information is being provided for
  fhir:Schedule.planningHorizon [ Period ]; # 0..1 Period of time covered by schedule
  fhir:Schedule.comment [ string ]; # 0..1 Comments on availability
]
```

Changes since R3

Schedule
Schedule.active | Default Value "true" removed
Schedule.serviceCategory | Max Cardinality changed from 1 to *

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 3 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid.)

Structure

Name | Flags | Card. | Type | Description & Constraints
Schedule | TU | | DomainResource | A container for slots of time that may be available for booking appointments Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | External Ids for this item
active | ?!Σ | 0..1 | boolean | Whether this schedule is in active use
serviceCategory | Σ | 0..* | CodeableConcept | High-level category Service category (Example)
serviceType | Σ | 0..* | CodeableConcept | Specific service Service type (Example)
specialty | Σ | 0..* | CodeableConcept | Type of specialty needed Practice Setting Code Value Set (Preferred)
actor | Σ | 1..* | Reference(Patient | Practitioner | PractitionerRole | RelatedPerson | Device | HealthcareService | Location) | Resource(s) that availability information is being provided for
planningHorizon | Σ | 0..1 | Period | Period of time covered by schedule
comment | | 0..1 | string | Comments on availability
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Schedule xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier External Ids for this item --></identifier>
 <active value="[boolean]"/><!-- 0..1 Whether this schedule is in active use -->
 <serviceCategory><!-- 0..* CodeableConcept High-level category --></serviceCategory>
 <serviceType><!-- 0..* CodeableConcept Specific service --></serviceType>
 <specialty><!-- 0..* CodeableConcept Type of specialty needed --></specialty>
 <actor><!-- 1..* Reference(Patient|Practitioner|PractitionerRole|RelatedPerson|
   Device|HealthcareService|Location) Resource(s) that availability information is being provided for --></actor>
 <planningHorizon><!-- 0..1 Period Period of time covered by schedule --></planningHorizon>
 <comment value="[string]"/><!-- 0..1 Comments on availability -->
</Schedule>
```

JSON Template

```
{
  "resourceType" : "Schedule",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // External Ids for this item
  "active" : <boolean>, // Whether this schedule is in active use
  "serviceCategory" : [{ CodeableConcept }], // High-level category
  "serviceType" : [{ CodeableConcept }], // Specific service
  "specialty" : [{ CodeableConcept }], // Type of specialty needed
  "actor" : [{ Reference(Patient|Practitioner|PractitionerRole|RelatedPerson|
   Device|HealthcareService|Location) }], // R!  Resource(s) that availability information is being provided for
  "planningHorizon" : { Period }, // Period of time covered by schedule
  "comment" : "<string>" // Comments on availability
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Schedule;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Schedule.identifier [ Identifier ], ... ; # 0..* External Ids for this item
  fhir:Schedule.active [ boolean ]; # 0..1 Whether this schedule is in active use
  fhir:Schedule.serviceCategory [ CodeableConcept ], ... ; # 0..* High-level category
  fhir:Schedule.serviceType [ CodeableConcept ], ... ; # 0..* Specific service
  fhir:Schedule.specialty [ CodeableConcept ], ... ; # 0..* Type of specialty needed
  fhir:Schedule.actor [ Reference(Patient|Practitioner|PractitionerRole|RelatedPerson|Device|HealthcareService|
  Location) ], ... ; # 1..* Resource(s) that availability information is being provided for
  fhir:Schedule.planningHorizon [ Period ]; # 0..1 Period of time covered by schedule
  fhir:Schedule.comment [ string ]; # 0..1 Comments on availability
]
```

Changes since Release 3

Schedule
Schedule.active | Default Value "true" removed
Schedule.serviceCategory | Max Cardinality changed from 1 to *

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 3 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid.)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

12.12.2.1 Terminology Bindings

Path | Definition | Type | Reference
Schedule.serviceCategory | | Example | ServiceCategory
Schedule.serviceType | | Example | ServiceType
Schedule.specialty | Additional details about where the content was created (e.g. clinical specialty). | Preferred | PracticeSettingCodeValueSet

12.12.3 Notes:

12.12.3.1 Interaction with other Standards

There is a strong desire that implementers of this resource should consider providing the resource in the iCalendar format as an alternative representation. Many 3rd party applications and component providers have parsers and user interface controls to display this information. This may lower the entry point to integrate outside the health-care specific applications, and into the consumer space. This would permit the easier creation of a mobile application that creates appointments in the devices native calendar. The iCalendar specification can be found at http://www.ietf.org/rfc/rfc2445.txt .

12.12.3.2 Slots - not a contained resource

Due to the dynamic nature of slots they are not included as a part of this resource.

It is anticipated that this resource is likely to be updated intermittently when the scope of slots is changed, i.e. to change the period of slots in the planning horizon. This could be performed each night to move the start and end date forward to keep the planning horizon as exactly 4 weeks.

The slot resource however is anticipated to be updated very regularly as the appointments that reference them are created/updated/cancelled.

12.12.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
active | token | Is the schedule in active use | Schedule.active
actor | reference | The individual(HealthcareService, Practitioner, Location, ...) to find a Schedule for | Schedule.actor (Practitioner, Device, Patient, HealthcareService, PractitionerRole, RelatedPerson, Location)
date | date | Search for Schedule resources that have a period that contains this date specified | Schedule.planningHorizon
identifier | token | A Schedule Identifier | Schedule.identifier
service-category | token | High-level category | Schedule.serviceCategory
service-type | token | The type of appointments that can be booked into associated slot(s) | Schedule.serviceType
specialty | token | Type of specialty needed | Schedule.specialty

12.12.5 Resource Schedule - Examples

URL: https://hl7.org/fhir/R4/schedule-examples.html

12.12.5 Resource Schedule - Examples

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Device, Patient, Practitioner, RelatedPerson

Example Name | id | Format
Example of an Schedule resource | example | XML | JSON | Turtle
Example 1 of a Schedule resource with a composite schedulable resource | exampleloc1 | XML | JSON | Turtle
Example 2 of a Schedule resource with a composite schedulable resource | exampleloc2 | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

12.12.6 Resource Schedule - Detailed Descriptions

URL: https://hl7.org/fhir/R4/schedule-definitions.html

12.12.6 Resource Schedule - Detailed Descriptions

Patient Administration Work Group | Maturity Level: 3 | Trial Use | Security Category: Business | Compartments: Device, Patient, Practitioner, RelatedPerson

Detailed Descriptions for the elements in the Schedule resource.

Schedule
Element Id | Schedule
Definition | A container for slots of time that may be available for booking appointments.

Cardinality | 0..*
Type | DomainResource
Alternate Names | Availability
Schedule.identifier
Element Id | Schedule.identifier
Definition | External Ids for this item.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Summary | true
Schedule.active
Element Id | Schedule.active
Definition | Whether this schedule record is in active use or should not be used (such as was entered in error).
Cardinality | 0..1
Type | boolean
Is Modifier | true (Reason: This element is labelled as a modifier because it is a status element that can indicate that a record should not be treated as valid)
Meaning if Missing | This resource is generally assumed to be active if no value is provided for the active element
Summary | true
Comments | This element is labeled as a modifier because it may be used to mark that the resource was created in error.
Schedule.serviceCategory
Element Id | Schedule.serviceCategory
Definition | A broad categorization of the service that is to be performed during this appointment.
Cardinality | 0..*
Terminology Binding | Service category (Example)
Type | CodeableConcept
Summary | true
Schedule.serviceType
Element Id | Schedule.serviceType
Definition | The specific service that is to be performed during this appointment.
Cardinality | 0..*
Terminology Binding | Service type (Example)
Type | CodeableConcept
Summary | true
Schedule.specialty
Element Id | Schedule.specialty
Definition | The specialty of a practitioner that would be required to perform the service requested in this appointment.
Cardinality | 0..*
Terminology Binding | Practice Setting Code Value Set (Preferred)
Type | CodeableConcept
Summary | true
Schedule.actor
Element Id | Schedule.actor
Definition | Slots that reference this schedule resource provide the availability details to these referenced resource(s).
Cardinality | 1..*
Type | Reference(Patient | Practitioner | PractitionerRole | RelatedPerson | Device | HealthcareService | Location)
Summary | true
Comments | The capacity to support multiple referenced resource types should be used in cases where the specific resources themselves cannot be scheduled without the other, and thus only make sense to the system exposing them as a group. Common examples of this are where the combination of a practitioner and a room (Location) are always required by a system.
Schedule.planningHorizon
Element Id | Schedule.planningHorizon
Definition | The period of time that the slots that reference this Schedule resource cover (even if none exist). These cover the amount of time that an organization's planning horizon; the interval for which they are currently accepting appointments. This does not define a "template" for planning outside these dates.
Cardinality | 0..1
Type | Period
Summary | true
Schedule.comment
Element Id | Schedule.comment
Definition | Comments on the availability to describe any extended information. Such as custom constraints on the slots that may be associated.
Cardinality | 0..1
Type | string

12.12.7 Resource Schedule - Mappings

URL: https://hl7.org/fhir/R4/schedule-mappings.html

12.12.7 Resource Schedule - Mappings

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Device, Patient, Practitioner, RelatedPerson

Mappings:

FiveWs Pattern Mapping

RIM Mapping

iCalendar

Mappings for the schedule resource (see Mappings to Other Standards for further information & status).

12.12.7.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

Schedule
identifier | FiveWs.identifier
active | FiveWs.status
serviceCategory | FiveWs.class
serviceType | FiveWs.class
specialty | FiveWs.class
actor | FiveWs.subject[x]
planningHorizon | FiveWs.planned

12.12.7.2 RIM Mapping (http://hl7.org/v3)

Schedule | n/a

12.12.7.3 iCalendar (http://ietf.org/rfc/2445)

Schedule | BEGIN:VFREEBUSY ??? END:VFREEBUSY (If your fhir server provides this information via calDAV, then you may want to include addressing information in a URL section, and with the search, a DTSTART and DTEND component can be included to indicate the range of time that has been covered in the collection of slots, where these overlap the resources period values. For all slots that reference this resource, include a FREEBUSY row for each slot)
identifier | UID
active
serviceCategory
serviceType | n/a
specialty
actor | ATTENDEE:MAILTO:john_public@host2.com This can be populated with a value from the referenced resource
planningHorizon | DTSTART:20131201T003000Z DTEND:2014030101T003000Z
comment | COMMENT:(comment) If there is no comment, then the line can be excluded from the iCalendar entry

12.12.8 Resource Schedule - Extensions & Profiles

URL: https://hl7.org/fhir/R4/schedule-profiles.html

12.12.8 Resource Schedule - Extensions & Profiles

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Device, Patient, Practitioner, RelatedPerson

This table lists profiles and extensions for the Schedule resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

12.12.8.1 Profiles

No Profiles defined for this resource

12.12.8.2 Extensions

No Extensions defined for this resource
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

12.12.8.3 Search Extensions

No Search Extensions defined for this resource

12.12.9 Resource Schedule - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/schedule-version-maps.html

No validation errors - all conversions are clean

12.12.9 Resource Schedule - R3/R4 Conversions

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Device, Patient, Practitioner, RelatedPerson

R3 : R4 Conversion maps for Schedule.

Functional status for this map: 3 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid. (see documentation)

12.12.9.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/Schedule3to4" = "R3 to R4 Conversions for Schedule"

uses "http://hl7.org/fhir/3.0/StructureDefinition/Schedule" alias ScheduleR3 as source
uses "http://hl7.org/fhir/StructureDefinition/Schedule" alias Schedule as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group Schedule(source src : ScheduleR3, target tgt : Schedule) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.active -> tgt.active;
  src.serviceCategory -> tgt.serviceCategory;
  src.serviceType -> tgt.serviceType;
  src.specialty -> tgt.specialty;
  src.actor -> tgt.actor;
  src.planningHorizon -> tgt.planningHorizon;
  src.comment -> tgt.comment;
}
```

12.12.9.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/Schedule4to3" = "R4 to R3 Conversion for Schedule"

uses "http://hl7.org/fhir/StructureDefinition/Schedule" alias Schedule as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/Schedule" alias ScheduleR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group Schedule(source src : Schedule, target tgt : ScheduleR3) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.active -> tgt.active;
  src.serviceCategory -> tgt.serviceCategory;
  src.serviceType -> tgt.serviceType;
  src.specialty -> tgt.specialty;
  src.actor -> tgt.actor;
  src.planningHorizon -> tgt.planningHorizon;
  src.comment -> tgt.comment;
}
```

12.12.9.3 R4 Validation Errors

12.12.10 JSON Schema for Schedule

URL: https://hl7.org/fhir/R4/schedule.schema.json.html

12.12.10 JSON Schema for Schedule

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Device, Patient, Practitioner, RelatedPerson

This is a representation of the json schema for Schedule, which is just a part of the full JSON Schema.


```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/Schedule",
  "$ref": "#/definitions/Schedule",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "Schedule": {
      "description": "A container for slots of time that may be available for booking appointments.",
      "properties": {
        "resourceType": {
          "description": "This is a Schedule resource",
          "const": "Schedule"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
```

```
    "implicitRules": {
      "description": "A reference to a set of rules that were followed when the resource was constructed, and which
must be understood when processing the content. Often, this is a reference to an implementation guide that defines
the special rules along with other profiles etc.",
      "$ref": "#/definitions/uri"
    },
    "_implicitRules": {
      "description": "Extensions for implicitRules",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "language": {
      "description": "The base language in which the resource is written.",
      "$ref": "#/definitions/code"
    },
    "_language": {
      "description": "Extensions for language",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "text": {
      "description": "A human-readable narrative that contains a summary of the resource and can be used to
represent the content of the resource to a human. The narrative need not encode all the structured data, but is
required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
      "$ref": "Narrative.schema.json#/definitions/Narrative"
    },
    "contained": {
      "description": "These resources do not have an independent existence apart from the resource that contains
them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
      "items": {
        "$ref": "ResourceList.schema.json#/definitions/ResourceList"
      },
      "type": "array"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
 definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource and that modifies the understanding of the element that contains it and/or the understanding of the
containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
met as part of the definition of the extension. Applications processing a resource are required to check for
modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "identifier": {
      "description": "External Ids for this item.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "active": {
      "description": "Whether this schedule record is in active use or should not be used (such as was entered in
error).",
      "$ref": "#/definitions/boolean"
    },
    "_active": {
      "description": "Extensions for active",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "serviceCategory": {
      "description": "A broad categorization of the service that is to be performed during this appointment.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "serviceType": {
      "description": "The specific service that is to be performed during this appointment.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "specialty": {
      "description": "The specialty of a practitioner that would be required to perform the service requested in
```

```
    this appointment.",
          "items": {
            "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
          },
          "type": "array"
        },
        "actor": {
          "description": "Slots that reference this schedule resource provide the availability details to these
    referenced resource(s).",
          "items": {
            "$ref": "Reference.schema.json#/definitions/Reference"
          },
          "type": "array"
        },
        "planningHorizon": {
          "description": "The period of time that the slots that reference this Schedule resource cover (even if none
    exist). These  cover the amount of time that an organization\u0027s planning horizon; the interval for which they
    are currently accepting appointments. This does not define a \"template\" for planning outside these dates.",
          "$ref": "Period.schema.json#/definitions/Period"
        },
        "comment": {
          "description": "Comments on the availability to describe any extended information. Such as custom constraints
    on the slots that may be associated.",
          "$ref": "#/definitions/string"
        },
        "_comment": {
          "description": "Extensions for comment",
          "$ref": "Element.schema.json#/definitions/Element"
        }
      },
      "additionalProperties": false,
      "required": [
        "actor",
        "resourceType"
      ]
    }
  }
}
```


12.12.11 Resource Schedule - Maturity Dependency View

URL: https://hl7.org/fhir/R4/schedule-dependencies.html

12.12.11 Resource Schedule - Maturity Dependency View

Patient Administration Work Group | Maturity Level: 3 | Trial Use | Compartments: Device, Patient, Practitioner, RelatedPerson

Dependency Graph for Schedule FMM level 3

Name | Card. | Type | Dependency Analysis
Schedule | | DomainResource
identifier | 0..* | Identifier
active | 0..1 | boolean
serviceCategory | 0..* | CodeableConcept
serviceType | 0..* | CodeableConcept
specialty | 0..* | CodeableConcept
actor | 1..* | Reference(Patient | Practitioner | PractitionerRole | RelatedPerson | Device | HealthcareService | Location) | Type Warning: (PractitionerRole = FMM2-Trial Use vs. Element = FMM3-Trial Use) Type Warning: (RelatedPerson = FMM2-Trial Use vs. Element = FMM3-Trial Use) Type Warning: (Device = FMM2-Trial Use vs. Element = FMM3-Trial Use) Type Warning: (HealthcareService = FMM2-Trial Use vs. Element = FMM3-Trial Use)
planningHorizon | 0..1 | Period
comment | 0..1 | string

12.13 Resource Slot - Content

URL: https://hl7.org/fhir/R4/slot.html
Patient Administration Work Group | Maturity Level: 3 | Trial Use | Security Category: Business | Compartments: Not linked to any defined compartments

A slot of time on a schedule that may be available for booking appointments.

12.13.1 Scope and Usage

Slot resources are used to provide time-slots that can be booked using an appointment. They do not provide any information about appointments that are available, just the time, and optionally what the time can be used for. These are effectively spaces of free/busy time. Slots can also be marked as busy without having appointments associated.

A slot can have more than one appointment allocated to it. A scheduling system may permit multiple allocations up to a specific number of places. An example of this type of usage could be where the slot is being used for a group service that permits 5 participants at the same time.

A slot can be marked as over-booked indicating that there are too many appointments allocated to it.

In some situations a service may have a specific set of slots reserved for specific uses, such as "walk-ins" or a specific organization has a "standing booking" for Thursday mornings. These should be represented using the appointmentType field with a specified and agreed value. Security Permissions or specific business rules on the system could enforce that only eligible appointments are allocated to them.

If a service had a weekly schedule created that permitted eight 1 hour appointments each day of a working week (Monday - Friday), this would be constructed by a single Schedule resource with the dates for the start and end of the week set, and then 40 (5x8) Slot resources associated with it. As appointments fill up the schedule, these slots would individually be marked as busy as the appointments are filled into the slots. The slots in a schedule do not need to be the same size, and can be different for different days of the week.

Slot instances do not have any recurrence information included. If recurring information is desired, this will be managed outside these resources, or included as extensions.

Note that booking an appointment does not necessarily require that slot resources be identified. When attempting to book an appointment, if the requestor knows ahead of time which schedulable resources are required, then identifying individual slots from the resources' schedules prior to creating the appointment is appropriate. However, in some medical scheduling scenarios, determining which resources are required for an appointment is very complex, and options other than using schedule+slot may be a better solution.

This resource is referenced by Appointment

12.13.2 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
Slot | TU | | DomainResource | A slot of time on a schedule that may be available for booking appointments Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | External Ids for this item
serviceCategory | Σ | 0..* | CodeableConcept | A broad categorization of the service that is to be performed during this appointment Service category (Example)
serviceType | Σ | 0..* | CodeableConcept | The type of appointments that can be booked into this slot (ideally this would be an identifiable service - which is at a location, rather than the location itself). If provided then this overrides the value provided on the availability resource Service type (Example)
specialty | Σ | 0..* | CodeableConcept | The specialty of a practitioner that would be required to perform the service requested in this appointment Practice Setting Code Value Set (Preferred)
appointmentType | Σ | 0..1 | CodeableConcept | The style of appointment or patient that may be booked in the slot (not service type) v2 Appointment Reason Codes (Preferred)
schedule | Σ | 1..1 | Reference(Schedule) | The schedule resource that this slot defines an interval of status information
status | Σ | 1..1 | code | busy | free | busy-unavailable | busy-tentative | entered-in-error SlotStatus (Required)
start | Σ | 1..1 | instant | Date/Time that the slot is to begin
end | Σ | 1..1 | instant | Date/Time that the slot is to conclude
overbooked | | 0..1 | boolean | This slot has already been overbooked, appointments are unlikely to be accepted for this time
comment | | 0..1 | string | Comments on the slot to describe any extended information. Such as custom constraints on the slot
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Slot xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier External Ids for this item --></identifier>
 <serviceCategory><!-- 0..* CodeableConcept A broad categorization of the service that is to be performed during this
    appointment --></serviceCategory>
 <serviceType><!-- 0..* CodeableConcept The type of appointments that can be booked into this slot (ideally this would
    be an identifiable service - which is at a location, rather than the location itself). If provided then this
    overrides the value provided on the availability resource --></serviceType>
 <specialty><!-- 0..* CodeableConcept The specialty of a practitioner that would be required to perform the service
    requested in this appointment --></specialty>
 <appointmentType><!-- 0..1 CodeableConcept The style of appointment or patient that may be booked in the slot (not
    service type) --></appointmentType>
 <schedule><!-- 1..1 Reference(Schedule) The schedule resource that this slot defines an interval of status information
    --></schedule>
 <status value="[code]"/><!-- 1..1 busy | free | busy-unavailable | busy-tentative | entered-in-error -->
 <start value="[instant]"/><!-- 1..1 Date/Time that the slot is to begin -->
 <end value="[instant]"/><!-- 1..1 Date/Time that the slot is to conclude -->
 <overbooked value="[boolean]"/><!-- 0..1 This slot has already been overbooked, appointments are unlikely to be
```

```
    accepted for this time -->
 <comment value="[string]"/><!-- 0..1 Comments on the slot to describe any extended information. Such as custom
    constraints on the slot -->
</Slot>
```

JSON Template

```
{
  "resourceType" : "Slot",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // External Ids for this item
  "serviceCategory" : [{ CodeableConcept }], // A broad categorization of the service that is to be performed during
    this appointment
  "serviceType" : [{ CodeableConcept }], // The type of appointments that can be booked into this slot (ideally this
    would be an identifiable service - which is at a location, rather than the location itself). If provided then this
    overrides the value provided on the availability resource
  "specialty" : [{ CodeableConcept }], // The specialty of a practitioner that would be required to perform the service
    requested in this appointment
  "appointmentType" : { CodeableConcept }, // The style of appointment or patient that may be booked in the slot (not
    service type)
  "schedule" : { Reference(Schedule) }, // R!  The schedule resource that this slot defines an interval of status
    information
  "status" : "<code>", // R!  busy | free | busy-unavailable | busy-tentative | entered-in-error
  "start" : "<instant>", // R!  Date/Time that the slot is to begin
  "end" : "<instant>", // R!  Date/Time that the slot is to conclude
  "overbooked" : <boolean>, // This slot has already been overbooked, appointments are unlikely to be accepted for this
    time
  "comment" : "<string>" // Comments on the slot to describe any extended information. Such as custom constraints on the
     slot
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Slot;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Slot.identifier [ Identifier ], ... ; # 0..* External Ids for this item
  fhir:Slot.serviceCategory [ CodeableConcept ], ... ; # 0..* A broad categorization of the service that is to be
    performed during this appointment
  fhir:Slot.serviceType [ CodeableConcept ], ... ; # 0..* The type of appointments that can be booked into this slot
    (ideally this would be an identifiable service - which is at a location, rather than the location itself). If
    provided then this overrides the value provided on the availability resource
  fhir:Slot.specialty [ CodeableConcept ], ... ; # 0..* The specialty of a practitioner that would be required to
    perform the service requested in this appointment
  fhir:Slot.appointmentType [ CodeableConcept ]; # 0..1 The style of appointment or patient that may be booked in the
    slot (not service type)
  fhir:Slot.schedule [ Reference(Schedule) ]; # 1..1 The schedule resource that this slot defines an interval of status
    information
  fhir:Slot.status [ code ]; # 1..1 busy | free | busy-unavailable | busy-tentative | entered-in-error
  fhir:Slot.start [ instant ]; # 1..1 Date/Time that the slot is to begin
  fhir:Slot.end [ instant ]; # 1..1 Date/Time that the slot is to conclude
  fhir:Slot.overbooked [ boolean ]; # 0..1 This slot has already been overbooked, appointments are unlikely to be
    accepted for this time
  fhir:Slot.comment [ string ]; # 0..1 Comments on the slot to describe any extended information. Such as custom
    constraints on the slot
]
```

Changes since R3

Slot

Slot.serviceCategory | Max Cardinality changed from 1 to *

Slot.status | Change value set from http://hl7.org/fhir/ValueSet/slotstatus to http://hl7.org/fhir/ValueSet/slotstatus|4.0.1

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 4 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid.)

Structure

Name | Flags | Card. | Type | Description & Constraints

Slot | TU | | DomainResource | A slot of time on a schedule that may be available for booking appointments Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

identifier | Σ | 0..* | Identifier | External Ids for this item

serviceCategory | Σ | 0..* | CodeableConcept | A broad categorization of the service that is to be performed during this appointment Service category (Example)

serviceType | Σ | 0..* | CodeableConcept | The type of appointments that can be booked into this slot (ideally this would be an identifiable service - which is at a location, rather than the location itself). If provided then this overrides the value provided on the availability resource Service type (Example)

specialty | Σ | 0..* | CodeableConcept | The specialty of a practitioner that would be required to perform the service requested in this appointment Practice Setting Code Value Set (Preferred)

appointmentType | Σ | 0..1 | CodeableConcept | The style of appointment or patient that may be booked in the slot (not service type) v2 Appointment Reason Codes (Preferred)
schedule | Σ | 1..1 | Reference(Schedule) | The schedule resource that this slot defines an interval of status information
status | Σ | 1..1 | code | busy | free | busy-unavailable | busy-tentative | entered-in-error SlotStatus (Required)
start | Σ | 1..1 | instant | Date/Time that the slot is to begin
end | Σ | 1..1 | instant | Date/Time that the slot is to conclude
overbooked | | 0..1 | boolean | This slot has already been overbooked, appointments are unlikely to be accepted for this time
comment | | 0..1 | string | Comments on the slot to describe any extended information. Such as custom constraints on the slot
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Slot xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier External Ids for this item --></identifier>
 <serviceCategory><!-- 0..* CodeableConcept A broad categorization of the service that is to be performed during this
    appointment --></serviceCategory>
 <serviceType><!-- 0..* CodeableConcept The type of appointments that can be booked into this slot (ideally this would
    be an identifiable service - which is at a location, rather than the location itself). If provided then this
    overrides the value provided on the availability resource --></serviceType>
 <specialty><!-- 0..* CodeableConcept The specialty of a practitioner that would be required to perform the service
    requested in this appointment --></specialty>
 <appointmentType><!-- 0..1 CodeableConcept The style of appointment or patient that may be booked in the slot (not
    service type) --></appointmentType>
 <schedule><!-- 1..1 Reference(Schedule) The schedule resource that this slot defines an interval of status information
    --></schedule>
 <status value="[code]"/><!-- 1..1 busy | free | busy-unavailable | busy-tentative | entered-in-error -->
 <start value="[instant]"/><!-- 1..1 Date/Time that the slot is to begin -->
 <end value="[instant]"/><!-- 1..1 Date/Time that the slot is to conclude -->
 <overbooked value="[boolean]"/><!-- 0..1 This slot has already been overbooked, appointments are unlikely to be
    accepted for this time -->
 <comment value="[string]"/><!-- 0..1 Comments on the slot to describe any extended information. Such as custom
    constraints on the slot -->
</Slot>
```

JSON Template

```
{
  "resourceType" : "Slot",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // External Ids for this item
  "serviceCategory" : [{ CodeableConcept }], // A broad categorization of the service that is to be performed during
    this appointment
  "serviceType" : [{ CodeableConcept }], // The type of appointments that can be booked into this slot (ideally this
    would be an identifiable service - which is at a location, rather than the location itself). If provided then this
    overrides the value provided on the availability resource
  "specialty" : [{ CodeableConcept }], // The specialty of a practitioner that would be required to perform the service
    requested in this appointment
  "appointmentType" : { CodeableConcept }, // The style of appointment or patient that may be booked in the slot (not
    service type)
  "schedule" : { Reference(Schedule) }, // R!  The schedule resource that this slot defines an interval of status
    information
  "status" : "<code>", // R!  busy | free | busy-unavailable | busy-tentative | entered-in-error
  "start" : "<instant>", // R!  Date/Time that the slot is to begin
  "end" : "<instant>", // R!  Date/Time that the slot is to conclude
  "overbooked" : <boolean>, // This slot has already been overbooked, appointments are unlikely to be accepted for this
    time
  "comment" : "<string>" // Comments on the slot to describe any extended information. Such as custom constraints on the
     slot
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Slot;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Slot.identifier [ Identifier ], ... ; # 0..* External Ids for this item
  fhir:Slot.serviceCategory [ CodeableConcept ], ... ; # 0..* A broad categorization of the service that is to be
    performed during this appointment
  fhir:Slot.serviceType [ CodeableConcept ], ... ; # 0..* The type of appointments that can be booked into this slot
    (ideally this would be an identifiable service - which is at a location, rather than the location itself). If
    provided then this overrides the value provided on the availability resource
  fhir:Slot.specialty [ CodeableConcept ], ... ; # 0..* The specialty of a practitioner that would be required to
    perform the service requested in this appointment
  fhir:Slot.appointmentType [ CodeableConcept ]; # 0..1 The style of appointment or patient that may be booked in the
    slot (not service type)
  fhir:Slot.schedule [ Reference(Schedule) ]; # 1..1 The schedule resource that this slot defines an interval of status
    information
  fhir:Slot.status [ code ]; # 1..1 busy | free | busy-unavailable | busy-tentative | entered-in-error
  fhir:Slot.start [ instant ]; # 1..1 Date/Time that the slot is to begin
  fhir:Slot.end [ instant ]; # 1..1 Date/Time that the slot is to conclude
```

```
  fhir:Slot.overbooked [ boolean ]; # 0..1 This slot has already been overbooked, appointments are unlikely to be
    accepted for this time
  fhir:Slot.comment [ string ]; # 0..1 Comments on the slot to describe any extended information. Such as custom
    constraints on the slot
]
```

Changes since Release 3

Slot

Slot.serviceCategory | Max Cardinality changed from 1 to *

Slot.status | Change value set from http://hl7.org/fhir/ValueSet/slotstatus to http://hl7.org/fhir/ValueSet/slotstatus|4.0.1

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 4 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid.)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

12.13.2.1 Terminology Bindings

Path | Definition | Type | Reference

Slot.serviceCategory | | Example | ServiceCategory

Slot.serviceType | | Example | ServiceType

Slot.specialty | Additional details about where the content was created (e.g. clinical specialty). | Preferred | PracticeSettingCodeValueSet

Slot.appointmentType | | Preferred | v2.0276

Slot.status | The free/busy status of the slot. | Required | SlotStatus

12.13.3 Notes:

• Interaction with other Standards

There is a strong desire that implementers of this resource should consider providing the resource in the iCalendar format as an alternative representation. Many 3rd party applications and component providers have parsers and user interface controls to display this information. This may lower the entry point to integrate outside the health-care specific applications, and into the consumer space. This would permit the easier creation of a mobile application that creates appointments in the devices native calendar. The iCalendar specification can be found at http://www.ietf.org/rfc/rfc2445.txt .

12.13.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common

appointment-type | token | The style of appointment or patient that may be booked in the slot (not service type) | Slot.appointmentType

identifier | token | A Slot Identifier | Slot.identifier

schedule | reference | The Schedule Resource that we are seeking a slot within | Slot.schedule (Schedule)

service-category | token | A broad categorization of the service that is to be performed during this appointment | Slot.serviceCategory

service-type | token | The type of appointments that can be booked into the slot | Slot.serviceType

specialty | token | The specialty of a practitioner that would be required to perform the service requested in this appointment | Slot.specialty

start | date | Appointment date/time. | Slot.start

status | token | The free/busy status of the appointment | Slot.status

12.13.5 Resource Slot - Examples

URL: https://hl7.org/fhir/R4/slot-examples.html

12.13.5 Resource Slot - Examples

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Not linked to any defined compartments

Example Name | id | Format

Example of an available slot (no appointments allocated) | example | XML | JSON | Turtle

Example for a slot that has an appointment allocated to it | 1 | XML | JSON | Turtle

Example for tentatively occupied slot | 2 | XML | JSON | Turtle

Example of a slot that has been marked as unavailable | 3 | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

12.13.6 Resource Slot - Detailed Descriptions

URL: https://hl7.org/fhir/R4/slot-definitions.html

12.13.6 Resource Slot - Detailed Descriptions

Patient Administration Work Group | Maturity Level: 3 | Trial Use | Security Category: Business | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the Slot resource.

Slot
Element Id | Slot
Definition | A slot of time on a schedule that may be available for booking appointments.
Cardinality | 0..*
Type | DomainResource
Slot.identifier
Element Id | Slot.identifier
Definition | External Ids for this item.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Summary | true
Slot.serviceCategory
Element Id | Slot.serviceCategory
Definition | A broad categorization of the service that is to be performed during this appointment.
Cardinality | 0..*
Terminology Binding | Service category (Example)
Type | CodeableConcept
Summary | true
Slot.serviceType
Element Id | Slot.serviceType
Definition | The type of appointments that can be booked into this slot (ideally this would be an identifiable service - which is at a location, rather than the location itself). If provided then this overrides the value provided on the availability resource.
Cardinality | 0..*
Terminology Binding | Service type (Example)
Type | CodeableConcept
Summary | true
Slot.specialty
Element Id | Slot.specialty
Definition | The specialty of a practitioner that would be required to perform the service requested in this appointment.
Cardinality | 0..*
Terminology Binding | Practice Setting Code Value Set (Preferred)
Type | CodeableConcept
Summary | true
Slot.appointmentType
Element Id | Slot.appointmentType
Definition | The style of appointment or patient that may be booked in the slot (not service type).
Cardinality | 0..1
Terminology Binding | v2 Appointment Reason Codes (Preferred)
Type | CodeableConcept
Summary | true
Slot.schedule
Element Id | Slot.schedule
Definition | The schedule resource that this slot defines an interval of status information.
Cardinality | 1..1
Type | Reference(Schedule)
Summary | true
Slot.status
Element Id | Slot.status
Definition | busy | free | busy-unavailable | busy-tentative | entered-in-error.
Cardinality | 1..1
Terminology Binding | SlotStatus (Required)
Type | code
Summary | true
Slot.start
Element Id | Slot.start
Definition | Date/Time that the slot is to begin.
Cardinality | 1..1
Type | instant
Summary | true
Slot.end
Element Id | Slot.end
Definition | Date/Time that the slot is to conclude.
Cardinality | 1..1
Type | instant

Summary | true
Slot.overbooked
Element Id | Slot.overbooked
Definition | This slot has already been overbooked, appointments are unlikely to be accepted for this time.
Cardinality | 0..1
Type | boolean
Meaning if Missing | If overbooked is missing, systems may assume that there are still appointments available
Slot.comment
Element Id | Slot.comment
Definition | Comments on the slot to describe any extended information. Such as custom constraints on the slot.
Cardinality | 0..1
Type | string

12.13.7 Resource Slot - Mappings

URL: https://hl7.org/fhir/R4/slot-mappings.html

12.13.7 Resource Slot - Mappings

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Not linked to any defined compartments

Mappings:

FiveWs Pattern Mapping

RIM Mapping

iCalendar

Mappings for the slot resource (see Mappings to Other Standards for further information & status).

12.13.7.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

Slot
identifier | FiveWs.identifier
serviceCategory | FiveWs.class
serviceType | FiveWs.class
specialty | FiveWs.class
appointmentType | FiveWs.class
start | FiveWs.planned
end | FiveWs.planned

12.13.7.2 RIM Mapping (http://hl7.org/v3)

Slot | n/a

12.13.7.3 iCalendar (http://ietf.org/rfc/2445)

Slot | (refer to the Availability resource for the header mapping details)
identifier
serviceCategory
serviceType
specialty
appointmentType
schedule
status | FREEBUSY;FBTYPE=(freeBusyType):19980314T233000Z/19980315T003000Z If the freeBusyType is BUSY, then this value can be excluded
start | FREEBUSY;FBTYPE=FREE:(start)/19980315T003000Z If the freeBusyType is BUSY, then this value can be excluded
end | FREEBUSY:19980314T233000Z/(end) If the freeBusyType is BUSY, then this value can be excluded (as shown)
overbooked
comment

12.13.8 Resource Slot - Extensions & Profiles

URL: https://hl7.org/fhir/R4/slot-profiles.html

12.13.8 Resource Slot - Extensions & Profiles

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Not linked to any defined compartments

This table lists profiles and extensions for the Slot resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

12.13.8.1 Profiles

No Profiles defined for this resource

12.13.8.2 Extensions

No Extensions defined for this resource
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

12.13.8.3 Search Extensions

No Search Extensions defined for this resource

12.13.9 Resource Slot - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/slot-version-maps.html

No validation errors - all conversions are clean

12.13.9 Resource Slot - R3/R4 Conversions

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Not linked to any defined compartments

R3 : R4 Conversion maps for Slot.

Functional status for this map: 4 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid. (see documentation)

12.13.9.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/Slot3to4" = "R3 to R4 Conversions for Slot"

uses "http://hl7.org/fhir/3.0/StructureDefinition/Slot" alias SlotR3 as source
uses "http://hl7.org/fhir/StructureDefinition/Slot" alias Slot as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group Slot(source src : SlotR3, target tgt : Slot) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.serviceCategory -> tgt.serviceCategory;
  src.serviceType -> tgt.serviceType;
  src.specialty -> tgt.specialty;
  src.appointmentType -> tgt.appointmentType;
  src.schedule -> tgt.schedule;
  src.status -> tgt.status;
  src.start -> tgt.start;
  src.end -> tgt.end;
  src.overbooked -> tgt.overbooked;
  src.comment -> tgt.comment;
}
```

12.13.9.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/Slot4to3" = "R4 to R3 Conversion for Slot"

uses "http://hl7.org/fhir/StructureDefinition/Slot" alias Slot as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/Slot" alias SlotR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group Slot(source src : Slot, target tgt : SlotR3) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.serviceCategory -> tgt.serviceCategory;
  src.serviceType -> tgt.serviceType;
  src.specialty -> tgt.specialty;
  src.appointmentType -> tgt.appointmentType;
  src.schedule -> tgt.schedule;
  src.status -> tgt.status;
  src.start -> tgt.start;
  src.end -> tgt.end;
  src.overbooked -> tgt.overbooked;
  src.comment -> tgt.comment;
}
```

12.13.9.3 R4 Validation Errors

12.13.10 JSON Schema for Slot

URL: https://hl7.org/fhir/R4/slot.schema.json.html

12.13.10 JSON Schema for Slot

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Not linked to any defined compartments

This is a representation of the json schema for Slot, which is just a part of the full JSON Schema.


```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/Slot",
  "$ref": "#/definitions/Slot",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "Slot": {
      "description": "A slot of time on a schedule that may be available for booking appointments.",
      "properties": {
        "resourceType": {
          "description": "This is a Slot resource",
          "const": "Slot"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
          "$ref": "#/definitions/code"
        },
        "_language": {
          "description": "Extensions for language",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "text": {
          "description": "A human-readable narrative that contains a summary of the resource and can be used to
    represent the content of the resource to a human. The narrative need not encode all the structured data, but is
    required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
    Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
          "$ref": "Narrative.schema.json#/definitions/Narrative"
        },
        "contained": {
          "description": "These resources do not have an independent existence apart from the resource that contains
    them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
          "items": {
            "$ref": "ResourceList.schema.json#/definitions/ResourceList"
          },
          "type": "array"
        },
        "extension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
     definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
    that SHALL be met as part of the definition of the extension.",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "modifierExtension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the resource and that modifies the understanding of the element that contains it and/or the understanding of the
    containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
    of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
    extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
    met as part of the definition of the extension. Applications processing a resource are required to check for
    modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
    DomainResource (including cannot change the meaning of modifierExtension itself).",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
```

```
      "type": "array"
    },
    "identifier": {
      "description": "External Ids for this item.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "serviceCategory": {
      "description": "A broad categorization of the service that is to be performed during this appointment.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "serviceType": {
      "description": "The type of appointments that can be booked into this slot (ideally this would be an
identifiable service - which is at a location, rather than the location itself). If provided then this overrides the
 value provided on the availability resource.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "specialty": {
      "description": "The specialty of a practitioner that would be required to perform the service requested in
this appointment.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "appointmentType": {
      "description": "The style of appointment or patient that may be booked in the slot (not service type).",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "schedule": {
      "description": "The schedule resource that this slot defines an interval of status information.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "status": {
      "description": "busy | free | busy-unavailable | busy-tentative | entered-in-error.",
      "enum": [
        "busy",
        "free",
        "busy-unavailable",
        "busy-tentative",
        "entered-in-error"
      ]
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "start": {
      "description": "Date/Time that the slot is to begin.",
      "$ref": "#/definitions/instant"
    },
    "_start": {
      "description": "Extensions for start",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "end": {
      "description": "Date/Time that the slot is to conclude.",
      "$ref": "#/definitions/instant"
    },
    "_end": {
      "description": "Extensions for end",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "overbooked": {
      "description": "This slot has already been overbooked, appointments are unlikely to be accepted for this
time.",
      "$ref": "#/definitions/boolean"
    },
    "_overbooked": {
      "description": "Extensions for overbooked",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "comment": {
      "description": "Comments on the slot to describe any extended information. Such as custom constraints on the
slot.",
      "$ref": "#/definitions/string"
    },
    "_comment": {
      "description": "Extensions for comment",
      "$ref": "Element.schema.json#/definitions/Element"
    }
```

```
      },
      "additionalProperties": false,
      "required": [
        "schedule",
        "resourceType"
      ]
    }
  }
}
```

12.13.11 Resource Slot - Maturity Dependency View

URL: https://hl7.org/fhir/R4/slot-dependencies.html

12.13.11 Resource Slot - Maturity Dependency View

Patient Administration Work Group | Maturity Level: 3 | Trial Use | Compartments: Not linked to any defined compartments

Dependency Graph for Slot FMM level 3

Name | Card. | Type | Dependency Analysis
Slot | | DomainResource
identifier | 0..* | Identifier
serviceCategory | 0..* | CodeableConcept
serviceType | 0..* | CodeableConcept
specialty | 0..* | CodeableConcept
appointmentType | 0..1 | CodeableConcept
schedule | 1..1 | Reference(Schedule)
status | 1..1 | code
start | 1..1 | instant
end | 1..1 | instant
overbooked | 0..1 | boolean
comment | 0..1 | string

12.13.12 Value Set Sc-valueset-slotstatus - Status Code Mapping

URL: https://hl7.org/fhir/R4/sc-valueset-slotstatus.html

12.13.12 Value Set Sc-valueset-slotstatus - Status Code Mapping

Patient Administration Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Not linked to any defined compartments

This is the narrative for the resource. See also the XML, JSON or Turtle format.

12.13.13 SlotStatusCanonicalMap (http://hl7.org/fhir/ConceptMap/sc-slotstatus)

Mapping from http://hl7.org/fhir/ValueSet/slotstatus to http://hl7.org/fhir/ValueSet/resource-status

DRAFT. Published on 01/11/2019 9:29:23 AM by HL7 (FHIR Project) (http://hl7.org/fhir , fhir@lists.hl7.org).

Canonical Mapping for "The free/busy status of the slot."

Source Code | Equivalence | Destination Code
entered-in-error | equivalent | error
busy | equivalent | accepted
busy-tentative (Busy (Tentative)) | equivalent | unconfirmed
busy-unavailable (Busy (Unavailable)) | equivalent | busy-unavailable
free | equivalent | free

12.14 Resource ServiceRequest - Content

URL: https://hl7.org/fhir/R4/servicerequest.html
Orders and Observations Work Group | Maturity Level: 2 | Trial Use | Security Category: Patient | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

A record of a request for service such as diagnostic investigations, treatments, or operations to be performed.

12.14.1 Scope and Usage

This resource is a request resource from a FHIR workflow perspective - see Workflow.

ServiceRequest is a record of a request for a procedure or diagnostic or other service to be planned, proposed, or performed, as distinguished by the ServiceRequest.intent field value, with or on a patient. The procedure will lead to either a Procedure or DiagnosticReport, which in turn may reference one or more Observations, which summarize the performance of the procedures and associated documentation such as observations, images, findings that are relevant to the treatment/management of the subject. This resource

may be used to share relevant information required to support a referral or a transfer of care request from one practitioner or organization to another when a patient is required to be referred to another provider for a consultation /second opinion and/or for short term or longer term management of one or more health issues or problems.

Examples include:

• diagnostic tests/studies
• endoscopic procedures
• counseling
• biopsies
• therapies (e.g., physio-, social-, psychological-)
• (exploratory) surgeries or procedures
• exercises
• specialist consultation and assessments
• community services
• nursing services
• pharmacist medication review, and
• other clinical interventions.

Procedures may be performed by a healthcare professional, a friend or relative or in some cases by the patient themselves.

The principal intention of ServiceRequest is to support ordering procedures for one patient (which includes non-human patients in veterinary medicine). However, in many contexts, healthcare related processes include performing diagnostic investigations on groups of subjects, devices involved in the provision of healthcare, and even environmental locations such as ducts, bodies of water, etc. ServiceRequest supports all these usages. The service request may represent an order that is entered by a practitioner in a CPOE system as well as a proposal made by a clinical decision support (CDS) system based on a patient's clinical record and context of care. Planned procedures referenced by a CarePlan may also be represented by this resource.

The general work flow that this resource facilitates is that a clinical system creates a service request. The service request is then accessed by or exchanged with a system, perhaps via intermediaries, that represents an organization (e.g., diagnostic or imaging service, surgical team, physical therapy department) that can perform the procedure. The organization receiving the service request will, after it accepts the request, update the request as the work is performed, and then finally issue a report that references the requests that it fulfilled.

The ServiceRequest resource allows requesting only a single procedure. If a workflow requires requesting multiple procedures simultaneously, this is done using multiple instances of this resource. These instances can be linked in different ways, depending on the needs of the workflow. For guidance, refer to the Request pattern

12.14.2 Boundaries and Relationships

ServiceRequest is a record of a proposal/plan or order for a service to be performed that would result in a Procedure, Observation, DiagnosticReport, ImagingStudy or similar resource. In contrast to ServiceRequest, Task which spans both intent and event and tracks the execution through to completion and is intended for "administrative" actions like requesting and tracking things to be done to a record, or keeping track of a checklist of steps such to be performed as part of a fulfilment process. A ServiceRequest can be higher-level authorization that triggered the creation of Task, or it can be the "request" resource Task is seeking to fulfill.

ServiceRequest and CommunicationRequest are related. A CommunicationRequest is a request to merely disclose information. Whereas a ServiceRequest would be used to request information as part of training or counseling - i.e. when the process will involve verification of the patient's comprehension or an attempt to change the patient's mental state. In some workflows both may exist. For example, upon receiving a CommunicationRequest a practitioner might initiate a ServiceRequest.

This resource is referenced by Appointment, BiologicallyDerivedProduct, CarePlan, Claim, DeviceUseStatement, DiagnosticReport, Encounter, EpisodeOfCare, ExplanationOfBenefit, Goal, ImagingStudy, Media, MedicationRequest, MedicationStatement, Observation, Procedure, QuestionnaireResponse, itself and Specimen

12.14.3 Resource Content

• Structure
• UML
• XML
• JSON

- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
ServiceRequest | ITU | | DomainResource | A request for a service to be performed + Rule: orderDetail SHALL only be present if code is present Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | Identifiers assigned to this order
instantiatesCanonical | Σ | 0..* | canonical(ActivityDefinition | PlanDefinition) | Instantiates FHIR protocol or definition
instantiatesUri | Σ | 0..* | uri | Instantiates external protocol or definition
basedOn | Σ | 0..* | Reference(CarePlan | ServiceRequest | MedicationRequest) | What request fulfills
replaces | Σ | 0..* | Reference(ServiceRequest) | What request replaces
requisition | Σ | 0..1 | Identifier | Composite Request ID
status | ?!Σ | 1..1 | code | draft | active | on-hold | revoked | completed | entered-in-error | unknown RequestStatus (Required)
intent | ?!Σ | 1..1 | code | proposal | plan | directive | order | original-order | reflex-order | filler-order | instance-order | option RequestIntent (Required)
category | Σ | 0..* | CodeableConcept | Classification of service Service Request Category Codes (Example)
priority | Σ | 0..1 | code | routine | urgent | asap | stat Request priority (Required)
doNotPerform | ?!Σ | 0..1 | boolean | True if service/procedure should not be performed
code | Σ | 0..1 | CodeableConcept | What is being requested/ordered Procedure Codes (SNOMED CT) (Example)
orderDetail | ΣI | 0..* | CodeableConcept | Additional order information Service Request Order Details Codes (Example)
quantity[x] | Σ | 0..1 | | Service amount
quantityQuantity | | | Quantity
quantityRatio | | | Ratio
quantityRange | | | Range
subject | Σ | 1..1 | Reference(Patient | Group | Location | Device) | Individual or Entity the service is ordered for
encounter | Σ | 0..1 | Reference(Encounter) | Encounter in which the request was created
occurrence[x] | Σ | 0..1 | | When service should occur
occurrenceDateTime | | | dateTime
occurrencePeriod | | | Period
occurrenceTiming | | | Timing
asNeeded[x] | Σ | 0..1 | | Preconditions for service SNOMED CT Medication As Needed Reason Codes (Example)
asNeededBoolean | | | boolean
asNeededCodeableConcept | | | CodeableConcept
authoredOn | Σ | 0..1 | dateTime | Date request signed
requester | Σ | 0..1 | Reference(Practitioner | PractitionerRole | Organization | Patient | RelatedPerson | Device) | Who/what is requesting service
performerType | Σ | 0..1 | CodeableConcept | Performer role Participant Roles (Example)
performer | Σ | 0..* | Reference(Practitioner | PractitionerRole | Organization | CareTeam | HealthcareService | Patient | Device | RelatedPerson) | Requested performer
locationCode | Σ | 0..* | CodeableConcept | Requested location V3 Value SetServiceDeliveryLocationRoleType (Example)
locationReference | Σ | 0..* | Reference(Location) | Requested location
reasonCode | Σ | 0..* | CodeableConcept | Explanation/Justification for procedure or service Procedure Reason Codes (Example)
reasonReference | Σ | 0..* | Reference(Condition | Observation | DiagnosticReport | DocumentReference) | Explanation/Justification for service or service
insurance | | 0..* | Reference(Coverage | ClaimResponse) | Associated insurance coverage
supportingInfo | | 0..* | Reference(Any) | Additional clinical information
specimen | Σ | 0..* | Reference(Specimen) | Procedure Samples
bodySite | Σ | 0..* | CodeableConcept | Location on Body SNOMED CT Body Structures (Example)
note | | 0..* | Annotation | Comments
patientInstruction | Σ | 0..1 | string | Patient or consumer-oriented instructions
relevantHistory | | 0..* | Reference(Provenance) | Request provenance
Documentation for this format

UML Diagram (Legend)

XML Template

```
<ServiceRequest xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Identifiers assigned to this order --></identifier>
 <instantiatesCanonical><!-- 0..* canonical(ActivityDefinition|PlanDefinition) Instantiates FHIR protocol or definition
    --></instantiatesCanonical>
 <instantiatesUri value="[uri]"/><!-- 0..* Instantiates external protocol or definition -->
 <basedOn><!-- 0..* Reference(CarePlan|ServiceRequest|MedicationRequest) What request fulfills --></basedOn>
 <replaces><!-- 0..* Reference(ServiceRequest) What request replaces --></replaces>
 <requisition><!-- 0..1 Identifier Composite Request ID --></requisition>
 <status value="[code]"/><!-- 1..1 draft | active | on-hold | revoked | completed | entered-in-error | unknown -->
 <intent value="[code]"/><!-- 1..1 proposal | plan | directive | order | original-order | reflex-order | filler-order |
    instance-order | option -->
 <category><!-- 0..* CodeableConcept Classification of service --></category>
 <priority value="[code]"/><!-- 0..1 routine | urgent | asap | stat -->
 <doNotPerform value="[boolean]"/><!-- 0..1 True if service/procedure should not be performed -->
 <code><!-- 0..1 CodeableConcept What is being requested/ordered --></code>
 <orderDetail><!--  0..* CodeableConcept Additional order information --></orderDetail>
 <quantity[x]><!-- 0..1 Quantity|Ratio|Range Service amount --></quantity[x]>
```

```
 <subject><!-- 1..1 Reference(Patient|Group|Location|Device) Individual or Entity the service is ordered for
    --></subject>
 <encounter><!-- 0..1 Reference(Encounter) Encounter in which the request was created --></encounter>
 <occurrence[x]><!-- 0..1 dateTime|Period|Timing When service should occur --></occurrence[x]>
 <asNeeded[x]><!-- 0..1 boolean|CodeableConcept Preconditions for service --></asNeeded[x]>
 <authoredOn value="[dateTime]"/><!-- 0..1 Date request signed -->
 <requester><!-- 0..1 Reference(Practitioner|PractitionerRole|Organization|
   Patient|RelatedPerson|Device) Who/what is requesting service --></requester>
 <performerType><!-- 0..1 CodeableConcept Performer role --></performerType>
 <performer><!-- 0..* Reference(Practitioner|PractitionerRole|Organization|
   CareTeam|HealthcareService|Patient|Device|RelatedPerson) Requested performer --></performer>
 <locationCode><!-- 0..* CodeableConcept Requested location --></locationCode>
 <locationReference><!-- 0..* Reference(Location) Requested location --></locationReference>
 <reasonCode><!-- 0..* CodeableConcept Explanation/Justification for procedure or service --></reasonCode>
 <reasonReference><!-- 0..* Reference(Condition|Observation|DiagnosticReport|
   DocumentReference) Explanation/Justification for service or service --></reasonReference>
 <insurance><!-- 0..* Reference(Coverage|ClaimResponse) Associated insurance coverage --></insurance>
 <supportingInfo><!-- 0..* Reference(Any) Additional clinical information --></supportingInfo>
 <specimen><!-- 0..* Reference(Specimen) Procedure Samples --></specimen>
 <bodySite><!-- 0..* CodeableConcept Location on Body --></bodySite>
 <note><!-- 0..* Annotation Comments --></note>
 <patientInstruction value="[string]"/><!-- 0..1 Patient or consumer-oriented instructions -->
 <relevantHistory><!-- 0..* Reference(Provenance) Request provenance --></relevantHistory>
</ServiceRequest>
```

JSON Template

```
{
  "resourceType" : "ServiceRequest",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Identifiers assigned to this order
  "instantiatesCanonical" : [{ canonical(ActivityDefinition|PlanDefinition) }], // Instantiates FHIR protocol or
    definition
  "instantiatesUri" : ["<uri>"], // Instantiates external protocol or definition
  "basedOn" : [{ Reference(CarePlan|ServiceRequest|MedicationRequest) }], // What request fulfills
  "replaces" : [{ Reference(ServiceRequest) }], // What request replaces
  "requisition" : { Identifier }, // Composite Request ID
  "status" : "<code>", // R!  draft | active | on-hold | revoked | completed | entered-in-error | unknown
  "intent" : "<code>", // R!  proposal | plan | directive | order | original-order | reflex-order | filler-order |
    instance-order | option
  "category" : [{ CodeableConcept }], // Classification of service
  "priority" : "<code>", // routine | urgent | asap | stat
  "doNotPerform" : <boolean>, // True if service/procedure should not be performed
  "code" : { CodeableConcept }, // What is being requested/ordered
  "orderDetail" : [{ CodeableConcept }], // C? Additional order information
  // quantity[x]: Service amount. One of these 3:
  "quantityQuantity" : { Quantity },
  "quantityRatio" : { Ratio },
  "quantityRange" : { Range },
  "subject" : { Reference(Patient|Group|Location|Device) }, // R!  Individual or Entity the service is ordered for
  "encounter" : { Reference(Encounter) }, // Encounter in which the request was created
  // occurrence[x]: When service should occur. One of these 3:
  "occurrenceDateTime" : "<dateTime>",
  "occurrencePeriod" : { Period },
  "occurrenceTiming" : { Timing },
  // asNeeded[x]: Preconditions for service. One of these 2:
  "asNeededBoolean" : <boolean>,
  "asNeededCodeableConcept" : { CodeableConcept },
  "authoredOn" : "<dateTime>", // Date request signed
  "requester" : { Reference(Practitioner|PractitionerRole|Organization|
   Patient|RelatedPerson|Device) }, // Who/what is requesting service
  "performerType" : { CodeableConcept }, // Performer role
  "performer" : [{ Reference(Practitioner|PractitionerRole|Organization|
   CareTeam|HealthcareService|Patient|Device|RelatedPerson) }], // Requested performer
  "locationCode" : [{ CodeableConcept }], // Requested location
  "locationReference" : [{ Reference(Location) }], // Requested location
  "reasonCode" : [{ CodeableConcept }], // Explanation/Justification for procedure or service
  "reasonReference" : [{ Reference(Condition|Observation|DiagnosticReport|
   DocumentReference) }], // Explanation/Justification for service or service
  "insurance" : [{ Reference(Coverage|ClaimResponse) }], // Associated insurance coverage
  "supportingInfo" : [{ Reference(Any) }], // Additional clinical information
  "specimen" : [{ Reference(Specimen) }], // Procedure Samples
  "bodySite" : [{ CodeableConcept }], // Location on Body
  "note" : [{ Annotation }], // Comments
  "patientInstruction" : "<string>", // Patient or consumer-oriented instructions
  "relevantHistory" : [{ Reference(Provenance) }] // Request provenance
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:ServiceRequest;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:ServiceRequest.identifier [ Identifier ], ... ; # 0..* Identifiers assigned to this order
  fhir:ServiceRequest.instantiatesCanonical [ canonical(ActivityDefinition|PlanDefinition) ], ... ; # 0..* Instantiates
```

```
    FHIR protocol or definition
  fhir:ServiceRequest.instantiatesUri [ uri ], ... ; # 0..* Instantiates external protocol or definition
  fhir:ServiceRequest.basedOn [ Reference(CarePlan|ServiceRequest|MedicationRequest) ], ... ; # 0..* What request
    fulfills
  fhir:ServiceRequest.replaces [ Reference(ServiceRequest) ], ... ; # 0..* What request replaces
  fhir:ServiceRequest.requisition [ Identifier ]; # 0..1 Composite Request ID
  fhir:ServiceRequest.status [ code ]; # 1..1 draft | active | on-hold | revoked | completed | entered-in-error |
    unknown
  fhir:ServiceRequest.intent [ code ]; # 1..1 proposal | plan | directive | order | original-order | reflex-order |
    filler-order | instance-order | option
  fhir:ServiceRequest.category [ CodeableConcept ], ... ; # 0..* Classification of service
  fhir:ServiceRequest.priority [ code ]; # 0..1 routine | urgent | asap | stat
  fhir:ServiceRequest.doNotPerform [ boolean ]; # 0..1 True if service/procedure should not be performed
  fhir:ServiceRequest.code [ CodeableConcept ]; # 0..1 What is being requested/ordered
  fhir:ServiceRequest.orderDetail [ CodeableConcept ], ... ; # 0..* Additional order information
  # ServiceRequest.quantity[x] : 0..1 Service amount. One of these 3
    fhir:ServiceRequest.quantityQuantity [ Quantity ]
    fhir:ServiceRequest.quantityRatio [ Ratio ]
    fhir:ServiceRequest.quantityRange [ Range ]
  fhir:ServiceRequest.subject [ Reference(Patient|Group|Location|Device) ]; # 1..1 Individual or Entity the service is
    ordered for
  fhir:ServiceRequest.encounter [ Reference(Encounter) ]; # 0..1 Encounter in which the request was created
  # ServiceRequest.occurrence[x] : 0..1 When service should occur. One of these 3
    fhir:ServiceRequest.occurrenceDateTime [ dateTime ]
    fhir:ServiceRequest.occurrencePeriod [ Period ]
    fhir:ServiceRequest.occurrenceTiming [ Timing ]
  # ServiceRequest.asNeeded[x] : 0..1 Preconditions for service. One of these 2
    fhir:ServiceRequest.asNeededBoolean [ boolean ]
    fhir:ServiceRequest.asNeededCodeableConcept [ CodeableConcept ]
  fhir:ServiceRequest.authoredOn [ dateTime ]; # 0..1 Date request signed
  fhir:ServiceRequest.requester [ Reference(Practitioner|PractitionerRole|Organization|Patient|RelatedPerson|Device) ];
    # 0..1 Who/what is requesting service
  fhir:ServiceRequest.performerType [ CodeableConcept ]; # 0..1 Performer role
  fhir:ServiceRequest.performer [
    Reference(Practitioner|PractitionerRole|Organization|CareTeam|HealthcareService|Patient|
  Device|RelatedPerson) ], ... ; # 0..* Requested performer
  fhir:ServiceRequest.locationCode [ CodeableConcept ], ... ; # 0..* Requested location
  fhir:ServiceRequest.locationReference [ Reference(Location) ], ... ; # 0..* Requested location
  fhir:ServiceRequest.reasonCode [ CodeableConcept ], ... ; # 0..* Explanation/Justification for procedure or service
  fhir:ServiceRequest.reasonReference [ Reference(Condition|Observation|DiagnosticReport|DocumentReference) ], ... ; #
    0..* Explanation/Justification for service or service
  fhir:ServiceRequest.insurance [ Reference(Coverage|ClaimResponse) ], ... ; # 0..* Associated insurance coverage
  fhir:ServiceRequest.supportingInfo [ Reference(Any) ], ... ; # 0..* Additional clinical information
  fhir:ServiceRequest.specimen [ Reference(Specimen) ], ... ; # 0..* Procedure Samples
  fhir:ServiceRequest.bodySite [ CodeableConcept ], ... ; # 0..* Location on Body
  fhir:ServiceRequest.note [ Annotation ], ... ; # 0..* Comments
  fhir:ServiceRequest.patientInstruction [ string ]; # 0..1 Patient or consumer-oriented instructions
  fhir:ServiceRequest.relevantHistory [ Reference(Provenance) ], ... ; # 0..* Request provenance
]
```

Changes since R3

This resource did not exist in Release 2

This analysis is available as XML or JSON.

Structure

Name | Flags | Card. | Type | Description & Constraints

ServiceRequest | ITU | | DomainResource | A request for a service to be performed + Rule: orderDetail SHALL only be present if code is present Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

identifier | Σ | 0..* | Identifier | Identifiers assigned to this order

instantiatesCanonical | Σ | 0..* | canonical(ActivityDefinition | PlanDefinition) | Instantiates FHIR protocol or definition

instantiatesUri | Σ | 0..* | uri | Instantiates external protocol or definition

basedOn | Σ | 0..* | Reference(CarePlan | ServiceRequest | MedicationRequest) | What request fulfills

replaces | Σ | 0..* | Reference(ServiceRequest) | What request replaces

requisition | Σ | 0..1 | Identifier | Composite Request ID

status | ?!Σ | 1..1 | code | draft | active | on-hold | revoked | completed | entered-in-error | unknown RequestStatus (Required)

intent | ?!Σ | 1..1 | code | proposal | plan | directive | order | original-order | reflex-order | filler-order | instance-order | option RequestIntent (Required)

category | Σ | 0..* | CodeableConcept | Classification of service Service Request Category Codes (Example)

priority | Σ | 0..1 | code | routine | urgent | asap | stat Request priority (Required)

doNotPerform | ?!Σ | 0..1 | boolean | True if service/procedure should not be performed

code | Σ | 0..1 | CodeableConcept | What is being requested/ordered Procedure Codes (SNOMED CT) (Example)

orderDetail | ΣI | 0..* | CodeableConcept | Additional order information Service Request Order Details Codes (Example)

quantity[x] | Σ | 0..1 | | Service amount

quantityQuantity | | | Quantity

quantityRatio | | | Ratio

quantityRange | | | Range

subject | Σ | 1..1 | Reference(Patient | Group | Location | Device) | Individual or Entity the service is ordered for

encounter | Σ | 0..1 | Reference(Encounter) | Encounter in which the request was created

occurrence[x] | Σ | 0..1 | | When service should occur

occurrenceDateTime | | | dateTime

occurrencePeriod | | | Period

occurrenceTiming | | | Timing

asNeeded[x] | Σ | 0..1 | | Preconditions for service SNOMED CT Medication As Needed Reason Codes (Example)
asNeededBoolean | | | boolean
asNeededCodeableConcept | | | CodeableConcept
authoredOn | Σ | 0..1 | dateTime | Date request signed
requester | Σ | 0..1 | Reference(Practitioner | PractitionerRole | Organization | Patient | RelatedPerson | Device) | Who/what is requesting service
performerType | Σ | 0..1 | CodeableConcept | Performer role Participant Roles (Example)
performer | Σ | 0..* | Reference(Practitioner | PractitionerRole | Organization | CareTeam | HealthcareService | Patient | Device | RelatedPerson) | Requested performer
locationCode | Σ | 0..* | CodeableConcept | Requested location V3 Value SetServiceDeliveryLocationRoleType (Example)
locationReference | Σ | 0..* | Reference(Location) | Requested location
reasonCode | Σ | 0..* | CodeableConcept | Explanation/Justification for procedure or service Procedure Reason Codes (Example)
reasonReference | Σ | 0..* | Reference(Condition | Observation | DiagnosticReport | DocumentReference) | Explanation/Justification for service or service
insurance | | 0..* | Reference(Coverage | ClaimResponse) | Associated insurance coverage
supportingInfo | | 0..* | Reference(Any) | Additional clinical information
specimen | Σ | 0..* | Reference(Specimen) | Procedure Samples
bodySite | Σ | 0..* | CodeableConcept | Location on Body SNOMED CT Body Structures (Example)
note | | 0..* | Annotation | Comments
patientInstruction | Σ | 0..1 | string | Patient or consumer-oriented instructions
relevantHistory | | 0..* | Reference(Provenance) | Request provenance
Documentation for this format

UML Diagram (Legend)

XML Template

```
<ServiceRequest xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Identifiers assigned to this order --></identifier>
 <instantiatesCanonical><!-- 0..* canonical(ActivityDefinition|PlanDefinition) Instantiates FHIR protocol or definition
    --></instantiatesCanonical>
 <instantiatesUri value="[uri]"/><!-- 0..* Instantiates external protocol or definition -->
 <basedOn><!-- 0..* Reference(CarePlan|ServiceRequest|MedicationRequest) What request fulfills --></basedOn>
 <replaces><!-- 0..* Reference(ServiceRequest) What request replaces --></replaces>
 <requisition><!-- 0..1 Identifier Composite Request ID --></requisition>
 <status value="[code]"/><!-- 1..1 draft | active | on-hold | revoked | completed | entered-in-error | unknown -->
 <intent value="[code]"/><!-- 1..1 proposal | plan | directive | order | original-order | reflex-order | filler-order |
    instance-order | option -->
 <category><!-- 0..* CodeableConcept Classification of service --></category>
 <priority value="[code]"/><!-- 0..1 routine | urgent | asap | stat -->
 <doNotPerform value="[boolean]"/><!-- 0..1 True if service/procedure should not be performed -->
 <code><!-- 0..1 CodeableConcept What is being requested/ordered --></code>
 <orderDetail><!--  0..* CodeableConcept Additional order information --></orderDetail>
 <quantity[x]><!-- 0..1 Quantity|Ratio|Range Service amount --></quantity[x]>
 <subject><!-- 1..1 Reference(Patient|Group|Location|Device) Individual or Entity the service is ordered for
    --></subject>
 <encounter><!-- 0..1 Reference(Encounter) Encounter in which the request was created --></encounter>
 <occurrence[x]><!-- 0..1 dateTime|Period|Timing When service should occur --></occurrence[x]>
 <asNeeded[x]><!-- 0..1 boolean|CodeableConcept Preconditions for service --></asNeeded[x]>
 <authoredOn value="[dateTime]"/><!-- 0..1 Date request signed -->
 <requester><!-- 0..1 Reference(Practitioner|PractitionerRole|Organization|
   Patient|RelatedPerson|Device) Who/what is requesting service --></requester>
 <performerType><!-- 0..1 CodeableConcept Performer role --></performerType>
 <performer><!-- 0..* Reference(Practitioner|PractitionerRole|Organization|
   CareTeam|HealthcareService|Patient|Device|RelatedPerson) Requested performer --></performer>
 <locationCode><!-- 0..* CodeableConcept Requested location --></locationCode>
 <locationReference><!-- 0..* Reference(Location) Requested location --></locationReference>
 <reasonCode><!-- 0..* CodeableConcept Explanation/Justification for procedure or service --></reasonCode>
 <reasonReference><!-- 0..* Reference(Condition|Observation|DiagnosticReport|
   DocumentReference) Explanation/Justification for service or service --></reasonReference>
 <insurance><!-- 0..* Reference(Coverage|ClaimResponse) Associated insurance coverage --></insurance>
 <supportingInfo><!-- 0..* Reference(Any) Additional clinical information --></supportingInfo>
 <specimen><!-- 0..* Reference(Specimen) Procedure Samples --></specimen>
 <bodySite><!-- 0..* CodeableConcept Location on Body --></bodySite>
 <note><!-- 0..* Annotation Comments --></note>
 <patientInstruction value="[string]"/><!-- 0..1 Patient or consumer-oriented instructions -->
 <relevantHistory><!-- 0..* Reference(Provenance) Request provenance --></relevantHistory>
</ServiceRequest>
```

JSON Template

```
{
  "resourceType" : "ServiceRequest",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Identifiers assigned to this order
  "instantiatesCanonical" : [{ canonical(ActivityDefinition|PlanDefinition) }], // Instantiates FHIR protocol or
    definition
  "instantiatesUri" : ["<uri>"], // Instantiates external protocol or definition
  "basedOn" : [{ Reference(CarePlan|ServiceRequest|MedicationRequest) }], // What request fulfills
  "replaces" : [{ Reference(ServiceRequest) }], // What request replaces
  "requisition" : { Identifier }, // Composite Request ID
  "status" : "<code>", // R!  draft | active | on-hold | revoked | completed | entered-in-error | unknown
  "intent" : "<code>", // R!  proposal | plan | directive | order | original-order | reflex-order | filler-order |
```

```
    instance-order | option
  "category" : [{ CodeableConcept }], // Classification of service
  "priority" : "<code>", // routine | urgent | asap | stat
  "doNotPerform" : <boolean>, // True if service/procedure should not be performed
  "code" : { CodeableConcept }, // What is being requested/ordered
  "orderDetail" : [{ CodeableConcept }], // C? Additional order information
  // quantity[x]: Service amount. One of these 3:
  "quantityQuantity" : { Quantity },
  "quantityRatio" : { Ratio },
  "quantityRange" : { Range },
  "subject" : { Reference(Patient|Group|Location|Device) }, // R!  Individual or Entity the service is ordered for
  "encounter" : { Reference(Encounter) }, // Encounter in which the request was created
  // occurrence[x]: When service should occur. One of these 3:
  "occurrenceDateTime" : "<dateTime>",
  "occurrencePeriod" : { Period },
  "occurrenceTiming" : { Timing },
  // asNeeded[x]: Preconditions for service. One of these 2:
  "asNeededBoolean" : <boolean>,
  "asNeededCodeableConcept" : { CodeableConcept },
  "authoredOn" : "<dateTime>", // Date request signed
  "requester" : { Reference(Practitioner|PractitionerRole|Organization|
   Patient|RelatedPerson|Device) }, // Who/what is requesting service
  "performerType" : { CodeableConcept }, // Performer role
  "performer" : [{ Reference(Practitioner|PractitionerRole|Organization|
   CareTeam|HealthcareService|Patient|Device|RelatedPerson) }], // Requested performer
  "locationCode" : [{ CodeableConcept }], // Requested location
  "locationReference" : [{ Reference(Location) }], // Requested location
  "reasonCode" : [{ CodeableConcept }], // Explanation/Justification for procedure or service
  "reasonReference" : [{ Reference(Condition|Observation|DiagnosticReport|
   DocumentReference) }], // Explanation/Justification for service or service
  "insurance" : [{ Reference(Coverage|ClaimResponse) }], // Associated insurance coverage
  "supportingInfo" : [{ Reference(Any) }], // Additional clinical information
  "specimen" : [{ Reference(Specimen) }], // Procedure Samples
  "bodySite" : [{ CodeableConcept }], // Location on Body
  "note" : [{ Annotation }], // Comments
  "patientInstruction" : "<string>", // Patient or consumer-oriented instructions
  "relevantHistory" : [{ Reference(Provenance) }] // Request provenance
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:ServiceRequest;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:ServiceRequest.identifier [ Identifier ], ... ; # 0..* Identifiers assigned to this order
  fhir:ServiceRequest.instantiatesCanonical [ canonical(ActivityDefinition|PlanDefinition) ], ... ; # 0..* Instantiates
    FHIR protocol or definition
  fhir:ServiceRequest.instantiatesUri [ uri ], ... ; # 0..* Instantiates external protocol or definition
  fhir:ServiceRequest.basedOn [ Reference(CarePlan|ServiceRequest|MedicationRequest) ], ... ; # 0..* What request
    fulfills
  fhir:ServiceRequest.replaces [ Reference(ServiceRequest) ], ... ; # 0..* What request replaces
  fhir:ServiceRequest.requisition [ Identifier ]; # 0..1 Composite Request ID
  fhir:ServiceRequest.status [ code ]; # 1..1 draft | active | on-hold | revoked | completed | entered-in-error |
    unknown
  fhir:ServiceRequest.intent [ code ]; # 1..1 proposal | plan | directive | order | original-order | reflex-order |
    filler-order | instance-order | option
  fhir:ServiceRequest.category [ CodeableConcept ], ... ; # 0..* Classification of service
  fhir:ServiceRequest.priority [ code ]; # 0..1 routine | urgent | asap | stat
  fhir:ServiceRequest.doNotPerform [ boolean ]; # 0..1 True if service/procedure should not be performed
  fhir:ServiceRequest.code [ CodeableConcept ]; # 0..1 What is being requested/ordered
  fhir:ServiceRequest.orderDetail [ CodeableConcept ], ... ; # 0..* Additional order information
  # ServiceRequest.quantity[x] : 0..1 Service amount. One of these 3
    fhir:ServiceRequest.quantityQuantity [ Quantity ]
    fhir:ServiceRequest.quantityRatio [ Ratio ]
    fhir:ServiceRequest.quantityRange [ Range ]
  fhir:ServiceRequest.subject [ Reference(Patient|Group|Location|Device) ]; # 1..1 Individual or Entity the service is
    ordered for
  fhir:ServiceRequest.encounter [ Reference(Encounter) ]; # 0..1 Encounter in which the request was created
  # ServiceRequest.occurrence[x] : 0..1 When service should occur. One of these 3
    fhir:ServiceRequest.occurrenceDateTime [ dateTime ]
    fhir:ServiceRequest.occurrencePeriod [ Period ]
    fhir:ServiceRequest.occurrenceTiming [ Timing ]
  # ServiceRequest.asNeeded[x] : 0..1 Preconditions for service. One of these 2
    fhir:ServiceRequest.asNeededBoolean [ boolean ]
    fhir:ServiceRequest.asNeededCodeableConcept [ CodeableConcept ]
  fhir:ServiceRequest.authoredOn [ dateTime ]; # 0..1 Date request signed
  fhir:ServiceRequest.requester [ Reference(Practitioner|PractitionerRole|Organization|Patient|RelatedPerson|Device) ];
    # 0..1 Who/what is requesting service
  fhir:ServiceRequest.performerType [ CodeableConcept ]; # 0..1 Performer role
  fhir:ServiceRequest.performer [
    Reference(Practitioner|PractitionerRole|Organization|CareTeam|HealthcareService|Patient|
  Device|RelatedPerson) ], ... ; # 0..* Requested performer
  fhir:ServiceRequest.locationCode [ CodeableConcept ], ... ; # 0..* Requested location
  fhir:ServiceRequest.locationReference [ Reference(Location) ], ... ; # 0..* Requested location
  fhir:ServiceRequest.reasonCode [ CodeableConcept ], ... ; # 0..* Explanation/Justification for procedure or service
```

```
  fhir:ServiceRequest.reasonReference [ Reference(Condition|Observation|DiagnosticReport|DocumentReference) ], ... ; #
    0..* Explanation/Justification for service or service
  fhir:ServiceRequest.insurance [ Reference(Coverage|ClaimResponse) ], ... ; # 0..* Associated insurance coverage
  fhir:ServiceRequest.supportingInfo [ Reference(Any) ], ... ; # 0..* Additional clinical information
  fhir:ServiceRequest.specimen [ Reference(Specimen) ], ... ; # 0..* Procedure Samples
  fhir:ServiceRequest.bodySite [ CodeableConcept ], ... ; # 0..* Location on Body
  fhir:ServiceRequest.note [ Annotation ], ... ; # 0..* Comments
  fhir:ServiceRequest.patientInstruction [ string ]; # 0..1 Patient or consumer-oriented instructions
  fhir:ServiceRequest.relevantHistory [ Reference(Provenance) ], ... ; # 0..* Request provenance
]
```

Changes since Release 3

This resource did not exist in Release 2

This analysis is available as XML or JSON.

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

12.14.3.1 Terminology Bindings

Path | Definition | Type | Reference

ServiceRequest.status | The status of a service order. | Required | RequestStatus

ServiceRequest.intent | The kind of service request. | Required | RequestIntent

ServiceRequest.category | Classification of the requested service. | Example | ServiceRequestCategoryCodes

ServiceRequest.priority | Identifies the level of importance to be assigned to actioning the request. | Required | RequestPriority

ServiceRequest.code | Codes for tests or services that can be carried out by a designated individual, organization or healthcare service. For laboratory, LOINC is (preferred)[http://build.fhir.org/terminologies.html#preferred] and a valueset using LOINC Order codes is available [here](valueset-diagnostic-requests.html). | Example | ProcedureCodes(SNOMEDCT)

ServiceRequest.orderDetail | Codified order entry details which are based on order context. | Example | ServiceRequestOrderDetailsCodes

ServiceRequest.asNeeded[x] | A coded concept identifying the pre-condition that should hold prior to performing a procedure. For example "pain", "on flare-up", etc. | Example | SNOMEDCTMedicationAsNeededReasonCodes

ServiceRequest.performerType | Indicates specific responsibility of an individual within the care team, such as "Primary physician", "Team coordinator", "Caregiver", etc. | Example | ParticipantRoles

ServiceRequest.locationCode | A location type where services are delivered. | Example | v3.ServiceDeliveryLocationRoleType

ServiceRequest.reasonCode | Diagnosis or problem codes justifying the reason for requesting the service investigation. | Example | ProcedureReasonCodes

ServiceRequest.bodySite | Codes describing anatomical locations. May include laterality. | Example | SNOMEDCTBodyStructures

12.14.3.2 Constraints

id | Level | Location | Description | Expression

prr-1 | Rule | (base) | orderDetail SHALL only be present if code is present | orderDetail.empty() or code.exists()

12.14.4 Notes:

• Many service requests will create a need to specify a specimen, body site, or body system. The request code will often have this information embedded in it - for example, 'serum glucose' or 'chest x-ray'. Alternatively, the specimen or bodysite element may be used to specify it.

• The ServiceRequest should only reference the Specimen resource directly when the diagnostic investigation is requested on already existing specimens. Conversely, if the request is entered first with an uncollected specimen, the Specimen resource will reference the ServiceRequest resource when it is created.

• The reasonCode element is often for billing purposes. It may relate to the resources referred to in supportingInfo element and may be used to decide how a procedure or diagnostic investigation will be performed, or even if it will be performed at all

12.14.5 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common

authored | date | Date request signed | ServiceRequest.authoredOn

based-on | reference | What request fulfills | ServiceRequest.basedOn (CarePlan, MedicationRequest, ServiceRequest)

body-site | token | Where procedure is going to be done | ServiceRequest.bodySite

category | token | Classification of service | ServiceRequest.category

code | token | What is being requested/ordered | ServiceRequest.code | 13 Resources

encounter | reference | An encounter in which this request is made | ServiceRequest.encounter (Encounter) | 11 Resources

identifier | token | Identifiers assigned to this order | ServiceRequest.identifier | 29 Resources

instantiates-canonical | reference | Instantiates FHIR protocol or definition | ServiceRequest.instantiatesCanonical (PlanDefinition, ActivityDefinition)

instantiates-uri | uri | Instantiates external protocol or definition | ServiceRequest.instantiatesUri

intent | token | proposal | plan | directive | order | original-order | reflex-order | filler-order | instance-order | option | ServiceRequest.intent

occurrence | date | When service should occur | ServiceRequest.occurrence

patient | reference | Search by subject - a patient | ServiceRequest.subject.where(resolve() is Patient) (Patient) | 32 Resources

performer | reference | Requested performer | ServiceRequest.performer (Practitioner, Organization, CareTeam, Device, Patient, HealthcareService, PractitionerRole, RelatedPerson)
performer-type | token | Performer role | ServiceRequest.performerType
priority | token | routine | urgent | asap | stat | ServiceRequest.priority
replaces | reference | What request replaces | ServiceRequest.replaces (ServiceRequest)
requester | reference | Who/what is requesting service | ServiceRequest.requester (Practitioner, Organization, Device, Patient, PractitionerRole, RelatedPerson)
requisition | token | Composite Request ID | ServiceRequest.requisition
specimen | reference | Specimen to be tested | ServiceRequest.specimen (Specimen)
status | token | draft | active | on-hold | revoked | completed | entered-in-error | unknown | ServiceRequest.status
subject | reference | Search by subject | ServiceRequest.subject (Group, Device, Patient, Location)

12.14.6 Resource ServiceRequest - Examples

URL: https://hl7.org/fhir/R4/servicerequest-examples.html

12.14.6 Resource ServiceRequest - Examples

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

An introduction to this page as well as proper examples are to be submitted shortly.

Example Name | id | Format
Lipid Panel Order | lipid | XML | JSON | Turtle
CT Scan order | di | XML | JSON | Turtle
Free T4 Add on Order | ft4 | XML | JSON | Turtle
An example of a Head CT procedure | example | XML | JSON | Turtle
Example with sub-requests derived from the request | subrequest | XML | JSON | Turtle
An example of an order for home physiotherapy | physiotherapy | XML | JSON | Turtle
An example of an order to not turn a patient | do-not-turn | XML | JSON | Turtle
Part of an exercise plan | benchpress | XML | JSON | Turtle
Example of an order for an implant | example-implant | XML | JSON | Turtle
Example of an order for anambulation procedure | ambulation | XML | JSON | Turtle
Example of an order for a biopsy procedure that was part of a colonoscopy | colon-biopsy | XML | JSON | Turtle
Example of an order for a colonoscopy procedure with complication | colonoscopy | XML | JSON | Turtle
Example of an order for an appendectomy procedure that is primarily narrative | appendectomy-narrative | XML | JSON | Turtle
Example of an order for an obstetric procedure | ob | XML | JSON | Turtle
Example of an order for an physical therapy evaluation procedure | physical-therapy | XML | JSON | Turtle
Example of an order for patient education | education | XML | JSON | Turtle
An example of a Myringotomy referral request | myringotomy | XML | JSON | Turtle
Order for mechanical ventilation with order details | vent | XML | JSON | Turtle
PGx Phenotype example | example-pgx | XML | JSON | Turtle | for Profile ServiceRequest-Genetics
Order for ABCB4 gene mutation analysis | og-example1 | XML | JSON | Turtle | for Profile ServiceRequest-Genetics

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

12.14.7 Resource ServiceRequest - Detailed Descriptions

URL: https://hl7.org/fhir/R4/servicerequest-definitions.html

12.14.7 Resource ServiceRequest - Detailed Descriptions

Orders and Observations Work Group | Maturity Level: 2 | Trial Use | Security Category: Patient | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

Detailed Descriptions for the elements in the ServiceRequest resource.

ServiceRequest
Element Id | ServiceRequest
Definition | A record of a request for service such as diagnostic investigations, treatments, or operations to be performed.
Cardinality | 0..*
Type | DomainResource
Alternate Names | diagnostic request; referral; referral request; transfer of care request
Invariants | Defined on this element prr-1 Rule orderDetail SHALL only be present if code is present orderDetail.empty() or code.exists()
Defined on this element
prr-1 | Rule | orderDetail SHALL only be present if code is present | orderDetail.empty() or code.exists()
ServiceRequest.identifier
Element Id | ServiceRequest.identifier
Definition | Identifiers assigned to this order instance by the orderer and/or the receiver and/or order fulfiller.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Summary | true

Comments | The identifier.type element is used to distinguish between the identifiers assigned by the orderer (known as the 'Placer' in HL7 v2) and the producer of the observations in response to the order (known as the 'Filler' in HL7 v2). For further discussion and examples see the resource notes section below.
ServiceRequest.instantiatesCanonical
Element Id | ServiceRequest.instantiatesCanonical
Definition | The URL pointing to a FHIR-defined protocol, guideline, orderset or other definition that is adhered to in whole or in part by this ServiceRequest.
Cardinality | 0..*
Type | canonical(ActivityDefinition | PlanDefinition)
Summary | true
Comments | Note: This is a business identifier, not a resource identifier (see discussion). It is best practice for the identifier to only appear on a single resource instance, however business practices may occasionally dictate that multiple resource instances with the same identifier can exist - possibly even with different resource types. For example, multiple Patient and a Person resource instance might share the same social insurance number.
ServiceRequest.instantiatesUri
Element Id | ServiceRequest.instantiatesUri
Definition | The URL pointing to an externally maintained protocol, guideline, orderset or other definition that is adhered to in whole or in part by this ServiceRequest.
Cardinality | 0..*
Type | uri
Summary | true
Comments | This might be an HTML page, PDF, etc. or could just be a non-resolvable URI identifier.
ServiceRequest.basedOn
Element Id | ServiceRequest.basedOn
Definition | Plan/proposal/order fulfilled by this request.
Cardinality | 0..*
Type | Reference(CarePlan | ServiceRequest | MedicationRequest)
Alternate Names | fulfills
Summary | true
ServiceRequest.replaces
Element Id | ServiceRequest.replaces
Definition | The request takes the place of the referenced completed or terminated request(s).
Cardinality | 0..*
Type | Reference(ServiceRequest)
Alternate Names | supersedes; prior; renewed order
Summary | true
ServiceRequest.requisition
Element Id | ServiceRequest.requisition
Definition | A shared identifier common to all service requests that were authorized more or less simultaneously by a single author, representing the composite or group identifier.
Cardinality | 0..1
Type | Identifier
Requirements | Some business processes need to know if multiple items were ordered as part of the same "requisition" for billing or other purposes.
Alternate Names | grouperId; groupIdentifier
Summary | true
Comments | Requests are linked either by a "basedOn" relationship (i.e. one request is fulfilling another) or by having a common requisition. Requests that are part of the same requisition are generally treated independently from the perspective of changing their state or maintaining them after initial creation.
ServiceRequest.status
Element Id | ServiceRequest.status
Definition | The status of the order.
Cardinality | 1..1
Terminology Binding | RequestStatus (Required)
Type | code
Is Modifier | true (Reason: This element is labeled as a modifier because it is a status element that contains status entered-in-error which means that the resource should not be treated as valid)
Summary | true
Comments | The status is generally fully in the control of the requester - they determine whether the order is draft or active and, after it has been activated, competed, cancelled or suspended. States relating to the activities of the performer are reflected on either the corresponding event (see Event Pattern for general discussion) or using the Task resource.
ServiceRequest.intent
Element Id | ServiceRequest.intent
Definition | Whether the request is a proposal, plan, an original order or a reflex order.
Cardinality | 1..1
Terminology Binding | RequestIntent (Required)
Type | code
Is Modifier | true (Reason: This element changes the interpretation of all descriptive attributes. For example "the time the request is recommended to occur" vs. "the time the request is authorized to occur" or "who is recommended to perform the request" vs. "who is authorized to perform the request)
Summary | true
Comments | This element is labeled as a modifier because the intent alters when and how the resource is actually applicable.
ServiceRequest.category
Element Id | ServiceRequest.category

Definition | A code that classifies the service for searching, sorting and display purposes (e.g. "Surgical Procedure").
Cardinality | 0..*
Terminology Binding | Service Request Category Codes (Example)
Type | CodeableConcept
Requirements | Used for filtering what service request are retrieved and displayed.
Summary | true
Comments | There may be multiple axis of categorization depending on the context or use case for retrieving or displaying the resource. The level of granularity is defined by the category concepts in the value set.
ServiceRequest.priority
Element Id | ServiceRequest.priority
Definition | Indicates how quickly the ServiceRequest should be addressed with respect to other requests.
Cardinality | 0..1
Terminology Binding | Request priority (Required)
Type | code
Meaning if Missing | If missing, this task should be performed with normal priority
Summary | true
ServiceRequest.doNotPerform
Element Id | ServiceRequest.doNotPerform
Definition | Set this to true if the record is saying that the service/procedure should NOT be performed.
Cardinality | 0..1
Type | boolean
Is Modifier | true (Reason: If true this element negates the specified action. For Example, instead of a request for a procedure, it is a request for the procedure to not occur.)
Meaning if Missing | If missing, the request is a positive request e.g. "do perform"
Requirements | Used for do not ambulate, do not elevate head of bed, do not flush NG tube, do not take blood pressure on a certain arm, etc.
Summary | true
Comments | In general, only the code and timeframe will be present, though occasional additional qualifiers such as body site or even performer could be included to narrow the scope of the prohibition. If the ServiceRequest.code and ServiceRequest.doNotPerform both contain negation, that will reinforce prohibition and should not have a double negative interpretation.
ServiceRequest.code
Element Id | ServiceRequest.code
Definition | A code that identifies a particular service (i.e., procedure, diagnostic investigation, or panel of investigations) that have been requested.
Cardinality | 0..1
Terminology Binding | Procedure Codes (SNOMED CT) (Example)
Type | CodeableConcept
Alternate Names | service requested
Summary | true
Comments | Many laboratory and radiology procedure codes embed the specimen/organ system in the test order name, for example, serum or serum/plasma glucose, or a chest x-ray. The specimen might not be recorded separately from the test code.
ServiceRequest.orderDetail
Element Id | ServiceRequest.orderDetail
Definition | Additional details and instructions about the how the services are to be delivered. For example, and order for a urinary catheter may have an order detail for an external or indwelling catheter, or an order for a bandage may require additional instructions specifying how the bandage should be applied.
Cardinality | 0..*
Terminology Binding | Service Request Order Details Codes (Example)
Type | CodeableConcept
Alternate Names | detailed instructions
Summary | true
Comments | For information from the medical record intended to support the delivery of the requested services, use the supportingInformation element.
Invariants | Affect this element prr-1 Rule orderDetail SHALL only be present if code is present orderDetail.empty() or code.exists()
Affect this element
prr-1 | Rule | orderDetail SHALL only be present if code is present | orderDetail.empty() or code.exists()
ServiceRequest.quantity[x]
Element Id | ServiceRequest.quantity[x]
Definition | An amount of service being requested which can be a quantity (for example $1,500 home modification), a ratio (for example, 20 half day visits per month), or a range (2.0 to 1.8 Gy per fraction).
Cardinality | 0..1
Type | Quantity|Ratio|Range
[x] Note | See Choice of Data Types for further information about how to use [x]
Requirements | When ordering a service the number of service items may need to be specified separately from the the service item.
Summary | true
ServiceRequest.subject
Element Id | ServiceRequest.subject
Definition | On whom or what the service is to be performed. This is usually a human patient, but can also be requested on animals, groups of humans or animals, devices such as dialysis machines, or even locations (typically for environmental scans).
Cardinality | 1..1
Type | Reference(Patient | Group | Location | Device)

Summary | true
ServiceRequest.encounter
Element Id | ServiceRequest.encounter
Definition | An encounter that provides additional information about the healthcare context in which this request is made.
Cardinality | 0..1
Type | Reference(Encounter)
Alternate Names | context
Summary | true
ServiceRequest.occurrence[x]
Element Id | ServiceRequest.occurrence[x]
Definition | The date/time at which the requested service should occur.
Cardinality | 0..1
Type | dateTime|Period|Timing
[x] Note | See Choice of Data Types for further information about how to use [x]
Alternate Names | schedule
Summary | true
ServiceRequest.asNeeded[x]
Element Id | ServiceRequest.asNeeded[x]
Definition | If a CodeableConcept is present, it indicates the pre-condition for performing the service. For example "pain", "on flare-up", etc.
Cardinality | 0..1
Terminology Binding | SNOMED CT Medication As Needed Reason Codes (Example)
Type | boolean|CodeableConcept
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true
ServiceRequest.authoredOn
Element Id | ServiceRequest.authoredOn
Definition | When the request transitioned to being actionable.
Cardinality | 0..1
Type | dateTime
Alternate Names | orderedOn
Summary | true
ServiceRequest.requester
Element Id | ServiceRequest.requester
Definition | The individual who initiated the request and has responsibility for its activation.
Cardinality | 0..1
Type | Reference(Practitioner | PractitionerRole | Organization | Patient | RelatedPerson | Device)
Alternate Names | author; orderer
Summary | true
Comments | This not the dispatcher, but rather who is the authorizer. This element is not intended to handle delegation which would generally be managed through the Provenance resource.
ServiceRequest.performerType
Element Id | ServiceRequest.performerType
Definition | Desired type of performer for doing the requested service.
Cardinality | 0..1
Terminology Binding | Participant Roles (Example)
Type | CodeableConcept
Alternate Names | specialty
Summary | true
Comments | This is a role, not a participation type. In other words, does not describe the task but describes the capacity. For example, “compounding pharmacy”, “psychiatrist” or “internal referral”.
ServiceRequest.performer
Element Id | ServiceRequest.performer
Definition | The desired performer for doing the requested service. For example, the surgeon, dermatopathologist, endoscopist, etc.
Cardinality | 0..*
Type | Reference(Practitioner | PractitionerRole | Organization | CareTeam | HealthcareService | Patient | Device | RelatedPerson)
Alternate Names | request recipient
Summary | true
Comments | If multiple performers are present, it is interpreted as a list of alternative performers without any preference regardless of order. If order of preference is needed use the request-performerOrder extension. Use CareTeam to represent a group of performers (for example, Practitioner A and Practitioner B).
ServiceRequest.locationCode
Element Id | ServiceRequest.locationCode
Definition | The preferred location(s) where the procedure should actually happen in coded or free text form. E.g. at home or nursing day care center.
Cardinality | 0..*
Terminology Binding | V3 Value SetServiceDeliveryLocationRoleType (Example)
Type | CodeableConcept
Summary | true
ServiceRequest.locationReference
Element Id | ServiceRequest.locationReference

Definition | A reference to the the preferred location(s) where the procedure should actually happen. E.g. at home or nursing day care center.
Cardinality | 0..*
Type | Reference(Location)
Summary | true
ServiceRequest.reasonCode
Element Id | ServiceRequest.reasonCode
Definition | An explanation or justification for why this service is being requested in coded or textual form. This is often for billing purposes. May relate to the resources referred to in supportingInfo.
Cardinality | 0..*
Terminology Binding | Procedure Reason Codes (Example)
Type | CodeableConcept
Summary | true
Comments | This element represents why the referral is being made and may be used to decide how the service will be performed, or even if it will be performed at all. Use CodeableConcept.text element if the data is free (uncoded) text as shown in the CT Scan example.
ServiceRequest.reasonReference
Element Id | ServiceRequest.reasonReference
Definition | Indicates another resource that provides a justification for why this service is being requested. May relate to the resources referred to in supportingInfo.
Cardinality | 0..*
Type | Reference(Condition | Observation | DiagnosticReport | DocumentReference)
Summary | true
Comments | This element represents why the referral is being made and may be used to decide how the service will be performed, or even if it will be performed at all. To be as specific as possible, a reference to Observation or Condition should be used if available. Otherwise when referencing DiagnosticReport it should contain a finding in DiagnosticReport.conclusion and/or DiagnosticReport.conclusionCode. When using a reference to DocumentReference, the target document should contain clear findings language providing the relevant reason for this service request. Use the CodeableConcept text element in ServiceRequest.reasonCode if the data is free (uncoded) text as shown in the CT Scan example.
ServiceRequest.insurance
Element Id | ServiceRequest.insurance
Definition | Insurance plans, coverage extensions, pre-authorizations and/or pre-determinations that may be needed for delivering the requested service.
Cardinality | 0..*
Type | Reference(Coverage | ClaimResponse)
ServiceRequest.supportingInfo
Element Id | ServiceRequest.supportingInfo
Definition | Additional clinical information about the patient or specimen that may influence the services or their interpretations. This information includes diagnosis, clinical findings and other observations. In laboratory ordering these are typically referred to as "ask at order entry questions (AOEs)". This includes observations explicitly requested by the producer (filler) to provide context or supporting information needed to complete the order. For example, reporting the amount of inspired oxygen for blood gas measurements.
Cardinality | 0..*
Type | Reference(Any)
Alternate Names | Ask at order entry question; AOE
Comments | To represent information about how the services are to be delivered use the instructions element.
ServiceRequest.specimen
Element Id | ServiceRequest.specimen
Definition | One or more specimens that the laboratory procedure will use.
Cardinality | 0..*
Type | Reference(Specimen)
Summary | true
Comments | Many diagnostic procedures need a specimen, but the request itself is not actually about the specimen. This element is for when the diagnostic is requested on already existing specimens and the request points to the specimen it applies to. Conversely, if the request is entered first with an unknown specimen, then the Specimen resource points to the ServiceRequest.
ServiceRequest.bodySite
Element Id | ServiceRequest.bodySite
Definition | Anatomic location where the procedure should be performed. This is the target site.
Cardinality | 0..*
Terminology Binding | SNOMED CT Body Structures (Example)
Type | CodeableConcept
Requirements | Knowing where the procedure is performed is important for tracking if multiple sites are possible.
Alternate Names | location
Summary | true
Comments | Only used if not implicit in the code found in ServiceRequest.code. If the use case requires BodySite to be handled as a separate resource instead of an inline coded element (e.g. to identify and track separately) then use the standard extension procedure-targetBodyStructure.
ServiceRequest.note
Element Id | ServiceRequest.note
Definition | Any other notes and comments made about the service request. For example, internal billing notes.
Cardinality | 0..*
Type | Annotation
ServiceRequest.patientInstruction
Element Id | ServiceRequest.patientInstruction

Definition | Instructions in terms that are understood by the patient or consumer.
Cardinality | 0..1
Type | string
Summary | true
ServiceRequest.relevantHistory
Element Id | ServiceRequest.relevantHistory
Definition | Key events in the history of the request.
Cardinality | 0..*
Type | Reference(Provenance)
Comments | This might not include provenances for all versions of the request – only those deemed “relevant” or important. This SHALL NOT include the Provenance associated with this current version of the resource. (If that provenance is deemed to be a “relevant” change, it will need to be added as part of a later update. Until then, it can be queried directly as the Provenance that points to this version using _revinclude All Provenances should have some historical version of this Request as their subject.

12.14.8 Resource ServiceRequest - Mappings

URL: https://hl7.org/fhir/R4/servicerequest-mappings.html

12.14.8 Resource ServiceRequest - Mappings

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

Mappings:

FiveWs Pattern Mapping

Workflow Pattern

HL7 v2 Mapping

RIM Mapping

Quality Improvement and Clinical Knowledge (QUICK)

Mappings for the servicerequest resource (see Mappings to Other Standards for further information & status).

12.14.8.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

ServiceRequest
identifier | FiveWs.identifier
status | FiveWs.status
intent | FiveWs.class
category | FiveWs.class
priority | FiveWs.grade
code | FiveWs.what[x]
subject | FiveWs.subject[x]
encounter | FiveWs.context
occurrence[x] | FiveWs.planned
authoredOn | FiveWs.recorded
requester | FiveWs.author
performerType | FiveWs.actor
performer | FiveWs.actor
locationCode | FiveWs.actor
locationReference | FiveWs.actor
reasonCode | FiveWs.why[x]
reasonReference | FiveWs.why[x]

12.14.8.2 Workflow Pattern (http://hl7.org/fhir/workflow)

ServiceRequest | Request
identifier | Request.identifier
instantiatesCanonical | Request.instantiatesCanonical
instantiatesUri | Request.instantiatesUri
basedOn | Request.basedOn
replaces | Request.replaces
requisition | Request.groupIdentifier
status | Request.status
intent | Request.intent
priority | Request.priority
doNotPerform | Request.doNotPerform
code | Request.code
subject | Request.subject
encounter | Request.encounter
occurrence[x] | Request.occurrence[x]
authoredOn | Request.authoredOn
requester | Request.requester
performerType | Request.performerType
performer | Request.performer

reasonCode | Request.reasonCode
reasonReference | Request.reasonReference
insurance | Request.insurance
supportingInfo | Request.supportingInfo
note | Request.note
relevantHistory | Request.relevantHistory

12.14.8.3 HL7 v2 Mapping (http://hl7.org/v2)

ServiceRequest | ORC
identifier | ORC.2, ORC.3, RF1-6 / RF1-11,
instantiatesCanonical | Varies by domain
instantiatesUri | Varies by domain
basedOn | ORC.8 (plus others)
replaces | Handled by message location of ORC (ORC.1=RO or RU)
requisition | ORC.4
status | ORC.5,RF1-1
intent | N/A
category | RF1-5
priority | TQ1.9, RF1-2
doNotPerform
code | PR1-3 / OBR-4 (varies by domain)
orderDetail | NTE
quantity[x] | NTE
subject | PID
encounter | PV1
occurrence[x] | TQ1/TQ2, OBR-7/OBR-8
asNeeded[x]
authoredOn | ORC.9, RF1-7 / RF1-9
requester | ORC.12, PRT
performerType | PRT, RF!-3
performer | PRT, Practitioner: PRD-2/PRD-7 where PRD-3 = RT; Organization: PRD-10 where PRD-3 = RT
locationCode
locationReference
reasonCode | ORC.16, RF1-10
reasonReference | ORC.16
insurance | IN1/IN2
supportingInfo | Accompanying segments
specimen | SPM
bodySite | SPM
note | NTE
patientInstruction | NTE
relevantHistory | N/A

12.14.8.4 RIM Mapping (http://hl7.org/v3)

ServiceRequest | Act[moodCode<=INT]
identifier | .identifier
instantiatesCanonical | .outboundRelationship[typeCode=DEFN].target
instantiatesUri | .outboundRelationship[typeCode=DEFN].target
basedOn | .outboundRelationship[typeCode=FLFS].target
replaces | .outboundRelationship[typeCode=RPLC].target
requisition | .inboundRelationship(typeCode=COMP].source[moodCode=INT].identifier
status | .status
intent | .moodCode (nuances beyond PRP/PLAN/RQO would need to be elsewhere)
category | .outboundRelationship[typeCode="COMP].target[classCode="LIST", moodCode="INT"].code
priority | .priorityCode
doNotPerform | .actionNegationInd
code | .code
orderDetail | .code
quantity[x] | .quantity
subject | .participation[typeCode=SBJ].role
encounter | .inboundRelationship(typeCode=COMP].source[classCode<=PCPR, moodCode=EVN]
occurrence[x] | .effectiveTime
asNeeded[x] | boolean: precondition.negationInd (inversed - so negationInd = true means asNeeded=false CodeableConcept: precondition.observationEventCriterion[code="Assertion"].value
authoredOn | .participation[typeCode=AUT].time
requester | .participation[typeCode=AUT].role
performerType | .participation[typeCode=PRF].role[scoper.determinerCode=KIND].code
performer | .participation[typeCode=PRF].role[scoper.determinerCode=INSTANCE]
locationCode | .participation[typeCode=LOC].role[scoper.determinerCode=KIND].code
locationReference | .participation[typeCode=LOC].role[scoper.determinerCode=INSTANCE]
reasonCode | .reasonCode
reasonReference | .outboundRelationship[typeCode=RSON].target

insurance | .outboundRelationship[typeCode=COVBY].target
supportingInfo | .outboundRelationship[typeCode=PERT].target
specimen | .participation[typeCode=SPC].role
bodySite | targetSiteCode
note | .inboundRelationship(typeCode=SUBJ].source[classCode=ANNGEN, moodCode=EVN].value[xsi:type=ST]
patientInstruction | .text
relevantHistory | .inboundRelationship(typeCode=SUBJ].source[classCode=CACT, moodCode=EVN]

12.14.8.5 Quality Improvement and Clinical Knowledge (QUICK) (http://siframework.org/cqf)

ServiceRequest
identifier | ClinicalStatement.identifier
instantiatesCanonical
instantiatesUri
basedOn
replaces
requisition
status | Action.currentStatus
intent
category
priority
doNotPerform
code | Procedure.procedureCode
orderDetail | Procedure.procedureCode
quantity[x]
subject | ClinicalStatement.subject
encounter | ClinicalStatement.encounter
occurrence[x] | Procedure.procedureSchedule
asNeeded[x] | Proposal.prnReason.reason
authoredOn | Proposal.proposedAtTime
requester | ClinicalStatement.statementAuthor
performerType
performer
locationCode
locationReference
reasonCode
reasonReference
insurance
supportingInfo
specimen
bodySite | Procedure.targetBodySite
note | ClinicalStatement.additionalText
patientInstruction
relevantHistory

12.14.9 Resource ServiceRequest - Extensions & Profiles

URL: https://hl7.org/fhir/R4/servicerequest-profiles.html

12.14.9 Resource ServiceRequest - Extensions & Profiles

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

This table lists profiles and extensions for the ServiceRequest resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

12.14.9.1 Profiles

Profile | Description | Context
ServiceRequest-Genetics | Describes how the ServiceRequest resource is used to for genetics | ServiceRequest-Genetics

12.14.9.2 Extensions

procedure-approachBodyStructure | ServiceRequest | approachBodyStructure | for Procedure HL7 Extensions
procedure-directedBy | ServiceRequest | directedBy | for Procedure HL7 Extensions
procedure-targetBodyStructure | ServiceRequest | targetBodyStructure | for Procedure HL7 Extensions
request-performerOrder | ServiceRequest.performer | performerOrder | for Request Pattern HL7 Extensions
request-statusReason | ServiceRequest | statusReason | for Request Pattern HL7 Extensions
servicerequest-precondition | ServiceRequest | precondition | for Service Request HL7 Extensions
servicerequest-questionnaireRequest | ServiceRequest | questionnaireRequest | for Service Request HL7 Extensions
workflow-episodeOfCare | ServiceRequest | episodeOfCare | for Workflow Pattern HL7 Extensions
Extensions for all resources or elements

resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

12.14.9.3 Search Extensions

No Search Extensions defined for this resource

12.14.10 ServiceRequest-Genetics

URL: https://hl7.org/fhir/R4/servicerequest-genetic.html

12.14.10 ServiceRequest-Genetics

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative

The material on this page is currently undergoing work to be refactored or potentially removed in a future release as further analysis is done as part of the Genomics Reporting Implementation Guide .

12.14.10.1 Content

Profiles:
ServiceRequest-Genetics | Describes how the ServiceRequest resource is used to for genetics
Extensions:
servicerequest-geneticsItem | Item : The specific diagnostic investigations that are requested as part of this request. Sometimes, there can only be one item per request, but in most contexts, more than one investigation can be requested.
Examples:
PGx | PGx Phenotype example
Order-genetics-1 | Order for ABCB4 gene mutation analysis

12.14.10.2 StructureDefinition: ServiceRequest-Genetics

URL: https://hl7.org/fhir/R4/servicerequest-genetics.html

12.14.10.2 StructureDefinition: ServiceRequest-Genetics

Orders and Observations Work Group | Maturity Level: 1 | Trial Use | Use Context: Any

The official URL for this profile is:

`http://hl7.org/fhir/StructureDefinition/servicerequest-genetics`

Describes how the ServiceRequest resource is used to for genetics

This profile was published on Tue, Oct 20, 2015 00:00+1100 as a draft by Health Level Seven International (Clinical Genomics).

12.14.10.2.1 Formal Views of Profile Content

Description of Profiles, Differentials, Snapshots, and how the XML and JSON presentations work.

• Text Summary
• Differential Table
• Snapshot Table
• XML Template
• JSON Template
• All

This structure is derived from ServiceRequest.

Summary

Extensions

This structure refers to these extensions:

• http://hl7.org/fhir/StructureDefinition/servicerequest-geneticsItem

This structure is derived from ServiceRequest.

Name | Flags | Card. | Type | Description & Constraints
ServiceRequest | | 0..*

servicerequest-geneticsItem | | 0..* | (Complex) | The items the orderer requested URL: http://hl7.org/fhir/StructureDefinition/servicerequest-geneticsItem
Documentation for this format

Name | Flags | Card. | Type | Description & Constraints
ServiceRequest | I | 0..* | | A request for a service to be performed
id | Σ | 0..1 | string | Logical id of this artifact
meta | ΣI | 0..1 | Meta | Metadata about the resource
implicitRules | ?!ΣI | 0..1 | uri | A set of rules under which this content was created
language | I | 0..1 | code | Language of the resource content Binding: Common Languages (preferred) Max Binding: All Languages
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | I | 0..* | Extension | Extension Slice: Unordered, Open by value:url
servicerequest-geneticsItem | I | 0..* | (Complex) | The items the orderer requested URL: http://hl7.org/fhir/StructureDefinition/servicerequest-geneticsItem
modifierExtension | ?!I | 0..* | Extension | Extensions that cannot be ignored
identifier | ΣI | 0..* | Identifier | Identifiers assigned to this order
instantiatesCanonical | ΣI | 0..* | canonical(ActivityDefinition | PlanDefinition) | Instantiates FHIR protocol or definition
instantiatesUri | ΣI | 0..* | uri | Instantiates external protocol or definition
basedOn | ΣI | 0..* | Reference(CarePlan | ServiceRequest | MedicationRequest) | What request fulfills
replaces | ΣI | 0..* | Reference(ServiceRequest) | What request replaces
requisition | ΣI | 0..1 | Identifier | Composite Request ID
status | ?!ΣI | 1..1 | code | draft | active | on-hold | revoked | completed | entered-in-error | unknown Binding: RequestStatus (required)
intent | ?!ΣI | 1..1 | code | proposal | plan | directive | order | original-order | reflex-order | filler-order | instance-order | option Binding: RequestIntent (required)
category | ΣI | 0..* | CodeableConcept | Classification of service Binding: Service Request Category Codes (example)
priority | ΣI | 0..1 | code | routine | urgent | asap | stat Binding: Request priority (required)
doNotPerform | ?!ΣI | 0..1 | boolean | True if service/procedure should not be performed
code | ΣI | 0..1 | CodeableConcept | What is being requested/ordered Binding: Procedure Codes (SNOMED CT) (example)
orderDetail | ΣI | 0..* | CodeableConcept | Additional order information Binding: Service Request Order Details Codes (example)
quantity[x] | ΣI | 0..1 | | Service amount
quantityQuantity | | | Quantity
quantityRatio | | | Ratio
quantityRange | | | Range
subject | ΣI | 1..1 | Reference(Patient | Group | Location | Device) | Individual or Entity the service is ordered for
encounter | ΣI | 0..1 | Reference(Encounter) | Encounter in which the request was created
occurrence[x] | ΣI | 0..1 | | When service should occur
occurrenceDateTime | | | dateTime
occurrencePeriod | | | Period
occurrenceTiming | | | Timing
asNeeded[x] | ΣI | 0..1 | | Preconditions for service Binding: SNOMED CT Medication As Needed Reason Codes (example)
asNeededBoolean | | | boolean
asNeededCodeableConcept | | | CodeableConcept
authoredOn | ΣI | 0..1 | dateTime | Date request signed
requester | ΣI | 0..1 | Reference(Practitioner | PractitionerRole | Organization | Patient | RelatedPerson | Device) | Who/what is requesting service
performerType | ΣI | 0..1 | CodeableConcept | Performer role Binding: Participant Roles (example)
performer | ΣI | 0..* | Reference(Practitioner | PractitionerRole | Organization | CareTeam | HealthcareService | Patient | Device | RelatedPerson) | Requested performer
locationCode | ΣI | 0..* | CodeableConcept | Requested location Binding: ServiceDeliveryLocationRoleType (example)
locationReference | ΣI | 0..* | Reference(Location) | Requested location
reasonCode | ΣI | 0..* | CodeableConcept | Explanation/Justification for procedure or service Binding: Procedure Reason Codes (example)
reasonReference | ΣI | 0..* | Reference(Condition | Observation | DiagnosticReport | DocumentReference) | Explanation/Justification for service or service
insurance | I | 0..* | Reference(Coverage | ClaimResponse) | Associated insurance coverage
supportingInfo | I | 0..* | Reference(Resource) | Additional clinical information
specimen | ΣI | 0..* | Reference(Specimen) | Procedure Samples
bodySite | ΣI | 0..* | CodeableConcept | Location on Body Binding: SNOMED CT Body Structures (example)
note | I | 0..* | Annotation | Comments
patientInstruction | ΣI | 0..1 | string | Patient or consumer-oriented instructions
relevantHistory | I | 0..* | Reference(Provenance) | Request provenance
Documentation for this format

```
<!-- ServiceRequest-Genetics -->

<ServiceRequest xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id><!-- 0..1 string Logical id of this artifact --></id>
 <meta><!--  0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!--  0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!--  0..1 Language of the resource content   -->
 <text><!--  0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
```

```
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!--  0..* Identifier
     Identifiers assigned to this order --></identifier>
 <instantiatesCanonical value="[canonical]"/><!--  0..* Instantiates FHIR protocol or definition -->
 <instantiatesUri value="[uri]"/><!--  0..* Instantiates external protocol or definition -->
 <basedOn><!--  0..* Reference(CarePlan) What request fulfills --></basedOn>
 <replaces><!--  0..* Reference(ServiceRequest) What request replaces --></replaces>
 <requisition><!--  0..1 Identifier Composite Request ID --></requisition>
 <status value="[code]"/><!--  1..1 draft | active | on-hold | revoked | completed | entered-in-error | unknown   -->
 <intent value="[code]"/><!--  1..1 proposal | plan | directive | order | original-order | reflex-order | filler-order |
     instance-order | option   -->
 <category><!--  0..* CodeableConcept Classification of service   --></category>
 <priority value="[code]"/><!--  0..1 routine | urgent | asap | stat   -->
 <doNotPerform value="[boolean]"/><!--  0..1 True if service/procedure should not be performed -->
 <code><!--  0..1 CodeableConcept What is being requested/ordered   --></code>
 <orderDetail><!--  0..* CodeableConcept Additional order information   --></orderDetail>
 <quantity[x]><!--  0..1 Quantity|Ratio|Range
     Service amount --></quantity[x]>
 <subject><!--  1..1 Reference(Patient)
     Individual or Entity the service is ordered for --></subject>
 <encounter><!--  0..1 Reference(Encounter)
     Encounter in which the request was created --></encounter>
 <occurrence[x]><!--  0..1 dateTime|Period|Timing
     When service should occur --></occurrence[x]>
 <asNeeded[x]><!--  0..1 boolean|CodeableConcept Preconditions for service   --></asNeeded[x]>
 <authoredOn value="[dateTime]"/><!--  0..1 Date request signed -->
 <requester><!--  0..1 Reference(Practitioner)
     Who/what is requesting service --></requester>
 <performerType><!--  0..1 CodeableConcept Performer role   --></performerType>
 <performer><!--  0..* Reference(Practitioner) Requested performer --></performer>
 <locationCode><!--  0..* CodeableConcept Requested location   --></locationCode>
 <locationReference><!--  0..* Reference(Location)
     Requested location --></locationReference>
 <reasonCode><!--  0..* CodeableConcept Explanation/Justification for procedure or service   --></reasonCode>
 <reasonReference><!--  0..* Reference(Condition)
     Explanation/Justification for service or service --></reasonReference>
 <insurance><!--  0..* Reference(Coverage)
     Associated insurance coverage --></insurance>
 <supportingInfo><!--  0..* Reference(Resource)
     Additional clinical information --></supportingInfo>
 <specimen><!--  0..* Reference(Specimen) Procedure Samples --></specimen>
 <bodySite><!--  0..* CodeableConcept Location on Body   --></bodySite>
 <note><!--  0..* Annotation Comments --></note>
 <patientInstruction value="[string]"/><!--  0..1 Patient or consumer-oriented instructions -->
 <relevantHistory><!--  0..* Reference(Provenance)
     Request provenance --></relevantHistory>
</ServiceRequest>
```

```
{ // ServiceRequest-Genetics
  // from Element: extension
    "meta" : { Meta }, // C?Metadata about the resource
    "implicitRules" : "<uri>", // C?A set of rules under which this content was created
    "language" : "<code>", // C?Language of the resource content
    "text" : { Narrative }, // C?Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "extension" : [ //  sliced by value:url  in the specified order, Open
      { // The items the orderer requested // C?
        // from Element: extension
      }
    ],
    "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored
    "identifier" : [{ Identifier }], // C?Identifiers assigned to this order
    "instantiatesCanonical" : ["<canonical>"], // C?Instantiates FHIR protocol or definition
    "instantiatesUri" : ["<uri>"], // C?Instantiates external protocol or definition
    "basedOn" : [{ Reference(CarePlan) }], // C?What request fulfills
    "replaces" : [{ Reference(ServiceRequest) }], // C?What request replaces
    "requisition" : { Identifier }, // C?Composite Request ID
    "status" : "<code>", // C? R! draft | active | on-hold | revoked | completed | entered-in-error | unknown
    "intent" : "<code>", // C? R! proposal | plan | directive | order | original-order | reflex-order | filler-order |
    instance-order | option
    "category" : [{ CodeableConcept }], // C?Classification of service
    "priority" : "<code>", // C?routine | urgent | asap | stat
    "doNotPerform" : <boolean>, // C?True if service/procedure should not be performed
    "code" : { CodeableConcept }, // C?What is being requested/ordered
    "orderDetail" : [{ CodeableConcept }], // C?Additional order information
// value[x]: Service amount. One of these 3:
    "quantityQuantity" : { Quantity }, // C?Service amount
    "quantityRatio" : { Ratio }, // C?Service amount
    "quantityRange" : { Range }, // C?Service amount
    "subject" : { Reference(Patient) }, // C? R! Individual or Entity the service is ordered for
    "encounter" : { Reference(Encounter) }, // C?Encounter in which the request was created
// value[x]: When service should occur. One of these 3:
    "occurrenceDateTime" : "<dateTime>", // C?When service should occur
    "occurrencePeriod" : { Period }, // C?When service should occur
    "occurrenceTiming" : { Timing }, // C?When service should occur
// value[x]: Preconditions for service. One of these 2:
    "asNeededBoolean" : <boolean>, // C?Preconditions for service
    "asNeededCodeableConcept" : { CodeableConcept }, // C?Preconditions for service
    "authoredOn" : "<dateTime>", // C?Date request signed
    "requester" : { Reference(Practitioner) }, // C?Who/what is requesting service
```

```
  "performerType" : { CodeableConcept }, // C?Performer role
  "performer" : [{ Reference(Practitioner) }], // C?Requested performer
  "locationCode" : [{ CodeableConcept }], // C?Requested location
  "locationReference" : [{ Reference(Location) }], // C?Requested location
  "reasonCode" : [{ CodeableConcept }], // C?Explanation/Justification for procedure or service
  "reasonReference" : [{ Reference(Condition) }], // C?Explanation/Justification for service or service
  "insurance" : [{ Reference(Coverage) }], // C?Associated insurance coverage
  "supportingInfo" : [{ Reference(Resource) }], // C?Additional clinical information
  "specimen" : [{ Reference(Specimen) }] // C?Procedure Samples
  "bodySite" : [{ CodeableConcept }], // C?Location on Body
  "note" : [{ Annotation }], // C?Comments
  "patientInstruction" : "<string>", // C?Patient or consumer-oriented instructions
  "relevantHistory" : [{ Reference(Provenance) }], // C?Request provenance
}
```

This structure is derived from ServiceRequest.

Summary

Extensions

This structure refers to these extensions:

• http://hl7.org/fhir/StructureDefinition/servicerequest-geneticsItem

Differential View

This structure is derived from ServiceRequest.

Name	Flags	Card.	Type	Description & Constraints
ServiceRequest		0..*		
servicerequest-geneticsItem		0..*	(Complex)	The items the orderer requested URL: http://hl7.org/fhir/StructureDefinition/servicerequest-geneticsItem
Documentation for this format				

Snapshot View

Name	Flags	Card.	Type	Description & Constraints
ServiceRequest	I	0..*		A request for a service to be performed
id	Σ	0..1	string	Logical id of this artifact
meta	ΣI	0..1	Meta	Metadata about the resource
implicitRules	?!ΣI	0..1	uri	A set of rules under which this content was created
language	I	0..1	code	Language of the resource content Binding: Common Languages (preferred) Max Binding: All Languages
text	I	0..1	Narrative	Text summary of the resource, for human interpretation
contained		0..*	Resource	Contained, inline Resources
extension	I	0..*	Extension	Extension Slice: Unordered, Open by value:url
servicerequest-geneticsItem	I	0..*	(Complex)	The items the orderer requested URL: http://hl7.org/fhir/StructureDefinition/servicerequest-geneticsItem
modifierExtension	?!I	0..*	Extension	Extensions that cannot be ignored
identifier	ΣI	0..*	Identifier	Identifiers assigned to this order
instantiatesCanonical	ΣI	0..*	canonical(ActivityDefinition | PlanDefinition)	Instantiates FHIR protocol or definition
instantiatesUri	ΣI	0..*	uri	Instantiates external protocol or definition
basedOn	ΣI	0..*	Reference(CarePlan | ServiceRequest | MedicationRequest)	What request fulfills
replaces	ΣI	0..*	Reference(ServiceRequest)	What request replaces
requisition	ΣI	0..1	Identifier	Composite Request ID
status	?!ΣI	1..1	code	draft | active | on-hold | revoked | completed | entered-in-error | unknown Binding: RequestStatus (required)
intent	?!ΣI	1..1	code	proposal | plan | directive | order | original-order | reflex-order | filler-order | instance-order | option Binding: RequestIntent (required)
category	ΣI	0..*	CodeableConcept	Classification of service Binding: Service Request Category Codes (example)
priority	ΣI	0..1	code	routine | urgent | asap | stat Binding: Request priority (required)
doNotPerform	?!ΣI	0..1	boolean	True if service/procedure should not be performed
code	ΣI	0..1	CodeableConcept	What is being requested/ordered Binding: Procedure Codes (SNOMED CT) (example)
orderDetail	ΣI	0..*	CodeableConcept	Additional order information Binding: Service Request Order Details Codes (example)
quantity[x]	ΣI	0..1		Service amount
quantityQuantity			Quantity	
quantityRatio			Ratio	
quantityRange			Range	
subject	ΣI	1..1	Reference(Patient | Group | Location | Device)	Individual or Entity the service is ordered for
encounter	ΣI	0..1	Reference(Encounter)	Encounter in which the request was created
occurrence[x]	ΣI	0..1		When service should occur
occurrenceDateTime			dateTime	
occurrencePeriod			Period	
occurrenceTiming			Timing	
asNeeded[x]	ΣI	0..1		Preconditions for service Binding: SNOMED CT Medication As Needed Reason Codes (example)
asNeededBoolean			boolean	
asNeededCodeableConcept			CodeableConcept	
authoredOn	ΣI	0..1	dateTime	Date request signed
requester	ΣI	0..1	Reference(Practitioner | PractitionerRole | Organization | Patient | RelatedPerson | Device)	Who/what is requesting service
performerType	ΣI	0..1	CodeableConcept	Performer role Binding: Participant Roles (example)

performer | ΣI | 0..* | Reference(Practitioner | PractitionerRole | Organization | CareTeam | HealthcareService | Patient | Device | RelatedPerson) | Requested performer
locationCode | ΣI | 0..* | CodeableConcept | Requested location Binding: ServiceDeliveryLocationRoleType (example)
locationReference | ΣI | 0..* | Reference(Location) | Requested location
reasonCode | ΣI | 0..* | CodeableConcept | Explanation/Justification for procedure or service Binding: Procedure Reason Codes (example)
reasonReference | ΣI | 0..* | Reference(Condition | Observation | DiagnosticReport | DocumentReference) | Explanation/Justification for service or service
insurance | I | 0..* | Reference(Coverage | ClaimResponse) | Associated insurance coverage
supportingInfo | I | 0..* | Reference(Resource) | Additional clinical information
specimen | ΣI | 0..* | Reference(Specimen) | Procedure Samples
bodySite | ΣI | 0..* | CodeableConcept | Location on Body Binding: SNOMED CT Body Structures (example)
note | I | 0..* | Annotation | Comments
patientInstruction | ΣI | 0..1 | string | Patient or consumer-oriented instructions
relevantHistory | I | 0..* | Reference(Provenance) | Request provenance
Documentation for this format

XML Template

```
<!-- ServiceRequest-Genetics -->

<ServiceRequest xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id><!-- 0..1 string Logical id of this artifact --></id>
 <meta><!--  0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!--  0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!--  0..1 Language of the resource content   -->
 <text><!--  0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!--  0..* Identifier
     Identifiers assigned to this order --></identifier>
 <instantiatesCanonical value="[canonical]"/><!--  0..* Instantiates FHIR protocol or definition -->
 <instantiatesUri value="[uri]"/><!--  0..* Instantiates external protocol or definition -->
 <basedOn><!--  0..* Reference(CarePlan) What request fulfills --></basedOn>
 <replaces><!--  0..* Reference(ServiceRequest) What request replaces --></replaces>
 <requisition><!--  0..1 Identifier Composite Request ID --></requisition>
 <status value="[code]"/><!--  1..1 draft | active | on-hold | revoked | completed | entered-in-error | unknown   -->
 <intent value="[code]"/><!--  1..1 proposal | plan | directive | order | original-order | reflex-order | filler-order |
     instance-order | option   -->
 <category><!--  0..* CodeableConcept Classification of service   --></category>
 <priority value="[code]"/><!--  0..1 routine | urgent | asap | stat   -->
 <doNotPerform value="[boolean]"/><!--  0..1 True if service/procedure should not be performed -->
 <code><!--  0..1 CodeableConcept What is being requested/ordered   --></code>
 <orderDetail><!--  0..* CodeableConcept Additional order information   --></orderDetail>
 <quantity[x]><!--  0..1 Quantity|Ratio|Range
     Service amount --></quantity[x]>
 <subject><!--  1..1 Reference(Patient)
     Individual or Entity the service is ordered for --></subject>
 <encounter><!--  0..1 Reference(Encounter)
     Encounter in which the request was created --></encounter>
 <occurrence[x]><!--  0..1 dateTime|Period|Timing
     When service should occur --></occurrence[x]>
 <asNeeded[x]><!--  0..1 boolean|CodeableConcept Preconditions for service   --></asNeeded[x]>
 <authoredOn value="[dateTime]"/><!--  0..1 Date request signed -->
 <requester><!--  0..1 Reference(Practitioner)
     Who/what is requesting service --></requester>
 <performerType><!--  0..1 CodeableConcept Performer role   --></performerType>
 <performer><!--  0..* Reference(Practitioner) Requested performer --></performer>
 <locationCode><!--  0..* CodeableConcept Requested location   --></locationCode>
 <locationReference><!--  0..* Reference(Location)
     Requested location --></locationReference>
 <reasonCode><!--  0..* CodeableConcept Explanation/Justification for procedure or service   --></reasonCode>
 <reasonReference><!--  0..* Reference(Condition)
     Explanation/Justification for service or service --></reasonReference>
 <insurance><!--  0..* Reference(Coverage)
     Associated insurance coverage --></insurance>
 <supportingInfo><!--  0..* Reference(Resource)
     Additional clinical information --></supportingInfo>
 <specimen><!--  0..* Reference(Specimen) Procedure Samples --></specimen>
 <bodySite><!--  0..* CodeableConcept Location on Body   --></bodySite>
 <note><!--  0..* Annotation Comments --></note>
 <patientInstruction value="[string]"/><!--  0..1 Patient or consumer-oriented instructions -->
 <relevantHistory><!--  0..* Reference(Provenance)
     Request provenance --></relevantHistory>
</ServiceRequest>
```

JSON Template

```
{ // ServiceRequest-Genetics
  // from Element: extension
    "meta" : { Meta }, // C?Metadata about the resource
    "implicitRules" : "<uri>", // C?A set of rules under which this content was created
    "language" : "<code>", // C?Language of the resource content
    "text" : { Narrative }, // C?Text summary of the resource, for human interpretation
```

```
    "contained" : [{ Resource }], //Contained, inline Resources
    "extension" : [ //  sliced by value:url  in the specified order, Open
      { // The items the orderer requested // C?
        // from Element: extension
      }
    ],
    "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored
    "identifier" : [{ Identifier }], // C?Identifiers assigned to this order
    "instantiatesCanonical" : ["<canonical>"], // C?Instantiates FHIR protocol or definition
    "instantiatesUri" : ["<uri>"], // C?Instantiates external protocol or definition
    "basedOn" : [{ Reference(CarePlan) }], // C?What request fulfills
    "replaces" : [{ Reference(ServiceRequest) }], // C?What request replaces
    "requisition" : { Identifier }, // C?Composite Request ID
    "status" : "<code>", // C? R! draft | active | on-hold | revoked | completed | entered-in-error | unknown
    "intent" : "<code>", // C? R! proposal | plan | directive | order | original-order | reflex-order | filler-order |
    instance-order | option
    "category" : [{ CodeableConcept }], // C?Classification of service
    "priority" : "<code>", // C?routine | urgent | asap | stat
    "doNotPerform" : <boolean>, // C?True if service/procedure should not be performed
    "code" : { CodeableConcept }, // C?What is being requested/ordered
    "orderDetail" : [{ CodeableConcept }], // C?Additional order information
// value[x]: Service amount. One of these 3:
    "quantityQuantity" : { Quantity }, // C?Service amount
    "quantityRatio" : { Ratio }, // C?Service amount
    "quantityRange" : { Range }, // C?Service amount
    "subject" : { Reference(Patient) }, // C? R! Individual or Entity the service is ordered for
    "encounter" : { Reference(Encounter) }, // C?Encounter in which the request was created
// value[x]: When service should occur. One of these 3:
    "occurrenceDateTime" : "<dateTime>", // C?When service should occur
    "occurrencePeriod" : { Period }, // C?When service should occur
    "occurrenceTiming" : { Timing }, // C?When service should occur
// value[x]: Preconditions for service. One of these 2:
    "asNeededBoolean" : <boolean>, // C?Preconditions for service
    "asNeededCodeableConcept" : { CodeableConcept }, // C?Preconditions for service
    "authoredOn" : "<dateTime>", // C?Date request signed
    "requester" : { Reference(Practitioner) }, // C?Who/what is requesting service
    "performerType" : { CodeableConcept }, // C?Performer role
    "performer" : [{ Reference(Practitioner) }], // C?Requested performer
    "locationCode" : [{ CodeableConcept }], // C?Requested location
    "locationReference" : [{ Reference(Location) }], // C?Requested location
    "reasonCode" : [{ CodeableConcept }], // C?Explanation/Justification for procedure or service
    "reasonReference" : [{ Reference(Condition) }], // C?Explanation/Justification for service or service
    "insurance" : [{ Reference(Coverage) }], // C?Associated insurance coverage
    "supportingInfo" : [{ Reference(Resource) }], // C?Additional clinical information
    "specimen" : [{ Reference(Specimen) }] // C?Procedure Samples
    "bodySite" : [{ CodeableConcept }], // C?Location on Body
    "note" : [{ Annotation }], // C?Comments
    "patientInstruction" : "<string>", // C?Patient or consumer-oriented instructions
    "relevantHistory" : [{ Reference(Provenance) }], // C?Request provenance
  }
```

Alternate definitions: Master Definition (XML, JSON), Schematron

12.14.10.2.2 Terminology Bindings

Path | Name | Conformance | ValueSet
ServiceRequest.language | Common Languages | preferred | Common Languages
ServiceRequest.status | RequestStatus | required | RequestStatus
ServiceRequest.intent | RequestIntent | required | RequestIntent
ServiceRequest.category | Service Request Category Codes | example | Service Request Category Codes
ServiceRequest.priority | Request priority | required | Request priority
ServiceRequest.code | Procedure Codes (SNOMED CT) | example | Procedure Codes (SNOMED CT)
ServiceRequest.orderDetail | Service Request Order Details Codes | example | Service Request Order Details Codes
ServiceRequest.asNeeded[x] | SNOMED CT Medication As Needed Reason Codes | example | SNOMED CT Medication As Needed Reason Codes
ServiceRequest.performerType | Participant Roles | example | Participant Roles
ServiceRequest.locationCode | V3 Value SetServiceDeliveryLocationRoleType | example | V3 Value SetServiceDeliveryLocationRoleType
ServiceRequest.reasonCode | Procedure Reason Codes | example | Procedure Reason Codes
ServiceRequest.bodySite | SNOMED CT Body Structures | example | SNOMED CT Body Structures

12.14.11 Service Request HL7 Extensions

URL: https://hl7.org/fhir/R4/servicerequest-extensions.html

Defines common extensions used with or related to the Service Request resource

12.14.11 Service Request HL7 Extensions

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative

12.14.11.1 Content

Extensions:

servicerequest-precondition | precondition : The condition or state of the patient, prior or during the diagnostic procedure or test, for example, fasting, at-rest, or post-operative. This captures circumstances that may influence the measured value and have bearing on the interpretation of the result.
servicerequest-questionnaireRequest | questionnaireRequest : Reference to a specific Questionnaire Resource as an ordered item. Allows for ordering a specific questionnaire to be completed.

12.14.12 JSON Schema for ServiceRequest

URL: https://hl7.org/fhir/R4/servicerequest.schema.json.html

12.14.12 JSON Schema for ServiceRequest

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

This is a representation of the json schema for ServiceRequest, which is just a part of the full JSON Schema.


```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/ServiceRequest",
  "$ref": "#/definitions/ServiceRequest",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "ServiceRequest": {
      "description": "A record of a request for service such as diagnostic investigations, treatments, or operations to
    be performed.",
      "properties": {
        "resourceType": {
          "description": "This is a ServiceRequest resource",
          "const": "ServiceRequest"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
          "$ref": "#/definitions/code"
        },
        "_language": {
          "description": "Extensions for language",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "text": {
          "description": "A human-readable narrative that contains a summary of the resource and can be used to
    represent the content of the resource to a human. The narrative need not encode all the structured data, but is
    required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
    Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
          "$ref": "Narrative.schema.json#/definitions/Narrative"
        },
        "contained": {
          "description": "These resources do not have an independent existence apart from the resource that contains
    them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
          "items": {
            "$ref": "ResourceList.schema.json#/definitions/ResourceList"
          },
          "type": "array"
        },
        "extension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
     definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
    that SHALL be met as part of the definition of the extension.",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "modifierExtension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the resource and that modifies the understanding of the element that contains it and/or the understanding of the
```

```
containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
met as part of the definition of the extension. Applications processing a resource are required to check for
modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "identifier": {
      "description": "Identifiers assigned to this order instance by the orderer and/or the receiver and/or order
fulfiller.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "instantiatesCanonical": {
      "description": "The URL pointing to a FHIR-defined protocol, guideline, orderset or other definition that is
adhered to in whole or in part by this ServiceRequest.",
      "items": {
        "$ref": "canonical.schema.json#/definitions/canonical"
      },
      "type": "array"
    },
    "instantiatesUri": {
      "description": "The URL pointing to an externally maintained protocol, guideline, orderset or other definition
 that is adhered to in whole or in part by this ServiceRequest.",
      "items": {
        "$ref": "#/definitions/uri"
      },
      "type": "array"
    },
    "_instantiatesUri": {
      "description": "Extensions for instantiatesUri",
      "items": {
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "type": "array"
    },
    "basedOn": {
      "description": "Plan/proposal/order fulfilled by this request.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "replaces": {
      "description": "The request takes the place of the referenced completed or terminated request(s).",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "requisition": {
      "description": "A shared identifier common to all service requests that were authorized more or less
simultaneously by a single author, representing the composite or group identifier.",
      "$ref": "Identifier.schema.json#/definitions/Identifier"
    },
    "status": {
      "description": "The status of the order.",
      "$ref": "#/definitions/code"
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "intent": {
      "description": "Whether the request is a proposal, plan, an original order or a reflex order.",
      "$ref": "#/definitions/code"
    },
    "_intent": {
      "description": "Extensions for intent",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "category": {
      "description": "A code that classifies the service for searching, sorting and display purposes (e.g.
\"Surgical Procedure\").",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "priority": {
      "description": "Indicates how quickly the ServiceRequest should be addressed with respect to other requests.",
      "$ref": "#/definitions/code"
    },
```

```
    "_priority": {
      "description": "Extensions for priority",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "doNotPerform": {
      "description": "Set this to true if the record is saying that the service/procedure should NOT be performed.",
      "$ref": "#/definitions/boolean"
    },
    "_doNotPerform": {
      "description": "Extensions for doNotPerform",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "code": {
      "description": "A code that identifies a particular service (i.e., procedure, diagnostic investigation, or
panel of investigations) that have been requested.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "orderDetail": {
      "description": "Additional details and instructions about the how the services are to be delivered.   For
example, and order for a urinary catheter may have an order detail for an external or indwelling catheter, or an
order for a bandage may require additional instructions specifying how the bandage should be applied.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "quantityQuantity": {
      "description": "An amount of service being requested which can be a quantity ( for example $1,500 home
modification), a ratio ( for example, 20 half day visits per month), or a range (2.0 to 1.8 Gy per fraction).",
      "$ref": "Quantity.schema.json#/definitions/Quantity"
    },
    "quantityRatio": {
      "description": "An amount of service being requested which can be a quantity ( for example $1,500 home
modification), a ratio ( for example, 20 half day visits per month), or a range (2.0 to 1.8 Gy per fraction).",
      "$ref": "Ratio.schema.json#/definitions/Ratio"
    },
    "quantityRange": {
      "description": "An amount of service being requested which can be a quantity ( for example $1,500 home
modification), a ratio ( for example, 20 half day visits per month), or a range (2.0 to 1.8 Gy per fraction).",
      "$ref": "Range.schema.json#/definitions/Range"
    },
    "subject": {
      "description": "On whom or what the service is to be performed. This is usually a human patient, but can also
be requested on animals, groups of humans or animals, devices such as dialysis machines, or even locations
(typically for environmental scans).",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "encounter": {
      "description": "An encounter that provides additional information about the healthcare context in which this
request is made.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "occurrenceDateTime": {
      "description": "The date/time at which the requested service should occur.",
      "pattern": "^([0-9]([0-9]([0-9][1-9]|[1-9]0)|[1-9]00)|[1-9]000)(-(0[1-9]|1[0-2])(-(0[1-9]|[1-2][0-9]|3[0-
1])(T([01][0-9]|2[0-3]):[0-5][0-9]:([0-5][0-9]|60)(\\.[0-9]+)?(Z|(\\+|-)((0[0-9]|1[0-3]):[0-5][0-9]|14:00)))?)?)?$",
      "type": "string"
    },
    "_occurrenceDateTime": {
      "description": "Extensions for occurrenceDateTime",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "occurrencePeriod": {
      "description": "The date/time at which the requested service should occur.",
      "$ref": "Period.schema.json#/definitions/Period"
    },
    "occurrenceTiming": {
      "description": "The date/time at which the requested service should occur.",
      "$ref": "Timing.schema.json#/definitions/Timing"
    },
    "asNeededBoolean": {
      "description": "If a CodeableConcept is present, it indicates the pre-condition for performing the service.
For example \"pain\", \"on flare-up\", etc.",
      "pattern": "^true|false$",
      "type": "boolean"
    },
    "_asNeededBoolean": {
      "description": "Extensions for asNeededBoolean",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "asNeededCodeableConcept": {
      "description": "If a CodeableConcept is present, it indicates the pre-condition for performing the service.
For example \"pain\", \"on flare-up\", etc.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "authoredOn": {
      "description": "When the request transitioned to being actionable.",
      "$ref": "#/definitions/dateTime"
    },
```

```
    "_authoredOn": {
      "description": "Extensions for authoredOn",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "requester": {
      "description": "The individual who initiated the request and has responsibility for its activation.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "performerType": {
      "description": "Desired type of performer for doing the requested service.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "performer": {
      "description": "The desired performer for doing the requested service.  For example, the surgeon,
dermatopathologist, endoscopist, etc.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "locationCode": {
      "description": "The preferred location(s) where the procedure should actually happen in coded or free text
form. E.g. at home or nursing day care center.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "locationReference": {
      "description": "A reference to the the preferred location(s) where the procedure should actually happen. E.g.
at home or nursing day care center.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "reasonCode": {
      "description": "An explanation or justification for why this service is being requested in coded or textual
form.   This is often for billing purposes.  May relate to the resources referred to in `supportingInfo`.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "reasonReference": {
      "description": "Indicates another resource that provides a justification for why this service is being
requested.   May relate to the resources referred to in `supportingInfo`.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "insurance": {
      "description": "Insurance plans, coverage extensions, pre-authorizations and/or pre-determinations that may be
 needed for delivering the requested service.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "supportingInfo": {
      "description": "Additional clinical information about the patient or specimen that may influence the services
or their interpretations.     This information includes diagnosis, clinical findings and other observations.  In
laboratory ordering these are typically referred to as \"ask at order entry questions (AOEs)\".  This includes
observations explicitly requested by the producer (filler) to provide context or supporting information needed to
complete the order. For example,  reporting the amount of inspired oxygen for blood gas measurements.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "specimen": {
      "description": "One or more specimens that the laboratory procedure will use.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "bodySite": {
      "description": "Anatomic location where the procedure should be performed. This is the target site.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "note": {
      "description": "Any other notes and comments made about the service request. For example, internal billing
notes.",
      "items": {
```

```
            "$ref": "Annotation.schema.json#/definitions/Annotation"
          },
          "type": "array"
        },
        "patientInstruction": {
          "description": "Instructions in terms that are understood by the patient or consumer.",
          "$ref": "#/definitions/string"
        },
        "_patientInstruction": {
          "description": "Extensions for patientInstruction",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "relevantHistory": {
          "description": "Key events in the history of the request.",
          "items": {
            "$ref": "Reference.schema.json#/definitions/Reference"
          },
          "type": "array"
        }
      },
      "additionalProperties": false,
      "required": [
        "subject",
        "resourceType"
      ]
    }
  }
}
```

12.14.13 Resource ServiceRequest - Maturity Dependency View

URL: https://hl7.org/fhir/R4/servicerequest-dependencies.html

12.14.13 Resource ServiceRequest - Maturity Dependency View

Orders and Observations Work Group | Maturity Level: 2 | Trial Use | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

Dependency Graph for ServiceRequest FMM level 2

Name | Card. | Type | Dependency Analysis
ServiceRequest | | DomainResource
identifier | 0..* | Identifier
instantiatesCanonical | 0..* | canonical(ActivityDefinition | PlanDefinition)
instantiatesUri | 0..* | uri
basedOn | 0..* | Reference(CarePlan | ServiceRequest | MedicationRequest)
replaces | 0..* | Reference(ServiceRequest)
requisition | 0..1 | Identifier
status | 1..1 | code
intent | 1..1 | code
category | 0..* | CodeableConcept
priority | 0..1 | code
doNotPerform | 0..1 | boolean
code | 0..1 | CodeableConcept
orderDetail | 0..* | CodeableConcept
quantity[x] | 0..1
quantityQuantity | | Quantity
quantityRatio | | Ratio
quantityRange | | Range
subject | 1..1 | Reference(Patient | Group | Location | Device) | Type Warning: (Group = FMM1-Trial Use vs. Element = FMM2-Trial Use)
encounter | 0..1 | Reference(Encounter)
occurrence[x] | 0..1
occurrenceDateTime | | dateTime
occurrencePeriod | | Period
occurrenceTiming | | Timing
asNeeded[x] | 0..1
asNeededBoolean | | boolean
asNeededCodeableConcept | | CodeableConcept
authoredOn | 0..1 | dateTime
requester | 0..1 | Reference(Practitioner | PractitionerRole | Organization | Patient | RelatedPerson | Device)
performerType | 0..1 | CodeableConcept
performer | 0..* | Reference(Practitioner | PractitionerRole | Organization | CareTeam | HealthcareService | Patient | Device | RelatedPerson)
locationCode | 0..* | CodeableConcept
locationReference | 0..* | Reference(Location)
reasonCode | 0..* | CodeableConcept
reasonReference | 0..* | Reference(Condition | Observation | DiagnosticReport | DocumentReference)

insurance | 0..* | Reference(Coverage | ClaimResponse)
supportingInfo | 0..* | Reference(Any) | Type Warning: (Any = FMM1-Trial Use vs. Element = FMM2-Trial Use)
specimen | 0..* | Reference(Specimen)
bodySite | 0..* | CodeableConcept
note | 0..* | Annotation
patientInstruction | 0..1 | string
relevantHistory | 0..* | Reference(Provenance)

12.14.14 Value Set Sc-valueset-request-status - Status Code Mapping

URL: https://hl7.org/fhir/R4/sc-valueset-request-status.html

12.14.14 Value Set Sc-valueset-request-status - Status Code Mapping

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

This is the narrative for the resource. See also the XML, JSON or Turtle format.

12.14.15 RequestStatusCanonicalMap (http://hl7.org/fhir/ConceptMap/sc-request-status)

Mapping from http://hl7.org/fhir/ValueSet/request-status to http://hl7.org/fhir/ValueSet/resource-status

DRAFT. Published on 01/11/2019 9:29:23 AM by HL7 (FHIR Project) (http://hl7.org/fhir , fhir@lists.hl7.org).

Canonical Mapping for "Codes identifying the lifecycle stage of a request."

Source Code | Equivalence | Destination Code
entered-in-error | equivalent | error
draft | equivalent | draft
active | equivalent | active
on-hold | equivalent | suspended
revoked | equivalent | failed
completed | equivalent | complete
unknown | equivalent | unknown

12.15 Resource NutritionOrder - Content

URL: https://hl7.org/fhir/R4/nutritionorder.html
Orders and Observations Work Group | Maturity Level: 2 | Trial Use | Security Category: Patient | Compartments: Encounter, Patient, Practitioner

A request to supply a diet, formula feeding (enteral) or oral nutritional supplement to a patient/resident.

12.15.1 Scope and Usage

This resource is a request resource from a FHIR workflow perspective - see Workflow. It is the intent of the Orders and Observation Workgroup to align this resource with the workflow pattern for request resources.

The NutritionOrder resource describes a request for oral diets (including general diets such as General Healthy diet, or therapeutic diets such as Consistent Carbohydrate, 2 gram Sodium, or Fluid Restricted), oral nutrition supplements (such as nutritionally complete pre-packed drinks), enteral nutrition (tube feedings) and infant formula which govern the distribution of food and nutritional products used to feed patients within an in-patient setting. It does not cover orders for parenteral (IV) nutrition which are typically filled by pharmacy. These nutrition orders are combined with information on a patient's food allergies and intolerances, and ethnic or cultural food preferences (e.g. Kosher or Vegetarian) to inform healthcare personnel about the type, texture, and/or quantity of foods that the patient should receive or consume.

Enteral orders are distinguished from supplements because they have some unique attributes and typically include administration information whereas oral nutritional supplements may simply be supplied (e.g. home health or outpatient settings). In a simple case, the requestor may designate type of product, product name and the route of administration along with free text instructions without a having to complete the additional structured details.

This resource is intended to be used by providers from a variety of specialties such as physicians, dietitian/nutritionists, or speech therapists. One provider may simply order a base element oral diet such as General Healthful diet. Another provider, based on scope of practice, may use other elements to communicate additional therapeutic needs or patient preferences. The optionality included gives an ordering provider the capability to write a simple order for an oral diet, nutritional supplement or formula with minimal requirements beyond that of specifying the diet, supplement or formula product, but also supports the ability to provide more detailed information that may be further augmented by a dietitian or nutrition specialist. For example, a physician may order a 2 g sodium diet. A speech therapist, based on the results of a swallowing evaluation, then orders a mechanically altered texture with nectar thick liquids.

12.15.2 Boundaries and Relationships

The NutritionOrder resource is used for requesting oral diets, oral nutrition supplements and enteral feedings in an in-patient setting. The MedicationRequest resource should be used for requesting parenteral (IV) nutrition and prescribing dietary supplements such as vitamin or mineral supplements.

The Nutrition Order is a record of the request for the supply of a diet, oral supplement or enteral formulas for a patient. However, to initiate the request requires the use of the Task resource and its associated workflow with the Nutrition Order referenced from Task.basedOn, or by using the Nutrition Task resource in the context of a messaging or service workflow where the request is explicit or implicit.

This resource is referenced by CarePlan, DiagnosticReport, Goal and Observation

12.15.3 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
NutritionOrder | ITU | | DomainResource | Diet, formula or nutritional supplement request + Warning: Nutrition Order SHALL contain either Oral Diet , Supplement, or Enteral Formula class Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | | 0..* | Identifier | Identifiers assigned to this order
instantiatesCanonical | Σ | 0..* | canonical(ActivityDefinition | PlanDefinition) | Instantiates FHIR protocol or definition
instantiatesUri | Σ | 0..* | uri | Instantiates external protocol or definition
instantiates | | 0..* | uri | Instantiates protocol or definition
status | ?!Σ | 1..1 | code | draft | active | on-hold | revoked | completed | entered-in-error | unknown RequestStatus (Required)
intent | ?!Σ | 1..1 | code | proposal | plan | directive | order | original-order | reflex-order | filler-order | instance-order | option RequestIntent (Required)
patient | Σ | 1..1 | Reference(Patient) | The person who requires the diet, formula or nutritional supplement
encounter | | 0..1 | Reference(Encounter) | The encounter associated with this nutrition order
dateTime | Σ | 1..1 | dateTime | Date and time the nutrition order was requested
orderer | Σ | 0..1 | Reference(Practitioner | PractitionerRole) | Who ordered the diet, formula or nutritional supplement
allergyIntolerance | | 0..* | Reference(AllergyIntolerance) | List of the patient's food and nutrition-related allergies and intolerances
foodPreferenceModifier | | 0..* | CodeableConcept | Order-specific modifier about the type of food that should be given Diet (Example)
excludeFoodModifier | | 0..* | CodeableConcept | Order-specific modifier about the type of food that should not be given Food Type Codes (Example)
oralDiet | I | 0..1 | BackboneElement | Oral diet components
type | Σ | 0..* | CodeableConcept | Type of oral diet or diet restrictions that describe what can be consumed orally Diet Codes (Example)
schedule | | 0..* | Timing | Scheduled frequency of diet
nutrient | | 0..* | BackboneElement | Required nutrient modifications
modifier | | 0..1 | CodeableConcept | Type of nutrient that is being modified Nutrient Modifier Codes (Example)
amount | | 0..1 | SimpleQuantity | Quantity of the specified nutrient
texture | | 0..* | BackboneElement | Required texture modifications
modifier | | 0..1 | CodeableConcept | Code to indicate how to alter the texture of the foods, e.g. pureed Texture Modifier Codes (Example)
foodType | | 0..1 | CodeableConcept | Concepts that are used to identify an entity that is ingested for nutritional purposes Texture Modified Food Type Codes (Example)
fluidConsistencyType | | 0..* | CodeableConcept | The required consistency of fluids and liquids provided to the patient Fluid Consistency Type Codes (Example)
instruction | Σ | 0..1 | string | Instructions or additional information about the oral diet
supplement | I | 0..* | BackboneElement | Supplement components
type | Σ | 0..1 | CodeableConcept | Type of supplement product requested Supplement Type Codes (Example)
productName | | 0..1 | string | Product or brand name of the nutritional supplement
schedule | | 0..* | Timing | Scheduled frequency of supplement
quantity | | 0..1 | SimpleQuantity | Amount of the nutritional supplement
instruction | Σ | 0..1 | string | Instructions or additional information about the oral supplement
enteralFormula | I | 0..1 | BackboneElement | Enteral formula components
baseFormulaType | Σ | 0..1 | CodeableConcept | Type of enteral or infant formula Enteral Formula Type Codes (Example)
baseFormulaProductName | | 0..1 | string | Product or brand name of the enteral or infant formula
additiveType | | 0..1 | CodeableConcept | Type of modular component to add to the feeding Enteral Formula Additive Type Code (Example)
additiveProductName | | 0..1 | string | Product or brand name of the modular additive
caloricDensity | | 0..1 | SimpleQuantity | Amount of energy per specified volume that is required

routeofAdministration | | 0..1 | CodeableConcept | How the formula should enter the patient's gastrointestinal tract Enteral Route Codes (Extensible)
administration | | 0..* | BackboneElement | Formula feeding instruction as structured data
schedule | | 0..1 | Timing | Scheduled frequency of enteral feeding
quantity | | 0..1 | SimpleQuantity | The volume of formula to provide
rate[x] | | 0..1 | | Speed with which the formula is provided per period of time
rateQuantity | | | SimpleQuantity
rateRatio | | | Ratio
maxVolumeToDeliver | | 0..1 | SimpleQuantity | Upper limit on formula volume per unit of time
administrationInstruction | Σ | 0..1 | string | Formula feeding instructions expressed as text
note | | 0..* | Annotation | Comments
Documentation for this format

UML Diagram (Legend)

XML Template

```
<NutritionOrder xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Identifiers assigned to this order --></identifier>
 <instantiatesCanonical><!-- 0..* canonical(ActivityDefinition|PlanDefinition) Instantiates FHIR protocol or definition
    --></instantiatesCanonical>
 <instantiatesUri value="[uri]"/><!-- 0..* Instantiates external protocol or definition -->
 <instantiates value="[uri]"/><!-- 0..* Instantiates protocol or definition -->
 <status value="[code]"/><!-- 1..1 draft | active | on-hold | revoked | completed | entered-in-error | unknown -->
 <intent value="[code]"/><!-- 1..1 proposal | plan | directive | order | original-order | reflex-order | filler-order |
    instance-order | option -->
 <patient><!-- 1..1 Reference(Patient) The person who requires the diet, formula or nutritional supplement --></patient>
 <encounter><!-- 0..1 Reference(Encounter) The encounter associated with this nutrition order --></encounter>
 <dateTime value="[dateTime]"/><!-- 1..1 Date and time the nutrition order was requested -->
 <orderer><!-- 0..1 Reference(Practitioner|PractitionerRole) Who ordered the diet, formula or nutritional supplement
    --></orderer>
 <allergyIntolerance><!-- 0..* Reference(AllergyIntolerance) List of the patient's food and nutrition-related allergies
    and intolerances --></allergyIntolerance>
 <foodPreferenceModifier><!-- 0..* CodeableConcept Order-specific modifier about the type of food that should be given
    --></foodPreferenceModifier>
 <excludeFoodModifier><!-- 0..* CodeableConcept Order-specific modifier about the type of food that should not be given
    --></excludeFoodModifier>
 <oralDiet>  <!--  0..1 Oral diet components -->
  <type><!-- 0..* CodeableConcept Type of oral diet or diet restrictions that describe what can be consumed orally
    --></type>
  <schedule><!-- 0..* Timing Scheduled frequency of diet --></schedule>
  <nutrient>  <!-- 0..* Required  nutrient modifications -->
   <modifier><!-- 0..1 CodeableConcept Type of nutrient that is being modified --></modifier>
   <amount><!-- 0..1 Quantity(SimpleQuantity) Quantity of the specified nutrient --></amount>
  </nutrient>
  <texture>  <!-- 0..* Required  texture modifications -->
   <modifier><!-- 0..1 CodeableConcept Code to indicate how to alter the texture of the foods, e.g. pureed
    --></modifier>
   <foodType><!-- 0..1 CodeableConcept Concepts that are used to identify an entity that is ingested for nutritional
    purposes --></foodType>
  </texture>
  <fluidConsistencyType><!-- 0..* CodeableConcept The required consistency of fluids and liquids provided to the patient
     --></fluidConsistencyType>
  <instruction value="[string]"/><!-- 0..1 Instructions or additional information about the oral diet -->
 </oralDiet>
 <supplement>  <!--  0..* Supplement components -->
  <type><!-- 0..1 CodeableConcept Type of supplement product requested --></type>
  <productName value="[string]"/><!-- 0..1 Product or brand name of the nutritional supplement -->
  <schedule><!-- 0..* Timing Scheduled frequency of supplement --></schedule>
  <quantity><!-- 0..1 Quantity(SimpleQuantity) Amount of the nutritional supplement --></quantity>
  <instruction value="[string]"/><!-- 0..1 Instructions or additional information about the oral supplement -->
 </supplement>
 <enteralFormula>  <!--  0..1 Enteral formula components -->
  <baseFormulaType><!-- 0..1 CodeableConcept Type of enteral or infant formula --></baseFormulaType>
  <baseFormulaProductName value="[string]"/><!-- 0..1 Product or brand name of the enteral or infant formula -->
  <additiveType><!-- 0..1 CodeableConcept Type of modular component to add to the feeding --></additiveType>
  <additiveProductName value="[string]"/><!-- 0..1 Product or brand name of the modular additive -->
  <caloricDensity><!-- 0..1 Quantity(SimpleQuantity) Amount of energy per specified volume that is required
    --></caloricDensity>
  <routeofAdministration><!-- 0..1 CodeableConcept How the formula should enter the patient's gastrointestinal tract
    --></routeofAdministration>
  <administration>  <!-- 0..* Formula feeding instruction as structured data -->
   <schedule><!-- 0..1 Timing Scheduled frequency of enteral feeding --></schedule>
   <quantity><!-- 0..1 Quantity(SimpleQuantity) The volume of formula to provide --></quantity>
   <rate[x]><!-- 0..1 Quantity(SimpleQuantity)|Ratio Speed with which the formula is provided per period of time
    --></rate[x]>
  </administration>
  <maxVolumeToDeliver><!-- 0..1 Quantity(SimpleQuantity) Upper limit on formula volume per unit of time
    --></maxVolumeToDeliver>
  <administrationInstruction value="[string]"/><!-- 0..1 Formula feeding instructions expressed as text -->
 </enteralFormula>
 <note><!-- 0..* Annotation Comments --></note>
</NutritionOrder>
```

JSON Template

```
{
  "resourceType" : "NutritionOrder",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Identifiers assigned to this order
  "instantiatesCanonical" : [{ canonical(ActivityDefinition|PlanDefinition) }], // Instantiates FHIR protocol or
    definition
  "instantiatesUri" : ["<uri>"], // Instantiates external protocol or definition
  "instantiates" : ["<uri>"], // Instantiates protocol or definition
  "status" : "<code>", // R!  draft | active | on-hold | revoked | completed | entered-in-error | unknown
  "intent" : "<code>", // R!  proposal | plan | directive | order | original-order | reflex-order | filler-order |
    instance-order | option
  "patient" : { Reference(Patient) }, // R!  The person who requires the diet, formula or nutritional supplement
  "encounter" : { Reference(Encounter) }, // The encounter associated with this nutrition order
  "dateTime" : "<dateTime>", // R!  Date and time the nutrition order was requested
  "orderer" : { Reference(Practitioner|PractitionerRole) }, // Who ordered the diet, formula or nutritional supplement
  "allergyIntolerance" : [{ Reference(AllergyIntolerance) }], // List of the patient's food and nutrition-related
    allergies and intolerances
  "foodPreferenceModifier" : [{ CodeableConcept }], // Order-specific modifier about the type of food that should be
    given
  "excludeFoodModifier" : [{ CodeableConcept }], // Order-specific modifier about the type of food that should not be
    given
  "oralDiet" : { // C? Oral diet components
    "type" : [{ CodeableConcept }], // Type of oral diet or diet restrictions that describe what can be consumed orally
    "schedule" : [{ Timing }], // Scheduled frequency of diet
    "nutrient" : [{ // Required  nutrient modifications
      "modifier" : { CodeableConcept }, // Type of nutrient that is being modified
      "amount" : { Quantity(SimpleQuantity) } // Quantity of the specified nutrient
    }],
    "texture" : [{ // Required  texture modifications
      "modifier" : { CodeableConcept }, // Code to indicate how to alter the texture of the foods, e.g. pureed
      "foodType" : { CodeableConcept } // Concepts that are used to identify an entity that is ingested for nutritional
    purposes
    }],
    "fluidConsistencyType" : [{ CodeableConcept }], // The required consistency of fluids and liquids provided to the
    patient
    "instruction" : "<string>" // Instructions or additional information about the oral diet
  },
  "supplement" : [{ // C? Supplement components
    "type" : { CodeableConcept }, // Type of supplement product requested
    "productName" : "<string>", // Product or brand name of the nutritional supplement
    "schedule" : [{ Timing }], // Scheduled frequency of supplement
    "quantity" : { Quantity(SimpleQuantity) }, // Amount of the nutritional supplement
    "instruction" : "<string>" // Instructions or additional information about the oral supplement
  }],
  "enteralFormula" : { // C? Enteral formula components
    "baseFormulaType" : { CodeableConcept }, // Type of enteral or infant formula
    "baseFormulaProductName" : "<string>", // Product or brand name of the enteral or infant formula
    "additiveType" : { CodeableConcept }, // Type of modular component to add to the feeding
    "additiveProductName" : "<string>", // Product or brand name of the modular additive
    "caloricDensity" : { Quantity(SimpleQuantity) }, // Amount of energy per specified volume that is required
    "routeofAdministration" : { CodeableConcept }, // How the formula should enter the patient's gastrointestinal tract
    "administration" : [{ // Formula feeding instruction as structured data
      "schedule" : { Timing }, // Scheduled frequency of enteral feeding
      "quantity" : { Quantity(SimpleQuantity) }, // The volume of formula to provide
      // rate[x]: Speed with which the formula is provided per period of time. One of these 2:
      "rateQuantity" : { Quantity(SimpleQuantity) }
      "rateRatio" : { Ratio }
    }],
    "maxVolumeToDeliver" : { Quantity(SimpleQuantity) }, // Upper limit on formula volume per unit of time
    "administrationInstruction" : "<string>" // Formula feeding instructions expressed as text
  },
  "note" : [{ Annotation }] // Comments
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:NutritionOrder;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:NutritionOrder.identifier [ Identifier ], ... ; # 0..* Identifiers assigned to this order
  fhir:NutritionOrder.instantiatesCanonical [ canonical(ActivityDefinition|PlanDefinition) ], ... ; # 0..* Instantiates
    FHIR protocol or definition
  fhir:NutritionOrder.instantiatesUri [ uri ], ... ; # 0..* Instantiates external protocol or definition
  fhir:NutritionOrder.instantiates [ uri ], ... ; # 0..* Instantiates protocol or definition
  fhir:NutritionOrder.status [ code ]; # 1..1 draft | active | on-hold | revoked | completed | entered-in-error |
    unknown
  fhir:NutritionOrder.intent [ code ]; # 1..1 proposal | plan | directive | order | original-order | reflex-order |
    filler-order | instance-order | option
  fhir:NutritionOrder.patient [ Reference(Patient) ]; # 1..1 The person who requires the diet, formula or nutritional
    supplement
  fhir:NutritionOrder.encounter [ Reference(Encounter) ]; # 0..1 The encounter associated with this nutrition order
  fhir:NutritionOrder.dateTime [ dateTime ]; # 1..1 Date and time the nutrition order was requested
  fhir:NutritionOrder.orderer [ Reference(Practitioner|PractitionerRole) ]; # 0..1 Who ordered the diet, formula or
    nutritional supplement
```

```
  fhir:NutritionOrder.allergyIntolerance [ Reference(AllergyIntolerance) ], ... ; # 0..* List of the patient's food and
    nutrition-related allergies and intolerances
  fhir:NutritionOrder.foodPreferenceModifier [ CodeableConcept ], ... ; # 0..* Order-specific modifier about the type of
     food that should be given
  fhir:NutritionOrder.excludeFoodModifier [ CodeableConcept ], ... ; # 0..* Order-specific modifier about the type of
    food that should not be given
  fhir:NutritionOrder.oralDiet [ # 0..1 Oral diet components
    fhir:NutritionOrder.oralDiet.type [ CodeableConcept ], ... ; # 0..* Type of oral diet or diet restrictions that
    describe what can be consumed orally
    fhir:NutritionOrder.oralDiet.schedule [ Timing ], ... ; # 0..* Scheduled frequency of diet
    fhir:NutritionOrder.oralDiet.nutrient [ # 0..* Required  nutrient modifications
      fhir:NutritionOrder.oralDiet.nutrient.modifier [ CodeableConcept ]; # 0..1 Type of nutrient that is being modified
      fhir:NutritionOrder.oralDiet.nutrient.amount [ Quantity(SimpleQuantity) ]; # 0..1 Quantity of the specified
    nutrient
    ], ...;
    fhir:NutritionOrder.oralDiet.texture [ # 0..* Required  texture modifications
      fhir:NutritionOrder.oralDiet.texture.modifier [ CodeableConcept ]; # 0..1 Code to indicate how to alter the
    texture of the foods, e.g. pureed
      fhir:NutritionOrder.oralDiet.texture.foodType [ CodeableConcept ]; # 0..1 Concepts that are used to identify an
    entity that is ingested for nutritional purposes
    ], ...;
    fhir:NutritionOrder.oralDiet.fluidConsistencyType [ CodeableConcept ], ... ; # 0..* The required consistency of
    fluids and liquids provided to the patient
    fhir:NutritionOrder.oralDiet.instruction [ string ]; # 0..1 Instructions or additional information about the oral
    diet
  ];
  fhir:NutritionOrder.supplement [ # 0..* Supplement components
    fhir:NutritionOrder.supplement.type [ CodeableConcept ]; # 0..1 Type of supplement product requested
    fhir:NutritionOrder.supplement.productName [ string ]; # 0..1 Product or brand name of the nutritional supplement
    fhir:NutritionOrder.supplement.schedule [ Timing ], ... ; # 0..* Scheduled frequency of supplement
    fhir:NutritionOrder.supplement.quantity [ Quantity(SimpleQuantity) ]; # 0..1 Amount of the nutritional supplement
    fhir:NutritionOrder.supplement.instruction [ string ]; # 0..1 Instructions or additional information about the oral
    supplement
  ], ...;
  fhir:NutritionOrder.enteralFormula [ # 0..1 Enteral formula components
    fhir:NutritionOrder.enteralFormula.baseFormulaType [ CodeableConcept ]; # 0..1 Type of enteral or infant formula
    fhir:NutritionOrder.enteralFormula.baseFormulaProductName [ string ]; # 0..1 Product or brand name of the enteral or
     infant formula
    fhir:NutritionOrder.enteralFormula.additiveType [ CodeableConcept ]; # 0..1 Type of modular component to add to the
    feeding
    fhir:NutritionOrder.enteralFormula.additiveProductName [ string ]; # 0..1 Product or brand name of the modular
    additive
    fhir:NutritionOrder.enteralFormula.caloricDensity [ Quantity(SimpleQuantity) ]; # 0..1 Amount of energy per
    specified volume that is required
    fhir:NutritionOrder.enteralFormula.routeofAdministration [ CodeableConcept ]; # 0..1 How the formula should enter
    the patient's gastrointestinal tract
    fhir:NutritionOrder.enteralFormula.administration [ # 0..* Formula feeding instruction as structured data
      fhir:NutritionOrder.enteralFormula.administration.schedule [ Timing ]; # 0..1 Scheduled frequency of enteral
    feeding
      fhir:NutritionOrder.enteralFormula.administration.quantity [ Quantity(SimpleQuantity) ]; # 0..1 The volume of
    formula to provide
      # NutritionOrder.enteralFormula.administration.rate[x] : 0..1 Speed with which the formula is provided per period
    of time. One of these 2
        fhir:NutritionOrder.enteralFormula.administration.rateSimpleQuantity [ Quantity(SimpleQuantity) ]
        fhir:NutritionOrder.enteralFormula.administration.rateRatio [ Ratio ]
    ], ...;
    fhir:NutritionOrder.enteralFormula.maxVolumeToDeliver [ Quantity(SimpleQuantity) ]; # 0..1 Upper limit on formula
    volume per unit of time
    fhir:NutritionOrder.enteralFormula.administrationInstruction [ string ]; # 0..1 Formula feeding instructions
    expressed as text
  ];
  fhir:NutritionOrder.note [ Annotation ], ... ; # 0..* Comments
]
```

Changes since R3

NutritionOrder

NutritionOrder.instantiatesCanonical | Added Element

NutritionOrder.instantiatesUri | Added Element

NutritionOrder.instantiates | Added Element

NutritionOrder.status | Min Cardinality changed from 0 to 1 Change value set from http://hl7.org/fhir/ValueSet/nutrition-request-status to http://hl7.org/fhir/ValueSet/request-status|4.0.1

NutritionOrder.intent | Added Mandatory Element

NutritionOrder.orderer | Type Reference: Added Target Type PractitionerRole

NutritionOrder.enteralFormula.administration.rate[x] | Add Type Quantity() Remove Type Quantity()

NutritionOrder.note | Added Element

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 13 tests that all execute ok. All tests pass round-trip testing and 13 r3 resources are invalid (0 errors).)

Structure

Name | Flags | Card. | Type | Description & Constraints

NutritionOrder | ITU | | DomainResource | Diet, formula or nutritional supplement request + Warning: Nutrition Order SHALL contain either Oral Diet , Supplement, or Enteral Formula class Elements defined in Ancestors: id, meta, implicitRules, language,

text, contained, extension, modifierExtension
identifier | | 0..* | Identifier | Identifiers assigned to this order
instantiatesCanonical | Σ | 0..* | canonical(ActivityDefinition | PlanDefinition) | Instantiates FHIR protocol or definition
instantiatesUri | Σ | 0..* | uri | Instantiates external protocol or definition
instantiates | | 0..* | uri | Instantiates protocol or definition
status | ?!Σ | 1..1 | code | draft | active | on-hold | revoked | completed | entered-in-error | unknown RequestStatus (Required)
intent | ?!Σ | 1..1 | code | proposal | plan | directive | order | original-order | reflex-order | filler-order | instance-order | option RequestIntent (Required)
patient | Σ | 1..1 | Reference(Patient) | The person who requires the diet, formula or nutritional supplement
encounter | | 0..1 | Reference(Encounter) | The encounter associated with this nutrition order
dateTime | Σ | 1..1 | dateTime | Date and time the nutrition order was requested
orderer | Σ | 0..1 | Reference(Practitioner | PractitionerRole) | Who ordered the diet, formula or nutritional supplement
allergyIntolerance | | 0..* | Reference(AllergyIntolerance) | List of the patient's food and nutrition-related allergies and intolerances
foodPreferenceModifier | | 0..* | CodeableConcept | Order-specific modifier about the type of food that should be given Diet (Example)
excludeFoodModifier | | 0..* | CodeableConcept | Order-specific modifier about the type of food that should not be given Food Type Codes (Example)
oralDiet | I | 0..1 | BackboneElement | Oral diet components
type | Σ | 0..* | CodeableConcept | Type of oral diet or diet restrictions that describe what can be consumed orally Diet Codes (Example)
schedule | | 0..* | Timing | Scheduled frequency of diet
nutrient | | 0..* | BackboneElement | Required nutrient modifications
modifier | | 0..1 | CodeableConcept | Type of nutrient that is being modified Nutrient Modifier Codes (Example)
amount | | 0..1 | SimpleQuantity | Quantity of the specified nutrient
texture | | 0..* | BackboneElement | Required texture modifications
modifier | | 0..1 | CodeableConcept | Code to indicate how to alter the texture of the foods, e.g. pureed Texture Modifier Codes (Example)
foodType | | 0..1 | CodeableConcept | Concepts that are used to identify an entity that is ingested for nutritional purposes Texture Modified Food Type Codes (Example)
fluidConsistencyType | | 0..* | CodeableConcept | The required consistency of fluids and liquids provided to the patient Fluid Consistency Type Codes (Example)
instruction | Σ | 0..1 | string | Instructions or additional information about the oral diet
supplement | I | 0..* | BackboneElement | Supplement components
type | Σ | 0..1 | CodeableConcept | Type of supplement product requested Supplement Type Codes (Example)
productName | | 0..1 | string | Product or brand name of the nutritional supplement
schedule | | 0..* | Timing | Scheduled frequency of supplement
quantity | | 0..1 | SimpleQuantity | Amount of the nutritional supplement
instruction | Σ | 0..1 | string | Instructions or additional information about the oral supplement
enteralFormula | I | 0..1 | BackboneElement | Enteral formula components
baseFormulaType | Σ | 0..1 | CodeableConcept | Type of enteral or infant formula Enteral Formula Type Codes (Example)
baseFormulaProductName | | 0..1 | string | Product or brand name of the enteral or infant formula
additiveType | | 0..1 | CodeableConcept | Type of modular component to add to the feeding Enteral Formula Additive Type Code (Example)
additiveProductName | | 0..1 | string | Product or brand name of the modular additive
caloricDensity | | 0..1 | SimpleQuantity | Amount of energy per specified volume that is required
routeofAdministration | | 0..1 | CodeableConcept | How the formula should enter the patient's gastrointestinal tract Enteral Route Codes (Extensible)
administration | | 0..* | BackboneElement | Formula feeding instruction as structured data
schedule | | 0..1 | Timing | Scheduled frequency of enteral feeding
quantity | | 0..1 | SimpleQuantity | The volume of formula to provide
rate[x] | | 0..1 | | Speed with which the formula is provided per period of time
rateQuantity | | | SimpleQuantity
rateRatio | | | Ratio
maxVolumeToDeliver | | 0..1 | SimpleQuantity | Upper limit on formula volume per unit of time
administrationInstruction | Σ | 0..1 | string | Formula feeding instructions expressed as text
note | | 0..* | Annotation | Comments
Documentation for this format

UML Diagram (Legend)

XML Template

```
<NutritionOrder xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Identifiers assigned to this order --></identifier>
 <instantiatesCanonical><!-- 0..* canonical(ActivityDefinition|PlanDefinition) Instantiates FHIR protocol or definition
    --></instantiatesCanonical>
 <instantiatesUri value="[uri]"/><!-- 0..* Instantiates external protocol or definition -->
 <instantiates value="[uri]"/><!-- 0..* Instantiates protocol or definition -->
 <status value="[code]"/><!-- 1..1 draft | active | on-hold | revoked | completed | entered-in-error | unknown -->
 <intent value="[code]"/><!-- 1..1 proposal | plan | directive | order | original-order | reflex-order | filler-order |
    instance-order | option -->
 <patient><!-- 1..1 Reference(Patient) The person who requires the diet, formula or nutritional supplement --></patient>
 <encounter><!-- 0..1 Reference(Encounter) The encounter associated with this nutrition order --></encounter>
 <dateTime value="[dateTime]"/><!-- 1..1 Date and time the nutrition order was requested -->
 <orderer><!-- 0..1 Reference(Practitioner|PractitionerRole) Who ordered the diet, formula or nutritional supplement
    --></orderer>
```

```
 <allergyIntolerance><!-- 0..* Reference(AllergyIntolerance) List of the patient's food and nutrition-related allergies
    and intolerances --></allergyIntolerance>
 <foodPreferenceModifier><!-- 0..* CodeableConcept Order-specific modifier about the type of food that should be given
    --></foodPreferenceModifier>
 <excludeFoodModifier><!-- 0..* CodeableConcept Order-specific modifier about the type of food that should not be given
    --></excludeFoodModifier>
 <oralDiet>  <!--  0..1 Oral diet components -->
  <type><!-- 0..* CodeableConcept Type of oral diet or diet restrictions that describe what can be consumed orally
    --></type>
  <schedule><!-- 0..* Timing Scheduled frequency of diet --></schedule>
  <nutrient>  <!-- 0..* Required  nutrient modifications -->
   <modifier><!-- 0..1 CodeableConcept Type of nutrient that is being modified --></modifier>
   <amount><!-- 0..1 Quantity(SimpleQuantity) Quantity of the specified nutrient --></amount>
  </nutrient>
  <texture>  <!-- 0..* Required  texture modifications -->
   <modifier><!-- 0..1 CodeableConcept Code to indicate how to alter the texture of the foods, e.g. pureed
    --></modifier>
   <foodType><!-- 0..1 CodeableConcept Concepts that are used to identify an entity that is ingested for nutritional
    purposes --></foodType>
  </texture>
  <fluidConsistencyType><!-- 0..* CodeableConcept The required consistency of fluids and liquids provided to the patient
     --></fluidConsistencyType>
  <instruction value="[string]"/><!-- 0..1 Instructions or additional information about the oral diet -->
 </oralDiet>
 <supplement>  <!--  0..* Supplement components -->
  <type><!-- 0..1 CodeableConcept Type of supplement product requested --></type>
  <productName value="[string]"/><!-- 0..1 Product or brand name of the nutritional supplement -->
  <schedule><!-- 0..* Timing Scheduled frequency of supplement --></schedule>
  <quantity><!-- 0..1 Quantity(SimpleQuantity) Amount of the nutritional supplement --></quantity>
  <instruction value="[string]"/><!-- 0..1 Instructions or additional information about the oral supplement -->
 </supplement>
 <enteralFormula>  <!--  0..1 Enteral formula components -->
  <baseFormulaType><!-- 0..1 CodeableConcept Type of enteral or infant formula --></baseFormulaType>
  <baseFormulaProductName value="[string]"/><!-- 0..1 Product or brand name of the enteral or infant formula -->
  <additiveType><!-- 0..1 CodeableConcept Type of modular component to add to the feeding --></additiveType>
  <additiveProductName value="[string]"/><!-- 0..1 Product or brand name of the modular additive -->
  <caloricDensity><!-- 0..1 Quantity(SimpleQuantity) Amount of energy per specified volume that is required
    --></caloricDensity>
  <routeofAdministration><!-- 0..1 CodeableConcept How the formula should enter the patient's gastrointestinal tract
    --></routeofAdministration>
  <administration>  <!-- 0..* Formula feeding instruction as structured data -->
   <schedule><!-- 0..1 Timing Scheduled frequency of enteral feeding --></schedule>
   <quantity><!-- 0..1 Quantity(SimpleQuantity) The volume of formula to provide --></quantity>
   <rate[x]><!-- 0..1 Quantity(SimpleQuantity)|Ratio Speed with which the formula is provided per period of time
    --></rate[x]>
  </administration>
  <maxVolumeToDeliver><!-- 0..1 Quantity(SimpleQuantity) Upper limit on formula volume per unit of time
    --></maxVolumeToDeliver>
  <administrationInstruction value="[string]"/><!-- 0..1 Formula feeding instructions expressed as text -->
 </enteralFormula>
 <note><!-- 0..* Annotation Comments --></note>
</NutritionOrder>
```

JSON Template

```
{
  "resourceType" : "NutritionOrder",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Identifiers assigned to this order
  "instantiatesCanonical" : [{ canonical(ActivityDefinition|PlanDefinition) }], // Instantiates FHIR protocol or
    definition
  "instantiatesUri" : ["<uri>"], // Instantiates external protocol or definition
  "instantiates" : ["<uri>"], // Instantiates protocol or definition
  "status" : "<code>", // R!  draft | active | on-hold | revoked | completed | entered-in-error | unknown
  "intent" : "<code>", // R!  proposal | plan | directive | order | original-order | reflex-order | filler-order |
    instance-order | option
  "patient" : { Reference(Patient) }, // R!  The person who requires the diet, formula or nutritional supplement
  "encounter" : { Reference(Encounter) }, // The encounter associated with this nutrition order
  "dateTime" : "<dateTime>", // R!  Date and time the nutrition order was requested
  "orderer" : { Reference(Practitioner|PractitionerRole) }, // Who ordered the diet, formula or nutritional supplement
  "allergyIntolerance" : [{ Reference(AllergyIntolerance) }], // List of the patient's food and nutrition-related
    allergies and intolerances
  "foodPreferenceModifier" : [{ CodeableConcept }], // Order-specific modifier about the type of food that should be
    given
  "excludeFoodModifier" : [{ CodeableConcept }], // Order-specific modifier about the type of food that should not be
    given
  "oralDiet" : { // C? Oral diet components
    "type" : [{ CodeableConcept }], // Type of oral diet or diet restrictions that describe what can be consumed orally
    "schedule" : [{ Timing }], // Scheduled frequency of diet
    "nutrient" : [{ // Required  nutrient modifications
      "modifier" : { CodeableConcept }, // Type of nutrient that is being modified
      "amount" : { Quantity(SimpleQuantity) } // Quantity of the specified nutrient
    }],
    "texture" : [{ // Required  texture modifications
      "modifier" : { CodeableConcept }, // Code to indicate how to alter the texture of the foods, e.g. pureed
      "foodType" : { CodeableConcept } // Concepts that are used to identify an entity that is ingested for nutritional
    purposes
    }],
    "fluidConsistencyType" : [{ CodeableConcept }], // The required consistency of fluids and liquids provided to the
```

```
    patient
    "instruction" : "<string>" // Instructions or additional information about the oral diet
  },
  "supplement" : [{ // C? Supplement components
    "type" : { CodeableConcept }, // Type of supplement product requested
    "productName" : "<string>", // Product or brand name of the nutritional supplement
    "schedule" : [{ Timing }], // Scheduled frequency of supplement
    "quantity" : { Quantity(SimpleQuantity) }, // Amount of the nutritional supplement
    "instruction" : "<string>" // Instructions or additional information about the oral supplement
  }],
  "enteralFormula" : { // C? Enteral formula components
    "baseFormulaType" : { CodeableConcept }, // Type of enteral or infant formula
    "baseFormulaProductName" : "<string>", // Product or brand name of the enteral or infant formula
    "additiveType" : { CodeableConcept }, // Type of modular component to add to the feeding
    "additiveProductName" : "<string>", // Product or brand name of the modular additive
    "caloricDensity" : { Quantity(SimpleQuantity) }, // Amount of energy per specified volume that is required
    "routeofAdministration" : { CodeableConcept }, // How the formula should enter the patient's gastrointestinal tract
    "administration" : [{ // Formula feeding instruction as structured data
      "schedule" : { Timing }, // Scheduled frequency of enteral feeding
      "quantity" : { Quantity(SimpleQuantity) }, // The volume of formula to provide
      // rate[x]: Speed with which the formula is provided per period of time. One of these 2:
      "rateQuantity" : { Quantity(SimpleQuantity) }
      "rateRatio" : { Ratio }
    }],
    "maxVolumeToDeliver" : { Quantity(SimpleQuantity) }, // Upper limit on formula volume per unit of time
    "administrationInstruction" : "<string>" // Formula feeding instructions expressed as text
  },
  "note" : [{ Annotation }] // Comments
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:NutritionOrder;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:NutritionOrder.identifier [ Identifier ], ... ; # 0..* Identifiers assigned to this order
  fhir:NutritionOrder.instantiatesCanonical [ canonical(ActivityDefinition|PlanDefinition) ], ... ; # 0..* Instantiates
    FHIR protocol or definition
  fhir:NutritionOrder.instantiatesUri [ uri ], ... ; # 0..* Instantiates external protocol or definition
  fhir:NutritionOrder.instantiates [ uri ], ... ; # 0..* Instantiates protocol or definition
  fhir:NutritionOrder.status [ code ]; # 1..1 draft | active | on-hold | revoked | completed | entered-in-error |
    unknown
  fhir:NutritionOrder.intent [ code ]; # 1..1 proposal | plan | directive | order | original-order | reflex-order |
    filler-order | instance-order | option
  fhir:NutritionOrder.patient [ Reference(Patient) ]; # 1..1 The person who requires the diet, formula or nutritional
    supplement
  fhir:NutritionOrder.encounter [ Reference(Encounter) ]; # 0..1 The encounter associated with this nutrition order
  fhir:NutritionOrder.dateTime [ dateTime ]; # 1..1 Date and time the nutrition order was requested
  fhir:NutritionOrder.orderer [ Reference(Practitioner|PractitionerRole) ]; # 0..1 Who ordered the diet, formula or
    nutritional supplement
  fhir:NutritionOrder.allergyIntolerance [ Reference(AllergyIntolerance) ], ... ; # 0..* List of the patient's food and
    nutrition-related allergies and intolerances
  fhir:NutritionOrder.foodPreferenceModifier [ CodeableConcept ], ... ; # 0..* Order-specific modifier about the type of
     food that should be given
  fhir:NutritionOrder.excludeFoodModifier [ CodeableConcept ], ... ; # 0..* Order-specific modifier about the type of
    food that should not be given
  fhir:NutritionOrder.oralDiet [ # 0..1 Oral diet components
    fhir:NutritionOrder.oralDiet.type [ CodeableConcept ], ... ; # 0..* Type of oral diet or diet restrictions that
    describe what can be consumed orally
    fhir:NutritionOrder.oralDiet.schedule [ Timing ], ... ; # 0..* Scheduled frequency of diet
    fhir:NutritionOrder.oralDiet.nutrient [ # 0..* Required  nutrient modifications
      fhir:NutritionOrder.oralDiet.nutrient.modifier [ CodeableConcept ]; # 0..1 Type of nutrient that is being modified
      fhir:NutritionOrder.oralDiet.nutrient.amount [ Quantity(SimpleQuantity) ]; # 0..1 Quantity of the specified
    nutrient
    ], ...;
    fhir:NutritionOrder.oralDiet.texture [ # 0..* Required  texture modifications
      fhir:NutritionOrder.oralDiet.texture.modifier [ CodeableConcept ]; # 0..1 Code to indicate how to alter the
    texture of the foods, e.g. pureed
      fhir:NutritionOrder.oralDiet.texture.foodType [ CodeableConcept ]; # 0..1 Concepts that are used to identify an
    entity that is ingested for nutritional purposes
    ], ...;
    fhir:NutritionOrder.oralDiet.fluidConsistencyType [ CodeableConcept ], ... ; # 0..* The required consistency of
    fluids and liquids provided to the patient
    fhir:NutritionOrder.oralDiet.instruction [ string ]; # 0..1 Instructions or additional information about the oral
    diet
  ];
  fhir:NutritionOrder.supplement [ # 0..* Supplement components
    fhir:NutritionOrder.supplement.type [ CodeableConcept ]; # 0..1 Type of supplement product requested
    fhir:NutritionOrder.supplement.productName [ string ]; # 0..1 Product or brand name of the nutritional supplement
    fhir:NutritionOrder.supplement.schedule [ Timing ], ... ; # 0..* Scheduled frequency of supplement
    fhir:NutritionOrder.supplement.quantity [ Quantity(SimpleQuantity) ]; # 0..1 Amount of the nutritional supplement
    fhir:NutritionOrder.supplement.instruction [ string ]; # 0..1 Instructions or additional information about the oral
    supplement
  ], ...;
  fhir:NutritionOrder.enteralFormula [ # 0..1 Enteral formula components
```

```
    fhir:NutritionOrder.enteralFormula.baseFormulaType [ CodeableConcept ]; # 0..1 Type of enteral or infant formula
    fhir:NutritionOrder.enteralFormula.baseFormulaProductName [ string ]; # 0..1 Product or brand name of the enteral or
     infant formula
    fhir:NutritionOrder.enteralFormula.additiveType [ CodeableConcept ]; # 0..1 Type of modular component to add to the
    feeding
    fhir:NutritionOrder.enteralFormula.additiveProductName [ string ]; # 0..1 Product or brand name of the modular
    additive
    fhir:NutritionOrder.enteralFormula.caloricDensity [ Quantity(SimpleQuantity) ]; # 0..1 Amount of energy per
    specified volume that is required
    fhir:NutritionOrder.enteralFormula.routeofAdministration [ CodeableConcept ]; # 0..1 How the formula should enter
    the patient's gastrointestinal tract
    fhir:NutritionOrder.enteralFormula.administration [ # 0..* Formula feeding instruction as structured data
      fhir:NutritionOrder.enteralFormula.administration.schedule [ Timing ]; # 0..1 Scheduled frequency of enteral
    feeding
      fhir:NutritionOrder.enteralFormula.administration.quantity [ Quantity(SimpleQuantity) ]; # 0..1 The volume of
    formula to provide
      # NutritionOrder.enteralFormula.administration.rate[x] : 0..1 Speed with which the formula is provided per period
    of time. One of these 2
        fhir:NutritionOrder.enteralFormula.administration.rateSimpleQuantity [ Quantity(SimpleQuantity) ]
        fhir:NutritionOrder.enteralFormula.administration.rateRatio [ Ratio ]
    ], ...;
    fhir:NutritionOrder.enteralFormula.maxVolumeToDeliver [ Quantity(SimpleQuantity) ]; # 0..1 Upper limit on formula
    volume per unit of time
    fhir:NutritionOrder.enteralFormula.administrationInstruction [ string ]; # 0..1 Formula feeding instructions
    expressed as text
  ];
  fhir:NutritionOrder.note [ Annotation ], ... ; # 0..* Comments
]
```

Changes since Release 3

NutritionOrder
NutritionOrder.instantiatesCanonical | Added Element
NutritionOrder.instantiatesUri | Added Element
NutritionOrder.instantiates | Added Element
NutritionOrder.status | Min Cardinality changed from 0 to 1 Change value set from http://hl7.org/fhir/ValueSet/nutrition-request-status to http://hl7.org/fhir/ValueSet/request-status|4.0.1
NutritionOrder.intent | Added Mandatory Element
NutritionOrder.orderer | Type Reference: Added Target Type PractitionerRole
NutritionOrder.enteralFormula.administration.rate[x] | Add Type Quantity() Remove Type Quantity()
NutritionOrder.note | Added Element

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 13 tests that all execute ok. All tests pass round-trip testing and 13 r3 resources are invalid (0 errors).)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

12.15.3.1 Terminology Bindings

Path | Definition | Type | Reference
NutritionOrder.status | Codes identifying the lifecycle stage of the nutrition order. | Required | RequestStatus
NutritionOrder.intent | Codes indicating the degree of authority/intentionality associated with a nutrition order. | Required | RequestIntent
NutritionOrder.foodPreferenceModifier | Medical, cultural or ethical food preferences to help with catering requirements. | Example | Diet
NutritionOrder.excludeFoodModifier | Codes used to indicate the type of food that should NOT be given to the patient. | Example | FoodTypeCodes
NutritionOrder.oralDiet.type | Codes used to indicate the type of diet being ordered for a patient. | Example | DietCodes
NutritionOrder.oralDiet.nutrient.modifier | Codes for types of nutrients that are being modified such as carbohydrate or sodium. | Example | NutrientModifierCodes
NutritionOrder.oralDiet.texture.modifier | Codes for food consistency types or texture modifications to apply to foods. | Example | TextureModifierCodes
NutritionOrder.oralDiet.texture.foodType | Codes for types of foods that are texture-modified. | Example | TextureModifiedFoodTypeCodes
NutritionOrder.oralDiet.fluidConsistencyType | Codes used to represent the consistency of fluids and liquids provided to the patient. | Example | FluidConsistencyTypeCodes
NutritionOrder.supplement.type | Codes for nutritional supplements to be provided to the patient. | Example | SupplementTypeCodes
NutritionOrder.enteralFormula.baseFormulaType | Codes for type of enteral formula to be administered to patient. | Example | EnteralFormulaTypeCodes
NutritionOrder.enteralFormula.additiveType | Codes for the type of modular component such as protein, carbohydrate or fiber to be provided in addition to or mixed with the base formula. | Example | EnteralFormulaAdditiveTypeCode
NutritionOrder.enteralFormula.routeofAdministration | Codes specifying the route of administration of enteral formula. | Extensible | EnteralRouteCodes

12.15.3.2 Constraints

id | Level | Location | Description | Expression

nor-1 | Warning | (base) | Nutrition Order SHALL contain either Oral Diet , Supplement, or Enteral Formula class | oralDiet.exists() or supplement.exists() or enteralFormula.exists()

12.15.4 Notes:

12.15.4.1 Enteral continuous vs. intermittent tube feedings

Tube feedings can be administered via continuous drip using a pump or via intermittent feedings, using gravity drip or a pump. The examples Nutrition Order Enteral Bolus Feeding Example and Nutrition Order Enteral Continuous Feeding Example show how this resource can be used to order both kinds of enteral feeding using the structured data elements. The continuous feeding typically specifies rate of administration and a maximum volume of delivery using the enteralFormula.administration.rate and enteralFormula.maxVolumeToDeliver elements. On the other hand, the intermittent feeding typically specifies the amount and frequency of administration using the enteralFormula.administration.quantity and enteralFormula.schedule elements. In both cases, to vary the rate or quantity over time the enteralFormula.administration element can be repeated.

12.15.4.2 Note about the examples

The examples associated with this resource demonstrate the core elements and do not necessarily reflect real world implementations that may be constrained by future profiles for a given implementation or setting. For example, a renal diet is often comprised of pre-coordinated components including common nutrient modifications such as protein, potassium and phosphorus to assist with the speed of entry of complex diet orders.

12.15.5 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
additive | token | Type of module component to add to the feeding | NutritionOrder.enteralFormula.additiveType
datetime | date | Return nutrition orders requested on this date | NutritionOrder.dateTime
encounter | reference | Return nutrition orders with this encounter identifier | NutritionOrder.encounter (Encounter) | 12 Resources
formula | token | Type of enteral or infant formula | NutritionOrder.enteralFormula.baseFormulaType
identifier | token | Return nutrition orders with this external identifier | NutritionOrder.identifier | 30 Resources
instantiates-canonical | reference | Instantiates FHIR protocol or definition | NutritionOrder.instantiatesCanonical (PlanDefinition, ActivityDefinition)
instantiates-uri | uri | Instantiates external protocol or definition | NutritionOrder.instantiatesUri
oraldiet | token | Type of diet that can be consumed orally (i.e., take via the mouth). | NutritionOrder.oralDiet.type
patient | reference | The identity of the person who requires the diet, formula or nutritional supplement | NutritionOrder.patient (Patient) | 33 Resources
provider | reference | The identity of the provider who placed the nutrition order | NutritionOrder.orderer (Practitioner, PractitionerRole)
status | token | Status of the nutrition order. | NutritionOrder.status
supplement | token | Type of supplement product requested | NutritionOrder.supplement.type

12.15.6 Resource NutritionOrder - Examples

URL: https://hl7.org/fhir/R4/nutritionorder-examples.html

12.15.6 Resource NutritionOrder - Examples

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Encounter, Patient, Practitioner

Example Name | id | Format
Nutrition Order Pureed Diet and Energy/Protein Supplement Simple Example | pureeddiet-simple | XML | JSON | Turtle
Nutrition Order Low Sodium and Fluid Restricted Diet Example | cardiacdiet | XML | JSON | Turtle
Nutrition Order Diabetic Diet Example | diabeticdiet | XML | JSON | Turtle
Nutrition Order Texture Modified Diet Example | texturemodified | XML | JSON | Turtle
Nutrition Order Pureed Diet Example | pureeddiet | XML | JSON | Turtle
Nutrition Order Renal Diet Example | renaldiet | XML | JSON | Turtle
Nutrition Order Low Fat and Fiber Restricted Diet Example | fiberrestricteddiet | XML | JSON | Turtle
Nutrition Order Diabetic Supplement Example | diabeticsupplement | XML | JSON | Turtle
Nutrition Order Protein Supplement Example | proteinsupplement | XML | JSON | Turtle
Nutrition Order Energy/Protein Supplement Example | energysupplement | XML | JSON | Turtle
Nutrition Order Enteral Bolus Feeding Example | enteralbolus | XML | JSON | Turtle
Nutrition Order Enteral Continuous Feeding Example | enteralcontinuous | XML | JSON | Turtle
Nutrition Order Infant Enteral Feeding Example | infantenteral | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

12.15.7 Resource NutritionOrder - Detailed Descriptions

URL: https://hl7.org/fhir/R4/nutritionorder-definitions.html

12.15.7 Resource NutritionOrder - Detailed Descriptions

Orders and Observations Work Group | Maturity Level: 2 | Trial Use | Security Category: Patient | Compartments: Encounter, Patient, Practitioner

Detailed Descriptions for the elements in the NutritionOrder resource.

NutritionOrder
Element Id | NutritionOrder
Definition | A request to supply a diet, formula feeding (enteral) or oral nutritional supplement to a patient/resident.
Cardinality | 0..*
Type | DomainResource
Alternate Names | Nutrition Order; Diet Order; Diet; Nutritional Supplement; Enteral Nutrition
Comments | Referenced by an Order Request (workflow).
Invariants | Defined on this element nor-1 Warning Nutrition Order SHALL contain either Oral Diet , Supplement, or Enteral Formula class oralDiet.exists() or supplement.exists() or enteralFormula.exists()
Defined on this element
nor-1 | Warning | Nutrition Order SHALL contain either Oral Diet , Supplement, or Enteral Formula class | oralDiet.exists() or supplement.exists() or enteralFormula.exists()
NutritionOrder.identifier
Element Id | NutritionOrder.identifier
Definition | Identifiers assigned to this order by the order sender or by the order receiver.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Comments | The Identifier.type element can be to indicate filler vs. placer if needed. This is explained in further detail here.
NutritionOrder.instantiatesCanonical
Element Id | NutritionOrder.instantiatesCanonical
Definition | The URL pointing to a FHIR-defined protocol, guideline, orderset or other definition that is adhered to in whole or in part by this NutritionOrder.
Cardinality | 0..*
Type | canonical(ActivityDefinition | PlanDefinition)
Summary | true
Comments | Note: This is a business identifier, not a resource identifier (see discussion). It is best practice for the identifier to only appear on a single resource instance, however business practices may occasionally dictate that multiple resource instances with the same identifier can exist - possibly even with different resource types. For example, multiple Patient and a Person resource instance might share the same social insurance number.
NutritionOrder.instantiatesUri
Element Id | NutritionOrder.instantiatesUri
Definition | The URL pointing to an externally maintained protocol, guideline, orderset or other definition that is adhered to in whole or in part by this NutritionOrder.
Cardinality | 0..*
Type | uri
Summary | true
Comments | This might be an HTML page, PDF, etc. or could just be a non-resolvable URI identifier.
NutritionOrder.instantiates
Element Id | NutritionOrder.instantiates
Definition | The URL pointing to a protocol, guideline, orderset or other definition that is adhered to in whole or in part by this NutritionOrder.
Cardinality | 0..*
Type | uri
NutritionOrder.status
Element Id | NutritionOrder.status
Definition | The workflow status of the nutrition order/request.
Cardinality | 1..1
Terminology Binding | RequestStatus (Required)
Type | code
Is Modifier | true (Reason: This element is labeled as a modifier because it is a status element that contains status entered-in-error which means that the resource should not be treated as valid)
Summary | true
Comments | Typically the system placing the order sets the status to "requested". Thereafter, the order is maintained by the receiver that updates the status as the request is handled. This element is labeled as a modifier because the status contains codes that mark the resource as not currently valid.
NutritionOrder.intent
Element Id | NutritionOrder.intent
Definition | Indicates the level of authority/intentionality associated with the NutrionOrder and where the request fits into the workflow chain.
Cardinality | 1..1
Terminology Binding | RequestIntent (Required)
Type | code
Is Modifier | true (Reason: This element changes the interpretation of all descriptive attributes. For example "the time the request is recommended to occur" vs. "the time the request is authorized to occur" or "who is recommended to perform the request" vs. "who is authorized to perform the request)

Requirements | Proposals/recommendations, plans and orders all use the same structure and can exist in the same fulfillment chain.
Alternate Names | category
Summary | true
Comments | When resources map to this element, they are free to define as many codes as necessary to cover their space and will map to "proposal, plan or order". Can have multiple codes that map to one of these. E.g. "original order", "encoded order", "reflex order" would all map to "order". Expectation is that the set of codes is mutually exclusive or a strict all-encompassing hierarchy.
NutritionOrder.patient
Element Id | NutritionOrder.patient
Definition | The person (patient) who needs the nutrition order for an oral diet, nutritional supplement and/or enteral or formula feeding.
Cardinality | 1..1
Type | Reference(Patient)
Summary | true
NutritionOrder.encounter
Element Id | NutritionOrder.encounter
Definition | An encounter that provides additional information about the healthcare context in which this request is made.
Cardinality | 0..1
Type | Reference(Encounter)
NutritionOrder.dateTime
Element Id | NutritionOrder.dateTime
Definition | The date and time that this nutrition order was requested.
Cardinality | 1..1
Type | dateTime
Summary | true
NutritionOrder.orderer
Element Id | NutritionOrder.orderer
Definition | The practitioner that holds legal responsibility for ordering the diet, nutritional supplement, or formula feedings.
Cardinality | 0..1
Type | Reference(Practitioner | PractitionerRole)
Summary | true
NutritionOrder.allergyIntolerance
Element Id | NutritionOrder.allergyIntolerance
Definition | A link to a record of allergies or intolerances which should be included in the nutrition order.
Cardinality | 0..*
Type | Reference(AllergyIntolerance)
Comments | Information on a patient's food allergies and intolerances to inform healthcare personnel about the type of foods that the patient shouldn't receive or consume.
NutritionOrder.foodPreferenceModifier
Element Id | NutritionOrder.foodPreferenceModifier
Definition | This modifier is used to convey order-specific modifiers about the type of food that should be given. These can be derived from patient allergies, intolerances, or preferences such as Halal, Vegan or Kosher. This modifier applies to the entire nutrition order inclusive of the oral diet, nutritional supplements and enteral formula feedings.
Cardinality | 0..*
Terminology Binding | Diet (Example)
Type | CodeableConcept
Comments | Information on a patient's food preferences that inform healthcare personnel about the food that the patient should receive or consume.
NutritionOrder.excludeFoodModifier
Element Id | NutritionOrder.excludeFoodModifier
Definition | This modifier is used to convey Order-specific modifier about the type of oral food or oral fluids that should not be given. These can be derived from patient allergies, intolerances, or preferences such as No Red Meat, No Soy or No Wheat or Gluten-Free. While it should not be necessary to repeat allergy or intolerance information captured in the referenced AllergyIntolerance resource in the excludeFoodModifier, this element may be used to convey additional specificity related to foods that should be eliminated from the patient’s diet for any reason. This modifier applies to the entire nutrition order inclusive of the oral diet, nutritional supplements and enteral formula feedings.
Cardinality | 0..*
Terminology Binding | Food Type Codes (Example)
Type | CodeableConcept
Comments | Information on a patient's food allergies, intolerances and preferences to inform healthcare personnel about the type of foods that the patient shouldn't receive or consume.
NutritionOrder.oralDiet
Element Id | NutritionOrder.oralDiet
Definition | Diet given orally in contrast to enteral (tube) feeding.
Cardinality | 0..1
Invariants | Affect this element nor-1 Warning Nutrition Order SHALL contain either Oral Diet , Supplement, or Enteral Formula class oralDiet.exists() or supplement.exists() or enteralFormula.exists()
Affect this element
nor-1 | Warning | Nutrition Order SHALL contain either Oral Diet , Supplement, or Enteral Formula class | oralDiet.exists() or supplement.exists() or enteralFormula.exists()
NutritionOrder.oralDiet.type
Element Id | NutritionOrder.oralDiet.type
Definition | The kind of diet or dietary restriction such as fiber restricted diet or diabetic diet.
Cardinality | 0..*

Terminology Binding | Diet Codes (Example)
Type | CodeableConcept
Summary | true
NutritionOrder.oralDiet.schedule
Element Id | NutritionOrder.oralDiet.schedule
Definition | The time period and frequency at which the diet should be given. The diet should be given for the combination of all schedules if more than one schedule is present.
Cardinality | 0..*
Type | Timing
Alternate Names | Frequency
NutritionOrder.oralDiet.nutrient
Element Id | NutritionOrder.oralDiet.nutrient
Definition | Class that defines the quantity and type of nutrient modifications (for example carbohydrate, fiber or sodium) required for the oral diet.
Cardinality | 0..*
NutritionOrder.oralDiet.nutrient.modifier
Element Id | NutritionOrder.oralDiet.nutrient.modifier
Definition | The nutrient that is being modified such as carbohydrate or sodium.
Cardinality | 0..1
Terminology Binding | Nutrient Modifier Codes (Example)
Type | CodeableConcept
NutritionOrder.oralDiet.nutrient.amount
Element Id | NutritionOrder.oralDiet.nutrient.amount
Definition | The quantity of the specified nutrient to include in diet.
Cardinality | 0..1
Type | SimpleQuantity
NutritionOrder.oralDiet.texture
Element Id | NutritionOrder.oralDiet.texture
Definition | Class that describes any texture modifications required for the patient to safely consume various types of solid foods.
Cardinality | 0..*
NutritionOrder.oralDiet.texture.modifier
Element Id | NutritionOrder.oralDiet.texture.modifier
Definition | Any texture modifications (for solid foods) that should be made, e.g. easy to chew, chopped, ground, and pureed.
Cardinality | 0..1
Terminology Binding | Texture Modifier Codes (Example)
Type | CodeableConcept
Comments | Coupled with the foodType (Meat).
NutritionOrder.oralDiet.texture.foodType
Element Id | NutritionOrder.oralDiet.texture.foodType
Definition | The food type(s) (e.g. meats, all foods) that the texture modification applies to. This could be all foods types.
Cardinality | 0..1
Terminology Binding | Texture Modified Food Type Codes (Example)
Type | CodeableConcept
Comments | Coupled with the texture.modifier; could be (All Foods).
NutritionOrder.oralDiet.fluidConsistencyType
Element Id | NutritionOrder.oralDiet.fluidConsistencyType
Definition | The required consistency (e.g. honey-thick, nectar-thick, thin, thickened.) of liquids or fluids served to the patient.
Cardinality | 0..*
Terminology Binding | Fluid Consistency Type Codes (Example)
Type | CodeableConcept
NutritionOrder.oralDiet.instruction
Element Id | NutritionOrder.oralDiet.instruction
Definition | Free text or additional instructions or information pertaining to the oral diet.
Cardinality | 0..1
Type | string
Summary | true
Comments | Free text dosage instructions can be used for cases where the instructions are too complex to code.
NutritionOrder.supplement
Element Id | NutritionOrder.supplement
Definition | Oral nutritional products given in order to add further nutritional value to the patient's diet.
Cardinality | 0..*
Invariants | Affect this element nor-1 Warning Nutrition Order SHALL contain either Oral Diet , Supplement, or Enteral Formula class oralDiet.exists() or supplement.exists() or enteralFormula.exists()
Affect this element
nor-1 | Warning | Nutrition Order SHALL contain either Oral Diet , Supplement, or Enteral Formula class | oralDiet.exists() or supplement.exists() or enteralFormula.exists()
NutritionOrder.supplement.type
Element Id | NutritionOrder.supplement.type
Definition | The kind of nutritional supplement product required such as a high protein or pediatric clear liquid supplement.
Cardinality | 0..1
Terminology Binding | Supplement Type Codes (Example)
Type | CodeableConcept
Summary | true

NutritionOrder.supplement.productName
Element Id | NutritionOrder.supplement.productName
Definition | The product or brand name of the nutritional supplement such as "Acme Protein Shake".
Cardinality | 0..1
Type | string
NutritionOrder.supplement.schedule
Element Id | NutritionOrder.supplement.schedule
Definition | The time period and frequency at which the supplement(s) should be given. The supplement should be given for the combination of all schedules if more than one schedule is present.
Cardinality | 0..*
Type | Timing
Alternate Names | Frequency
NutritionOrder.supplement.quantity
Element Id | NutritionOrder.supplement.quantity
Definition | The amount of the nutritional supplement to be given.
Cardinality | 0..1
Type | SimpleQuantity
NutritionOrder.supplement.instruction
Element Id | NutritionOrder.supplement.instruction
Definition | Free text or additional instructions or information pertaining to the oral supplement.
Cardinality | 0..1
Type | string
Summary | true
Comments | Free text dosage instructions can be used for cases where the instructions are too complex to code.
NutritionOrder.enteralFormula
Element Id | NutritionOrder.enteralFormula
Definition | Feeding provided through the gastrointestinal tract via a tube, catheter, or stoma that delivers nutrition distal to the oral cavity.
Cardinality | 0..1
Invariants | Affect this element nor-1 Warning Nutrition Order SHALL contain either Oral Diet , Supplement, or Enteral Formula class oralDiet.exists() or supplement.exists() or enteralFormula.exists()
Affect this element
nor-1 | Warning | Nutrition Order SHALL contain either Oral Diet , Supplement, or Enteral Formula class | oralDiet.exists() or supplement.exists() or enteralFormula.exists()
NutritionOrder.enteralFormula.baseFormulaType
Element Id | NutritionOrder.enteralFormula.baseFormulaType
Definition | The type of enteral or infant formula such as an adult standard formula with fiber or a soy-based infant formula.
Cardinality | 0..1
Terminology Binding | Enteral Formula Type Codes (Example)
Type | CodeableConcept
Summary | true
NutritionOrder.enteralFormula.baseFormulaProductName
Element Id | NutritionOrder.enteralFormula.baseFormulaProductName
Definition | The product or brand name of the enteral or infant formula product such as "ACME Adult Standard Formula".
Cardinality | 0..1
Type | string
NutritionOrder.enteralFormula.additiveType
Element Id | NutritionOrder.enteralFormula.additiveType
Definition | Indicates the type of modular component such as protein, carbohydrate, fat or fiber to be provided in addition to or mixed with the base formula.
Cardinality | 0..1
Terminology Binding | Enteral Formula Additive Type Code (Example)
Type | CodeableConcept
NutritionOrder.enteralFormula.additiveProductName
Element Id | NutritionOrder.enteralFormula.additiveProductName
Definition | The product or brand name of the type of modular component to be added to the formula.
Cardinality | 0..1
Type | string
NutritionOrder.enteralFormula.caloricDensity
Element Id | NutritionOrder.enteralFormula.caloricDensity
Definition | The amount of energy (calories) that the formula should provide per specified volume, typically per mL or fluid oz. For example, an infant may require a formula that provides 24 calories per fluid ounce or an adult may require an enteral formula that provides 1.5 calorie/mL.
Cardinality | 0..1
Type | SimpleQuantity
NutritionOrder.enteralFormula.routeofAdministration
Element Id | NutritionOrder.enteralFormula.routeofAdministration
Definition | The route or physiological path of administration into the patient's gastrointestinal tract for purposes of providing the formula feeding, e.g. nasogastric tube.
Cardinality | 0..1
Terminology Binding | Enteral Route Codes (Extensible)
Type | CodeableConcept
NutritionOrder.enteralFormula.administration

Element Id | NutritionOrder.enteralFormula.administration
Definition | Formula administration instructions as structured data. This repeating structure allows for changing the administration rate or volume over time for both bolus and continuous feeding. An example of this would be an instruction to increase the rate of continuous feeding every 2 hours.
Cardinality | 0..*
Comments | See implementation notes below for further discussion on how to order continuous vs bolus enteral feeding using this resource.
NutritionOrder.enteralFormula.administration.schedule
Element Id | NutritionOrder.enteralFormula.administration.schedule
Definition | The time period and frequency at which the enteral formula should be delivered to the patient.
Cardinality | 0..1
Type | Timing
Alternate Names | Frequency
NutritionOrder.enteralFormula.administration.quantity
Element Id | NutritionOrder.enteralFormula.administration.quantity
Definition | The volume of formula to provide to the patient per the specified administration schedule.
Cardinality | 0..1
Type | SimpleQuantity
NutritionOrder.enteralFormula.administration.rate[x]
Element Id | NutritionOrder.enteralFormula.administration.rate[x]
Definition | The rate of administration of formula via a feeding pump, e.g. 60 mL per hour, according to the specified schedule.
Cardinality | 0..1
Type | SimpleQuantity|Ratio
[x] Note | See Choice of Data Types for further information about how to use [x]
Comments | Ratio is used when the quantity value in the denominator is not "1", otherwise use Quantity. For example, the Ratio datatype is used for "200 mL/4 hrs" versus the Quantity datatype for "50 mL/hr".
NutritionOrder.enteralFormula.maxVolumeToDeliver
Element Id | NutritionOrder.enteralFormula.maxVolumeToDeliver
Definition | The maximum total quantity of formula that may be administered to a subject over the period of time, e.g. 1440 mL over 24 hours.
Cardinality | 0..1
Type | SimpleQuantity
NutritionOrder.enteralFormula.administrationInstruction
Element Id | NutritionOrder.enteralFormula.administrationInstruction
Definition | Free text formula administration, feeding instructions or additional instructions or information.
Cardinality | 0..1
Type | string
Summary | true
Comments | Free text dosage instructions can be used for cases where the instructions are too complex to code.
NutritionOrder.note
Element Id | NutritionOrder.note
Definition | Comments made about the {{title}} by the requester, performer, subject or other participants.
Cardinality | 0..*
Type | Annotation
Comments | This element SHALL NOT be used to supply free text instructions for the diet which are represented in the .oralDiet.instruction, supplement.instruction, or enteralFormula.administrationInstruction elements.

12.15.8 Resource NutritionOrder - Mappings

URL: https://hl7.org/fhir/R4/nutritionorder-mappings.html

12.15.8 Resource NutritionOrder - Mappings

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Encounter, Patient, Practitioner

Mappings:

FiveWs Pattern Mapping

Workflow Pattern

HL7 v2 Mapping

RIM Mapping

Mappings for the nutritionorder resource (see Mappings to Other Standards for further information & status).

12.15.8.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

NutritionOrder
identifier | FiveWs.identifier
status | FiveWs.status
intent | FiveWs.class
patient | FiveWs.subject[x]
encounter | FiveWs.context
dateTime | FiveWs.recorded
orderer | FiveWs.author

type | FiveWs.what[x]
modifier | FiveWs.what[x]
type | FiveWs.what[x]
baseFormulaType | FiveWs.what[x]
additiveType | FiveWs.what[x]

12.15.8.2 Workflow Pattern (http://hl7.org/fhir/workflow)

NutritionOrder | Request
identifier | Request.identifier
instantiatesCanonical | Request.instantiatesCanonical
instantiatesUri | Request.instantiatesUri
instantiates | Request.instantiates
status | Request.status
intent | Request.intent
patient | Request.subject
encounter | Request.encounter
dateTime | Request.authoredOn
orderer | Request.requester
type | Request.code
schedule | Request.occurrence[x]
modifier | Request.code
modifier | Request.code
foodType | Request.code
fluidConsistencyType | Request.code
instruction | Request.supportingInfo
type | Request.code
schedule | Request.occurrence[x]
instruction | Request.supportingInfo
baseFormulaType | Request.code
additiveType | Request.code
schedule | Request.occurrence[x]
administrationInstruction | Request.supportingInfo
note | Request.note

12.15.8.3 HL7 v2 Mapping (http://hl7.org/v2)

NutritionOrder | ORC / ODS
identifier | ORC-2/ORC-3
instantiatesCanonical | Varies by domain
instantiatesUri | Varies by domain
instantiates | Varies by domain
status | ORC-1
intent | N/A
patient | PID
encounter | PV1
dateTime | ORC-9
orderer | ORC-12
allergyIntolerance
foodPreferenceModifier | ODS-1 Value = "P",ODS-3
excludeFoodModifier
oralDiet | ODS-1 Value = "D"
type | ODS-3
schedule | ODS-2, TQ1-7/8
nutrient | ODS-4
modifier | ODS-4
amount | ODS-4
texture | ODS-4
modifier | ODS-4
foodType | ODS-4
fluidConsistencyType | ODS-4
instruction | ODS-4
supplement | ODS-1 Value = "P"
type | ODS-3
productName | ODS-3
schedule | ODS-2,TQ1-7/8
quantity | TQ1-2
instruction | ODS-4
enteralFormula | ODS-1 Value coded for enteral
baseFormulaType | ODS-3
baseFormulaProductName | ODS-4
additiveType | ODS-3
additiveProductName | ODS-4

caloricDensity | TQ1-2
routeofAdministration | ODS-4
administration | ODS-4
schedule | ODS-2,TQ1-7/8
quantity | TQ1-2
rate[x] | ODS-4
maxVolumeToDeliver | ODS-4
administrationInstruction | ODS-4
note

12.15.8.4 RIM Mapping (http://hl7.org/v3)

NutritionOrder | SupplyRequest
identifier | .id
instantiatesCanonical | .outboundRelationship[typeCode=DEFN].target
instantiatesUri | .outboundRelationship[typeCode=DEFN].target
instantiates | .outboundRelationship[typeCode=DEFN].target
status | .statusCode
intent | .moodCode (nuances beyond PRP/PLAN/RQO would need to be elsewhere)
patient | .participation[typeCode=SBJ].role
encounter | .inboundRelationship[typeCode=COMP].source[classCode=ENC, moodCode=EVN]
dateTime | author.time
orderer | .participation[typeCode=AUT].role
allergyIntolerance | .outboundRelationship[typeCode=RSON].target[classCode=OBS, moodCode=EVN]
foodPreferenceModifier | .outboundRelationship [typeCode =COMP], target [classCode=PROC, moodCode=EVN, code=NutritionModificationType]
excludeFoodModifier | N/A
oralDiet | oralDiet .outboundRelationship [typeCode =COMP], target [classCode=SPLY, moodCode=RQO, code=OralDietSupplyType]
type | .code
schedule | .expectedUseTime
nutrient | .participation[typeCode=PRD].role[classCode=MANU], manufacturedFoodKind
modifier | .code
amount | .quantity
texture | .inboundRelationship [typeCode =COMP], target [classCode=PROC, moodCode=EVN, code=DietTextureModificationType]
modifier | .code
foodType | .participation[typeCode=PRD].role[classCode=MANU], manufacturedFoodKind; .code
fluidConsistencyType | .code (Not differentiated from dietTextureModificationType in HL7 v3)
instruction | .text
supplement | .outboundRelationship [typeCode =COMP], target [classCode=SPLY, moodCode=RQO, code=SUPPLEMENT]
type | .participation[typeCode=PRD].role[classCode=MANU], manufacturedFoodKind; .code (NutritionalSupplementType)
productName | .id
schedule | .effectiveTime
quantity | .quantity
instruction | .text
enteralFormula | .outboundRelationship [typeCode =COMP], target [classCode=SPLY, moodCode=RQO, code=FORMULA]
baseFormulaType | .participation[typeCode=PRD].role[classCode=MANU], manufacturedFoodKind; .code (NutritionalFormulaType)
baseFormulaProductName | .id
additiveType | N/A (not differentiated from NutritionalFormulaType in HL7 v3)
additiveProductName | N/A
caloricDensity | .quantity (caloricDensity)
routeofAdministration | .routeCode
administration | .outboundRelationship [typeCode =COMP], target [classCode=SBADM, moodCode=RQO]
schedule | .effectiveUseTime
quantity | .doseQuantity
rate[x] | .rateQuantity
maxVolumeToDeliver | .maxDoseQuantity
administrationInstruction | N/A

12.15.9 Resource NutritionOrder - Extensions & Profiles

URL: https://hl7.org/fhir/R4/nutritionorder-profiles.html

12.15.9 Resource NutritionOrder - Extensions & Profiles

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Encounter, Patient, Practitioner

This table lists profiles and extensions for the NutritionOrder resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

12.15.9.1 Profiles

No Profiles defined for this resource

12.15.9.2 Extensions

nutritionorder-adaptiveFeedingDevice | NutritionOrder.oralDiet | adaptiveFeedingDevice | for NutritionOrder HL7 Extensions
request-doNotPerform | NutritionOrder | doNotPerform | for Request Pattern HL7 Extensions
request-insurance | NutritionOrder | insurance | for Request Pattern HL7 Extensions
request-relevantHistory | NutritionOrder | relevantHistory | for Request Pattern HL7 Extensions
request-replaces | NutritionOrder | replaces | for Request Pattern HL7 Extensions
request-statusReason | NutritionOrder | statusReason | for Request Pattern HL7 Extensions
workflow-episodeOfCare | NutritionOrder | episodeOfCare | for Workflow Pattern HL7 Extensions
workflow-reasonCode | NutritionOrder | reasonCode | for Workflow Pattern HL7 Extensions
workflow-reasonReference | NutritionOrder | reasonReference | for Workflow Pattern HL7 Extensions
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

12.15.9.3 Search Extensions

No Search Extensions defined for this resource

12.15.10 Resource NutritionOrder - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/nutritionorder-version-maps.html

No validation errors - all conversions are clean

12.15.10 Resource NutritionOrder - R3/R4 Conversions

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Encounter, Patient, Practitioner

R3 : R4 Conversion maps for NutritionOrder.

Functional status for this map: 13 tests that all execute ok. All tests pass round-trip testing and 13 r3 resources are invalid (0 errors). (see documentation)

12.15.10.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/NutritionOrder3to4" = "R3 to R4 Conversions for NutritionOrder"

uses "http://hl7.org/fhir/3.0/StructureDefinition/NutritionOrder" alias NutritionOrderR3 as source
uses "http://hl7.org/fhir/StructureDefinition/NutritionOrder" alias NutritionOrder as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group NutritionOrder(source src : NutritionOrderR3, target tgt : NutritionOrder) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.status -> tgt.status;
  src.patient -> tgt.patient;
  src.encounter -> tgt.encounter;
  src.dateTime -> tgt.dateTime;
  src.orderer -> tgt.orderer;
  src.allergyIntolerance -> tgt.allergyIntolerance;
  src.foodPreferenceModifier -> tgt.foodPreferenceModifier;
  src.excludeFoodModifier -> tgt.excludeFoodModifier;
  src.oralDiet as s -> tgt.oralDiet as t then oralDiet(s, t);
  src.supplement as s -> tgt.supplement as t then supplement(s, t);
  src.enteralFormula as s -> tgt.enteralFormula as t then enteralFormula(s, t);
}

group oralDiet(source src, target tgt) extends BackboneElement {
  src.type -> tgt.type;
  src.schedule -> tgt.schedule;
  src.nutrient as s -> tgt.nutrient as t then nutrient(s, t);
  src.texture as s -> tgt.texture as t then texture(s, t);
  src.fluidConsistencyType -> tgt.fluidConsistencyType;
  src.instruction -> tgt.instruction;
}

group nutrient(source src, target tgt) extends BackboneElement {
  src.modifier -> tgt.modifier;
  src.amount -> tgt.amount;
```

```
}

group texture(source src, target tgt) extends BackboneElement {
  src.modifier -> tgt.modifier;
  src.foodType -> tgt.foodType;
}

group supplement(source src, target tgt) extends BackboneElement {
  src.type -> tgt.type;
  src.productName -> tgt.productName;
  src.schedule -> tgt.schedule;
  src.quantity -> tgt.quantity;
  src.instruction -> tgt.instruction;
}

group enteralFormula(source src, target tgt) extends BackboneElement {
  src.baseFormulaType -> tgt.baseFormulaType;
  src.baseFormulaProductName -> tgt.baseFormulaProductName;
  src.additiveType -> tgt.additiveType;
  src.additiveProductName -> tgt.additiveProductName;
  src.caloricDensity -> tgt.caloricDensity;
  src.routeofAdministration -> tgt.routeofAdministration;
  src.administration as s -> tgt.administration as t then administration(s, t);
  src.maxVolumeToDeliver -> tgt.maxVolumeToDeliver;
  src.administrationInstruction -> tgt.administrationInstruction;
}

group administration(source src, target tgt) extends BackboneElement {
  src.schedule -> tgt.schedule;
  src.quantity -> tgt.quantity;
  src.rate : Quantity as vs -> tgt.rate = create('Quantity') as vt then Quantity(vs, vt);
  src.rate : Ratio as vs -> tgt.rate = create('Ratio') as vt then Ratio(vs, vt);
}
```

12.15.10.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/NutritionOrder4to3" = "R4 to R3 Conversion for NutritionOrder"

uses "http://hl7.org/fhir/StructureDefinition/NutritionOrder" alias NutritionOrder as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/NutritionOrder" alias NutritionOrderR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group NutritionOrder(source src : NutritionOrderR3, target tgt : NutritionOrder) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.status -> tgt.status;
  src.patient -> tgt.patient;
  src.encounter -> tgt.encounter;
  src.dateTime -> tgt.dateTime;
  src.orderer -> tgt.orderer;
  src.allergyIntolerance -> tgt.allergyIntolerance;
  src.foodPreferenceModifier -> tgt.foodPreferenceModifier;
  src.excludeFoodModifier -> tgt.excludeFoodModifier;
  src.oralDiet as s -> tgt.oralDiet as t then oralDiet(s, t);
  src.supplement as s -> tgt.supplement as t then supplement(s, t);
  src.enteralFormula as s -> tgt.enteralFormula as t then enteralFormula(s, t);
}

group oralDiet(source src, target tgt) extends BackboneElement {
  src.type -> tgt.type;
  src.schedule -> tgt.schedule;
  src.nutrient as s -> tgt.nutrient as t then nutrient(s, t);
  src.texture as s -> tgt.texture as t then texture(s, t);
  src.fluidConsistencyType -> tgt.fluidConsistencyType;
  src.instruction -> tgt.instruction;
}

group nutrient(source src, target tgt) extends BackboneElement {
  src.modifier -> tgt.modifier;
  src.amount -> tgt.amount;
}

group texture(source src, target tgt) extends BackboneElement {
  src.modifier -> tgt.modifier;
  src.foodType -> tgt.foodType;
}

group supplement(source src, target tgt) extends BackboneElement {
  src.type -> tgt.type;
  src.productName -> tgt.productName;
  src.schedule -> tgt.schedule;
  src.quantity -> tgt.quantity;
  src.instruction -> tgt.instruction;
}

group enteralFormula(source src, target tgt) extends BackboneElement {
  src.baseFormulaType -> tgt.baseFormulaType;
  src.baseFormulaProductName -> tgt.baseFormulaProductName;
  src.additiveType -> tgt.additiveType;
  src.additiveProductName -> tgt.additiveProductName;
```

```
  src.caloricDensity -> tgt.caloricDensity;
  src.routeofAdministration -> tgt.routeofAdministration;
  src.administration as s -> tgt.administration as t then administration(s, t);
  src.maxVolumeToDeliver -> tgt.maxVolumeToDeliver;
  src.administrationInstruction -> tgt.administrationInstruction;
}

group administration(source src, target tgt) extends BackboneElement {
  src.schedule -> tgt.schedule;
  src.quantity -> tgt.quantity;
  src.rate : Quantity as vs -> tgt.rate = create('Quantity') as vt then Quantity(vs, vt);
  src.rate : Ratio as vs -> tgt.rate = create('Ratio') as vt then Ratio(vs, vt);
}
```

12.15.10.3 R4 Validation Errors

12.15.11 NutritionOrder HL7 Extensions

URL: https://hl7.org/fhir/R4/nutritionorder-extensions.html

Defines common extensions used with or related to the NutritionOrder resource

12.15.11 NutritionOrder HL7 Extensions

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative

12.15.11.1 Content

Extensions:

nutritionorder-adaptiveFeedingDevice | adaptiveFeedingDevice : Materials used or needed to feed the patient.

12.15.12 JSON Schema for NutritionOrder

URL: https://hl7.org/fhir/R4/nutritionorder.schema.json.html

12.15.12 JSON Schema for NutritionOrder

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Encounter, Patient, Practitioner

This is a representation of the json schema for NutritionOrder, which is just a part of the full JSON Schema.

```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/NutritionOrder",
  "$ref": "#/definitions/NutritionOrder",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "NutritionOrder": {
      "description": "A request to supply a diet, formula feeding (enteral) or oral nutritional supplement to a
    patient/resident.",
      "properties": {
        "resourceType": {
          "description": "This is a NutritionOrder resource",
          "const": "NutritionOrder"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
          "$ref": "#/definitions/code"
        },
        "_language": {
          "description": "Extensions for language",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "text": {
          "description": "A human-readable narrative that contains a summary of the resource and can be used to
    represent the content of the resource to a human. The narrative need not encode all the structured data, but is
```

```
required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative. Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
      "$ref": "Narrative.schema.json#/definitions/Narrative"
    },
    "contained": {
      "description": "These resources do not have an independent existence apart from the resource that contains them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
      "items": {
        "$ref": "ResourceList.schema.json#/definitions/ResourceList"
      },
      "type": "array"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the resource and that modifies the understanding of the element that contains it and/or the understanding of the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension. Applications processing a resource are required to check for modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "identifier": {
      "description": "Identifiers assigned to this order by the order sender or by the order receiver.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "instantiatesCanonical": {
      "description": "The URL pointing to a FHIR-defined protocol, guideline, orderset or other definition that is adhered to in whole or in part by this NutritionOrder.",
      "items": {
        "$ref": "canonical.schema.json#/definitions/canonical"
      },
      "type": "array"
    },
    "instantiatesUri": {
      "description": "The URL pointing to an externally maintained protocol, guideline, orderset or other definition that is adhered to in whole or in part by this NutritionOrder.",
      "items": {
        "$ref": "#/definitions/uri"
      },
      "type": "array"
    },
    "_instantiatesUri": {
      "description": "Extensions for instantiatesUri",
      "items": {
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "type": "array"
    },
    "instantiates": {
      "description": "The URL pointing to a protocol, guideline, orderset or other definition that is adhered to in whole or in part by this NutritionOrder.",
      "items": {
        "$ref": "#/definitions/uri"
      },
      "type": "array"
    },
    "_instantiates": {
      "description": "Extensions for instantiates",
      "items": {
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "type": "array"
    },
    "status": {
      "description": "The workflow status of the nutrition order/request.",
      "$ref": "#/definitions/code"
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
```

```
    "intent": {
      "description": "Indicates the level of authority/intentionality associated with the NutrionOrder and where the
 request fits into the workflow chain.",
      "$ref": "#/definitions/code"
    },
    "_intent": {
      "description": "Extensions for intent",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "patient": {
      "description": "The person (patient) who needs the nutrition order for an oral diet, nutritional supplement
and/or enteral or formula feeding.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "encounter": {
      "description": "An encounter that provides additional information about the healthcare context in which this
request is made.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "dateTime": {
      "description": "The date and time that this nutrition order was requested.",
      "$ref": "#/definitions/dateTime"
    },
    "_dateTime": {
      "description": "Extensions for dateTime",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "orderer": {
      "description": "The practitioner that holds legal responsibility for ordering the diet, nutritional
supplement, or formula feedings.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "allergyIntolerance": {
      "description": "A link to a record of allergies or intolerances  which should be included in the nutrition
order.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "foodPreferenceModifier": {
      "description": "This modifier is used to convey order-specific modifiers about the type of food that should be
 given. These can be derived from patient allergies, intolerances, or preferences such as Halal, Vegan or Kosher.
This modifier applies to the entire nutrition order inclusive of the oral diet, nutritional supplements and enteral
formula feedings.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "excludeFoodModifier": {
      "description": "This modifier is used to convey Order-specific modifier about the type of oral food or oral
fluids that should not be given. These can be derived from patient allergies, intolerances, or preferences such as
No Red Meat, No Soy or No Wheat or  Gluten-Free.  While it should not be necessary to repeat allergy or intolerance
information captured in the referenced AllergyIntolerance resource in the excludeFoodModifier, this element may be
used to convey additional specificity related to foods that should be eliminated from the patient’s diet for any
reason.  This modifier applies to the entire nutrition order inclusive of the oral diet, nutritional supplements and
 enteral formula feedings.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "oralDiet": {
      "description": "Diet given orally in contrast to enteral (tube) feeding.",
      "$ref": "#/definitions/NutritionOrder_OralDiet"
    },
    "supplement": {
      "description": "Oral nutritional products given in order to add further nutritional value to the
patient\u0027s diet.",
      "items": {
        "$ref": "#/definitions/NutritionOrder_Supplement"
      },
      "type": "array"
    },
    "enteralFormula": {
      "description": "Feeding provided through the gastrointestinal tract via a tube, catheter, or stoma that
delivers nutrition distal to the oral cavity.",
      "$ref": "#/definitions/NutritionOrder_EnteralFormula"
    },
    "note": {
      "description": "Comments made about the {{title}} by the requester, performer, subject or other
participants.",
      "items": {
        "$ref": "Annotation.schema.json#/definitions/Annotation"
      },
      "type": "array"
    }
  },
```

```
  "additionalProperties": false,
  "required": [
    "patient",
    "resourceType"
  ]
},
"NutritionOrder_OralDiet": {
  "description": "A request to supply a diet, formula feeding (enteral) or oral nutritional supplement to a
patient/resident.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "type": {
      "description": "The kind of diet or dietary restriction such as fiber restricted diet or diabetic diet.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "schedule": {
      "description": "The time period and frequency at which the diet should be given.  The diet should be given for
 the combination of all schedules if more than one schedule is present.",
      "items": {
        "$ref": "Timing.schema.json#/definitions/Timing"
      },
      "type": "array"
    },
    "nutrient": {
      "description": "Class that defines the quantity and type of nutrient modifications (for example carbohydrate,
fiber or sodium) required for the oral diet.",
      "items": {
        "$ref": "#/definitions/NutritionOrder_Nutrient"
      },
      "type": "array"
    },
    "texture": {
      "description": "Class that describes any texture modifications required for the patient to safely consume
various types of solid foods.",
      "items": {
        "$ref": "#/definitions/NutritionOrder_Texture"
      },
      "type": "array"
    },
    "fluidConsistencyType": {
      "description": "The required consistency (e.g. honey-thick, nectar-thick, thin, thickened.) of liquids or
fluids served to the patient.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "instruction": {
      "description": "Free text or additional instructions or information pertaining to the oral diet.",
      "$ref": "#/definitions/string"
    },
    "_instruction": {
      "description": "Extensions for instruction",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
  "additionalProperties": false
},
```

```
"NutritionOrder_Nutrient": {
  "description": "A request to supply a diet, formula feeding (enteral) or oral nutritional supplement to a
patient/resident.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifier": {
      "description": "The nutrient that is being modified such as carbohydrate or sodium.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "amount": {
      "description": "The quantity of the specified nutrient to include in diet.",
      "$ref": "Quantity.schema.json#/definitions/Quantity"
    }
  },
  "additionalProperties": false
},
"NutritionOrder_Texture": {
  "description": "A request to supply a diet, formula feeding (enteral) or oral nutritional supplement to a
patient/resident.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifier": {
      "description": "Any texture modifications (for solid foods) that should be made, e.g. easy to chew, chopped,
ground, and pureed.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "foodType": {
      "description": "The food type(s) (e.g. meats, all foods)  that the texture modification applies to.  This
could be all foods types.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    }
  },
  "additionalProperties": false
```

```
},
"NutritionOrder_Supplement": {
  "description": "A request to supply a diet, formula feeding (enteral) or oral nutritional supplement to a
patient/resident.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "type": {
      "description": "The kind of nutritional supplement product required such as a high protein or pediatric clear
liquid supplement.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "productName": {
      "description": "The product or brand name of the nutritional supplement such as \"Acme Protein Shake\".",
      "$ref": "#/definitions/string"
    },
    "_productName": {
      "description": "Extensions for productName",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "schedule": {
      "description": "The time period and frequency at which the supplement(s) should be given.  The supplement
should be given for the combination of all schedules if more than one schedule is present.",
      "items": {
        "$ref": "Timing.schema.json#/definitions/Timing"
      },
      "type": "array"
    },
    "quantity": {
      "description": "The amount of the nutritional supplement to be given.",
      "$ref": "Quantity.schema.json#/definitions/Quantity"
    },
    "instruction": {
      "description": "Free text or additional instructions or information pertaining to the oral supplement.",
      "$ref": "#/definitions/string"
    },
    "_instruction": {
      "description": "Extensions for instruction",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
  "additionalProperties": false
},
"NutritionOrder_EnteralFormula": {
  "description": "A request to supply a diet, formula feeding (enteral) or oral nutritional supplement to a
patient/resident.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
```

```
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "baseFormulaType": {
      "description": "The type of enteral or infant formula such as an adult standard formula with fiber or a soy-
based infant formula.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "baseFormulaProductName": {
      "description": "The product or brand name of the enteral or infant formula product such as \"ACME Adult
Standard Formula\".",
      "$ref": "#/definitions/string"
    },
    "_baseFormulaProductName": {
      "description": "Extensions for baseFormulaProductName",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "additiveType": {
      "description": "Indicates the type of modular component such as protein, carbohydrate, fat or fiber to be
provided in addition to or mixed with the base formula.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "additiveProductName": {
      "description": "The product or brand name of the type of modular component to be added to the formula.",
      "$ref": "#/definitions/string"
    },
    "_additiveProductName": {
      "description": "Extensions for additiveProductName",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "caloricDensity": {
      "description": "The amount of energy (calories) that the formula should provide per specified volume,
typically per mL or fluid oz.  For example, an infant may require a formula that provides 24 calories per fluid
ounce or an adult may require an enteral formula that provides 1.5 calorie/mL.",
      "$ref": "Quantity.schema.json#/definitions/Quantity"
    },
    "routeofAdministration": {
      "description": "The route or physiological path of administration into the patient\u0027s gastrointestinal
tract for purposes of providing the formula feeding, e.g. nasogastric tube.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "administration": {
      "description": "Formula administration instructions as structured data.  This repeating structure allows for
changing the administration rate or volume over time for both bolus and continuous feeding.  An example of this
would be an instruction to increase the rate of continuous feeding every 2 hours.",
      "items": {
        "$ref": "#/definitions/NutritionOrder_Administration"
      },
      "type": "array"
    },
    "maxVolumeToDeliver": {
      "description": "The maximum total quantity of formula that may be administered to a subject over the period of
 time, e.g. 1440 mL over 24 hours.",
      "$ref": "Quantity.schema.json#/definitions/Quantity"
    },
    "administrationInstruction": {
      "description": "Free text formula administration, feeding instructions or additional instructions or
information.",
      "$ref": "#/definitions/string"
    },
    "_administrationInstruction": {
      "description": "Extensions for administrationInstruction",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
  "additionalProperties": false
},
"NutritionOrder_Administration": {
  "description": "A request to supply a diet, formula feeding (enteral) or oral nutritional supplement to a
patient/resident.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
```

```
    the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
    definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
    that SHALL be met as part of the definition of the extension.",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "modifierExtension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the element and that modifies the understanding of the element in which it is contained and/or the understanding of
    the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
    use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
    extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
     of the definition of the extension. Applications processing a resource are required to check for modifier
    extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
    (including cannot change the meaning of modifierExtension itself).",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "schedule": {
          "description": "The time period and frequency at which the enteral formula should be delivered to the
    patient.",
          "$ref": "Timing.schema.json#/definitions/Timing"
        },
        "quantity": {
          "description": "The volume of formula to provide to the patient per the specified administration schedule.",
          "$ref": "Quantity.schema.json#/definitions/Quantity"
        },
        "rateQuantity": {
          "description": "The rate of administration of formula via a feeding pump, e.g. 60 mL per hour, according to
    the specified schedule.",
          "$ref": "Quantity.schema.json#/definitions/Quantity"
        },
        "rateRatio": {
          "description": "The rate of administration of formula via a feeding pump, e.g. 60 mL per hour, according to
    the specified schedule.",
          "$ref": "Ratio.schema.json#/definitions/Ratio"
        }
      },
      "additionalProperties": false
    }
  }
}
```


12.15.13 Resource NutritionOrder - Maturity Dependency View

URL: https://hl7.org/fhir/R4/nutritionorder-dependencies.html

12.15.13 Resource NutritionOrder - Maturity Dependency View

Orders and Observations Work Group | Maturity Level: 2 | Trial Use | Compartments: Encounter, Patient, Practitioner

Dependency Graph for NutritionOrder FMM level 2

Name | Card. | Type | Dependency Analysis
NutritionOrder | | DomainResource
identifier | 0..* | Identifier
instantiatesCanonical | 0..* | canonical(ActivityDefinition | PlanDefinition)
instantiatesUri | 0..* | uri
instantiates | 0..* | uri
status | 1..1 | code
intent | 1..1 | code
patient | 1..1 | Reference(Patient)
encounter | 0..1 | Reference(Encounter)
dateTime | 1..1 | dateTime
orderer | 0..1 | Reference(Practitioner | PractitionerRole)
allergyIntolerance | 0..* | Reference(AllergyIntolerance)
foodPreferenceModifier | 0..* | CodeableConcept
excludeFoodModifier | 0..* | CodeableConcept
oralDiet | 0..1 | BackboneElement
type | 0..* | CodeableConcept
schedule | 0..* | Timing
nutrient | 0..* | BackboneElement
modifier | 0..1 | CodeableConcept
amount | 0..1 | SimpleQuantity
texture | 0..* | BackboneElement
modifier | 0..1 | CodeableConcept
foodType | 0..1 | CodeableConcept
fluidConsistencyType | 0..* | CodeableConcept

instruction | 0..1 | string
supplement | 0..* | BackboneElement
type | 0..1 | CodeableConcept
productName | 0..1 | string
schedule | 0..* | Timing
quantity | 0..1 | SimpleQuantity
instruction | 0..1 | string
enteralFormula | 0..1 | BackboneElement
baseFormulaType | 0..1 | CodeableConcept
baseFormulaProductName | 0..1 | string
additiveType | 0..1 | CodeableConcept
additiveProductName | 0..1 | string
caloricDensity | 0..1 | SimpleQuantity
routeofAdministration | 0..1 | CodeableConcept
administration | 0..* | BackboneElement
schedule | 0..1 | Timing
quantity | 0..1 | SimpleQuantity
rate[x] | 0..1
rateQuantity | | SimpleQuantity
rateRatio | | Ratio
maxVolumeToDeliver | 0..1 | SimpleQuantity
administrationInstruction | 0..1 | string
note | 0..* | Annotation

12.16 Resource VisionPrescription - Content

URL: https://hl7.org/fhir/R4/visionprescription.html
Financial Management Work Group | Maturity Level: 2 | Trial Use | Security Category: Patient | Compartments: Encounter, Patient, Practitioner

An authorization for the provision of glasses and/or contact lenses to a patient.

12.16.1 Scope and Usage

The VisionPrescription resource is intended to support the information requirements for a prescription for glasses and contact lenses for a patient. Corrective optical lenses are considered a controlled substance and therefore a prescription is typically required for the provision of patient-specific lenses.

Vision prescriptions may only be issued by providers who have optical diagnosis within their scope of practice, typically ophthalmologists and optometrists. The prescription is a time-limited, typically two years, authorization for the provision of an unlimited number of lenses, glasses or contacts, which conform to the authorized specification by providers which have lens provision within their scope of practice, typically ophthalmologists , optometrists and opticians.

Health insurers typically require a copy of the prescription to ensure that a vision professional has examined the patient and that the power of the lenses exceeds some minimum, such as +- 0.75 diopters, to qualify for reimbursement under the insurance coverage.

The VisionPrescription resource is an "request" resource from a FHIR workflow perspective - see Workflow Request.

12.16.1.1 Additional Information

Additional information regarding resources conveying electronic claims content and usage may be found at:

• Financial Resource Status Lifecycle: how .status is used in the financial resources.

12.16.2 Boundaries and Relationships

The VisionPrescription resource is used to provide the authorized specification of lenses for glasses and contacts which have well known and internationally consistent requirements, which is distinct from other types of health care authorizations detailed below.

The health care domain includes a number of related resources

VisionPrescription | The VisionPrescription resource is used to provide the authorized specification of lenses for glasses and contacts.
DeviceRequest | This resource is used to prescribe, authorize, and request provision of a health care device such as a halter monitor or a wheelchair services.
MedicationRequest | This resource is used to prescribe, authorize, and request provision of medications such a pharmaceuticals and vaccines.
ServiceRequest | This resource is used to prescribe, authorize, and request provision of health care services such as consultative or rehabilitative services.

This resource is referenced by CarePlan, Claim and ExplanationOfBenefit

12.16.3 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
VisionPrescription | TU | | DomainResource | Prescription for vision correction products for a patient Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | | 0..* | Identifier | Business Identifier for vision prescription
status | ?!Σ | 1..1 | code | active | cancelled | draft | entered-in-error Financial Resource Status Codes (Required)
created | Σ | 1..1 | dateTime | Response creation date
patient | Σ | 1..1 | Reference(Patient) | Who prescription is for
encounter | | 0..1 | Reference(Encounter) | Created during encounter / admission / stay
dateWritten | Σ | 1..1 | dateTime | When prescription was authorized
prescriber | Σ | 1..1 | Reference(Practitioner | PractitionerRole) | Who authorized the vision prescription
lensSpecification | Σ | 1..* | BackboneElement | Vision lens authorization
product | Σ | 1..1 | CodeableConcept | Product to be supplied Example Vision Prescription Product Codes (Example)
eye | Σ | 1..1 | code | right | left VisionEyes (Required)
sphere | | 0..1 | decimal | Power of the lens
cylinder | | 0..1 | decimal | Lens power for astigmatism
axis | | 0..1 | integer | Lens meridian which contain no power for astigmatism
prism | | 0..* | BackboneElement | Eye alignment compensation
amount | | 1..1 | decimal | Amount of adjustment
base | | 1..1 | code | up | down | in | out VisionBase (Required)
add | | 0..1 | decimal | Added power for multifocal levels
power | | 0..1 | decimal | Contact lens power
backCurve | | 0..1 | decimal | Contact lens back curvature
diameter | | 0..1 | decimal | Contact lens diameter
duration | | 0..1 | SimpleQuantity | Lens wear duration
color | | 0..1 | string | Color required
brand | | 0..1 | string | Brand required
note | | 0..* | Annotation | Notes for coatings
Documentation for this format

UML Diagram (Legend)

XML Template

```
<VisionPrescription xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Business Identifier for vision prescription --></identifier>
 <status value="[code]"/><!-- 1..1 active | cancelled | draft | entered-in-error -->
 <created value="[dateTime]"/><!-- 1..1 Response creation date -->
 <patient><!-- 1..1 Reference(Patient) Who prescription is for --></patient>
 <encounter><!-- 0..1 Reference(Encounter) Created during encounter / admission / stay --></encounter>
 <dateWritten value="[dateTime]"/><!-- 1..1 When prescription was authorized -->
 <prescriber><!-- 1..1 Reference(Practitioner|PractitionerRole) Who authorized the vision prescription --></prescriber>
 <lensSpecification>  <!-- 1..* Vision lens authorization -->
  <product><!-- 1..1 CodeableConcept Product to be supplied --></product>
  <eye value="[code]"/><!-- 1..1 right | left -->
  <sphere value="[decimal]"/><!-- 0..1 Power of the lens -->
  <cylinder value="[decimal]"/><!-- 0..1 Lens power for astigmatism -->
  <axis value="[integer]"/><!-- 0..1 Lens meridian which contain no power for astigmatism -->
  <prism>  <!-- 0..* Eye alignment compensation -->
   <amount value="[decimal]"/><!-- 1..1 Amount of adjustment -->
   <base value="[code]"/><!-- 1..1 up | down | in | out -->
  </prism>
  <add value="[decimal]"/><!-- 0..1 Added power for multifocal levels -->
  <power value="[decimal]"/><!-- 0..1 Contact lens power -->
  <backCurve value="[decimal]"/><!-- 0..1 Contact lens back curvature -->
  <diameter value="[decimal]"/><!-- 0..1 Contact lens diameter -->
  <duration><!-- 0..1 Quantity(SimpleQuantity) Lens wear duration --></duration>
  <color value="[string]"/><!-- 0..1 Color required -->
  <brand value="[string]"/><!-- 0..1 Brand required -->
  <note><!-- 0..* Annotation Notes for coatings --></note>
 </lensSpecification>
</VisionPrescription>
```

JSON Template

```
{
  "resourceType" : "VisionPrescription",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Business Identifier for vision prescription
```

```
  "status" : "<code>", // R!  active | cancelled | draft | entered-in-error
  "created" : "<dateTime>", // R!  Response creation date
  "patient" : { Reference(Patient) }, // R!  Who prescription is for
  "encounter" : { Reference(Encounter) }, // Created during encounter / admission / stay
  "dateWritten" : "<dateTime>", // R!  When prescription was authorized
  "prescriber" : { Reference(Practitioner|PractitionerRole) }, // R!  Who authorized the vision prescription
  "lensSpecification" : [{ // R!  Vision lens authorization
    "product" : { CodeableConcept }, // R!  Product to be supplied
    "eye" : "<code>", // R!  right | left
    "sphere" : <decimal>, // Power of the lens
    "cylinder" : <decimal>, // Lens power for astigmatism
    "axis" : <integer>, // Lens meridian which contain no power for astigmatism
    "prism" : [{ // Eye alignment compensation
      "amount" : <decimal>, // R!  Amount of adjustment
      "base" : "<code>" // R!  up | down | in | out
    }],
    "add" : <decimal>, // Added power for multifocal levels
    "power" : <decimal>, // Contact lens power
    "backCurve" : <decimal>, // Contact lens back curvature
    "diameter" : <decimal>, // Contact lens diameter
    "duration" : { Quantity(SimpleQuantity) }, // Lens wear duration
    "color" : "<string>", // Color required
    "brand" : "<string>", // Brand required
    "note" : [{ Annotation }] // Notes for coatings
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:VisionPrescription;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:VisionPrescription.identifier [ Identifier ], ... ; # 0..* Business Identifier for vision prescription
  fhir:VisionPrescription.status [ code ]; # 1..1 active | cancelled | draft | entered-in-error
  fhir:VisionPrescription.created [ dateTime ]; # 1..1 Response creation date
  fhir:VisionPrescription.patient [ Reference(Patient) ]; # 1..1 Who prescription is for
  fhir:VisionPrescription.encounter [ Reference(Encounter) ]; # 0..1 Created during encounter / admission / stay
  fhir:VisionPrescription.dateWritten [ dateTime ]; # 1..1 When prescription was authorized
  fhir:VisionPrescription.prescriber [ Reference(Practitioner|PractitionerRole) ]; # 1..1 Who authorized the vision
    prescription
  fhir:VisionPrescription.lensSpecification [ # 1..* Vision lens authorization
    fhir:VisionPrescription.lensSpecification.product [ CodeableConcept ]; # 1..1 Product to be supplied
    fhir:VisionPrescription.lensSpecification.eye [ code ]; # 1..1 right | left
    fhir:VisionPrescription.lensSpecification.sphere [ decimal ]; # 0..1 Power of the lens
    fhir:VisionPrescription.lensSpecification.cylinder [ decimal ]; # 0..1 Lens power for astigmatism
    fhir:VisionPrescription.lensSpecification.axis [ integer ]; # 0..1 Lens meridian which contain no power for
    astigmatism
    fhir:VisionPrescription.lensSpecification.prism [ # 0..* Eye alignment compensation
      fhir:VisionPrescription.lensSpecification.prism.amount [ decimal ]; # 1..1 Amount of adjustment
      fhir:VisionPrescription.lensSpecification.prism.base [ code ]; # 1..1 up | down | in | out
    ], ...;
    fhir:VisionPrescription.lensSpecification.add [ decimal ]; # 0..1 Added power for multifocal levels
    fhir:VisionPrescription.lensSpecification.power [ decimal ]; # 0..1 Contact lens power
    fhir:VisionPrescription.lensSpecification.backCurve [ decimal ]; # 0..1 Contact lens back curvature
    fhir:VisionPrescription.lensSpecification.diameter [ decimal ]; # 0..1 Contact lens diameter
    fhir:VisionPrescription.lensSpecification.duration [ Quantity(SimpleQuantity) ]; # 0..1 Lens wear duration
    fhir:VisionPrescription.lensSpecification.color [ string ]; # 0..1 Color required
    fhir:VisionPrescription.lensSpecification.brand [ string ]; # 0..1 Brand required
    fhir:VisionPrescription.lensSpecification.note [ Annotation ], ... ; # 0..* Notes for coatings
  ], ...;
]
```

Changes since R3

VisionPrescription

VisionPrescription.status | Min Cardinality changed from 0 to 1 Change value set from http://hl7.org/fhir/ValueSet/fm-status to http://hl7.org/fhir/ValueSet/fm-status|4.0.1

VisionPrescription.created | Added Mandatory Element

VisionPrescription.patient | Min Cardinality changed from 0 to 1

VisionPrescription.dateWritten | Min Cardinality changed from 0 to 1

VisionPrescription.prescriber | Min Cardinality changed from 0 to 1 Type Reference: Added Target Type PractitionerRole

VisionPrescription.lensSpecification | Added Mandatory Element

VisionPrescription.lensSpecification.product | Added Mandatory Element

VisionPrescription.lensSpecification.eye | Added Mandatory Element

VisionPrescription.lensSpecification.sphere | Added Element

VisionPrescription.lensSpecification.cylinder | Added Element

VisionPrescription.lensSpecification.axis | Added Element

VisionPrescription.lensSpecification.prism | Added Element

VisionPrescription.lensSpecification.prism.amount | Added Mandatory Element

VisionPrescription.lensSpecification.prism.base | Added Mandatory Element

VisionPrescription.lensSpecification.add | Added Element

VisionPrescription.lensSpecification.power | Added Element

VisionPrescription.lensSpecification.backCurve | Added Element
VisionPrescription.lensSpecification.diameter | Added Element
VisionPrescription.lensSpecification.duration | Added Element
VisionPrescription.lensSpecification.color | Added Element
VisionPrescription.lensSpecification.brand | Added Element
VisionPrescription.lensSpecification.note | Added Element
VisionPrescription.reason[x] | deleted
VisionPrescription.dispense | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 2 tests that all execute ok. 1 fail round-trip testing and 2 r3 resources are invalid (0 errors).)

Structure

Name | Flags | Card. | Type | Description & Constraints
VisionPrescription | TU | | DomainResource | Prescription for vision correction products for a patient Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | | 0..* | Identifier | Business Identifier for vision prescription
status | ?!Σ | 1..1 | code | active | cancelled | draft | entered-in-error Financial Resource Status Codes (Required)
created | Σ | 1..1 | dateTime | Response creation date
patient | Σ | 1..1 | Reference(Patient) | Who prescription is for
encounter | | 0..1 | Reference(Encounter) | Created during encounter / admission / stay
dateWritten | Σ | 1..1 | dateTime | When prescription was authorized
prescriber | Σ | 1..1 | Reference(Practitioner | PractitionerRole) | Who authorized the vision prescription
lensSpecification | Σ | 1..* | BackboneElement | Vision lens authorization
product | Σ | 1..1 | CodeableConcept | Product to be supplied Example Vision Prescription Product Codes (Example)
eye | Σ | 1..1 | code | right | left VisionEyes (Required)
sphere | | 0..1 | decimal | Power of the lens
cylinder | | 0..1 | decimal | Lens power for astigmatism
axis | | 0..1 | integer | Lens meridian which contain no power for astigmatism
prism | | 0..* | BackboneElement | Eye alignment compensation
amount | | 1..1 | decimal | Amount of adjustment
base | | 1..1 | code | up | down | in | out VisionBase (Required)
add | | 0..1 | decimal | Added power for multifocal levels
power | | 0..1 | decimal | Contact lens power
backCurve | | 0..1 | decimal | Contact lens back curvature
diameter | | 0..1 | decimal | Contact lens diameter
duration | | 0..1 | SimpleQuantity | Lens wear duration
color | | 0..1 | string | Color required
brand | | 0..1 | string | Brand required
note | | 0..* | Annotation | Notes for coatings
Documentation for this format

UML Diagram (Legend)

XML Template

```
<VisionPrescription xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Business Identifier for vision prescription --></identifier>
 <status value="[code]"/><!-- 1..1 active | cancelled | draft | entered-in-error -->
 <created value="[dateTime]"/><!-- 1..1 Response creation date -->
 <patient><!-- 1..1 Reference(Patient) Who prescription is for --></patient>
 <encounter><!-- 0..1 Reference(Encounter) Created during encounter / admission / stay --></encounter>
 <dateWritten value="[dateTime]"/><!-- 1..1 When prescription was authorized -->
 <prescriber><!-- 1..1 Reference(Practitioner|PractitionerRole) Who authorized the vision prescription --></prescriber>
 <lensSpecification>  <!-- 1..* Vision lens authorization -->
  <product><!-- 1..1 CodeableConcept Product to be supplied --></product>
  <eye value="[code]"/><!-- 1..1 right | left -->
  <sphere value="[decimal]"/><!-- 0..1 Power of the lens -->
  <cylinder value="[decimal]"/><!-- 0..1 Lens power for astigmatism -->
  <axis value="[integer]"/><!-- 0..1 Lens meridian which contain no power for astigmatism -->
  <prism>  <!-- 0..* Eye alignment compensation -->
   <amount value="[decimal]"/><!-- 1..1 Amount of adjustment -->
   <base value="[code]"/><!-- 1..1 up | down | in | out -->
  </prism>
  <add value="[decimal]"/><!-- 0..1 Added power for multifocal levels -->
  <power value="[decimal]"/><!-- 0..1 Contact lens power -->
  <backCurve value="[decimal]"/><!-- 0..1 Contact lens back curvature -->
  <diameter value="[decimal]"/><!-- 0..1 Contact lens diameter -->
  <duration><!-- 0..1 Quantity(SimpleQuantity) Lens wear duration --></duration>
  <color value="[string]"/><!-- 0..1 Color required -->
  <brand value="[string]"/><!-- 0..1 Brand required -->
  <note><!-- 0..* Annotation Notes for coatings --></note>
 </lensSpecification>
</VisionPrescription>
```

JSON Template

```
{
  "resourceType" : "VisionPrescription",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Business Identifier for vision prescription
  "status" : "<code>", // R!  active | cancelled | draft | entered-in-error
  "created" : "<dateTime>", // R!  Response creation date
  "patient" : { Reference(Patient) }, // R!  Who prescription is for
  "encounter" : { Reference(Encounter) }, // Created during encounter / admission / stay
  "dateWritten" : "<dateTime>", // R!  When prescription was authorized
  "prescriber" : { Reference(Practitioner|PractitionerRole) }, // R!  Who authorized the vision prescription
  "lensSpecification" : [{ // R!  Vision lens authorization
    "product" : { CodeableConcept }, // R!  Product to be supplied
    "eye" : "<code>", // R!  right | left
    "sphere" : <decimal>, // Power of the lens
    "cylinder" : <decimal>, // Lens power for astigmatism
    "axis" : <integer>, // Lens meridian which contain no power for astigmatism
    "prism" : [{ // Eye alignment compensation
      "amount" : <decimal>, // R!  Amount of adjustment
      "base" : "<code>" // R!  up | down | in | out
    }],
    "add" : <decimal>, // Added power for multifocal levels
    "power" : <decimal>, // Contact lens power
    "backCurve" : <decimal>, // Contact lens back curvature
    "diameter" : <decimal>, // Contact lens diameter
    "duration" : { Quantity(SimpleQuantity) }, // Lens wear duration
    "color" : "<string>", // Color required
    "brand" : "<string>", // Brand required
    "note" : [{ Annotation }] // Notes for coatings
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:VisionPrescription;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:VisionPrescription.identifier [ Identifier ], ... ; # 0..* Business Identifier for vision prescription
  fhir:VisionPrescription.status [ code ]; # 1..1 active | cancelled | draft | entered-in-error
  fhir:VisionPrescription.created [ dateTime ]; # 1..1 Response creation date
  fhir:VisionPrescription.patient [ Reference(Patient) ]; # 1..1 Who prescription is for
  fhir:VisionPrescription.encounter [ Reference(Encounter) ]; # 0..1 Created during encounter / admission / stay
  fhir:VisionPrescription.dateWritten [ dateTime ]; # 1..1 When prescription was authorized
  fhir:VisionPrescription.prescriber [ Reference(Practitioner|PractitionerRole) ]; # 1..1 Who authorized the vision
    prescription
  fhir:VisionPrescription.lensSpecification [ # 1..* Vision lens authorization
    fhir:VisionPrescription.lensSpecification.product [ CodeableConcept ]; # 1..1 Product to be supplied
    fhir:VisionPrescription.lensSpecification.eye [ code ]; # 1..1 right | left
    fhir:VisionPrescription.lensSpecification.sphere [ decimal ]; # 0..1 Power of the lens
    fhir:VisionPrescription.lensSpecification.cylinder [ decimal ]; # 0..1 Lens power for astigmatism
    fhir:VisionPrescription.lensSpecification.axis [ integer ]; # 0..1 Lens meridian which contain no power for
    astigmatism
    fhir:VisionPrescription.lensSpecification.prism [ # 0..* Eye alignment compensation
      fhir:VisionPrescription.lensSpecification.prism.amount [ decimal ]; # 1..1 Amount of adjustment
      fhir:VisionPrescription.lensSpecification.prism.base [ code ]; # 1..1 up | down | in | out
    ], ...;
    fhir:VisionPrescription.lensSpecification.add [ decimal ]; # 0..1 Added power for multifocal levels
    fhir:VisionPrescription.lensSpecification.power [ decimal ]; # 0..1 Contact lens power
    fhir:VisionPrescription.lensSpecification.backCurve [ decimal ]; # 0..1 Contact lens back curvature
    fhir:VisionPrescription.lensSpecification.diameter [ decimal ]; # 0..1 Contact lens diameter
    fhir:VisionPrescription.lensSpecification.duration [ Quantity(SimpleQuantity) ]; # 0..1 Lens wear duration
    fhir:VisionPrescription.lensSpecification.color [ string ]; # 0..1 Color required
    fhir:VisionPrescription.lensSpecification.brand [ string ]; # 0..1 Brand required
    fhir:VisionPrescription.lensSpecification.note [ Annotation ], ... ; # 0..* Notes for coatings
  ], ...;
]
```

Changes since Release 3

VisionPrescription

VisionPrescription.status | Min Cardinality changed from 0 to 1 Change value set from http://hl7.org/fhir/ValueSet/fm-status to http://hl7.org/fhir/ValueSet/fm-status|4.0.1

VisionPrescription.created | Added Mandatory Element

VisionPrescription.patient | Min Cardinality changed from 0 to 1

VisionPrescription.dateWritten | Min Cardinality changed from 0 to 1

VisionPrescription.prescriber | Min Cardinality changed from 0 to 1 Type Reference: Added Target Type PractitionerRole

VisionPrescription.lensSpecification | Added Mandatory Element

VisionPrescription.lensSpecification.product | Added Mandatory Element

VisionPrescription.lensSpecification.eye | Added Mandatory Element

VisionPrescription.lensSpecification.sphere | Added Element

VisionPrescription.lensSpecification.cylinder | Added Element

VisionPrescription.lensSpecification.axis | Added Element

VisionPrescription.lensSpecification.prism | Added Element

VisionPrescription.lensSpecification.prism.amount | Added Mandatory Element
VisionPrescription.lensSpecification.prism.base | Added Mandatory Element
VisionPrescription.lensSpecification.add | Added Element
VisionPrescription.lensSpecification.power | Added Element
VisionPrescription.lensSpecification.backCurve | Added Element
VisionPrescription.lensSpecification.diameter | Added Element
VisionPrescription.lensSpecification.duration | Added Element
VisionPrescription.lensSpecification.color | Added Element
VisionPrescription.lensSpecification.brand | Added Element
VisionPrescription.lensSpecification.note | Added Element
VisionPrescription.reason[x] | deleted
VisionPrescription.dispense | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 2 tests that all execute ok. 1 fail round-trip testing and 2 r3 resources are invalid (0 errors).)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

12.16.3.1 Terminology Bindings

Path | Definition | Type | Reference
VisionPrescription.status | A code specifying the state of the resource instance. | Required | FinancialResourceStatusCodes
VisionPrescription.lensSpecification.product | A coded concept describing the vision products. | Example | ExampleVisionPrescriptionProductCodes
VisionPrescription.lensSpecification.eye | A coded concept listing the eye codes. | Required | VisionEyes
VisionPrescription.lensSpecification.prism.base | A coded concept listing the base codes. | Required | VisionBase

12.16.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
datewritten | date | Return prescriptions written on this date | VisionPrescription.dateWritten
encounter | reference | Return prescriptions with this encounter identifier | VisionPrescription.encounter (Encounter) | 12 Resources
identifier | token | Return prescriptions with this external identifier | VisionPrescription.identifier | 30 Resources
patient | reference | The identity of a patient to list dispenses for | VisionPrescription.patient (Patient) | 33 Resources
prescriber | reference | Who authorized the vision prescription | VisionPrescription.prescriber (Practitioner, PractitionerRole)
status | token | The status of the vision prescription | VisionPrescription.status

12.16.5 Resource VisionPrescription - Examples

URL: https://hl7.org/fhir/R4/visionprescription-examples.html

12.16.5 Resource VisionPrescription - Examples

Financial Management Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Encounter, Patient, Practitioner

Example Name | id | Format
General Glasses Example | 33123 | XML | JSON | Turtle
General Contacts Example | 33124 | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

12.16.6 Resource VisionPrescription - Detailed Descriptions

URL: https://hl7.org/fhir/R4/visionprescription-definitions.html

12.16.6 Resource VisionPrescription - Detailed Descriptions

Financial Management Work Group | Maturity Level: 2 | Trial Use | Security Category: Patient | Compartments: Encounter, Patient, Practitioner

Detailed Descriptions for the elements in the VisionPrescription resource.

VisionPrescription
Element Id | VisionPrescription
Definition | An authorization for the provision of glasses and/or contact lenses to a patient.
Cardinality | 0..*
Type | DomainResource
VisionPrescription.identifier
Element Id | VisionPrescription.identifier

Definition | A unique identifier assigned to this vision prescription.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Requirements | Allows vision prescriptions to be distinguished and referenced.
VisionPrescription.status
Element Id | VisionPrescription.status
Definition | The status of the resource instance.
Cardinality | 1..1
Terminology Binding | Financial Resource Status Codes (Required)
Type | code
Is Modifier | true (Reason: This element is labelled as a modifier because it is a status element that contains status entered-in-error which means that the resource should not be treated as valid)
Requirements | Need to track the status of the resource as 'draft' resources may undergo further edits while 'active' resources are immutable and may only have their status changed to 'cancelled'.
Summary | true
Comments | This element is labeled as a modifier because the status contains codes that mark the resource as not currently valid.
VisionPrescription.created
Element Id | VisionPrescription.created
Definition | The date this resource was created.
Cardinality | 1..1
Type | dateTime
Requirements | Need to record a timestamp for use by both the recipient and the issuer.
Summary | true
VisionPrescription.patient
Element Id | VisionPrescription.patient
Definition | A resource reference to the person to whom the vision prescription applies.
Cardinality | 1..1
Type | Reference(Patient)
Requirements | Needed to identify the patient for whom the vision prescription has been issued.
Summary | true
VisionPrescription.encounter
Element Id | VisionPrescription.encounter
Definition | A reference to a resource that identifies the particular occurrence of contact between patient and health care provider during which the prescription was issued.
Cardinality | 0..1
Type | Reference(Encounter)
Requirements | Used in some jurisdictions to link clinical events to claim items.
VisionPrescription.dateWritten
Element Id | VisionPrescription.dateWritten
Definition | The date (and perhaps time) when the prescription was written.
Cardinality | 1..1
Type | dateTime
Requirements | Need to record a timestamp for use by both the recipient and the issuer.
Summary | true
Comments | Jurisdictions determine the valid lifetime of a prescription. Typically vision prescriptions are valid for two years from the date written.
VisionPrescription.prescriber
Element Id | VisionPrescription.prescriber
Definition | The healthcare professional responsible for authorizing the prescription.
Cardinality | 1..1
Type | Reference(Practitioner | PractitionerRole)
Requirements | Need to record the authorizing practitioner.
Summary | true
VisionPrescription.lensSpecification
Element Id | VisionPrescription.lensSpecification
Definition | Contain the details of the individual lens specifications and serves as the authorization for the fullfillment by certified professionals.
Cardinality | 1..*
Summary | true
VisionPrescription.lensSpecification.product
Element Id | VisionPrescription.lensSpecification.product
Definition | Identifies the type of vision correction product which is required for the patient.
Cardinality | 1..1
Terminology Binding | Example Vision Prescription Product Codes (Example)
Type | CodeableConcept
Requirements | Need to identify whether glasses or contacts.
Summary | true
VisionPrescription.lensSpecification.eye
Element Id | VisionPrescription.lensSpecification.eye
Definition | The eye for which the lens specification applies.
Cardinality | 1..1
Terminology Binding | VisionEyes (Required)

Type | code
Requirements | Needed to identify which eye the lens specification applies to.
Summary | true
Comments | May also appear as OD (oculus dexter) for the right eye and OS (oculus siniter) for the left eye.
VisionPrescription.lensSpecification.sphere
Element Id | VisionPrescription.lensSpecification.sphere
Definition | Lens power measured in dioptres (0.25 units).
Cardinality | 0..1
Type | decimal
Requirements | Standard lens prescription measures.
Comments | The value is negative for near-sighted and positive for far sighted. Often insurance will not cover a lens with power between +75 and -75.
VisionPrescription.lensSpecification.cylinder
Element Id | VisionPrescription.lensSpecification.cylinder
Definition | Power adjustment for astigmatism measured in dioptres (0.25 units).
Cardinality | 0..1
Type | decimal
Requirements | Standard lens prescription measures.
VisionPrescription.lensSpecification.axis
Element Id | VisionPrescription.lensSpecification.axis
Definition | Adjustment for astigmatism measured in integer degrees.
Cardinality | 0..1
Type | integer
Requirements | Standard lens prescription measures.
Comments | The limits are +180 and -180 degrees.
VisionPrescription.lensSpecification.prism
Element Id | VisionPrescription.lensSpecification.prism
Definition | Allows for adjustment on two axis.
Cardinality | 0..*
Requirements | Standard lens prescription measures.
VisionPrescription.lensSpecification.prism.amount
Element Id | VisionPrescription.lensSpecification.prism.amount
Definition | Amount of prism to compensate for eye alignment in fractional units.
Cardinality | 1..1
Type | decimal
Requirements | Standard lens prescription measures.
VisionPrescription.lensSpecification.prism.base
Element Id | VisionPrescription.lensSpecification.prism.base
Definition | The relative base, or reference lens edge, for the prism.
Cardinality | 1..1
Terminology Binding | VisionBase (Required)
Type | code
Requirements | Standard lens prescription measures.
VisionPrescription.lensSpecification.add
Element Id | VisionPrescription.lensSpecification.add
Definition | Power adjustment for multifocal lenses measured in dioptres (0.25 units).
Cardinality | 0..1
Type | decimal
Requirements | Standard lens prescription measures.
VisionPrescription.lensSpecification.power
Element Id | VisionPrescription.lensSpecification.power
Definition | Contact lens power measured in dioptres (0.25 units).
Cardinality | 0..1
Type | decimal
Requirements | Standard contact lens prescription measures.
VisionPrescription.lensSpecification.backCurve
Element Id | VisionPrescription.lensSpecification.backCurve
Definition | Back curvature measured in millimetres.
Cardinality | 0..1
Type | decimal
Requirements | Standard contact lens prescription measures.
VisionPrescription.lensSpecification.diameter
Element Id | VisionPrescription.lensSpecification.diameter
Definition | Contact lens diameter measured in millimetres.
Cardinality | 0..1
Type | decimal
Requirements | Standard contact lens prescription measures.
VisionPrescription.lensSpecification.duration
Element Id | VisionPrescription.lensSpecification.duration
Definition | The recommended maximum wear period for the lens.
Cardinality | 0..1
Type | SimpleQuantity

Requirements | Standard contact lens prescription measures.
VisionPrescription.lensSpecification.color
Element Id | VisionPrescription.lensSpecification.color
Definition | Special color or pattern.
Cardinality | 0..1
Type | string
Requirements | Standard contact lens prescription measures.
VisionPrescription.lensSpecification.brand
Element Id | VisionPrescription.lensSpecification.brand
Definition | Brand recommendations or restrictions.
Cardinality | 0..1
Type | string
Requirements | Standard lens prescription measures.
VisionPrescription.lensSpecification.note
Element Id | VisionPrescription.lensSpecification.note
Definition | Notes for special requirements such as coatings and lens materials.
Cardinality | 0..*
Type | Annotation
Requirements | Standard lens prescription measures.

12.16.7 Resource VisionPrescription - Mappings

URL: https://hl7.org/fhir/R4/visionprescription-mappings.html

12.16.7 Resource VisionPrescription - Mappings

Financial Management Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Encounter, Patient, Practitioner

Mappings:

FiveWs Pattern Mapping

Workflow Pattern

HL7 v2 Mapping

RIM Mapping

Mappings for the visionprescription resource (see Mappings to Other Standards for further information & status).

12.16.7.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

VisionPrescription
identifier | FiveWs.identifier
status | FiveWs.status
patient | FiveWs.subject[x]
encounter | FiveWs.context
dateWritten | FiveWs.recorded
prescriber | FiveWs.author

12.16.7.2 Workflow Pattern (http://hl7.org/fhir/workflow)

VisionPrescription | Request
identifier | Request.identifier
status | Request.status
patient | Request.subject
encounter | Request.context
dateWritten | Request.authoredOn
prescriber | Request.requester

12.16.7.3 HL7 v2 Mapping (http://hl7.org/v2)

VisionPrescription
identifier | ORC-2-Placer Order Number / ORC-3-Filler Order Number
status
created
patient | PID-3-Patient ID List
encounter | PV1-19-Visit Number
dateWritten | RXE-32-Original Order Date/Time / ORC-9-Date/Time of Transaction
prescriber | RXE-13-Ordering Provider's DEA Number / RXO-14-Ordering Provider's DEA Number / RXE-14-Pharmacist/Treatment Supplier's Verifier ID / RXO-15-Pharmacist/Treatment Supplier's Verifier ID / ORC-12-Ordering Provider / PRT-5-Participation Person: PRT-4-Participation='OP' (all but last deprecated)
lensSpecification
product | RXE-2-Give Code / RXO-1-Requested Give Code / RXD-2-Dispense/Give Code
eye
sphere
cylinder

axis
prism
amount
base
add
power
backCurve
diameter
duration
color
brand
note

12.16.7.4 RIM Mapping (http://hl7.org/v3)

VisionPrescription | CombinedVisionRequest
identifier | id
status | Act.status
patient | subject.role
encounter | componentOf.patientEncounter
dateWritten | author.time
prescriber | author.role
lensSpecification | component.supplyEvent
product | consumable.role
eye | n/a
sphere | n/a
cylinder | n/a
axis | n/a
amount | n/a
base | n/a
add | n/a
power | n/a
backCurve | n/a
diameter | n/a
duration | n/a
color | n/a
brand | n/a
note | n/a

12.16.8 Resource VisionPrescription - Extensions & Profiles

URL: https://hl7.org/fhir/R4/visionprescription-profiles.html

12.16.8 Resource VisionPrescription - Extensions & Profiles

Financial Management Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Encounter, Patient, Practitioner

This table lists profiles and extensions for the VisionPrescription resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

12.16.8.1 Profiles

No Profiles defined for this resource

12.16.8.2 Extensions

No Extensions defined for this resource
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

12.16.8.3 Search Extensions

No Search Extensions defined for this resource

12.16.9 Resource VisionPrescription - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/visionprescription-version-maps.html

No validation errors - all conversions are clean

12.16.9 Resource VisionPrescription - R3/R4 Conversions

Financial Management Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Encounter, Patient, Practitioner

R3 : R4 Conversion maps for VisionPrescription.

Functional status for this map: 2 tests that all execute ok. 1 fail round-trip testing and 2 r3 resources are invalid (0 errors). (see documentation)

12.16.9.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/VisionPrescription3to4" = "R3 to R4 Conversions for VisionPrescription"

uses "http://hl7.org/fhir/3.0/StructureDefinition/VisionPrescription" alias VisionPrescriptionR3 as source
uses "http://hl7.org/fhir/StructureDefinition/VisionPrescription" alias VisionPrescription as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group VisionPrescription(source src : VisionPrescriptionR3, target tgt : VisionPrescription) extends DomainResource
    <<type+>> {
  src.identifier -> tgt.identifier;
  src.status -> tgt.status;
  src.patient -> tgt.patient;
  src.encounter -> tgt.encounter;
  src.dateWritten -> tgt.dateWritten;
  src.prescriber -> tgt.prescriber;
  src.reason as v ->  tgt.extension as vt,  vt.url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-
    VisionPrescription.Reason',  vt.value = v;
  src.dispense as s -> tgt.lensSpecification as t then dispense(s, t);
}

group dispense(source src, target tgt) extends BackboneElement {
  src.product -> tgt.product;
  src.eye -> tgt.eye;
  src.sphere -> tgt.sphere;
  src.cylinder -> tgt.cylinder;
  src.axis -> tgt.axis;
  src where src.prism.exists() or src.base.exists() -> tgt.prism as t then {
    src.prism -> t.amount;
    src.base -> t.base;
  } "prism";
  src.add -> tgt.add;
  src.power -> tgt.power;
  src.backCurve -> tgt.backCurve;
  src.diameter -> tgt.diameter;
  src.duration -> tgt.duration;
  src.color -> tgt.color;
  src.brand -> tgt.brand;
  src.note -> tgt.note;
}
```

12.16.9.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/VisionPrescription4to3" = "R4 to R3 Conversion for VisionPrescription"

uses "http://hl7.org/fhir/StructureDefinition/VisionPrescription" alias VisionPrescription as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/VisionPrescription" alias VisionPrescriptionR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group VisionPrescription(source src : VisionPrescription, target tgt : VisionPrescriptionR3) extends DomainResource
    <<type+>> {
  src.identifier -> tgt.identifier;
  src.status -> tgt.status;
  src.patient -> tgt.patient;
  src.encounter -> tgt.encounter;
  src.dateWritten -> tgt.dateWritten;
  src.prescriber -> tgt.prescriber;
  src.extension as e where url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-VisionPrescription.reason' then
    {
    e.value -> tgt.reason;
  };
  src.lensSpecification as s -> tgt.dispense as t then dispense(s, t);
}

group dispense(source src, target tgt) extends BackboneElement {
  src.product -> tgt.product;
  src.eye -> tgt.eye;
  src.sphere -> tgt.sphere;
  src.cylinder -> tgt.cylinder;
```

```
  src.axis -> tgt.axis;
  src.prism as s then {
    s.amount -> tgt.prism;
    s.base -> tgt.base;
  };
  src.add -> tgt.add;
  src.power -> tgt.power;
  src.backCurve -> tgt.backCurve;
  src.diameter -> tgt.diameter;
  src.duration -> tgt.duration;
  src.color -> tgt.color;
  src.brand -> tgt.brand;
  src.note -> tgt.note;
}
```

12.16.9.3 R4 Validation Errors

12.16.10 JSON Schema for VisionPrescription

URL: https://hl7.org/fhir/R4/visionprescription.schema.json.html

12.16.10 JSON Schema for VisionPrescription

Financial Management Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Encounter, Patient, Practitioner

This is a representation of the json schema for VisionPrescription, which is just a part of the full JSON Schema.


```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/VisionPrescription",
  "$ref": "#/definitions/VisionPrescription",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "VisionPrescription": {
      "description": "An authorization for the provision of glasses and/or contact lenses to a patient.",
      "properties": {
        "resourceType": {
          "description": "This is a VisionPrescription resource",
          "const": "VisionPrescription"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
          "$ref": "#/definitions/code"
        },
        "_language": {
          "description": "Extensions for language",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "text": {
          "description": "A human-readable narrative that contains a summary of the resource and can be used to
    represent the content of the resource to a human. The narrative need not encode all the structured data, but is
    required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
    Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
          "$ref": "Narrative.schema.json#/definitions/Narrative"
        },
        "contained": {
          "description": "These resources do not have an independent existence apart from the resource that contains
    them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
          "items": {
            "$ref": "ResourceList.schema.json#/definitions/ResourceList"
          },
          "type": "array"
        },
        "extension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
     definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
```

```
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource and that modifies the understanding of the element that contains it and/or the understanding of the
containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
met as part of the definition of the extension. Applications processing a resource are required to check for
modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "identifier": {
      "description": "A unique identifier assigned to this vision prescription.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "status": {
      "description": "The status of the resource instance.",
      "$ref": "#/definitions/code"
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "created": {
      "description": "The date this resource was created.",
      "$ref": "#/definitions/dateTime"
    },
    "_created": {
      "description": "Extensions for created",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "patient": {
      "description": "A resource reference to the person to whom the vision prescription applies.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "encounter": {
      "description": "A reference to a resource that identifies the particular occurrence of contact between patient
 and health care provider during which the prescription was issued.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "dateWritten": {
      "description": "The date (and perhaps time) when the prescription was written.",
      "$ref": "#/definitions/dateTime"
    },
    "_dateWritten": {
      "description": "Extensions for dateWritten",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "prescriber": {
      "description": "The healthcare professional responsible for authorizing the prescription.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "lensSpecification": {
      "description": "Contain the details of  the individual lens specifications and serves as the authorization for
 the fullfillment by certified professionals.",
      "items": {
        "$ref": "#/definitions/VisionPrescription_LensSpecification"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "prescriber",
    "patient",
    "lensSpecification",
    "resourceType"
  ]
},
"VisionPrescription_LensSpecification": {
  "description": "An authorization for the provision of glasses and/or contact lenses to a patient.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
```

```
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "product": {
      "description": "Identifies the type of vision correction product which is required for the patient.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "eye": {
      "description": "The eye for which the lens specification applies.",
      "enum": [
        "right",
        "left"
      ]
    },
    "_eye": {
      "description": "Extensions for eye",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "sphere": {
      "description": "Lens power measured in dioptres (0.25 units).",
      "$ref": "#/definitions/decimal"
    },
    "_sphere": {
      "description": "Extensions for sphere",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "cylinder": {
      "description": "Power adjustment for astigmatism measured in dioptres (0.25 units).",
      "$ref": "#/definitions/decimal"
    },
    "_cylinder": {
      "description": "Extensions for cylinder",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "axis": {
      "description": "Adjustment for astigmatism measured in integer degrees.",
      "$ref": "#/definitions/integer"
    },
    "_axis": {
      "description": "Extensions for axis",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "prism": {
      "description": "Allows for adjustment on two axis.",
      "items": {
        "$ref": "#/definitions/VisionPrescription_Prism"
      },
      "type": "array"
    },
    "add": {
      "description": "Power adjustment for multifocal lenses measured in dioptres (0.25 units).",
      "$ref": "#/definitions/decimal"
    },
    "_add": {
      "description": "Extensions for add",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "power": {
      "description": "Contact lens power measured in dioptres (0.25 units).",
      "$ref": "#/definitions/decimal"
    },
    "_power": {
      "description": "Extensions for power",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "backCurve": {
      "description": "Back curvature measured in millimetres.",
      "$ref": "#/definitions/decimal"
```

```
    },
    "_backCurve": {
      "description": "Extensions for backCurve",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "diameter": {
      "description": "Contact lens diameter measured in millimetres.",
      "$ref": "#/definitions/decimal"
    },
    "_diameter": {
      "description": "Extensions for diameter",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "duration": {
      "description": "The recommended maximum wear period for the lens.",
      "$ref": "Quantity.schema.json#/definitions/Quantity"
    },
    "color": {
      "description": "Special color or pattern.",
      "$ref": "#/definitions/string"
    },
    "_color": {
      "description": "Extensions for color",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "brand": {
      "description": "Brand recommendations or restrictions.",
      "$ref": "#/definitions/string"
    },
    "_brand": {
      "description": "Extensions for brand",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "note": {
      "description": "Notes for special requirements such as coatings and lens materials.",
      "items": {
        "$ref": "Annotation.schema.json#/definitions/Annotation"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "product"
  ]
},
"VisionPrescription_Prism": {
  "description": "An authorization for the provision of glasses and/or contact lenses to a patient.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "amount": {
      "description": "Amount of prism to compensate for eye alignment in fractional units.",
      "$ref": "#/definitions/decimal"
    },
    "_amount": {
      "description": "Extensions for amount",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "base": {
      "description": "The relative base, or reference lens edge, for the prism.",
      "enum": [
```

```
            "up",
            "down",
            "in",
            "out"
          ]
        },
        "_base": {
          "description": "Extensions for base",
          "$ref": "Element.schema.json#/definitions/Element"
        }
      },
      "additionalProperties": false
    }
  }
}
```

12.16.11 Resource VisionPrescription - Maturity Dependency View

URL: https://hl7.org/fhir/R4/visionprescription-dependencies.html

12.16.11 Resource VisionPrescription - Maturity Dependency View

Financial Management Work Group | Maturity Level: 2 | Trial Use | Compartments: Encounter, Patient, Practitioner

Dependency Graph for VisionPrescription FMM level 2

Name | Card. | Type | Dependency Analysis
VisionPrescription | | DomainResource
identifier | 0..* | Identifier
status | 1..1 | code
created | 1..1 | dateTime
patient | 1..1 | Reference(Patient)
encounter | 0..1 | Reference(Encounter)
dateWritten | 1..1 | dateTime
prescriber | 1..1 | Reference(Practitioner | PractitionerRole)
lensSpecification | 1..* | BackboneElement
product | 1..1 | CodeableConcept
eye | 1..1 | code
sphere | 0..1 | decimal
cylinder | 0..1 | decimal
axis | 0..1 | integer
prism | 0..* | BackboneElement
amount | 1..1 | decimal
base | 1..1 | code
add | 0..1 | decimal
power | 0..1 | decimal
backCurve | 0..1 | decimal
diameter | 0..1 | decimal
duration | 0..1 | SimpleQuantity
color | 0..1 | string
brand | 0..1 | string
note | 0..* | Annotation

12.17 Resource ActivityDefinition - Content

URL: https://hl7.org/fhir/R4/activitydefinition.html
Clinical Decision Support Work Group | Maturity Level: 2 | Trial Use | Security Category: Anonymous | Compartments: Not linked to any defined compartments

This resource allows for the definition of some activity to be performed, independent of a particular patient, practitioner, or other performance context.

12.17.1 Scope and Usage

This resource is a definition resource from a FHIR workflow perspective - see Workflow, specifically Definition.

An ActivityDefinition is a shareable, consumable description of some activity to be performed. It may be used to specify actions to be taken as part of a workflow, order set, or protocol, or it may be used independently as part of a catalog of activities such as orderables.

For more information on how activity definitions can be used to construct request resources, see the Applying an ActivityDefinition topic below.

12.17.2 Boundaries and Relationships

In general, an activity definition is simply a conceptual description of some specific action that should be taken. An instance of an ActivityDefinition does not indicate that any action has been performed (as an event

resource does), nor does it indicate the actual intent to carry out any particular action (as a request resource does). Instead, an activity definition provides a reusable template that can be used to construct specific request resources such as ServiceRequest and MedicationRequest.

Note that this is conceptually similar to the Task resource as well, with the distinction being that ActivityDefinition represents the description of a task in the abstract, while the Task resource is used to track a specific instance of a task as it moves through the steps of a workflow.

This resource is referenced by CarePlan, CatalogEntry, Communication, DeviceRequest, EvidenceVariable, FamilyMemberHistory, MessageDefinition, NutritionOrder, PlanDefinition, Procedure, ServiceRequest and Task

12.17.3 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
ActivityDefinition | ITU | | DomainResource | The definition of a specific activity to be taken, independent of any particular patient or context + Warning: Name should be usable as an identifier for the module by machine processing applications such as code generation Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
url | Σ | 0..1 | uri | Canonical identifier for this activity definition, represented as a URI (globally unique)
identifier | Σ | 0..* | Identifier | Additional identifier for the activity definition
version | Σ | 0..1 | string | Business version of the activity definition
name | ΣI | 0..1 | string | Name for this activity definition (computer friendly)
title | Σ | 0..1 | string | Name for this activity definition (human friendly)
subtitle | | 0..1 | string | Subordinate title of the activity definition
status | ?!Σ | 1..1 | code | draft | active | retired | unknown PublicationStatus (Required)
experimental | Σ | 0..1 | boolean | For testing purposes, not real usage
subject[x] | | 0..1 | | Type of individual the activity definition is intended for Subject type (Extensible)
subjectCodeableConcept | | | CodeableConcept
subjectReference | | | Reference(Group)
date | Σ | 0..1 | dateTime | Date last changed
publisher | Σ | 0..1 | string | Name of the publisher (organization or individual)
contact | Σ | 0..* | ContactDetail | Contact details for the publisher
description | Σ | 0..1 | markdown | Natural language description of the activity definition
useContext | Σ | 0..* | UsageContext | The context that the content is intended to support
jurisdiction | Σ | 0..* | CodeableConcept | Intended jurisdiction for activity definition (if applicable) Jurisdiction (Extensible)
purpose | | 0..1 | markdown | Why this activity definition is defined
usage | | 0..1 | string | Describes the clinical usage of the activity definition
copyright | | 0..1 | markdown | Use and/or publishing restrictions
approvalDate | | 0..1 | date | When the activity definition was approved by publisher
lastReviewDate | | 0..1 | date | When the activity definition was last reviewed
effectivePeriod | Σ | 0..1 | Period | When the activity definition is expected to be used
topic | | 0..* | CodeableConcept | E.g. Education, Treatment, Assessment, etc. DefinitionTopic (Example)
author | | 0..* | ContactDetail | Who authored the content
editor | | 0..* | ContactDetail | Who edited the content
reviewer | | 0..* | ContactDetail | Who reviewed the content
endorser | | 0..* | ContactDetail | Who endorsed the content
relatedArtifact | | 0..* | RelatedArtifact | Additional documentation, citations, etc.
library | | 0..* | canonical(Library) | Logic used by the activity definition
kind | Σ | 0..1 | code | Kind of resource RequestResourceType (Required)
profile | | 0..1 | canonical(StructureDefinition) | What profile the resource needs to conform to
code | Σ | 0..1 | CodeableConcept | Detail type of activity Procedure Codes (SNOMED CT) (Example)
intent | | 0..1 | code | proposal | plan | directive | order | original-order | reflex-order | filler-order | instance-order | option RequestIntent (Required)
priority | | 0..1 | code | routine | urgent | asap | stat Request priority (Required)
doNotPerform | ?!Σ | 0..1 | boolean | True if the activity should not be performed
timing[x] | | 0..1 | | When activity is to occur
timingTiming | | | Timing
timingDateTime | | | dateTime
timingAge | | | Age
timingPeriod | | | Period
timingRange | | | Range
timingDuration | | | Duration
location | | 0..1 | Reference(Location) | Where it should happen
participant | | 0..* | BackboneElement | Who should participate in the action

type | | 1..1 | code | patient | practitioner | related-person | device ActionParticipantType (Required)
role | | 0..1 | CodeableConcept | E.g. Nurse, Surgeon, Parent, etc. Action participant role (Example)
product[x] | | 0..1 | | What's administered/supplied SNOMED CT Medication Codes (Example)
productReference | | | Reference(Medication | Substance)
productCodeableConcept | | | CodeableConcept
quantity | | 0..1 | SimpleQuantity | How much is administered/consumed/supplied
dosage | | 0..* | Dosage | Detailed dosage instructions
bodySite | | 0..* | CodeableConcept | What part of body to perform on SNOMED CT Body Structures (Example)
specimenRequirement | | 0..* | Reference(SpecimenDefinition) | What specimens are required to perform this action
observationRequirement | | 0..* | Reference(ObservationDefinition) | What observations are required to perform this action
observationResultRequirement | | 0..* | Reference(ObservationDefinition) | What observations must be produced by this action
transform | | 0..1 | canonical(StructureMap) | Transform to apply the template
dynamicValue | | 0..* | BackboneElement | Dynamic aspects of the definition
path | | 1..1 | string | The path to the element to be set dynamically
expression | | 1..1 | Expression | An expression that provides the dynamic value for the customization
Documentation for this format

UML Diagram (Legend)

XML Template

```
<ActivityDefinition xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <url value="[uri]"/><!-- 0..1 Canonical identifier for this activity definition, represented as a URI (globally unique)
     -->
 <identifier><!-- 0..* Identifier Additional identifier for the activity definition --></identifier>
 <version value="[string]"/><!-- 0..1 Business version of the activity definition -->
 <name value="[string]"/><!--  0..1 Name for this activity definition (computer friendly) -->
 <title value="[string]"/><!-- 0..1 Name for this activity definition (human friendly) -->
 <subtitle value="[string]"/><!-- 0..1 Subordinate title of the activity definition -->
 <status value="[code]"/><!-- 1..1 draft | active | retired | unknown -->
 <experimental value="[boolean]"/><!-- 0..1 For testing purposes, not real usage -->
 <subject[x]><!-- 0..1 CodeableConcept|Reference(Group) Type of individual the activity definition is intended for
    --></subject[x]>
 <date value="[dateTime]"/><!-- 0..1 Date last changed -->
 <publisher value="[string]"/><!-- 0..1 Name of the publisher (organization or individual) -->
 <contact><!-- 0..* ContactDetail Contact details for the publisher --></contact>
 <description value="[markdown]"/><!-- 0..1 Natural language description of the activity definition -->
 <useContext><!-- 0..* UsageContext The context that the content is intended to support --></useContext>
 <jurisdiction><!-- 0..* CodeableConcept Intended jurisdiction for activity definition (if applicable)
    --></jurisdiction>
 <purpose value="[markdown]"/><!-- 0..1 Why this activity definition is defined -->
 <usage value="[string]"/><!-- 0..1 Describes the clinical usage of the activity definition -->
 <copyright value="[markdown]"/><!-- 0..1 Use and/or publishing restrictions -->
 <approvalDate value="[date]"/><!-- 0..1 When the activity definition was approved by publisher -->
 <lastReviewDate value="[date]"/><!-- 0..1 When the activity definition was last reviewed -->
 <effectivePeriod><!-- 0..1 Period When the activity definition is expected to be used --></effectivePeriod>
 <topic><!-- 0..* CodeableConcept E.g. Education, Treatment, Assessment, etc. --></topic>
 <author><!-- 0..* ContactDetail Who authored the content --></author>
 <editor><!-- 0..* ContactDetail Who edited the content --></editor>
 <reviewer><!-- 0..* ContactDetail Who reviewed the content --></reviewer>
 <endorser><!-- 0..* ContactDetail Who endorsed the content --></endorser>
 <relatedArtifact><!-- 0..* RelatedArtifact Additional documentation, citations, etc. --></relatedArtifact>
 <library><!-- 0..* canonical(Library) Logic used by the activity definition --></library>
 <kind value="[code]"/><!-- 0..1 Kind of resource -->
 <profile><!-- 0..1 canonical(StructureDefinition) What profile the resource needs to conform to --></profile>
 <code><!-- 0..1 CodeableConcept Detail type of activity --></code>
 <intent value="[code]"/><!-- 0..1 proposal | plan | directive | order | original-order | reflex-order | filler-order |
    instance-order | option -->
 <priority value="[code]"/><!-- 0..1 routine | urgent | asap | stat -->
 <doNotPerform value="[boolean]"/><!-- 0..1 True if the activity should not be performed -->
 <timing[x]><!-- 0..1 Timing|dateTime|Age|Period|Range|Duration When activity is to occur --></timing[x]>
 <location><!-- 0..1 Reference(Location) Where it should happen --></location>
 <participant>  <!-- 0..* Who should participate in the action -->
  <type value="[code]"/><!-- 1..1 patient | practitioner | related-person | device -->
  <role><!-- 0..1 CodeableConcept E.g. Nurse, Surgeon, Parent, etc. --></role>
 </participant>
 <product[x]><!-- 0..1 Reference(Medication|Substance)|CodeableConcept What's administered/supplied --></product[x]>
 <quantity><!-- 0..1 Quantity(SimpleQuantity) How much is administered/consumed/supplied --></quantity>
 <dosage><!-- 0..* Dosage Detailed dosage instructions --></dosage>
 <bodySite><!-- 0..* CodeableConcept What part of body to perform on --></bodySite>
 <specimenRequirement><!-- 0..* Reference(SpecimenDefinition) What specimens are required to perform this action
    --></specimenRequirement>
 <observationRequirement><!-- 0..* Reference(ObservationDefinition) What observations are required to perform this
    action --></observationRequirement>
 <observationResultRequirement><!-- 0..* Reference(ObservationDefinition) What observations must be produced by this
    action --></observationResultRequirement>
 <transform><!-- 0..1 canonical(StructureMap) Transform to apply the template --></transform>
 <dynamicValue>  <!-- 0..* Dynamic aspects of the definition -->
  <path value="[string]"/><!-- 1..1 The path to the element to be set dynamically -->
  <expression><!-- 1..1 Expression An expression that provides the dynamic value for the customization --></expression>
 </dynamicValue>
</ActivityDefinition>
```

JSON Template

```
{
  "resourceType" : "ActivityDefinition",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "url" : "<uri>", // Canonical identifier for this activity definition, represented as a URI (globally unique)
  "identifier" : [{ Identifier }], // Additional identifier for the activity definition
  "version" : "<string>", // Business version of the activity definition
  "name" : "<string>", // C? Name for this activity definition (computer friendly)
  "title" : "<string>", // Name for this activity definition (human friendly)
  "subtitle" : "<string>", // Subordinate title of the activity definition
  "status" : "<code>", // R!  draft | active | retired | unknown
  "experimental" : <boolean>, // For testing purposes, not real usage
  // subject[x]: Type of individual the activity definition is intended for. One of these 2:
  "subjectCodeableConcept" : { CodeableConcept },
  "subjectReference" : { Reference(Group) },
  "date" : "<dateTime>", // Date last changed
  "publisher" : "<string>", // Name of the publisher (organization or individual)
  "contact" : [{ ContactDetail }], // Contact details for the publisher
  "description" : "<markdown>", // Natural language description of the activity definition
  "useContext" : [{ UsageContext }], // The context that the content is intended to support
  "jurisdiction" : [{ CodeableConcept }], // Intended jurisdiction for activity definition (if applicable)
  "purpose" : "<markdown>", // Why this activity definition is defined
  "usage" : "<string>", // Describes the clinical usage of the activity definition
  "copyright" : "<markdown>", // Use and/or publishing restrictions
  "approvalDate" : "<date>", // When the activity definition was approved by publisher
  "lastReviewDate" : "<date>", // When the activity definition was last reviewed
  "effectivePeriod" : { Period }, // When the activity definition is expected to be used
  "topic" : [{ CodeableConcept }], // E.g. Education, Treatment, Assessment, etc.
  "author" : [{ ContactDetail }], // Who authored the content
  "editor" : [{ ContactDetail }], // Who edited the content
  "reviewer" : [{ ContactDetail }], // Who reviewed the content
  "endorser" : [{ ContactDetail }], // Who endorsed the content
  "relatedArtifact" : [{ RelatedArtifact }], // Additional documentation, citations, etc.
  "library" : [{ canonical(Library) }], // Logic used by the activity definition
  "kind" : "<code>", // Kind of resource
  "profile" : { canonical(StructureDefinition) }, // What profile the resource needs to conform to
  "code" : { CodeableConcept }, // Detail type of activity
  "intent" : "<code>", // proposal | plan | directive | order | original-order | reflex-order | filler-order | instance-
    order | option
  "priority" : "<code>", // routine | urgent | asap | stat
  "doNotPerform" : <boolean>, // True if the activity should not be performed
  // timing[x]: When activity is to occur. One of these 6:
  "timingTiming" : { Timing },
  "timingDateTime" : "<dateTime>",
  "timingAge" : { Age },
  "timingPeriod" : { Period },
  "timingRange" : { Range },
  "timingDuration" : { Duration },
  "location" : { Reference(Location) }, // Where it should happen
  "participant" : [{ // Who should participate in the action
    "type" : "<code>", // R!  patient | practitioner | related-person | device
    "role" : { CodeableConcept } // E.g. Nurse, Surgeon, Parent, etc.
  }],
  // product[x]: What's administered/supplied. One of these 2:
  "productReference" : { Reference(Medication|Substance) },
  "productCodeableConcept" : { CodeableConcept },
  "quantity" : { Quantity(SimpleQuantity) }, // How much is administered/consumed/supplied
  "dosage" : [{ Dosage }], // Detailed dosage instructions
  "bodySite" : [{ CodeableConcept }], // What part of body to perform on
  "specimenRequirement" : [{ Reference(SpecimenDefinition) }], // What specimens are required to perform this action
  "observationRequirement" : [{ Reference(ObservationDefinition) }], // What observations are required to perform this
    action
  "observationResultRequirement" : [{ Reference(ObservationDefinition) }], // What observations must be produced by this
     action
  "transform" : { canonical(StructureMap) }, // Transform to apply the template
  "dynamicValue" : [{ // Dynamic aspects of the definition
    "path" : "<string>", // R!  The path to the element to be set dynamically
    "expression" : { Expression } // R!  An expression that provides the dynamic value for the customization
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:ActivityDefinition;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:ActivityDefinition.url [ uri ]; # 0..1 Canonical identifier for this activity definition, represented as a URI
    (globally unique)
  fhir:ActivityDefinition.identifier [ Identifier ], ... ; # 0..* Additional identifier for the activity definition
  fhir:ActivityDefinition.version [ string ]; # 0..1 Business version of the activity definition
  fhir:ActivityDefinition.name [ string ]; # 0..1 Name for this activity definition (computer friendly)
  fhir:ActivityDefinition.title [ string ]; # 0..1 Name for this activity definition (human friendly)
  fhir:ActivityDefinition.subtitle [ string ]; # 0..1 Subordinate title of the activity definition
  fhir:ActivityDefinition.status [ code ]; # 1..1 draft | active | retired | unknown
  fhir:ActivityDefinition.experimental [ boolean ]; # 0..1 For testing purposes, not real usage
```

```
  # ActivityDefinition.subject[x] : 0..1 Type of individual the activity definition is intended for. One of these 2
    fhir:ActivityDefinition.subjectCodeableConcept [ CodeableConcept ]
    fhir:ActivityDefinition.subjectReference [ Reference(Group) ]
  fhir:ActivityDefinition.date [ dateTime ]; # 0..1 Date last changed
  fhir:ActivityDefinition.publisher [ string ]; # 0..1 Name of the publisher (organization or individual)
  fhir:ActivityDefinition.contact [ ContactDetail ], ... ; # 0..* Contact details for the publisher
  fhir:ActivityDefinition.description [ markdown ]; # 0..1 Natural language description of the activity definition
  fhir:ActivityDefinition.useContext [ UsageContext ], ... ; # 0..* The context that the content is intended to support
  fhir:ActivityDefinition.jurisdiction [ CodeableConcept ], ... ; # 0..* Intended jurisdiction for activity definition
    (if applicable)
  fhir:ActivityDefinition.purpose [ markdown ]; # 0..1 Why this activity definition is defined
  fhir:ActivityDefinition.usage [ string ]; # 0..1 Describes the clinical usage of the activity definition
  fhir:ActivityDefinition.copyright [ markdown ]; # 0..1 Use and/or publishing restrictions
  fhir:ActivityDefinition.approvalDate [ date ]; # 0..1 When the activity definition was approved by publisher
  fhir:ActivityDefinition.lastReviewDate [ date ]; # 0..1 When the activity definition was last reviewed
  fhir:ActivityDefinition.effectivePeriod [ Period ]; # 0..1 When the activity definition is expected to be used
  fhir:ActivityDefinition.topic [ CodeableConcept ], ... ; # 0..* E.g. Education, Treatment, Assessment, etc.
  fhir:ActivityDefinition.author [ ContactDetail ], ... ; # 0..* Who authored the content
  fhir:ActivityDefinition.editor [ ContactDetail ], ... ; # 0..* Who edited the content
  fhir:ActivityDefinition.reviewer [ ContactDetail ], ... ; # 0..* Who reviewed the content
  fhir:ActivityDefinition.endorser [ ContactDetail ], ... ; # 0..* Who endorsed the content
  fhir:ActivityDefinition.relatedArtifact [ RelatedArtifact ], ... ; # 0..* Additional documentation, citations, etc.
  fhir:ActivityDefinition.library [ canonical(Library) ], ... ; # 0..* Logic used by the activity definition
  fhir:ActivityDefinition.kind [ code ]; # 0..1 Kind of resource
  fhir:ActivityDefinition.profile [ canonical(StructureDefinition) ]; # 0..1 What profile the resource needs to conform
    to
  fhir:ActivityDefinition.code [ CodeableConcept ]; # 0..1 Detail type of activity
  fhir:ActivityDefinition.intent [ code ]; # 0..1 proposal | plan | directive | order | original-order | reflex-order |
    filler-order | instance-order | option
  fhir:ActivityDefinition.priority [ code ]; # 0..1 routine | urgent | asap | stat
  fhir:ActivityDefinition.doNotPerform [ boolean ]; # 0..1 True if the activity should not be performed
  # ActivityDefinition.timing[x] : 0..1 When activity is to occur. One of these 6
    fhir:ActivityDefinition.timingTiming [ Timing ]
    fhir:ActivityDefinition.timingDateTime [ dateTime ]
    fhir:ActivityDefinition.timingAge [ Age ]
    fhir:ActivityDefinition.timingPeriod [ Period ]
    fhir:ActivityDefinition.timingRange [ Range ]
    fhir:ActivityDefinition.timingDuration [ Duration ]
  fhir:ActivityDefinition.location [ Reference(Location) ]; # 0..1 Where it should happen
  fhir:ActivityDefinition.participant [ # 0..* Who should participate in the action
    fhir:ActivityDefinition.participant.type [ code ]; # 1..1 patient | practitioner | related-person | device
    fhir:ActivityDefinition.participant.role [ CodeableConcept ]; # 0..1 E.g. Nurse, Surgeon, Parent, etc.
  ], ...;
  # ActivityDefinition.product[x] : 0..1 What's administered/supplied. One of these 2
    fhir:ActivityDefinition.productReference [ Reference(Medication|Substance) ]
    fhir:ActivityDefinition.productCodeableConcept [ CodeableConcept ]
  fhir:ActivityDefinition.quantity [ Quantity(SimpleQuantity) ]; # 0..1 How much is administered/consumed/supplied
  fhir:ActivityDefinition.dosage [ Dosage ], ... ; # 0..* Detailed dosage instructions
  fhir:ActivityDefinition.bodySite [ CodeableConcept ], ... ; # 0..* What part of body to perform on
  fhir:ActivityDefinition.specimenRequirement [ Reference(SpecimenDefinition) ], ... ; # 0..* What specimens are
    required to perform this action
  fhir:ActivityDefinition.observationRequirement [ Reference(ObservationDefinition) ], ... ; # 0..* What observations
    are required to perform this action
  fhir:ActivityDefinition.observationResultRequirement [ Reference(ObservationDefinition) ], ... ; # 0..* What
    observations must be produced by this action
  fhir:ActivityDefinition.transform [ canonical(StructureMap) ]; # 0..1 Transform to apply the template
  fhir:ActivityDefinition.dynamicValue [ # 0..* Dynamic aspects of the definition
    fhir:ActivityDefinition.dynamicValue.path [ string ]; # 1..1 The path to the element to be set dynamically
    fhir:ActivityDefinition.dynamicValue.expression [ Expression ]; # 1..1 An expression that provides the dynamic value
     for the customization
  ], ...;
]
```

Changes since R3

ActivityDefinition

ActivityDefinition.subtitle | Added Element

ActivityDefinition.status | Change value set from http://hl7.org/fhir/ValueSet/publication-status to http://hl7.org/fhir/ValueSet/publication-status|4.0.1

ActivityDefinition.experimental | No longer marked as Modifier

ActivityDefinition.subject[x] | Added Element

ActivityDefinition.author | Added Element

ActivityDefinition.editor | Added Element

ActivityDefinition.reviewer | Added Element

ActivityDefinition.endorser | Added Element

ActivityDefinition.library | Type changed from Reference(Library) to canonical(Library)

ActivityDefinition.kind | Change value set from http://hl7.org/fhir/ValueSet/resource-types to http://hl7.org/fhir/ValueSet/request-resource-types|4.0.1

ActivityDefinition.profile | Added Element

ActivityDefinition.intent | Added Element

ActivityDefinition.priority | Added Element

ActivityDefinition.doNotPerform | Added Element

ActivityDefinition.timing[x] | Add Types Age, Duration

ActivityDefinition.participant.type | Change value set from http://hl7.org/fhir/ValueSet/action-participant-type to http://hl7.org/fhir/ValueSet/action-participant-type|4.0.1

ActivityDefinition.specimenRequirement | Added Element
ActivityDefinition.observationRequirement | Added Element
ActivityDefinition.observationResultRequirement | Added Element
ActivityDefinition.transform | Type changed from Reference(StructureMap) to canonical(StructureMap)
ActivityDefinition.dynamicValue.path | Min Cardinality changed from 0 to 1
ActivityDefinition.dynamicValue.expression | Min Cardinality changed from 0 to 1 Type changed from string to Expression
ActivityDefinition.contributor | deleted
ActivityDefinition.dynamicValue.description | deleted
ActivityDefinition.dynamicValue.language | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 9 tests that all execute ok. 2 fail round-trip testing and 9 r3 resources are invalid (0 errors).)

Structure

Name | Flags | Card. | Type | Description & Constraints
ActivityDefinition | ITU | | DomainResource | The definition of a specific activity to be taken, independent of any particular patient or context + Warning: Name should be usable as an identifier for the module by machine processing applications such as code generation Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
url | Σ | 0..1 | uri | Canonical identifier for this activity definition, represented as a URI (globally unique)
identifier | Σ | 0..* | Identifier | Additional identifier for the activity definition
version | Σ | 0..1 | string | Business version of the activity definition
name | ΣI | 0..1 | string | Name for this activity definition (computer friendly)
title | Σ | 0..1 | string | Name for this activity definition (human friendly)
subtitle | | 0..1 | string | Subordinate title of the activity definition
status | ?!Σ | 1..1 | code | draft | active | retired | unknown PublicationStatus (Required)
experimental | Σ | 0..1 | boolean | For testing purposes, not real usage
subject[x] | | 0..1 | | Type of individual the activity definition is intended for Subject type (Extensible)
subjectCodeableConcept | | | CodeableConcept
subjectReference | | | Reference(Group)
date | Σ | 0..1 | dateTime | Date last changed
publisher | Σ | 0..1 | string | Name of the publisher (organization or individual)
contact | Σ | 0..* | ContactDetail | Contact details for the publisher
description | Σ | 0..1 | markdown | Natural language description of the activity definition
useContext | Σ | 0..* | UsageContext | The context that the content is intended to support
jurisdiction | Σ | 0..* | CodeableConcept | Intended jurisdiction for activity definition (if applicable) Jurisdiction (Extensible)
purpose | | 0..1 | markdown | Why this activity definition is defined
usage | | 0..1 | string | Describes the clinical usage of the activity definition
copyright | | 0..1 | markdown | Use and/or publishing restrictions
approvalDate | | 0..1 | date | When the activity definition was approved by publisher
lastReviewDate | | 0..1 | date | When the activity definition was last reviewed
effectivePeriod | Σ | 0..1 | Period | When the activity definition is expected to be used
topic | | 0..* | CodeableConcept | E.g. Education, Treatment, Assessment, etc. DefinitionTopic (Example)
author | | 0..* | ContactDetail | Who authored the content
editor | | 0..* | ContactDetail | Who edited the content
reviewer | | 0..* | ContactDetail | Who reviewed the content
endorser | | 0..* | ContactDetail | Who endorsed the content
relatedArtifact | | 0..* | RelatedArtifact | Additional documentation, citations, etc.
library | | 0..* | canonical(Library) | Logic used by the activity definition
kind | Σ | 0..1 | code | Kind of resource RequestResourceType (Required)
profile | | 0..1 | canonical(StructureDefinition) | What profile the resource needs to conform to
code | Σ | 0..1 | CodeableConcept | Detail type of activity Procedure Codes (SNOMED CT) (Example)
intent | | 0..1 | code | proposal | plan | directive | order | original-order | reflex-order | filler-order | instance-order | option RequestIntent (Required)
priority | | 0..1 | code | routine | urgent | asap | stat Request priority (Required)
doNotPerform | ?!Σ | 0..1 | boolean | True if the activity should not be performed
timing[x] | | 0..1 | | When activity is to occur
timingTiming | | | Timing
timingDateTime | | | dateTime
timingAge | | | Age
timingPeriod | | | Period
timingRange | | | Range
timingDuration | | | Duration
location | | 0..1 | Reference(Location) | Where it should happen
participant | | 0..* | BackboneElement | Who should participate in the action
type | | 1..1 | code | patient | practitioner | related-person | device ActionParticipantType (Required)
role | | 0..1 | CodeableConcept | E.g. Nurse, Surgeon, Parent, etc. Action participant role (Example)
product[x] | | 0..1 | | What's administered/supplied SNOMED CT Medication Codes (Example)
productReference | | | Reference(Medication | Substance)
productCodeableConcept | | | CodeableConcept
quantity | | 0..1 | SimpleQuantity | How much is administered/consumed/supplied
dosage | | 0..* | Dosage | Detailed dosage instructions

bodySite | | 0..* | CodeableConcept | What part of body to perform on SNOMED CT Body Structures (Example)
specimenRequirement | | 0..* | Reference(SpecimenDefinition) | What specimens are required to perform this action
observationRequirement | | 0..* | Reference(ObservationDefinition) | What observations are required to perform this action
observationResultRequirement | | 0..* | Reference(ObservationDefinition) | What observations must be produced by this action
transform | | 0..1 | canonical(StructureMap) | Transform to apply the template
dynamicValue | | 0..* | BackboneElement | Dynamic aspects of the definition
path | | 1..1 | string | The path to the element to be set dynamically
expression | | 1..1 | Expression | An expression that provides the dynamic value for the customization
Documentation for this format

UML Diagram (Legend)

XML Template

```
<ActivityDefinition xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <url value="[uri]"/><!-- 0..1 Canonical identifier for this activity definition, represented as a URI (globally unique)
     -->
 <identifier><!-- 0..* Identifier Additional identifier for the activity definition --></identifier>
 <version value="[string]"/><!-- 0..1 Business version of the activity definition -->
 <name value="[string]"/><!--  0..1 Name for this activity definition (computer friendly) -->
 <title value="[string]"/><!-- 0..1 Name for this activity definition (human friendly) -->
 <subtitle value="[string]"/><!-- 0..1 Subordinate title of the activity definition -->
 <status value="[code]"/><!-- 1..1 draft | active | retired | unknown -->
 <experimental value="[boolean]"/><!-- 0..1 For testing purposes, not real usage -->
 <subject[x]><!-- 0..1 CodeableConcept|Reference(Group) Type of individual the activity definition is intended for
    --></subject[x]>
 <date value="[dateTime]"/><!-- 0..1 Date last changed -->
 <publisher value="[string]"/><!-- 0..1 Name of the publisher (organization or individual) -->
 <contact><!-- 0..* ContactDetail Contact details for the publisher --></contact>
 <description value="[markdown]"/><!-- 0..1 Natural language description of the activity definition -->
 <useContext><!-- 0..* UsageContext The context that the content is intended to support --></useContext>
 <jurisdiction><!-- 0..* CodeableConcept Intended jurisdiction for activity definition (if applicable)
    --></jurisdiction>
 <purpose value="[markdown]"/><!-- 0..1 Why this activity definition is defined -->
 <usage value="[string]"/><!-- 0..1 Describes the clinical usage of the activity definition -->
 <copyright value="[markdown]"/><!-- 0..1 Use and/or publishing restrictions -->
 <approvalDate value="[date]"/><!-- 0..1 When the activity definition was approved by publisher -->
 <lastReviewDate value="[date]"/><!-- 0..1 When the activity definition was last reviewed -->
 <effectivePeriod><!-- 0..1 Period When the activity definition is expected to be used --></effectivePeriod>
 <topic><!-- 0..* CodeableConcept E.g. Education, Treatment, Assessment, etc. --></topic>
 <author><!-- 0..* ContactDetail Who authored the content --></author>
 <editor><!-- 0..* ContactDetail Who edited the content --></editor>
 <reviewer><!-- 0..* ContactDetail Who reviewed the content --></reviewer>
 <endorser><!-- 0..* ContactDetail Who endorsed the content --></endorser>
 <relatedArtifact><!-- 0..* RelatedArtifact Additional documentation, citations, etc. --></relatedArtifact>
 <library><!-- 0..* canonical(Library) Logic used by the activity definition --></library>
 <kind value="[code]"/><!-- 0..1 Kind of resource -->
 <profile><!-- 0..1 canonical(StructureDefinition) What profile the resource needs to conform to --></profile>
 <code><!-- 0..1 CodeableConcept Detail type of activity --></code>
 <intent value="[code]"/><!-- 0..1 proposal | plan | directive | order | original-order | reflex-order | filler-order |
    instance-order | option -->
 <priority value="[code]"/><!-- 0..1 routine | urgent | asap | stat -->
 <doNotPerform value="[boolean]"/><!-- 0..1 True if the activity should not be performed -->
 <timing[x]><!-- 0..1 Timing|dateTime|Age|Period|Range|Duration When activity is to occur --></timing[x]>
 <location><!-- 0..1 Reference(Location) Where it should happen --></location>
 <participant>  <!-- 0..* Who should participate in the action -->
  <type value="[code]"/><!-- 1..1 patient | practitioner | related-person | device -->
  <role><!-- 0..1 CodeableConcept E.g. Nurse, Surgeon, Parent, etc. --></role>
 </participant>
 <product[x]><!-- 0..1 Reference(Medication|Substance)|CodeableConcept What's administered/supplied --></product[x]>
 <quantity><!-- 0..1 Quantity(SimpleQuantity) How much is administered/consumed/supplied --></quantity>
 <dosage><!-- 0..* Dosage Detailed dosage instructions --></dosage>
 <bodySite><!-- 0..* CodeableConcept What part of body to perform on --></bodySite>
 <specimenRequirement><!-- 0..* Reference(SpecimenDefinition) What specimens are required to perform this action
    --></specimenRequirement>
 <observationRequirement><!-- 0..* Reference(ObservationDefinition) What observations are required to perform this
    action --></observationRequirement>
 <observationResultRequirement><!-- 0..* Reference(ObservationDefinition) What observations must be produced by this
    action --></observationResultRequirement>
 <transform><!-- 0..1 canonical(StructureMap) Transform to apply the template --></transform>
 <dynamicValue>  <!-- 0..* Dynamic aspects of the definition -->
  <path value="[string]"/><!-- 1..1 The path to the element to be set dynamically -->
  <expression><!-- 1..1 Expression An expression that provides the dynamic value for the customization --></expression>
 </dynamicValue>
</ActivityDefinition>
```

JSON Template

```
{
  "resourceType" : "ActivityDefinition",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "url" : "<uri>", // Canonical identifier for this activity definition, represented as a URI (globally unique)
  "identifier" : [{ Identifier }], // Additional identifier for the activity definition
  "version" : "<string>", // Business version of the activity definition
  "name" : "<string>", // C? Name for this activity definition (computer friendly)
  "title" : "<string>", // Name for this activity definition (human friendly)
```

```
  "subtitle" : "<string>", // Subordinate title of the activity definition
  "status" : "<code>", // R!  draft | active | retired | unknown
  "experimental" : <boolean>, // For testing purposes, not real usage
  // subject[x]: Type of individual the activity definition is intended for. One of these 2:
  "subjectCodeableConcept" : { CodeableConcept },
  "subjectReference" : { Reference(Group) },
  "date" : "<dateTime>", // Date last changed
  "publisher" : "<string>", // Name of the publisher (organization or individual)
  "contact" : [{ ContactDetail }], // Contact details for the publisher
  "description" : "<markdown>", // Natural language description of the activity definition
  "useContext" : [{ UsageContext }], // The context that the content is intended to support
  "jurisdiction" : [{ CodeableConcept }], // Intended jurisdiction for activity definition (if applicable)
  "purpose" : "<markdown>", // Why this activity definition is defined
  "usage" : "<string>", // Describes the clinical usage of the activity definition
  "copyright" : "<markdown>", // Use and/or publishing restrictions
  "approvalDate" : "<date>", // When the activity definition was approved by publisher
  "lastReviewDate" : "<date>", // When the activity definition was last reviewed
  "effectivePeriod" : { Period }, // When the activity definition is expected to be used
  "topic" : [{ CodeableConcept }], // E.g. Education, Treatment, Assessment, etc.
  "author" : [{ ContactDetail }], // Who authored the content
  "editor" : [{ ContactDetail }], // Who edited the content
  "reviewer" : [{ ContactDetail }], // Who reviewed the content
  "endorser" : [{ ContactDetail }], // Who endorsed the content
  "relatedArtifact" : [{ RelatedArtifact }], // Additional documentation, citations, etc.
  "library" : [{ canonical(Library) }], // Logic used by the activity definition
  "kind" : "<code>", // Kind of resource
  "profile" : { canonical(StructureDefinition) }, // What profile the resource needs to conform to
  "code" : { CodeableConcept }, // Detail type of activity
  "intent" : "<code>", // proposal | plan | directive | order | original-order | reflex-order | filler-order | instance-
    order | option
  "priority" : "<code>", // routine | urgent | asap | stat
  "doNotPerform" : <boolean>, // True if the activity should not be performed
  // timing[x]: When activity is to occur. One of these 6:
  "timingTiming" : { Timing },
  "timingDateTime" : "<dateTime>",
  "timingAge" : { Age },
  "timingPeriod" : { Period },
  "timingRange" : { Range },
  "timingDuration" : { Duration },
  "location" : { Reference(Location) }, // Where it should happen
  "participant" : [{ // Who should participate in the action
    "type" : "<code>", // R!  patient | practitioner | related-person | device
    "role" : { CodeableConcept } // E.g. Nurse, Surgeon, Parent, etc.
  }],
  // product[x]: What's administered/supplied. One of these 2:
  "productReference" : { Reference(Medication|Substance) },
  "productCodeableConcept" : { CodeableConcept },
  "quantity" : { Quantity(SimpleQuantity) }, // How much is administered/consumed/supplied
  "dosage" : [{ Dosage }], // Detailed dosage instructions
  "bodySite" : [{ CodeableConcept }], // What part of body to perform on
  "specimenRequirement" : [{ Reference(SpecimenDefinition) }], // What specimens are required to perform this action
  "observationRequirement" : [{ Reference(ObservationDefinition) }], // What observations are required to perform this
    action
  "observationResultRequirement" : [{ Reference(ObservationDefinition) }], // What observations must be produced by this
     action
  "transform" : { canonical(StructureMap) }, // Transform to apply the template
  "dynamicValue" : [{ // Dynamic aspects of the definition
    "path" : "<string>", // R!  The path to the element to be set dynamically
    "expression" : { Expression } // R!  An expression that provides the dynamic value for the customization
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:ActivityDefinition;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:ActivityDefinition.url [ uri ]; # 0..1 Canonical identifier for this activity definition, represented as a URI
    (globally unique)
  fhir:ActivityDefinition.identifier [ Identifier ], ... ; # 0..* Additional identifier for the activity definition
  fhir:ActivityDefinition.version [ string ]; # 0..1 Business version of the activity definition
  fhir:ActivityDefinition.name [ string ]; # 0..1 Name for this activity definition (computer friendly)
  fhir:ActivityDefinition.title [ string ]; # 0..1 Name for this activity definition (human friendly)
  fhir:ActivityDefinition.subtitle [ string ]; # 0..1 Subordinate title of the activity definition
  fhir:ActivityDefinition.status [ code ]; # 1..1 draft | active | retired | unknown
  fhir:ActivityDefinition.experimental [ boolean ]; # 0..1 For testing purposes, not real usage
  # ActivityDefinition.subject[x] : 0..1 Type of individual the activity definition is intended for. One of these 2
    fhir:ActivityDefinition.subjectCodeableConcept [ CodeableConcept ]
    fhir:ActivityDefinition.subjectReference [ Reference(Group) ]
  fhir:ActivityDefinition.date [ dateTime ]; # 0..1 Date last changed
  fhir:ActivityDefinition.publisher [ string ]; # 0..1 Name of the publisher (organization or individual)
  fhir:ActivityDefinition.contact [ ContactDetail ], ... ; # 0..* Contact details for the publisher
  fhir:ActivityDefinition.description [ markdown ]; # 0..1 Natural language description of the activity definition
  fhir:ActivityDefinition.useContext [ UsageContext ], ... ; # 0..* The context that the content is intended to support
  fhir:ActivityDefinition.jurisdiction [ CodeableConcept ], ... ; # 0..* Intended jurisdiction for activity definition
```

```
    (if applicable)
  fhir:ActivityDefinition.purpose [ markdown ]; # 0..1 Why this activity definition is defined
  fhir:ActivityDefinition.usage [ string ]; # 0..1 Describes the clinical usage of the activity definition
  fhir:ActivityDefinition.copyright [ markdown ]; # 0..1 Use and/or publishing restrictions
  fhir:ActivityDefinition.approvalDate [ date ]; # 0..1 When the activity definition was approved by publisher
  fhir:ActivityDefinition.lastReviewDate [ date ]; # 0..1 When the activity definition was last reviewed
  fhir:ActivityDefinition.effectivePeriod [ Period ]; # 0..1 When the activity definition is expected to be used
  fhir:ActivityDefinition.topic [ CodeableConcept ], ... ; # 0..* E.g. Education, Treatment, Assessment, etc.
  fhir:ActivityDefinition.author [ ContactDetail ], ... ; # 0..* Who authored the content
  fhir:ActivityDefinition.editor [ ContactDetail ], ... ; # 0..* Who edited the content
  fhir:ActivityDefinition.reviewer [ ContactDetail ], ... ; # 0..* Who reviewed the content
  fhir:ActivityDefinition.endorser [ ContactDetail ], ... ; # 0..* Who endorsed the content
  fhir:ActivityDefinition.relatedArtifact [ RelatedArtifact ], ... ; # 0..* Additional documentation, citations, etc.
  fhir:ActivityDefinition.library [ canonical(Library) ], ... ; # 0..* Logic used by the activity definition
  fhir:ActivityDefinition.kind [ code ]; # 0..1 Kind of resource
  fhir:ActivityDefinition.profile [ canonical(StructureDefinition) ]; # 0..1 What profile the resource needs to conform
    to
  fhir:ActivityDefinition.code [ CodeableConcept ]; # 0..1 Detail type of activity
  fhir:ActivityDefinition.intent [ code ]; # 0..1 proposal | plan | directive | order | original-order | reflex-order |
    filler-order | instance-order | option
  fhir:ActivityDefinition.priority [ code ]; # 0..1 routine | urgent | asap | stat
  fhir:ActivityDefinition.doNotPerform [ boolean ]; # 0..1 True if the activity should not be performed
  # ActivityDefinition.timing[x] : 0..1 When activity is to occur. One of these 6
    fhir:ActivityDefinition.timingTiming [ Timing ]
    fhir:ActivityDefinition.timingDateTime [ dateTime ]
    fhir:ActivityDefinition.timingAge [ Age ]
    fhir:ActivityDefinition.timingPeriod [ Period ]
    fhir:ActivityDefinition.timingRange [ Range ]
    fhir:ActivityDefinition.timingDuration [ Duration ]
  fhir:ActivityDefinition.location [ Reference(Location) ]; # 0..1 Where it should happen
  fhir:ActivityDefinition.participant [ # 0..* Who should participate in the action
    fhir:ActivityDefinition.participant.type [ code ]; # 1..1 patient | practitioner | related-person | device
    fhir:ActivityDefinition.participant.role [ CodeableConcept ]; # 0..1 E.g. Nurse, Surgeon, Parent, etc.
  ], ...;
  # ActivityDefinition.product[x] : 0..1 What's administered/supplied. One of these 2
    fhir:ActivityDefinition.productReference [ Reference(Medication|Substance) ]
    fhir:ActivityDefinition.productCodeableConcept [ CodeableConcept ]
  fhir:ActivityDefinition.quantity [ Quantity(SimpleQuantity) ]; # 0..1 How much is administered/consumed/supplied
  fhir:ActivityDefinition.dosage [ Dosage ], ... ; # 0..* Detailed dosage instructions
  fhir:ActivityDefinition.bodySite [ CodeableConcept ], ... ; # 0..* What part of body to perform on
  fhir:ActivityDefinition.specimenRequirement [ Reference(SpecimenDefinition) ], ... ; # 0..* What specimens are
    required to perform this action
  fhir:ActivityDefinition.observationRequirement [ Reference(ObservationDefinition) ], ... ; # 0..* What observations
    are required to perform this action
  fhir:ActivityDefinition.observationResultRequirement [ Reference(ObservationDefinition) ], ... ; # 0..* What
    observations must be produced by this action
  fhir:ActivityDefinition.transform [ canonical(StructureMap) ]; # 0..1 Transform to apply the template
  fhir:ActivityDefinition.dynamicValue [ # 0..* Dynamic aspects of the definition
    fhir:ActivityDefinition.dynamicValue.path [ string ]; # 1..1 The path to the element to be set dynamically
    fhir:ActivityDefinition.dynamicValue.expression [ Expression ]; # 1..1 An expression that provides the dynamic value
     for the customization
  ], ...;
]
```

Changes since Release 3

ActivityDefinition
ActivityDefinition.subtitle | Added Element
ActivityDefinition.status | Change value set from http://hl7.org/fhir/ValueSet/publication-status to http://hl7.org/fhir/ValueSet/publication-status|4.0.1
ActivityDefinition.experimental | No longer marked as Modifier
ActivityDefinition.subject[x] | Added Element
ActivityDefinition.author | Added Element
ActivityDefinition.editor | Added Element
ActivityDefinition.reviewer | Added Element
ActivityDefinition.endorser | Added Element
ActivityDefinition.library | Type changed from Reference(Library) to canonical(Library)
ActivityDefinition.kind | Change value set from http://hl7.org/fhir/ValueSet/resource-types to http://hl7.org/fhir/ValueSet/request-resource-types|4.0.1
ActivityDefinition.profile | Added Element
ActivityDefinition.intent | Added Element
ActivityDefinition.priority | Added Element
ActivityDefinition.doNotPerform | Added Element
ActivityDefinition.timing[x] | Add Types Age, Duration
ActivityDefinition.participant.type | Change value set from http://hl7.org/fhir/ValueSet/action-participant-type to http://hl7.org/fhir/ValueSet/action-participant-type|4.0.1
ActivityDefinition.specimenRequirement | Added Element
ActivityDefinition.observationRequirement | Added Element
ActivityDefinition.observationResultRequirement | Added Element
ActivityDefinition.transform | Type changed from Reference(StructureMap) to canonical(StructureMap)
ActivityDefinition.dynamicValue.path | Min Cardinality changed from 0 to 1
ActivityDefinition.dynamicValue.expression | Min Cardinality changed from 0 to 1 Type changed from string to Expression
ActivityDefinition.contributor | deleted
ActivityDefinition.dynamicValue.description | deleted

ActivityDefinition.dynamicValue.language | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 9 tests that all execute ok. 2 fail round-trip testing and 9 r3 resources are invalid (0 errors).)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

12.17.3.1 Terminology Bindings

Path | Definition | Type | Reference
ActivityDefinition.status | The lifecycle status of an artifact. | Required | PublicationStatus
ActivityDefinition.subject[x] | The possible types of subjects for an activity (E.g. Patient, Practitioner, Organization, Location, etc.). | Extensible | SubjectType
ActivityDefinition.jurisdiction | Countries and regions within which this artifact is targeted for use. | Extensible | Jurisdiction ValueSet
ActivityDefinition.topic | High-level categorization of the definition, used for searching, sorting, and filtering. | Example | DefinitionTopic
ActivityDefinition.kind | The kind of activity the definition is describing. | Required | RequestResourceType
ActivityDefinition.code | Detailed type of the activity; e.g. CBC. | Example | ProcedureCodes(SNOMEDCT)
ActivityDefinition.intent | Codes indicating the degree of authority/intentionality associated with a request. | Required | RequestIntent
ActivityDefinition.priority | Identifies the level of importance to be assigned to actioning the request. | Required | RequestPriority
ActivityDefinition.participant.type | The type of participant in the activity. | Required | ActionParticipantType
ActivityDefinition.participant.role | Defines roles played by participants for the action. | Example | ActionParticipantRole
ActivityDefinition.product[x] | Code describing the type of substance or medication. | Example | SNOMEDCTMedicationCodes
ActivityDefinition.bodySite | A code that identifies the anatomical location. | Example | SNOMEDCTBodyStructures

12.17.3.2 Constraints

id | Level | Location | Description | Expression
adf-0 | Warning | (base) | Name should be usable as an identifier for the module by machine processing applications such as code generation | name.matches('[A-Z]([A-Za-z0-9_]){0,254}')

12.17.3.3 Applying an ActivityDefinition

An ActivityDefinition resource provides a description, or template, of an action to performed. These actions can be purely text-based descriptions of the action to be performed, only interpretable by a human user, or they can be structured definitions with enough information to construct a resource to represent the request or activity directly. This process of converting the ActivityDefinition into a specific resource in a particular context is performed with the $apply operation.

In the simplest case, the process maps the elements specified in the ActivityDefinition to the corresponding elements on a resource of the appropriate type, using the kind element of the definition to determine the type of resource to be created.

More dynamic scenarios can be achieved in one of two ways, either by providing dynamicValue expressions, or by specifying a StructureMap that transforms the definition into the appropriate request resource.

Note that systems will likely vary widely in the degree of support they provide for the $apply operation. In particular, the fact that a system supports applying definitions of one category, does not imply that it supports applying definitions for all categories. For example, a service focused on medication order sets may have sophisticated support for MedicationRequest activities, but no support at all for the ServiceRequest activities.

The following diagram illustrates the relationship between the PlanDefinition and ActivityDefinition resources, as well as a typical application to CarePlan, RequestGroup, and Request resources:

To accommodate the variability in systems and use cases, this specification is not prescriptive about exactly how an ActivityDefinition is applied to produce target request or event resources. However, the approach depicted above provides a general framework for the process, and the following steps provide more detail on the potential approach:

• Create the target resource of the type specified by the kind element and focused on the Patient in context

• Set the status of the target resource to draft

• Apply the structural elements of the ActivityDefinition to the target resource such as code, timing, doNotPerform, product, quantity, dosage, and so on

• Resolve the participant element based on the user in context

• Resolve the location element based on the location in context

• If the transform element is specified, apply the transform to the resource. Note that the referenced StructureMap may actually construct the resource, rather than taking an instance. See the StructureMap for more information

• Apply any dynamicValue elements (in the order in which they appear in the ActivityDefinition resource) by evaluating the expression and setting the value of the appropriate element of the target resource (as

specified by the dynamicValue.path element)

Note that the parameters to the $apply operation are available within dynamicValue CQL and FHIRPath expressions as context variables, accessible by the name of the parameter prefixed with a percent (%) symbol. For example, to access the subject given to the apply, use the expression %subject.

12.17.3.4 Profiling ActivityDefinition

Because the ActivityDefinition resource can be used to describe many different types of request resources, profiles of the resource will be useful in communicating additional constraints and expectations about how the resource should be used in a particular context. For example, a profile of ActivityDefinition may specify that it is intended to describe medication orders within an order set. Such a profile would indicate that the quantity element should not be used, but the dosage element must be.

12.17.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
composed-of | reference | What resource is being referenced | ActivityDefinition.relatedArtifact.where(type='composed-of').resource (Any)
context | token | A use context assigned to the activity definition | (ActivityDefinition.useContext.value as CodeableConcept)
context-quantity | quantity | A quantity- or range-valued use context assigned to the activity definition | (ActivityDefinition.useContext.value as Quantity) | (ActivityDefinition.useContext.value as Range)
context-type | token | A type of use context assigned to the activity definition | ActivityDefinition.useContext.code
context-type-quantity | composite | A use context type and quantity- or range-based value assigned to the activity definition | On ActivityDefinition.useContext: context-type: code context-quantity: value.as(Quantity) | value.as(Range)
context-type-value | composite | A use context type and value assigned to the activity definition | On ActivityDefinition.useContext: context-type: code context: value.as(CodeableConcept)
date | date | The activity definition publication date | ActivityDefinition.date
depends-on | reference | What resource is being referenced | ActivityDefinition.relatedArtifact.where(type='depends-on').resource | ActivityDefinition.library (Library, Any)
derived-from | reference | What resource is being referenced | ActivityDefinition.relatedArtifact.where(type='derived-from').resource (Any)
description | string | The description of the activity definition | ActivityDefinition.description
effective | date | The time during which the activity definition is intended to be in use | ActivityDefinition.effectivePeriod
identifier | token | External identifier for the activity definition | ActivityDefinition.identifier
jurisdiction | token | Intended jurisdiction for the activity definition | ActivityDefinition.jurisdiction
name | string | Computationally friendly name of the activity definition | ActivityDefinition.name
predecessor | reference | What resource is being referenced | ActivityDefinition.relatedArtifact.where(type='predecessor').resource (Any)
publisher | string | Name of the publisher of the activity definition | ActivityDefinition.publisher
status | token | The current status of the activity definition | ActivityDefinition.status
successor | reference | What resource is being referenced | ActivityDefinition.relatedArtifact.where(type='successor').resource (Any)
title | string | The human-friendly name of the activity definition | ActivityDefinition.title
topic | token | Topics associated with the module | ActivityDefinition.topic
url | uri | The uri that identifies the activity definition | ActivityDefinition.url
version | token | The business version of the activity definition | ActivityDefinition.version

12.17.5 Resource ActivityDefinition - Examples

URL: https://hl7.org/fhir/R4/activitydefinition-examples.html

These examples illustrate the use of the ActivityDefinition resource to represent activities on various artifacts includes a Suicide Risk Assessment Order Set, and a Zika Virus Infection Management workflow.

12.17.5 Resource ActivityDefinition - Examples

Clinical Decision Support Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Example Name | id | Format
Activity definition to refer a patient to a mental health integrated care program | referralPrimaryCareMentalHealth | XML | JSON | Turtle
Activity definition to prescribe citalopram to a patient | citalopramPrescription | XML | JSON | Turtle
Definition for administering a Zika Virus Exposure assessment | administer-zika-virus-exposure-assessment | XML | JSON | Turtle
Definition for ordering a Serum Zika and Dengue Virus IgM Antibody Diagnostic | serum-zika-dengue-virus-igm | XML | JSON | Turtle
Definition for providing mosquito prevention advice | provide-mosquito-prevention-advice | XML | JSON | Turtle
Definition for heart valve replacement service request | heart-valve-replacement | XML | JSON | Turtle
Definition for blood collection tubes supply | blood-tubes-supply | XML | JSON | Turtle
Example of a predecessor/successor relationship | referralPrimaryCareMentalHealth-initial | XML | JSON | Turtle
Definition for ordering a Serum Dengue Virus IgM Antibody Diagnostic | serum-dengue-virus-igm | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

12.17.6 Resource ActivityDefinition - Detailed Descriptions

URL: https://hl7.org/fhir/R4/activitydefinition-definitions.html

12.17.6 Resource ActivityDefinition - Detailed Descriptions

Clinical Decision Support Work Group | Maturity Level: 2 | Trial Use | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the ActivityDefinition resource.

ActivityDefinition
Element Id | ActivityDefinition
Definition | This resource allows for the definition of some activity to be performed, independent of a particular patient, practitioner, or other performance context.
Cardinality | 0..*
Type | DomainResource
Invariants | Defined on this element adf-0 Warning Name should be usable as an identifier for the module by machine processing applications such as code generation name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
Defined on this element
adf-0 | Warning | Name should be usable as an identifier for the module by machine processing applications such as code generation | name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
ActivityDefinition.url
Element Id | ActivityDefinition.url
Definition | An absolute URI that is used to identify this activity definition when it is referenced in a specification, model, design or an instance; also called its canonical identifier. This SHOULD be globally unique and SHOULD be a literal address at which at which an authoritative instance of this activity definition is (or will be) published. This URL can be the target of a canonical reference. It SHALL remain the same when the activity definition is stored on different servers.
Cardinality | 0..1
Type | uri
Requirements | Allows the activity definition to be referenced by a single globally unique identifier.
Summary | true
Comments | Can be a urn:uuid: or a urn:oid: but real http: addresses are preferred. Multiple instances may share the same URL if they have a distinct version. The determination of when to create a new version of a resource (same url, new version) vs. defining a new artifact is up to the author. Considerations for making this decision are found in Technical and Business Versions. In some cases, the resource can no longer be found at the stated url, but the url itself cannot change. Implementations can use the meta.source element to indicate where the current master source of the resource can be found.
ActivityDefinition.identifier
Element Id | ActivityDefinition.identifier
Definition | A formal identifier that is used to identify this activity definition when it is represented in other formats, or referenced in a specification, model, design or an instance.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Requirements | Allows externally provided and/or usable business identifiers to be easily associated with the module.
Summary | true
Comments | Typically, this is used for identifiers that can go in an HL7 V3 II (instance identifier) data type, and can then identify this activity definition outside of FHIR, where it is not possible to use the logical URI.
ActivityDefinition.version
Element Id | ActivityDefinition.version
Definition | The identifier that is used to identify this version of the activity definition when it is referenced in a specification, model, design or instance. This is an arbitrary value managed by the activity definition author and is not expected to be globally unique. For example, it might be a timestamp (e.g. yyyymmdd) if a managed version is not available. There is also no expectation that versions can be placed in a lexicographical sequence. To provide a version consistent with the Decision Support Service specification, use the format Major.Minor.Revision (e.g. 1.0.0). For more information on versioning knowledge assets, refer to the Decision Support Service specification. Note that a version is required for non-experimental active assets.
Note | This is a business versionId, not a resource version id (see discussion)
Cardinality | 0..1
Type | string
Summary | true
Comments | There may be different activity definition instances that have the same identifier but different versions. The version can be appended to the url in a reference to allow a reference to a particular business version of the activity definition with the format [url]|[version].
ActivityDefinition.name
Element Id | ActivityDefinition.name
Definition | A natural language name identifying the activity definition. This name should be usable as an identifier for the module by machine processing applications such as code generation.
Cardinality | 0..1
Type | string
Requirements | Support human navigation and code generation.
Summary | true
Comments | The name is not expected to be globally unique. The name should be a simple alphanumeric type name to ensure that it is machine-processing friendly.
Invariants | Affect this element inv-0 Warning Name should be usable as an identifier for the module by machine processing applications such as code generation name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
Affect this element

inv-0 | Warning | Name should be usable as an identifier for the module by machine processing applications such as code generation | name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
ActivityDefinition.title
Element Id | ActivityDefinition.title
Definition | A short, descriptive, user-friendly title for the activity definition.
Cardinality | 0..1
Type | string
Summary | true
Comments | This name does not need to be machine-processing friendly and may contain punctuation, white-space, etc.
ActivityDefinition.subtitle
Element Id | ActivityDefinition.subtitle
Definition | An explanatory or alternate title for the activity definition giving additional information about its content.
Cardinality | 0..1
Type | string
ActivityDefinition.status
Element Id | ActivityDefinition.status
Definition | The status of this activity definition. Enables tracking the life-cycle of the content.
Cardinality | 1..1
Terminology Binding | PublicationStatus (Required)
Type | code
Is Modifier | true (Reason: This is labeled as "Is Modifier" because applications should not use a retired {{title}} without due consideration)
Summary | true
Comments | Allows filtering of activity definitions that are appropriate for use versus not.
ActivityDefinition.experimental
Element Id | ActivityDefinition.experimental
Definition | A Boolean value to indicate that this activity definition is authored for testing purposes (or education/evaluation/marketing) and is not intended to be used for genuine usage.
Cardinality | 0..1
Type | boolean
Requirements | Enables experimental content to be developed following the same lifecycle that would be used for a production-level activity definition.
Summary | true
Comments | Allows filtering of activity definitions that are appropriate for use versus not.
ActivityDefinition.subject[x]
Element Id | ActivityDefinition.subject[x]
Definition | A code or group definition that describes the intended subject of the activity being defined.
Cardinality | 0..1
Terminology Binding | Subject type (Extensible)
Type | CodeableConcept|Reference(Group)
[x] Note | See Choice of Data Types for further information about how to use [x]
Meaning if Missing | Patient
ActivityDefinition.date
Element Id | ActivityDefinition.date
Definition | The date (and optionally time) when the activity definition was published. The date must change when the business version changes and it must change if the status code changes. In addition, it should change when the substantive content of the activity definition changes.
Cardinality | 0..1
Type | dateTime
Alternate Names | Revision Date
Summary | true
Comments | Note that this is not the same as the resource last-modified-date, since the resource may be a secondary representation of the activity definition. Additional specific dates may be added as extensions or be found by consulting Provenances associated with past versions of the resource.
ActivityDefinition.publisher
Element Id | ActivityDefinition.publisher
Definition | The name of the organization or individual that published the activity definition.
Cardinality | 0..1
Type | string
Requirements | Helps establish the "authority/credibility" of the activity definition. May also allow for contact.
Summary | true
Comments | Usually an organization but may be an individual. The publisher (or steward) of the activity definition is the organization or individual primarily responsible for the maintenance and upkeep of the activity definition. This is not necessarily the same individual or organization that developed and initially authored the content. The publisher is the primary point of contact for questions or issues with the activity definition. This item SHOULD be populated unless the information is available from context.
ActivityDefinition.contact
Element Id | ActivityDefinition.contact
Definition | Contact details to assist a user in finding and communicating with the publisher.
Cardinality | 0..*
Type | ContactDetail
Summary | true
Comments | May be a web site, an email address, a telephone number, etc.

ActivityDefinition.description
Element Id | ActivityDefinition.description
Definition | A free text natural language description of the activity definition from a consumer's perspective.
Cardinality | 0..1
Type | markdown
Summary | true
Comments | This description can be used to capture details such as why the activity definition was built, comments about misuse, instructions for clinical use and interpretation, literature references, examples from the paper world, etc. It is not a rendering of the activity definition as conveyed in the 'text' field of the resource itself. This item SHOULD be populated unless the information is available from context (e.g. the language of the activity definition is presumed to be the predominant language in the place the activity definition was created).
ActivityDefinition.useContext
Element Id | ActivityDefinition.useContext
Definition | The content was developed with a focus and intent of supporting the contexts that are listed. These contexts may be general categories (gender, age, ...) or may be references to specific programs (insurance plans, studies, ...) and may be used to assist with indexing and searching for appropriate activity definition instances.
Cardinality | 0..*
Type | UsageContext
Requirements | Assist in searching for appropriate content.
Summary | true
Comments | When multiple useContexts are specified, there is no expectation that all or any of the contexts apply.
ActivityDefinition.jurisdiction
Element Id | ActivityDefinition.jurisdiction
Definition | A legal or geographic region in which the activity definition is intended to be used.
Cardinality | 0..*
Terminology Binding | Jurisdiction (Extensible)
Type | CodeableConcept
Summary | true
Comments | It may be possible for the activity definition to be used in jurisdictions other than those for which it was originally designed or intended.
ActivityDefinition.purpose
Element Id | ActivityDefinition.purpose
Definition | Explanation of why this activity definition is needed and why it has been designed as it has.
Cardinality | 0..1
Type | markdown
Comments | This element does not describe the usage of the activity definition. Instead, it provides traceability of ''why'' the resource is either needed or ''why'' it is defined as it is. This may be used to point to source materials or specifications that drove the structure of this activity definition.
ActivityDefinition.usage
Element Id | ActivityDefinition.usage
Definition | A detailed description of how the activity definition is used from a clinical perspective.
Cardinality | 0..1
Type | string
ActivityDefinition.copyright
Element Id | ActivityDefinition.copyright
Definition | A copyright statement relating to the activity definition and/or its contents. Copyright statements are generally legal restrictions on the use and publishing of the activity definition.
Cardinality | 0..1
Type | markdown
Requirements | Consumers must be able to determine any legal restrictions on the use of the activity definition and/or its content.
Alternate Names | License; Restrictions
ActivityDefinition.approvalDate
Element Id | ActivityDefinition.approvalDate
Definition | The date on which the resource content was approved by the publisher. Approval happens once when the content is officially approved for usage.
Cardinality | 0..1
Type | date
Comments | The 'date' element may be more recent than the approval date because of minor changes or editorial corrections.
ActivityDefinition.lastReviewDate
Element Id | ActivityDefinition.lastReviewDate
Definition | The date on which the resource content was last reviewed. Review happens periodically after approval but does not change the original approval date.
Cardinality | 0..1
Type | date
Requirements | Gives a sense of how "current" the content is. Resources that have not been reviewed in a long time may have a risk of being less appropriate/relevant.
Comments | If specified, this date follows the original approval date.
ActivityDefinition.effectivePeriod
Element Id | ActivityDefinition.effectivePeriod
Definition | The period during which the activity definition content was or is planned to be in active use.
Cardinality | 0..1
Type | Period

Requirements | Allows establishing a transition before a resource comes into effect and also allows for a sunsetting process when new versions of the activity definition are or are expected to be used instead.
Summary | true
Comments | The effective period for a activity definition determines when the content is applicable for usage and is independent of publication and review dates. For example, a measure intended to be used for the year 2016 might be published in 2015.
ActivityDefinition.topic
Element Id | ActivityDefinition.topic
Definition | Descriptive topics related to the content of the activity. Topics provide a high-level categorization of the activity that can be useful for filtering and searching.
Cardinality | 0..*
Terminology Binding | DefinitionTopic (Example)
Type | CodeableConcept
Requirements | Repositories must be able to determine how to categorize the activity definition so that it can be found by topical searches.
ActivityDefinition.author
Element Id | ActivityDefinition.author
Definition | An individiual or organization primarily involved in the creation and maintenance of the content.
Cardinality | 0..*
Type | ContactDetail
ActivityDefinition.editor
Element Id | ActivityDefinition.editor
Definition | An individual or organization primarily responsible for internal coherence of the content.
Cardinality | 0..*
Type | ContactDetail
ActivityDefinition.reviewer
Element Id | ActivityDefinition.reviewer
Definition | An individual or organization primarily responsible for review of some aspect of the content.
Cardinality | 0..*
Type | ContactDetail
ActivityDefinition.endorser
Element Id | ActivityDefinition.endorser
Definition | An individual or organization responsible for officially endorsing the content for use in some setting.
Cardinality | 0..*
Type | ContactDetail
ActivityDefinition.relatedArtifact
Element Id | ActivityDefinition.relatedArtifact
Definition | Related artifacts such as additional documentation, justification, or bibliographic references.
Cardinality | 0..*
Type | RelatedArtifact
Requirements | Activity definitions must be able to provide enough information for consumers of the content (and/or interventions or results produced by the content) to be able to determine and understand the justification for and evidence in support of the content.
Comments | Each related artifact is either an attachment, or a reference to another resource, but not both.
ActivityDefinition.library
Element Id | ActivityDefinition.library
Definition | A reference to a Library resource containing any formal logic used by the activity definition.
Cardinality | 0..*
Type | canonical(Library)
ActivityDefinition.kind
Element Id | ActivityDefinition.kind
Definition | A description of the kind of resource the activity definition is representing. For example, a MedicationRequest, a ServiceRequest, or a CommunicationRequest. Typically, but not always, this is a Request resource.
Cardinality | 0..1
Terminology Binding | RequestResourceType (Required)
Type | code
Summary | true
Comments | May determine what types of extensions are permitted.
ActivityDefinition.profile
Element Id | ActivityDefinition.profile
Definition | A profile to which the target of the activity definition is expected to conform.
Cardinality | 0..1
Type | canonical(StructureDefinition)
Requirements | Allows profiles to be used to describe the types of activities that can be performed within a workflow, protocol, or order set.
ActivityDefinition.code
Element Id | ActivityDefinition.code
Definition | Detailed description of the type of activity; e.g. What lab test, what procedure, what kind of encounter.
Cardinality | 0..1
Terminology Binding | Procedure Codes (SNOMED CT) (Example)
Type | CodeableConcept
Requirements | Allows matching performed to planned as well as validation against protocols.
Summary | true
Comments | Tends to be less relevant for activities involving particular products.

ActivityDefinition.intent
Element Id | ActivityDefinition.intent
Definition | Indicates the level of authority/intentionality associated with the activity and where the request should fit into the workflow chain.
Cardinality | 0..1
Terminology Binding | RequestIntent (Required)
Type | code
ActivityDefinition.priority
Element Id | ActivityDefinition.priority
Definition | Indicates how quickly the activity should be addressed with respect to other requests.
Cardinality | 0..1
Terminology Binding | Request priority (Required)
Type | code
ActivityDefinition.doNotPerform
Element Id | ActivityDefinition.doNotPerform
Definition | Set this to true if the definition is to indicate that a particular activity should NOT be performed. If true, this element should be interpreted to reinforce a negative coding. For example NPO as a code with a doNotPerform of true would still indicate to NOT perform the action.
Cardinality | 0..1
Type | boolean
Is Modifier | true (Reason: If true this element negates the specified action. For Example, instead of a request for a procedure, it is a request for the procedure to not occur.)
Summary | true
Comments | This element is not intended to be used to communicate a decision support response to cancel an order in progress. That should be done with the "remove" type of a PlanDefinition or RequestGroup.
ActivityDefinition.timing[x]
Element Id | ActivityDefinition.timing[x]
Definition | The period, timing or frequency upon which the described activity is to occur.
Cardinality | 0..1
Type | Timing|dateTime|Age|Period|Range|Duration
[x] Note | See Choice of Data Types for further information about how to use [x]
Requirements | Allows prompting for activities and detection of missed planned activities.
ActivityDefinition.location
Element Id | ActivityDefinition.location
Definition | Identifies the facility where the activity will occur; e.g. home, hospital, specific clinic, etc.
Cardinality | 0..1
Type | Reference(Location)
Requirements | Helps in planning of activity.
Alternate Names | facility
Comments | May reference a specific clinical location or may just identify a type of location.
ActivityDefinition.participant
Element Id | ActivityDefinition.participant
Definition | Indicates who should participate in performing the action described.
Cardinality | 0..*
ActivityDefinition.participant.type
Element Id | ActivityDefinition.participant.type
Definition | The type of participant in the action.
Cardinality | 1..1
Terminology Binding | ActionParticipantType (Required)
Type | code
ActivityDefinition.participant.role
Element Id | ActivityDefinition.participant.role
Definition | The role the participant should play in performing the described action.
Cardinality | 0..1
Terminology Binding | Action participant role (Example)
Type | CodeableConcept
ActivityDefinition.product[x]
Element Id | ActivityDefinition.product[x]
Definition | Identifies the food, drug or other product being consumed or supplied in the activity.
Cardinality | 0..1
Terminology Binding | SNOMED CT Medication Codes (Example)
Type | Reference(Medication | Substance)|CodeableConcept
[x] Note | See Choice of Data Types for further information about how to use [x]
ActivityDefinition.quantity
Element Id | ActivityDefinition.quantity
Definition | Identifies the quantity expected to be consumed at once (per dose, per meal, etc.).
Cardinality | 0..1
Type | SimpleQuantity
Alternate Names | dose
ActivityDefinition.dosage
Element Id | ActivityDefinition.dosage
Definition | Provides detailed dosage instructions in the same way that they are described for MedicationRequest resources.
Cardinality | 0..*

Type | Dosage
Comments | If a dosage instruction is used, the definition should not specify timing or quantity.
ActivityDefinition.bodySite
Element Id | ActivityDefinition.bodySite
Definition | Indicates the sites on the subject's body where the procedure should be performed (I.e. the target sites).
Cardinality | 0..*
Terminology Binding | SNOMED CT Body Structures (Example)
Type | CodeableConcept
Requirements | Knowing where the procedure is made is important for tracking if multiple sites are possible.
Comments | Only used if not implicit in the code found in ServiceRequest.type.
ActivityDefinition.specimenRequirement
Element Id | ActivityDefinition.specimenRequirement
Definition | Defines specimen requirements for the action to be performed, such as required specimens for a lab test.
Cardinality | 0..*
Type | Reference(SpecimenDefinition)
Requirements | Needed to represent lab order definitions.
ActivityDefinition.observationRequirement
Element Id | ActivityDefinition.observationRequirement
Definition | Defines observation requirements for the action to be performed, such as body weight or surface area.
Cardinality | 0..*
Type | Reference(ObservationDefinition)
Requirements | Needed to represent observation definitions.
ActivityDefinition.observationResultRequirement
Element Id | ActivityDefinition.observationResultRequirement
Definition | Defines the observations that are expected to be produced by the action.
Cardinality | 0..*
Type | Reference(ObservationDefinition)
ActivityDefinition.transform
Element Id | ActivityDefinition.transform
Definition | A reference to a StructureMap resource that defines a transform that can be executed to produce the intent resource using the ActivityDefinition instance as the input.
Cardinality | 0..1
Type | canonical(StructureMap)
Comments | Note that if both a transform and dynamic values are specified, the dynamic values will be applied to the result of the transform.
ActivityDefinition.dynamicValue
Element Id | ActivityDefinition.dynamicValue
Definition | Dynamic values that will be evaluated to produce values for elements of the resulting resource. For example, if the dosage of a medication must be computed based on the patient's weight, a dynamic value would be used to specify an expression that calculated the weight, and the path on the request resource that would contain the result.
Cardinality | 0..*
Comments | Dynamic values are applied in the order in which they are defined in the ActivityDefinition. Note that if both a transform and dynamic values are specified, the dynamic values will be applied to the result of the transform.
ActivityDefinition.dynamicValue.path
Element Id | ActivityDefinition.dynamicValue.path
Definition | The path to the element to be customized. This is the path on the resource that will hold the result of the calculation defined by the expression. The specified path SHALL be a FHIRPath resolveable on the specified target type of the ActivityDefinition, and SHALL consist only of identifiers, constant indexers, and a restricted subset of functions. The path is allowed to contain qualifiers (.) to traverse sub-elements, as well as indexers ([x]) to traverse multiple-cardinality sub-elements (see the Simple FHIRPath Profile for full details).
Cardinality | 1..1
Type | string
Comments | The path attribute contains a Simple FHIRPath Subset that allows path traversal, but not calculation.
ActivityDefinition.dynamicValue.expression
Element Id | ActivityDefinition.dynamicValue.expression
Definition | An expression specifying the value of the customized element.
Cardinality | 1..1
Type | Expression
Comments | The expression may be inlined, or may be a reference to a named expression within a logic library referenced by the library element.

12.17.7 Resource ActivityDefinition - Mappings

URL: https://hl7.org/fhir/R4/activitydefinition-mappings.html

12.17.7 Resource ActivityDefinition - Mappings

Clinical Decision Support Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Mappings:

FiveWs Pattern Mapping

Workflow Pattern

RIM Mapping

Mappings for the activitydefinition resource (see Mappings to Other Standards for further information & status).

12.17.7.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

ActivityDefinition
url | FiveWs.identifier
identifier | FiveWs.identifier
version | FiveWs.version
status | FiveWs.status
experimental | FiveWs.class
date | FiveWs.recorded
publisher | FiveWs.witness
purpose | FiveWs.why[x]

12.17.7.2 Workflow Pattern (http://hl7.org/fhir/workflow)

ActivityDefinition | Definition
url | Definition.url
identifier | Definition.identifier
version | Definition.version
title | Definition.title
status | Definition.status {different ValueSet}
experimental | Definition.experimental
subject[x] | Definition.subject
date | Definition.date
publisher | Definition.publisher {as string instead of CodeableConcept}
contact | Definition.contact
description | Definition.description
useContext | Definition.useContext
jurisdiction | Definition.jurisdiction
purpose | Definition.purpose
copyright | Definition.copyright
approvalDate | Definition.approvalDate
lastReviewDate | Definition.lastReviewDate
effectivePeriod | Definition.effectivePeriod

12.17.7.3 RIM Mapping (http://hl7.org/v3)

ActivityDefinition | Act[classCode=ACT; moodCode=DEFN]
url | .identifier[scope=BUSN;reliability=ISS]
identifier | .identifier
version | N/A (to add?)
name | N/A
title | .title
status | .status
experimental | N/A (to add?)
date | .participation[typeCode=AUT].time
publisher | .participation[typeCode=AUT].role
contact | .participation[typeCode=CALLBCK].role
description | .text
useContext | N/A (to add?)
jurisdiction | N/A (to add?)
purpose | .reasonCode.text
usage | N/A (to add?)
copyright | N/A (to add?)
approvalDate | .outboundRelationship[typeCode="SUBJ"].act[classCode=CACT;moodCode=EVN;code="approval"].effectiveTime
lastReviewDate | .outboundRelationship[typeCode="SUBJ";
subsetCode="RECENT"].act[classCode=CACT;moodCode=EVN;code="review"].effectiveTime
effectivePeriod | N/A (to add?)
topic | N/A (to add?)
author | .participation[typeCode=AUT]
reviewer | .participation[typeCode=VRF] {not clear whether VRF best corresponds to reviewer or endorser}
endorser | .participation[typeCode=VRF] {not clear whether VRF best corresponds to reviewer or endorser}
relatedArtifact | .outboundRelationship[typeCode=DOC,RSON,PREV, DRIV, USE, COMP] {successor would be PREV w/ inversionInd=true; No support for citation}
library | .outboundRelationship[typeCode=USE].target[classCode=GROUPER,moodCode=EVN]
kind | .classCode
code | .code
timing[x] | .effectiveTime
location | .participation[typeCode=LOC].role
participant | .participation[typeCode=PFM]
type | .role.classCode

role | .role.code
product[x] | .participation[typeCode=CSM].role
quantity | .quantity {for classCode=SPLY}
dosage | .outboundRelationship[typeCode=COMP].target[classCode=SBADM; moodCode=DEFN]
bodySite | .targetSiteCode {for classCode=PROC, SBADM}
transform | N/A
dynamicValue | N/A
path | N/A
expression | N/A

12.17.8 Resource ActivityDefinition - Extensions & Profiles

URL: https://hl7.org/fhir/R4/activitydefinition-profiles.html

12.17.8 Resource ActivityDefinition - Extensions & Profiles

Clinical Decision Support Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

This table lists profiles and extensions for the ActivityDefinition resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

12.17.8.1 Profiles

Profile | Description | Context
ShareableActivityDefinition | Enforces the minimum information set for the activity definition metadata required by HL7 and other organizations that share and publish activity definitions | Shareable ActivityDefinition

12.17.8.2 Extensions

replaces | ActivityDefinition | replaces | for General Extensions for use by FHIR Implementers
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

workflow-instantiatesCanonical | instantiatesCanonical | for Workflow Pattern HL7 Extensions
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

12.17.8.3 Search Extensions

No Search Extensions defined for this resource

12.17.9 Resource ActivityDefinition - Operations

URL: https://hl7.org/fhir/R4/activitydefinition-operations.html

12.17.9 Resource ActivityDefinition - Operations

Clinical Decision Support Work Group | Maturity Level: 2 | Trial Use | Security Category: Anonymous | Compartments: Not linked to any defined compartments

This resource has 2 operations operations associated with it:

$apply | Apply
$data-requirements | Data Requirements

For more information about operations, including how they are invoked, see Operations.

12.17.9.1 Operation $apply on ActivityDefinition

URL: https://hl7.org/fhir/R4/activitydefinition-operation-apply.html

12.17.9.1 Operation $apply on ActivityDefinition

Clinical Decision Support Work Group | Maturity Level: 2 | Trial Use | Compartments: N/A

The apply operation applies a definition in a specific context.

The official URL for this operation definition is

```
http://hl7.org/fhir/OperationDefinition/ActivityDefinition-apply
```

Formal Definition (as a OperationDefinition).

URL: [base]/ActivityDefinition/$apply

URL: [base]/ActivityDefinition/[id]/$apply

This is an idempotent operation

In Parameters:

Name | Cardinality | Type | Binding | Profile | Documentation

activityDefinition | 0..1 | ActivityDefinition | | | The activity definition to apply. If the operation is invoked on an instance, this parameter is not allowed. If the operation is invoked at the type level, this parameter is required

subject | 1..* | string (reference) | | | The subject(s) that is/are the target of the activity definition to be applied. The subject may be a Patient, Practitioner, Organization, Location, Device, or Group. Subjects provided in this parameter will be resolved as the subject of the PlanDefinition based on the type of the subject. If multiple subjects of the same type are provided, the behavior is implementation-defined

encounter | 0..1 | string (reference) | | | The encounter in context, if any

practitioner | 0..1 | string (reference) | | | The practitioner in context

organization | 0..1 | string (reference) | | | The organization in context

userType | 0..1 | CodeableConcept | | | The type of user initiating the request, e.g. patient, healthcare provider, or specific type of healthcare provider (physician, nurse, etc.)

userLanguage | 0..1 | CodeableConcept | | | Preferred language of the person using the system

userTaskContext | 0..1 | CodeableConcept | | | The task the system user is performing, e.g. laboratory results review, medication list review, etc. This information can be used to tailor decision support outputs, such as recommended information resources

setting | 0..1 | CodeableConcept | | | The current setting of the request (inpatient, outpatient, etc.)

settingContext | 0..1 | CodeableConcept | | | Additional detail about the setting of the request, if any

Out Parameters:

Name | Cardinality | Type | Binding | Profile | Documentation

return | 1..1 | Any | | | The resource that is the result of applying the definition Note: as this is the only out parameter, it is a resource, and it has the name 'return', the result of this operation is returned directly as a resource

The result of invoking this operation is a resource of the type specified by the activity definition, with all the definitions resolved as appropriate for the type of resource. Any dynamicValue elements will be evaluated (in the order in which they appear in the resource) and the results applied to the returned resource. If the ActivityDefinition includes library references, those libraries will be available to the evaluated expressions. If those libraries have parameters, those parameters will be bound by name to the parameters given to the operation. In addition, parameters to the $apply operation are available within dynamicValue expressions as context variables, accessible by the name of the parameter, prefixed with a percent (%) symbol. For a more detailed description, refer to the ActivityDefinition resource

12.17.9.1.1 Examples

Request: Apply a citalopram prescription activity definition to a patient

```
GET [base]/ActivityDefinition/citalopramPrescription/$apply?subject=Patient/124
```

Response: Returned on successful application

```
HTTP/1.1 200 OK

<?xml version="1.0" encoding="UTF-8"?>
<MedicationRequest xmlns="http://hl7.org/fhir">
  <!-- This example demonstrates the result of an $apply operation for the citalopramPrescription activity definition.
    -->
  <id value="apply-operation-response"/>
  <text>
  <status value="generated"/>
    <div xmlns="http://www.w3.org/1999/xhtml">
    Generated...
    </div>
  </text>
  <contained>
    <Medication>
      <id value="citalopramMedication"/>
      <code>
        <coding>
          <system value="http://www.nlm.nih.gov/research/umls/rxnorm"/>
          <code value="200371"/>
        </coding>
        <text value="citalopram"/>
      </code>
        <form>
          <coding>
            <system value="http://snomed.info/sct"/>
            <code value="385055001"/>
            <display value="Tablet dose form"/>
          </coding>
          <text value="Tablet dose form"/>
        </form>
        <ingredient>
          <itemReference>
            <reference value="#citalopramSubstance"/>
          </itemReference>
          <amount>
            <numerator>
```

```
            <value value="20"/>
            <unit value="mg"/>
          </numerator>
          <denominator>
            <value value="1"/>
            <unit value="{tbl}"/>
          </denominator>
        </amount>
      </ingredient>
    </Medication>
  </contained>
  <contained>
    <Substance>
      <id value="citalopramSubstance"/>
      <code>
        <coding>
          <system value="http://www.nlm.nih.gov/research/umls/rxnorm"/>
          <code value="2556"/>
        </coding>
        <text value="citalopram"/>
      </code>
    </Substance>
  </contained>
  <status value="draft"/>
  <intent value="proposal"/>
  <medicationReference>
    <reference value="#citalopramMedication"/>
  </medicationReference>
  <subject>
    <reference value="Patient/124"/>
  </subject>
  <!--definition>
    <reference value="ActivityDefinition/citalopramPrescription"/>
  </definition-->
  <instantiates value="ActivityDefinition/citalopramPrescription"/>
  <dosageInstruction>
  <text value="1 tablet oral 1 time daily"/>
  <timing>
    <repeat>
      <frequency value="1"/>
      <period value="1"/>
      <periodUnit value="d"/>
    </repeat>
  </timing>
  <route>
    <coding>
      <code value="26643006"/>
      <display value="Oral route (qualifier value)"/>
    </coding>
    <text value="Oral route (qualifier value)"/>
  </route>
  <doseAndRate>
      <type>
        <text value="ordered"/>
      </type>
      <doseQuantity>
      <value value="1"/>
      <unit value="{tbl}"/>
      </doseQuantity>
  </doseAndRate>
  </dosageInstruction>
  <dispenseRequest>
    <numberOfRepeatsAllowed value="3"/>
    <quantity>
      <value value="30"/>
      <unit value="{tbl}"/>
    </quantity>
  </dispenseRequest>
</MedicationRequest>
```

For more information about operations, including how they are invoked, see Operations.

12.17.9.2 Operation $data-requirements on ActivityDefinition

URL: https://hl7.org/fhir/R4/activitydefinition-operation-data-requirements.html

12.17.9.2 Operation $data-requirements on ActivityDefinition

Clinical Decision Support Work Group | Maturity Level: 2 | Trial Use | Compartments: N/A

The data-requirements operation aggregates and returns the parameters and data requirements for the activity definition and all its dependencies as a single module definition library.

The official URL for this operation definition is

```
http://hl7.org/fhir/OperationDefinition/ActivityDefinition-data-requirements
```

Formal Definition (as a OperationDefinition).

URL: [base]/ActivityDefinition/[id]/$data-requirements

This is an idempotent operation

Out Parameters:
Name | Cardinality | Type | Binding | Profile | Documentation
return | 1..1 | Library | | | The result of the requirements gathering represented as a module-definition Library that describes the aggregate parameters, data requirements, and dependencies of the activity definition Note: as this is the only out parameter, it is a resource, and it has the name 'return', the result of this operation is returned directly as a resource

The effect of invoking this operation is to determine the aggregate set of data requirements and dependencies for the activity definition. The result is a Library resource with a type of module-definition that contains all the parameter definitions and data requirements of the activity definition and any libraries referenced by it. Implementations SHOULD aggregate data requirements intelligently (i.e. by collapsing overlapping data requirements)

12.17.9.2.1 Examples

Request: Request data requirements for the administer zika virus exposure assessment activity definition

```
GET [base]/ActivityDefinition/administer-zika-virus-exposure-assessment/$data-requirements
```

Response: Returned on successful evaluation

```
HTTP/1.1 200 OK
```

```
{
  "resourceType": "Library",
  "type": {
    "coding": [
      {
        "code": "module-definition"
      }
    ]
  },
  "relatedArtifact": [
    {
      "type": "depends-on",
      "display": "Zika Affected Areas",
      "resource": {
        "reference": "ValueSet/zika-affected-areas"
      }
    },
    {
      "type": "depends-on",
      "display": "Pregnancy",
      "resource": {
        "identifier": "urn:oid:X.Y.Z",
        "display": "Pregnancy value set"
      }
    },
    {
      "type": "depends-on",
      "display": "Zika Exposure Assessment",
      "resource": {
        "identifier": "urn:oid:X.Y.Z",
        "display": "Zika Virus Exposure Assessment value set"
      }
    },
    {
      "type": "depends-on",
      "display": "Zika Virus Signs and Symptoms",
      "url": "https://phinvads.cdc.gov/vads/ViewValueSet.action?oid=2.16.840.1.114222.4.11.7459",
      "resource": {
        "identifier": "urn:oid:2.16.840.1.114222.4.11.7459",
        "display": "Zika Virus Signs and Symptoms value set"
      }
    },
  ]
  "dataRequirement": [
    {
      "type": "Condition",
      "codeFilter": [
        {
          "path": "code",
          "valueSetString": "Pregnancy"
        }
      ]
    },
    {
      "type": "Patient"
    },
    {
      "type": "Observation",
      "codeFilter": [
        {
          "path": "code",
          "valueSetString": [
            "Zika Exposure Assessment"
          ]
        },
```

```
        {
          "path": "status",
          "valueCode": [
            "final"
          ]
        }
      ]
    },
    {
      "type": "Condition",
      "codeFilter": [
        {
          "path": "category",
          "valueCode": [
            "diagnosis"
          ]
        },
        {
          "path": "clinicalStatus",
          "valueCode": [
            "confirmed"
          ]
        },
        {
          "path": "code",
          "valueSetString": "Zika Virus Signs and Symptoms"
        }
      ]
    }
  ]
}
```

For more information about operations, including how they are invoked, see Operations.

12.17.10 Resource ActivityDefinition - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/activitydefinition-version-maps.html

No validation errors - all conversions are clean

12.17.10 Resource ActivityDefinition - R3/R4 Conversions

Clinical Decision Support Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

R3 : R4 Conversion maps for ActivityDefinition.

Functional status for this map: 9 tests that all execute ok. 2 fail round-trip testing and 9 r3 resources are invalid (0 errors). (see documentation)

12.17.10.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/ActivityDefinition3to4" = "R3 to R4 Conversions for ActivityDefinition"

uses "http://hl7.org/fhir/3.0/StructureDefinition/ActivityDefinition" alias ActivityDefinitionR3 as source
uses "http://hl7.org/fhir/StructureDefinition/ActivityDefinition" alias ActivityDefinition as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group ActivityDefinition(source src : ActivityDefinitionR3, target tgt : ActivityDefinition) extends DomainResource
    <<type+>> {
  src.url -> tgt.url;
  src.identifier -> tgt.identifier;
  src.version -> tgt.version;
  src.name -> tgt.name;
  src.title -> tgt.title;
  src.subtitle -> tgt.subtitle;
  src.status -> tgt.status;
  src.experimental -> tgt.experimental;
  src.subject -> tgt.subject;
  src.date -> tgt.date;
  src.publisher -> tgt.publisher;
  src.contact -> tgt.contact;
  src.description -> tgt.description;
  src.useContext -> tgt.useContext;
  src.jurisdiction -> tgt.jurisdiction;
  src.purpose -> tgt.purpose;
  src.usage -> tgt.usage;
  src.copyright -> tgt.copyright;
  src.approvalDate -> tgt.approvalDate;
  src.lastReviewDate -> tgt.lastReviewDate;
  src.effectivePeriod -> tgt.effectivePeriod;
  src.topic -> tgt.topic;
  src.contributor as vs where type = 'author' -> tgt.author as vt then Contributor(vs, vt);
  src.contributor as vs where type = 'editor' -> tgt.editor as vt then Contributor(vs, vt);
  src.contributor as vs where type = 'reviewer' -> tgt.reviewer as vt then Contributor(vs, vt);
  src.contributor as vs where type = 'endorser' -> tgt.endorser as vt then Contributor(vs, vt);
  src.relatedArtifact -> tgt.relatedArtifact;
  src.library -> tgt.library;
```

```
  src.kind as v -> tgt.kind = translate(v, 'http://hl7.org/fhir/StructureMap/ConceptMaps3to4#ResourceTypeMap', 'code');
  src.code -> tgt.code;
  src.doNotPerform -> tgt.doNotPerform;
  src.timing -> tgt.timing;
  src.location -> tgt.location;
  src.participant as s -> tgt.participant as t then ActivityDefinitionParticipant(s, t);
  src.product -> tgt.product;
  src.quantity -> tgt.quantity;
  src.dosage -> tgt.dosage;
  src.bodySite -> tgt.bodySite;
  src.specimenRequirement -> tgt.specimenRequirement;
  src.transform -> tgt.transform;
  src.dynamicValue as s -> tgt.dynamicValue as t then ActivityDefinitionDynamicValue(s, t);
}

group ActivityDefinitionParticipant(source src, target tgt) extends BackboneElement {
  src.type -> tgt.type;
  src.role -> tgt.role;
}

group ActivityDefinitionDynamicValue(source src, target tgt) extends BackboneElement {
  src.path -> tgt.path;
  src as vs where vs.type.exists().not() -> tgt.expression = create('Expression') as vt then {
    vs.description -> vt.description;
    vs.language -> vt.language;
    vs.expression -> vt.expression;
  } "expression";
}
```

12.17.10.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/ActivityDefinition4to3" = "R4 to R3 Conversion for ActivityDefinition"

uses "http://hl7.org/fhir/StructureDefinition/ActivityDefinition" alias ActivityDefinition as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/ActivityDefinition" alias ActivityDefinitionR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group ActivityDefinition(source src : ActivityDefinitionR3, target tgt : ActivityDefinition) extends DomainResource
    <<type+>> {
  src.url -> tgt.url;
  src.identifier -> tgt.identifier;
  src.version -> tgt.version;
  src.name -> tgt.name;
  src.title -> tgt.title;
  src.subtitle -> tgt.subtitle;
  src.status -> tgt.status;
  src.experimental -> tgt.experimental;
  src.subject -> tgt.subject;
  src.date -> tgt.date;
  src.publisher -> tgt.publisher;
  src.contact -> tgt.contact;
  src.description -> tgt.description;
  src.useContext -> tgt.useContext;
  src.jurisdiction -> tgt.jurisdiction;
  src.purpose -> tgt.purpose;
  src.usage -> tgt.usage;
  src.copyright -> tgt.copyright;
  src.approvalDate -> tgt.approvalDate;
  src.lastReviewDate -> tgt.lastReviewDate;
  src.effectivePeriod -> tgt.effectivePeriod;
  src.topic -> tgt.topic;
  src.author as vs ->  tgt.contributor as vt,  vt.type = 'author' then Contributor(vs, vt);
  src.editor as vs ->  tgt.contributor as vt,  vt.type = 'editor' then Contributor(vs, vt);
  src.reviewer as vs ->  tgt.contributor as vt,  vt.type = 'reviewer' then Contributor(vs, vt);
  src.endorser as vs ->  tgt.contributor as vt,  vt.type = 'endorser' then Contributor(vs, vt);
  src.relatedArtifact -> tgt.relatedArtifact;
  src.library -> tgt.library;
  src.kind as v -> tgt.kind = translate(v, 'http://hl7.org/fhir/StructureMap/ConceptMaps4to3#ResourceTypeMap', 'code');
  src.code -> tgt.code;
  src.doNotPerform -> tgt.doNotPerform;
  src.timing -> tgt.timing;
  src.location -> tgt.location;
  src.participant as s -> tgt.participant as t then ActivityDefinitionParticipant(s, t);
  src.product -> tgt.product;
  src.quantity -> tgt.quantity;
  src.dosage -> tgt.dosage;
  src.bodySite -> tgt.bodySite;
  src.specimenRequirement -> tgt.specimenRequirement;
  src.transform -> tgt.transform;
  src.dynamicValue as s -> tgt.dynamicValue as t then ActivityDefinitionDynamicValue(s, t);
}

group ActivityDefinitionParticipant(source src, target tgt) extends BackboneElement {
  src.type -> tgt.type;
  src.role -> tgt.role;
}

group ActivityDefinitionDynamicValue(source src, target tgt) extends BackboneElement {
  src.path -> tgt.path;
```

```
  src.expression as vs where vs.type.exists().not() -> tgt as vt then {
    vs.description -> vt.description;
    vs.language -> vt.language;
    vs.expression -> vt.expression;
  };
}
```

12.17.10.3 R4 Validation Errors

12.17.11 Shareable ActivityDefinition

URL: https://hl7.org/fhir/R4/activitydefinition-shareable.html

Enforces the minimum information set for the activity definition metadata required by HL7 and other organizations that share and publish activity definitions

12.17.11 Shareable ActivityDefinition

Clinical Decision Support Work Group | Maturity Level: N/A | Standards Status: Informative

12.17.11.1 Content

Profiles:

ShareableActivityDefinition | Enforces the minimum information set for the activity definition metadata required by HL7 and other organizations that share and publish activity definitions

12.17.11.2 StructureDefinition: ShareableActivityDefinition

URL: https://hl7.org/fhir/R4/shareableactivitydefinition.html

12.17.11.2 StructureDefinition: ShareableActivityDefinition

Clinical Decision Support Work Group | Maturity Level: 1 | Trial Use | Use Context: Any

The official URL for this profile is:

`http://hl7.org/fhir/StructureDefinition/shareableactivitydefinition`

Enforces the minimum information set for the activity definition metadata required by HL7 and other organizations that share and publish activity definitions

This profile was published on [no date] as a draft by HL7.

12.17.11.2.1 Formal Views of Profile Content

Description of Profiles, Differentials, Snapshots, and how the XML and JSON presentations work.

- Text Summary
- Differential Table
- Snapshot Table
- XML Template
- JSON Template
- All

This structure is derived from ActivityDefinition.

Summary

Mandatory: 6 elements Must-Support: 6 elements

This structure is derived from ActivityDefinition.

Name | Flags | Card. | Type | Description & Constraints
ActivityDefinition | | 1..1
url | | 1..1 | uri
identifier | S | 0..* | Identifier
version | | 1..1 | string
name | | 1..1 | string
title | S | 0..1 | string
experimental | | 1..1 | boolean
date | S | 0..1 | dateTime
publisher | | 1..1 | string
contact | S | 0..* | ContactDetail
description | | 1..1 | markdown
useContext | S | 0..* | UsageContext
jurisdiction | S | 0..* | CodeableConcept
Documentation for this format

Name | Flags | Card. | Type | Description & Constraints
ActivityDefinition | I | 1..1 | | The definition of a specific activity to be taken, independent of any particular patient or context
id | Σ | 0..1 | string | Logical id of this artifact
meta | ΣI | 0..1 | Meta | Metadata about the resource

implicitRules | ?!ΣI | 0..1 | uri | A set of rules under which this content was created
language | I | 0..1 | code | Language of the resource content Binding: Common Languages (preferred) Max Binding: All Languages
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!I | 0..* | Extension | Extensions that cannot be ignored
url | ΣI | 1..1 | uri | Canonical identifier for this activity definition, represented as a URI (globally unique)
identifier | SΣI | 0..* | Identifier | Additional identifier for the activity definition
version | ΣI | 1..1 | string | Business version of the activity definition
name | ΣI | 1..1 | string | Name for this activity definition (computer friendly)
title | SΣI | 0..1 | string | Name for this activity definition (human friendly)
subtitle | I | 0..1 | string | Subordinate title of the activity definition
status | ?!ΣI | 1..1 | code | draft | active | retired | unknown Binding: PublicationStatus (required)
experimental | ΣI | 1..1 | boolean | For testing purposes, not real usage
subject[x] | I | 0..1 | | Type of individual the activity definition is intended for Binding: Subject type (extensible)
subjectCodeableConcept | | | CodeableConcept
subjectReference | | | Reference(group.html)
date | SΣI | 0..1 | dateTime | Date last changed
publisher | ΣI | 1..1 | string | Name of the publisher (organization or individual)
contact | SΣI | 0..* | ContactDetail | Contact details for the publisher
description | ΣI | 1..1 | markdown | Natural language description of the activity definition
useContext | SΣI | 0..* | UsageContext | The context that the content is intended to support
jurisdiction | SΣI | 0..* | CodeableConcept | Intended jurisdiction for activity definition (if applicable) Binding: Jurisdiction (extensible)
purpose | I | 0..1 | markdown | Why this activity definition is defined
usage | I | 0..1 | string | Describes the clinical usage of the activity definition
copyright | I | 0..1 | markdown | Use and/or publishing restrictions
approvalDate | I | 0..1 | date | When the activity definition was approved by publisher
lastReviewDate | I | 0..1 | date | When the activity definition was last reviewed
effectivePeriod | ΣI | 0..1 | Period | When the activity definition is expected to be used
topic | I | 0..* | CodeableConcept | E.g. Education, Treatment, Assessment, etc. Binding: DefinitionTopic (example)
author | I | 0..* | ContactDetail | Who authored the content
editor | I | 0..* | ContactDetail | Who edited the content
reviewer | I | 0..* | ContactDetail | Who reviewed the content
endorser | I | 0..* | ContactDetail | Who endorsed the content
relatedArtifact | I | 0..* | RelatedArtifact | Additional documentation, citations, etc.
library | I | 0..* | canonical(Library) | Logic used by the activity definition
kind | ΣI | 0..1 | code | Kind of resource Binding: RequestResourceType (required)
profile | I | 0..1 | canonical(StructureDefinition) | What profile the resource needs to conform to
code | ΣI | 0..1 | CodeableConcept | Detail type of activity Binding: Procedure Codes (SNOMED CT) (example)
intent | I | 0..1 | code | proposal | plan | directive | order | original-order | reflex-order | filler-order | instance-order | option Binding: RequestIntent (required)
priority | I | 0..1 | code | routine | urgent | asap | stat Binding: Request priority (required)
doNotPerform | ?!ΣI | 0..1 | boolean | True if the activity should not be performed
timing[x] | I | 0..1 | | When activity is to occur
timingTiming | | | Timing
timingDateTime | | | dateTime
timingAge | | | Age
timingPeriod | | | Period
timingRange | | | Range
timingDuration | | | Duration
location | I | 0..1 | Reference(Location) | Where it should happen
participant | I | 0..* | BackboneElement | Who should participate in the action
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
type | I | 1..1 | code | patient | practitioner | related-person | device Binding: ActionParticipantType (required)
role | I | 0..1 | CodeableConcept | E.g. Nurse, Surgeon, Parent, etc. Binding: Action participant role (example)
product[x] | I | 0..1 | | What's administered/supplied Binding: SNOMED CT Medication Codes (example)
productReference | | | Reference(medication.html | substance.html)
productCodeableConcept | | | CodeableConcept
quantity | I | 0..1 | SimpleQuantity | How much is administered/consumed/supplied
dosage | I | 0..* | Dosage | Detailed dosage instructions
bodySite | I | 0..* | CodeableConcept | What part of body to perform on Binding: SNOMED CT Body Structures (example)
specimenRequirement | I | 0..* | Reference(SpecimenDefinition) | What specimens are required to perform this action
observationRequirement | I | 0..* | Reference(ObservationDefinition) | What observations are required to perform this action
observationResultRequirement | I | 0..* | Reference(ObservationDefinition) | What observations must be produced by this action
transform | I | 0..1 | canonical(StructureMap) | Transform to apply the template
dynamicValue | I | 0..* | BackboneElement | Dynamic aspects of the definition
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized

path | I | 1..1 | string | The path to the element to be set dynamically
expression | I | 1..1 | Expression | An expression that provides the dynamic value for the customization
Documentation for this format

```
<!-- Shareable ActivityDefinition -->

<ActivityDefinition xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id><!-- 0..1 string Logical id of this artifact --></id>
 <meta><!--  0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!--  0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!--  0..1 Language of the resource content   -->
 <text><!--  0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource
     Contained, inline Resources --></contained>
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <url value="[uri]"/><!--  1..1
     Canonical identifier for this activity definition, represented as a URI (globally unique) -->
 <identifier><!--  0..* Identifier
     Additional identifier for the activity definition --></identifier>
 <version value="[string]"/><!--  1..1 Business version of the activity definition -->
 <name value="[string]"/><!--  1..1 Name for this activity definition (computer friendly) -->
 <title value="[string]"/><!--  0..1 Name for this activity definition (human friendly) -->
 <subtitle value="[string]"/><!--  0..1 Subordinate title of the activity definition -->
 <status value="[code]"/><!--  1..1 draft | active | retired | unknown   -->
 <experimental value="[boolean]"/><!--  1..1 For testing purposes, not real usage -->
 <subject[x]><!--  0..1 CodeableConcept|Reference(Group) Type of individual the activity definition is intended for
    --></subject[x]>
 <date value="[dateTime]"/><!--  0..1 Date last changed -->
 <publisher value="[string]"/><!--  1..1 Name of the publisher (organization or individual) -->
 <contact><!--  0..* ContactDetail
     Contact details for the publisher --></contact>
 <description value="[markdown]"/><!--  1..1 Natural language description of the activity definition -->
 <useContext><!--  0..* UsageContext
     The context that the content is intended to support --></useContext>
 <jurisdiction><!--  0..* CodeableConcept Intended jurisdiction for activity definition (if applicable)
    --></jurisdiction>
 <purpose value="[markdown]"/><!--  0..1 Why this activity definition is defined -->
 <usage value="[string]"/><!--  0..1 Describes the clinical usage of the activity definition -->
 <copyright value="[markdown]"/><!--  0..1 Use and/or publishing restrictions -->
 <approvalDate value="[date]"/><!--  0..1 When the activity definition was approved by publisher -->
 <lastReviewDate value="[date]"/><!--  0..1 When the activity definition was last reviewed -->
 <effectivePeriod><!--  0..1 Period
     When the activity definition is expected to be used --></effectivePeriod>
 <topic><!--  0..* CodeableConcept E.g. Education, Treatment, Assessment, etc.   --></topic>
 <author><!--  0..* ContactDetail
     Who authored the content --></author>
 <editor><!--  0..* ContactDetail
     Who edited the content --></editor>
 <reviewer><!--  0..* ContactDetail
     Who reviewed the content --></reviewer>
 <endorser><!--  0..* ContactDetail
     Who endorsed the content --></endorser>
 <relatedArtifact><!--  0..* RelatedArtifact
     Additional documentation, citations, etc. --></relatedArtifact>
 <library value="[canonical]"/><!--  0..* Logic used by the activity definition -->
 <kind value="[code]"/><!--  0..1 Kind of resource   -->
 <profile value="[canonical]"/><!--  0..1 What profile the resource needs to conform to -->
 <code><!--  0..1 CodeableConcept Detail type of activity   --></code>
 <intent value="[code]"/><!--  0..1 proposal | plan | directive | order | original-order | reflex-order | filler-order |
     instance-order | option   -->
 <priority value="[code]"/><!--  0..1 routine | urgent | asap | stat   -->
 <doNotPerform value="[boolean]"/><!--  0..1 True if the activity should not be performed -->
 <timing[x]><!--  0..1 Timing|dateTime|Age|Period|Range|
   Duration When activity is to occur --></timing[x]>
 <location><!--  0..1 Reference(Location)
     Where it should happen --></location>
 <participant>  0..* BackboneElement  <!--  0..* Who should participate in the action -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <type value="[code]"/><!--  1..1 patient | practitioner | related-person | device   -->
  <role><!--  0..1 CodeableConcept E.g. Nurse, Surgeon, Parent, etc.   --></role>
 </participant>
 <product[x]><!--  0..1 Reference(Medication)|CodeableConcept What's administered/supplied   --></product[x]>
 <quantity><!--  0..1 Quantity
     How much is administered/consumed/supplied --></quantity>
 <dosage><!--  0..* Dosage Detailed dosage instructions --></dosage>
 <bodySite><!--  0..* CodeableConcept What part of body to perform on   --></bodySite>
 <specimenRequirement><!--  0..* Reference(SpecimenDefinition)
     What specimens are required to perform this action --></specimenRequirement>
 <observationRequirement><!--  0..* Reference(ObservationDefinition)
     What observations are required to perform this action --></observationRequirement>
 <observationResultRequirement><!--  0..* Reference(ObservationDefinition)
     What observations must be produced by this action --></observationResultRequirement>
 <transform value="[canonical]"/><!--  0..1 Transform to apply the template -->
 <dynamicValue>  0..* BackboneElement  <!--  0..* Dynamic aspects of the definition -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
```

```
    --></modifierExtension>
  <path value="[string]"/><!--  1..1 The path to the element to be set dynamically -->
  <expression><!--  1..1 Expression
      An expression that provides the dynamic value for the customization --></expression>
 </dynamicValue>
</ActivityDefinition>
```

```
{ // Shareable ActivityDefinition
  // from Element: extension
    "meta" : { Meta }, // C?Metadata about the resource
    "implicitRules" : "<uri>", // C?A set of rules under which this content was created
    "language" : "<code>", // C?Language of the resource content
    "text" : { Narrative }, // C?Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored
    "url" : "<uri>", // C? R! Canonical identifier for this activity definition, represented as a URI (globally unique)
    "identifier" : [{ Identifier }], // C?Additional identifier for the activity definition
    "version" : "<string>", // C? R! Business version of the activity definition
    "name" : "<string>", // C? R! Name for this activity definition (computer friendly)
    "title" : "<string>", // C?Name for this activity definition (human friendly)
    "subtitle" : "<string>", // C?Subordinate title of the activity definition
    "status" : "<code>", // C? R! draft | active | retired | unknown
    "experimental" : <boolean>, // C? R! For testing purposes, not real usage
// value[x]: Type of individual the activity definition is intended for. One of these 2:
    "subjectCodeableConcept" : { CodeableConcept }, // C?Type of individual the activity definition is intended for
    "subjectReference" : { Reference(Group) }, // C?Type of individual the activity definition is intended for
    "date" : "<dateTime>", // C?Date last changed
    "publisher" : "<string>", // C? R! Name of the publisher (organization or individual)
    "contact" : [{ ContactDetail }], // C?Contact details for the publisher
    "description" : "<markdown>", // C? R! Natural language description of the activity definition
    "useContext" : [{ UsageContext }], // C?The context that the content is intended to support
    "jurisdiction" : [{ CodeableConcept }], // C?Intended jurisdiction for activity definition (if applicable)
    "purpose" : "<markdown>", // C?Why this activity definition is defined
    "usage" : "<string>", // C?Describes the clinical usage of the activity definition
    "copyright" : "<markdown>", // C?Use and/or publishing restrictions
    "approvalDate" : "<date>", // C?When the activity definition was approved by publisher
    "lastReviewDate" : "<date>", // C?When the activity definition was last reviewed
    "effectivePeriod" : { Period }, // C?When the activity definition is expected to be used
    "topic" : [{ CodeableConcept }], // C?E.g. Education, Treatment, Assessment, etc.
    "author" : [{ ContactDetail }], // C?Who authored the content
    "editor" : [{ ContactDetail }], // C?Who edited the content
    "reviewer" : [{ ContactDetail }], // C?Who reviewed the content
    "endorser" : [{ ContactDetail }], // C?Who endorsed the content
    "relatedArtifact" : [{ RelatedArtifact }], // C?Additional documentation, citations, etc.
    "library" : ["<canonical>"], // C?Logic used by the activity definition
    "kind" : "<code>", // C?Kind of resource
    "profile" : "<canonical>", // C?What profile the resource needs to conform to
    "code" : { CodeableConcept }, // C?Detail type of activity
    "intent" : "<code>", // C?proposal | plan | directive | order | original-order | reflex-order | filler-order |
    instance-order | option
    "priority" : "<code>", // C?routine | urgent | asap | stat
    "doNotPerform" : <boolean>, // C?True if the activity should not be performed
// value[x]: When activity is to occur. One of these 6:
    "timingTiming" : { Timing }, // C?When activity is to occur
    "timingDateTime" : "<dateTime>", // C?When activity is to occur
    "timingAge" : { Age }, // C?When activity is to occur
    "timingPeriod" : { Period }, // C?When activity is to occur
    "timingRange" : { Range }, // C?When activity is to occur
    "timingDuration" : { Duration }, // C?When activity is to occur
    "location" : { Reference(Location) }, // C?Where it should happen
    "participant" : [{ BackboneElement }], // C?Who should participate in the action
// value[x]: What's administered/supplied. One of these 2:
    "productReference" : { Reference(Medication) }, // C?What's administered/supplied
    "productCodeableConcept" : { CodeableConcept }, // C?What's administered/supplied
    "quantity" : { Quantity(SimpleQuantity) } // C?How much is administered/consumed/supplied
    "dosage" : [{ Dosage }], // C?Detailed dosage instructions
    "bodySite" : [{ CodeableConcept }], // C?What part of body to perform on
    "specimenRequirement" : [{ Reference(SpecimenDefinition) }], // C?What specimens are required to perform this action
    "observationRequirement" : [{ Reference(ObservationDefinition) }], // C?What observations are required to perform
    this action
    "observationResultRequirement" : [{ Reference(ObservationDefinition) }], // C?What observations must be produced by
    this action
    "transform" : "<canonical>", // C?Transform to apply the template
    "dynamicValue" : [{ BackboneElement }], // C?Dynamic aspects of the definition
  }
```

This structure is derived from ActivityDefinition.

Summary

Mandatory: 6 elements Must-Support: 6 elements

Differential View

This structure is derived from ActivityDefinition.

Name	Flags	Card.	Type	Description & Constraints
ActivityDefinition		1..1		
url		1..1	uri	
identifier	S	0..*	Identifier	
version		1..1	string	

name | | 1..1 | string
title | S | 0..1 | string
experimental | | 1..1 | boolean
date | S | 0..1 | dateTime
publisher | | 1..1 | string
contact | S | 0..* | ContactDetail
description | | 1..1 | markdown
useContext | S | 0..* | UsageContext
jurisdiction | S | 0..* | CodeableConcept
Documentation for this format

Snapshot View

Name | Flags | Card. | Type | Description & Constraints
ActivityDefinition | I | 1..1 | | The definition of a specific activity to be taken, independent of any particular patient or context
id | Σ | 0..1 | string | Logical id of this artifact
meta | ΣI | 0..1 | Meta | Metadata about the resource
implicitRules | ?!ΣI | 0..1 | uri | A set of rules under which this content was created
language | I | 0..1 | code | Language of the resource content Binding: Common Languages (preferred) Max Binding: All Languages
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!I | 0..* | Extension | Extensions that cannot be ignored
url | ΣI | 1..1 | uri | Canonical identifier for this activity definition, represented as a URI (globally unique)
identifier | SΣI | 0..* | Identifier | Additional identifier for the activity definition
version | ΣI | 1..1 | string | Business version of the activity definition
name | ΣI | 1..1 | string | Name for this activity definition (computer friendly)
title | SΣI | 0..1 | string | Name for this activity definition (human friendly)
subtitle | I | 0..1 | string | Subordinate title of the activity definition
status | ?!ΣI | 1..1 | code | draft | active | retired | unknown Binding: PublicationStatus (required)
experimental | ΣI | 1..1 | boolean | For testing purposes, not real usage
subject[x] | I | 0..1 | | Type of individual the activity definition is intended for Binding: Subject type (extensible)
subjectCodeableConcept | | | CodeableConcept
subjectReference | | | Reference(group.html)
date | SΣI | 0..1 | dateTime | Date last changed
publisher | ΣI | 1..1 | string | Name of the publisher (organization or individual)
contact | SΣI | 0..* | ContactDetail | Contact details for the publisher
description | ΣI | 1..1 | markdown | Natural language description of the activity definition
useContext | SΣI | 0..* | UsageContext | The context that the content is intended to support
jurisdiction | SΣI | 0..* | CodeableConcept | Intended jurisdiction for activity definition (if applicable) Binding: Jurisdiction (extensible)
purpose | I | 0..1 | markdown | Why this activity definition is defined
usage | I | 0..1 | string | Describes the clinical usage of the activity definition
copyright | I | 0..1 | markdown | Use and/or publishing restrictions
approvalDate | I | 0..1 | date | When the activity definition was approved by publisher
lastReviewDate | I | 0..1 | date | When the activity definition was last reviewed
effectivePeriod | ΣI | 0..1 | Period | When the activity definition is expected to be used
topic | I | 0..* | CodeableConcept | E.g. Education, Treatment, Assessment, etc. Binding: DefinitionTopic (example)
author | I | 0..* | ContactDetail | Who authored the content
editor | I | 0..* | ContactDetail | Who edited the content
reviewer | I | 0..* | ContactDetail | Who reviewed the content
endorser | I | 0..* | ContactDetail | Who endorsed the content
relatedArtifact | I | 0..* | RelatedArtifact | Additional documentation, citations, etc.
library | I | 0..* | canonical(Library) | Logic used by the activity definition
kind | ΣI | 0..1 | code | Kind of resource Binding: RequestResourceType (required)
profile | I | 0..1 | canonical(StructureDefinition) | What profile the resource needs to conform to
code | ΣI | 0..1 | CodeableConcept | Detail type of activity Binding: Procedure Codes (SNOMED CT) (example)
intent | I | 0..1 | code | proposal | plan | directive | order | original-order | reflex-order | filler-order | instance-order | option Binding: RequestIntent (required)
priority | I | 0..1 | code | routine | urgent | asap | stat Binding: Request priority (required)
doNotPerform | ?!ΣI | 0..1 | boolean | True if the activity should not be performed
timing[x] | I | 0..1 | | When activity is to occur
timingTiming | | | Timing
timingDateTime | | | dateTime
timingAge | | | Age
timingPeriod | | | Period
timingRange | | | Range
timingDuration | | | Duration
location | I | 0..1 | Reference(Location) | Where it should happen
participant | I | 0..* | BackboneElement | Who should participate in the action
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
type | I | 1..1 | code | patient | practitioner | related-person | device Binding: ActionParticipantType (required)

role | I | 0..1 | CodeableConcept | E.g. Nurse, Surgeon, Parent, etc. Binding: Action participant role (example)
product[x] | I | 0..1 | | What's administered/supplied Binding: SNOMED CT Medication Codes (example)
productReference | | | Reference(medication.html | substance.html)
productCodeableConcept | | | CodeableConcept
quantity | I | 0..1 | SimpleQuantity | How much is administered/consumed/supplied
dosage | I | 0..* | Dosage | Detailed dosage instructions
bodySite | I | 0..* | CodeableConcept | What part of body to perform on Binding: SNOMED CT Body Structures (example)
specimenRequirement | I | 0..* | Reference(SpecimenDefinition) | What specimens are required to perform this action
observationRequirement | I | 0..* | Reference(ObservationDefinition) | What observations are required to perform this action
observationResultRequirement | I | 0..* | Reference(ObservationDefinition) | What observations must be produced by this action
transform | I | 0..1 | canonical(StructureMap) | Transform to apply the template
dynamicValue | I | 0..* | BackboneElement | Dynamic aspects of the definition
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
path | I | 1..1 | string | The path to the element to be set dynamically
expression | I | 1..1 | Expression | An expression that provides the dynamic value for the customization
Documentation for this format

XML Template

```
<!-- Shareable ActivityDefinition -->

<ActivityDefinition xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id><!-- 0..1 string Logical id of this artifact --></id>
 <meta><!--  0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!--  0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!--  0..1 Language of the resource content   -->
 <text><!--  0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource
     Contained, inline Resources --></contained>
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <url value="[uri]"/><!--  1..1
     Canonical identifier for this activity definition, represented as a URI (globally unique) -->
 <identifier><!--  0..* Identifier
     Additional identifier for the activity definition --></identifier>
 <version value="[string]"/><!--  1..1 Business version of the activity definition -->
 <name value="[string]"/><!--  1..1 Name for this activity definition (computer friendly) -->
 <title value="[string]"/><!--  0..1 Name for this activity definition (human friendly) -->
 <subtitle value="[string]"/><!--  0..1 Subordinate title of the activity definition -->
 <status value="[code]"/><!--  1..1 draft | active | retired | unknown   -->
 <experimental value="[boolean]"/><!--  1..1 For testing purposes, not real usage -->
 <subject[x]><!--  0..1 CodeableConcept|Reference(Group) Type of individual the activity definition is intended for
    --></subject[x]>
 <date value="[dateTime]"/><!--  0..1 Date last changed -->
 <publisher value="[string]"/><!--  1..1 Name of the publisher (organization or individual) -->
 <contact><!--  0..* ContactDetail
     Contact details for the publisher --></contact>
 <description value="[markdown]"/><!--  1..1 Natural language description of the activity definition -->
 <useContext><!--  0..* UsageContext
     The context that the content is intended to support --></useContext>
 <jurisdiction><!--  0..* CodeableConcept Intended jurisdiction for activity definition (if applicable)
    --></jurisdiction>
 <purpose value="[markdown]"/><!--  0..1 Why this activity definition is defined -->
 <usage value="[string]"/><!--  0..1 Describes the clinical usage of the activity definition -->
 <copyright value="[markdown]"/><!--  0..1 Use and/or publishing restrictions -->
 <approvalDate value="[date]"/><!--  0..1 When the activity definition was approved by publisher -->
 <lastReviewDate value="[date]"/><!--  0..1 When the activity definition was last reviewed -->
 <effectivePeriod><!--  0..1 Period
     When the activity definition is expected to be used --></effectivePeriod>
 <topic><!--  0..* CodeableConcept E.g. Education, Treatment, Assessment, etc.   --></topic>
 <author><!--  0..* ContactDetail
     Who authored the content --></author>
 <editor><!--  0..* ContactDetail
     Who edited the content --></editor>
 <reviewer><!--  0..* ContactDetail
     Who reviewed the content --></reviewer>
 <endorser><!--  0..* ContactDetail
     Who endorsed the content --></endorser>
 <relatedArtifact><!--  0..* RelatedArtifact
     Additional documentation, citations, etc. --></relatedArtifact>
 <library value="[canonical]"/><!--  0..* Logic used by the activity definition -->
 <kind value="[code]"/><!--  0..1 Kind of resource   -->
 <profile value="[canonical]"/><!--  0..1 What profile the resource needs to conform to -->
 <code><!--  0..1 CodeableConcept Detail type of activity   --></code>
 <intent value="[code]"/><!--  0..1 proposal | plan | directive | order | original-order | reflex-order | filler-order |
     instance-order | option   -->
 <priority value="[code]"/><!--  0..1 routine | urgent | asap | stat   -->
 <doNotPerform value="[boolean]"/><!--  0..1 True if the activity should not be performed -->
 <timing[x]><!--  0..1 Timing|dateTime|Age|Period|Range|
   Duration When activity is to occur --></timing[x]>
 <location><!--  0..1 Reference(Location)
     Where it should happen --></location>
```

```
 <participant>  0..* BackboneElement  <!--  0..* Who should participate in the action -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <type value="[code]"/><!--  1..1 patient | practitioner | related-person | device   -->
  <role><!--  0..1 CodeableConcept E.g. Nurse, Surgeon, Parent, etc.   --></role>
 </participant>
 <product[x]><!--  0..1 Reference(Medication)|CodeableConcept What's administered/supplied   --></product[x]>
 <quantity><!--  0..1 Quantity
     How much is administered/consumed/supplied --></quantity>
 <dosage><!--  0..* Dosage Detailed dosage instructions --></dosage>
 <bodySite><!--  0..* CodeableConcept What part of body to perform on   --></bodySite>
 <specimenRequirement><!--  0..* Reference(SpecimenDefinition)
     What specimens are required to perform this action --></specimenRequirement>
 <observationRequirement><!--  0..* Reference(ObservationDefinition)
     What observations are required to perform this action --></observationRequirement>
 <observationResultRequirement><!--  0..* Reference(ObservationDefinition)
     What observations must be produced by this action --></observationResultRequirement>
 <transform value="[canonical]"/><!--  0..1 Transform to apply the template -->
 <dynamicValue>  0..* BackboneElement  <!--  0..* Dynamic aspects of the definition -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <path value="[string]"/><!--  1..1 The path to the element to be set dynamically -->
  <expression><!--  1..1 Expression
      An expression that provides the dynamic value for the customization --></expression>
 </dynamicValue>
</ActivityDefinition>
```

JSON Template

```
{ // Shareable ActivityDefinition
  // from Element: extension
    "meta" : { Meta }, // C?Metadata about the resource
    "implicitRules" : "<uri>", // C?A set of rules under which this content was created
    "language" : "<code>", // C?Language of the resource content
    "text" : { Narrative }, // C?Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored
    "url" : "<uri>", // C? R! Canonical identifier for this activity definition, represented as a URI (globally unique)
    "identifier" : [{ Identifier }], // C?Additional identifier for the activity definition
    "version" : "<string>", // C? R! Business version of the activity definition
    "name" : "<string>", // C? R! Name for this activity definition (computer friendly)
    "title" : "<string>", // C?Name for this activity definition (human friendly)
    "subtitle" : "<string>", // C?Subordinate title of the activity definition
    "status" : "<code>", // C? R! draft | active | retired | unknown
    "experimental" : <boolean>, // C? R! For testing purposes, not real usage
// value[x]: Type of individual the activity definition is intended for. One of these 2:
    "subjectCodeableConcept" : { CodeableConcept }, // C?Type of individual the activity definition is intended for
    "subjectReference" : { Reference(Group) }, // C?Type of individual the activity definition is intended for
    "date" : "<dateTime>", // C?Date last changed
    "publisher" : "<string>", // C? R! Name of the publisher (organization or individual)
    "contact" : [{ ContactDetail }], // C?Contact details for the publisher
    "description" : "<markdown>", // C? R! Natural language description of the activity definition
    "useContext" : [{ UsageContext }], // C?The context that the content is intended to support
    "jurisdiction" : [{ CodeableConcept }], // C?Intended jurisdiction for activity definition (if applicable)
    "purpose" : "<markdown>", // C?Why this activity definition is defined
    "usage" : "<string>", // C?Describes the clinical usage of the activity definition
    "copyright" : "<markdown>", // C?Use and/or publishing restrictions
    "approvalDate" : "<date>", // C?When the activity definition was approved by publisher
    "lastReviewDate" : "<date>", // C?When the activity definition was last reviewed
    "effectivePeriod" : { Period }, // C?When the activity definition is expected to be used
    "topic" : [{ CodeableConcept }], // C?E.g. Education, Treatment, Assessment, etc.
    "author" : [{ ContactDetail }], // C?Who authored the content
    "editor" : [{ ContactDetail }], // C?Who edited the content
    "reviewer" : [{ ContactDetail }], // C?Who reviewed the content
    "endorser" : [{ ContactDetail }], // C?Who endorsed the content
    "relatedArtifact" : [{ RelatedArtifact }], // C?Additional documentation, citations, etc.
    "library" : ["<canonical>"], // C?Logic used by the activity definition
    "kind" : "<code>", // C?Kind of resource
    "profile" : "<canonical>", // C?What profile the resource needs to conform to
    "code" : { CodeableConcept }, // C?Detail type of activity
    "intent" : "<code>", // C?proposal | plan | directive | order | original-order | reflex-order | filler-order |
    instance-order | option
    "priority" : "<code>", // C?routine | urgent | asap | stat
    "doNotPerform" : <boolean>, // C?True if the activity should not be performed
// value[x]: When activity is to occur. One of these 6:
    "timingTiming" : { Timing }, // C?When activity is to occur
    "timingDateTime" : "<dateTime>", // C?When activity is to occur
    "timingAge" : { Age }, // C?When activity is to occur
    "timingPeriod" : { Period }, // C?When activity is to occur
    "timingRange" : { Range }, // C?When activity is to occur
    "timingDuration" : { Duration }, // C?When activity is to occur
    "location" : { Reference(Location) }, // C?Where it should happen
    "participant" : [{ BackboneElement }], // C?Who should participate in the action
// value[x]: What's administered/supplied. One of these 2:
    "productReference" : { Reference(Medication) }, // C?What's administered/supplied
    "productCodeableConcept" : { CodeableConcept }, // C?What's administered/supplied
    "quantity" : { Quantity(SimpleQuantity) } // C?How much is administered/consumed/supplied
    "dosage" : [{ Dosage }], // C?Detailed dosage instructions
    "bodySite" : [{ CodeableConcept }], // C?What part of body to perform on
    "specimenRequirement" : [{ Reference(SpecimenDefinition) }], // C?What specimens are required to perform this action
```

```
    "observationRequirement" : [{ Reference(ObservationDefinition) }], // C?What observations are required to perform
    this action
    "observationResultRequirement" : [{ Reference(ObservationDefinition) }], // C?What observations must be produced by
    this action
    "transform" : "<canonical>", // C?Transform to apply the template
    "dynamicValue" : [{ BackboneElement }], // C?Dynamic aspects of the definition
  }
```

Alternate definitions: Master Definition (XML, JSON), Schematron

12.17.11.2.2 Terminology Bindings

Path	Name	Conformance	ValueSet
ActivityDefinition.language	Common Languages	preferred	Common Languages
ActivityDefinition.status	PublicationStatus	required	PublicationStatus
ActivityDefinition.subject[x]	Subject type	extensible	Subject type
ActivityDefinition.jurisdiction	Jurisdiction	extensible	Jurisdiction
ActivityDefinition.topic	DefinitionTopic	example	DefinitionTopic
ActivityDefinition.kind	RequestResourceType	required	RequestResourceType
ActivityDefinition.code	Procedure Codes (SNOMED CT)	example	Procedure Codes (SNOMED CT)
ActivityDefinition.intent	RequestIntent	required	RequestIntent
ActivityDefinition.priority	Request priority	required	Request priority
ActivityDefinition.participant.type	ActionParticipantType	required	ActionParticipantType
ActivityDefinition.participant.role	Action participant role	example	Action participant role
ActivityDefinition.product[x]	SNOMED CT Medication Codes	example	SNOMED CT Medication Codes
ActivityDefinition.bodySite	SNOMED CT Body Structures	example	SNOMED CT Body Structures

12.17.12 JSON Schema for ActivityDefinition

URL: https://hl7.org/fhir/R4/activitydefinition.schema.json.html

12.17.12 JSON Schema for ActivityDefinition

Clinical Decision Support Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Not linked to any defined compartments

This is a representation of the json schema for ActivityDefinition, which is just a part of the full JSON Schema.

```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/ActivityDefinition",
  "$ref": "#/definitions/ActivityDefinition",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "ActivityDefinition": {
      "description": "This resource allows for the definition of some activity to be performed, independent of a
    particular patient, practitioner, or other performance context.",
      "properties": {
        "resourceType": {
          "description": "This is a ActivityDefinition resource",
          "const": "ActivityDefinition"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
          "$ref": "#/definitions/code"
        },
        "_language": {
          "description": "Extensions for language",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "text": {
          "description": "A human-readable narrative that contains a summary of the resource and can be used to
    represent the content of the resource to a human. The narrative need not encode all the structured data, but is
    required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
    Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
          "$ref": "Narrative.schema.json#/definitions/Narrative"
```

```
    },
    "contained": {
      "description": "These resources do not have an independent existence apart from the resource that contains
them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
      "items": {
        "$ref": "ResourceList.schema.json#/definitions/ResourceList"
      },
      "type": "array"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
 definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource and that modifies the understanding of the element that contains it and/or the understanding of the
containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
met as part of the definition of the extension. Applications processing a resource are required to check for
modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "url": {
      "description": "An absolute URI that is used to identify this activity definition when it is referenced in a
specification, model, design or an instance; also called its canonical identifier. This SHOULD be globally unique
and SHOULD be a literal address at which at which an authoritative instance of this activity definition is (or will
be) published. This URL can be the target of a canonical reference. It SHALL remain the same when the activity
definition is stored on different servers.",
      "$ref": "#/definitions/uri"
    },
    "_url": {
      "description": "Extensions for url",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "identifier": {
      "description": "A formal identifier that is used to identify this activity definition when it is represented
in other formats, or referenced in a specification, model, design or an instance.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "version": {
      "description": "The identifier that is used to identify this version of the activity definition when it is
referenced in a specification, model, design or instance. This is an arbitrary value managed by the activity
definition author and is not expected to be globally unique. For example, it might be a timestamp (e.g. yyyymmdd) if
 a managed version is not available. There is also no expectation that versions can be placed in a lexicographical
sequence. To provide a version consistent with the Decision Support Service specification, use the format
Major.Minor.Revision (e.g. 1.0.0). For more information on versioning knowledge assets, refer to the Decision
Support Service specification. Note that a version is required for non-experimental active assets.",
      "$ref": "#/definitions/string"
    },
    "_version": {
      "description": "Extensions for version",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "name": {
      "description": "A natural language name identifying the activity definition. This name should be usable as an
identifier for the module by machine processing applications such as code generation.",
      "$ref": "#/definitions/string"
    },
    "_name": {
      "description": "Extensions for name",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "title": {
      "description": "A short, descriptive, user-friendly title for the activity definition.",
      "$ref": "#/definitions/string"
    },
    "_title": {
      "description": "Extensions for title",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "subtitle": {
      "description": "An explanatory or alternate title for the activity definition giving additional information
about its content.",
      "$ref": "#/definitions/string"
    },
```

```
    "_subtitle": {
      "description": "Extensions for subtitle",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "status": {
      "description": "The status of this activity definition. Enables tracking the life-cycle of the content.",
      "enum": [
        "draft",
        "active",
        "retired",
        "unknown"
      ]
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "experimental": {
      "description": "A Boolean value to indicate that this activity definition is authored for testing purposes (or
 education/evaluation/marketing) and is not intended to be used for genuine usage.",
      "$ref": "#/definitions/boolean"
    },
    "_experimental": {
      "description": "Extensions for experimental",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "subjectCodeableConcept": {
      "description": "A code or group definition that describes the intended subject of the activity being
defined.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "subjectReference": {
      "description": "A code or group definition that describes the intended subject of the activity being
defined.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "date": {
      "description": "The date  (and optionally time) when the activity definition was published. The date must
change when the business version changes and it must change if the status code changes. In addition, it should
change when the substantive content of the activity definition changes.",
      "$ref": "#/definitions/dateTime"
    },
    "_date": {
      "description": "Extensions for date",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "publisher": {
      "description": "The name of the organization or individual that published the activity definition.",
      "$ref": "#/definitions/string"
    },
    "_publisher": {
      "description": "Extensions for publisher",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "contact": {
      "description": "Contact details to assist a user in finding and communicating with the publisher.",
      "items": {
        "$ref": "ContactDetail.schema.json#/definitions/ContactDetail"
      },
      "type": "array"
    },
    "description": {
      "description": "A free text natural language description of the activity definition from a consumer\u0027s
perspective.",
      "$ref": "#/definitions/markdown"
    },
    "_description": {
      "description": "Extensions for description",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "useContext": {
      "description": "The content was developed with a focus and intent of supporting the contexts that are listed.
These contexts may be general categories (gender, age, ...) or may be references to specific programs (insurance
plans, studies, ...) and may be used to assist with indexing and searching for appropriate activity definition
instances.",
      "items": {
        "$ref": "UsageContext.schema.json#/definitions/UsageContext"
      },
      "type": "array"
    },
    "jurisdiction": {
      "description": "A legal or geographic region in which the activity definition is intended to be used.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "purpose": {
      "description": "Explanation of why this activity definition is needed and why it has been designed as it
```

```
has.",
      "$ref": "#/definitions/markdown"
    },
    "_purpose": {
      "description": "Extensions for purpose",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "usage": {
      "description": "A detailed description of how the activity definition is used from a clinical perspective.",
      "$ref": "#/definitions/string"
    },
    "_usage": {
      "description": "Extensions for usage",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "copyright": {
      "description": "A copyright statement relating to the activity definition and/or its contents. Copyright
statements are generally legal restrictions on the use and publishing of the activity definition.",
      "$ref": "#/definitions/markdown"
    },
    "_copyright": {
      "description": "Extensions for copyright",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "approvalDate": {
      "description": "The date on which the resource content was approved by the publisher. Approval happens once
when the content is officially approved for usage.",
      "$ref": "#/definitions/date"
    },
    "_approvalDate": {
      "description": "Extensions for approvalDate",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "lastReviewDate": {
      "description": "The date on which the resource content was last reviewed. Review happens periodically after
approval but does not change the original approval date.",
      "$ref": "#/definitions/date"
    },
    "_lastReviewDate": {
      "description": "Extensions for lastReviewDate",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "effectivePeriod": {
      "description": "The period during which the activity definition content was or is planned to be in active
use.",
      "$ref": "Period.schema.json#/definitions/Period"
    },
    "topic": {
      "description": "Descriptive topics related to the content of the activity. Topics provide a high-level
categorization of the activity that can be useful for filtering and searching.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "author": {
      "description": "An individiual or organization primarily involved in the creation and maintenance of the
content.",
      "items": {
        "$ref": "ContactDetail.schema.json#/definitions/ContactDetail"
      },
      "type": "array"
    },
    "editor": {
      "description": "An individual or organization primarily responsible for internal coherence of the content.",
      "items": {
        "$ref": "ContactDetail.schema.json#/definitions/ContactDetail"
      },
      "type": "array"
    },
    "reviewer": {
      "description": "An individual or organization primarily responsible for review of some aspect of the
content.",
      "items": {
        "$ref": "ContactDetail.schema.json#/definitions/ContactDetail"
      },
      "type": "array"
    },
    "endorser": {
      "description": "An individual or organization responsible for officially endorsing the content for use in some
 setting.",
      "items": {
        "$ref": "ContactDetail.schema.json#/definitions/ContactDetail"
      },
      "type": "array"
    },
    "relatedArtifact": {
      "description": "Related artifacts such as additional documentation, justification, or bibliographic
references.",
```

```
      "items": {
        "$ref": "RelatedArtifact.schema.json#/definitions/RelatedArtifact"
      },
      "type": "array"
    },
    "library": {
      "description": "A reference to a Library resource containing any formal logic used by the activity
definition.",
      "items": {
        "$ref": "canonical.schema.json#/definitions/canonical"
      },
      "type": "array"
    },
    "kind": {
      "description": "A description of the kind of resource the activity definition is representing. For example, a
MedicationRequest, a ServiceRequest, or a CommunicationRequest. Typically, but not always, this is a Request
resource.",
      "$ref": "#/definitions/code"
    },
    "_kind": {
      "description": "Extensions for kind",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "profile": {
      "description": "A profile to which the target of the activity definition is expected to conform.",
      "$ref": "canonical.schema.json#/definitions/canonical"
    },
    "code": {
      "description": "Detailed description of the type of activity; e.g. What lab test, what procedure, what kind of
 encounter.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "intent": {
      "description": "Indicates the level of authority/intentionality associated with the activity and where the
request should fit into the workflow chain.",
      "$ref": "#/definitions/code"
    },
    "_intent": {
      "description": "Extensions for intent",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "priority": {
      "description": "Indicates how quickly the activity  should be addressed with respect to other requests.",
      "$ref": "#/definitions/code"
    },
    "_priority": {
      "description": "Extensions for priority",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "doNotPerform": {
      "description": "Set this to true if the definition is to indicate that a particular activity should NOT be
performed. If true, this element should be interpreted to reinforce a negative coding. For example NPO as a code
with a doNotPerform of true would still indicate to NOT perform the action.",
      "$ref": "#/definitions/boolean"
    },
    "_doNotPerform": {
      "description": "Extensions for doNotPerform",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "timingTiming": {
      "description": "The period, timing or frequency upon which the described activity is to occur.",
      "$ref": "Timing.schema.json#/definitions/Timing"
    },
    "timingDateTime": {
      "description": "The period, timing or frequency upon which the described activity is to occur.",
      "pattern": "^([0-9]([0-9]([0-9][1-9]|[1-9]0)|[1-9]00)|[1-9]000)(-(0[1-9]|1[0-2])(-(0[1-9]|[1-2][0-9]|3[0-
1])(T([01][0-9]|2[0-3]):[0-5][0-9]:([0-5][0-9]|60)(\\.[0-9]+)?(Z|(\\+|-)((0[0-9]|1[0-3]):[0-5][0-9]|14:00)))?)?)?$",
      "type": "string"
    },
    "_timingDateTime": {
      "description": "Extensions for timingDateTime",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "timingAge": {
      "description": "The period, timing or frequency upon which the described activity is to occur.",
      "$ref": "Age.schema.json#/definitions/Age"
    },
    "timingPeriod": {
      "description": "The period, timing or frequency upon which the described activity is to occur.",
      "$ref": "Period.schema.json#/definitions/Period"
    },
    "timingRange": {
      "description": "The period, timing or frequency upon which the described activity is to occur.",
      "$ref": "Range.schema.json#/definitions/Range"
    },
    "timingDuration": {
      "description": "The period, timing or frequency upon which the described activity is to occur.",
      "$ref": "Duration.schema.json#/definitions/Duration"
    },
```

```
    "location": {
      "description": "Identifies the facility where the activity will occur; e.g. home, hospital, specific clinic,
etc.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "participant": {
      "description": "Indicates who should participate in performing the action described.",
      "items": {
        "$ref": "#/definitions/ActivityDefinition_Participant"
      },
      "type": "array"
    },
    "productReference": {
      "description": "Identifies the food, drug or other product being consumed or supplied in the activity.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "productCodeableConcept": {
      "description": "Identifies the food, drug or other product being consumed or supplied in the activity.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "quantity": {
      "description": "Identifies the quantity expected to be consumed at once (per dose, per meal, etc.).",
      "$ref": "Quantity.schema.json#/definitions/Quantity"
    },
    "dosage": {
      "description": "Provides detailed dosage instructions in the same way that they are described for
MedicationRequest resources.",
      "items": {
        "$ref": "Dosage.schema.json#/definitions/Dosage"
      },
      "type": "array"
    },
    "bodySite": {
      "description": "Indicates the sites on the subject\u0027s body where the procedure should be performed (I.e.
the target sites).",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "specimenRequirement": {
      "description": "Defines specimen requirements for the action to be performed, such as required specimens for a
 lab test.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "observationRequirement": {
      "description": "Defines observation requirements for the action to be performed, such as body weight or
surface area.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "observationResultRequirement": {
      "description": "Defines the observations that are expected to be produced by the action.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "transform": {
      "description": "A reference to a StructureMap resource that defines a transform that can be executed to
produce the intent resource using the ActivityDefinition instance as the input.",
      "$ref": "canonical.schema.json#/definitions/canonical"
    },
    "dynamicValue": {
      "description": "Dynamic values that will be evaluated to produce values for elements of the resulting
resource. For example, if the dosage of a medication must be computed based on the patient\u0027s weight, a dynamic
value would be used to specify an expression that calculated the weight, and the path on the request resource that
would contain the result.",
      "items": {
        "$ref": "#/definitions/ActivityDefinition_DynamicValue"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "resourceType"
  ]
},
"ActivityDefinition_Participant": {
  "description": "This resource allows for the definition of some activity to be performed, independent of a
particular patient, practitioner, or other performance context.",
  "properties": {
    "id": {
```

```
      "description": "Unique id for the element within a resource (for internal references). This may be any string value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the element and that modifies the understanding of the element in which it is contained and/or the understanding of the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension. Applications processing a resource are required to check for modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "type": {
      "description": "The type of participant in the action.",
      "$ref": "#/definitions/code"
    },
    "_type": {
      "description": "Extensions for type",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "role": {
      "description": "The role the participant should play in performing the described action.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    }
  },
  "additionalProperties": false
},
"ActivityDefinition_DynamicValue": {
  "description": "This resource allows for the definition of some activity to be performed, independent of a particular patient, practitioner, or other performance context.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the element and that modifies the understanding of the element in which it is contained and/or the understanding of the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension. Applications processing a resource are required to check for modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "path": {
      "description": "The path to the element to be customized. This is the path on the resource that will hold the result of the calculation defined by the expression. The specified path SHALL be a FHIRPath resolveable on the specified target type of the ActivityDefinition, and SHALL consist only of identifiers, constant indexers, and a restricted subset of functions. The path is allowed to contain qualifiers (.) to traverse sub-elements, as well as indexers ([x]) to traverse multiple-cardinality sub-elements (see the [Simple FHIRPath Profile](fhirpath.html#simple) for full details).",
      "$ref": "#/definitions/string"
    },
    "_path": {
      "description": "Extensions for path",
      "$ref": "Element.schema.json#/definitions/Element"
    },
```

```
        "expression": {
          "description": "An expression specifying the value of the customized element.",
          "$ref": "Expression.schema.json#/definitions/Expression"
        }
      },
      "additionalProperties": false,
      "required": [
        "expression"
      ]
    }
  }
}
```

12.17.13 Resource ActivityDefinition - Maturity Dependency View

URL: https://hl7.org/fhir/R4/activitydefinition-dependencies.html

12.17.13 Resource ActivityDefinition - Maturity Dependency View

Clinical Decision Support Work Group | Maturity Level: 2 | Trial Use | Compartments: Not linked to any defined compartments

Dependency Graph for ActivityDefinition FMM level 2

Name | Card. | Type | Dependency Analysis
ActivityDefinition | | DomainResource
url | 0..1 | uri
identifier | 0..* | Identifier
version | 0..1 | string
name | 0..1 | string
title | 0..1 | string
subtitle | 0..1 | string
status | 1..1 | code
experimental | 0..1 | boolean
subject[x] | 0..1
subjectCodeableConcept | | CodeableConcept
subjectReference | | Reference(Group
date | 0..1 | dateTime
publisher | 0..1 | string
contact | 0..* | ContactDetail
description | 0..1 | markdown
useContext | 0..* | UsageContext
jurisdiction | 0..* | CodeableConcept
purpose | 0..1 | markdown
usage | 0..1 | string
copyright | 0..1 | markdown
approvalDate | 0..1 | date
lastReviewDate | 0..1 | date
effectivePeriod | 0..1 | Period
topic | 0..* | CodeableConcept
author | 0..* | ContactDetail
editor | 0..* | ContactDetail
reviewer | 0..* | ContactDetail
endorser | 0..* | ContactDetail
relatedArtifact | 0..* | RelatedArtifact
library | 0..* | canonical(Library)
kind | 0..1 | code
profile | 0..1 | canonical(StructureDefinition)
code | 0..1 | CodeableConcept
intent | 0..1 | code
priority | 0..1 | code
doNotPerform | 0..1 | boolean
timing[x] | 0..1
timingTiming | | Timing
timingDateTime | | dateTime
timingAge | | Age
timingPeriod | | Period
timingRange | | Range
timingDuration | | Duration
location | 0..1 | Reference(Location)
participant | 0..* | BackboneElement
type | 1..1 | code
role | 0..1 | CodeableConcept
product[x] | 0..1
productReference | | Reference(Medication | Substance
productCodeableConcept | | CodeableConcept

quantity | 0..1 | SimpleQuantity
dosage | 0..* | Dosage
bodySite | 0..* | CodeableConcept
specimenRequirement | 0..* | Reference(SpecimenDefinition) | Type Warning: (SpecimenDefinition = FMM0-Trial Use vs. Element = FMM2-Trial Use)
observationRequirement | 0..* | Reference(ObservationDefinition) | Type Warning: (ObservationDefinition = FMM0-Trial Use vs. Element = FMM2-Trial Use)
observationResultRequirement | 0..* | Reference(ObservationDefinition) | Type Warning: (ObservationDefinition = FMM0-Trial Use vs. Element = FMM2-Trial Use)
transform | 0..1 | canonical(StructureMap)
dynamicValue | 0..* | BackboneElement
path | 1..1 | string
expression | 1..1 | Expression

12.18 Resource PlanDefinition - Content

URL: https://hl7.org/fhir/R4/plandefinition.html
Clinical Decision Support Work Group | Maturity Level: 2 | Trial Use | Security Category: Business | Compartments: Not linked to any defined compartments

This resource allows for the definition of various types of plans as a sharable, consumable, and executable artifact. The resource is general enough to support the description of a broad range of clinical artifacts such as clinical decision support rules, order sets and protocols.

12.18.1 Scope and Usage

This resource is a definition resource from a FHIR workflow perspective - see Workflow, specifically Definition.

A plan definition is a pre-defined group of actions to be taken in particular circumstances, often including conditional elements, options, and other decision points. The resource is flexible enough to be used to represent a variety of workflows, as well as clinical decision support and quality improvement assets, including order sets, protocols, and decision support rules.

PlanDefinitions can contain hierarchical groups of action definitions, where each action definition describes an activity to be performed (often in terms of an ActivityDefinition resource), and each group defines additional behavior, relationships, and applicable conditions between the actions in the overall definition.

In addition to describing what should take place, each action in a plan definition can specify when and whether the action should take place. For when the action should be taken, the trigger element specifies the action should be taken in response to some trigger occurring (such as a particular point in a workflow being reached, or as the result of a prescription being ordered). For whether the action should be taken, the condition element can be used to provide an expression that evaluates to true or false to indicate the applicability of the action to the specific context.

The process of applying a PlanDefinition to a particular context typically produces request resources representing the actions that should be performed, typically grouped within a CarePlan and/or RequestGroup to capture relationships between the resulting request resources.

Each ActivityDefinition is used to construct a specific resource, based on the definition of the activity and combined with contextual information for the particular patient that the plan definition is being applied to.

As with the ActivityDefinition, a PlanDefinition may provide information about how to transform the activity to a specific intent resource, either by specifying a StructureMap that can be used to perform the transformation completely, or by specifying values for specific elements of the resulting resource using dynamicValue elements in the action.

Note that these mechanisms are provided on both the ActivityDefinition and the PlanDefinition to allow both reusable transformation descriptions, as well as customization of those descriptions within specific contexts. As such, the transform descriptions specified on the PlanDefinition override transform descriptions defined on the ActivityDefinition.

Dynamic values within the definitions can be provided by specifying the expression directly, or by referencing an expression defined within a library. For more information on how to reference expressions within resources, refer to the Using Expressions topic.

As an example, the Low Suicide Risk example order set from the Clinical Decision Support Knowledge Artifact Specification can be represented using the PlanDefinition and ActivityDefinition structures: Low Suicide Risk Example Order Set.

In addition to the representation of PlanDefinitions, the $apply operation allows PlanDefinitions to be applied to a specific context such as a patient, practitioner, or institution. For Order Sets specifically, this operation is expected to place the orders defined by the order set, consistent with the service functional requirements defined by the Order Set specification .

12.18.2 Boundaries and Relationships

The PlanDefinition resource is used to describe series, sequences, or groups of actions to be taken, while the ActivityDefinition resource is used to define each specific step or activity to be performed.

As the name implies, the PlanDefinition resource is strictly definitional. It does not represent the intention to take any action, nor does it represent that any actions have been taken. Rather, the resource provides a definition that can be applied in the appropriate circumstances. When the plan definition is applied, the result will in general be a set of actions that should be (or potentially even have been) performed.

Note that the PlanDefinition still has action-level information, as well as a reference to an ActivityDefinition. The action-level information defined in the PlanDefinition itself is used to describe how the actions are related to each other within the plan, where the ActivityDefinition contains only information about the activity itself. In addition, there is some overlapping information that allows the resources to be used independently, or in combination. See the Applying a PlanDefinition section for more information.

This resource is referenced by UsageContext, CarePlan, CatalogEntry, Communication, DeviceRequest, FamilyMemberHistory, Flag, MessageDefinition, NutritionOrder, itself, Procedure, ResearchStudy and ServiceRequest

12.18.3 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
PlanDefinition | ITU | | DomainResource | The definition of a plan for a series of actions, independent of any specific patient or context + Warning: Name should be usable as an identifier for the module by machine processing applications such as code generation Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
url | Σ | 0..1 | uri | Canonical identifier for this plan definition, represented as a URI (globally unique)
identifier | Σ | 0..* | Identifier | Additional identifier for the plan definition
version | Σ | 0..1 | string | Business version of the plan definition
name | ΣI | 0..1 | string | Name for this plan definition (computer friendly)
title | Σ | 0..1 | string | Name for this plan definition (human friendly)
subtitle | | 0..1 | string | Subordinate title of the plan definition
type | Σ | 0..1 | CodeableConcept | order-set | clinical-protocol | eca-rule | workflow-definition PlanDefinitionType (Extensible)
status | ?!Σ | 1..1 | code | draft | active | retired | unknown PublicationStatus (Required)
experimental | Σ | 0..1 | boolean | For testing purposes, not real usage
subject[x] | | 0..1 | | Type of individual the plan definition is focused on Subject type (Extensible)
subjectCodeableConcept | | | CodeableConcept
subjectReference | | | Reference(Group)
date | Σ | 0..1 | dateTime | Date last changed
publisher | Σ | 0..1 | string | Name of the publisher (organization or individual)
contact | Σ | 0..* | ContactDetail | Contact details for the publisher
description | Σ | 0..1 | markdown | Natural language description of the plan definition
useContext | Σ | 0..* | UsageContext | The context that the content is intended to support
jurisdiction | Σ | 0..* | CodeableConcept | Intended jurisdiction for plan definition (if applicable) Jurisdiction (Extensible)
purpose | | 0..1 | markdown | Why this plan definition is defined
usage | | 0..1 | string | Describes the clinical usage of the plan
copyright | | 0..1 | markdown | Use and/or publishing restrictions
approvalDate | | 0..1 | date | When the plan definition was approved by publisher
lastReviewDate | | 0..1 | date | When the plan definition was last reviewed
effectivePeriod | Σ | 0..1 | Period | When the plan definition is expected to be used
topic | | 0..* | CodeableConcept | E.g. Education, Treatment, Assessment DefinitionTopic (Example)
author | | 0..* | ContactDetail | Who authored the content
editor | | 0..* | ContactDetail | Who edited the content
reviewer | | 0..* | ContactDetail | Who reviewed the content
endorser | | 0..* | ContactDetail | Who endorsed the content
relatedArtifact | | 0..* | RelatedArtifact | Additional documentation, citations
library | | 0..* | canonical(Library) | Logic used by the plan definition
goal | | 0..* | BackboneElement | What the plan is trying to accomplish
category | | 0..1 | CodeableConcept | E.g. Treatment, dietary, behavioral Goal category (Example)
description | | 1..1 | CodeableConcept | Code or text describing the goal SNOMED CT Clinical Findings (Example)
priority | | 0..1 | CodeableConcept | high-priority | medium-priority | low-priority Goal priority (Preferred)
start | | 0..1 | CodeableConcept | When goal pursuit begins Goal start event (Example)
addresses | | 0..* | CodeableConcept | What does the goal address Condition/Problem/Diagnosis Codes (Example)
documentation | | 0..* | RelatedArtifact | Supporting documentation for the goal
target | | 0..* | BackboneElement | Target outcome for the goal

measure | | 0..1 | CodeableConcept | The parameter whose value is to be tracked LOINC Codes (Example)
detail[x] | | 0..1 | | The target value to be achieved
detailQuantity | | | Quantity
detailRange | | | Range
detailCodeableConcept | | | CodeableConcept
due | | 0..1 | Duration | Reach goal within
action | | 0..* | BackboneElement | Action defined by the plan
prefix | | 0..1 | string | User-visible prefix for the action (e.g. 1. or A.)
title | | 0..1 | string | User-visible title
description | | 0..1 | string | Brief description of the action
textEquivalent | | 0..1 | string | Static text equivalent of the action, used if the dynamic aspects cannot be interpreted by the receiving system
priority | | 0..1 | code | routine | urgent | asap | stat Request priority (Required)
code | | 0..* | CodeableConcept | Code representing the meaning of the action or sub-actions
reason | | 0..* | CodeableConcept | Why the action should be performed
documentation | | 0..* | RelatedArtifact | Supporting documentation for the intended performer of the action
goalId | | 0..* | id | What goals this action supports
subject[x] | | 0..1 | | Type of individual the action is focused on Subject type (Extensible)
subjectCodeableConcept | | | CodeableConcept
subjectReference | | | Reference(Group)
trigger | | 0..* | TriggerDefinition | When the action should be triggered
condition | | 0..* | BackboneElement | Whether or not the action is applicable
kind | | 1..1 | code | applicability | start | stop ActionConditionKind (Required)
expression | | 0..1 | Expression | Boolean-valued expression
input | | 0..* | DataRequirement | Input data requirements
output | | 0..* | DataRequirement | Output data definition
relatedAction | | 0..* | BackboneElement | Relationship to another action
actionId | | 1..1 | id | What action is this related to
relationship | | 1..1 | code | before-start | before | before-end | concurrent-with-start | concurrent | concurrent-with-end | after-start | after | after-end ActionRelationshipType (Required)
offset[x] | | 0..1 | | Time offset for the relationship
offsetDuration | | | Duration
offsetRange | | | Range
timing[x] | | 0..1 | | When the action should take place
timingDateTime | | | dateTime
timingAge | | | Age
timingPeriod | | | Period
timingDuration | | | Duration
timingRange | | | Range
timingTiming | | | Timing
participant | | 0..* | BackboneElement | Who should participate in the action
type | | 1..1 | code | patient | practitioner | related-person | device ActionParticipantType (Required)
role | | 0..1 | CodeableConcept | E.g. Nurse, Surgeon, Parent Action participant role (Example)
type | | 0..1 | CodeableConcept | create | update | remove | fire-event ActionType (Extensible)
groupingBehavior | | 0..1 | code | visual-group | logical-group | sentence-group ActionGroupingBehavior (Required)
selectionBehavior | | 0..1 | code | any | all | all-or-none | exactly-one | at-most-one | one-or-more ActionSelectionBehavior (Required)
requiredBehavior | | 0..1 | code | must | could | must-unless-documented ActionRequiredBehavior (Required)
precheckBehavior | | 0..1 | code | yes | no ActionPrecheckBehavior (Required)
cardinalityBehavior | | 0..1 | code | single | multiple ActionCardinalityBehavior (Required)
definition[x] | | 0..1 | | Description of the activity to be performed
definitionCanonical | | | canonical(ActivityDefinition | PlanDefinition | Questionnaire)
definitionUri | | | uri
transform | | 0..1 | canonical(StructureMap) | Transform to apply the template
dynamicValue | | 0..* | BackboneElement | Dynamic aspects of the definition
path | | 0..1 | string | The path to the element to be set dynamically
expression | | 0..1 | Expression | An expression that provides the dynamic value for the customization
action | | 0..* | see action | A sub-action
Documentation for this format

UML Diagram (Legend)

XML Template

```
<PlanDefinition xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <url value="[uri]"/><!-- 0..1 Canonical identifier for this plan definition, represented as a URI (globally unique) -->
 <identifier><!-- 0..* Identifier Additional identifier for the plan definition --></identifier>
 <version value="[string]"/><!-- 0..1 Business version of the plan definition -->
 <name value="[string]"/><!--  0..1 Name for this plan definition (computer friendly) -->
 <title value="[string]"/><!-- 0..1 Name for this plan definition (human friendly) -->
 <subtitle value="[string]"/><!-- 0..1 Subordinate title of the plan definition -->
 <type><!-- 0..1 CodeableConcept order-set | clinical-protocol | eca-rule | workflow-definition --></type>
 <status value="[code]"/><!-- 1..1 draft | active | retired | unknown -->
 <experimental value="[boolean]"/><!-- 0..1 For testing purposes, not real usage -->
 <subject[x]><!-- 0..1 CodeableConcept|Reference(Group) Type of individual the plan definition is focused on
```

```
    --></subject[x]>
 <date value="[dateTime]"/><!-- 0..1 Date last changed -->
 <publisher value="[string]"/><!-- 0..1 Name of the publisher (organization or individual) -->
 <contact><!-- 0..* ContactDetail Contact details for the publisher --></contact>
 <description value="[markdown]"/><!-- 0..1 Natural language description of the plan definition -->
 <useContext><!-- 0..* UsageContext The context that the content is intended to support --></useContext>
 <jurisdiction><!-- 0..* CodeableConcept Intended jurisdiction for plan definition (if applicable) --></jurisdiction>
 <purpose value="[markdown]"/><!-- 0..1 Why this plan definition is defined -->
 <usage value="[string]"/><!-- 0..1 Describes the clinical usage of the plan -->
 <copyright value="[markdown]"/><!-- 0..1 Use and/or publishing restrictions -->
 <approvalDate value="[date]"/><!-- 0..1 When the plan definition was approved by publisher -->
 <lastReviewDate value="[date]"/><!-- 0..1 When the plan definition was last reviewed -->
 <effectivePeriod><!-- 0..1 Period When the plan definition is expected to be used --></effectivePeriod>
 <topic><!-- 0..* CodeableConcept E.g. Education, Treatment, Assessment --></topic>
 <author><!-- 0..* ContactDetail Who authored the content --></author>
 <editor><!-- 0..* ContactDetail Who edited the content --></editor>
 <reviewer><!-- 0..* ContactDetail Who reviewed the content --></reviewer>
 <endorser><!-- 0..* ContactDetail Who endorsed the content --></endorser>
 <relatedArtifact><!-- 0..* RelatedArtifact Additional documentation, citations --></relatedArtifact>
 <library><!-- 0..* canonical(Library) Logic used by the plan definition --></library>
 <goal>  <!-- 0..* What the plan is trying to accomplish -->
  <category><!-- 0..1 CodeableConcept E.g. Treatment, dietary, behavioral --></category>
  <description><!-- 1..1 CodeableConcept Code or text describing the goal --></description>
  <priority><!-- 0..1 CodeableConcept high-priority | medium-priority | low-priority --></priority>
  <start><!-- 0..1 CodeableConcept When goal pursuit begins --></start>
  <addresses><!-- 0..* CodeableConcept What does the goal address --></addresses>
  <documentation><!-- 0..* RelatedArtifact Supporting documentation for the goal --></documentation>
  <target>  <!-- 0..* Target outcome for the goal -->
   <measure><!-- 0..1 CodeableConcept The parameter whose value is to be tracked --></measure>
   <detail[x]><!-- 0..1 Quantity|Range|CodeableConcept The target value to be achieved --></detail[x]>
   <due><!-- 0..1 Duration Reach goal within --></due>
  </target>
 </goal>
 <action>  <!-- 0..* Action defined by the plan -->
  <prefix value="[string]"/><!-- 0..1 User-visible prefix for the action (e.g. 1. or A.) -->
  <title value="[string]"/><!-- 0..1 User-visible title -->
  <description value="[string]"/><!-- 0..1 Brief description of the action -->
  <textEquivalent value="[string]"/><!-- 0..1 Static text equivalent of the action, used if the dynamic aspects cannot
    be interpreted by the receiving system -->
  <priority value="[code]"/><!-- 0..1 routine | urgent | asap | stat -->
  <code><!-- 0..* CodeableConcept Code representing the meaning of the action or sub-actions --></code>
  <reason><!-- 0..* CodeableConcept Why the action should be performed --></reason>
  <documentation><!-- 0..* RelatedArtifact Supporting documentation for the intended performer of the action
    --></documentation>
  <goalId value="[id]"/><!-- 0..* What goals this action supports -->
  <subject[x]><!-- 0..1 CodeableConcept|Reference(Group) Type of individual the action is focused on --></subject[x]>
  <trigger><!-- 0..* TriggerDefinition When the action should be triggered --></trigger>
  <condition>  <!-- 0..* Whether or not the action is applicable -->
   <kind value="[code]"/><!-- 1..1 applicability | start | stop -->
   <expression><!-- 0..1 Expression Boolean-valued expression --></expression>
  </condition>
  <input><!-- 0..* DataRequirement Input data requirements --></input>
  <output><!-- 0..* DataRequirement Output data definition --></output>
  <relatedAction>  <!-- 0..* Relationship to another action -->
   <actionId value="[id]"/><!-- 1..1 What action is this related to -->
   <relationship value="[code]"/><!-- 1..1 before-start | before | before-end | concurrent-with-start | concurrent |
    concurrent-with-end | after-start | after | after-end -->
   <offset[x]><!-- 0..1 Duration|Range Time offset for the relationship --></offset[x]>
  </relatedAction>
  <timing[x]><!-- 0..1 dateTime|Age|Period|Duration|Range|Timing When the action should take place --></timing[x]>
  <participant>  <!-- 0..* Who should participate in the action -->
   <type value="[code]"/><!-- 1..1 patient | practitioner | related-person | device -->
   <role><!-- 0..1 CodeableConcept E.g. Nurse, Surgeon, Parent --></role>
  </participant>
  <type><!-- 0..1 CodeableConcept create | update | remove | fire-event --></type>
  <groupingBehavior value="[code]"/><!-- 0..1 visual-group | logical-group | sentence-group -->
  <selectionBehavior value="[code]"/><!-- 0..1 any | all | all-or-none | exactly-one | at-most-one | one-or-more -->
  <requiredBehavior value="[code]"/><!-- 0..1 must | could | must-unless-documented -->
  <precheckBehavior value="[code]"/><!-- 0..1 yes | no -->
  <cardinalityBehavior value="[code]"/><!-- 0..1 single | multiple -->
  <definition[x]><!-- 0..1 canonical(ActivityDefinition|PlanDefinition|
    Questionnaire)|uri Description of the activity to be performed --></definition[x]>
  <transform><!-- 0..1 canonical(StructureMap) Transform to apply the template --></transform>
  <dynamicValue>  <!-- 0..* Dynamic aspects of the definition -->
   <path value="[string]"/><!-- 0..1 The path to the element to be set dynamically -->
   <expression><!-- 0..1 Expression An expression that provides the dynamic value for the customization --></expression>
  </dynamicValue>
  <action><!-- 0..* Content as for PlanDefinition.action A sub-action --></action>
 </action>
</PlanDefinition>
```

JSON Template

```
{
  "resourceType" : "PlanDefinition",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "url" : "<uri>", // Canonical identifier for this plan definition, represented as a URI (globally unique)
  "identifier" : [{ Identifier }], // Additional identifier for the plan definition
  "version" : "<string>", // Business version of the plan definition
```

```
"name" : "<string>", // C? Name for this plan definition (computer friendly)
"title" : "<string>", // Name for this plan definition (human friendly)
"subtitle" : "<string>", // Subordinate title of the plan definition
"type" : { CodeableConcept }, // order-set | clinical-protocol | eca-rule | workflow-definition
"status" : "<code>", // R!  draft | active | retired | unknown
"experimental" : <boolean>, // For testing purposes, not real usage
// subject[x]: Type of individual the plan definition is focused on. One of these 2:
"subjectCodeableConcept" : { CodeableConcept },
"subjectReference" : { Reference(Group) },
"date" : "<dateTime>", // Date last changed
"publisher" : "<string>", // Name of the publisher (organization or individual)
"contact" : [{ ContactDetail }], // Contact details for the publisher
"description" : "<markdown>", // Natural language description of the plan definition
"useContext" : [{ UsageContext }], // The context that the content is intended to support
"jurisdiction" : [{ CodeableConcept }], // Intended jurisdiction for plan definition (if applicable)
"purpose" : "<markdown>", // Why this plan definition is defined
"usage" : "<string>", // Describes the clinical usage of the plan
"copyright" : "<markdown>", // Use and/or publishing restrictions
"approvalDate" : "<date>", // When the plan definition was approved by publisher
"lastReviewDate" : "<date>", // When the plan definition was last reviewed
"effectivePeriod" : { Period }, // When the plan definition is expected to be used
"topic" : [{ CodeableConcept }], // E.g. Education, Treatment, Assessment
"author" : [{ ContactDetail }], // Who authored the content
"editor" : [{ ContactDetail }], // Who edited the content
"reviewer" : [{ ContactDetail }], // Who reviewed the content
"endorser" : [{ ContactDetail }], // Who endorsed the content
"relatedArtifact" : [{ RelatedArtifact }], // Additional documentation, citations
"library" : [{ canonical(Library) }], // Logic used by the plan definition
"goal" : [{ // What the plan is trying to accomplish
  "category" : { CodeableConcept }, // E.g. Treatment, dietary, behavioral
  "description" : { CodeableConcept }, // R!  Code or text describing the goal
  "priority" : { CodeableConcept }, // high-priority | medium-priority | low-priority
  "start" : { CodeableConcept }, // When goal pursuit begins
  "addresses" : [{ CodeableConcept }], // What does the goal address
  "documentation" : [{ RelatedArtifact }], // Supporting documentation for the goal
  "target" : [{ // Target outcome for the goal
    "measure" : { CodeableConcept }, // The parameter whose value is to be tracked
    // detail[x]: The target value to be achieved. One of these 3:
    "detailQuantity" : { Quantity },
    "detailRange" : { Range },
    "detailCodeableConcept" : { CodeableConcept },
    "due" : { Duration } // Reach goal within
  }]
}],
"action" : [{ // Action defined by the plan
  "prefix" : "<string>", // User-visible prefix for the action (e.g. 1. or A.)
  "title" : "<string>", // User-visible title
  "description" : "<string>", // Brief description of the action
  "textEquivalent" : "<string>", // Static text equivalent of the action, used if the dynamic aspects cannot be
  interpreted by the receiving system
  "priority" : "<code>", // routine | urgent | asap | stat
  "code" : [{ CodeableConcept }], // Code representing the meaning of the action or sub-actions
  "reason" : [{ CodeableConcept }], // Why the action should be performed
  "documentation" : [{ RelatedArtifact }], // Supporting documentation for the intended performer of the action
  "goalId" : ["<id>"], // What goals this action supports
  // subject[x]: Type of individual the action is focused on. One of these 2:
  "subjectCodeableConcept" : { CodeableConcept },
  "subjectReference" : { Reference(Group) },
  "trigger" : [{ TriggerDefinition }], // When the action should be triggered
  "condition" : [{ // Whether or not the action is applicable
    "kind" : "<code>", // R!  applicability | start | stop
    "expression" : { Expression } // Boolean-valued expression
  }],
  "input" : [{ DataRequirement }], // Input data requirements
  "output" : [{ DataRequirement }], // Output data definition
  "relatedAction" : [{ // Relationship to another action
    "actionId" : "<id>", // R!  What action is this related to
    "relationship" : "<code>", // R!  before-start | before | before-end | concurrent-with-start | concurrent |
  concurrent-with-end | after-start | after | after-end
    // offset[x]: Time offset for the relationship. One of these 2:
    "offsetDuration" : { Duration }
    "offsetRange" : { Range }
  }],
  // timing[x]: When the action should take place. One of these 6:
  "timingDateTime" : "<dateTime>",
  "timingAge" : { Age },
  "timingPeriod" : { Period },
  "timingDuration" : { Duration },
  "timingRange" : { Range },
  "timingTiming" : { Timing },
  "participant" : [{ // Who should participate in the action
    "type" : "<code>", // R!  patient | practitioner | related-person | device
    "role" : { CodeableConcept } // E.g. Nurse, Surgeon, Parent
  }],
  "type" : { CodeableConcept }, // create | update | remove | fire-event
  "groupingBehavior" : "<code>", // visual-group | logical-group | sentence-group
  "selectionBehavior" : "<code>", // any | all | all-or-none | exactly-one | at-most-one | one-or-more
  "requiredBehavior" : "<code>", // must | could | must-unless-documented
  "precheckBehavior" : "<code>", // yes | no
```

```
    "cardinalityBehavior" : "<code>", // single | multiple
    // definition[x]: Description of the activity to be performed. One of these 2:
    "definitionCanonical" : { canonical(ActivityDefinition|PlanDefinition|
    Questionnaire) },
    "definitionUri" : "<uri>",
    "transform" : { canonical(StructureMap) }, // Transform to apply the template
    "dynamicValue" : [{ // Dynamic aspects of the definition
      "path" : "<string>", // The path to the element to be set dynamically
      "expression" : { Expression } // An expression that provides the dynamic value for the customization
    }],
    "action" : [{ Content as for PlanDefinition.action }] // A sub-action
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:PlanDefinition;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:PlanDefinition.url [ uri ]; # 0..1 Canonical identifier for this plan definition, represented as a URI (globally
    unique)
  fhir:PlanDefinition.identifier [ Identifier ], ... ; # 0..* Additional identifier for the plan definition
  fhir:PlanDefinition.version [ string ]; # 0..1 Business version of the plan definition
  fhir:PlanDefinition.name [ string ]; # 0..1 Name for this plan definition (computer friendly)
  fhir:PlanDefinition.title [ string ]; # 0..1 Name for this plan definition (human friendly)
  fhir:PlanDefinition.subtitle [ string ]; # 0..1 Subordinate title of the plan definition
  fhir:PlanDefinition.type [ CodeableConcept ]; # 0..1 order-set | clinical-protocol | eca-rule | workflow-definition
  fhir:PlanDefinition.status [ code ]; # 1..1 draft | active | retired | unknown
  fhir:PlanDefinition.experimental [ boolean ]; # 0..1 For testing purposes, not real usage
  # PlanDefinition.subject[x] : 0..1 Type of individual the plan definition is focused on. One of these 2
    fhir:PlanDefinition.subjectCodeableConcept [ CodeableConcept ]
    fhir:PlanDefinition.subjectReference [ Reference(Group) ]
  fhir:PlanDefinition.date [ dateTime ]; # 0..1 Date last changed
  fhir:PlanDefinition.publisher [ string ]; # 0..1 Name of the publisher (organization or individual)
  fhir:PlanDefinition.contact [ ContactDetail ], ... ; # 0..* Contact details for the publisher
  fhir:PlanDefinition.description [ markdown ]; # 0..1 Natural language description of the plan definition
  fhir:PlanDefinition.useContext [ UsageContext ], ... ; # 0..* The context that the content is intended to support
  fhir:PlanDefinition.jurisdiction [ CodeableConcept ], ... ; # 0..* Intended jurisdiction for plan definition (if
    applicable)
  fhir:PlanDefinition.purpose [ markdown ]; # 0..1 Why this plan definition is defined
  fhir:PlanDefinition.usage [ string ]; # 0..1 Describes the clinical usage of the plan
  fhir:PlanDefinition.copyright [ markdown ]; # 0..1 Use and/or publishing restrictions
  fhir:PlanDefinition.approvalDate [ date ]; # 0..1 When the plan definition was approved by publisher
  fhir:PlanDefinition.lastReviewDate [ date ]; # 0..1 When the plan definition was last reviewed
  fhir:PlanDefinition.effectivePeriod [ Period ]; # 0..1 When the plan definition is expected to be used
  fhir:PlanDefinition.topic [ CodeableConcept ], ... ; # 0..* E.g. Education, Treatment, Assessment
  fhir:PlanDefinition.author [ ContactDetail ], ... ; # 0..* Who authored the content
  fhir:PlanDefinition.editor [ ContactDetail ], ... ; # 0..* Who edited the content
  fhir:PlanDefinition.reviewer [ ContactDetail ], ... ; # 0..* Who reviewed the content
  fhir:PlanDefinition.endorser [ ContactDetail ], ... ; # 0..* Who endorsed the content
  fhir:PlanDefinition.relatedArtifact [ RelatedArtifact ], ... ; # 0..* Additional documentation, citations
  fhir:PlanDefinition.library [ canonical(Library) ], ... ; # 0..* Logic used by the plan definition
  fhir:PlanDefinition.goal [ # 0..* What the plan is trying to accomplish
    fhir:PlanDefinition.goal.category [ CodeableConcept ]; # 0..1 E.g. Treatment, dietary, behavioral
    fhir:PlanDefinition.goal.description [ CodeableConcept ]; # 1..1 Code or text describing the goal
    fhir:PlanDefinition.goal.priority [ CodeableConcept ]; # 0..1 high-priority | medium-priority | low-priority
    fhir:PlanDefinition.goal.start [ CodeableConcept ]; # 0..1 When goal pursuit begins
    fhir:PlanDefinition.goal.addresses [ CodeableConcept ], ... ; # 0..* What does the goal address
    fhir:PlanDefinition.goal.documentation [ RelatedArtifact ], ... ; # 0..* Supporting documentation for the goal
    fhir:PlanDefinition.goal.target [ # 0..* Target outcome for the goal
      fhir:PlanDefinition.goal.target.measure [ CodeableConcept ]; # 0..1 The parameter whose value is to be tracked
      # PlanDefinition.goal.target.detail[x] : 0..1 The target value to be achieved. One of these 3
        fhir:PlanDefinition.goal.target.detailQuantity [ Quantity ]
        fhir:PlanDefinition.goal.target.detailRange [ Range ]
        fhir:PlanDefinition.goal.target.detailCodeableConcept [ CodeableConcept ]
      fhir:PlanDefinition.goal.target.due [ Duration ]; # 0..1 Reach goal within
    ], ...;
  ], ...;
  fhir:PlanDefinition.action [ # 0..* Action defined by the plan
    fhir:PlanDefinition.action.prefix [ string ]; # 0..1 User-visible prefix for the action (e.g. 1. or A.)
    fhir:PlanDefinition.action.title [ string ]; # 0..1 User-visible title
    fhir:PlanDefinition.action.description [ string ]; # 0..1 Brief description of the action
    fhir:PlanDefinition.action.textEquivalent [ string ]; # 0..1 Static text equivalent of the action, used if the
    dynamic aspects cannot be interpreted by the receiving system
    fhir:PlanDefinition.action.priority [ code ]; # 0..1 routine | urgent | asap | stat
    fhir:PlanDefinition.action.code [ CodeableConcept ], ... ; # 0..* Code representing the meaning of the action or
    sub-actions
    fhir:PlanDefinition.action.reason [ CodeableConcept ], ... ; # 0..* Why the action should be performed
    fhir:PlanDefinition.action.documentation [ RelatedArtifact ], ... ; # 0..* Supporting documentation for the intended
     performer of the action
    fhir:PlanDefinition.action.goalId [ id ], ... ; # 0..* What goals this action supports
    # PlanDefinition.action.subject[x] : 0..1 Type of individual the action is focused on. One of these 2
      fhir:PlanDefinition.action.subjectCodeableConcept [ CodeableConcept ]
      fhir:PlanDefinition.action.subjectReference [ Reference(Group) ]
    fhir:PlanDefinition.action.trigger [ TriggerDefinition ], ... ; # 0..* When the action should be triggered
```

```
    fhir:PlanDefinition.action.condition [ # 0..* Whether or not the action is applicable
      fhir:PlanDefinition.action.condition.kind [ code ]; # 1..1 applicability | start | stop
      fhir:PlanDefinition.action.condition.expression [ Expression ]; # 0..1 Boolean-valued expression
    ], ...;
    fhir:PlanDefinition.action.input [ DataRequirement ], ... ; # 0..* Input data requirements
    fhir:PlanDefinition.action.output [ DataRequirement ], ... ; # 0..* Output data definition
    fhir:PlanDefinition.action.relatedAction [ # 0..* Relationship to another action
      fhir:PlanDefinition.action.relatedAction.actionId [ id ]; # 1..1 What action is this related to
      fhir:PlanDefinition.action.relatedAction.relationship [ code ]; # 1..1 before-start | before | before-end |
    concurrent-with-start | concurrent | concurrent-with-end | after-start | after | after-end
      # PlanDefinition.action.relatedAction.offset[x] : 0..1 Time offset for the relationship. One of these 2
        fhir:PlanDefinition.action.relatedAction.offsetDuration [ Duration ]
        fhir:PlanDefinition.action.relatedAction.offsetRange [ Range ]
    ], ...;
    # PlanDefinition.action.timing[x] : 0..1 When the action should take place. One of these 6
      fhir:PlanDefinition.action.timingDateTime [ dateTime ]
      fhir:PlanDefinition.action.timingAge [ Age ]
      fhir:PlanDefinition.action.timingPeriod [ Period ]
      fhir:PlanDefinition.action.timingDuration [ Duration ]
      fhir:PlanDefinition.action.timingRange [ Range ]
      fhir:PlanDefinition.action.timingTiming [ Timing ]
    fhir:PlanDefinition.action.participant [ # 0..* Who should participate in the action
      fhir:PlanDefinition.action.participant.type [ code ]; # 1..1 patient | practitioner | related-person | device
      fhir:PlanDefinition.action.participant.role [ CodeableConcept ]; # 0..1 E.g. Nurse, Surgeon, Parent
    ], ...;
    fhir:PlanDefinition.action.type [ CodeableConcept ]; # 0..1 create | update | remove | fire-event
    fhir:PlanDefinition.action.groupingBehavior [ code ]; # 0..1 visual-group | logical-group | sentence-group
    fhir:PlanDefinition.action.selectionBehavior [ code ]; # 0..1 any | all | all-or-none | exactly-one | at-most-one |
    one-or-more
    fhir:PlanDefinition.action.requiredBehavior [ code ]; # 0..1 must | could | must-unless-documented
    fhir:PlanDefinition.action.precheckBehavior [ code ]; # 0..1 yes | no
    fhir:PlanDefinition.action.cardinalityBehavior [ code ]; # 0..1 single | multiple
    # PlanDefinition.action.definition[x] : 0..1 Description of the activity to be performed. One of these 2
      fhir:PlanDefinition.action.definitionCanonical [ canonical(ActivityDefinition|PlanDefinition|Questionnaire) ]
      fhir:PlanDefinition.action.definitionUri [ uri ]
    fhir:PlanDefinition.action.transform [ canonical(StructureMap) ]; # 0..1 Transform to apply the template
    fhir:PlanDefinition.action.dynamicValue [ # 0..* Dynamic aspects of the definition
      fhir:PlanDefinition.action.dynamicValue.path [ string ]; # 0..1 The path to the element to be set dynamically
      fhir:PlanDefinition.action.dynamicValue.expression [ Expression ]; # 0..1 An expression that provides the dynamic
    value for the customization
    ], ...;
    fhir:PlanDefinition.action.action [ See PlanDefinition.action ], ... ; # 0..* A sub-action
  ], ...;
]
```

Changes since R3

PlanDefinition

PlanDefinition.subtitle | Added Element

PlanDefinition.type | Change code system for extensibly bound codes from "http://hl7.org/fhir/plan-definition-type" to "http://terminology.hl7.org/CodeSystem/plan-definition-type"

PlanDefinition.status | Change value set from http://hl7.org/fhir/ValueSet/publication-status to http://hl7.org/fhir/ValueSet/publication-status|4.0.1

PlanDefinition.experimental | No longer marked as Modifier

PlanDefinition.subject[x] | Added Element

PlanDefinition.author | Added Element

PlanDefinition.editor | Added Element

PlanDefinition.reviewer | Added Element

PlanDefinition.endorser | Added Element

PlanDefinition.library | Type changed from Reference(Library) to canonical(Library)

PlanDefinition.action.prefix | Added Element

PlanDefinition.action.priority | Added Element

PlanDefinition.action.subject[x] | Added Element

PlanDefinition.action.trigger | Added Element

PlanDefinition.action.condition.kind | Change value set from http://hl7.org/fhir/ValueSet/action-condition-kind to http://hl7.org/fhir/ValueSet/action-condition-kind|4.0.1

PlanDefinition.action.condition.expression | Type changed from string to Expression

PlanDefinition.action.relatedAction.relationship | Change value set from http://hl7.org/fhir/ValueSet/action-relationship-type to http://hl7.org/fhir/ValueSet/action-relationship-type|4.0.1

PlanDefinition.action.timing[x] | Add Type Age

PlanDefinition.action.participant.type | Change value set from http://hl7.org/fhir/ValueSet/action-participant-type to http://hl7.org/fhir/ValueSet/action-participant-type|4.0.1

PlanDefinition.action.type | Type changed from Coding to CodeableConcept Change binding strength from required to extensible

PlanDefinition.action.groupingBehavior | Change value set from http://hl7.org/fhir/ValueSet/action-grouping-behavior to http://hl7.org/fhir/ValueSet/action-grouping-behavior|4.0.1

PlanDefinition.action.selectionBehavior | Change value set from http://hl7.org/fhir/ValueSet/action-selection-behavior to http://hl7.org/fhir/ValueSet/action-selection-behavior|4.0.1

PlanDefinition.action.requiredBehavior | Change value set from http://hl7.org/fhir/ValueSet/action-required-behavior to http://hl7.org/fhir/ValueSet/action-required-behavior|4.0.1

PlanDefinition.action.precheckBehavior | Change value set from http://hl7.org/fhir/ValueSet/action-precheck-behavior to http://hl7.org/fhir/ValueSet/action-precheck-behavior|4.0.1

PlanDefinition.action.cardinalityBehavior | Change value set from http://hl7.org/fhir/ValueSet/action-cardinality-behavior to http://hl7.org/fhir/ValueSet/action-cardinality-behavior|4.0.1

PlanDefinition.action.definition[x] | Renamed from definition to definition[x] Add Types canonical(ActivityDefinition | PlanDefinition | Questionnaire), uri Remove Type Reference(ActivityDefinition | PlanDefinition)
PlanDefinition.action.transform | Type changed from Reference(StructureMap) to canonical(StructureMap)
PlanDefinition.action.dynamicValue.expression | Type changed from string to Expression
PlanDefinition.contributor | deleted
PlanDefinition.action.label | deleted
PlanDefinition.action.triggerDefinition | deleted
PlanDefinition.action.condition.description | deleted
PlanDefinition.action.condition.language | deleted
PlanDefinition.action.dynamicValue.description | deleted
PlanDefinition.action.dynamicValue.language | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 11 tests of which 6 fail to execute. All tests pass round-trip testing and 5 r3 resources are invalid (0 errors).)

Structure

Name | Flags | Card. | Type | Description & Constraints
PlanDefinition | ITU | | DomainResource | The definition of a plan for a series of actions, independent of any specific patient or context + Warning: Name should be usable as an identifier for the module by machine processing applications such as code generation Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
url | Σ | 0..1 | uri | Canonical identifier for this plan definition, represented as a URI (globally unique)
identifier | Σ | 0..* | Identifier | Additional identifier for the plan definition
version | Σ | 0..1 | string | Business version of the plan definition
name | ΣI | 0..1 | string | Name for this plan definition (computer friendly)
title | Σ | 0..1 | string | Name for this plan definition (human friendly)
subtitle | | 0..1 | string | Subordinate title of the plan definition
type | Σ | 0..1 | CodeableConcept | order-set | clinical-protocol | eca-rule | workflow-definition PlanDefinitionType (Extensible)
status | ?!Σ | 1..1 | code | draft | active | retired | unknown PublicationStatus (Required)
experimental | Σ | 0..1 | boolean | For testing purposes, not real usage
subject[x] | | 0..1 | | Type of individual the plan definition is focused on Subject type (Extensible)
subjectCodeableConcept | | | CodeableConcept
subjectReference | | | Reference(Group)
date | Σ | 0..1 | dateTime | Date last changed
publisher | Σ | 0..1 | string | Name of the publisher (organization or individual)
contact | Σ | 0..* | ContactDetail | Contact details for the publisher
description | Σ | 0..1 | markdown | Natural language description of the plan definition
useContext | Σ | 0..* | UsageContext | The context that the content is intended to support
jurisdiction | Σ | 0..* | CodeableConcept | Intended jurisdiction for plan definition (if applicable) Jurisdiction (Extensible)
purpose | | 0..1 | markdown | Why this plan definition is defined
usage | | 0..1 | string | Describes the clinical usage of the plan
copyright | | 0..1 | markdown | Use and/or publishing restrictions
approvalDate | | 0..1 | date | When the plan definition was approved by publisher
lastReviewDate | | 0..1 | date | When the plan definition was last reviewed
effectivePeriod | Σ | 0..1 | Period | When the plan definition is expected to be used
topic | | 0..* | CodeableConcept | E.g. Education, Treatment, Assessment DefinitionTopic (Example)
author | | 0..* | ContactDetail | Who authored the content
editor | | 0..* | ContactDetail | Who edited the content
reviewer | | 0..* | ContactDetail | Who reviewed the content
endorser | | 0..* | ContactDetail | Who endorsed the content
relatedArtifact | | 0..* | RelatedArtifact | Additional documentation, citations
library | | 0..* | canonical(Library) | Logic used by the plan definition
goal | | 0..* | BackboneElement | What the plan is trying to accomplish
category | | 0..1 | CodeableConcept | E.g. Treatment, dietary, behavioral Goal category (Example)
description | | 1..1 | CodeableConcept | Code or text describing the goal SNOMED CT Clinical Findings (Example)
priority | | 0..1 | CodeableConcept | high-priority | medium-priority | low-priority Goal priority (Preferred)
start | | 0..1 | CodeableConcept | When goal pursuit begins Goal start event (Example)
addresses | | 0..* | CodeableConcept | What does the goal address Condition/Problem/Diagnosis Codes (Example)
documentation | | 0..* | RelatedArtifact | Supporting documentation for the goal
target | | 0..* | BackboneElement | Target outcome for the goal
measure | | 0..1 | CodeableConcept | The parameter whose value is to be tracked LOINC Codes (Example)
detail[x] | | 0..1 | | The target value to be achieved
detailQuantity | | | Quantity
detailRange | | | Range
detailCodeableConcept | | | CodeableConcept
due | | 0..1 | Duration | Reach goal within
action | | 0..* | BackboneElement | Action defined by the plan
prefix | | 0..1 | string | User-visible prefix for the action (e.g. 1. or A.)
title | | 0..1 | string | User-visible title
description | | 0..1 | string | Brief description of the action
textEquivalent | | 0..1 | string | Static text equivalent of the action, used if the dynamic aspects cannot be interpreted by the receiving system

priority | | 0..1 | code | routine | urgent | asap | stat Request priority (Required)
code | | 0..* | CodeableConcept | Code representing the meaning of the action or sub-actions
reason | | 0..* | CodeableConcept | Why the action should be performed
documentation | | 0..* | RelatedArtifact | Supporting documentation for the intended performer of the action
goalId | | 0..* | id | What goals this action supports
subject[x] | | 0..1 | | Type of individual the action is focused on Subject type (Extensible)
subjectCodeableConcept | | | CodeableConcept
subjectReference | | | Reference(Group)
trigger | | 0..* | TriggerDefinition | When the action should be triggered
condition | | 0..* | BackboneElement | Whether or not the action is applicable
kind | | 1..1 | code | applicability | start | stop ActionConditionKind (Required)
expression | | 0..1 | Expression | Boolean-valued expression
input | | 0..* | DataRequirement | Input data requirements
output | | 0..* | DataRequirement | Output data definition
relatedAction | | 0..* | BackboneElement | Relationship to another action
actionId | | 1..1 | id | What action is this related to
relationship | | 1..1 | code | before-start | before | before-end | concurrent-with-start | concurrent | concurrent-with-end | after-start | after | after-end ActionRelationshipType (Required)
offset[x] | | 0..1 | | Time offset for the relationship
offsetDuration | | | Duration
offsetRange | | | Range
timing[x] | | 0..1 | | When the action should take place
timingDateTime | | | dateTime
timingAge | | | Age
timingPeriod | | | Period
timingDuration | | | Duration
timingRange | | | Range
timingTiming | | | Timing
participant | | 0..* | BackboneElement | Who should participate in the action
type | | 1..1 | code | patient | practitioner | related-person | device ActionParticipantType (Required)
role | | 0..1 | CodeableConcept | E.g. Nurse, Surgeon, Parent Action participant role (Example)
type | | 0..1 | CodeableConcept | create | update | remove | fire-event ActionType (Extensible)
groupingBehavior | | 0..1 | code | visual-group | logical-group | sentence-group ActionGroupingBehavior (Required)
selectionBehavior | | 0..1 | code | any | all | all-or-none | exactly-one | at-most-one | one-or-more ActionSelectionBehavior (Required)
requiredBehavior | | 0..1 | code | must | could | must-unless-documented ActionRequiredBehavior (Required)
precheckBehavior | | 0..1 | code | yes | no ActionPrecheckBehavior (Required)
cardinalityBehavior | | 0..1 | code | single | multiple ActionCardinalityBehavior (Required)
definition[x] | | 0..1 | | Description of the activity to be performed
definitionCanonical | | | canonical(ActivityDefinition | PlanDefinition | Questionnaire)
definitionUri | | | uri
transform | | 0..1 | canonical(StructureMap) | Transform to apply the template
dynamicValue | | 0..* | BackboneElement | Dynamic aspects of the definition
path | | 0..1 | string | The path to the element to be set dynamically
expression | | 0..1 | Expression | An expression that provides the dynamic value for the customization
action | | 0..* | see action | A sub-action
Documentation for this format

UML Diagram (Legend)

XML Template

```
<PlanDefinition xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <url value="[uri]"/><!-- 0..1 Canonical identifier for this plan definition, represented as a URI (globally unique) -->
 <identifier><!-- 0..* Identifier Additional identifier for the plan definition --></identifier>
 <version value="[string]"/><!-- 0..1 Business version of the plan definition -->
 <name value="[string]"/><!--  0..1 Name for this plan definition (computer friendly) -->
 <title value="[string]"/><!-- 0..1 Name for this plan definition (human friendly) -->
 <subtitle value="[string]"/><!-- 0..1 Subordinate title of the plan definition -->
 <type><!-- 0..1 CodeableConcept order-set | clinical-protocol | eca-rule | workflow-definition --></type>
 <status value="[code]"/><!-- 1..1 draft | active | retired | unknown -->
 <experimental value="[boolean]"/><!-- 0..1 For testing purposes, not real usage -->
 <subject[x]><!-- 0..1 CodeableConcept|Reference(Group) Type of individual the plan definition is focused on
    --></subject[x]>
 <date value="[dateTime]"/><!-- 0..1 Date last changed -->
 <publisher value="[string]"/><!-- 0..1 Name of the publisher (organization or individual) -->
 <contact><!-- 0..* ContactDetail Contact details for the publisher --></contact>
 <description value="[markdown]"/><!-- 0..1 Natural language description of the plan definition -->
 <useContext><!-- 0..* UsageContext The context that the content is intended to support --></useContext>
 <jurisdiction><!-- 0..* CodeableConcept Intended jurisdiction for plan definition (if applicable) --></jurisdiction>
 <purpose value="[markdown]"/><!-- 0..1 Why this plan definition is defined -->
 <usage value="[string]"/><!-- 0..1 Describes the clinical usage of the plan -->
 <copyright value="[markdown]"/><!-- 0..1 Use and/or publishing restrictions -->
 <approvalDate value="[date]"/><!-- 0..1 When the plan definition was approved by publisher -->
 <lastReviewDate value="[date]"/><!-- 0..1 When the plan definition was last reviewed -->
 <effectivePeriod><!-- 0..1 Period When the plan definition is expected to be used --></effectivePeriod>
 <topic><!-- 0..* CodeableConcept E.g. Education, Treatment, Assessment --></topic>
 <author><!-- 0..* ContactDetail Who authored the content --></author>
```

```
 <editor><!-- 0..* ContactDetail Who edited the content --></editor>
 <reviewer><!-- 0..* ContactDetail Who reviewed the content --></reviewer>
 <endorser><!-- 0..* ContactDetail Who endorsed the content --></endorser>
 <relatedArtifact><!-- 0..* RelatedArtifact Additional documentation, citations --></relatedArtifact>
 <library><!-- 0..* canonical(Library) Logic used by the plan definition --></library>
 <goal>  <!-- 0..* What the plan is trying to accomplish -->
  <category><!-- 0..1 CodeableConcept E.g. Treatment, dietary, behavioral --></category>
  <description><!-- 1..1 CodeableConcept Code or text describing the goal --></description>
  <priority><!-- 0..1 CodeableConcept high-priority | medium-priority | low-priority --></priority>
  <start><!-- 0..1 CodeableConcept When goal pursuit begins --></start>
  <addresses><!-- 0..* CodeableConcept What does the goal address --></addresses>
  <documentation><!-- 0..* RelatedArtifact Supporting documentation for the goal --></documentation>
  <target>  <!-- 0..* Target outcome for the goal -->
   <measure><!-- 0..1 CodeableConcept The parameter whose value is to be tracked --></measure>
   <detail[x]><!-- 0..1 Quantity|Range|CodeableConcept The target value to be achieved --></detail[x]>
   <due><!-- 0..1 Duration Reach goal within --></due>
  </target>
 </goal>
 <action>  <!-- 0..* Action defined by the plan -->
  <prefix value="[string]"/><!-- 0..1 User-visible prefix for the action (e.g. 1. or A.) -->
  <title value="[string]"/><!-- 0..1 User-visible title -->
  <description value="[string]"/><!-- 0..1 Brief description of the action -->
  <textEquivalent value="[string]"/><!-- 0..1 Static text equivalent of the action, used if the dynamic aspects cannot
    be interpreted by the receiving system -->
  <priority value="[code]"/><!-- 0..1 routine | urgent | asap | stat -->
  <code><!-- 0..* CodeableConcept Code representing the meaning of the action or sub-actions --></code>
  <reason><!-- 0..* CodeableConcept Why the action should be performed --></reason>
  <documentation><!-- 0..* RelatedArtifact Supporting documentation for the intended performer of the action
    --></documentation>
  <goalId value="[id]"/><!-- 0..* What goals this action supports -->
  <subject[x]><!-- 0..1 CodeableConcept|Reference(Group) Type of individual the action is focused on --></subject[x]>
  <trigger><!-- 0..* TriggerDefinition When the action should be triggered --></trigger>
  <condition>  <!-- 0..* Whether or not the action is applicable -->
   <kind value="[code]"/><!-- 1..1 applicability | start | stop -->
   <expression><!-- 0..1 Expression Boolean-valued expression --></expression>
  </condition>
  <input><!-- 0..* DataRequirement Input data requirements --></input>
  <output><!-- 0..* DataRequirement Output data definition --></output>
  <relatedAction>  <!-- 0..* Relationship to another action -->
   <actionId value="[id]"/><!-- 1..1 What action is this related to -->
   <relationship value="[code]"/><!-- 1..1 before-start | before | before-end | concurrent-with-start | concurrent |
    concurrent-with-end | after-start | after | after-end -->
   <offset[x]><!-- 0..1 Duration|Range Time offset for the relationship --></offset[x]>
  </relatedAction>
  <timing[x]><!-- 0..1 dateTime|Age|Period|Duration|Range|Timing When the action should take place --></timing[x]>
  <participant>  <!-- 0..* Who should participate in the action -->
   <type value="[code]"/><!-- 1..1 patient | practitioner | related-person | device -->
   <role><!-- 0..1 CodeableConcept E.g. Nurse, Surgeon, Parent --></role>
  </participant>
  <type><!-- 0..1 CodeableConcept create | update | remove | fire-event --></type>
  <groupingBehavior value="[code]"/><!-- 0..1 visual-group | logical-group | sentence-group -->
  <selectionBehavior value="[code]"/><!-- 0..1 any | all | all-or-none | exactly-one | at-most-one | one-or-more -->
  <requiredBehavior value="[code]"/><!-- 0..1 must | could | must-unless-documented -->
  <precheckBehavior value="[code]"/><!-- 0..1 yes | no -->
  <cardinalityBehavior value="[code]"/><!-- 0..1 single | multiple -->
  <definition[x]><!-- 0..1 canonical(ActivityDefinition|PlanDefinition|
    Questionnaire)|uri Description of the activity to be performed --></definition[x]>
  <transform><!-- 0..1 canonical(StructureMap) Transform to apply the template --></transform>
  <dynamicValue>  <!-- 0..* Dynamic aspects of the definition -->
   <path value="[string]"/><!-- 0..1 The path to the element to be set dynamically -->
   <expression><!-- 0..1 Expression An expression that provides the dynamic value for the customization --></expression>
  </dynamicValue>
  <action><!-- 0..* Content as for PlanDefinition.action A sub-action --></action>
 </action>
</PlanDefinition>
```

JSON Template

```
{
  "resourceType" : "PlanDefinition",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "url" : "<uri>", // Canonical identifier for this plan definition, represented as a URI (globally unique)
  "identifier" : [{ Identifier }], // Additional identifier for the plan definition
  "version" : "<string>", // Business version of the plan definition
  "name" : "<string>", // C? Name for this plan definition (computer friendly)
  "title" : "<string>", // Name for this plan definition (human friendly)
  "subtitle" : "<string>", // Subordinate title of the plan definition
  "type" : { CodeableConcept }, // order-set | clinical-protocol | eca-rule | workflow-definition
  "status" : "<code>", // R!  draft | active | retired | unknown
  "experimental" : <boolean>, // For testing purposes, not real usage
  // subject[x]: Type of individual the plan definition is focused on. One of these 2:
  "subjectCodeableConcept" : { CodeableConcept },
  "subjectReference" : { Reference(Group) },
  "date" : "<dateTime>", // Date last changed
  "publisher" : "<string>", // Name of the publisher (organization or individual)
  "contact" : [{ ContactDetail }], // Contact details for the publisher
  "description" : "<markdown>", // Natural language description of the plan definition
  "useContext" : [{ UsageContext }], // The context that the content is intended to support
  "jurisdiction" : [{ CodeableConcept }], // Intended jurisdiction for plan definition (if applicable)
```

```
  "purpose" : "<markdown>", // Why this plan definition is defined
  "usage" : "<string>", // Describes the clinical usage of the plan
  "copyright" : "<markdown>", // Use and/or publishing restrictions
  "approvalDate" : "<date>", // When the plan definition was approved by publisher
  "lastReviewDate" : "<date>", // When the plan definition was last reviewed
  "effectivePeriod" : { Period }, // When the plan definition is expected to be used
  "topic" : [{ CodeableConcept }], // E.g. Education, Treatment, Assessment
  "author" : [{ ContactDetail }], // Who authored the content
  "editor" : [{ ContactDetail }], // Who edited the content
  "reviewer" : [{ ContactDetail }], // Who reviewed the content
  "endorser" : [{ ContactDetail }], // Who endorsed the content
  "relatedArtifact" : [{ RelatedArtifact }], // Additional documentation, citations
  "library" : [{ canonical(Library) }], // Logic used by the plan definition
  "goal" : [{ // What the plan is trying to accomplish
    "category" : { CodeableConcept }, // E.g. Treatment, dietary, behavioral
    "description" : { CodeableConcept }, // R!  Code or text describing the goal
    "priority" : { CodeableConcept }, // high-priority | medium-priority | low-priority
    "start" : { CodeableConcept }, // When goal pursuit begins
    "addresses" : [{ CodeableConcept }], // What does the goal address
    "documentation" : [{ RelatedArtifact }], // Supporting documentation for the goal
    "target" : [{ // Target outcome for the goal
      "measure" : { CodeableConcept }, // The parameter whose value is to be tracked
      // detail[x]: The target value to be achieved. One of these 3:
      "detailQuantity" : { Quantity },
      "detailRange" : { Range },
      "detailCodeableConcept" : { CodeableConcept },
      "due" : { Duration } // Reach goal within
    }]
  }],
  "action" : [{ // Action defined by the plan
    "prefix" : "<string>", // User-visible prefix for the action (e.g. 1. or A.)
    "title" : "<string>", // User-visible title
    "description" : "<string>", // Brief description of the action
    "textEquivalent" : "<string>", // Static text equivalent of the action, used if the dynamic aspects cannot be
    interpreted by the receiving system
    "priority" : "<code>", // routine | urgent | asap | stat
    "code" : [{ CodeableConcept }], // Code representing the meaning of the action or sub-actions
    "reason" : [{ CodeableConcept }], // Why the action should be performed
    "documentation" : [{ RelatedArtifact }], // Supporting documentation for the intended performer of the action
    "goalId" : ["<id>"], // What goals this action supports
    // subject[x]: Type of individual the action is focused on. One of these 2:
    "subjectCodeableConcept" : { CodeableConcept },
    "subjectReference" : { Reference(Group) },
    "trigger" : [{ TriggerDefinition }], // When the action should be triggered
    "condition" : [{ // Whether or not the action is applicable
      "kind" : "<code>", // R!  applicability | start | stop
      "expression" : { Expression } // Boolean-valued expression
    }],
    "input" : [{ DataRequirement }], // Input data requirements
    "output" : [{ DataRequirement }], // Output data definition
    "relatedAction" : [{ // Relationship to another action
      "actionId" : "<id>", // R!  What action is this related to
      "relationship" : "<code>", // R!  before-start | before | before-end | concurrent-with-start | concurrent |
    concurrent-with-end | after-start | after | after-end
      // offset[x]: Time offset for the relationship. One of these 2:
      "offsetDuration" : { Duration }
      "offsetRange" : { Range }
    }],
    // timing[x]: When the action should take place. One of these 6:
    "timingDateTime" : "<dateTime>",
    "timingAge" : { Age },
    "timingPeriod" : { Period },
    "timingDuration" : { Duration },
    "timingRange" : { Range },
    "timingTiming" : { Timing },
    "participant" : [{ // Who should participate in the action
      "type" : "<code>", // R!  patient | practitioner | related-person | device
      "role" : { CodeableConcept } // E.g. Nurse, Surgeon, Parent
    }],
    "type" : { CodeableConcept }, // create | update | remove | fire-event
    "groupingBehavior" : "<code>", // visual-group | logical-group | sentence-group
    "selectionBehavior" : "<code>", // any | all | all-or-none | exactly-one | at-most-one | one-or-more
    "requiredBehavior" : "<code>", // must | could | must-unless-documented
    "precheckBehavior" : "<code>", // yes | no
    "cardinalityBehavior" : "<code>", // single | multiple
    // definition[x]: Description of the activity to be performed. One of these 2:
    "definitionCanonical" : { canonical(ActivityDefinition|PlanDefinition|
    Questionnaire) },
    "definitionUri" : "<uri>",
    "transform" : { canonical(StructureMap) }, // Transform to apply the template
    "dynamicValue" : [{ // Dynamic aspects of the definition
      "path" : "<string>", // The path to the element to be set dynamically
      "expression" : { Expression } // An expression that provides the dynamic value for the customization
    }],
    "action" : [{ Content as for PlanDefinition.action }] // A sub-action
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:PlanDefinition;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:PlanDefinition.url [ uri ]; # 0..1 Canonical identifier for this plan definition, represented as a URI (globally
    unique)
  fhir:PlanDefinition.identifier [ Identifier ], ... ; # 0..* Additional identifier for the plan definition
  fhir:PlanDefinition.version [ string ]; # 0..1 Business version of the plan definition
  fhir:PlanDefinition.name [ string ]; # 0..1 Name for this plan definition (computer friendly)
  fhir:PlanDefinition.title [ string ]; # 0..1 Name for this plan definition (human friendly)
  fhir:PlanDefinition.subtitle [ string ]; # 0..1 Subordinate title of the plan definition
  fhir:PlanDefinition.type [ CodeableConcept ]; # 0..1 order-set | clinical-protocol | eca-rule | workflow-definition
  fhir:PlanDefinition.status [ code ]; # 1..1 draft | active | retired | unknown
  fhir:PlanDefinition.experimental [ boolean ]; # 0..1 For testing purposes, not real usage
  # PlanDefinition.subject[x] : 0..1 Type of individual the plan definition is focused on. One of these 2
    fhir:PlanDefinition.subjectCodeableConcept [ CodeableConcept ]
    fhir:PlanDefinition.subjectReference [ Reference(Group) ]
  fhir:PlanDefinition.date [ dateTime ]; # 0..1 Date last changed
  fhir:PlanDefinition.publisher [ string ]; # 0..1 Name of the publisher (organization or individual)
  fhir:PlanDefinition.contact [ ContactDetail ], ... ; # 0..* Contact details for the publisher
  fhir:PlanDefinition.description [ markdown ]; # 0..1 Natural language description of the plan definition
  fhir:PlanDefinition.useContext [ UsageContext ], ... ; # 0..* The context that the content is intended to support
  fhir:PlanDefinition.jurisdiction [ CodeableConcept ], ... ; # 0..* Intended jurisdiction for plan definition (if
    applicable)
  fhir:PlanDefinition.purpose [ markdown ]; # 0..1 Why this plan definition is defined
  fhir:PlanDefinition.usage [ string ]; # 0..1 Describes the clinical usage of the plan
  fhir:PlanDefinition.copyright [ markdown ]; # 0..1 Use and/or publishing restrictions
  fhir:PlanDefinition.approvalDate [ date ]; # 0..1 When the plan definition was approved by publisher
  fhir:PlanDefinition.lastReviewDate [ date ]; # 0..1 When the plan definition was last reviewed
  fhir:PlanDefinition.effectivePeriod [ Period ]; # 0..1 When the plan definition is expected to be used
  fhir:PlanDefinition.topic [ CodeableConcept ], ... ; # 0..* E.g. Education, Treatment, Assessment
  fhir:PlanDefinition.author [ ContactDetail ], ... ; # 0..* Who authored the content
  fhir:PlanDefinition.editor [ ContactDetail ], ... ; # 0..* Who edited the content
  fhir:PlanDefinition.reviewer [ ContactDetail ], ... ; # 0..* Who reviewed the content
  fhir:PlanDefinition.endorser [ ContactDetail ], ... ; # 0..* Who endorsed the content
  fhir:PlanDefinition.relatedArtifact [ RelatedArtifact ], ... ; # 0..* Additional documentation, citations
  fhir:PlanDefinition.library [ canonical(Library) ], ... ; # 0..* Logic used by the plan definition
  fhir:PlanDefinition.goal [ # 0..* What the plan is trying to accomplish
    fhir:PlanDefinition.goal.category [ CodeableConcept ]; # 0..1 E.g. Treatment, dietary, behavioral
    fhir:PlanDefinition.goal.description [ CodeableConcept ]; # 1..1 Code or text describing the goal
    fhir:PlanDefinition.goal.priority [ CodeableConcept ]; # 0..1 high-priority | medium-priority | low-priority
    fhir:PlanDefinition.goal.start [ CodeableConcept ]; # 0..1 When goal pursuit begins
    fhir:PlanDefinition.goal.addresses [ CodeableConcept ], ... ; # 0..* What does the goal address
    fhir:PlanDefinition.goal.documentation [ RelatedArtifact ], ... ; # 0..* Supporting documentation for the goal
    fhir:PlanDefinition.goal.target [ # 0..* Target outcome for the goal
      fhir:PlanDefinition.goal.target.measure [ CodeableConcept ]; # 0..1 The parameter whose value is to be tracked
      # PlanDefinition.goal.target.detail[x] : 0..1 The target value to be achieved. One of these 3
        fhir:PlanDefinition.goal.target.detailQuantity [ Quantity ]
        fhir:PlanDefinition.goal.target.detailRange [ Range ]
        fhir:PlanDefinition.goal.target.detailCodeableConcept [ CodeableConcept ]
      fhir:PlanDefinition.goal.target.due [ Duration ]; # 0..1 Reach goal within
    ], ...;
  ], ...;
  fhir:PlanDefinition.action [ # 0..* Action defined by the plan
    fhir:PlanDefinition.action.prefix [ string ]; # 0..1 User-visible prefix for the action (e.g. 1. or A.)
    fhir:PlanDefinition.action.title [ string ]; # 0..1 User-visible title
    fhir:PlanDefinition.action.description [ string ]; # 0..1 Brief description of the action
    fhir:PlanDefinition.action.textEquivalent [ string ]; # 0..1 Static text equivalent of the action, used if the
    dynamic aspects cannot be interpreted by the receiving system
    fhir:PlanDefinition.action.priority [ code ]; # 0..1 routine | urgent | asap | stat
    fhir:PlanDefinition.action.code [ CodeableConcept ], ... ; # 0..* Code representing the meaning of the action or
    sub-actions
    fhir:PlanDefinition.action.reason [ CodeableConcept ], ... ; # 0..* Why the action should be performed
    fhir:PlanDefinition.action.documentation [ RelatedArtifact ], ... ; # 0..* Supporting documentation for the intended
     performer of the action
    fhir:PlanDefinition.action.goalId [ id ], ... ; # 0..* What goals this action supports
    # PlanDefinition.action.subject[x] : 0..1 Type of individual the action is focused on. One of these 2
      fhir:PlanDefinition.action.subjectCodeableConcept [ CodeableConcept ]
      fhir:PlanDefinition.action.subjectReference [ Reference(Group) ]
    fhir:PlanDefinition.action.trigger [ TriggerDefinition ], ... ; # 0..* When the action should be triggered
    fhir:PlanDefinition.action.condition [ # 0..* Whether or not the action is applicable
      fhir:PlanDefinition.action.condition.kind [ code ]; # 1..1 applicability | start | stop
      fhir:PlanDefinition.action.condition.expression [ Expression ]; # 0..1 Boolean-valued expression
    ], ...;
    fhir:PlanDefinition.action.input [ DataRequirement ], ... ; # 0..* Input data requirements
    fhir:PlanDefinition.action.output [ DataRequirement ], ... ; # 0..* Output data definition
    fhir:PlanDefinition.action.relatedAction [ # 0..* Relationship to another action
      fhir:PlanDefinition.action.relatedAction.actionId [ id ]; # 1..1 What action is this related to
      fhir:PlanDefinition.action.relatedAction.relationship [ code ]; # 1..1 before-start | before | before-end |
    concurrent-with-start | concurrent | concurrent-with-end | after-start | after | after-end
      # PlanDefinition.action.relatedAction.offset[x] : 0..1 Time offset for the relationship. One of these 2
        fhir:PlanDefinition.action.relatedAction.offsetDuration [ Duration ]
        fhir:PlanDefinition.action.relatedAction.offsetRange [ Range ]
    ], ...;
    # PlanDefinition.action.timing[x] : 0..1 When the action should take place. One of these 6
```

```
      fhir:PlanDefinition.action.timingDateTime [ dateTime ]
      fhir:PlanDefinition.action.timingAge [ Age ]
      fhir:PlanDefinition.action.timingPeriod [ Period ]
      fhir:PlanDefinition.action.timingDuration [ Duration ]
      fhir:PlanDefinition.action.timingRange [ Range ]
      fhir:PlanDefinition.action.timingTiming [ Timing ]
    fhir:PlanDefinition.action.participant [ # 0..* Who should participate in the action
      fhir:PlanDefinition.action.participant.type [ code ]; # 1..1 patient | practitioner | related-person | device
      fhir:PlanDefinition.action.participant.role [ CodeableConcept ]; # 0..1 E.g. Nurse, Surgeon, Parent
    ], ...;
    fhir:PlanDefinition.action.type [ CodeableConcept ]; # 0..1 create | update | remove | fire-event
    fhir:PlanDefinition.action.groupingBehavior [ code ]; # 0..1 visual-group | logical-group | sentence-group
    fhir:PlanDefinition.action.selectionBehavior [ code ]; # 0..1 any | all | all-or-none | exactly-one | at-most-one |
    one-or-more
    fhir:PlanDefinition.action.requiredBehavior [ code ]; # 0..1 must | could | must-unless-documented
    fhir:PlanDefinition.action.precheckBehavior [ code ]; # 0..1 yes | no
    fhir:PlanDefinition.action.cardinalityBehavior [ code ]; # 0..1 single | multiple
    # PlanDefinition.action.definition[x] : 0..1 Description of the activity to be performed. One of these 2
      fhir:PlanDefinition.action.definitionCanonical [ canonical(ActivityDefinition|PlanDefinition|Questionnaire) ]
      fhir:PlanDefinition.action.definitionUri [ uri ]
    fhir:PlanDefinition.action.transform [ canonical(StructureMap) ]; # 0..1 Transform to apply the template
    fhir:PlanDefinition.action.dynamicValue [ # 0..* Dynamic aspects of the definition
      fhir:PlanDefinition.action.dynamicValue.path [ string ]; # 0..1 The path to the element to be set dynamically
      fhir:PlanDefinition.action.dynamicValue.expression [ Expression ]; # 0..1 An expression that provides the dynamic
    value for the customization
    ], ...;
    fhir:PlanDefinition.action.action [ See PlanDefinition.action ], ... ; # 0..* A sub-action
  ], ...;
]
```

Changes since Release 3

PlanDefinition

PlanDefinition.subtitle | Added Element

PlanDefinition.type | Change code system for extensibly bound codes from "http://hl7.org/fhir/plan-definition-type" to "http://terminology.hl7.org/CodeSystem/plan-definition-type"

PlanDefinition.status | Change value set from http://hl7.org/fhir/ValueSet/publication-status to http://hl7.org/fhir/ValueSet/publication-status|4.0.1

PlanDefinition.experimental | No longer marked as Modifier

PlanDefinition.subject[x] | Added Element

PlanDefinition.author | Added Element

PlanDefinition.editor | Added Element

PlanDefinition.reviewer | Added Element

PlanDefinition.endorser | Added Element

PlanDefinition.library | Type changed from Reference(Library) to canonical(Library)

PlanDefinition.action.prefix | Added Element

PlanDefinition.action.priority | Added Element

PlanDefinition.action.subject[x] | Added Element

PlanDefinition.action.trigger | Added Element

PlanDefinition.action.condition.kind | Change value set from http://hl7.org/fhir/ValueSet/action-condition-kind to http://hl7.org/fhir/ValueSet/action-condition-kind|4.0.1

PlanDefinition.action.condition.expression | Type changed from string to Expression

PlanDefinition.action.relatedAction.relationship | Change value set from http://hl7.org/fhir/ValueSet/action-relationship-type to http://hl7.org/fhir/ValueSet/action-relationship-type|4.0.1

PlanDefinition.action.timing[x] | Add Type Age

PlanDefinition.action.participant.type | Change value set from http://hl7.org/fhir/ValueSet/action-participant-type to http://hl7.org/fhir/ValueSet/action-participant-type|4.0.1

PlanDefinition.action.type | Type changed from Coding to CodeableConcept Change binding strength from required to extensible

PlanDefinition.action.groupingBehavior | Change value set from http://hl7.org/fhir/ValueSet/action-grouping-behavior to http://hl7.org/fhir/ValueSet/action-grouping-behavior|4.0.1

PlanDefinition.action.selectionBehavior | Change value set from http://hl7.org/fhir/ValueSet/action-selection-behavior to http://hl7.org/fhir/ValueSet/action-selection-behavior|4.0.1

PlanDefinition.action.requiredBehavior | Change value set from http://hl7.org/fhir/ValueSet/action-required-behavior to http://hl7.org/fhir/ValueSet/action-required-behavior|4.0.1

PlanDefinition.action.precheckBehavior | Change value set from http://hl7.org/fhir/ValueSet/action-precheck-behavior to http://hl7.org/fhir/ValueSet/action-precheck-behavior|4.0.1

PlanDefinition.action.cardinalityBehavior | Change value set from http://hl7.org/fhir/ValueSet/action-cardinality-behavior to http://hl7.org/fhir/ValueSet/action-cardinality-behavior|4.0.1

PlanDefinition.action.definition[x] | Renamed from definition to definition[x] Add Types canonical(ActivityDefinition | PlanDefinition | Questionnaire), uri Remove Type Reference(ActivityDefinition | PlanDefinition)

PlanDefinition.action.transform | Type changed from Reference(StructureMap) to canonical(StructureMap)

PlanDefinition.action.dynamicValue.expression | Type changed from string to Expression

PlanDefinition.contributor | deleted

PlanDefinition.action.label | deleted

PlanDefinition.action.triggerDefinition | deleted

PlanDefinition.action.condition.description | deleted

PlanDefinition.action.condition.language | deleted

PlanDefinition.action.dynamicValue.description | deleted

PlanDefinition.action.dynamicValue.language | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 11 tests of which 6 fail to execute. All tests pass round-trip testing and 5 r3 resources are invalid (0 errors).)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

12.18.3.1 Terminology Bindings

Path | Definition | Type | Reference
PlanDefinition.type | The type of PlanDefinition. | Extensible | PlanDefinitionType
PlanDefinition.status | The lifecycle status of an artifact. | Required | PublicationStatus
PlanDefinition.subject[x] PlanDefinition.action.subject[x] | The possible types of subjects for a plan definition (E.g. Patient, Practitioner, Organization, Location, etc.). | Extensible | SubjectType
PlanDefinition.jurisdiction | Countries and regions within which this artifact is targeted for use. | Extensible | Jurisdiction ValueSet
PlanDefinition.topic | High-level categorization of the definition, used for searching, sorting, and filtering. | Example | DefinitionTopic
PlanDefinition.goal.category | Example codes for grouping goals for filtering or presentation. | Example | GoalCategory
PlanDefinition.goal.description | Describes goals that can be achieved. | Example | SNOMEDCTClinicalFindings
PlanDefinition.goal.priority | Indicates the level of importance associated with reaching or sustaining a goal. | Preferred | GoalPriority
PlanDefinition.goal.start | Identifies the types of events that might trigger the start of a goal. | Example | GoalStartEvent
PlanDefinition.goal.addresses | Identifies problems, conditions, issues, or concerns that goals may address. | Example | Condition/Problem/DiagnosisCodes
PlanDefinition.goal.target.measure | Identifies types of parameters that can be tracked to determine goal achievement. | Example | LOINCCodes
PlanDefinition.action.priority | Identifies the level of importance to be assigned to actioning the request. | Required | RequestPriority
PlanDefinition.action.condition.kind | Defines the kinds of conditions that can appear on actions. | Required | ActionConditionKind
PlanDefinition.action.relatedAction.relationship | Defines the types of relationships between actions. | Required | ActionRelationshipType
PlanDefinition.action.participant.type | The type of participant for the action. | Required | ActionParticipantType
PlanDefinition.action.participant.role | Defines roles played by participants for the action. | Example | ActionParticipantRole
PlanDefinition.action.type | The type of action to be performed. | Extensible | ActionType
PlanDefinition.action.groupingBehavior | Defines organization behavior of a group. | Required | ActionGroupingBehavior
PlanDefinition.action.selectionBehavior | Defines selection behavior of a group. | Required | ActionSelectionBehavior
PlanDefinition.action.requiredBehavior | Defines expectations around whether an action or action group is required. | Required | ActionRequiredBehavior
PlanDefinition.action.precheckBehavior | Defines selection frequency behavior for an action or group. | Required | ActionPrecheckBehavior
PlanDefinition.action.cardinalityBehavior | Defines behavior for an action or a group for how many times that item may be repeated. | Required | ActionCardinalityBehavior

12.18.3.2 Constraints

id | Level | Location | Description | Expression
pdf-0 | Warning | (base) | Name should be usable as an identifier for the module by machine processing applications such as code generation | name.matches('[A-Z]([A-Za-z0-9_]){0,254}')

12.18.3.3 Applying a PlanDefinition

The following diagram illustrates the relationship between the PlanDefinition and ActivityDefinition resources, as well as a typical application to CarePlan, RequestGroup, and Request resources. The resources depicted on the left side of the arrow are definition resources, while the ones on the right side of the arrow are request resources, with the arrow representing the $apply operation:

The PlanDefinition and ActivityDefinition resources support the representation of a broad range of use cases including order sets, flow sheets, event-condition-action rules, clinical protocols, and research trials. To support this range of use cases, as well as the variability in capabilities of systems that use these types of artifacts, this specification is not prescriptive about exactly how these definitions are applied to produce request and event resources. However, the approach depicted above provides a general framework for the process, and the following steps provide more detail on the potential approach:

• Create a CarePlan resource focused on the Patient in context and linked to the PlanDefinition using the instantiates element

• Create goal elements in the CarePlan based on the goal definitions in the plan

• Create a RequestGroup resource focused on the Patient in context and linked to the PlanDefinition using the instantiatesCanonical element

• Create an activity in the CarePlan to reference the RequestGroup

• Process each action element of the PlanDefinition

Processing for each action proceeds according to the following steps:

• Determine applicability by evaluating the applicability conditions defined for the element

• If the action is applicable, determine whether the action is a group or a single, atomic activity (does the action have child actions?)

• If the action is atomic, process according to the following steps: Create an action element in the RequestGroup with the same id as the action being processed Apply the elements of the action to the corresponding elements of the newly created action in the RequestGroup such as title, description, textEquivalent, timing, and so on Carry any start and stop conditions defined in the plan action forward to the request group action. There are three possibilities for the definition element: ActivityDefinition: Create the target resource as described in the Applying an ActivityDefinition topic Reference the resulting resource in the resource element of the action. Note that the target resource can be set as a contained resource in the RequestGroup, or it can be persisted independently, as appropriate for the environment Set the status of the target resource to option so that it is clearly indicated as part of a RequestGroup. Note that the ActivityDefinition/$apply operation will not necessarily produce resource with this status, so this is an important step. Apply any overrides based on the elements of the action (see the section on Overlap below for details) PlanDefinition: Create a CarePlan by applying the target PlanDefinition Reference the resulting resource in the resource element of the action. Note that the resulting CarePlan can be set as a contained resource in the RequestGroup, but doing so would require expanding any potentially contained resources. Set the status of the CarePlan to option so that it is clearly indicated as part of a RequestGroup. Apply any overrides based on the elements of the action such as title, description, and dynamicValue. Questionnaire: Set the resource element of the action to the Questionnaire, indicating that the activity to be performed is filling out the given questionnaire.

• If the action is a group, determine which actions to process based on the behaviors specified in the group. Note that this aspect of the process may require input from a user. In these cases, either the choices made by the user can be provided as input to the process, or the process can be performed as part of a user-entry workflow that enables user input to be provided as necessary.

The parameters to the $apply operation are available within dynamicValue CQL and FHIRPath expressions as context variables, accessible by the name of the parameter prefixed with a percent (%) symbol. For example, to access the subject given to the apply, use the expression %subject. The value of the %subject context variable in a dynamicValue expression is determined using the current subject, as specified by the subject element on the PlanDefinition, current PlanDefinition.action, or ActivityDefinition.

In addition to the $apply operation parameters, the context variable %action can be used within the path element of a dynamicValue to specify the current action target. For example, to specify the path to the description element of the current action, use %action.description.

12.18.3.4 Overlap with ActivityDefinition

As noted in the Boundaries section, there is some overlap between the content that can be represented within the action element of a PlanDefinition, and the elements of the ActivityDefinition resource. This overlap allows for both resources to be used independently, as well as in combination. For example, a PlanDefinition may be used without any supporting ActivityDefinitions to describe a particular workflow, where it is sufficient to describe the actions simply as textual descriptions of what needs to take place. On the other hand, the PlanDefinition may be used together with ActivityDefinition to provide a detailed structural representation of the activities to be performed.

In general, where there is overlap, the elements from the PlanDefinition provide overriding behavior. Specifically, the following elements of action overlap with ActivityDefinition:

Element | Behavior

title | The title element in ActivityDefinition is the title of the activity "as defined", where the title element in PlanDefinition reflects the title in the scope of the plan.

description | The description element in ActivityDefinition is the description of the activity "as defined", where the description element in PlanDefinition reflects the description in the scope of the plan.

code | The code element in ActivityDefinition represents the meaning of the activity "as defined", where the code element in PlanDefinition represents the meaning in the scope of the plan.

documentation | The documentation element in PlanDefinition represents additional documentation for the action specific to the scope of the plan, where the relatedArtifact element in ActivityDefinition provides documentation specific to the activity itself.

timing | The timing element in ActivityDefinition represents timings associated within the activity itself, where the timing element in PlanDefinition represents the timing of the activity with respect to the plan and its other actions. When a timing is provided on both, the timing defined in the plan generally takes precedence.

participant | The participant element in ActivityDefinition represents what type of actor is expected to perform the activity generally, where the participant element in PlanDefinition represents the type of actor that is expected to perform the activity in the context of the plan.

transform | The transform element in ActivityDefinition describes the transformation of the definition to a request or event resource in general, where the transform element in PlanDefinition describes the transformation within the scope of the plan. When a transform is present in both, the transform in the plan takes precedence.

dynamicValue | The dynamicValue element in ActivityDefinition defines values for elements of the target request or event resource in general, where the dynamicValue element in PlanDefinition defines values within the scope of the plan. When dynamic values are present in both, the dynamic values from the ActivityDefinition are applied first (in the order in which they appear on the ActivityDefinition), followed by the dynamic values from the PlanDefinition (in the order in which they appear on the PlanDefinition).

12.18.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
composed-of | reference | What resource is being referenced | PlanDefinition.relatedArtifact.where(type='composed-of').resource (Any)
context | token | A use context assigned to the plan definition | (PlanDefinition.useContext.value as CodeableConcept)
context-quantity | quantity | A quantity- or range-valued use context assigned to the plan definition | (PlanDefinition.useContext.value as Quantity) | (PlanDefinition.useContext.value as Range)
context-type | token | A type of use context assigned to the plan definition | PlanDefinition.useContext.code
context-type-quantity | composite | A use context type and quantity- or range-based value assigned to the plan definition | On PlanDefinition.useContext: context-type: code context-quantity: value.as(Quantity) | value.as(Range)
context-type-value | composite | A use context type and value assigned to the plan definition | On PlanDefinition.useContext: context-type: code context: value.as(CodeableConcept)
date | date | The plan definition publication date | PlanDefinition.date
definition | reference | Activity or plan definitions used by plan definition | PlanDefinition.action.definition (Questionnaire, PlanDefinition, ActivityDefinition)
depends-on | reference | What resource is being referenced | PlanDefinition.relatedArtifact.where(type='depends-on').resource | PlanDefinition.library (Library, Any)
derived-from | reference | What resource is being referenced | PlanDefinition.relatedArtifact.where(type='derived-from').resource (Any)
description | string | The description of the plan definition | PlanDefinition.description
effective | date | The time during which the plan definition is intended to be in use | PlanDefinition.effectivePeriod
identifier | token | External identifier for the plan definition | PlanDefinition.identifier
jurisdiction | token | Intended jurisdiction for the plan definition | PlanDefinition.jurisdiction
name | string | Computationally friendly name of the plan definition | PlanDefinition.name
predecessor | reference | What resource is being referenced | PlanDefinition.relatedArtifact.where(type='predecessor').resource (Any)
publisher | string | Name of the publisher of the plan definition | PlanDefinition.publisher
status | token | The current status of the plan definition | PlanDefinition.status
successor | reference | What resource is being referenced | PlanDefinition.relatedArtifact.where(type='successor').resource (Any)
title | string | The human-friendly name of the plan definition | PlanDefinition.title
topic | token | Topics associated with the module | PlanDefinition.topic
type | token | The type of artifact the plan (e.g. order-set, eca-rule, protocol) | PlanDefinition.type
url | uri | The uri that identifies the plan definition | PlanDefinition.url
version | token | The business version of the plan definition | PlanDefinition.version

12.18.5 Resource PlanDefinition - Examples

URL: https://hl7.org/fhir/R4/plandefinition-examples.html

This example illustrates the use of the PlanDefinition resource to represent a Suicide Risk Assessment order set.

12.18.5 Resource PlanDefinition - Examples

Clinical Decision Support Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Not linked to any defined compartments

Example Name | id | Format
Low Suicide Risk Order Set | low-suicide-risk-order-set | XML | JSON | Turtle
Chlamydia Screening CDS example | chlamydia-screening-intervention | XML | JSON | Turtle
Exclusive Breastfeeding Intervention 01 | exclusive-breastfeeding-intervention-01 | XML | JSON | Turtle
Exclusive Breastfeeding Intervention 02 | exclusive-breastfeeding-intervention-02 | XML | JSON | Turtle
Exclusive Breastfeeding Intervention 03 | exclusive-breastfeeding-intervention-03 | XML | JSON | Turtle
Exclusive Breastfeeding Intervention 04 | exclusive-breastfeeding-intervention-04 | XML | JSON | Turtle
Obesity Assessment Protocol | protocol-example | XML | JSON | Turtle
Chemotherapy Regimen | KDN5 | XML | JSON | Turtle
Zika Virus Infection Management Process | zika-virus-intervention | XML | JSON | Turtle
Plan Definition Illustrating Related Actions | options-example | XML | JSON | Turtle
Plan Definition Illustrating Predecessor Relationship | zika-virus-intervention-initial | XML | JSON | Turtle
Opioid CDS Recommendation 4 ECA Rule | opioidcds-04 | XML | JSON | Turtle
Opioid CDS Recommendation 5 ECA Rule | opioidcds-05 | XML | JSON | Turtle
Opioid CDS Recommendation 7 ECA Rule | opioidcds-07 | XML | JSON | Turtle
Opioid CDS Recommendation 8 ECA Rule | opioidcds-08 | XML | JSON | Turtle
Opioid CDS Recommendation 10 ECA Rule | opioidcds-10 | XML | JSON | Turtle
Opioid CDS Recommendation 11 ECA Rule | opioidcds-11 | XML | JSON | Turtle
Chest Pain Coronary Artery Disease Order Set KNART | example-cardiology-os | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

12.18.6 Resource PlanDefinition - Detailed Descriptions

URL: https://hl7.org/fhir/R4/plandefinition-definitions.html

12.18.6 Resource PlanDefinition - Detailed Descriptions

Clinical Decision Support Work Group | Maturity Level: 2 | Trial Use | Security Category: Business | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the PlanDefinition resource.

PlanDefinition

Element Id | PlanDefinition

Definition | This resource allows for the definition of various types of plans as a sharable, consumable, and executable artifact. The resource is general enough to support the description of a broad range of clinical artifacts such as clinical decision support rules, order sets and protocols.

Cardinality | 0..*

Type | DomainResource

Invariants | Defined on this element pdf-0 Warning Name should be usable as an identifier for the module by machine processing applications such as code generation name.matches('[A-Z]([A-Za-z0-9_]){0,254}')

Defined on this element

pdf-0 | Warning | Name should be usable as an identifier for the module by machine processing applications such as code generation | name.matches('[A-Z]([A-Za-z0-9_]){0,254}')

PlanDefinition.url

Element Id | PlanDefinition.url

Definition | An absolute URI that is used to identify this plan definition when it is referenced in a specification, model, design or an instance; also called its canonical identifier. This SHOULD be globally unique and SHOULD be a literal address at which at which an authoritative instance of this plan definition is (or will be) published. This URL can be the target of a canonical reference. It SHALL remain the same when the plan definition is stored on different servers.

Cardinality | 0..1

Type | uri

Requirements | Allows the plan definition to be referenced by a single globally unique identifier.

Summary | true

Comments | Can be a urn:uuid: or a urn:oid: but real http: addresses are preferred. Multiple instances may share the same URL if they have a distinct version. The determination of when to create a new version of a resource (same url, new version) vs. defining a new artifact is up to the author. Considerations for making this decision are found in Technical and Business Versions. In some cases, the resource can no longer be found at the stated url, but the url itself cannot change. Implementations can use the meta.source element to indicate where the current master source of the resource can be found.

PlanDefinition.identifier

Element Id | PlanDefinition.identifier

Definition | A formal identifier that is used to identify this plan definition when it is represented in other formats, or referenced in a specification, model, design or an instance.

Note | This is a business identifier, not a resource identifier (see discussion)

Cardinality | 0..*

Type | Identifier

Requirements | Allows externally provided and/or usable business identifiers to be easily associated with the module.

Summary | true

Comments | Typically, this is used for identifiers that can go in an HL7 V3 II (instance identifier) data type, and can then identify this plan definition outside of FHIR, where it is not possible to use the logical URI.

PlanDefinition.version

Element Id | PlanDefinition.version

Definition | The identifier that is used to identify this version of the plan definition when it is referenced in a specification, model, design or instance. This is an arbitrary value managed by the plan definition author and is not expected to be globally unique. For example, it might be a timestamp (e.g. yyyymmdd) if a managed version is not available. There is also no expectation that versions can be placed in a lexicographical sequence. To provide a version consistent with the Decision Support Service specification, use the format Major.Minor.Revision (e.g. 1.0.0). For more information on versioning knowledge assets, refer to the Decision Support Service specification. Note that a version is required for non-experimental active artifacts.

Note | This is a business versionId, not a resource version id (see discussion)

Cardinality | 0..1

Type | string

Summary | true

Comments | There may be different plan definition instances that have the same identifier but different versions. The version can be appended to the url in a reference to allow a reference to a particular business version of the plan definition with the format [url]|[version].

PlanDefinition.name

Element Id | PlanDefinition.name

Definition | A natural language name identifying the plan definition. This name should be usable as an identifier for the module by machine processing applications such as code generation.

Cardinality | 0..1

Type | string

Requirements | Support human navigation and code generation.

Summary | true

Comments | The name is not expected to be globally unique. The name should be a simple alphanumeric type name to ensure that it is machine-processing friendly.

Invariants | Affect this element inv-0 Warning Name should be usable as an identifier for the module by machine processing applications such as code generation name.matches('[A-Z]([A-Za-z0-9_]){0,254}')

Affect this element

inv-0 | Warning | Name should be usable as an identifier for the module by machine processing applications such as code generation | name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
PlanDefinition.title
Element Id | PlanDefinition.title
Definition | A short, descriptive, user-friendly title for the plan definition.
Cardinality | 0..1
Type | string
Summary | true
Comments | This name does not need to be machine-processing friendly and may contain punctuation, white-space, etc.
PlanDefinition.subtitle
Element Id | PlanDefinition.subtitle
Definition | An explanatory or alternate title for the plan definition giving additional information about its content.
Cardinality | 0..1
Type | string
PlanDefinition.type
Element Id | PlanDefinition.type
Definition | A high-level category for the plan definition that distinguishes the kinds of systems that would be interested in the plan definition.
Cardinality | 0..1
Terminology Binding | PlanDefinitionType (Extensible)
Type | CodeableConcept
Summary | true
PlanDefinition.status
Element Id | PlanDefinition.status
Definition | The status of this plan definition. Enables tracking the life-cycle of the content.
Cardinality | 1..1
Terminology Binding | PublicationStatus (Required)
Type | code
Is Modifier | true (Reason: This is labeled as "Is Modifier" because applications should not use a retired {{title}} without due consideration)
Summary | true
Comments | Allows filtering of plan definitions that are appropriate for use versus not.
PlanDefinition.experimental
Element Id | PlanDefinition.experimental
Definition | A Boolean value to indicate that this plan definition is authored for testing purposes (or education/evaluation/marketing) and is not intended to be used for genuine usage.
Cardinality | 0..1
Type | boolean
Requirements | Enables experimental content to be developed following the same lifecycle that would be used for a production-level plan definition.
Summary | true
Comments | Allows filtering of plan definitions that are appropriate for use versus not.
PlanDefinition.subject[x]
Element Id | PlanDefinition.subject[x]
Definition | A code or group definition that describes the intended subject of the plan definition.
Cardinality | 0..1
Terminology Binding | Subject type (Extensible)
Type | CodeableConcept|Reference(Group)
[x] Note | See Choice of Data Types for further information about how to use [x]
Meaning if Missing | Patient
PlanDefinition.date
Element Id | PlanDefinition.date
Definition | The date (and optionally time) when the plan definition was published. The date must change when the business version changes and it must change if the status code changes. In addition, it should change when the substantive content of the plan definition changes.
Cardinality | 0..1
Type | dateTime
Alternate Names | Revision Date
Summary | true
Comments | Note that this is not the same as the resource last-modified-date, since the resource may be a secondary representation of the plan definition. Additional specific dates may be added as extensions or be found by consulting Provenances associated with past versions of the resource.
PlanDefinition.publisher
Element Id | PlanDefinition.publisher
Definition | The name of the organization or individual that published the plan definition.
Cardinality | 0..1
Type | string
Requirements | Helps establish the "authority/credibility" of the plan definition. May also allow for contact.
Summary | true
Comments | Usually an organization but may be an individual. The publisher (or steward) of the plan definition is the organization or individual primarily responsible for the maintenance and upkeep of the plan definition. This is not necessarily the same individual or organization that developed and initially authored the content. The publisher is the primary point of contact for questions or issues with the plan definition. This item SHOULD be populated unless the information is available from context.

PlanDefinition.contact
Element Id | PlanDefinition.contact
Definition | Contact details to assist a user in finding and communicating with the publisher.
Cardinality | 0..*
Type | ContactDetail
Summary | true
Comments | May be a web site, an email address, a telephone number, etc.
PlanDefinition.description
Element Id | PlanDefinition.description
Definition | A free text natural language description of the plan definition from a consumer's perspective.
Cardinality | 0..1
Type | markdown
Summary | true
Comments | This description can be used to capture details such as why the plan definition was built, comments about misuse, instructions for clinical use and interpretation, literature references, examples from the paper world, etc. It is not a rendering of the plan definition as conveyed in the 'text' field of the resource itself. This item SHOULD be populated unless the information is available from context (e.g. the language of the plan definition is presumed to be the predominant language in the place the plan definition was created).
PlanDefinition.useContext
Element Id | PlanDefinition.useContext
Definition | The content was developed with a focus and intent of supporting the contexts that are listed. These contexts may be general categories (gender, age, ...) or may be references to specific programs (insurance plans, studies, ...) and may be used to assist with indexing and searching for appropriate plan definition instances.
Cardinality | 0..*
Type | UsageContext
Requirements | Assist in searching for appropriate content.
Summary | true
Comments | When multiple useContexts are specified, there is no expectation that all or any of the contexts apply.
PlanDefinition.jurisdiction
Element Id | PlanDefinition.jurisdiction
Definition | A legal or geographic region in which the plan definition is intended to be used.
Cardinality | 0..*
Terminology Binding | Jurisdiction (Extensible)
Type | CodeableConcept
Summary | true
Comments | It may be possible for the plan definition to be used in jurisdictions other than those for which it was originally designed or intended.
PlanDefinition.purpose
Element Id | PlanDefinition.purpose
Definition | Explanation of why this plan definition is needed and why it has been designed as it has.
Cardinality | 0..1
Type | markdown
Comments | This element does not describe the usage of the plan definition. Instead, it provides traceability of ''why'' the resource is either needed or ''why'' it is defined as it is. This may be used to point to source materials or specifications that drove the structure of this plan definition.
PlanDefinition.usage
Element Id | PlanDefinition.usage
Definition | A detailed description of how the plan definition is used from a clinical perspective.
Cardinality | 0..1
Type | string
PlanDefinition.copyright
Element Id | PlanDefinition.copyright
Definition | A copyright statement relating to the plan definition and/or its contents. Copyright statements are generally legal restrictions on the use and publishing of the plan definition.
Cardinality | 0..1
Type | markdown
Requirements | Consumers must be able to determine any legal restrictions on the use of the plan definition and/or its content.
Alternate Names | License; Restrictions
PlanDefinition.approvalDate
Element Id | PlanDefinition.approvalDate
Definition | The date on which the resource content was approved by the publisher. Approval happens once when the content is officially approved for usage.
Cardinality | 0..1
Type | date
Comments | The 'date' element may be more recent than the approval date because of minor changes or editorial corrections.
PlanDefinition.lastReviewDate
Element Id | PlanDefinition.lastReviewDate
Definition | The date on which the resource content was last reviewed. Review happens periodically after approval but does not change the original approval date.
Cardinality | 0..1
Type | date
Requirements | Gives a sense of how "current" the content is. Resources that have not been reviewed in a long time may have a risk of being less appropriate/relevant.

Comments | If specified, this date follows the original approval date.
PlanDefinition.effectivePeriod
Element Id | PlanDefinition.effectivePeriod
Definition | The period during which the plan definition content was or is planned to be in active use.
Cardinality | 0..1
Type | Period
Requirements | Allows establishing a transition before a resource comes into effect and also allows for a sunsetting process when new versions of the plan definition are or are expected to be used instead.
Summary | true
Comments | The effective period for a plan definition determines when the content is applicable for usage and is independent of publication and review dates. For example, a measure intended to be used for the year 2016 might be published in 2015.
PlanDefinition.topic
Element Id | PlanDefinition.topic
Definition | Descriptive topics related to the content of the plan definition. Topics provide a high-level categorization of the definition that can be useful for filtering and searching.
Cardinality | 0..*
Terminology Binding | DefinitionTopic (Example)
Type | CodeableConcept
Requirements | Repositories must be able to determine how to categorize the plan definition so that it can be found by topical searches.
PlanDefinition.author
Element Id | PlanDefinition.author
Definition | An individiual or organization primarily involved in the creation and maintenance of the content.
Cardinality | 0..*
Type | ContactDetail
PlanDefinition.editor
Element Id | PlanDefinition.editor
Definition | An individual or organization primarily responsible for internal coherence of the content.
Cardinality | 0..*
Type | ContactDetail
PlanDefinition.reviewer
Element Id | PlanDefinition.reviewer
Definition | An individual or organization primarily responsible for review of some aspect of the content.
Cardinality | 0..*
Type | ContactDetail
PlanDefinition.endorser
Element Id | PlanDefinition.endorser
Definition | An individual or organization responsible for officially endorsing the content for use in some setting.
Cardinality | 0..*
Type | ContactDetail
PlanDefinition.relatedArtifact
Element Id | PlanDefinition.relatedArtifact
Definition | Related artifacts such as additional documentation, justification, or bibliographic references.
Cardinality | 0..*
Type | RelatedArtifact
Requirements | Plan definitions must be able to provide enough information for consumers of the content (and/or interventions or results produced by the content) to be able to determine and understand the justification for and evidence in support of the content.
Comments | Each related artifact is either an attachment, or a reference to another resource, but not both.
PlanDefinition.library
Element Id | PlanDefinition.library
Definition | A reference to a Library resource containing any formal logic used by the plan definition.
Cardinality | 0..*
Type | canonical(Library)
PlanDefinition.goal
Element Id | PlanDefinition.goal
Definition | Goals that describe what the activities within the plan are intended to achieve. For example, weight loss, restoring an activity of daily living, obtaining herd immunity via immunization, meeting a process improvement objective, etc.
Cardinality | 0..*
Requirements | Goal information needs to be captured for order sets, protocols, and care plan definitions to better describe the objectives of the protocol activities and to guide the creation of specific goals within the derived care plans and orders.
PlanDefinition.goal.category
Element Id | PlanDefinition.goal.category
Definition | Indicates a category the goal falls within.
Cardinality | 0..1
Terminology Binding | Goal category (Example)
Type | CodeableConcept
PlanDefinition.goal.description
Element Id | PlanDefinition.goal.description
Definition | Human-readable and/or coded description of a specific desired objective of care, such as "control blood pressure" or "negotiate an obstacle course" or "dance with child at wedding".
Cardinality | 1..1
Terminology Binding | SNOMED CT Clinical Findings (Example)

Type | CodeableConcept
Comments | If no code is available, use CodeableConcept.text.
PlanDefinition.goal.priority
Element Id | PlanDefinition.goal.priority
Definition | Identifies the expected level of importance associated with reaching/sustaining the defined goal.
Cardinality | 0..1
Terminology Binding | Goal priority (Preferred)
Type | CodeableConcept
PlanDefinition.goal.start
Element Id | PlanDefinition.goal.start
Definition | The event after which the goal should begin being pursued.
Cardinality | 0..1
Terminology Binding | Goal start event (Example)
Type | CodeableConcept
PlanDefinition.goal.addresses
Element Id | PlanDefinition.goal.addresses
Definition | Identifies problems, conditions, issues, or concerns the goal is intended to address.
Cardinality | 0..*
Terminology Binding | Condition/Problem/Diagnosis Codes (Example)
Type | CodeableConcept
PlanDefinition.goal.documentation
Element Id | PlanDefinition.goal.documentation
Definition | Didactic or other informational resources associated with the goal that provide further supporting information about the goal. Information resources can include inline text commentary and links to web resources.
Cardinality | 0..*
Type | RelatedArtifact
PlanDefinition.goal.target
Element Id | PlanDefinition.goal.target
Definition | Indicates what should be done and within what timeframe.
Cardinality | 0..*
PlanDefinition.goal.target.measure
Element Id | PlanDefinition.goal.target.measure
Definition | The parameter whose value is to be tracked, e.g. body weight, blood pressure, or hemoglobin A1c level.
Cardinality | 0..1
Terminology Binding | LOINC Codes (Example)
Type | CodeableConcept
PlanDefinition.goal.target.detail[x]
Element Id | PlanDefinition.goal.target.detail[x]
Definition | The target value of the measure to be achieved to signify fulfillment of the goal, e.g. 150 pounds or 7.0%. Either the high or low or both values of the range can be specified. When a low value is missing, it indicates that the goal is achieved at any value at or below the high value. Similarly, if the high value is missing, it indicates that the goal is achieved at any value at or above the low value.
Cardinality | 0..1
Type | Quantity|Range|CodeableConcept
[x] Note | See Choice of Data Types for further information about how to use [x]
PlanDefinition.goal.target.due
Element Id | PlanDefinition.goal.target.due
Definition | Indicates the timeframe after the start of the goal in which the goal should be met.
Cardinality | 0..1
Type | Duration
PlanDefinition.action
Element Id | PlanDefinition.action
Definition | An action or group of actions to be taken as part of the plan.
Cardinality | 0..*
Comments | Note that there is overlap between many of the elements defined here and the ActivityDefinition resource. When an ActivityDefinition is referenced (using the definition element), the overlapping elements in the plan override the content of the referenced ActivityDefinition unless otherwise documented in the specific elements. See the PlanDefinition resource for more detailed information.
PlanDefinition.action.prefix
Element Id | PlanDefinition.action.prefix
Definition | A user-visible prefix for the action.
Cardinality | 0..1
Type | string
PlanDefinition.action.title
Element Id | PlanDefinition.action.title
Definition | The title of the action displayed to a user.
Cardinality | 0..1
Type | string
PlanDefinition.action.description
Element Id | PlanDefinition.action.description
Definition | A brief description of the action used to provide a summary to display to the user.
Cardinality | 0..1
Type | string

PlanDefinition.action.textEquivalent
Element Id | PlanDefinition.action.textEquivalent
Definition | A text equivalent of the action to be performed. This provides a human-interpretable description of the action when the definition is consumed by a system that might not be capable of interpreting it dynamically.
Cardinality | 0..1
Type | string
PlanDefinition.action.priority
Element Id | PlanDefinition.action.priority
Definition | Indicates how quickly the action should be addressed with respect to other actions.
Cardinality | 0..1
Terminology Binding | Request priority (Required)
Type | code
PlanDefinition.action.code
Element Id | PlanDefinition.action.code
Definition | A code that provides meaning for the action or action group. For example, a section may have a LOINC code for the section of a documentation template.
Cardinality | 0..*
Type | CodeableConcept
PlanDefinition.action.reason
Element Id | PlanDefinition.action.reason
Definition | A description of why this action is necessary or appropriate.
Cardinality | 0..*
Type | CodeableConcept
Comments | This is different than the clinical evidence documentation, it's an actual business description of the reason for performing the action.
PlanDefinition.action.documentation
Element Id | PlanDefinition.action.documentation
Definition | Didactic or other informational resources associated with the action that can be provided to the CDS recipient. Information resources can include inline text commentary and links to web resources.
Cardinality | 0..*
Type | RelatedArtifact
PlanDefinition.action.goalId
Element Id | PlanDefinition.action.goalId
Definition | Identifies goals that this action supports. The reference must be to a goal element defined within this plan definition.
Cardinality | 0..*
Type | id
PlanDefinition.action.subject[x]
Element Id | PlanDefinition.action.subject[x]
Definition | A code or group definition that describes the intended subject of the action and its children, if any.
Cardinality | 0..1
Terminology Binding | Subject type (Extensible)
Type | CodeableConcept|Reference(Group)
[x] Note | See Choice of Data Types for further information about how to use [x]
Meaning if Missing | Patient
Requirements | Multiple steps in a protocol often have different groups of steps that are focused on testing different things. The subject of an action specifies the focus of the action and any child actions.
Comments | The subject of an action overrides the subject at a parent action or on the root of the PlanDefinition if specified. In addition, because the subject needs to be resolved during realization, use of subjects in actions (or in the ActivityDefinition referenced by the action) resolves based on the set of subjects supplied in context and by type (i.e. the patient subject would resolve to a resource of type Patient).
PlanDefinition.action.trigger
Element Id | PlanDefinition.action.trigger
Definition | A description of when the action should be triggered.
Cardinality | 0..*
Type | TriggerDefinition
PlanDefinition.action.condition
Element Id | PlanDefinition.action.condition
Definition | An expression that describes applicability criteria or start/stop conditions for the action.
Cardinality | 0..*
Comments | When multiple conditions of the same kind are present, the effects are combined using AND semantics, so the overall condition is true only if all the conditions are true.
PlanDefinition.action.condition.kind
Element Id | PlanDefinition.action.condition.kind
Definition | The kind of condition.
Cardinality | 1..1
Terminology Binding | ActionConditionKind (Required)
Type | code
Comments | Applicability criteria are used to determine immediate applicability when a plan definition is applied to a given context. Start and stop criteria are carried through application and used to describe enter/exit criteria for an action.
PlanDefinition.action.condition.expression
Element Id | PlanDefinition.action.condition.expression
Definition | An expression that returns true or false, indicating whether the condition is satisfied.
Cardinality | 0..1

Type | Expression
Comments | The expression may be inlined or may be a reference to a named expression within a logic library referenced by the library element.
PlanDefinition.action.input
Element Id | PlanDefinition.action.input
Definition | Defines input data requirements for the action.
Cardinality | 0..*
Type | DataRequirement
PlanDefinition.action.output
Element Id | PlanDefinition.action.output
Definition | Defines the outputs of the action, if any.
Cardinality | 0..*
Type | DataRequirement
PlanDefinition.action.relatedAction
Element Id | PlanDefinition.action.relatedAction
Definition | A relationship to another action such as "before" or "30-60 minutes after start of".
Cardinality | 0..*
Comments | When an action depends on multiple actions, the meaning is that all actions are dependencies, rather than that any of the actions are a dependency.
PlanDefinition.action.relatedAction.actionId
Element Id | PlanDefinition.action.relatedAction.actionId
Definition | The element id of the related action.
Cardinality | 1..1
Type | id
PlanDefinition.action.relatedAction.relationship
Element Id | PlanDefinition.action.relatedAction.relationship
Definition | The relationship of this action to the related action.
Cardinality | 1..1
Terminology Binding | ActionRelationshipType (Required)
Type | code
PlanDefinition.action.relatedAction.offset[x]
Element Id | PlanDefinition.action.relatedAction.offset[x]
Definition | A duration or range of durations to apply to the relationship. For example, 30-60 minutes before.
Cardinality | 0..1
Type | Duration|Range
[x] Note | See Choice of Data Types for further information about how to use [x]
PlanDefinition.action.timing[x]
Element Id | PlanDefinition.action.timing[x]
Definition | An optional value describing when the action should be performed.
Cardinality | 0..1
Type | dateTime|Age|Period|Duration|Range|Timing
[x] Note | See Choice of Data Types for further information about how to use [x]
PlanDefinition.action.participant
Element Id | PlanDefinition.action.participant
Definition | Indicates who should participate in performing the action described.
Cardinality | 0..*
PlanDefinition.action.participant.type
Element Id | PlanDefinition.action.participant.type
Definition | The type of participant in the action.
Cardinality | 1..1
Terminology Binding | ActionParticipantType (Required)
Type | code
PlanDefinition.action.participant.role
Element Id | PlanDefinition.action.participant.role
Definition | The role the participant should play in performing the described action.
Cardinality | 0..1
Terminology Binding | Action participant role (Example)
Type | CodeableConcept
PlanDefinition.action.type
Element Id | PlanDefinition.action.type
Definition | The type of action to perform (create, update, remove).
Cardinality | 0..1
Terminology Binding | ActionType (Extensible)
Type | CodeableConcept
PlanDefinition.action.groupingBehavior
Element Id | PlanDefinition.action.groupingBehavior
Definition | Defines the grouping behavior for the action and its children.
Cardinality | 0..1
Terminology Binding | ActionGroupingBehavior (Required)
Type | code
PlanDefinition.action.selectionBehavior
Element Id | PlanDefinition.action.selectionBehavior

Definition | Defines the selection behavior for the action and its children.
Cardinality | 0..1
Terminology Binding | ActionSelectionBehavior (Required)
Type | code
PlanDefinition.action.requiredBehavior
Element Id | PlanDefinition.action.requiredBehavior
Definition | Defines the required behavior for the action.
Cardinality | 0..1
Terminology Binding | ActionRequiredBehavior (Required)
Type | code
PlanDefinition.action.precheckBehavior
Element Id | PlanDefinition.action.precheckBehavior
Definition | Defines whether the action should usually be preselected.
Cardinality | 0..1
Terminology Binding | ActionPrecheckBehavior (Required)
Type | code
PlanDefinition.action.cardinalityBehavior
Element Id | PlanDefinition.action.cardinalityBehavior
Definition | Defines whether the action can be selected multiple times.
Cardinality | 0..1
Terminology Binding | ActionCardinalityBehavior (Required)
Type | code
PlanDefinition.action.definition[x]
Element Id | PlanDefinition.action.definition[x]
Definition | A reference to an ActivityDefinition that describes the action to be taken in detail, or a PlanDefinition that describes a series of actions to be taken.
Cardinality | 0..1
Type | canonical(ActivityDefinition | PlanDefinition | Questionnaire)|uri
[x] Note | See Choice of Data Types for further information about how to use [x]
Comments | Note that the definition is optional, and if no definition is specified, a dynamicValue with a root ($this) path can be used to define the entire resource dynamically.
PlanDefinition.action.transform
Element Id | PlanDefinition.action.transform
Definition | A reference to a StructureMap resource that defines a transform that can be executed to produce the intent resource using the ActivityDefinition instance as the input.
Cardinality | 0..1
Type | canonical(StructureMap)
Comments | Note that when a referenced ActivityDefinition also defines a transform, the transform specified here generally takes precedence. In addition, if both a transform and dynamic values are specific, the dynamic values are applied to the result of the transform.
PlanDefinition.action.dynamicValue
Element Id | PlanDefinition.action.dynamicValue
Definition | Customizations that should be applied to the statically defined resource. For example, if the dosage of a medication must be computed based on the patient's weight, a customization would be used to specify an expression that calculated the weight, and the path on the resource that would contain the result.
Cardinality | 0..*
Comments | Dynamic values are applied in the order in which they are defined in the PlanDefinition resource. Note that when dynamic values are also specified by a referenced ActivityDefinition, the dynamicValues from the ActivityDefinition are applied first, followed by the dynamicValues specified here. In addition, if both a transform and dynamic values are specific, the dynamic values are applied to the result of the transform.
PlanDefinition.action.dynamicValue.path
Element Id | PlanDefinition.action.dynamicValue.path
Definition | The path to the element to be customized. This is the path on the resource that will hold the result of the calculation defined by the expression. The specified path SHALL be a FHIRPath resolveable on the specified target type of the ActivityDefinition, and SHALL consist only of identifiers, constant indexers, and a restricted subset of functions. The path is allowed to contain qualifiers (.) to traverse sub-elements, as well as indexers ([x]) to traverse multiple-cardinality sub-elements (see the Simple FHIRPath Profile for full details).
Cardinality | 0..1
Type | string
Comments | To specify the path to the current action being realized, the %action environment variable is available in this path. For example, to specify the description element of the target action, the path would be %action.description. The path attribute contains a Simple FHIRPath Subset that allows path traversal, but not calculation.
PlanDefinition.action.dynamicValue.expression
Element Id | PlanDefinition.action.dynamicValue.expression
Definition | An expression specifying the value of the customized element.
Cardinality | 0..1
Type | Expression
Comments | The expression may be inlined or may be a reference to a named expression within a logic library referenced by the library element.
PlanDefinition.action.action
Element Id | PlanDefinition.action.action
Definition | Sub actions that are contained within the action. The behavior of this action determines the functionality of the sub-actions. For example, a selection behavior of at-most-one indicates that of the sub-actions, at most one may be chosen as part of realizing the action definition.

Cardinality | 0..*
Type | See PlanDefinition.action

12.18.7 Resource PlanDefinition - Mappings

URL: https://hl7.org/fhir/R4/plandefinition-mappings.html

12.18.7 Resource PlanDefinition - Mappings

Clinical Decision Support Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Not linked to any defined compartments

Mappings:

FiveWs Pattern Mapping

Workflow Pattern

RIM Mapping

Mappings for the plandefinition resource (see Mappings to Other Standards for further information & status).

12.18.7.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

PlanDefinition
url | FiveWs.identifier
identifier | FiveWs.identifier
version | FiveWs.version
status | FiveWs.status
experimental | FiveWs.class
date | FiveWs.recorded
publisher | FiveWs.witness
purpose | FiveWs.why[x]

12.18.7.2 Workflow Pattern (http://hl7.org/fhir/workflow)

PlanDefinition | Definition
url | Definition.url
identifier | Definition.identifier
version | Definition.version
title | Definition.title
status | Definition.status {different ValueSet}
experimental | Definition.experimental
subject[x] | Definition.subject
date | Definition.date
publisher | Definition.publisher
contact | Definition.contact
description | Definition.description
useContext | Definition.useContext
jurisdiction | Definition.jurisdiction
purpose | Definition.purpose
copyright | Definition.copyright
approvalDate | Definition.approvalDate
lastReviewDate | Definition.lastReviewDate
effectivePeriod | Definition.effectivePeriod
topic | Definition.subject[x]
action | {Is a contained Definition}
title | Definition.title
description | Definition.description
textEquivalent | Definition.description
subject[x] | Definition.subject
definition[x] | Definition.derivedFrom
action | {InverseRelationship of Definition.partOf}

12.18.7.3 RIM Mapping (http://hl7.org/v3)

PlanDefinition
author | .participation[typeCode=AUT]
reviewer | .participation[typeCode=VRF] {not clear whether VRF best corresponds to reviewer or endorser}
endorser | .participation[typeCode=VRF] {not clear whether VRF best corresponds to reviewer or endorser}

12.18.8 Resource PlanDefinition - Extensions & Profiles

URL: https://hl7.org/fhir/R4/plandefinition-profiles.html

12.18.8 Resource PlanDefinition - Extensions & Profiles

Clinical Decision Support Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Not linked to any defined compartments

This table lists profiles and extensions for the PlanDefinition resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

12.18.8.1 Profiles

Profile | Description | Context
CDSHooksServicePlanDefinition | Defines a PlanDefinition that implements the behavior for a CDS Hooks service | CDS Hooks Service PlanDefinition
ComputablePlanDefinition | Defines a computable PlanDefinition that specifies a single library and requires all expressions referenced from the PlanDefinition to be definitions in that single library | Computable PlanDefinition
ShareablePlanDefinition | Enforces the minimum information set for the plan definition metadata required by HL7 and other organizations that share and publish plan definitions | Shareable PlanDefinition

12.18.8.2 Extensions

cqf-cdsHooksEndpoint | PlanDefinition | cdsHooksEndpoint | for Clinical Reasoning Extensions
replaces | PlanDefinition | replaces | for General Extensions for use by FHIR Implementers
timing-daysOfCycle | PlanDefinition.action | daysOfCycle | for Common extensions for Timing data type
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

workflow-instantiatesCanonical | instantiatesCanonical | for Workflow Pattern HL7 Extensions
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

12.18.8.3 Search Extensions

No Search Extensions defined for this resource

12.18.9 Resource PlanDefinition - Operations

URL: https://hl7.org/fhir/R4/plandefinition-operations.html

12.18.9 Resource PlanDefinition - Operations

Clinical Decision Support Work Group | Maturity Level: 2 | Trial Use | Security Category: Business | Compartments: Not linked to any defined compartments

This resource has 2 operations operations associated with it:

$apply | Apply
$data-requirements | Data Requirements

For more information about operations, including how they are invoked, see Operations.

12.18.9.1 Operation $apply on PlanDefinition

URL: https://hl7.org/fhir/R4/plandefinition-operation-apply.html

12.18.9.1 Operation $apply on PlanDefinition

Clinical Decision Support Work Group | Maturity Level: 2 | Trial Use | Compartments: N/A

The apply operation applies a PlanDefinition to a given context.

The official URL for this operation definition is

```
http://hl7.org/fhir/OperationDefinition/PlanDefinition-apply
```

Formal Definition (as a OperationDefinition).

URL: [base]/PlanDefinition/$apply

URL: [base]/PlanDefinition/[id]/$apply

This is an idempotent operation

In Parameters:
Name | Cardinality | Type | Binding | Profile | Documentation
planDefinition | 0..1 | PlanDefinition | | | The plan definition to be applied. If the operation is invoked at the instance level, this parameter is not allowed; if the operation is invoked at the type level, this parameter is required

subject | 1..* | string (reference) | | | The subject(s) that is/are the target of the plan to be applied. The subject may be a Patient, Practitioner, Organization, Location, Device, or Group. Subjects provided in this parameter will be resolved as the subject of the PlanDefinition based on the type of the subject. If multiple subjects of the same type are provided, the behavior is implementation-defined
encounter | 0..1 | string (reference) | | | The encounter in context, if any
practitioner | 0..1 | string (reference) | | | The practitioner applying the plan definition
organization | 0..1 | string (reference) | | | The organization applying the plan definition
userType | 0..1 | CodeableConcept | | | The type of user initiating the request, e.g. patient, healthcare provider, or specific type of healthcare provider (physician, nurse, etc.)
userLanguage | 0..1 | CodeableConcept | | | Preferred language of the person using the system
userTaskContext | 0..1 | CodeableConcept | | | The task the system user is performing, e.g. laboratory results review, medication list review, etc. This information can be used to tailor decision support outputs, such as recommended information resources
setting | 0..1 | CodeableConcept | | | The current setting of the request (inpatient, outpatient, etc.)
settingContext | 0..1 | CodeableConcept | | | Additional detail about the setting of the request, if any
Out Parameters:
Name | Cardinality | Type | Binding | Profile | Documentation
return | 1..1 | CarePlan | | | The CarePlan that is the result of applying the plan definition Note: as this is the only out parameter, it is a resource, and it has the name 'return', the result of this operation is returned directly as a resource

The result of this operation is a CarePlan resource with a single activity represented by a RequestGroup. The RequestGroup will have actions for each of the applicable actions in the plan based on evaluating the applicability condition in context. For each applicable action, the definition is applied as described in the $apply operation of the ActivityDefinition resource, and the resulting resource is added as an activity to the CarePlan. If the ActivityDefinition includes library references, those libraries will be available to the evaluated expressions. If those libraries have parameters, those parameters will be bound by name to the parameters given to the operation. In addition, parameters to the $apply operation are available within dynamicValue expressions as context variables, accessible by the name of the parameter, prefixed with a percent (%) symbol. For a more detailed description, refer to the PlanDefinition and ActivityDefinition resource documentation

12.18.9.1.1 Examples

Request: Apply the low suicide risk order set to patient 124

```
GET [base]/PlanDefinition/example/$apply?subject=Patient/124
```

Response: Returned on successful application of the order set

```
HTTP/1.1 200 OK

<?xml version="1.0" encoding="UTF-8"?>
<CarePlan xmlns="http://hl7.org/fhir">
  <!-- This example demonstrates the result of an $apply operation for the citalopramPrescription activity definition.
    -->
  <id value="apply-operation-response"/>
  <text>
  <status value="generated"/>
    <div xmlns="http://www.w3.org/1999/xhtml">
    Generated...
    </div>
  </text>
  <contained>
  <ServiceRequest>
    <id value="referralrequest"/>
    <status value="draft"/>
      <intent value="proposal"/>
      <code>
    <coding>
      <system value="http://snomed.info/sct"/>
      <code value="306206005"/>
    </coding>
    <text value="Referral to service (procedure)"/>
      </code>
      <subject>
        <reference value="Patient/124"/>
      </subject>
      <occurrenceDateTime value="2017-02-26T11:47:00.000Z"/>
      <requester>
    <reference value="Practitioner/123"/>
    </requester>
      <performerType>
      <coding>
        <system value="http://nucc.org/provider-taxonomy"/>
        <code value="261QM0850X"/>
        <display value="Adult Mental Health"/>
      </coding>
      </performerType>
    <reasonCode>
      <coding>
        <system value="http://example.org/CodeSystem/suicide-risk-assessment-score"/>
        <code value="Low"/>
        <display value="Low Suicide Risk"/>
      </coding>
      <text value="Low Suicide Risk"/>
```

```
  </reasonCode>
  <reasonReference>
    <reference value="RiskAssessment/suicide-risk-assessment"/>
  </reasonReference>
</ServiceRequest>
</contained>
<contained>
  <Medication>
    <id value="citalopramMedication"/>
    <code>
      <coding>
        <system value="http://www.nlm.nih.gov/research/umls/rxnorm"/>
        <code value="200371"/>
      </coding>
      <text value="citalopram"/>
    </code>
      <form>
        <coding>
          <system value="http://snomed.info/sct"/>
          <code value="385055001"/>
          <display value="Tablet dose form"/>
        </coding>
        <text value="Tablet dose form"/>
      </form>
      <ingredient>
        <itemReference>
          <reference value="#citalopramSubstance"/>
        </itemReference>
        <amount>
          <numerator>
            <value value="20"/>
            <unit value="mg"/>
          </numerator>
          <denominator>
            <value value="1"/>
            <unit value="{tbl}"/>
          </denominator>
        </amount>
      </ingredient>
  </Medication>
</contained>
<contained>
  <Substance>
    <id value="citalopramSubstance"/>
    <code>
      <coding>
        <system value="http://www.nlm.nih.gov/research/umls/rxnorm"/>
        <code value="2556"/>
      </coding>
      <text value="citalopram"/>
    </code>
  </Substance>
</contained>
<contained>
<MedicationRequest>
  <id value="citalopramPrescription"/>
  <status value="draft"/>
  <intent value="proposal"/>
  <medicationReference>
    <reference value="#citalopramMedication"/>
  </medicationReference>
  <subject>
    <reference value="Patient/124"/>
  </subject>
  <requester>
  <reference value="Practitioner/123"/>
  </requester>
  <reasonCode>
    <coding>
      <system value="http://example.org/CodeSystem/suicide-risk-assessment-score"/>
      <code value="Low"/>
      <display value="Low Suicide Risk"/>
    </coding>
    <text value="Low Suicide Risk"/>
  </reasonCode>
  <reasonReference>
    <reference value="RiskAssessment/suicide-risk-assessment"/>
  </reasonReference>
  <!--definition>
  <reference value="ActivityDefinition/citalopramPrescription"/>
  </definition-->
  <instantiates value="ActivityDefinition/citalopramPrescription"/>
  <dosageInstruction>
  <text value="1 tablet oral 1 time daily"/>
  <timing>
    <repeat>
      <frequency value="1"/>
      <period value="1"/>
      <periodUnit value="d"/>
```

```
      </repeat>
    </timing>
    <route>
      <coding>
        <code value="26643006"/>
        <display value="Oral route (qualifier value)"/>
      </coding>
      <text value="Oral route (qualifier value)"/>
    </route>
    <doseAndRate>
      <type><text value="ordered"/></type>
      <doseQuantity>
        <value value="1"/>
        <unit value="{tbl}"/>
      </doseQuantity>
    </doseAndRate>
    </dosageInstruction>
    <dispenseRequest>
      <numberOfRepeatsAllowed value="3"/>
      <quantity>
        <value value="30"/>
        <unit value="{tbl}"/>
      </quantity>
    </dispenseRequest>
  </MedicationRequest>
  </contained>
  <instantiatesCanonical value="PlanDefinition/example"/>
  <status value="draft"/>
  <intent value="proposal"/>
  <subject>
    <reference value="Patient/124"/>
  </subject>
  <activity>
  <reference>
    <reference value="#referralrequest"/>
  </reference>
  </activity>
  <activity>
    <reference>
      <reference value="#citalopramPrescription"/>
    </reference>
  </activity>
</CarePlan>
```

For more information about operations, including how they are invoked, see Operations.

12.18.9.2 Operation $data-requirements on PlanDefinition

URL: https://hl7.org/fhir/R4/plandefinition-operation-data-requirements.html

12.18.9.2 Operation $data-requirements on PlanDefinition

Clinical Decision Support Work Group | Maturity Level: 2 | Trial Use | Compartments: N/A

The data-requirements operation aggregates and returns the parameters and data requirements for the plan definition and all its dependencies as a single module definition library.

The official URL for this operation definition is

```
http://hl7.org/fhir/OperationDefinition/PlanDefinition-data-requirements
```

Formal Definition (as a OperationDefinition).

URL: [base]/PlanDefinition/[id]/$data-requirements

This is an idempotent operation

Out Parameters:

Name | Cardinality | Type | Binding | Profile | Documentation

return | 1..1 | Library | | | The result of the requirements gathering is a module-definition Library that describes the aggregate parameters, data requirements, and dependencies of the plan definition Note: as this is the only out parameter, it is a resource, and it has the name 'return', the result of this operation is returned directly as a resource

The effect of invoking this operation is to determine the aggregate set of data requirements and dependencies for the plan definition. The result is a Library resource with a type of module-definition that contains all the parameter definitions and data requirements of the plan definition and any libraries referenced by it. Implementations SHOULD aggregate data requirements intelligently (i.e. by collapsing overlapping data requirements)

12.18.9.2.1 Examples

Request: Request data requirements for the Zika Virus intervention plan definition

```
GET [base]/PlanDefinition/zika-virus-intervention/$data-requirements
```

Response: Returned on successful evaluation

```
HTTP/1.1 200 OK

{
```

```
  "resourceType": "Library",
  "type": {
    "coding": [
      {
        "code": "module-definition"
      }
    ]
  },
  "relatedArtifact": [
    {
      "type": "depends-on",
      "display": "Zika Affected Areas"
      "resource": "ValueSet/zika-affected-areas"
    },
    {
      "type": "depends-on",
      "display": "Pregnancy",
      "resource": "urn:oid:X.Y.Z"
    },
    {
      "type": "depends-on",
      "display": "Zika Exposure Assessment",
      "resource": "urn:oid:X.Y.Z"
    },
    {
      "type": "depends-on",
      "display": "Zika Virus Signs and Symptoms",
      "url": "https://phinvads.cdc.gov/vads/ViewValueSet.action?oid=2.16.840.1.114222.4.11.7459",
      "resource": "urn:oid:2.16.840.1.114222.4.11.7459"
    },
  ]
  "dataRequirement": [
    {
      "type": "Condition",
      "codeFilter": [
        {
          "path": "code",
          "valueSet": "http://example.org/valueset/pregnancy"
        }
      ]
    },
    {
      "type": "Patient"
    },
    {
      "type": "Observation",
      "codeFilter": [
        {
          "path": "code",
          "valueSet": "http://example.org/valueset/zika-exposure-assessment"
        },
        {
          "path": "status",
          "code": [
            {
              "code": "final"
            }
          ]
        }
      ]
    },
    {
      "type": "Condition",
      "codeFilter": [
        {
          "path": "category",
          "code": [
            {
              "code": "diagnosis"
            }
          ]
        },
        {
          "path": "clinicalStatus",
          "code": [
            {
              "code": "confirmed"
            }
          ]
        },
        {
          "path": "code",
          "valueSet": "urn:oid:2.16.840.1.114222.4.11.7459"
        }
      ]
    }
  ]
}
```

For more information about operations, including how they are invoked, see Operations.

12.18.10 Resource PlanDefinition - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/plandefinition-version-maps.html

No validation errors - all conversions are clean

12.18.10 Resource PlanDefinition - R3/R4 Conversions

Clinical Decision Support Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Not linked to any defined compartments

R3 : R4 Conversion maps for PlanDefinition.

Functional status for this map: 11 tests of which 6 fail to execute. All tests pass round-trip testing and 5 r3 resources are invalid (0 errors). (see documentation)

12.18.10.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/PlanDefinition3to4" = "R3 to R4 Conversions for PlanDefinition"

uses "http://hl7.org/fhir/3.0/StructureDefinition/PlanDefinition" alias PlanDefinitionR3 as source
uses "http://hl7.org/fhir/StructureDefinition/PlanDefinition" alias PlanDefinition as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group PlanDefinition(source src : PlanDefinitionR3, target tgt : PlanDefinition) extends DomainResource <<type+>> {
  src.url -> tgt.url;
  src.identifier -> tgt.identifier;
  src.version -> tgt.version;
  src.name -> tgt.name;
  src.title -> tgt.title;
  src.type -> tgt.type;
  src.status -> tgt.status;
  src.experimental -> tgt.experimental;
  src.date -> tgt.date;
  src.publisher -> tgt.publisher;
  src.description -> tgt.description;
  src.purpose -> tgt.purpose;
  src.usage -> tgt.usage;
  src.approvalDate -> tgt.approvalDate;
  src.lastReviewDate -> tgt.lastReviewDate;
  src.effectivePeriod -> tgt.effectivePeriod;
  src.useContext -> tgt.useContext;
  src.jurisdiction -> tgt.jurisdiction;
  src.topic -> tgt.topic;
  src.contributor as vs where type = 'author' -> tgt.author as vt then Contributor(vs, vt);
  src.contributor as vs where type = 'editor' -> tgt.editor as vt then Contributor(vs, vt);
  src.contributor as vs where type = 'reviewer' -> tgt.reviewer as vt then Contributor(vs, vt);
  src.contributor as vs where type = 'endorser' -> tgt.endorser as vt then Contributor(vs, vt);
  src.contact -> tgt.contact;
  src.copyright -> tgt.copyright;
  src.relatedArtifact -> tgt.relatedArtifact;
  src.library -> tgt.library;
  src.goal as vs0 -> tgt.goal as vt0 then goal(vs0, vt0);
  src.action as vs0 -> tgt.action as vt0 then action(vs0, vt0);
}

group goal(source src, target tgt) extends BackboneElement {
  src.category -> tgt.category;
  src.description -> tgt.description;
  src.priority -> tgt.priority;
  src.start -> tgt.start;
  src.addresses -> tgt.addresses;
  src.documentation -> tgt.documentation;
  src.target as vs0 -> tgt.target as vt0 then target(vs0, vt0);
}

group target(source src, target tgt) extends BackboneElement {
  src.measure -> tgt.measure;
  src.detail -> tgt.detail;
  src.due -> tgt.due;
}

group action(source src, target tgt) extends BackboneElement {
  src.label -> tgt.prefix;
  src.title -> tgt.title;
  src.description -> tgt.description;
  src.textEquivalent -> tgt.textEquivalent;
  src.code -> tgt.code;
  src.reason -> tgt.reason;
  src.documentation -> tgt.documentation;
  src.goalId -> tgt.goalId;
  src.triggerDefinition -> tgt.trigger;
  src.condition as vs0 -> tgt.condition as vt0 then condition(vs0, vt0);
  src.input -> tgt.input;
  src.output -> tgt.output;
  src.relatedAction as vs0 -> tgt.relatedAction as vt0 then relatedAction(vs0, vt0);
```

```
  src.timing : dateTime as vs -> tgt.timing = create('dateTime') as vt then dateTime(vs, vt);
  src.timing : Period as vs -> tgt.timing = create('Period') as vt then Period(vs, vt);
  src.timing : Duration as vs -> tgt.timing = create('Duration') as vt then Duration(vs, vt);
  src.timing : Range as vs -> tgt.timing = create('Range') as vt then Range(vs, vt);
  src.timing : Timing as vs -> tgt.timing = create('Timing') as vt then Timing(vs, vt);
  src.type -> tgt.type;
  src.groupingBehavior -> tgt.groupingBehavior;
  src.selectionBehavior -> tgt.selectionBehavior;
  src.requiredBehavior -> tgt.requiredBehavior;
  src.precheckBehavior -> tgt.precheckBehavior;
  src.cardinalityBehavior -> tgt.cardinalityBehavior;
  src.definition -> src.definition;
  src.transform -> src.transform;
  src.dynamicValue as vs0 -> tgt.dynamicValue as vt0 then dynamicValue(vs0, vt0);
  src.action as vs0 -> tgt.action as vt0 then action(vs0, vt0);
}

group condition(source src, target tgt) extends BackboneElement {
  src.kind -> tgt.kind;
  src.description as v ->  tgt.expression as e,  e.description = v;
  src.language as v ->  tgt.expression as e,  e.language = v;
  src.expression as v ->  tgt.expression as e,  e.expression = v;
}

group relatedAction(source src, target tgt) extends BackboneElement {
  src.actionid -> tgt.actionid;
  src.relationship -> tgt.relationship;
  src.offset : Duration as vs -> tgt.offset = create('Duration') as vt then Duration(vs, vt);
  src.offset : Range as vs -> tgt.offset = create('Range') as vt then Range(vs, vt);
}

group dynamicValue(source src, target tgt) extends BackboneElement {
  src.path -> tgt.path;
  src.description as v ->  tgt.expression as e,  e.description = v;
  src.language as v ->  tgt.expression as e,  e.language = v;
  src.expression as v ->  tgt.expression as e,  e.expression = v;
}
```

12.18.10.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/PlanDefinition4to3" = "R4 to R3 Conversion for PlanDefinition"

uses "http://hl7.org/fhir/StructureDefinition/PlanDefinition" alias PlanDefinition as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/PlanDefinition" alias PlanDefinitionR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group PlanDefinition(source src : PlanDefinition, target tgt : PlanDefinitionR3) extends DomainResource <<type+>> {
  src.url -> tgt.url;
  src.identifier -> tgt.identifier;
  src.version -> tgt.version;
  src.name -> tgt.name;
  src.title -> tgt.title;
  src.status -> tgt.status;
  src.experimental -> tgt.experimental;
  src.type -> tgt.type;
  src.date -> tgt.date;
  src.publisher -> tgt.publisher;
  src.contact -> tgt.contact;
  src.description -> tgt.description;
  src.useContext -> tgt.useContext;
  src.jurisdiction -> tgt.jurisdiction;
  src.purpose -> tgt.purpose;
  src.usage -> tgt.usage;
  src.copyright -> tgt.copyright;
  src.approvalDate -> tgt.approvalDate;
  src.lastReviewDate -> tgt.lastReviewDate;
  src.effectivePeriod -> tgt.effectivePeriod;
  src.topic -> tgt.topic;
  src.author as vs ->  tgt.contributor as vt,  vt.type = 'author' then Contributor(vs, vt);
  src.editor as vs ->  tgt.contributor as vt,  vt.type = 'editor',  vt.contact = vs;
  src.reviewer as vs ->  tgt.contributor as vt,  vt.type = 'reviewer',  vt.contact = vs;
  src.endorser as vs ->  tgt.contributor as vt,  vt.type = 'endorser',  vt.contact = vs;
  src.relatedArtifact -> tgt.relatedArtifact;
  src.library -> tgt.library;
  src.goal as vs0 -> tgt.goal as vt0 then goal(vs0, vt0);
  src.action as vs0 -> tgt.action as vt0 then action(vs0, vt0);
}

group goal(source src, target tgt) extends BackboneElement {
  src.category -> tgt.category;
  src.description -> tgt.description;
  src.priority -> tgt.priority;
  src.start -> tgt.start;
  src.addresses -> tgt.addresses;
  src.documentation -> tgt.documentation;
  src.target as vs0 -> tgt.target as vt0 then target(vs0, vt0);
}

group target(source src, target tgt) extends BackboneElement {
```

```
  src.measure -> tgt.measure;
  src.detail -> tgt.detail;
  src.due -> tgt.due;
}

group action(source src, target tgt) extends BackboneElement {
  src.prefix -> tgt.label;
  src.title -> tgt.title;
  src.description -> tgt.description;
  src.textEquivalent -> tgt.textEquivalent;
  src.code -> tgt.code;
  src.reason -> tgt.reason;
  src.documentation -> tgt.documentation;
  src.goalId -> tgt.goalId;
  src.trigger -> tgt.triggerDefinition;
  src.condition as vs0 -> tgt.condition as vt0 then condition(vs0, vt0);
  src.input -> tgt.input;
  src.output -> tgt.output;
  src.relatedAction as vs0 -> tgt.relatedAction as vt0 then relatedAction(vs0, vt0);
  src.timing : dateTime as vs -> tgt.timing = create('dateTime') as vt then dateTime(vs, vt);
  src.timing : Period as vs -> tgt.timing = create('Period') as vt then Period(vs, vt);
  src.timing : Duration as vs -> tgt.timing = create('Duration') as vt then Duration(vs, vt);
  src.timing : Range as vs -> tgt.timing = create('Range') as vt then Range(vs, vt);
  src.timing : Timing as vs -> tgt.timing = create('Timing') as vt then Timing(vs, vt);
  src.type -> tgt.type;
  src.groupingBehavior -> tgt.groupingBehavior;
  src.selectionBehavior -> tgt.selectionBehavior;
  src.requiredBehavior -> tgt.requiredBehavior;
  src.precheckBehavior -> tgt.precheckBehavior;
  src.cardinalityBehavior -> tgt.cardinalityBehavior;
  src.definition -> src.definition;
  src.transform -> src.transform;
  src.dynamicValue as vs0 -> tgt.dynamicValue as vt0 then dynamicValue(vs0, vt0);
  src.action as vs0 -> tgt.action as vt0 then action(vs0, vt0);
}

group condition(source src, target tgt) extends BackboneElement {
  src.kind -> tgt.kind;
  src.expression as e then {
    e.description -> tgt.description;
    e.language -> tgt.language;
    e.expression -> tgt.expression;
  };
}

group relatedAction(source src, target tgt) extends BackboneElement {
  src.actionid -> tgt.actionid;
  src.relationship -> tgt.relationship;
  src.offset : Duration as vs -> tgt.offset = create('Duration') as vt then Duration(vs, vt);
  src.offset : Range as vs -> tgt.offset = create('Range') as vt then Range(vs, vt);
}

group dynamicValue(source src, target tgt) extends BackboneElement {
  src.path -> tgt.path;
  src.expression as e then {
    e.description -> tgt.description;
    e.language -> tgt.language;
    e.expression -> tgt.expression;
  };
}
```

12.18.10.3 R4 Validation Errors

12.18.11 Shareable PlanDefinition

URL: https://hl7.org/fhir/R4/plandefinition-shareable.html

Enforces the minimum information set for the plan definition metadata required by HL7 and other organizations that share and publish plan definitions

12.18.11 Shareable PlanDefinition

Clinical Decision Support Work Group | Maturity Level: N/A | Standards Status: Informative

12.18.11.1 Content

Profiles:

ShareablePlanDefinition | Enforces the minimum information set for the plan definition metadata required by HL7 and other organizations that share and publish plan definitions

12.18.11.2 StructureDefinition: ShareablePlanDefinition

URL: https://hl7.org/fhir/R4/shareableplandefinition.html

12.18.11.2 StructureDefinition: ShareablePlanDefinition

Clinical Decision Support Work Group | Maturity Level: 1 | Trial Use | Use Context: Any

The official URL for this profile is:

`http://hl7.org/fhir/StructureDefinition/shareableplandefinition`

Enforces the minimum information set for the plan definition metadata required by HL7 and other organizations that share and publish plan definitions

This profile was published on [no date] as a draft by HL7.

12.18.11.2.1 Formal Views of Profile Content

Description of Profiles, Differentials, Snapshots, and how the XML and JSON presentations work.

- Text Summary
- Differential Table
- Snapshot Table
- XML Template
- JSON Template
- All

This structure is derived from PlanDefinition.

Summary

Mandatory: 6 elements Must-Support: 6 elements

This structure is derived from PlanDefinition.

Name | Flags | Card. | Type | Description & Constraints
PlanDefinition | | 1..1
url | | 1..1 | uri
identifier | S | 0..* | Identifier
version | | 1..1 | string
name | | 1..1 | string
title | S | 0..1 | string
experimental | | 1..1 | boolean
date | S | 0..1 | dateTime
publisher | | 1..1 | string
contact | S | 0..* | ContactDetail
description | | 1..1 | markdown
useContext | S | 0..* | UsageContext
jurisdiction | S | 0..* | CodeableConcept
Documentation for this format

Name | Flags | Card. | Type | Description & Constraints
PlanDefinition | I | 1..1 | | The definition of a plan for a series of actions, independent of any specific patient or context
id | Σ | 0..1 | string | Logical id of this artifact
meta | ΣI | 0..1 | Meta | Metadata about the resource
implicitRules | ?!ΣI | 0..1 | uri | A set of rules under which this content was created
language | I | 0..1 | code | Language of the resource content Binding: Common Languages (preferred) Max Binding: All Languages
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!I | 0..* | Extension | Extensions that cannot be ignored
url | ΣI | 1..1 | uri | Canonical identifier for this plan definition, represented as a URI (globally unique)
identifier | SΣI | 0..* | Identifier | Additional identifier for the plan definition
version | ΣI | 1..1 | string | Business version of the plan definition
name | ΣI | 1..1 | string | Name for this plan definition (computer friendly)
title | SΣI | 0..1 | string | Name for this plan definition (human friendly)
subtitle | I | 0..1 | string | Subordinate title of the plan definition
type | ΣI | 0..1 | CodeableConcept | order-set | clinical-protocol | eca-rule | workflow-definition Binding: PlanDefinitionType (extensible)
status | ?!ΣI | 1..1 | code | draft | active | retired | unknown Binding: PublicationStatus (required)
experimental | ΣI | 1..1 | boolean | For testing purposes, not real usage
subject[x] | I | 0..1 | | Type of individual the plan definition is focused on Binding: Subject type (extensible)
subjectCodeableConcept | | | CodeableConcept
subjectReference | | | Reference(group.html)
date | SΣI | 0..1 | dateTime | Date last changed
publisher | ΣI | 1..1 | string | Name of the publisher (organization or individual)
contact | SΣI | 0..* | ContactDetail | Contact details for the publisher
description | ΣI | 1..1 | markdown | Natural language description of the plan definition
useContext | SΣI | 0..* | UsageContext | The context that the content is intended to support
jurisdiction | SΣI | 0..* | CodeableConcept | Intended jurisdiction for plan definition (if applicable) Binding: Jurisdiction (extensible)
purpose | I | 0..1 | markdown | Why this plan definition is defined
usage | I | 0..1 | string | Describes the clinical usage of the plan
copyright | I | 0..1 | markdown | Use and/or publishing restrictions

approvalDate | I | 0..1 | date | When the plan definition was approved by publisher
lastReviewDate | I | 0..1 | date | When the plan definition was last reviewed
effectivePeriod | ΣI | 0..1 | Period | When the plan definition is expected to be used
topic | I | 0..* | CodeableConcept | E.g. Education, Treatment, Assessment Binding: DefinitionTopic (example)
author | I | 0..* | ContactDetail | Who authored the content
editor | I | 0..* | ContactDetail | Who edited the content
reviewer | I | 0..* | ContactDetail | Who reviewed the content
endorser | I | 0..* | ContactDetail | Who endorsed the content
relatedArtifact | I | 0..* | RelatedArtifact | Additional documentation, citations
library | I | 0..* | canonical(Library) | Logic used by the plan definition
goal | I | 0..* | BackboneElement | What the plan is trying to accomplish
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
category | I | 0..1 | CodeableConcept | E.g. Treatment, dietary, behavioral Binding: Goal category (example)
description | I | 1..1 | CodeableConcept | Code or text describing the goal Binding: SNOMED CT Clinical Findings (example)
priority | I | 0..1 | CodeableConcept | high-priority | medium-priority | low-priority Binding: Goal priority (preferred)
start | I | 0..1 | CodeableConcept | When goal pursuit begins Binding: Goal start event (example)
addresses | I | 0..* | CodeableConcept | What does the goal address Binding: Condition/Problem/Diagnosis Codes (example)
documentation | I | 0..* | RelatedArtifact | Supporting documentation for the goal
target | I | 0..* | BackboneElement | Target outcome for the goal
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
measure | I | 0..1 | CodeableConcept | The parameter whose value is to be tracked Binding: LOINC Codes (example)
detail[x] | I | 0..1 | | The target value to be achieved
detailQuantity | | | Quantity
detailRange | | | Range
detailCodeableConcept | | | CodeableConcept
due | I | 0..1 | Duration | Reach goal within
action | I | 0..* | BackboneElement | Action defined by the plan
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
prefix | I | 0..1 | string | User-visible prefix for the action (e.g. 1. or A.)
title | I | 0..1 | string | User-visible title
description | I | 0..1 | string | Brief description of the action
textEquivalent | I | 0..1 | string | Static text equivalent of the action, used if the dynamic aspects cannot be interpreted by the receiving system
priority | I | 0..1 | code | routine | urgent | asap | stat Binding: Request priority (required)
code | I | 0..* | CodeableConcept | Code representing the meaning of the action or sub-actions
reason | I | 0..* | CodeableConcept | Why the action should be performed
documentation | I | 0..* | RelatedArtifact | Supporting documentation for the intended performer of the action
goalId | I | 0..* | id | What goals this action supports
subject[x] | I | 0..1 | | Type of individual the action is focused on Binding: Subject type (extensible)
subjectCodeableConcept | | | CodeableConcept
subjectReference | | | Reference(group.html)
trigger | I | 0..* | TriggerDefinition | When the action should be triggered
condition | I | 0..* | BackboneElement | Whether or not the action is applicable
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
kind | I | 1..1 | code | applicability | start | stop Binding: ActionConditionKind (required)
expression | I | 0..1 | Expression | Boolean-valued expression
input | I | 0..* | DataRequirement | Input data requirements
output | I | 0..* | DataRequirement | Output data definition
relatedAction | I | 0..* | BackboneElement | Relationship to another action
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
actionId | I | 1..1 | id | What action is this related to
relationship | I | 1..1 | code | before-start | before | before-end | concurrent-with-start | concurrent | concurrent-with-end | after-start | after | after-end Binding: ActionRelationshipType (required)
offset[x] | I | 0..1 | | Time offset for the relationship
offsetDuration | | | Duration
offsetRange | | | Range
timing[x] | I | 0..1 | | When the action should take place
timingDateTime | | | dateTime
timingAge | | | Age
timingPeriod | | | Period
timingDuration | | | Duration
timingRange | | | Range

timingTiming | | | Timing
participant | I | 0..* | BackboneElement | Who should participate in the action
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
type | I | 1..1 | code | patient | practitioner | related-person | device Binding: ActionParticipantType (required)
role | I | 0..1 | CodeableConcept | E.g. Nurse, Surgeon, Parent Binding: Action participant role (example)
type | I | 0..1 | CodeableConcept | create | update | remove | fire-event Binding: ActionType (extensible)
groupingBehavior | I | 0..1 | code | visual-group | logical-group | sentence-group Binding: ActionGroupingBehavior (required)
selectionBehavior | I | 0..1 | code | any | all | all-or-none | exactly-one | at-most-one | one-or-more Binding: ActionSelectionBehavior (required)
requiredBehavior | I | 0..1 | code | must | could | must-unless-documented Binding: ActionRequiredBehavior (required)
precheckBehavior | I | 0..1 | code | yes | no Binding: ActionPrecheckBehavior (required)
cardinalityBehavior | I | 0..1 | code | single | multiple Binding: ActionCardinalityBehavior (required)
definition[x] | I | 0..1 | | Description of the activity to be performed
definitionCanonical | | | canonical(activitydefinition.html | plandefinition.html | questionnaire.html)
definitionUri | | | uri
transform | I | 0..1 | canonical(StructureMap) | Transform to apply the template
dynamicValue | I | 0..* | BackboneElement | Dynamic aspects of the definition
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
path | I | 0..1 | string | The path to the element to be set dynamically
expression | I | 0..1 | Expression | An expression that provides the dynamic value for the customization
action | I | 0..* | | Unknown reference to #PlanDefinition.action A sub-action
Documentation for this format

```
<!-- Shareable PlanDefinition -->

<PlanDefinition xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id><!-- 0..1 string Logical id of this artifact --></id>
 <meta><!--  0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!--  0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!--  0..1 Language of the resource content   -->
 <text><!--  0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <url value="[uri]"/><!--  1..1
     Canonical identifier for this plan definition, represented as a URI (globally unique) -->
 <identifier><!--  0..* Identifier
     Additional identifier for the plan definition --></identifier>
 <version value="[string]"/><!--  1..1 Business version of the plan definition -->
 <name value="[string]"/><!--  1..1 Name for this plan definition (computer friendly) -->
 <title value="[string]"/><!--  0..1 Name for this plan definition (human friendly) -->
 <subtitle value="[string]"/><!--  0..1 Subordinate title of the plan definition -->
 <type><!--  0..1 CodeableConcept order-set | clinical-protocol | eca-rule | workflow-definition   --></type>
 <status value="[code]"/><!--  1..1 draft | active | retired | unknown   -->
 <experimental value="[boolean]"/><!--  1..1 For testing purposes, not real usage -->
 <subject[x]><!--  0..1 CodeableConcept|Reference(Group) Type of individual the plan definition is focused on
    --></subject[x]>
 <date value="[dateTime]"/><!--  0..1 Date last changed -->
 <publisher value="[string]"/><!--  1..1 Name of the publisher (organization or individual) -->
 <contact><!--  0..* ContactDetail
     Contact details for the publisher --></contact>
 <description value="[markdown]"/><!--  1..1 Natural language description of the plan definition -->
 <useContext><!--  0..* UsageContext
     The context that the content is intended to support --></useContext>
 <jurisdiction><!--  0..* CodeableConcept Intended jurisdiction for plan definition (if applicable)   --></jurisdiction>
 <purpose value="[markdown]"/><!--  0..1 Why this plan definition is defined -->
 <usage value="[string]"/><!--  0..1 Describes the clinical usage of the plan -->
 <copyright value="[markdown]"/><!--  0..1 Use and/or publishing restrictions -->
 <approvalDate value="[date]"/><!--  0..1 When the plan definition was approved by publisher -->
 <lastReviewDate value="[date]"/><!--  0..1 When the plan definition was last reviewed -->
 <effectivePeriod><!--  0..1 Period
     When the plan definition is expected to be used --></effectivePeriod>
 <topic><!--  0..* CodeableConcept E.g. Education, Treatment, Assessment   --></topic>
 <author><!--  0..* ContactDetail Who authored the content --></author>
 <editor><!--  0..* ContactDetail Who edited the content --></editor>
 <reviewer><!--  0..* ContactDetail Who reviewed the content --></reviewer>
 <endorser><!--  0..* ContactDetail Who endorsed the content --></endorser>
 <relatedArtifact><!--  0..* RelatedArtifact
     Additional documentation, citations --></relatedArtifact>
 <library value="[canonical]"/><!--  0..* Logic used by the plan definition -->
 <goal>  0..* BackboneElement  <!--  0..* What the plan is trying to accomplish -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <category><!--  0..1 CodeableConcept E.g. Treatment, dietary, behavioral   --></category>
  <description><!--  1..1 CodeableConcept Code or text describing the goal   --></description>
  <priority><!--  0..1 CodeableConcept high-priority | medium-priority | low-priority   --></priority>
  <start><!--  0..1 CodeableConcept When goal pursuit begins   --></start>
  <addresses><!--  0..* CodeableConcept What does the goal address   --></addresses>
```

```
  <documentation><!--  0..* RelatedArtifact
      Supporting documentation for the goal --></documentation>
  <target>  0..* BackboneElement  <!--  0..* Target outcome for the goal -->
   <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
   <measure><!--  0..1 CodeableConcept The parameter whose value is to be tracked   --></measure>
   <detail[x]><!--  0..1 Quantity|Range|
     CodeableConcept The target value to be achieved --></detail[x]>
   <due><!--  0..1 Duration
       Reach goal within --></due>
  </target>
 </goal>
 <action>  0..* BackboneElement  <!--  0..* Action defined by the plan -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <prefix value="[string]"/><!--  0..1 User-visible prefix for the action (e.g. 1. or A.) -->
  <title value="[string]"/><!--  0..1 User-visible title -->
  <description value="[string]"/><!--  0..1 Brief description of the action -->
  <textEquivalent value="[string]"/><!--  0..1
      Static text equivalent of the action, used if the dynamic aspects cannot be interpreted by the receiving system
    -->
  <priority value="[code]"/><!--  0..1 routine | urgent | asap | stat   -->
  <code><!--  0..* CodeableConcept
      Code representing the meaning of the action or sub-actions --></code>
  <reason><!--  0..* CodeableConcept
      Why the action should be performed --></reason>
  <documentation><!--  0..* RelatedArtifact
      Supporting documentation for the intended performer of the action --></documentation>
  <goalId value="[id]"/><!--  0..* What goals this action supports -->
  <subject[x]><!--  0..1 CodeableConcept|Reference(Group) Type of individual the action is focused on   --></subject[x]>
  <trigger><!--  0..* TriggerDefinition
      When the action should be triggered --></trigger>
  <condition>  0..* BackboneElement  <!--  0..* Whether or not the action is applicable -->
   <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
   <kind value="[code]"/><!--  1..1 applicability | start | stop   -->
   <expression><!--  0..1 Expression
       Boolean-valued expression --></expression>
  </condition>
  <input><!--  0..* DataRequirement
      Input data requirements --></input>
  <output><!--  0..* DataRequirement
      Output data definition --></output>
  <relatedAction>  0..* BackboneElement  <!--  0..* Relationship to another action -->
   <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
   <actionId value="[id]"/><!--  1..1 What action is this related to -->
   <relationship value="[code]"/><!--  1..1 before-start | before | before-end | concurrent-with-start | concurrent |
    concurrent-with-end | after-start | after | after-end   -->
   <offset[x]><!--  0..1 Duration|Range
       Time offset for the relationship --></offset[x]>
  </relatedAction>
  <timing[x]><!--  0..1 dateTime|Age|Period|Duration|Range|
    Timing When the action should take place --></timing[x]>
  <participant>  0..* BackboneElement  <!--  0..* Who should participate in the action -->
   <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
   <type value="[code]"/><!--  1..1 patient | practitioner | related-person | device   -->
   <role><!--  0..1 CodeableConcept E.g. Nurse, Surgeon, Parent   --></role>
  </participant>
  <type><!--  0..1 CodeableConcept create | update | remove | fire-event   --></type>
  <groupingBehavior value="[code]"/><!--  0..1 visual-group | logical-group | sentence-group   -->
  <selectionBehavior value="[code]"/><!--  0..1 any | all | all-or-none | exactly-one | at-most-one | one-or-more   -->
  <requiredBehavior value="[code]"/><!--  0..1 must | could | must-unless-documented   -->
  <precheckBehavior value="[code]"/><!--  0..1 yes | no   -->
  <cardinalityBehavior value="[code]"/><!--  0..1 single | multiple   -->
  <definition[x]><!--  0..1 canonical|uri
      Description of the activity to be performed --></definition[x]>
  <transform value="[canonical]"/><!--  0..1 Transform to apply the template -->
  <dynamicValue>  0..* BackboneElement  <!--  0..* Dynamic aspects of the definition -->
   <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
   <path value="[string]"/><!--  0..1 The path to the element to be set dynamically -->
   <expression><!--  0..1 Expression
       An expression that provides the dynamic value for the customization --></expression>
  </dynamicValue>
  <action><!-- See #PlanDefinition.action  A sub-action --></action>
 </action>
</PlanDefinition>
```

```
{ // Shareable PlanDefinition
  // from Element: extension
    "meta" : { Meta }, // C?Metadata about the resource
    "implicitRules" : "<uri>", // C?A set of rules under which this content was created
    "language" : "<code>", // C?Language of the resource content
    "text" : { Narrative }, // C?Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored
    "url" : "<uri>", // C? R! Canonical identifier for this plan definition, represented as a URI (globally unique)
```

```
    "identifier" : [{ Identifier }], // C?Additional identifier for the plan definition
    "version" : "<string>", // C? R! Business version of the plan definition
    "name" : "<string>", // C? R! Name for this plan definition (computer friendly)
    "title" : "<string>", // C?Name for this plan definition (human friendly)
    "subtitle" : "<string>", // C?Subordinate title of the plan definition
    "type" : { CodeableConcept }, // C?order-set | clinical-protocol | eca-rule | workflow-definition
    "status" : "<code>", // C? R! draft | active | retired | unknown
    "experimental" : <boolean>, // C? R! For testing purposes, not real usage
// value[x]: Type of individual the plan definition is focused on. One of these 2:
    "subjectCodeableConcept" : { CodeableConcept }, // C?Type of individual the plan definition is focused on
    "subjectReference" : { Reference(Group) }, // C?Type of individual the plan definition is focused on
    "date" : "<dateTime>", // C?Date last changed
    "publisher" : "<string>", // C? R! Name of the publisher (organization or individual)
    "contact" : [{ ContactDetail }], // C?Contact details for the publisher
    "description" : "<markdown>", // C? R! Natural language description of the plan definition
    "useContext" : [{ UsageContext }], // C?The context that the content is intended to support
    "jurisdiction" : [{ CodeableConcept }], // C?Intended jurisdiction for plan definition (if applicable)
    "purpose" : "<markdown>", // C?Why this plan definition is defined
    "usage" : "<string>", // C?Describes the clinical usage of the plan
    "copyright" : "<markdown>", // C?Use and/or publishing restrictions
    "approvalDate" : "<date>", // C?When the plan definition was approved by publisher
    "lastReviewDate" : "<date>", // C?When the plan definition was last reviewed
    "effectivePeriod" : { Period }, // C?When the plan definition is expected to be used
    "topic" : [{ CodeableConcept }], // C?E.g. Education, Treatment, Assessment
    "author" : [{ ContactDetail }], // C?Who authored the content
    "editor" : [{ ContactDetail }], // C?Who edited the content
    "reviewer" : [{ ContactDetail }], // C?Who reviewed the content
    "endorser" : [{ ContactDetail }], // C?Who endorsed the content
    "relatedArtifact" : [{ RelatedArtifact }], // C?Additional documentation, citations
    "library" : ["<canonical>"], // C?Logic used by the plan definition
    "goal" : [{ BackboneElement }] // C?What the plan is trying to accomplish
    "action" : [{ BackboneElement }], // C?Action defined by the plan
  }
```

This structure is derived from PlanDefinition.

Summary

Mandatory: 6 elements Must-Support: 6 elements

Differential View

This structure is derived from PlanDefinition.

Name	Flags	Card.	Type	Description & Constraints
PlanDefinition		1..1		
url		1..1	uri	
identifier	S	0..*	Identifier	
version		1..1	string	
name		1..1	string	
title	S	0..1	string	
experimental		1..1	boolean	
date	S	0..1	dateTime	
publisher		1..1	string	
contact	S	0..*	ContactDetail	
description		1..1	markdown	
useContext	S	0..*	UsageContext	
jurisdiction	S	0..*	CodeableConcept	
Documentation for this format				

Snapshot View

Name	Flags	Card.	Type	Description & Constraints
PlanDefinition	I	1..1		The definition of a plan for a series of actions, independent of any specific patient or context
id	Σ	0..1	string	Logical id of this artifact
meta	ΣI	0..1	Meta	Metadata about the resource
implicitRules	?!ΣI	0..1	uri	A set of rules under which this content was created
language	I	0..1	code	Language of the resource content Binding: Common Languages (preferred) Max Binding: All Languages
text	I	0..1	Narrative	Text summary of the resource, for human interpretation
contained		0..*	Resource	Contained, inline Resources
extension	I	0..*	Extension	Additional content defined by implementations
modifierExtension	?!I	0..*	Extension	Extensions that cannot be ignored
url	ΣI	1..1	uri	Canonical identifier for this plan definition, represented as a URI (globally unique)
identifier	SΣI	0..*	Identifier	Additional identifier for the plan definition
version	ΣI	1..1	string	Business version of the plan definition
name	ΣI	1..1	string	Name for this plan definition (computer friendly)
title	SΣI	0..1	string	Name for this plan definition (human friendly)
subtitle	I	0..1	string	Subordinate title of the plan definition
type	ΣI	0..1	CodeableConcept	order-set | clinical-protocol | eca-rule | workflow-definition Binding: PlanDefinitionType (extensible)
status	?!ΣI	1..1	code	draft | active | retired | unknown Binding: PublicationStatus (required)
experimental	ΣI	1..1	boolean	For testing purposes, not real usage
subject[x]	I	0..1		Type of individual the plan definition is focused on Binding: Subject type (extensible)
subjectCodeableConcept			CodeableConcept	

subjectReference | | | Reference(group.html)
date | SΣI | 0..1 | dateTime | Date last changed
publisher | ΣI | 1..1 | string | Name of the publisher (organization or individual)
contact | SΣI | 0..* | ContactDetail | Contact details for the publisher
description | ΣI | 1..1 | markdown | Natural language description of the plan definition
useContext | SΣI | 0..* | UsageContext | The context that the content is intended to support
jurisdiction | SΣI | 0..* | CodeableConcept | Intended jurisdiction for plan definition (if applicable) Binding: Jurisdiction (extensible)
purpose | I | 0..1 | markdown | Why this plan definition is defined
usage | I | 0..1 | string | Describes the clinical usage of the plan
copyright | I | 0..1 | markdown | Use and/or publishing restrictions
approvalDate | I | 0..1 | date | When the plan definition was approved by publisher
lastReviewDate | I | 0..1 | date | When the plan definition was last reviewed
effectivePeriod | ΣI | 0..1 | Period | When the plan definition is expected to be used
topic | I | 0..* | CodeableConcept | E.g. Education, Treatment, Assessment Binding: DefinitionTopic (example)
author | I | 0..* | ContactDetail | Who authored the content
editor | I | 0..* | ContactDetail | Who edited the content
reviewer | I | 0..* | ContactDetail | Who reviewed the content
endorser | I | 0..* | ContactDetail | Who endorsed the content
relatedArtifact | I | 0..* | RelatedArtifact | Additional documentation, citations
library | I | 0..* | canonical(Library) | Logic used by the plan definition
goal | I | 0..* | BackboneElement | What the plan is trying to accomplish
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
category | I | 0..1 | CodeableConcept | E.g. Treatment, dietary, behavioral Binding: Goal category (example)
description | I | 1..1 | CodeableConcept | Code or text describing the goal Binding: SNOMED CT Clinical Findings (example)
priority | I | 0..1 | CodeableConcept | high-priority | medium-priority | low-priority Binding: Goal priority (preferred)
start | I | 0..1 | CodeableConcept | When goal pursuit begins Binding: Goal start event (example)
addresses | I | 0..* | CodeableConcept | What does the goal address Binding: Condition/Problem/Diagnosis Codes (example)
documentation | I | 0..* | RelatedArtifact | Supporting documentation for the goal
target | I | 0..* | BackboneElement | Target outcome for the goal
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
measure | I | 0..1 | CodeableConcept | The parameter whose value is to be tracked Binding: LOINC Codes (example)
detail[x] | I | 0..1 | | The target value to be achieved
detailQuantity | | | Quantity
detailRange | | | Range
detailCodeableConcept | | | CodeableConcept
due | I | 0..1 | Duration | Reach goal within
action | I | 0..* | BackboneElement | Action defined by the plan
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
prefix | I | 0..1 | string | User-visible prefix for the action (e.g. 1. or A.)
title | I | 0..1 | string | User-visible title
description | I | 0..1 | string | Brief description of the action
textEquivalent | I | 0..1 | string | Static text equivalent of the action, used if the dynamic aspects cannot be interpreted by the receiving system
priority | I | 0..1 | code | routine | urgent | asap | stat Binding: Request priority (required)
code | I | 0..* | CodeableConcept | Code representing the meaning of the action or sub-actions
reason | I | 0..* | CodeableConcept | Why the action should be performed
documentation | I | 0..* | RelatedArtifact | Supporting documentation for the intended performer of the action
goalId | I | 0..* | id | What goals this action supports
subject[x] | I | 0..1 | | Type of individual the action is focused on Binding: Subject type (extensible)
subjectCodeableConcept | | | CodeableConcept
subjectReference | | | Reference(group.html)
trigger | I | 0..* | TriggerDefinition | When the action should be triggered
condition | I | 0..* | BackboneElement | Whether or not the action is applicable
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
kind | I | 1..1 | code | applicability | start | stop Binding: ActionConditionKind (required)
expression | I | 0..1 | Expression | Boolean-valued expression
input | I | 0..* | DataRequirement | Input data requirements
output | I | 0..* | DataRequirement | Output data definition
relatedAction | I | 0..* | BackboneElement | Relationship to another action
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
actionId | I | 1..1 | id | What action is this related to

relationship | I | 1..1 | code | before-start | before | before-end | concurrent-with-start | concurrent | concurrent-with-end | after-start | after | after-end Binding: ActionRelationshipType (required)
offset[x] | I | 0..1 | | Time offset for the relationship
offsetDuration | | | Duration
offsetRange | | | Range
timing[x] | I | 0..1 | | When the action should take place
timingDateTime | | | dateTime
timingAge | | | Age
timingPeriod | | | Period
timingDuration | | | Duration
timingRange | | | Range
timingTiming | | | Timing
participant | I | 0..* | BackboneElement | Who should participate in the action
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
type | I | 1..1 | code | patient | practitioner | related-person | device Binding: ActionParticipantType (required)
role | I | 0..1 | CodeableConcept | E.g. Nurse, Surgeon, Parent Binding: Action participant role (example)
type | I | 0..1 | CodeableConcept | create | update | remove | fire-event Binding: ActionType (extensible)
groupingBehavior | I | 0..1 | code | visual-group | logical-group | sentence-group Binding: ActionGroupingBehavior (required)
selectionBehavior | I | 0..1 | code | any | all | all-or-none | exactly-one | at-most-one | one-or-more Binding: ActionSelectionBehavior (required)
requiredBehavior | I | 0..1 | code | must | could | must-unless-documented Binding: ActionRequiredBehavior (required)
precheckBehavior | I | 0..1 | code | yes | no Binding: ActionPrecheckBehavior (required)
cardinalityBehavior | I | 0..1 | code | single | multiple Binding: ActionCardinalityBehavior (required)
definition[x] | I | 0..1 | | Description of the activity to be performed
definitionCanonical | | | canonical(activitydefinition.html | plandefinition.html | questionnaire.html)
definitionUri | | | uri
transform | I | 0..1 | canonical(StructureMap) | Transform to apply the template
dynamicValue | I | 0..* | BackboneElement | Dynamic aspects of the definition
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
path | I | 0..1 | string | The path to the element to be set dynamically
expression | I | 0..1 | Expression | An expression that provides the dynamic value for the customization
action | I | 0..* | | Unknown reference to #PlanDefinition.action A sub-action
Documentation for this format

XML Template

```
<!-- Shareable PlanDefinition -->

<PlanDefinition xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id><!-- 0..1 string Logical id of this artifact --></id>
 <meta><!--  0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!--  0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!--  0..1 Language of the resource content   -->
 <text><!--  0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <url value="[uri]"/><!--  1..1
     Canonical identifier for this plan definition, represented as a URI (globally unique) -->
 <identifier><!--  0..* Identifier
     Additional identifier for the plan definition --></identifier>
 <version value="[string]"/><!--  1..1 Business version of the plan definition -->
 <name value="[string]"/><!--  1..1 Name for this plan definition (computer friendly) -->
 <title value="[string]"/><!--  0..1 Name for this plan definition (human friendly) -->
 <subtitle value="[string]"/><!--  0..1 Subordinate title of the plan definition -->
 <type><!--  0..1 CodeableConcept order-set | clinical-protocol | eca-rule | workflow-definition   --></type>
 <status value="[code]"/><!--  1..1 draft | active | retired | unknown   -->
 <experimental value="[boolean]"/><!--  1..1 For testing purposes, not real usage -->
 <subject[x]><!--  0..1 CodeableConcept|Reference(Group) Type of individual the plan definition is focused on
    --></subject[x]>
 <date value="[dateTime]"/><!--  0..1 Date last changed -->
 <publisher value="[string]"/><!--  1..1 Name of the publisher (organization or individual) -->
 <contact><!--  0..* ContactDetail
     Contact details for the publisher --></contact>
 <description value="[markdown]"/><!--  1..1 Natural language description of the plan definition -->
 <useContext><!--  0..* UsageContext
     The context that the content is intended to support --></useContext>
 <jurisdiction><!--  0..* CodeableConcept Intended jurisdiction for plan definition (if applicable)   --></jurisdiction>
 <purpose value="[markdown]"/><!--  0..1 Why this plan definition is defined -->
 <usage value="[string]"/><!--  0..1 Describes the clinical usage of the plan -->
 <copyright value="[markdown]"/><!--  0..1 Use and/or publishing restrictions -->
 <approvalDate value="[date]"/><!--  0..1 When the plan definition was approved by publisher -->
 <lastReviewDate value="[date]"/><!--  0..1 When the plan definition was last reviewed -->
 <effectivePeriod><!--  0..1 Period
     When the plan definition is expected to be used --></effectivePeriod>
 <topic><!--  0..* CodeableConcept E.g. Education, Treatment, Assessment   --></topic>
```

```
<author><!--  0..* ContactDetail Who authored the content --></author>
<editor><!--  0..* ContactDetail Who edited the content --></editor>
<reviewer><!--  0..* ContactDetail Who reviewed the content --></reviewer>
<endorser><!--  0..* ContactDetail Who endorsed the content --></endorser>
<relatedArtifact><!--  0..* RelatedArtifact
    Additional documentation, citations --></relatedArtifact>
<library value="[canonical]"/><!--  0..* Logic used by the plan definition -->
<goal>  0..* BackboneElement  <!--  0..* What the plan is trying to accomplish -->
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
   --></modifierExtension>
 <category><!--  0..1 CodeableConcept E.g. Treatment, dietary, behavioral   --></category>
 <description><!--  1..1 CodeableConcept Code or text describing the goal   --></description>
 <priority><!--  0..1 CodeableConcept high-priority | medium-priority | low-priority   --></priority>
 <start><!--  0..1 CodeableConcept When goal pursuit begins   --></start>
 <addresses><!--  0..* CodeableConcept What does the goal address   --></addresses>
 <documentation><!--  0..* RelatedArtifact
     Supporting documentation for the goal --></documentation>
 <target>  0..* BackboneElement  <!--  0..* Target outcome for the goal -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
   --></modifierExtension>
  <measure><!--  0..1 CodeableConcept The parameter whose value is to be tracked   --></measure>
  <detail[x]><!--  0..1 Quantity|Range|
    CodeableConcept The target value to be achieved --></detail[x]>
  <due><!--  0..1 Duration
      Reach goal within --></due>
 </target>
</goal>
<action>  0..* BackboneElement  <!--  0..* Action defined by the plan -->
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
   --></modifierExtension>
 <prefix value="[string]"/><!--  0..1 User-visible prefix for the action (e.g. 1. or A.) -->
 <title value="[string]"/><!--  0..1 User-visible title -->
 <description value="[string]"/><!--  0..1 Brief description of the action -->
 <textEquivalent value="[string]"/><!--  0..1
     Static text equivalent of the action, used if the dynamic aspects cannot be interpreted by the receiving system
   -->
 <priority value="[code]"/><!--  0..1 routine | urgent | asap | stat   -->
 <code><!--  0..* CodeableConcept
     Code representing the meaning of the action or sub-actions --></code>
 <reason><!--  0..* CodeableConcept
     Why the action should be performed --></reason>
 <documentation><!--  0..* RelatedArtifact
     Supporting documentation for the intended performer of the action --></documentation>
 <goalId value="[id]"/><!--  0..* What goals this action supports -->
 <subject[x]><!--  0..1 CodeableConcept|Reference(Group) Type of individual the action is focused on   --></subject[x]>
 <trigger><!--  0..* TriggerDefinition
     When the action should be triggered --></trigger>
 <condition>  0..* BackboneElement  <!--  0..* Whether or not the action is applicable -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
   --></modifierExtension>
  <kind value="[code]"/><!--  1..1 applicability | start | stop   -->
  <expression><!--  0..1 Expression
      Boolean-valued expression --></expression>
 </condition>
 <input><!--  0..* DataRequirement
     Input data requirements --></input>
 <output><!--  0..* DataRequirement
     Output data definition --></output>
 <relatedAction>  0..* BackboneElement  <!--  0..* Relationship to another action -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
   --></modifierExtension>
  <actionId value="[id]"/><!--  1..1 What action is this related to -->
  <relationship value="[code]"/><!--  1..1 before-start | before | before-end | concurrent-with-start | concurrent |
   concurrent-with-end | after-start | after | after-end   -->
  <offset[x]><!--  0..1 Duration|Range
      Time offset for the relationship --></offset[x]>
 </relatedAction>
 <timing[x]><!--  0..1 dateTime|Age|Period|Duration|Range|
   Timing When the action should take place --></timing[x]>
 <participant>  0..* BackboneElement  <!--  0..* Who should participate in the action -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
   --></modifierExtension>
  <type value="[code]"/><!--  1..1 patient | practitioner | related-person | device   -->
  <role><!--  0..1 CodeableConcept E.g. Nurse, Surgeon, Parent   --></role>
 </participant>
 <type><!--  0..1 CodeableConcept create | update | remove | fire-event   --></type>
 <groupingBehavior value="[code]"/><!--  0..1 visual-group | logical-group | sentence-group   -->
 <selectionBehavior value="[code]"/><!--  0..1 any | all | all-or-none | exactly-one | at-most-one | one-or-more   -->
 <requiredBehavior value="[code]"/><!--  0..1 must | could | must-unless-documented   -->
 <precheckBehavior value="[code]"/><!--  0..1 yes | no   -->
 <cardinalityBehavior value="[code]"/><!--  0..1 single | multiple   -->
 <definition[x]><!--  0..1 canonical|uri
     Description of the activity to be performed --></definition[x]>
 <transform value="[canonical]"/><!--  0..1 Transform to apply the template -->
 <dynamicValue>  0..* BackboneElement  <!--  0..* Dynamic aspects of the definition -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
   --></modifierExtension>
  <path value="[string]"/><!--  0..1 The path to the element to be set dynamically -->
  <expression><!--  0..1 Expression
```

```
      An expression that provides the dynamic value for the customization --></expression>
  </dynamicValue>
  <action><!-- See #PlanDefinition.action  A sub-action --></action>
 </action>
</PlanDefinition>
```

JSON Template

```
{ // Shareable PlanDefinition
  // from Element: extension
    "meta" : { Meta }, // C?Metadata about the resource
    "implicitRules" : "<uri>", // C?A set of rules under which this content was created
    "language" : "<code>", // C?Language of the resource content
    "text" : { Narrative }, // C?Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored
    "url" : "<uri>", // C? R! Canonical identifier for this plan definition, represented as a URI (globally unique)
    "identifier" : [{ Identifier }], // C?Additional identifier for the plan definition
    "version" : "<string>", // C? R! Business version of the plan definition
    "name" : "<string>", // C? R! Name for this plan definition (computer friendly)
    "title" : "<string>", // C?Name for this plan definition (human friendly)
    "subtitle" : "<string>", // C?Subordinate title of the plan definition
    "type" : { CodeableConcept }, // C?order-set | clinical-protocol | eca-rule | workflow-definition
    "status" : "<code>", // C? R! draft | active | retired | unknown
    "experimental" : <boolean>, // C? R! For testing purposes, not real usage
// value[x]: Type of individual the plan definition is focused on. One of these 2:
    "subjectCodeableConcept" : { CodeableConcept }, // C?Type of individual the plan definition is focused on
    "subjectReference" : { Reference(Group) }, // C?Type of individual the plan definition is focused on
    "date" : "<dateTime>", // C?Date last changed
    "publisher" : "<string>", // C? R! Name of the publisher (organization or individual)
    "contact" : [{ ContactDetail }], // C?Contact details for the publisher
    "description" : "<markdown>", // C? R! Natural language description of the plan definition
    "useContext" : [{ UsageContext }], // C?The context that the content is intended to support
    "jurisdiction" : [{ CodeableConcept }], // C?Intended jurisdiction for plan definition (if applicable)
    "purpose" : "<markdown>", // C?Why this plan definition is defined
    "usage" : "<string>", // C?Describes the clinical usage of the plan
    "copyright" : "<markdown>", // C?Use and/or publishing restrictions
    "approvalDate" : "<date>", // C?When the plan definition was approved by publisher
    "lastReviewDate" : "<date>", // C?When the plan definition was last reviewed
    "effectivePeriod" : { Period }, // C?When the plan definition is expected to be used
    "topic" : [{ CodeableConcept }], // C?E.g. Education, Treatment, Assessment
    "author" : [{ ContactDetail }], // C?Who authored the content
    "editor" : [{ ContactDetail }], // C?Who edited the content
    "reviewer" : [{ ContactDetail }], // C?Who reviewed the content
    "endorser" : [{ ContactDetail }], // C?Who endorsed the content
    "relatedArtifact" : [{ RelatedArtifact }], // C?Additional documentation, citations
    "library" : ["<canonical>"], // C?Logic used by the plan definition
    "goal" : [{ BackboneElement }] // C?What the plan is trying to accomplish
    "action" : [{ BackboneElement }], // C?Action defined by the plan
  }
```

Alternate definitions: Master Definition (XML, JSON), Schematron

12.18.11.2.2 Terminology Bindings

Path	Name	Conformance	ValueSet
PlanDefinition.language	Common Languages	preferred	Common Languages
PlanDefinition.type	PlanDefinitionType	extensible	PlanDefinitionType
PlanDefinition.status	PublicationStatus	required	PublicationStatus
PlanDefinition.subject[x]	Subject type	extensible	Subject type
PlanDefinition.jurisdiction	Jurisdiction	extensible	Jurisdiction
PlanDefinition.topic	DefinitionTopic	example	DefinitionTopic
PlanDefinition.goal.category	Goal category	example	Goal category
PlanDefinition.goal.description	SNOMED CT Clinical Findings	example	SNOMED CT Clinical Findings
PlanDefinition.goal.priority	Goal priority	preferred	Goal priority
PlanDefinition.goal.start	Goal start event	example	Goal start event
PlanDefinition.goal.addresses	Condition/Problem/Diagnosis Codes	example	Condition/Problem/Diagnosis Codes
PlanDefinition.goal.target.measure	LOINC Codes	example	LOINC Codes
PlanDefinition.action.priority	Request priority	required	Request priority
PlanDefinition.action.subject[x]	Subject type	extensible	Subject type
PlanDefinition.action.condition.kind	ActionConditionKind	required	ActionConditionKind
PlanDefinition.action.relatedAction.relationship	ActionRelationshipType	required	ActionRelationshipType
PlanDefinition.action.participant.type	ActionParticipantType	required	ActionParticipantType
PlanDefinition.action.participant.role	Action participant role	example	Action participant role
PlanDefinition.action.type	ActionType	extensible	ActionType
PlanDefinition.action.groupingBehavior	ActionGroupingBehavior	required	ActionGroupingBehavior
PlanDefinition.action.selectionBehavior	ActionSelectionBehavior	required	ActionSelectionBehavior
PlanDefinition.action.requiredBehavior	ActionRequiredBehavior	required	ActionRequiredBehavior
PlanDefinition.action.precheckBehavior	ActionPrecheckBehavior	required	ActionPrecheckBehavior
PlanDefinition.action.cardinalityBehavior	ActionCardinalityBehavior	required	ActionCardinalityBehavior

12.18.12 Computable PlanDefinition

URL: https://hl7.org/fhir/R4/plandefinition-computable.html

Defines a computable PlanDefinition that specifies a single library and requires all expressions referenced from the PlanDefinition to be definitions in that single library

12.18.12 Computable PlanDefinition

Clinical Decision Support Work Group | Maturity Level: N/A | Standards Status: Informative

12.18.12.1 Content

Profiles:

ComputablePlanDefinition | Defines a computable PlanDefinition that specifies a single library and requires all expressions referenced from the PlanDefinition to be definitions in that single library

12.18.12.2 StructureDefinition: ComputablePlanDefinition

URL: https://hl7.org/fhir/R4/computableplandefinition.html

12.18.12.2 StructureDefinition: ComputablePlanDefinition

Clinical Decision Support Work Group | Maturity Level: 1 | Trial Use | Use Context: Any

The official URL for this profile is:

`http://hl7.org/fhir/StructureDefinition/computableplandefinition`

Defines a computable PlanDefinition that specifies a single library and requires all expressions referenced from the PlanDefinition to be definitions in that single library

This profile was published on [no date] as a draft by HL7.

12.18.12.2.1 Formal Views of Profile Content

Description of Profiles, Differentials, Snapshots, and how the XML and JSON presentations work.

- Text Summary
- Differential Table
- Snapshot Table
- XML Template
- JSON Template
- All

This structure is derived from PlanDefinition.

Summary

Mandatory: 1 element

This structure is derived from PlanDefinition.

Name | Flags | Card. | Type | Description & Constraints
PlanDefinition | | 1..1
library | | 1..1 | canonical(Library)
Documentation for this format

Name | Flags | Card. | Type | Description & Constraints
PlanDefinition | I | 1..1 | | The definition of a plan for a series of actions, independent of any specific patient or context
id | Σ | 0..1 | string | Logical id of this artifact
meta | ΣI | 0..1 | Meta | Metadata about the resource
implicitRules | ?!ΣI | 0..1 | uri | A set of rules under which this content was created
language | I | 0..1 | code | Language of the resource content Binding: Common Languages (preferred) Max Binding: All Languages
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!I | 0..* | Extension | Extensions that cannot be ignored
url | ΣI | 0..1 | uri | Canonical identifier for this plan definition, represented as a URI (globally unique)
identifier | ΣI | 0..* | Identifier | Additional identifier for the plan definition
version | ΣI | 0..1 | string | Business version of the plan definition
name | ΣI | 0..1 | string | Name for this plan definition (computer friendly)
title | ΣI | 0..1 | string | Name for this plan definition (human friendly)
subtitle | I | 0..1 | string | Subordinate title of the plan definition
type | ΣI | 0..1 | CodeableConcept | order-set | clinical-protocol | eca-rule | workflow-definition Binding: PlanDefinitionType (extensible)
status | ?!ΣI | 1..1 | code | draft | active | retired | unknown Binding: PublicationStatus (required)
experimental | ΣI | 0..1 | boolean | For testing purposes, not real usage
subject[x] | I | 0..1 | | Type of individual the plan definition is focused on Binding: Subject type (extensible)
subjectCodeableConcept | | | CodeableConcept
subjectReference | | | Reference(group.html)
date | ΣI | 0..1 | dateTime | Date last changed
publisher | ΣI | 0..1 | string | Name of the publisher (organization or individual)

contact | ΣI | 0..* | ContactDetail | Contact details for the publisher
description | ΣI | 0..1 | markdown | Natural language description of the plan definition
useContext | ΣI | 0..* | UsageContext | The context that the content is intended to support
jurisdiction | ΣI | 0..* | CodeableConcept | Intended jurisdiction for plan definition (if applicable) Binding: Jurisdiction (extensible)
purpose | I | 0..1 | markdown | Why this plan definition is defined
usage | I | 0..1 | string | Describes the clinical usage of the plan
copyright | I | 0..1 | markdown | Use and/or publishing restrictions
approvalDate | I | 0..1 | date | When the plan definition was approved by publisher
lastReviewDate | I | 0..1 | date | When the plan definition was last reviewed
effectivePeriod | ΣI | 0..1 | Period | When the plan definition is expected to be used
topic | I | 0..* | CodeableConcept | E.g. Education, Treatment, Assessment Binding: DefinitionTopic (example)
author | I | 0..* | ContactDetail | Who authored the content
editor | I | 0..* | ContactDetail | Who edited the content
reviewer | I | 0..* | ContactDetail | Who reviewed the content
endorser | I | 0..* | ContactDetail | Who endorsed the content
relatedArtifact | I | 0..* | RelatedArtifact | Additional documentation, citations
library | I | 1..1 | canonical(Library) | Logic used by the plan definition
goal | I | 0..* | BackboneElement | What the plan is trying to accomplish
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
category | I | 0..1 | CodeableConcept | E.g. Treatment, dietary, behavioral Binding: Goal category (example)
description | I | 1..1 | CodeableConcept | Code or text describing the goal Binding: SNOMED CT Clinical Findings (example)
priority | I | 0..1 | CodeableConcept | high-priority | medium-priority | low-priority Binding: Goal priority (preferred)
start | I | 0..1 | CodeableConcept | When goal pursuit begins Binding: Goal start event (example)
addresses | I | 0..* | CodeableConcept | What does the goal address Binding: Condition/Problem/Diagnosis Codes (example)
documentation | I | 0..* | RelatedArtifact | Supporting documentation for the goal
target | I | 0..* | BackboneElement | Target outcome for the goal
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
measure | I | 0..1 | CodeableConcept | The parameter whose value is to be tracked Binding: LOINC Codes (example)
detail[x] | I | 0..1 | | The target value to be achieved
detailQuantity | | | Quantity
detailRange | | | Range
detailCodeableConcept | | | CodeableConcept
due | I | 0..1 | Duration | Reach goal within
action | I | 0..* | BackboneElement | Action defined by the plan
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
prefix | I | 0..1 | string | User-visible prefix for the action (e.g. 1. or A.)
title | I | 0..1 | string | User-visible title
description | I | 0..1 | string | Brief description of the action
textEquivalent | I | 0..1 | string | Static text equivalent of the action, used if the dynamic aspects cannot be interpreted by the receiving system
priority | I | 0..1 | code | routine | urgent | asap | stat Binding: Request priority (required)
code | I | 0..* | CodeableConcept | Code representing the meaning of the action or sub-actions
reason | I | 0..* | CodeableConcept | Why the action should be performed
documentation | I | 0..* | RelatedArtifact | Supporting documentation for the intended performer of the action
goalId | I | 0..* | id | What goals this action supports
subject[x] | I | 0..1 | | Type of individual the action is focused on Binding: Subject type (extensible)
subjectCodeableConcept | | | CodeableConcept
subjectReference | | | Reference(group.html)
trigger | I | 0..* | TriggerDefinition | When the action should be triggered
condition | I | 0..* | BackboneElement | Whether or not the action is applicable
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
kind | I | 1..1 | code | applicability | start | stop Binding: ActionConditionKind (required)
expression | I | 0..1 | Expression | Boolean-valued expression
input | I | 0..* | DataRequirement | Input data requirements
output | I | 0..* | DataRequirement | Output data definition
relatedAction | I | 0..* | BackboneElement | Relationship to another action
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
actionId | I | 1..1 | id | What action is this related to
relationship | I | 1..1 | code | before-start | before | before-end | concurrent-with-start | concurrent | concurrent-with-end | after-start | after | after-end Binding: ActionRelationshipType (required)
offset[x] | I | 0..1 | | Time offset for the relationship
offsetDuration | | | Duration

offsetRange | | | Range
timing[x] | I | 0..1 | | When the action should take place
timingDateTime | | | dateTime
timingAge | | | Age
timingPeriod | | | Period
timingDuration | | | Duration
timingRange | | | Range
timingTiming | | | Timing
participant | I | 0..* | BackboneElement | Who should participate in the action
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
type | I | 1..1 | code | patient | practitioner | related-person | device Binding: ActionParticipantType (required)
role | I | 0..1 | CodeableConcept | E.g. Nurse, Surgeon, Parent Binding: Action participant role (example)
type | I | 0..1 | CodeableConcept | create | update | remove | fire-event Binding: ActionType (extensible)
groupingBehavior | I | 0..1 | code | visual-group | logical-group | sentence-group Binding: ActionGroupingBehavior (required)
selectionBehavior | I | 0..1 | code | any | all | all-or-none | exactly-one | at-most-one | one-or-more Binding: ActionSelectionBehavior (required)
requiredBehavior | I | 0..1 | code | must | could | must-unless-documented Binding: ActionRequiredBehavior (required)
precheckBehavior | I | 0..1 | code | yes | no Binding: ActionPrecheckBehavior (required)
cardinalityBehavior | I | 0..1 | code | single | multiple Binding: ActionCardinalityBehavior (required)
definition[x] | I | 0..1 | | Description of the activity to be performed
definitionCanonical | | | canonical(activitydefinition.html | plandefinition.html | questionnaire.html)
definitionUri | | | uri
transform | I | 0..1 | canonical(StructureMap) | Transform to apply the template
dynamicValue | I | 0..* | BackboneElement | Dynamic aspects of the definition
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
path | I | 0..1 | string | The path to the element to be set dynamically
expression | I | 0..1 | Expression | An expression that provides the dynamic value for the customization
action | I | 0..* | | Unknown reference to #PlanDefinition.action A sub-action
Documentation for this format

```
<!-- Computable PlanDefinition -->

<PlanDefinition xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id><!-- 0..1 string Logical id of this artifact --></id>
 <meta><!--  0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!--  0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!--  0..1 Language of the resource content   -->
 <text><!--  0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <url value="[uri]"/><!--  0..1
     Canonical identifier for this plan definition, represented as a URI (globally unique) -->
 <identifier><!--  0..* Identifier
     Additional identifier for the plan definition --></identifier>
 <version value="[string]"/><!--  0..1 Business version of the plan definition -->
 <name value="[string]"/><!--  0..1 Name for this plan definition (computer friendly) -->
 <title value="[string]"/><!--  0..1 Name for this plan definition (human friendly) -->
 <subtitle value="[string]"/><!--  0..1 Subordinate title of the plan definition -->
 <type><!--  0..1 CodeableConcept order-set | clinical-protocol | eca-rule | workflow-definition   --></type>
 <status value="[code]"/><!--  1..1 draft | active | retired | unknown   -->
 <experimental value="[boolean]"/><!--  0..1 For testing purposes, not real usage -->
 <subject[x]><!--  0..1 CodeableConcept|Reference(Group) Type of individual the plan definition is focused on
    --></subject[x]>
 <date value="[dateTime]"/><!--  0..1 Date last changed -->
 <publisher value="[string]"/><!--  0..1 Name of the publisher (organization or individual) -->
 <contact><!--  0..* ContactDetail
     Contact details for the publisher --></contact>
 <description value="[markdown]"/><!--  0..1 Natural language description of the plan definition -->
 <useContext><!--  0..* UsageContext
     The context that the content is intended to support --></useContext>
 <jurisdiction><!--  0..* CodeableConcept Intended jurisdiction for plan definition (if applicable)   --></jurisdiction>
 <purpose value="[markdown]"/><!--  0..1 Why this plan definition is defined -->
 <usage value="[string]"/><!--  0..1 Describes the clinical usage of the plan -->
 <copyright value="[markdown]"/><!--  0..1 Use and/or publishing restrictions -->
 <approvalDate value="[date]"/><!--  0..1 When the plan definition was approved by publisher -->
 <lastReviewDate value="[date]"/><!--  0..1 When the plan definition was last reviewed -->
 <effectivePeriod><!--  0..1 Period
     When the plan definition is expected to be used --></effectivePeriod>
 <topic><!--  0..* CodeableConcept E.g. Education, Treatment, Assessment   --></topic>
 <author><!--  0..* ContactDetail Who authored the content --></author>
 <editor><!--  0..* ContactDetail Who edited the content --></editor>
 <reviewer><!--  0..* ContactDetail Who reviewed the content --></reviewer>
 <endorser><!--  0..* ContactDetail Who endorsed the content --></endorser>
 <relatedArtifact><!--  0..* RelatedArtifact
     Additional documentation, citations --></relatedArtifact>
```

```
 <library value="[canonical]"/><!--  1..1 Logic used by the plan definition -->
 <goal>  0..* BackboneElement  <!--  0..* What the plan is trying to accomplish -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <category><!--  0..1 CodeableConcept E.g. Treatment, dietary, behavioral   --></category>
  <description><!--  1..1 CodeableConcept Code or text describing the goal   --></description>
  <priority><!--  0..1 CodeableConcept high-priority | medium-priority | low-priority   --></priority>
  <start><!--  0..1 CodeableConcept When goal pursuit begins   --></start>
  <addresses><!--  0..* CodeableConcept What does the goal address   --></addresses>
  <documentation><!--  0..* RelatedArtifact
      Supporting documentation for the goal --></documentation>
  <target>  0..* BackboneElement  <!--  0..* Target outcome for the goal -->
   <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
   <measure><!--  0..1 CodeableConcept The parameter whose value is to be tracked   --></measure>
   <detail[x]><!--  0..1 Quantity|Range|
     CodeableConcept The target value to be achieved --></detail[x]>
   <due><!--  0..1 Duration
       Reach goal within --></due>
  </target>
 </goal>
 <action>  0..* BackboneElement  <!--  0..* Action defined by the plan -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <prefix value="[string]"/><!--  0..1 User-visible prefix for the action (e.g. 1. or A.) -->
  <title value="[string]"/><!--  0..1 User-visible title -->
  <description value="[string]"/><!--  0..1 Brief description of the action -->
  <textEquivalent value="[string]"/><!--  0..1
      Static text equivalent of the action, used if the dynamic aspects cannot be interpreted by the receiving system
    -->
  <priority value="[code]"/><!--  0..1 routine | urgent | asap | stat   -->
  <code><!--  0..* CodeableConcept
      Code representing the meaning of the action or sub-actions --></code>
  <reason><!--  0..* CodeableConcept
      Why the action should be performed --></reason>
  <documentation><!--  0..* RelatedArtifact
      Supporting documentation for the intended performer of the action --></documentation>
  <goalId value="[id]"/><!--  0..* What goals this action supports -->
  <subject[x]><!--  0..1 CodeableConcept|Reference(Group) Type of individual the action is focused on   --></subject[x]>
  <trigger><!--  0..* TriggerDefinition
      When the action should be triggered --></trigger>
  <condition>  0..* BackboneElement  <!--  0..* Whether or not the action is applicable -->
   <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
   <kind value="[code]"/><!--  1..1 applicability | start | stop   -->
   <expression><!--  0..1 Expression
       Boolean-valued expression --></expression>
  </condition>
  <input><!--  0..* DataRequirement
      Input data requirements --></input>
  <output><!--  0..* DataRequirement
      Output data definition --></output>
  <relatedAction>  0..* BackboneElement  <!--  0..* Relationship to another action -->
   <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
   <actionId value="[id]"/><!--  1..1 What action is this related to -->
   <relationship value="[code]"/><!--  1..1 before-start | before | before-end | concurrent-with-start | concurrent |
    concurrent-with-end | after-start | after | after-end   -->
   <offset[x]><!--  0..1 Duration|Range
       Time offset for the relationship --></offset[x]>
  </relatedAction>
  <timing[x]><!--  0..1 dateTime|Age|Period|Duration|Range|
    Timing When the action should take place --></timing[x]>
  <participant>  0..* BackboneElement  <!--  0..* Who should participate in the action -->
   <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
   <type value="[code]"/><!--  1..1 patient | practitioner | related-person | device   -->
   <role><!--  0..1 CodeableConcept E.g. Nurse, Surgeon, Parent   --></role>
  </participant>
  <type><!--  0..1 CodeableConcept create | update | remove | fire-event   --></type>
  <groupingBehavior value="[code]"/><!--  0..1 visual-group | logical-group | sentence-group   -->
  <selectionBehavior value="[code]"/><!--  0..1 any | all | all-or-none | exactly-one | at-most-one | one-or-more   -->
  <requiredBehavior value="[code]"/><!--  0..1 must | could | must-unless-documented   -->
  <precheckBehavior value="[code]"/><!--  0..1 yes | no   -->
  <cardinalityBehavior value="[code]"/><!--  0..1 single | multiple   -->
  <definition[x]><!--  0..1 canonical|uri
      Description of the activity to be performed --></definition[x]>
  <transform value="[canonical]"/><!--  0..1 Transform to apply the template -->
  <dynamicValue>  0..* BackboneElement  <!--  0..* Dynamic aspects of the definition -->
   <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
   <path value="[string]"/><!--  0..1 The path to the element to be set dynamically -->
   <expression><!--  0..1 Expression
       An expression that provides the dynamic value for the customization --></expression>
  </dynamicValue>
  <action><!-- See #PlanDefinition.action  A sub-action --></action>
 </action>
</PlanDefinition>
```

```
{ // Computable PlanDefinition
  // from Element: extension
    "meta" : { Meta }, // C?Metadata about the resource
    "implicitRules" : "<uri>", // C?A set of rules under which this content was created
    "language" : "<code>", // C?Language of the resource content
    "text" : { Narrative }, // C?Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored
    "url" : "<uri>", // C?Canonical identifier for this plan definition, represented as a URI (globally unique)
    "identifier" : [{ Identifier }], // C?Additional identifier for the plan definition
    "version" : "<string>", // C?Business version of the plan definition
    "name" : "<string>", // C?Name for this plan definition (computer friendly)
    "title" : "<string>", // C?Name for this plan definition (human friendly)
    "subtitle" : "<string>", // C?Subordinate title of the plan definition
    "type" : { CodeableConcept }, // C?order-set | clinical-protocol | eca-rule | workflow-definition
    "status" : "<code>", // C? R! draft | active | retired | unknown
    "experimental" : <boolean>, // C?For testing purposes, not real usage
// value[x]: Type of individual the plan definition is focused on. One of these 2:
    "subjectCodeableConcept" : { CodeableConcept }, // C?Type of individual the plan definition is focused on
    "subjectReference" : { Reference(Group) }, // C?Type of individual the plan definition is focused on
    "date" : "<dateTime>", // C?Date last changed
    "publisher" : "<string>", // C?Name of the publisher (organization or individual)
    "contact" : [{ ContactDetail }], // C?Contact details for the publisher
    "description" : "<markdown>", // C?Natural language description of the plan definition
    "useContext" : [{ UsageContext }], // C?The context that the content is intended to support
    "jurisdiction" : [{ CodeableConcept }], // C?Intended jurisdiction for plan definition (if applicable)
    "purpose" : "<markdown>", // C?Why this plan definition is defined
    "usage" : "<string>", // C?Describes the clinical usage of the plan
    "copyright" : "<markdown>", // C?Use and/or publishing restrictions
    "approvalDate" : "<date>", // C?When the plan definition was approved by publisher
    "lastReviewDate" : "<date>", // C?When the plan definition was last reviewed
    "effectivePeriod" : { Period }, // C?When the plan definition is expected to be used
    "topic" : [{ CodeableConcept }], // C?E.g. Education, Treatment, Assessment
    "author" : [{ ContactDetail }], // C?Who authored the content
    "editor" : [{ ContactDetail }], // C?Who edited the content
    "reviewer" : [{ ContactDetail }], // C?Who reviewed the content
    "endorser" : [{ ContactDetail }], // C?Who endorsed the content
    "relatedArtifact" : [{ RelatedArtifact }], // C?Additional documentation, citations
    "library" : "<canonical>", // C? R! Logic used by the plan definition
    "goal" : [{ BackboneElement }] // C?What the plan is trying to accomplish
    "action" : [{ BackboneElement }], // C?Action defined by the plan
  }
```

This structure is derived from PlanDefinition.

Summary

Mandatory: 1 element

Differential View

This structure is derived from PlanDefinition.

Name | Flags | Card. | Type | Description & Constraints
PlanDefinition | | 1..1
library | | 1..1 | canonical(Library)
Documentation for this format

Snapshot View

Name | Flags | Card. | Type | Description & Constraints
PlanDefinition | I | 1..1 | | The definition of a plan for a series of actions, independent of any specific patient or context
id | Σ | 0..1 | string | Logical id of this artifact
meta | ΣI | 0..1 | Meta | Metadata about the resource
implicitRules | ?!ΣI | 0..1 | uri | A set of rules under which this content was created
language | I | 0..1 | code | Language of the resource content Binding: Common Languages (preferred) Max Binding: All Languages
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!I | 0..* | Extension | Extensions that cannot be ignored
url | ΣI | 0..1 | uri | Canonical identifier for this plan definition, represented as a URI (globally unique)
identifier | ΣI | 0..* | Identifier | Additional identifier for the plan definition
version | ΣI | 0..1 | string | Business version of the plan definition
name | ΣI | 0..1 | string | Name for this plan definition (computer friendly)
title | ΣI | 0..1 | string | Name for this plan definition (human friendly)
subtitle | I | 0..1 | string | Subordinate title of the plan definition
type | ΣI | 0..1 | CodeableConcept | order-set | clinical-protocol | eca-rule | workflow-definition Binding: PlanDefinitionType (extensible)
status | ?!ΣI | 1..1 | code | draft | active | retired | unknown Binding: PublicationStatus (required)
experimental | ΣI | 0..1 | boolean | For testing purposes, not real usage
subject[x] | I | 0..1 | | Type of individual the plan definition is focused on Binding: Subject type (extensible)
subjectCodeableConcept | | | CodeableConcept
subjectReference | | | Reference(group.html)
date | ΣI | 0..1 | dateTime | Date last changed
publisher | ΣI | 0..1 | string | Name of the publisher (organization or individual)

contact | ΣI | 0..* | ContactDetail | Contact details for the publisher
description | ΣI | 0..1 | markdown | Natural language description of the plan definition
useContext | ΣI | 0..* | UsageContext | The context that the content is intended to support
jurisdiction | ΣI | 0..* | CodeableConcept | Intended jurisdiction for plan definition (if applicable) Binding: Jurisdiction (extensible)
purpose | I | 0..1 | markdown | Why this plan definition is defined
usage | I | 0..1 | string | Describes the clinical usage of the plan
copyright | I | 0..1 | markdown | Use and/or publishing restrictions
approvalDate | I | 0..1 | date | When the plan definition was approved by publisher
lastReviewDate | I | 0..1 | date | When the plan definition was last reviewed
effectivePeriod | ΣI | 0..1 | Period | When the plan definition is expected to be used
topic | I | 0..* | CodeableConcept | E.g. Education, Treatment, Assessment Binding: DefinitionTopic (example)
author | I | 0..* | ContactDetail | Who authored the content
editor | I | 0..* | ContactDetail | Who edited the content
reviewer | I | 0..* | ContactDetail | Who reviewed the content
endorser | I | 0..* | ContactDetail | Who endorsed the content
relatedArtifact | I | 0..* | RelatedArtifact | Additional documentation, citations
library | I | 1..1 | canonical(Library) | Logic used by the plan definition
goal | I | 0..* | BackboneElement | What the plan is trying to accomplish
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
category | I | 0..1 | CodeableConcept | E.g. Treatment, dietary, behavioral Binding: Goal category (example)
description | I | 1..1 | CodeableConcept | Code or text describing the goal Binding: SNOMED CT Clinical Findings (example)
priority | I | 0..1 | CodeableConcept | high-priority | medium-priority | low-priority Binding: Goal priority (preferred)
start | I | 0..1 | CodeableConcept | When goal pursuit begins Binding: Goal start event (example)
addresses | I | 0..* | CodeableConcept | What does the goal address Binding: Condition/Problem/Diagnosis Codes (example)
documentation | I | 0..* | RelatedArtifact | Supporting documentation for the goal
target | I | 0..* | BackboneElement | Target outcome for the goal
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
measure | I | 0..1 | CodeableConcept | The parameter whose value is to be tracked Binding: LOINC Codes (example)
detail[x] | I | 0..1 | | The target value to be achieved
detailQuantity | | | Quantity
detailRange | | | Range
detailCodeableConcept | | | CodeableConcept
due | I | 0..1 | Duration | Reach goal within
action | I | 0..* | BackboneElement | Action defined by the plan
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
prefix | I | 0..1 | string | User-visible prefix for the action (e.g. 1. or A.)
title | I | 0..1 | string | User-visible title
description | I | 0..1 | string | Brief description of the action
textEquivalent | I | 0..1 | string | Static text equivalent of the action, used if the dynamic aspects cannot be interpreted by the receiving system
priority | I | 0..1 | code | routine | urgent | asap | stat Binding: Request priority (required)
code | I | 0..* | CodeableConcept | Code representing the meaning of the action or sub-actions
reason | I | 0..* | CodeableConcept | Why the action should be performed
documentation | I | 0..* | RelatedArtifact | Supporting documentation for the intended performer of the action
goalId | I | 0..* | id | What goals this action supports
subject[x] | I | 0..1 | | Type of individual the action is focused on Binding: Subject type (extensible)
subjectCodeableConcept | | | CodeableConcept
subjectReference | | | Reference(group.html)
trigger | I | 0..* | TriggerDefinition | When the action should be triggered
condition | I | 0..* | BackboneElement | Whether or not the action is applicable
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
kind | I | 1..1 | code | applicability | start | stop Binding: ActionConditionKind (required)
expression | I | 0..1 | Expression | Boolean-valued expression
input | I | 0..* | DataRequirement | Input data requirements
output | I | 0..* | DataRequirement | Output data definition
relatedAction | I | 0..* | BackboneElement | Relationship to another action
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
actionId | I | 1..1 | id | What action is this related to
relationship | I | 1..1 | code | before-start | before | before-end | concurrent-with-start | concurrent | concurrent-with-end | after-start | after | after-end Binding: ActionRelationshipType (required)
offset[x] | I | 0..1 | | Time offset for the relationship
offsetDuration | | | Duration

offsetRange | | | Range
timing[x] | I | 0..1 | | When the action should take place
timingDateTime | | | dateTime
timingAge | | | Age
timingPeriod | | | Period
timingDuration | | | Duration
timingRange | | | Range
timingTiming | | | Timing
participant | I | 0..* | BackboneElement | Who should participate in the action
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
type | I | 1..1 | code | patient | practitioner | related-person | device Binding: ActionParticipantType (required)
role | I | 0..1 | CodeableConcept | E.g. Nurse, Surgeon, Parent Binding: Action participant role (example)
type | I | 0..1 | CodeableConcept | create | update | remove | fire-event Binding: ActionType (extensible)
groupingBehavior | I | 0..1 | code | visual-group | logical-group | sentence-group Binding: ActionGroupingBehavior (required)
selectionBehavior | I | 0..1 | code | any | all | all-or-none | exactly-one | at-most-one | one-or-more Binding: ActionSelectionBehavior (required)
requiredBehavior | I | 0..1 | code | must | could | must-unless-documented Binding: ActionRequiredBehavior (required)
precheckBehavior | I | 0..1 | code | yes | no Binding: ActionPrecheckBehavior (required)
cardinalityBehavior | I | 0..1 | code | single | multiple Binding: ActionCardinalityBehavior (required)
definition[x] | I | 0..1 | | Description of the activity to be performed
definitionCanonical | | | canonical(activitydefinition.html | plandefinition.html | questionnaire.html)
definitionUri | | | uri
transform | I | 0..1 | canonical(StructureMap) | Transform to apply the template
dynamicValue | I | 0..* | BackboneElement | Dynamic aspects of the definition
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
path | I | 0..1 | string | The path to the element to be set dynamically
expression | I | 0..1 | Expression | An expression that provides the dynamic value for the customization
action | I | 0..* | | Unknown reference to #PlanDefinition.action A sub-action
Documentation for this format

XML Template

```
<!-- Computable PlanDefinition -->

<PlanDefinition xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id><!-- 0..1 string Logical id of this artifact --></id>
 <meta><!--  0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!--  0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!--  0..1 Language of the resource content   -->
 <text><!--  0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <url value="[uri]"/><!--  0..1
     Canonical identifier for this plan definition, represented as a URI (globally unique) -->
 <identifier><!--  0..* Identifier
     Additional identifier for the plan definition --></identifier>
 <version value="[string]"/><!--  0..1 Business version of the plan definition -->
 <name value="[string]"/><!--  0..1 Name for this plan definition (computer friendly) -->
 <title value="[string]"/><!--  0..1 Name for this plan definition (human friendly) -->
 <subtitle value="[string]"/><!--  0..1 Subordinate title of the plan definition -->
 <type><!--  0..1 CodeableConcept order-set | clinical-protocol | eca-rule | workflow-definition   --></type>
 <status value="[code]"/><!--  1..1 draft | active | retired | unknown   -->
 <experimental value="[boolean]"/><!--  0..1 For testing purposes, not real usage -->
 <subject[x]><!--  0..1 CodeableConcept|Reference(Group) Type of individual the plan definition is focused on
    --></subject[x]>
 <date value="[dateTime]"/><!--  0..1 Date last changed -->
 <publisher value="[string]"/><!--  0..1 Name of the publisher (organization or individual) -->
 <contact><!--  0..* ContactDetail
     Contact details for the publisher --></contact>
 <description value="[markdown]"/><!--  0..1 Natural language description of the plan definition -->
 <useContext><!--  0..* UsageContext
     The context that the content is intended to support --></useContext>
 <jurisdiction><!--  0..* CodeableConcept Intended jurisdiction for plan definition (if applicable)   --></jurisdiction>
 <purpose value="[markdown]"/><!--  0..1 Why this plan definition is defined -->
 <usage value="[string]"/><!--  0..1 Describes the clinical usage of the plan -->
 <copyright value="[markdown]"/><!--  0..1 Use and/or publishing restrictions -->
 <approvalDate value="[date]"/><!--  0..1 When the plan definition was approved by publisher -->
 <lastReviewDate value="[date]"/><!--  0..1 When the plan definition was last reviewed -->
 <effectivePeriod><!--  0..1 Period
     When the plan definition is expected to be used --></effectivePeriod>
 <topic><!--  0..* CodeableConcept E.g. Education, Treatment, Assessment   --></topic>
 <author><!--  0..* ContactDetail Who authored the content --></author>
 <editor><!--  0..* ContactDetail Who edited the content --></editor>
 <reviewer><!--  0..* ContactDetail Who reviewed the content --></reviewer>
 <endorser><!--  0..* ContactDetail Who endorsed the content --></endorser>
```

```
<relatedArtifact><!--  0..* RelatedArtifact
    Additional documentation, citations --></relatedArtifact>
<library value="[canonical]"/><!--  1..1 Logic used by the plan definition -->
<goal>  0..* BackboneElement  <!--  0..* What the plan is trying to accomplish -->
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
   --></modifierExtension>
 <category><!--  0..1 CodeableConcept E.g. Treatment, dietary, behavioral   --></category>
 <description><!--  1..1 CodeableConcept Code or text describing the goal   --></description>
 <priority><!--  0..1 CodeableConcept high-priority | medium-priority | low-priority   --></priority>
 <start><!--  0..1 CodeableConcept When goal pursuit begins   --></start>
 <addresses><!--  0..* CodeableConcept What does the goal address   --></addresses>
 <documentation><!--  0..* RelatedArtifact
     Supporting documentation for the goal --></documentation>
 <target>  0..* BackboneElement  <!--  0..* Target outcome for the goal -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
   --></modifierExtension>
  <measure><!--  0..1 CodeableConcept The parameter whose value is to be tracked   --></measure>
  <detail[x]><!--  0..1 Quantity|Range|
    CodeableConcept The target value to be achieved --></detail[x]>
  <due><!--  0..1 Duration
      Reach goal within --></due>
 </target>
</goal>
<action>  0..* BackboneElement  <!--  0..* Action defined by the plan -->
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
   --></modifierExtension>
 <prefix value="[string]"/><!--  0..1 User-visible prefix for the action (e.g. 1. or A.) -->
 <title value="[string]"/><!--  0..1 User-visible title -->
 <description value="[string]"/><!--  0..1 Brief description of the action -->
 <textEquivalent value="[string]"/><!--  0..1
     Static text equivalent of the action, used if the dynamic aspects cannot be interpreted by the receiving system
   -->
 <priority value="[code]"/><!--  0..1 routine | urgent | asap | stat   -->
 <code><!--  0..* CodeableConcept
     Code representing the meaning of the action or sub-actions --></code>
 <reason><!--  0..* CodeableConcept
     Why the action should be performed --></reason>
 <documentation><!--  0..* RelatedArtifact
     Supporting documentation for the intended performer of the action --></documentation>
 <goalId value="[id]"/><!--  0..* What goals this action supports -->
 <subject[x]><!--  0..1 CodeableConcept|Reference(Group) Type of individual the action is focused on   --></subject[x]>
 <trigger><!--  0..* TriggerDefinition
     When the action should be triggered --></trigger>
 <condition>  0..* BackboneElement  <!--  0..* Whether or not the action is applicable -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
   --></modifierExtension>
  <kind value="[code]"/><!--  1..1 applicability | start | stop   -->
  <expression><!--  0..1 Expression
      Boolean-valued expression --></expression>
 </condition>
 <input><!--  0..* DataRequirement
     Input data requirements --></input>
 <output><!--  0..* DataRequirement
     Output data definition --></output>
 <relatedAction>  0..* BackboneElement  <!--  0..* Relationship to another action -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
   --></modifierExtension>
  <actionId value="[id]"/><!--  1..1 What action is this related to -->
  <relationship value="[code]"/><!--  1..1 before-start | before | before-end | concurrent-with-start | concurrent |
   concurrent-with-end | after-start | after | after-end   -->
  <offset[x]><!--  0..1 Duration|Range
      Time offset for the relationship --></offset[x]>
 </relatedAction>
 <timing[x]><!--  0..1 dateTime|Age|Period|Duration|Range|
   Timing When the action should take place --></timing[x]>
 <participant>  0..* BackboneElement  <!--  0..* Who should participate in the action -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
   --></modifierExtension>
  <type value="[code]"/><!--  1..1 patient | practitioner | related-person | device   -->
  <role><!--  0..1 CodeableConcept E.g. Nurse, Surgeon, Parent   --></role>
 </participant>
 <type><!--  0..1 CodeableConcept create | update | remove | fire-event   --></type>
 <groupingBehavior value="[code]"/><!--  0..1 visual-group | logical-group | sentence-group   -->
 <selectionBehavior value="[code]"/><!--  0..1 any | all | all-or-none | exactly-one | at-most-one | one-or-more   -->
 <requiredBehavior value="[code]"/><!--  0..1 must | could | must-unless-documented   -->
 <precheckBehavior value="[code]"/><!--  0..1 yes | no   -->
 <cardinalityBehavior value="[code]"/><!--  0..1 single | multiple   -->
 <definition[x]><!--  0..1 canonical|uri
     Description of the activity to be performed --></definition[x]>
 <transform value="[canonical]"/><!--  0..1 Transform to apply the template -->
 <dynamicValue>  0..* BackboneElement  <!--  0..* Dynamic aspects of the definition -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
   --></modifierExtension>
  <path value="[string]"/><!--  0..1 The path to the element to be set dynamically -->
  <expression><!--  0..1 Expression
      An expression that provides the dynamic value for the customization --></expression>
 </dynamicValue>
 <action><!-- See #PlanDefinition.action  A sub-action --></action>
</action>
```

```
</PlanDefinition>
```

JSON Template

```
{ // Computable PlanDefinition
  // from Element: extension
    "meta" : { Meta }, // C?Metadata about the resource
    "implicitRules" : "<uri>", // C?A set of rules under which this content was created
    "language" : "<code>", // C?Language of the resource content
    "text" : { Narrative }, // C?Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored
    "url" : "<uri>", // C?Canonical identifier for this plan definition, represented as a URI (globally unique)
    "identifier" : [{ Identifier }], // C?Additional identifier for the plan definition
    "version" : "<string>", // C?Business version of the plan definition
    "name" : "<string>", // C?Name for this plan definition (computer friendly)
    "title" : "<string>", // C?Name for this plan definition (human friendly)
    "subtitle" : "<string>", // C?Subordinate title of the plan definition
    "type" : { CodeableConcept }, // C?order-set | clinical-protocol | eca-rule | workflow-definition
    "status" : "<code>", // C? R! draft | active | retired | unknown
    "experimental" : <boolean>, // C?For testing purposes, not real usage
// value[x]: Type of individual the plan definition is focused on. One of these 2:
    "subjectCodeableConcept" : { CodeableConcept }, // C?Type of individual the plan definition is focused on
    "subjectReference" : { Reference(Group) }, // C?Type of individual the plan definition is focused on
    "date" : "<dateTime>", // C?Date last changed
    "publisher" : "<string>", // C?Name of the publisher (organization or individual)
    "contact" : [{ ContactDetail }], // C?Contact details for the publisher
    "description" : "<markdown>", // C?Natural language description of the plan definition
    "useContext" : [{ UsageContext }], // C?The context that the content is intended to support
    "jurisdiction" : [{ CodeableConcept }], // C?Intended jurisdiction for plan definition (if applicable)
    "purpose" : "<markdown>", // C?Why this plan definition is defined
    "usage" : "<string>", // C?Describes the clinical usage of the plan
    "copyright" : "<markdown>", // C?Use and/or publishing restrictions
    "approvalDate" : "<date>", // C?When the plan definition was approved by publisher
    "lastReviewDate" : "<date>", // C?When the plan definition was last reviewed
    "effectivePeriod" : { Period }, // C?When the plan definition is expected to be used
    "topic" : [{ CodeableConcept }], // C?E.g. Education, Treatment, Assessment
    "author" : [{ ContactDetail }], // C?Who authored the content
    "editor" : [{ ContactDetail }], // C?Who edited the content
    "reviewer" : [{ ContactDetail }], // C?Who reviewed the content
    "endorser" : [{ ContactDetail }], // C?Who endorsed the content
    "relatedArtifact" : [{ RelatedArtifact }], // C?Additional documentation, citations
    "library" : "<canonical>", // C? R! Logic used by the plan definition
    "goal" : [{ BackboneElement }] // C?What the plan is trying to accomplish
    "action" : [{ BackboneElement }], // C?Action defined by the plan
  }
```

Alternate definitions: Master Definition (XML, JSON), Schematron

12.18.12.2.2 Terminology Bindings

Path	Name	Conformance	ValueSet
PlanDefinition.language	Common Languages	preferred	Common Languages
PlanDefinition.type	PlanDefinitionType	extensible	PlanDefinitionType
PlanDefinition.status	PublicationStatus	required	PublicationStatus
PlanDefinition.subject[x]	Subject type	extensible	Subject type
PlanDefinition.jurisdiction	Jurisdiction	extensible	Jurisdiction
PlanDefinition.topic	DefinitionTopic	example	DefinitionTopic
PlanDefinition.goal.category	Goal category	example	Goal category
PlanDefinition.goal.description	SNOMED CT Clinical Findings	example	SNOMED CT Clinical Findings
PlanDefinition.goal.priority	Goal priority	preferred	Goal priority
PlanDefinition.goal.start	Goal start event	example	Goal start event
PlanDefinition.goal.addresses	Condition/Problem/Diagnosis Codes	example	Condition/Problem/Diagnosis Codes
PlanDefinition.goal.target.measure	LOINC Codes	example	LOINC Codes
PlanDefinition.action.priority	Request priority	required	Request priority
PlanDefinition.action.subject[x]	Subject type	extensible	Subject type
PlanDefinition.action.condition.kind	ActionConditionKind	required	ActionConditionKind
PlanDefinition.action.relatedAction.relationship	ActionRelationshipType	required	ActionRelationshipType
PlanDefinition.action.participant.type	ActionParticipantType	required	ActionParticipantType
PlanDefinition.action.participant.role	Action participant role	example	Action participant role
PlanDefinition.action.type	ActionType	extensible	ActionType
PlanDefinition.action.groupingBehavior	ActionGroupingBehavior	required	ActionGroupingBehavior
PlanDefinition.action.selectionBehavior	ActionSelectionBehavior	required	ActionSelectionBehavior
PlanDefinition.action.requiredBehavior	ActionRequiredBehavior	required	ActionRequiredBehavior
PlanDefinition.action.precheckBehavior	ActionPrecheckBehavior	required	ActionPrecheckBehavior
PlanDefinition.action.cardinalityBehavior	ActionCardinalityBehavior	required	ActionCardinalityBehavior

12.18.13 CDS Hooks Service PlanDefinition

URL: https://hl7.org/fhir/R4/plandefinition-cdshooks-service.html

Defines a PlanDefinition that implements the behavior for a CDS Hooks service

12.18.13 CDS Hooks Service PlanDefinition

Clinical Decision Support Work Group | Maturity Level: N/A | Standards Status: Informative

12.18.13.1 Content

Profiles:
CDSHooksServicePlanDefinition | Defines a PlanDefinition that implements the behavior for a CDS Hooks service

12.18.13.2 StructureDefinition: CDSHooksServicePlanDefinition

URL: https://hl7.org/fhir/R4/cdshooksserviceplandefinition.html

12.18.13.2 StructureDefinition: CDSHooksServicePlanDefinition

Clinical Decision Support Work Group | Maturity Level: 1 | Trial Use | Use Context: Any

The official URL for this profile is:

`http://hl7.org/fhir/StructureDefinition/cdshooksserviceplandefinition`

Defines a PlanDefinition that implements the behavior for a CDS Hooks service

This profile was published on [no date] as a draft by HL7.

12.18.13.2.1 Formal Views of Profile Content

Description of Profiles, Differentials, Snapshots, and how the XML and JSON presentations work.

• Text Summary
• Differential Table
• Snapshot Table
• XML Template
• JSON Template
• All

This structure is derived from PlanDefinition.

Summary

Mandatory: 1 element Must-Support: 10 elements

Extensions

This structure refers to these extensions:

• http://hl7.org/fhir/StructureDefinition/cqf-cdsHooksEndpoint

This structure is derived from PlanDefinition.

Name | Flags | Card. | Type | Description & Constraints
PlanDefinition | | 1..1
cqf-cdsHooksEndpoint | | 1..1 | uri | Service endpoint URL: http://hl7.org/fhir/StructureDefinition/cqf-cdsHooksEndpoint
action | | 0..* | BackboneElement
title | S | 0..1 | string
description | S | 0..1 | string
priority | S | 0..1 | code
trigger | S | 0..* | TriggerDefinition
condition | S | 0..* | BackboneElement
timing[x] | S | 0..1 | dateTime, Age, Period, Duration, Range, Timing
participant | S | 0..* | BackboneElement
type | S | 0..1 | CodeableConcept
selectionBehavior | S | 0..1 | code
definition[x] | S | 0..1 | canonical(ActivityDefinition | PlanDefinition | Questionnaire), uri
Documentation for this format

Name | Flags | Card. | Type | Description & Constraints
PlanDefinition | I | 1..1 | | The definition of a plan for a series of actions, independent of any specific patient or context
id | Σ | 0..1 | string | Logical id of this artifact
meta | ΣI | 0..1 | Meta | Metadata about the resource
implicitRules | ?!ΣI | 0..1 | uri | A set of rules under which this content was created
language | I | 0..1 | code | Language of the resource content Binding: Common Languages (preferred) Max Binding: All Languages
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | I | 0..* | Extension | Extension Slice: Unordered, Open by value:url
cqf-cdsHooksEndpoint | I | 1..1 | uri | Service endpoint URL: http://hl7.org/fhir/StructureDefinition/cqf-cdsHooksEndpoint
modifierExtension | ?!I | 0..* | Extension | Extensions that cannot be ignored
url | ΣI | 0..1 | uri | Canonical identifier for this plan definition, represented as a URI (globally unique)
identifier | ΣI | 0..* | Identifier | Additional identifier for the plan definition
version | ΣI | 0..1 | string | Business version of the plan definition

name | ΣI | 0..1 | string | Name for this plan definition (computer friendly)
title | ΣI | 0..1 | string | Name for this plan definition (human friendly)
subtitle | I | 0..1 | string | Subordinate title of the plan definition
type | ΣI | 0..1 | CodeableConcept | order-set | clinical-protocol | eca-rule | workflow-definition Binding: PlanDefinitionType (extensible)
status | ?!ΣI | 1..1 | code | draft | active | retired | unknown Binding: PublicationStatus (required)
experimental | ΣI | 0..1 | boolean | For testing purposes, not real usage
subject[x] | I | 0..1 | | Type of individual the plan definition is focused on Binding: Subject type (extensible)
subjectCodeableConcept | | | CodeableConcept
subjectReference | | | Reference(group.html)
date | ΣI | 0..1 | dateTime | Date last changed
publisher | ΣI | 0..1 | string | Name of the publisher (organization or individual)
contact | ΣI | 0..* | ContactDetail | Contact details for the publisher
description | ΣI | 0..1 | markdown | Natural language description of the plan definition
useContext | ΣI | 0..* | UsageContext | The context that the content is intended to support
jurisdiction | ΣI | 0..* | CodeableConcept | Intended jurisdiction for plan definition (if applicable) Binding: Jurisdiction (extensible)
purpose | I | 0..1 | markdown | Why this plan definition is defined
usage | I | 0..1 | string | Describes the clinical usage of the plan
copyright | I | 0..1 | markdown | Use and/or publishing restrictions
approvalDate | I | 0..1 | date | When the plan definition was approved by publisher
lastReviewDate | I | 0..1 | date | When the plan definition was last reviewed
effectivePeriod | ΣI | 0..1 | Period | When the plan definition is expected to be used
topic | I | 0..* | CodeableConcept | E.g. Education, Treatment, Assessment Binding: DefinitionTopic (example)
author | I | 0..* | ContactDetail | Who authored the content
editor | I | 0..* | ContactDetail | Who edited the content
reviewer | I | 0..* | ContactDetail | Who reviewed the content
endorser | I | 0..* | ContactDetail | Who endorsed the content
relatedArtifact | I | 0..* | RelatedArtifact | Additional documentation, citations
library | I | 0..* | canonical(Library) | Logic used by the plan definition
goal | I | 0..* | BackboneElement | What the plan is trying to accomplish
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
category | I | 0..1 | CodeableConcept | E.g. Treatment, dietary, behavioral Binding: Goal category (example)
description | I | 1..1 | CodeableConcept | Code or text describing the goal Binding: SNOMED CT Clinical Findings (example)
priority | I | 0..1 | CodeableConcept | high-priority | medium-priority | low-priority Binding: Goal priority (preferred)
start | I | 0..1 | CodeableConcept | When goal pursuit begins Binding: Goal start event (example)
addresses | I | 0..* | CodeableConcept | What does the goal address Binding: Condition/Problem/Diagnosis Codes (example)
documentation | I | 0..* | RelatedArtifact | Supporting documentation for the goal
target | I | 0..* | BackboneElement | Target outcome for the goal
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
measure | I | 0..1 | CodeableConcept | The parameter whose value is to be tracked Binding: LOINC Codes (example)
detail[x] | I | 0..1 | | The target value to be achieved
detailQuantity | | | Quantity
detailRange | | | Range
detailCodeableConcept | | | CodeableConcept
due | I | 0..1 | Duration | Reach goal within
action | I | 0..* | BackboneElement | Action defined by the plan
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
prefix | I | 0..1 | string | User-visible prefix for the action (e.g. 1. or A.)
title | SI | 0..1 | string | User-visible title
description | SI | 0..1 | string | Brief description of the action
textEquivalent | I | 0..1 | string | Static text equivalent of the action, used if the dynamic aspects cannot be interpreted by the receiving system
priority | SI | 0..1 | code | routine | urgent | asap | stat Binding: Request priority (required)
code | I | 0..* | CodeableConcept | Code representing the meaning of the action or sub-actions
reason | I | 0..* | CodeableConcept | Why the action should be performed
documentation | I | 0..* | RelatedArtifact | Supporting documentation for the intended performer of the action
goalId | I | 0..* | id | What goals this action supports
subject[x] | I | 0..1 | | Type of individual the action is focused on Binding: Subject type (extensible)
subjectCodeableConcept | | | CodeableConcept
subjectReference | | | Reference(group.html)
trigger | SI | 0..* | TriggerDefinition | When the action should be triggered
condition | SI | 0..* | BackboneElement | Whether or not the action is applicable
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
kind | I | 1..1 | code | applicability | start | stop Binding: ActionConditionKind (required)

expression | I | 0..1 | Expression | Boolean-valued expression
input | I | 0..* | DataRequirement | Input data requirements
output | I | 0..* | DataRequirement | Output data definition
relatedAction | I | 0..* | BackboneElement | Relationship to another action
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
actionId | I | 1..1 | id | What action is this related to
relationship | I | 1..1 | code | before-start | before | before-end | concurrent-with-start | concurrent | concurrent-with-end | after-start | after | after-end Binding: ActionRelationshipType (required)
offset[x] | I | 0..1 | | Time offset for the relationship
offsetDuration | | | Duration
offsetRange | | | Range
timing[x] | SI | 0..1 | | When the action should take place
timingDateTime | | | dateTime
timingAge | | | Age
timingPeriod | | | Period
timingDuration | | | Duration
timingRange | | | Range
timingTiming | | | Timing
participant | SI | 0..* | BackboneElement | Who should participate in the action
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
type | I | 1..1 | code | patient | practitioner | related-person | device Binding: ActionParticipantType (required)
role | I | 0..1 | CodeableConcept | E.g. Nurse, Surgeon, Parent Binding: Action participant role (example)
type | SI | 0..1 | CodeableConcept | create | update | remove | fire-event Binding: ActionType (extensible)
groupingBehavior | I | 0..1 | code | visual-group | logical-group | sentence-group Binding: ActionGroupingBehavior (required)
selectionBehavior | SI | 0..1 | code | any | all | all-or-none | exactly-one | at-most-one | one-or-more Binding: ActionSelectionBehavior (required)
requiredBehavior | I | 0..1 | code | must | could | must-unless-documented Binding: ActionRequiredBehavior (required)
precheckBehavior | I | 0..1 | code | yes | no Binding: ActionPrecheckBehavior (required)
cardinalityBehavior | I | 0..1 | code | single | multiple Binding: ActionCardinalityBehavior (required)
definition[x] | SI | 0..1 | | Description of the activity to be performed
definitionCanonical | | | canonical(activitydefinition.html | plandefinition.html | questionnaire.html)
definitionUri | | | uri
transform | I | 0..1 | canonical(StructureMap) | Transform to apply the template
dynamicValue | I | 0..* | BackboneElement | Dynamic aspects of the definition
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
path | I | 0..1 | string | The path to the element to be set dynamically
expression | I | 0..1 | Expression | An expression that provides the dynamic value for the customization
action | I | 0..* | | Unknown reference to #PlanDefinition.action A sub-action
Documentation for this format

```
<!-- CDS Hooks Service PlanDefinition -->

<PlanDefinition xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id><!-- 0..1 string Logical id of this artifact --></id>
 <meta><!--  0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!--  0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!--  0..1 Language of the resource content   -->
 <text><!--  0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <url value="[uri]"/><!--  0..1
     Canonical identifier for this plan definition, represented as a URI (globally unique) -->
 <identifier><!--  0..* Identifier
     Additional identifier for the plan definition --></identifier>
 <version value="[string]"/><!--  0..1 Business version of the plan definition -->
 <name value="[string]"/><!--  0..1 Name for this plan definition (computer friendly) -->
 <title value="[string]"/><!--  0..1 Name for this plan definition (human friendly) -->
 <subtitle value="[string]"/><!--  0..1 Subordinate title of the plan definition -->
 <type><!--  0..1 CodeableConcept order-set | clinical-protocol | eca-rule | workflow-definition   --></type>
 <status value="[code]"/><!--  1..1 draft | active | retired | unknown   -->
 <experimental value="[boolean]"/><!--  0..1 For testing purposes, not real usage -->
 <subject[x]><!--  0..1 CodeableConcept|Reference(Group) Type of individual the plan definition is focused on
    --></subject[x]>
 <date value="[dateTime]"/><!--  0..1 Date last changed -->
 <publisher value="[string]"/><!--  0..1 Name of the publisher (organization or individual) -->
 <contact><!--  0..* ContactDetail
     Contact details for the publisher --></contact>
 <description value="[markdown]"/><!--  0..1 Natural language description of the plan definition -->
 <useContext><!--  0..* UsageContext
     The context that the content is intended to support --></useContext>
```

```
<jurisdiction><!--  0..* CodeableConcept Intended jurisdiction for plan definition (if applicable)   --></jurisdiction>
<purpose value="[markdown]"/><!--  0..1 Why this plan definition is defined -->
<usage value="[string]"/><!--  0..1 Describes the clinical usage of the plan -->
<copyright value="[markdown]"/><!--  0..1 Use and/or publishing restrictions -->
<approvalDate value="[date]"/><!--  0..1 When the plan definition was approved by publisher -->
<lastReviewDate value="[date]"/><!--  0..1 When the plan definition was last reviewed -->
<effectivePeriod><!--  0..1 Period
    When the plan definition is expected to be used --></effectivePeriod>
<topic><!--  0..* CodeableConcept E.g. Education, Treatment, Assessment   --></topic>
<author><!--  0..* ContactDetail Who authored the content --></author>
<editor><!--  0..* ContactDetail Who edited the content --></editor>
<reviewer><!--  0..* ContactDetail Who reviewed the content --></reviewer>
<endorser><!--  0..* ContactDetail Who endorsed the content --></endorser>
<relatedArtifact><!--  0..* RelatedArtifact
    Additional documentation, citations --></relatedArtifact>
<library value="[canonical]"/><!--  0..* Logic used by the plan definition -->
<goal>  0..* BackboneElement  <!--  0..* What the plan is trying to accomplish -->
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
   --></modifierExtension>
 <category><!--  0..1 CodeableConcept E.g. Treatment, dietary, behavioral   --></category>
 <description><!--  1..1 CodeableConcept Code or text describing the goal   --></description>
 <priority><!--  0..1 CodeableConcept high-priority | medium-priority | low-priority   --></priority>
 <start><!--  0..1 CodeableConcept When goal pursuit begins   --></start>
 <addresses><!--  0..* CodeableConcept What does the goal address   --></addresses>
 <documentation><!--  0..* RelatedArtifact
     Supporting documentation for the goal --></documentation>
 <target>  0..* BackboneElement  <!--  0..* Target outcome for the goal -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
   --></modifierExtension>
  <measure><!--  0..1 CodeableConcept The parameter whose value is to be tracked   --></measure>
  <detail[x]><!--  0..1 Quantity|Range|
    CodeableConcept The target value to be achieved --></detail[x]>
  <due><!--  0..1 Duration
      Reach goal within --></due>
 </target>
</goal>
<action>  0..* BackboneElement  <!--  0..* Action defined by the plan -->
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
   --></modifierExtension>
 <prefix value="[string]"/><!--  0..1 User-visible prefix for the action (e.g. 1. or A.) -->
 <title value="[string]"/><!--  0..1 User-visible title -->
 <description value="[string]"/><!--  0..1 Brief description of the action -->
 <textEquivalent value="[string]"/><!--  0..1
     Static text equivalent of the action, used if the dynamic aspects cannot be interpreted by the receiving system
   -->
 <priority value="[code]"/><!--  0..1 routine | urgent | asap | stat   -->
 <code><!--  0..* CodeableConcept
     Code representing the meaning of the action or sub-actions --></code>
 <reason><!--  0..* CodeableConcept
     Why the action should be performed --></reason>
 <documentation><!--  0..* RelatedArtifact
     Supporting documentation for the intended performer of the action --></documentation>
 <goalId value="[id]"/><!--  0..* What goals this action supports -->
 <subject[x]><!--  0..1 CodeableConcept|Reference(Group) Type of individual the action is focused on   --></subject[x]>
 <trigger><!--  0..* TriggerDefinition
     When the action should be triggered --></trigger>
 <condition>  0..* BackboneElement  <!--  0..* Whether or not the action is applicable -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
   --></modifierExtension>
  <kind value="[code]"/><!--  1..1 applicability | start | stop   -->
  <expression><!--  0..1 Expression
      Boolean-valued expression --></expression>
 </condition>
 <input><!--  0..* DataRequirement
     Input data requirements --></input>
 <output><!--  0..* DataRequirement
     Output data definition --></output>
 <relatedAction>  0..* BackboneElement  <!--  0..* Relationship to another action -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
   --></modifierExtension>
  <actionId value="[id]"/><!--  1..1 What action is this related to -->
  <relationship value="[code]"/><!--  1..1 before-start | before | before-end | concurrent-with-start | concurrent |
   concurrent-with-end | after-start | after | after-end   -->
  <offset[x]><!--  0..1 Duration|Range
      Time offset for the relationship --></offset[x]>
 </relatedAction>
 <timing[x]><!--  0..1 dateTime|Age|Period|Duration|Range|
   Timing When the action should take place --></timing[x]>
 <participant>  0..* BackboneElement  <!--  0..* Who should participate in the action -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
   --></modifierExtension>
  <type value="[code]"/><!--  1..1 patient | practitioner | related-person | device   -->
  <role><!--  0..1 CodeableConcept E.g. Nurse, Surgeon, Parent   --></role>
 </participant>
 <type><!--  0..1 CodeableConcept create | update | remove | fire-event   --></type>
 <groupingBehavior value="[code]"/><!--  0..1 visual-group | logical-group | sentence-group   -->
 <selectionBehavior value="[code]"/><!--  0..1 any | all | all-or-none | exactly-one | at-most-one | one-or-more   -->
 <requiredBehavior value="[code]"/><!--  0..1 must | could | must-unless-documented   -->
 <precheckBehavior value="[code]"/><!--  0..1 yes | no   -->
```

```
  <cardinalityBehavior value="[code]"/><!--  0..1 single | multiple   -->
  <definition[x]><!--  0..1 canonical|uri
      Description of the activity to be performed --></definition[x]>
  <transform value="[canonical]"/><!--  0..1 Transform to apply the template -->
  <dynamicValue>  0..* BackboneElement  <!--  0..* Dynamic aspects of the definition -->
   <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
   <path value="[string]"/><!--  0..1 The path to the element to be set dynamically -->
   <expression><!--  0..1 Expression
       An expression that provides the dynamic value for the customization --></expression>
  </dynamicValue>
  <action><!-- See #PlanDefinition.action  A sub-action --></action>
 </action>
</PlanDefinition>
```

```
{ // CDS Hooks Service PlanDefinition
  // from Element: extension
    "meta" : { Meta }, // C?Metadata about the resource
    "implicitRules" : "<uri>", // C?A set of rules under which this content was created
    "language" : "<code>", // C?Language of the resource content
    "text" : { Narrative }, // C?Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "extension" : [ //  sliced by value:url  in the specified order, Open
      { // Service endpoint // C? R!
        // from Element: extension
      }
    ],
    "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored
    "url" : "<uri>", // C?Canonical identifier for this plan definition, represented as a URI (globally unique)
    "identifier" : [{ Identifier }], // C?Additional identifier for the plan definition
    "version" : "<string>", // C?Business version of the plan definition
    "name" : "<string>", // C?Name for this plan definition (computer friendly)
    "title" : "<string>", // C?Name for this plan definition (human friendly)
    "subtitle" : "<string>", // C?Subordinate title of the plan definition
    "type" : { CodeableConcept }, // C?order-set | clinical-protocol | eca-rule | workflow-definition
    "status" : "<code>", // C? R! draft | active | retired | unknown
    "experimental" : <boolean>, // C?For testing purposes, not real usage
// value[x]: Type of individual the plan definition is focused on. One of these 2:
    "subjectCodeableConcept" : { CodeableConcept }, // C?Type of individual the plan definition is focused on
    "subjectReference" : { Reference(Group) }, // C?Type of individual the plan definition is focused on
    "date" : "<dateTime>", // C?Date last changed
    "publisher" : "<string>", // C?Name of the publisher (organization or individual)
    "contact" : [{ ContactDetail }], // C?Contact details for the publisher
    "description" : "<markdown>", // C?Natural language description of the plan definition
    "useContext" : [{ UsageContext }], // C?The context that the content is intended to support
    "jurisdiction" : [{ CodeableConcept }], // C?Intended jurisdiction for plan definition (if applicable)
    "purpose" : "<markdown>", // C?Why this plan definition is defined
    "usage" : "<string>", // C?Describes the clinical usage of the plan
    "copyright" : "<markdown>", // C?Use and/or publishing restrictions
    "approvalDate" : "<date>", // C?When the plan definition was approved by publisher
    "lastReviewDate" : "<date>", // C?When the plan definition was last reviewed
    "effectivePeriod" : { Period }, // C?When the plan definition is expected to be used
    "topic" : [{ CodeableConcept }], // C?E.g. Education, Treatment, Assessment
    "author" : [{ ContactDetail }], // C?Who authored the content
    "editor" : [{ ContactDetail }], // C?Who edited the content
    "reviewer" : [{ ContactDetail }], // C?Who reviewed the content
    "endorser" : [{ ContactDetail }], // C?Who endorsed the content
    "relatedArtifact" : [{ RelatedArtifact }], // C?Additional documentation, citations
    "library" : ["<canonical>"], // C?Logic used by the plan definition
    "goal" : [{ BackboneElement }], // C?What the plan is trying to accomplish
    "action" : [{ BackboneElement }] // C?Action defined by the plan
  }
```

This structure is derived from PlanDefinition.

Summary

Mandatory: 1 element Must-Support: 10 elements

Extensions

This structure refers to these extensions:

• http://hl7.org/fhir/StructureDefinition/cqf-cdsHooksEndpoint

Differential View

This structure is derived from PlanDefinition.

Name	Flags	Card.	Type	Description & Constraints
PlanDefinition		1..1		
cqf-cdsHooksEndpoint		1..1	uri	Service endpoint URL: http://hl7.org/fhir/StructureDefinition/cqf-cdsHooksEndpoint
action		0..*	BackboneElement	
title	S	0..1	string	
description	S	0..1	string	
priority	S	0..1	code	
trigger	S	0..*	TriggerDefinition	
condition	S	0..*	BackboneElement	
timing[x]	S	0..1	dateTime, Age, Period, Duration, Range, Timing	
participant	S	0..*	BackboneElement	

type | S | 0..1 | CodeableConcept
selectionBehavior | S | 0..1 | code
definition[x] | S | 0..1 | canonical(ActivityDefinition | PlanDefinition | Questionnaire), uri
Documentation for this format

Snapshot View

Name | Flags | Card. | Type | Description & Constraints
PlanDefinition | I | 1..1 | | The definition of a plan for a series of actions, independent of any specific patient or context
id | Σ | 0..1 | string | Logical id of this artifact
meta | ΣI | 0..1 | Meta | Metadata about the resource
implicitRules | ?!ΣI | 0..1 | uri | A set of rules under which this content was created
language | I | 0..1 | code | Language of the resource content Binding: Common Languages (preferred) Max Binding: All Languages
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | I | 0..* | Extension | Extension Slice: Unordered, Open by value:url
cqf-cdsHooksEndpoint | I | 1..1 | uri | Service endpoint URL: http://hl7.org/fhir/StructureDefinition/cqf-cdsHooksEndpoint
modifierExtension | ?!I | 0..* | Extension | Extensions that cannot be ignored
url | ΣI | 0..1 | uri | Canonical identifier for this plan definition, represented as a URI (globally unique)
identifier | ΣI | 0..* | Identifier | Additional identifier for the plan definition
version | ΣI | 0..1 | string | Business version of the plan definition
name | ΣI | 0..1 | string | Name for this plan definition (computer friendly)
title | ΣI | 0..1 | string | Name for this plan definition (human friendly)
subtitle | I | 0..1 | string | Subordinate title of the plan definition
type | ΣI | 0..1 | CodeableConcept | order-set | clinical-protocol | eca-rule | workflow-definition Binding: PlanDefinitionType (extensible)
status | ?!ΣI | 1..1 | code | draft | active | retired | unknown Binding: PublicationStatus (required)
experimental | ΣI | 0..1 | boolean | For testing purposes, not real usage
subject[x] | I | 0..1 | | Type of individual the plan definition is focused on Binding: Subject type (extensible)
subjectCodeableConcept | | | CodeableConcept
subjectReference | | | Reference(group.html)
date | ΣI | 0..1 | dateTime | Date last changed
publisher | ΣI | 0..1 | string | Name of the publisher (organization or individual)
contact | ΣI | 0..* | ContactDetail | Contact details for the publisher
description | ΣI | 0..1 | markdown | Natural language description of the plan definition
useContext | ΣI | 0..* | UsageContext | The context that the content is intended to support
jurisdiction | ΣI | 0..* | CodeableConcept | Intended jurisdiction for plan definition (if applicable) Binding: Jurisdiction (extensible)
purpose | I | 0..1 | markdown | Why this plan definition is defined
usage | I | 0..1 | string | Describes the clinical usage of the plan
copyright | I | 0..1 | markdown | Use and/or publishing restrictions
approvalDate | I | 0..1 | date | When the plan definition was approved by publisher
lastReviewDate | I | 0..1 | date | When the plan definition was last reviewed
effectivePeriod | ΣI | 0..1 | Period | When the plan definition is expected to be used
topic | I | 0..* | CodeableConcept | E.g. Education, Treatment, Assessment Binding: DefinitionTopic (example)
author | I | 0..* | ContactDetail | Who authored the content
editor | I | 0..* | ContactDetail | Who edited the content
reviewer | I | 0..* | ContactDetail | Who reviewed the content
endorser | I | 0..* | ContactDetail | Who endorsed the content
relatedArtifact | I | 0..* | RelatedArtifact | Additional documentation, citations
library | I | 0..* | canonical(Library) | Logic used by the plan definition
goal | I | 0..* | BackboneElement | What the plan is trying to accomplish
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
category | I | 0..1 | CodeableConcept | E.g. Treatment, dietary, behavioral Binding: Goal category (example)
description | I | 1..1 | CodeableConcept | Code or text describing the goal Binding: SNOMED CT Clinical Findings (example)
priority | I | 0..1 | CodeableConcept | high-priority | medium-priority | low-priority Binding: Goal priority (preferred)
start | I | 0..1 | CodeableConcept | When goal pursuit begins Binding: Goal start event (example)
addresses | I | 0..* | CodeableConcept | What does the goal address Binding: Condition/Problem/Diagnosis Codes (example)
documentation | I | 0..* | RelatedArtifact | Supporting documentation for the goal
target | I | 0..* | BackboneElement | Target outcome for the goal
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
measure | I | 0..1 | CodeableConcept | The parameter whose value is to be tracked Binding: LOINC Codes (example)
detail[x] | I | 0..1 | | The target value to be achieved
detailQuantity | | | Quantity
detailRange | | | Range
detailCodeableConcept | | | CodeableConcept
due | I | 0..1 | Duration | Reach goal within
action | I | 0..* | BackboneElement | Action defined by the plan
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized

prefix | I | 0..1 | string | User-visible prefix for the action (e.g. 1. or A.)
title | SI | 0..1 | string | User-visible title
description | SI | 0..1 | string | Brief description of the action
textEquivalent | I | 0..1 | string | Static text equivalent of the action, used if the dynamic aspects cannot be interpreted by the receiving system
priority | SI | 0..1 | code | routine | urgent | asap | stat Binding: Request priority (required)
code | I | 0..* | CodeableConcept | Code representing the meaning of the action or sub-actions
reason | I | 0..* | CodeableConcept | Why the action should be performed
documentation | I | 0..* | RelatedArtifact | Supporting documentation for the intended performer of the action
goalId | I | 0..* | id | What goals this action supports
subject[x] | I | 0..1 | | Type of individual the action is focused on Binding: Subject type (extensible)
subjectCodeableConcept | | | CodeableConcept
subjectReference | | | Reference(group.html)
trigger | SI | 0..* | TriggerDefinition | When the action should be triggered
condition | SI | 0..* | BackboneElement | Whether or not the action is applicable
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
kind | I | 1..1 | code | applicability | start | stop Binding: ActionConditionKind (required)
expression | I | 0..1 | Expression | Boolean-valued expression
input | I | 0..* | DataRequirement | Input data requirements
output | I | 0..* | DataRequirement | Output data definition
relatedAction | I | 0..* | BackboneElement | Relationship to another action
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
actionId | I | 1..1 | id | What action is this related to
relationship | I | 1..1 | code | before-start | before | before-end | concurrent-with-start | concurrent | concurrent-with-end | after-start | after | after-end Binding: ActionRelationshipType (required)
offset[x] | I | 0..1 | | Time offset for the relationship
offsetDuration | | | Duration
offsetRange | | | Range
timing[x] | SI | 0..1 | | When the action should take place
timingDateTime | | | dateTime
timingAge | | | Age
timingPeriod | | | Period
timingDuration | | | Duration
timingRange | | | Range
timingTiming | | | Timing
participant | SI | 0..* | BackboneElement | Who should participate in the action
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
type | I | 1..1 | code | patient | practitioner | related-person | device Binding: ActionParticipantType (required)
role | I | 0..1 | CodeableConcept | E.g. Nurse, Surgeon, Parent Binding: Action participant role (example)
type | SI | 0..1 | CodeableConcept | create | update | remove | fire-event Binding: ActionType (extensible)
groupingBehavior | I | 0..1 | code | visual-group | logical-group | sentence-group Binding: ActionGroupingBehavior (required)
selectionBehavior | SI | 0..1 | code | any | all | all-or-none | exactly-one | at-most-one | one-or-more Binding: ActionSelectionBehavior (required)
requiredBehavior | I | 0..1 | code | must | could | must-unless-documented Binding: ActionRequiredBehavior (required)
precheckBehavior | I | 0..1 | code | yes | no Binding: ActionPrecheckBehavior (required)
cardinalityBehavior | I | 0..1 | code | single | multiple Binding: ActionCardinalityBehavior (required)
definition[x] | SI | 0..1 | | Description of the activity to be performed
definitionCanonical | | | canonical(activitydefinition.html | plandefinition.html | questionnaire.html)
definitionUri | | | uri
transform | I | 0..1 | canonical(StructureMap) | Transform to apply the template
dynamicValue | I | 0..* | BackboneElement | Dynamic aspects of the definition
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
path | I | 0..1 | string | The path to the element to be set dynamically
expression | I | 0..1 | Expression | An expression that provides the dynamic value for the customization
action | I | 0..* | | Unknown reference to #PlanDefinition.action A sub-action
Documentation for this format

XML Template

```
<!-- CDS Hooks Service PlanDefinition -->

<PlanDefinition xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id><!-- 0..1 string Logical id of this artifact --></id>
 <meta><!--  0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!--  0..1 A set of rules under which this content was created -->
```

```
<language value="[code]"/><!--  0..1 Language of the resource content   -->
<text><!--  0..1 Narrative
    Text summary of the resource, for human interpretation --></text>
<contained><!-- 0..* Resource Contained, inline Resources --></contained>
<modifierExtension><!--  0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
<url value="[uri]"/><!--  0..1
    Canonical identifier for this plan definition, represented as a URI (globally unique) -->
<identifier><!--  0..* Identifier
    Additional identifier for the plan definition --></identifier>
<version value="[string]"/><!--  0..1 Business version of the plan definition -->
<name value="[string]"/><!--  0..1 Name for this plan definition (computer friendly) -->
<title value="[string]"/><!--  0..1 Name for this plan definition (human friendly) -->
<subtitle value="[string]"/><!--  0..1 Subordinate title of the plan definition -->
<type><!--  0..1 CodeableConcept order-set | clinical-protocol | eca-rule | workflow-definition   --></type>
<status value="[code]"/><!--  1..1 draft | active | retired | unknown   -->
<experimental value="[boolean]"/><!--  0..1 For testing purposes, not real usage -->
<subject[x]><!--  0..1 CodeableConcept|Reference(Group) Type of individual the plan definition is focused on
   --></subject[x]>
<date value="[dateTime]"/><!--  0..1 Date last changed -->
<publisher value="[string]"/><!--  0..1 Name of the publisher (organization or individual) -->
<contact><!--  0..* ContactDetail
    Contact details for the publisher --></contact>
<description value="[markdown]"/><!--  0..1 Natural language description of the plan definition -->
<useContext><!--  0..* UsageContext
    The context that the content is intended to support --></useContext>
<jurisdiction><!--  0..* CodeableConcept Intended jurisdiction for plan definition (if applicable)   --></jurisdiction>
<purpose value="[markdown]"/><!--  0..1 Why this plan definition is defined -->
<usage value="[string]"/><!--  0..1 Describes the clinical usage of the plan -->
<copyright value="[markdown]"/><!--  0..1 Use and/or publishing restrictions -->
<approvalDate value="[date]"/><!--  0..1 When the plan definition was approved by publisher -->
<lastReviewDate value="[date]"/><!--  0..1 When the plan definition was last reviewed -->
<effectivePeriod><!--  0..1 Period
    When the plan definition is expected to be used --></effectivePeriod>
<topic><!--  0..* CodeableConcept E.g. Education, Treatment, Assessment   --></topic>
<author><!--  0..* ContactDetail Who authored the content --></author>
<editor><!--  0..* ContactDetail Who edited the content --></editor>
<reviewer><!--  0..* ContactDetail Who reviewed the content --></reviewer>
<endorser><!--  0..* ContactDetail Who endorsed the content --></endorser>
<relatedArtifact><!--  0..* RelatedArtifact
    Additional documentation, citations --></relatedArtifact>
<library value="[canonical]"/><!--  0..* Logic used by the plan definition -->
<goal>  0..* BackboneElement  <!--  0..* What the plan is trying to accomplish -->
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
   --></modifierExtension>
 <category><!--  0..1 CodeableConcept E.g. Treatment, dietary, behavioral   --></category>
 <description><!--  1..1 CodeableConcept Code or text describing the goal   --></description>
 <priority><!--  0..1 CodeableConcept high-priority | medium-priority | low-priority   --></priority>
 <start><!--  0..1 CodeableConcept When goal pursuit begins   --></start>
 <addresses><!--  0..* CodeableConcept What does the goal address   --></addresses>
 <documentation><!--  0..* RelatedArtifact
     Supporting documentation for the goal --></documentation>
 <target>  0..* BackboneElement  <!--  0..* Target outcome for the goal -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
   --></modifierExtension>
  <measure><!--  0..1 CodeableConcept The parameter whose value is to be tracked   --></measure>
  <detail[x]><!--  0..1 Quantity|Range|
    CodeableConcept The target value to be achieved --></detail[x]>
  <due><!--  0..1 Duration
      Reach goal within --></due>
 </target>
</goal>
<action>  0..* BackboneElement  <!--  0..* Action defined by the plan -->
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
   --></modifierExtension>
 <prefix value="[string]"/><!--  0..1 User-visible prefix for the action (e.g. 1. or A.) -->
 <title value="[string]"/><!--  0..1 User-visible title -->
 <description value="[string]"/><!--  0..1 Brief description of the action -->
 <textEquivalent value="[string]"/><!--  0..1
     Static text equivalent of the action, used if the dynamic aspects cannot be interpreted by the receiving system
   -->
 <priority value="[code]"/><!--  0..1 routine | urgent | asap | stat   -->
 <code><!--  0..* CodeableConcept
     Code representing the meaning of the action or sub-actions --></code>
 <reason><!--  0..* CodeableConcept
     Why the action should be performed --></reason>
 <documentation><!--  0..* RelatedArtifact
     Supporting documentation for the intended performer of the action --></documentation>
 <goalId value="[id]"/><!--  0..* What goals this action supports -->
 <subject[x]><!--  0..1 CodeableConcept|Reference(Group) Type of individual the action is focused on   --></subject[x]>
 <trigger><!--  0..* TriggerDefinition
     When the action should be triggered --></trigger>
 <condition>  0..* BackboneElement  <!--  0..* Whether or not the action is applicable -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
   --></modifierExtension>
  <kind value="[code]"/><!--  1..1 applicability | start | stop   -->
  <expression><!--  0..1 Expression
      Boolean-valued expression --></expression>
 </condition>
 <input><!--  0..* DataRequirement
```

```
      Input data requirements --></input>
  <output><!--  0..* DataRequirement
      Output data definition --></output>
  <relatedAction>  0..* BackboneElement  <!--  0..* Relationship to another action -->
   <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
   <actionId value="[id]"/><!--  1..1 What action is this related to -->
   <relationship value="[code]"/><!--  1..1 before-start | before | before-end | concurrent-with-start | concurrent |
    concurrent-with-end | after-start | after | after-end   -->
   <offset[x]><!--  0..1 Duration|Range
       Time offset for the relationship --></offset[x]>
  </relatedAction>
  <timing[x]><!--  0..1 dateTime|Age|Period|Duration|Range|
    Timing When the action should take place --></timing[x]>
  <participant>  0..* BackboneElement  <!--  0..* Who should participate in the action -->
   <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
   <type value="[code]"/><!--  1..1 patient | practitioner | related-person | device   -->
   <role><!--  0..1 CodeableConcept E.g. Nurse, Surgeon, Parent   --></role>
  </participant>
  <type><!--  0..1 CodeableConcept create | update | remove | fire-event   --></type>
  <groupingBehavior value="[code]"/><!--  0..1 visual-group | logical-group | sentence-group   -->
  <selectionBehavior value="[code]"/><!--  0..1 any | all | all-or-none | exactly-one | at-most-one | one-or-more   -->
  <requiredBehavior value="[code]"/><!--  0..1 must | could | must-unless-documented   -->
  <precheckBehavior value="[code]"/><!--  0..1 yes | no   -->
  <cardinalityBehavior value="[code]"/><!--  0..1 single | multiple   -->
  <definition[x]><!--  0..1 canonical|uri
      Description of the activity to be performed --></definition[x]>
  <transform value="[canonical]"/><!--  0..1 Transform to apply the template -->
  <dynamicValue>  0..* BackboneElement  <!--  0..* Dynamic aspects of the definition -->
   <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
   <path value="[string]"/><!--  0..1 The path to the element to be set dynamically -->
   <expression><!--  0..1 Expression
       An expression that provides the dynamic value for the customization --></expression>
  </dynamicValue>
  <action><!-- See #PlanDefinition.action  A sub-action --></action>
 </action>
</PlanDefinition>
```

JSON Template

```
{ // CDS Hooks Service PlanDefinition
  // from Element: extension
    "meta" : { Meta }, // C?Metadata about the resource
    "implicitRules" : "<uri>", // C?A set of rules under which this content was created
    "language" : "<code>", // C?Language of the resource content
    "text" : { Narrative }, // C?Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "extension" : [ //  sliced by value:url  in the specified order, Open
      { // Service endpoint // C? R!
        // from Element: extension
      }
    ],
    "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored
    "url" : "<uri>", // C?Canonical identifier for this plan definition, represented as a URI (globally unique)
    "identifier" : [{ Identifier }], // C?Additional identifier for the plan definition
    "version" : "<string>", // C?Business version of the plan definition
    "name" : "<string>", // C?Name for this plan definition (computer friendly)
    "title" : "<string>", // C?Name for this plan definition (human friendly)
    "subtitle" : "<string>", // C?Subordinate title of the plan definition
    "type" : { CodeableConcept }, // C?order-set | clinical-protocol | eca-rule | workflow-definition
    "status" : "<code>", // C? R! draft | active | retired | unknown
    "experimental" : <boolean>, // C?For testing purposes, not real usage
// value[x]: Type of individual the plan definition is focused on. One of these 2:
    "subjectCodeableConcept" : { CodeableConcept }, // C?Type of individual the plan definition is focused on
    "subjectReference" : { Reference(Group) }, // C?Type of individual the plan definition is focused on
    "date" : "<dateTime>", // C?Date last changed
    "publisher" : "<string>", // C?Name of the publisher (organization or individual)
    "contact" : [{ ContactDetail }], // C?Contact details for the publisher
    "description" : "<markdown>", // C?Natural language description of the plan definition
    "useContext" : [{ UsageContext }], // C?The context that the content is intended to support
    "jurisdiction" : [{ CodeableConcept }], // C?Intended jurisdiction for plan definition (if applicable)
    "purpose" : "<markdown>", // C?Why this plan definition is defined
    "usage" : "<string>", // C?Describes the clinical usage of the plan
    "copyright" : "<markdown>", // C?Use and/or publishing restrictions
    "approvalDate" : "<date>", // C?When the plan definition was approved by publisher
    "lastReviewDate" : "<date>", // C?When the plan definition was last reviewed
    "effectivePeriod" : { Period }, // C?When the plan definition is expected to be used
    "topic" : [{ CodeableConcept }], // C?E.g. Education, Treatment, Assessment
    "author" : [{ ContactDetail }], // C?Who authored the content
    "editor" : [{ ContactDetail }], // C?Who edited the content
    "reviewer" : [{ ContactDetail }], // C?Who reviewed the content
    "endorser" : [{ ContactDetail }], // C?Who endorsed the content
    "relatedArtifact" : [{ RelatedArtifact }], // C?Additional documentation, citations
    "library" : ["<canonical>"], // C?Logic used by the plan definition
    "goal" : [{ BackboneElement }], // C?What the plan is trying to accomplish
    "action" : [{ BackboneElement }] // C?Action defined by the plan
  }
```

Alternate definitions: Master Definition (XML, JSON), Schematron

12.18.13.2.2 Terminology Bindings

Path | Name | Conformance | ValueSet
PlanDefinition.language | Common Languages | preferred | Common Languages
PlanDefinition.type | PlanDefinitionType | extensible | PlanDefinitionType
PlanDefinition.status | PublicationStatus | required | PublicationStatus
PlanDefinition.subject[x] | Subject type | extensible | Subject type
PlanDefinition.jurisdiction | Jurisdiction | extensible | Jurisdiction
PlanDefinition.topic | DefinitionTopic | example | DefinitionTopic
PlanDefinition.goal.category | Goal category | example | Goal category
PlanDefinition.goal.description | SNOMED CT Clinical Findings | example | SNOMED CT Clinical Findings
PlanDefinition.goal.priority | Goal priority | preferred | Goal priority
PlanDefinition.goal.start | Goal start event | example | Goal start event
PlanDefinition.goal.addresses | Condition/Problem/Diagnosis Codes | example | Condition/Problem/Diagnosis Codes
PlanDefinition.goal.target.measure | LOINC Codes | example | LOINC Codes
PlanDefinition.action.priority | Request priority | required | Request priority
PlanDefinition.action.subject[x] | Subject type | extensible | Subject type
PlanDefinition.action.condition.kind | ActionConditionKind | required | ActionConditionKind
PlanDefinition.action.relatedAction.relationship | ActionRelationshipType | required | ActionRelationshipType
PlanDefinition.action.participant.type | ActionParticipantType | required | ActionParticipantType
PlanDefinition.action.participant.role | Action participant role | example | Action participant role
PlanDefinition.action.type | ActionType | extensible | ActionType
PlanDefinition.action.groupingBehavior | ActionGroupingBehavior | required | ActionGroupingBehavior
PlanDefinition.action.selectionBehavior | ActionSelectionBehavior | required | ActionSelectionBehavior
PlanDefinition.action.requiredBehavior | ActionRequiredBehavior | required | ActionRequiredBehavior
PlanDefinition.action.precheckBehavior | ActionPrecheckBehavior | required | ActionPrecheckBehavior
PlanDefinition.action.cardinalityBehavior | ActionCardinalityBehavior | required | ActionCardinalityBehavior

12.18.14 JSON Schema for PlanDefinition

URL: https://hl7.org/fhir/R4/plandefinition.schema.json.html

12.18.14 JSON Schema for PlanDefinition

Clinical Decision Support Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Not linked to any defined compartments

This is a representation of the json schema for PlanDefinition, which is just a part of the full JSON Schema.


```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/PlanDefinition",
  "$ref": "#/definitions/PlanDefinition",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "PlanDefinition": {
      "description": "This resource allows for the definition of various types of plans as a sharable, consumable, and
    executable artifact. The resource is general enough to support the description of a broad range of clinical
    artifacts such as clinical decision support rules, order sets and protocols.",
      "properties": {
        "resourceType": {
          "description": "This is a PlanDefinition resource",
          "const": "PlanDefinition"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
          "$ref": "#/definitions/code"
        },
        "_language": {
          "description": "Extensions for language",
          "$ref": "Element.schema.json#/definitions/Element"
```

```
    },
    "text": {
      "description": "A human-readable narrative that contains a summary of the resource and can be used to
represent the content of the resource to a human. The narrative need not encode all the structured data, but is
required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
      "$ref": "Narrative.schema.json#/definitions/Narrative"
    },
    "contained": {
      "description": "These resources do not have an independent existence apart from the resource that contains
them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
      "items": {
        "$ref": "ResourceList.schema.json#/definitions/ResourceList"
      },
      "type": "array"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
 definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource and that modifies the understanding of the element that contains it and/or the understanding of the
containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
met as part of the definition of the extension. Applications processing a resource are required to check for
modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "url": {
      "description": "An absolute URI that is used to identify this plan definition when it is referenced in a
specification, model, design or an instance; also called its canonical identifier. This SHOULD be globally unique
and SHOULD be a literal address at which at which an authoritative instance of this plan definition is (or will be)
published. This URL can be the target of a canonical reference. It SHALL remain the same when the plan definition is
 stored on different servers.",
      "$ref": "#/definitions/uri"
    },
    "_url": {
      "description": "Extensions for url",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "identifier": {
      "description": "A formal identifier that is used to identify this plan definition when it is represented in
other formats, or referenced in a specification, model, design or an instance.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "version": {
      "description": "The identifier that is used to identify this version of the plan definition when it is
referenced in a specification, model, design or instance. This is an arbitrary value managed by the plan definition
author and is not expected to be globally unique. For example, it might be a timestamp (e.g. yyyymmdd) if a managed
version is not available. There is also no expectation that versions can be placed in a lexicographical sequence. To
 provide a version consistent with the Decision Support Service specification, use the format Major.Minor.Revision
(e.g. 1.0.0). For more information on versioning knowledge assets, refer to the Decision Support Service
specification. Note that a version is required for non-experimental active artifacts.",
      "$ref": "#/definitions/string"
    },
    "_version": {
      "description": "Extensions for version",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "name": {
      "description": "A natural language name identifying the plan definition. This name should be usable as an
identifier for the module by machine processing applications such as code generation.",
      "$ref": "#/definitions/string"
    },
    "_name": {
      "description": "Extensions for name",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "title": {
      "description": "A short, descriptive, user-friendly title for the plan definition.",
      "$ref": "#/definitions/string"
    },
    "_title": {
      "description": "Extensions for title",
```

```
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "subtitle": {
      "description": "An explanatory or alternate title for the plan definition giving additional information about
its content.",
      "$ref": "#/definitions/string"
    },
    "_subtitle": {
      "description": "Extensions for subtitle",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "type": {
      "description": "A high-level category for the plan definition that distinguishes the kinds of systems that
would be interested in the plan definition.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "status": {
      "description": "The status of this plan definition. Enables tracking the life-cycle of the content.",
      "enum": [
        "draft",
        "active",
        "retired",
        "unknown"
      ]
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "experimental": {
      "description": "A Boolean value to indicate that this plan definition is authored for testing purposes (or
education/evaluation/marketing) and is not intended to be used for genuine usage.",
      "$ref": "#/definitions/boolean"
    },
    "_experimental": {
      "description": "Extensions for experimental",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "subjectCodeableConcept": {
      "description": "A code or group definition that describes the intended subject of the plan definition.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "subjectReference": {
      "description": "A code or group definition that describes the intended subject of the plan definition.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "date": {
      "description": "The date  (and optionally time) when the plan definition was published. The date must change
when the business version changes and it must change if the status code changes. In addition, it should change when
the substantive content of the plan definition changes.",
      "$ref": "#/definitions/dateTime"
    },
    "_date": {
      "description": "Extensions for date",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "publisher": {
      "description": "The name of the organization or individual that published the plan definition.",
      "$ref": "#/definitions/string"
    },
    "_publisher": {
      "description": "Extensions for publisher",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "contact": {
      "description": "Contact details to assist a user in finding and communicating with the publisher.",
      "items": {
        "$ref": "ContactDetail.schema.json#/definitions/ContactDetail"
      },
      "type": "array"
    },
    "description": {
      "description": "A free text natural language description of the plan definition from a consumer\u0027s
perspective.",
      "$ref": "#/definitions/markdown"
    },
    "_description": {
      "description": "Extensions for description",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "useContext": {
      "description": "The content was developed with a focus and intent of supporting the contexts that are listed.
These contexts may be general categories (gender, age, ...) or may be references to specific programs (insurance
plans, studies, ...) and may be used to assist with indexing and searching for appropriate plan definition
instances.",
      "items": {
        "$ref": "UsageContext.schema.json#/definitions/UsageContext"
      },
      "type": "array"
```

```
    },
    "jurisdiction": {
      "description": "A legal or geographic region in which the plan definition is intended to be used.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "purpose": {
      "description": "Explanation of why this plan definition is needed and why it has been designed as it has.",
      "$ref": "#/definitions/markdown"
    },
    "_purpose": {
      "description": "Extensions for purpose",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "usage": {
      "description": "A detailed description of how the plan definition is used from a clinical perspective.",
      "$ref": "#/definitions/string"
    },
    "_usage": {
      "description": "Extensions for usage",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "copyright": {
      "description": "A copyright statement relating to the plan definition and/or its contents. Copyright
statements are generally legal restrictions on the use and publishing of the plan definition.",
      "$ref": "#/definitions/markdown"
    },
    "_copyright": {
      "description": "Extensions for copyright",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "approvalDate": {
      "description": "The date on which the resource content was approved by the publisher. Approval happens once
when the content is officially approved for usage.",
      "$ref": "#/definitions/date"
    },
    "_approvalDate": {
      "description": "Extensions for approvalDate",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "lastReviewDate": {
      "description": "The date on which the resource content was last reviewed. Review happens periodically after
approval but does not change the original approval date.",
      "$ref": "#/definitions/date"
    },
    "_lastReviewDate": {
      "description": "Extensions for lastReviewDate",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "effectivePeriod": {
      "description": "The period during which the plan definition content was or is planned to be in active use.",
      "$ref": "Period.schema.json#/definitions/Period"
    },
    "topic": {
      "description": "Descriptive topics related to the content of the plan definition. Topics provide a high-level
categorization of the definition that can be useful for filtering and searching.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "author": {
      "description": "An individiual or organization primarily involved in the creation and maintenance of the
content.",
      "items": {
        "$ref": "ContactDetail.schema.json#/definitions/ContactDetail"
      },
      "type": "array"
    },
    "editor": {
      "description": "An individual or organization primarily responsible for internal coherence of the content.",
      "items": {
        "$ref": "ContactDetail.schema.json#/definitions/ContactDetail"
      },
      "type": "array"
    },
    "reviewer": {
      "description": "An individual or organization primarily responsible for review of some aspect of the
content.",
      "items": {
        "$ref": "ContactDetail.schema.json#/definitions/ContactDetail"
      },
      "type": "array"
    },
    "endorser": {
      "description": "An individual or organization responsible for officially endorsing the content for use in some
 setting.",
```

```
      "items": {
        "$ref": "ContactDetail.schema.json#/definitions/ContactDetail"
      },
      "type": "array"
    },
    "relatedArtifact": {
      "description": "Related artifacts such as additional documentation, justification, or bibliographic
references.",
      "items": {
        "$ref": "RelatedArtifact.schema.json#/definitions/RelatedArtifact"
      },
      "type": "array"
    },
    "library": {
      "description": "A reference to a Library resource containing any formal logic used by the plan definition.",
      "items": {
        "$ref": "canonical.schema.json#/definitions/canonical"
      },
      "type": "array"
    },
    "goal": {
      "description": "Goals that describe what the activities within the plan are intended to achieve. For example,
weight loss, restoring an activity of daily living, obtaining herd immunity via immunization, meeting a process
improvement objective, etc.",
      "items": {
        "$ref": "#/definitions/PlanDefinition_Goal"
      },
      "type": "array"
    },
    "action": {
      "description": "An action or group of actions to be taken as part of the plan.",
      "items": {
        "$ref": "#/definitions/PlanDefinition_Action"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "resourceType"
  ]
},
"PlanDefinition_Goal": {
  "description": "This resource allows for the definition of various types of plans as a sharable, consumable, and
executable artifact. The resource is general enough to support the description of a broad range of clinical
artifacts such as clinical decision support rules, order sets and protocols.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "category": {
      "description": "Indicates a category the goal falls within.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "description": {
      "description": "Human-readable and/or coded description of a specific desired objective of care, such as
\"control blood pressure\" or \"negotiate an obstacle course\" or \"dance with child at wedding\".",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "priority": {
      "description": "Identifies the expected level of importance associated with reaching/sustaining the defined
goal.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
```

```
    },
    "start": {
      "description": "The event after which the goal should begin being pursued.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "addresses": {
      "description": "Identifies problems, conditions, issues, or concerns the goal is intended to address.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "documentation": {
      "description": "Didactic or other informational resources associated with the goal that provide further
supporting information about the goal. Information resources can include inline text commentary and links to web
resources.",
      "items": {
        "$ref": "RelatedArtifact.schema.json#/definitions/RelatedArtifact"
      },
      "type": "array"
    },
    "target": {
      "description": "Indicates what should be done and within what timeframe.",
      "items": {
        "$ref": "#/definitions/PlanDefinition_Target"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "description"
  ]
},
"PlanDefinition_Target": {
  "description": "This resource allows for the definition of various types of plans as a sharable, consumable, and
executable artifact. The resource is general enough to support the description of a broad range of clinical
artifacts such as clinical decision support rules, order sets and protocols.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "measure": {
      "description": "The parameter whose value is to be tracked, e.g. body weight, blood pressure, or hemoglobin
A1c level.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "detailQuantity": {
      "description": "The target value of the measure to be achieved to signify fulfillment of the goal, e.g. 150
pounds or 7.0%. Either the high or low or both values of the range can be specified. When a low value is missing, it
 indicates that the goal is achieved at any value at or below the high value. Similarly, if the high value is
missing, it indicates that the goal is achieved at any value at or above the low value.",
      "$ref": "Quantity.schema.json#/definitions/Quantity"
    },
    "detailRange": {
      "description": "The target value of the measure to be achieved to signify fulfillment of the goal, e.g. 150
pounds or 7.0%. Either the high or low or both values of the range can be specified. When a low value is missing, it
 indicates that the goal is achieved at any value at or below the high value. Similarly, if the high value is
missing, it indicates that the goal is achieved at any value at or above the low value.",
      "$ref": "Range.schema.json#/definitions/Range"
    },
    "detailCodeableConcept": {
      "description": "The target value of the measure to be achieved to signify fulfillment of the goal, e.g. 150
```

```
pounds or 7.0%. Either the high or low or both values of the range can be specified. When a low value is missing, it
 indicates that the goal is achieved at any value at or below the high value. Similarly, if the high value is
missing, it indicates that the goal is achieved at any value at or above the low value.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "due": {
      "description": "Indicates the timeframe after the start of the goal in which the goal should be met.",
      "$ref": "Duration.schema.json#/definitions/Duration"
    }
  },
  "additionalProperties": false
},
"PlanDefinition_Action": {
  "description": "This resource allows for the definition of various types of plans as a sharable, consumable, and
executable artifact. The resource is general enough to support the description of a broad range of clinical
artifacts such as clinical decision support rules, order sets and protocols.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "prefix": {
      "description": "A user-visible prefix for the action.",
      "$ref": "#/definitions/string"
    },
    "_prefix": {
      "description": "Extensions for prefix",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "title": {
      "description": "The title of the action displayed to a user.",
      "$ref": "#/definitions/string"
    },
    "_title": {
      "description": "Extensions for title",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "description": {
      "description": "A brief description of the action used to provide a summary to display to the user.",
      "$ref": "#/definitions/string"
    },
    "_description": {
      "description": "Extensions for description",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "textEquivalent": {
      "description": "A text equivalent of the action to be performed. This provides a human-interpretable
description of the action when the definition is consumed by a system that might not be capable of interpreting it
dynamically.",
      "$ref": "#/definitions/string"
    },
    "_textEquivalent": {
      "description": "Extensions for textEquivalent",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "priority": {
      "description": "Indicates how quickly the action should be addressed with respect to other actions.",
      "$ref": "#/definitions/code"
    },
    "_priority": {
      "description": "Extensions for priority",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "code": {
```

```
      "description": "A code that provides meaning for the action or action group. For example, a section may have a
 LOINC code for the section of a documentation template.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "reason": {
      "description": "A description of why this action is necessary or appropriate.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "documentation": {
      "description": "Didactic or other informational resources associated with the action that can be provided to
the CDS recipient. Information resources can include inline text commentary and links to web resources.",
      "items": {
        "$ref": "RelatedArtifact.schema.json#/definitions/RelatedArtifact"
      },
      "type": "array"
    },
    "goalId": {
      "description": "Identifies goals that this action supports. The reference must be to a goal element defined
within this plan definition.",
      "items": {
        "$ref": "#/definitions/id"
      },
      "type": "array"
    },
    "_goalId": {
      "description": "Extensions for goalId",
      "items": {
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "type": "array"
    },
    "subjectCodeableConcept": {
      "description": "A code or group definition that describes the intended subject of the action and its children,
 if any.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "subjectReference": {
      "description": "A code or group definition that describes the intended subject of the action and its children,
 if any.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "trigger": {
      "description": "A description of when the action should be triggered.",
      "items": {
        "$ref": "TriggerDefinition.schema.json#/definitions/TriggerDefinition"
      },
      "type": "array"
    },
    "condition": {
      "description": "An expression that describes applicability criteria or start/stop conditions for the action.",
      "items": {
        "$ref": "#/definitions/PlanDefinition_Condition"
      },
      "type": "array"
    },
    "input": {
      "description": "Defines input data requirements for the action.",
      "items": {
        "$ref": "DataRequirement.schema.json#/definitions/DataRequirement"
      },
      "type": "array"
    },
    "output": {
      "description": "Defines the outputs of the action, if any.",
      "items": {
        "$ref": "DataRequirement.schema.json#/definitions/DataRequirement"
      },
      "type": "array"
    },
    "relatedAction": {
      "description": "A relationship to another action such as \"before\" or \"30-60 minutes after start of\".",
      "items": {
        "$ref": "#/definitions/PlanDefinition_RelatedAction"
      },
      "type": "array"
    },
    "timingDateTime": {
      "description": "An optional value describing when the action should be performed.",
      "pattern": "^([0-9]([0-9]([0-9][1-9]|[1-9]0)|[1-9]00)|[1-9]000)(-(0[1-9]|1[0-2])(-(0[1-9]|[1-2][0-9]|3[0-
1])(T([01][0-9]|2[0-3]):[0-5][0-9]:([0-5][0-9]|60)(\\.[0-9]+)?(Z|(\\+|-)((0[0-9]|1[0-3]):[0-5][0-9]|14:00)))?)?)?$",
      "type": "string"
    },
    "_timingDateTime": {
```

```
  "description": "Extensions for timingDateTime",
  "$ref": "Element.schema.json#/definitions/Element"
},
"timingAge": {
  "description": "An optional value describing when the action should be performed.",
  "$ref": "Age.schema.json#/definitions/Age"
},
"timingPeriod": {
  "description": "An optional value describing when the action should be performed.",
  "$ref": "Period.schema.json#/definitions/Period"
},
"timingDuration": {
  "description": "An optional value describing when the action should be performed.",
  "$ref": "Duration.schema.json#/definitions/Duration"
},
"timingRange": {
  "description": "An optional value describing when the action should be performed.",
  "$ref": "Range.schema.json#/definitions/Range"
},
"timingTiming": {
  "description": "An optional value describing when the action should be performed.",
  "$ref": "Timing.schema.json#/definitions/Timing"
},
"participant": {
  "description": "Indicates who should participate in performing the action described.",
  "items": {
    "$ref": "#/definitions/PlanDefinition_Participant"
  },
  "type": "array"
},
"type": {
  "description": "The type of action to perform (create, update, remove).",
  "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
},
"groupingBehavior": {
  "description": "Defines the grouping behavior for the action and its children.",
  "enum": [
    "visual-group",
    "logical-group",
    "sentence-group"
  ]
},
"_groupingBehavior": {
  "description": "Extensions for groupingBehavior",
  "$ref": "Element.schema.json#/definitions/Element"
},
"selectionBehavior": {
  "description": "Defines the selection behavior for the action and its children.",
  "enum": [
    "any",
    "all",
    "all-or-none",
    "exactly-one",
    "at-most-one",
    "one-or-more"
  ]
},
"_selectionBehavior": {
  "description": "Extensions for selectionBehavior",
  "$ref": "Element.schema.json#/definitions/Element"
},
"requiredBehavior": {
  "description": "Defines the required behavior for the action.",
  "enum": [
    "must",
    "could",
    "must-unless-documented"
  ]
},
"_requiredBehavior": {
  "description": "Extensions for requiredBehavior",
  "$ref": "Element.schema.json#/definitions/Element"
},
"precheckBehavior": {
  "description": "Defines whether the action should usually be preselected.",
  "enum": [
    "yes",
    "no"
  ]
},
"_precheckBehavior": {
  "description": "Extensions for precheckBehavior",
  "$ref": "Element.schema.json#/definitions/Element"
},
"cardinalityBehavior": {
  "description": "Defines whether the action can be selected multiple times.",
  "enum": [
    "single",
    "multiple"
```

```
      ]
    },
    "_cardinalityBehavior": {
      "description": "Extensions for cardinalityBehavior",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "definitionCanonical": {
      "description": "A reference to an ActivityDefinition that describes the action to be taken in detail, or a
PlanDefinition that describes a series of actions to be taken.",
      "pattern": "^\\S*$",
      "type": "string"
    },
    "_definitionCanonical": {
      "description": "Extensions for definitionCanonical",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "definitionUri": {
      "description": "A reference to an ActivityDefinition that describes the action to be taken in detail, or a
PlanDefinition that describes a series of actions to be taken.",
      "pattern": "^\\S*$",
      "type": "string"
    },
    "_definitionUri": {
      "description": "Extensions for definitionUri",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "transform": {
      "description": "A reference to a StructureMap resource that defines a transform that can be executed to
produce the intent resource using the ActivityDefinition instance as the input.",
      "$ref": "canonical.schema.json#/definitions/canonical"
    },
    "dynamicValue": {
      "description": "Customizations that should be applied to the statically defined resource. For example, if the
dosage of a medication must be computed based on the patient\u0027s weight, a customization would be used to specify
 an expression that calculated the weight, and the path on the resource that would contain the result.",
      "items": {
        "$ref": "#/definitions/PlanDefinition_DynamicValue"
      },
      "type": "array"
    },
    "action": {
      "description": "Sub actions that are contained within the action. The behavior of this action determines the
functionality of the sub-actions. For example, a selection behavior of at-most-one indicates that of the sub-
actions, at most one may be chosen as part of realizing the action definition.",
      "items": {
        "$ref": "#/definitions/PlanDefinition_Action"
      },
      "type": "array"
    }
  },
  "additionalProperties": false
},
"PlanDefinition_Condition": {
  "description": "This resource allows for the definition of various types of plans as a sharable, consumable, and
executable artifact. The resource is general enough to support the description of a broad range of clinical
artifacts such as clinical decision support rules, order sets and protocols.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "kind": {
      "description": "The kind of condition.",
      "enum": [
```

```
        "applicability",
        "start",
        "stop"
      ]
    },
    "_kind": {
      "description": "Extensions for kind",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "expression": {
      "description": "An expression that returns true or false, indicating whether the condition is satisfied.",
      "$ref": "Expression.schema.json#/definitions/Expression"
    }
  },
  "additionalProperties": false
},
"PlanDefinition_RelatedAction": {
  "description": "This resource allows for the definition of various types of plans as a sharable, consumable, and
executable artifact. The resource is general enough to support the description of a broad range of clinical
artifacts such as clinical decision support rules, order sets and protocols.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "actionId": {
      "description": "The element id of the related action.",
      "$ref": "#/definitions/id"
    },
    "_actionId": {
      "description": "Extensions for actionId",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "relationship": {
      "description": "The relationship of this action to the related action.",
      "enum": [
        "before-start",
        "before",
        "before-end",
        "concurrent-with-start",
        "concurrent",
        "concurrent-with-end",
        "after-start",
        "after",
        "after-end"
      ]
    },
    "_relationship": {
      "description": "Extensions for relationship",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "offsetDuration": {
      "description": "A duration or range of durations to apply to the relationship. For example, 30-60 minutes
before.",
      "$ref": "Duration.schema.json#/definitions/Duration"
    },
    "offsetRange": {
      "description": "A duration or range of durations to apply to the relationship. For example, 30-60 minutes
before.",
      "$ref": "Range.schema.json#/definitions/Range"
    }
  },
  "additionalProperties": false
},
```

```
"PlanDefinition_Participant": {
  "description": "This resource allows for the definition of various types of plans as a sharable, consumable, and executable artifact. The resource is general enough to support the description of a broad range of clinical artifacts such as clinical decision support rules, order sets and protocols.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the element and that modifies the understanding of the element in which it is contained and/or the understanding of the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension. Applications processing a resource are required to check for modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "type": {
      "description": "The type of participant in the action.",
      "enum": [
        "patient",
        "practitioner",
        "related-person",
        "device"
      ]
    },
    "_type": {
      "description": "Extensions for type",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "role": {
      "description": "The role the participant should play in performing the described action.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    }
  },
  "additionalProperties": false
},
"PlanDefinition_DynamicValue": {
  "description": "This resource allows for the definition of various types of plans as a sharable, consumable, and executable artifact. The resource is general enough to support the description of a broad range of clinical artifacts such as clinical decision support rules, order sets and protocols.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the element and that modifies the understanding of the element in which it is contained and/or the understanding of the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension. Applications processing a resource are required to check for modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "path": {
```

```
        "description": "The path to the element to be customized. This is the path on the resource that will hold the
    result of the calculation defined by the expression. The specified path SHALL be a FHIRPath resolveable on the
    specified target type of the ActivityDefinition, and SHALL consist only of identifiers, constant indexers, and a
    restricted subset of functions. The path is allowed to contain qualifiers (.) to traverse sub-elements, as well as
    indexers ([x]) to traverse multiple-cardinality sub-elements (see the [Simple FHIRPath
    Profile](fhirpath.html#simple) for full details).",
          "$ref": "#/definitions/string"
        },
        "_path": {
          "description": "Extensions for path",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "expression": {
          "description": "An expression specifying the value of the customized element.",
          "$ref": "Expression.schema.json#/definitions/Expression"
        }
      },
      "additionalProperties": false
    }
  }
}
```

12.18.15 Resource PlanDefinition - Maturity Dependency View

URL: https://hl7.org/fhir/R4/plandefinition-dependencies.html

12.18.15 Resource PlanDefinition - Maturity Dependency View

Clinical Decision Support Work Group | Maturity Level: 2 | Trial Use | Compartments: Not linked to any defined compartments

Dependency Graph for PlanDefinition FMM level 2

Name | Card. | Type | Dependency Analysis
PlanDefinition | | DomainResource
url | 0..1 | uri
identifier | 0..* | Identifier
version | 0..1 | string
name | 0..1 | string
title | 0..1 | string
subtitle | 0..1 | string
type | 0..1 | CodeableConcept
status | 1..1 | code
experimental | 0..1 | boolean
subject[x] | 0..1
subjectCodeableConcept | | CodeableConcept
subjectReference | | Reference(Group
date | 0..1 | dateTime
publisher | 0..1 | string
contact | 0..* | ContactDetail
description | 0..1 | markdown
useContext | 0..* | UsageContext
jurisdiction | 0..* | CodeableConcept
purpose | 0..1 | markdown
usage | 0..1 | string
copyright | 0..1 | markdown
approvalDate | 0..1 | date
lastReviewDate | 0..1 | date
effectivePeriod | 0..1 | Period
topic | 0..* | CodeableConcept
author | 0..* | ContactDetail
editor | 0..* | ContactDetail
reviewer | 0..* | ContactDetail
endorser | 0..* | ContactDetail
relatedArtifact | 0..* | RelatedArtifact
library | 0..* | canonical(Library)
goal | 0..* | BackboneElement
category | 0..1 | CodeableConcept
description | 1..1 | CodeableConcept
priority | 0..1 | CodeableConcept
start | 0..1 | CodeableConcept
addresses | 0..* | CodeableConcept
documentation | 0..* | RelatedArtifact
target | 0..* | BackboneElement
measure | 0..1 | CodeableConcept
detail[x] | 0..1
detailQuantity | | Quantity
detailRange | | Range
detailCodeableConcept | | CodeableConcept

due | 0..1 | Duration
action | 0..* | BackboneElement
prefix | 0..1 | string
title | 0..1 | string
description | 0..1 | string
textEquivalent | 0..1 | string
priority | 0..1 | code
code | 0..* | CodeableConcept
reason | 0..* | CodeableConcept
documentation | 0..* | RelatedArtifact
goalId | 0..* | id
subject[x] | 0..1
subjectCodeableConcept | | CodeableConcept
subjectReference | | Reference(Group
trigger | 0..* | TriggerDefinition
condition | 0..* | BackboneElement
kind | 1..1 | code
expression | 0..1 | Expression
input | 0..* | DataRequirement
output | 0..* | DataRequirement
relatedAction | 0..* | BackboneElement
actionId | 1..1 | id
relationship | 1..1 | code
offset[x] | 0..1
offsetDuration | | Duration
offsetRange | | Range
timing[x] | 0..1
timingDateTime | | dateTime
timingAge | | Age
timingPeriod | | Period
timingDuration | | Duration
timingRange | | Range
timingTiming | | Timing
participant | 0..* | BackboneElement
type | 1..1 | code
role | 0..1 | CodeableConcept
type | 0..1 | CodeableConcept
groupingBehavior | 0..1 | code
selectionBehavior | 0..1 | code
requiredBehavior | 0..1 | code
precheckBehavior | 0..1 | code
cardinalityBehavior | 0..1 | code
definition[x] | 0..1
definitionCanonical | | canonical
definitionUri | | uri
transform | 0..1 | canonical(StructureMap)
dynamicValue | 0..* | BackboneElement
path | 0..1 | string
expression | 0..1 | Expression
action | 0..* | see action

12.19 Resource DeviceRequest - Content

URL: https://hl7.org/fhir/R4/devicerequest.html
Orders and Observations Work Group | Maturity Level: 1 | Trial Use | Security Category: Patient | Compartments: Device, Encounter, Patient, Practitioner

Represents a request for a patient to employ a medical device. The device may be an implantable device, or an external assistive device, such as a walker.

12.19.1 Scope and Usage

Trial-Use Note: In terms of scope and usage, the Orders and Observations workgroups wish to draw the attention of reviewers and implementers to the following issue:

• We are seeking input from the implementation community on whether to combine DeviceRequest and VisionPrescription for R4. The parameters element was added to DeviceRequest for representing custom details for ordering devices. With this change, VisionPrescription significantly overlaps DeviceRequest. To illustrate this, the devicerequest-left-lens and devicerequest-right-lens examples are based on the general glasses example in VisionPrescription.

This resource is a request resource from a FHIR workflow perspective - see Workflow.

This resource describes the request for the use of a device by a patient. The device may be any pertinent device specified in the Device resource. Examples of devices that may be requested include wheelchair, hearing aids, or an insulin pump. The request may lead to the dispensing of the device to the patient or for use by the patient.

The device use request may represent an order or a prescription entered by a practitioner in a CPOE system or a proposal made by a clinical decision support (CDS) system based on a patient's clinical record and context of care.

12.19.2 Boundaries and Relationships

This resource deals with the allocation of a device to a patient and while it may contain instructions on how to use the device, the data about getting the device to the patient is addressed in other resources. For example, certain devices must be implanted via a surgical or other procedure and the implantation or explantation is represented in the [Procedure](procedure.html) or [ServiceRequest](servicerequest.html) resource.

The SupplyRequest resource is similar in that it deals with requesting a particular item for a specific patient. However, DeviceRequest is concerned with items where there is a patient focus, or instructions regarding their use, whereas SupplyRequest is concerned with inventory management.

This resource is referenced by CarePlan, Claim and Observation

12.19.3 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
DeviceRequest | TU | | DomainResource | Medical device request Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | External Request identifier
instantiatesCanonical | Σ | 0..* | canonical(ActivityDefinition | PlanDefinition) | Instantiates FHIR protocol or definition
instantiatesUri | Σ | 0..* | uri | Instantiates external protocol or definition
basedOn | Σ | 0..* | Reference(Any) | What request fulfills
priorRequest | Σ | 0..* | Reference(Any) | What request replaces
groupIdentifier | Σ | 0..1 | Identifier | Identifier of composite request
status | ?!Σ | 0..1 | code | draft | active | on-hold | revoked | completed | entered-in-error | unknown RequestStatus (Required)
intent | ?!Σ | 1..1 | code | proposal | plan | directive | order | original-order | reflex-order | filler-order | instance-order | option RequestIntent (Required)
priority | Σ | 0..1 | code | routine | urgent | asap | stat Request priority (Required)
code[x] | Σ | 1..1 | | Device requested FHIR Device Types (Example)
codeReference | | | Reference(Device)
codeCodeableConcept | | | CodeableConcept
parameter | | 0..* | BackboneElement | Device details
code | | 0..1 | CodeableConcept | Device detail
value[x] | | 0..1 | | Value of detail
valueCodeableConcept | | | CodeableConcept
valueQuantity | | | Quantity
valueRange | | | Range
valueBoolean | | | boolean
subject | Σ | 1..1 | Reference(Patient | Group | Location | Device) | Focus of request
encounter | Σ | 0..1 | Reference(Encounter) | Encounter motivating request
occurrence[x] | Σ | 0..1 | | Desired time or schedule for use
occurrenceDateTime | | | dateTime
occurrencePeriod | | | Period
occurrenceTiming | | | Timing
authoredOn | Σ | 0..1 | dateTime | When recorded
requester | Σ | 0..1 | Reference(Device | Practitioner | PractitionerRole | Organization) | Who/what is requesting diagnostics
performerType | Σ | 0..1 | CodeableConcept | Filler role Participant Roles (Example)
performer | Σ | 0..1 | Reference(Practitioner | PractitionerRole | Organization | CareTeam | HealthcareService | Patient | Device | RelatedPerson) | Requested Filler
reasonCode | Σ | 0..* | CodeableConcept | Coded Reason for request Condition/Problem/Diagnosis Codes (Example)
reasonReference | Σ | 0..* | Reference(Condition | Observation | DiagnosticReport | DocumentReference) | Linked Reason for request
insurance | | 0..* | Reference(Coverage | ClaimResponse) | Associated insurance coverage
supportingInfo | | 0..* | Reference(Any) | Additional clinical information

note | | 0..* | Annotation | Notes or comments
relevantHistory | | 0..* | Reference(Provenance) | Request provenance
Documentation for this format

UML Diagram (Legend)

XML Template

```
<DeviceRequest xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier External Request identifier --></identifier>
 <instantiatesCanonical><!-- 0..* canonical(ActivityDefinition|PlanDefinition) Instantiates FHIR protocol or definition
    --></instantiatesCanonical>
 <instantiatesUri value="[uri]"/><!-- 0..* Instantiates external protocol or definition -->
 <basedOn><!-- 0..* Reference(Any) What request fulfills --></basedOn>
 <priorRequest><!-- 0..* Reference(Any) What request replaces --></priorRequest>
 <groupIdentifier><!-- 0..1 Identifier Identifier of composite request --></groupIdentifier>
 <status value="[code]"/><!-- 0..1 draft | active | on-hold | revoked | completed | entered-in-error | unknown -->
 <intent value="[code]"/><!-- 1..1 proposal | plan | directive | order | original-order | reflex-order | filler-order |
    instance-order | option -->
 <priority value="[code]"/><!-- 0..1 routine | urgent | asap | stat -->
 <code[x]><!-- 1..1 Reference(Device)|CodeableConcept Device requested --></code[x]>
 <parameter>  <!-- 0..* Device details -->
  <code><!-- 0..1 CodeableConcept Device detail --></code>
  <value[x]><!-- 0..1 CodeableConcept|Quantity|Range|boolean Value of detail --></value[x]>
 </parameter>
 <subject><!-- 1..1 Reference(Patient|Group|Location|Device) Focus of request --></subject>
 <encounter><!-- 0..1 Reference(Encounter) Encounter motivating request --></encounter>
 <occurrence[x]><!-- 0..1 dateTime|Period|Timing Desired time or schedule for use --></occurrence[x]>
 <authoredOn value="[dateTime]"/><!-- 0..1 When recorded -->
 <requester><!-- 0..1 Reference(Device|Practitioner|PractitionerRole|Organization) Who/what is requesting diagnostics
    --></requester>
 <performerType><!-- 0..1 CodeableConcept Filler role --></performerType>
 <performer><!-- 0..1 Reference(Practitioner|PractitionerRole|Organization|
   CareTeam|HealthcareService|Patient|Device|RelatedPerson) Requested Filler --></performer>
 <reasonCode><!-- 0..* CodeableConcept Coded Reason for request --></reasonCode>
 <reasonReference><!-- 0..* Reference(Condition|Observation|DiagnosticReport|
   DocumentReference) Linked Reason for request --></reasonReference>
 <insurance><!-- 0..* Reference(Coverage|ClaimResponse) Associated insurance coverage --></insurance>
 <supportingInfo><!-- 0..* Reference(Any) Additional clinical information --></supportingInfo>
 <note><!-- 0..* Annotation Notes or comments --></note>
 <relevantHistory><!-- 0..* Reference(Provenance) Request provenance --></relevantHistory>
</DeviceRequest>
```

JSON Template

```
{
  "resourceType" : "DeviceRequest",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // External Request identifier
  "instantiatesCanonical" : [{ canonical(ActivityDefinition|PlanDefinition) }], // Instantiates FHIR protocol or
    definition
  "instantiatesUri" : ["<uri>"], // Instantiates external protocol or definition
  "basedOn" : [{ Reference(Any) }], // What request fulfills
  "priorRequest" : [{ Reference(Any) }], // What request replaces
  "groupIdentifier" : { Identifier }, // Identifier of composite request
  "status" : "<code>", // draft | active | on-hold | revoked | completed | entered-in-error | unknown
  "intent" : "<code>", // R!  proposal | plan | directive | order | original-order | reflex-order | filler-order |
    instance-order | option
  "priority" : "<code>", // routine | urgent | asap | stat
  // code[x]: Device requested. One of these 2:
  "codeReference" : { Reference(Device) },
  "codeCodeableConcept" : { CodeableConcept },
  "parameter" : [{ // Device details
    "code" : { CodeableConcept }, // Device detail
    // value[x]: Value of detail. One of these 4:
    "valueCodeableConcept" : { CodeableConcept }
    "valueQuantity" : { Quantity }
    "valueRange" : { Range }
    "valueBoolean" : <boolean>
  }],
  "subject" : { Reference(Patient|Group|Location|Device) }, // R!  Focus of request
  "encounter" : { Reference(Encounter) }, // Encounter motivating request
  // occurrence[x]: Desired time or schedule for use. One of these 3:
  "occurrenceDateTime" : "<dateTime>",
  "occurrencePeriod" : { Period },
  "occurrenceTiming" : { Timing },
  "authoredOn" : "<dateTime>", // When recorded
  "requester" : { Reference(Device|Practitioner|PractitionerRole|Organization) }, // Who/what is requesting diagnostics
  "performerType" : { CodeableConcept }, // Filler role
  "performer" : { Reference(Practitioner|PractitionerRole|Organization|
   CareTeam|HealthcareService|Patient|Device|RelatedPerson) }, // Requested Filler
  "reasonCode" : [{ CodeableConcept }], // Coded Reason for request
  "reasonReference" : [{ Reference(Condition|Observation|DiagnosticReport|
   DocumentReference) }], // Linked Reason for request
  "insurance" : [{ Reference(Coverage|ClaimResponse) }], // Associated insurance coverage
  "supportingInfo" : [{ Reference(Any) }], // Additional clinical information
  "note" : [{ Annotation }], // Notes or comments
```

```
  "relevantHistory" : [{ Reference(Provenance) }] // Request provenance
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:DeviceRequest;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:DeviceRequest.identifier [ Identifier ], ... ; # 0..* External Request identifier
  fhir:DeviceRequest.instantiatesCanonical [ canonical(ActivityDefinition|PlanDefinition) ], ... ; # 0..* Instantiates
    FHIR protocol or definition
  fhir:DeviceRequest.instantiatesUri [ uri ], ... ; # 0..* Instantiates external protocol or definition
  fhir:DeviceRequest.basedOn [ Reference(Any) ], ... ; # 0..* What request fulfills
  fhir:DeviceRequest.priorRequest [ Reference(Any) ], ... ; # 0..* What request replaces
  fhir:DeviceRequest.groupIdentifier [ Identifier ]; # 0..1 Identifier of composite request
  fhir:DeviceRequest.status [ code ]; # 0..1 draft | active | on-hold | revoked | completed | entered-in-error | unknown
  fhir:DeviceRequest.intent [ code ]; # 1..1 proposal | plan | directive | order | original-order | reflex-order |
    filler-order | instance-order | option
  fhir:DeviceRequest.priority [ code ]; # 0..1 routine | urgent | asap | stat
  # DeviceRequest.code[x] : 1..1 Device requested. One of these 2
    fhir:DeviceRequest.codeReference [ Reference(Device) ]
    fhir:DeviceRequest.codeCodeableConcept [ CodeableConcept ]
  fhir:DeviceRequest.parameter [ # 0..* Device details
    fhir:DeviceRequest.parameter.code [ CodeableConcept ]; # 0..1 Device detail
    # DeviceRequest.parameter.value[x] : 0..1 Value of detail. One of these 4
      fhir:DeviceRequest.parameter.valueCodeableConcept [ CodeableConcept ]
      fhir:DeviceRequest.parameter.valueQuantity [ Quantity ]
      fhir:DeviceRequest.parameter.valueRange [ Range ]
      fhir:DeviceRequest.parameter.valueBoolean [ boolean ]
  ], ...;
  fhir:DeviceRequest.subject [ Reference(Patient|Group|Location|Device) ]; # 1..1 Focus of request
  fhir:DeviceRequest.encounter [ Reference(Encounter) ]; # 0..1 Encounter motivating request
  # DeviceRequest.occurrence[x] : 0..1 Desired time or schedule for use. One of these 3
    fhir:DeviceRequest.occurrenceDateTime [ dateTime ]
    fhir:DeviceRequest.occurrencePeriod [ Period ]
    fhir:DeviceRequest.occurrenceTiming [ Timing ]
  fhir:DeviceRequest.authoredOn [ dateTime ]; # 0..1 When recorded
  fhir:DeviceRequest.requester [ Reference(Device|Practitioner|PractitionerRole|Organization) ]; # 0..1 Who/what is
    requesting diagnostics
  fhir:DeviceRequest.performerType [ CodeableConcept ]; # 0..1 Filler role
  fhir:DeviceRequest.performer [
    Reference(Practitioner|PractitionerRole|Organization|CareTeam|HealthcareService|Patient|
  Device|RelatedPerson) ]; # 0..1 Requested Filler
  fhir:DeviceRequest.reasonCode [ CodeableConcept ], ... ; # 0..* Coded Reason for request
  fhir:DeviceRequest.reasonReference [ Reference(Condition|Observation|DiagnosticReport|DocumentReference) ], ... ; #
    0..* Linked Reason for request
  fhir:DeviceRequest.insurance [ Reference(Coverage|ClaimResponse) ], ... ; # 0..* Associated insurance coverage
  fhir:DeviceRequest.supportingInfo [ Reference(Any) ], ... ; # 0..* Additional clinical information
  fhir:DeviceRequest.note [ Annotation ], ... ; # 0..* Notes or comments
  fhir:DeviceRequest.relevantHistory [ Reference(Provenance) ], ... ; # 0..* Request provenance
]
```

Changes since R3

DeviceRequest
DeviceRequest.instantiatesCanonical | Added Element
DeviceRequest.instantiatesUri | Added Element
DeviceRequest.status | Change value set from http://hl7.org/fhir/ValueSet/request-status to http://hl7.org/fhir/ValueSet/request-status|4.0.1
DeviceRequest.intent | Type changed from CodeableConcept to code Change value set from http://build.fhir.org/valueset-request-intent.html to http://hl7.org/fhir/ValueSet/request-intent|4.0.1
DeviceRequest.priority | Change value set from http://hl7.org/fhir/ValueSet/request-priority to http://hl7.org/fhir/ValueSet/request-priority|4.0.1 Default Value "If missing, normal priority" removed
DeviceRequest.parameter | Added Element
DeviceRequest.parameter.code | Added Element
DeviceRequest.parameter.value[x] | Added Element
DeviceRequest.encounter | Added Element
DeviceRequest.requester | Type changed from BackboneElement to Reference(Device | Practitioner | PractitionerRole | Organization)
DeviceRequest.performer | Type Reference: Added Target Types PractitionerRole, CareTeam
DeviceRequest.reasonReference | Type Reference: Added Target Types Condition, Observation, DiagnosticReport, DocumentReference Type Reference: Removed Target Type Resource
DeviceRequest.insurance | Added Element
DeviceRequest.definition | deleted
DeviceRequest.context | deleted
DeviceRequest.requester.agent | deleted
DeviceRequest.requester.onBehalfOf | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 2 tests that all execute ok. 2 fail round-trip testing and 1 r3 resources are invalid (0 errors).)

Structure

Name | Flags | Card. | Type | Description & Constraints
DeviceRequest | TU | | DomainResource | Medical device request Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | External Request identifier
instantiatesCanonical | Σ | 0..* | canonical(ActivityDefinition | PlanDefinition) | Instantiates FHIR protocol or definition
instantiatesUri | Σ | 0..* | uri | Instantiates external protocol or definition
basedOn | Σ | 0..* | Reference(Any) | What request fulfills
priorRequest | Σ | 0..* | Reference(Any) | What request replaces
groupIdentifier | Σ | 0..1 | Identifier | Identifier of composite request
status | ?!Σ | 0..1 | code | draft | active | on-hold | revoked | completed | entered-in-error | unknown RequestStatus (Required)
intent | ?!Σ | 1..1 | code | proposal | plan | directive | order | original-order | reflex-order | filler-order | instance-order | option RequestIntent (Required)
priority | Σ | 0..1 | code | routine | urgent | asap | stat Request priority (Required)
code[x] | Σ | 1..1 | | Device requested FHIR Device Types (Example)
codeReference | | | Reference(Device)
codeCodeableConcept | | | CodeableConcept
parameter | | 0..* | BackboneElement | Device details
code | | 0..1 | CodeableConcept | Device detail
value[x] | | 0..1 | | Value of detail
valueCodeableConcept | | | CodeableConcept
valueQuantity | | | Quantity
valueRange | | | Range
valueBoolean | | | boolean
subject | Σ | 1..1 | Reference(Patient | Group | Location | Device) | Focus of request
encounter | Σ | 0..1 | Reference(Encounter) | Encounter motivating request
occurrence[x] | Σ | 0..1 | | Desired time or schedule for use
occurrenceDateTime | | | dateTime
occurrencePeriod | | | Period
occurrenceTiming | | | Timing
authoredOn | Σ | 0..1 | dateTime | When recorded
requester | Σ | 0..1 | Reference(Device | Practitioner | PractitionerRole | Organization) | Who/what is requesting diagnostics
performerType | Σ | 0..1 | CodeableConcept | Filler role Participant Roles (Example)
performer | Σ | 0..1 | Reference(Practitioner | PractitionerRole | Organization | CareTeam | HealthcareService | Patient | Device | RelatedPerson) | Requested Filler
reasonCode | Σ | 0..* | CodeableConcept | Coded Reason for request Condition/Problem/Diagnosis Codes (Example)
reasonReference | Σ | 0..* | Reference(Condition | Observation | DiagnosticReport | DocumentReference) | Linked Reason for request
insurance | | 0..* | Reference(Coverage | ClaimResponse) | Associated insurance coverage
supportingInfo | | 0..* | Reference(Any) | Additional clinical information
note | | 0..* | Annotation | Notes or comments
relevantHistory | | 0..* | Reference(Provenance) | Request provenance
Documentation for this format

UML Diagram (Legend)

XML Template

```
<DeviceRequest xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier External Request identifier --></identifier>
 <instantiatesCanonical><!-- 0..* canonical(ActivityDefinition|PlanDefinition) Instantiates FHIR protocol or definition
    --></instantiatesCanonical>
 <instantiatesUri value="[uri]"/><!-- 0..* Instantiates external protocol or definition -->
 <basedOn><!-- 0..* Reference(Any) What request fulfills --></basedOn>
 <priorRequest><!-- 0..* Reference(Any) What request replaces --></priorRequest>
 <groupIdentifier><!-- 0..1 Identifier Identifier of composite request --></groupIdentifier>
 <status value="[code]"/><!-- 0..1 draft | active | on-hold | revoked | completed | entered-in-error | unknown -->
 <intent value="[code]"/><!-- 1..1 proposal | plan | directive | order | original-order | reflex-order | filler-order |
    instance-order | option -->
 <priority value="[code]"/><!-- 0..1 routine | urgent | asap | stat -->
 <code[x]><!-- 1..1 Reference(Device)|CodeableConcept Device requested --></code[x]>
 <parameter>  <!-- 0..* Device details -->
  <code><!-- 0..1 CodeableConcept Device detail --></code>
  <value[x]><!-- 0..1 CodeableConcept|Quantity|Range|boolean Value of detail --></value[x]>
 </parameter>
 <subject><!-- 1..1 Reference(Patient|Group|Location|Device) Focus of request --></subject>
 <encounter><!-- 0..1 Reference(Encounter) Encounter motivating request --></encounter>
 <occurrence[x]><!-- 0..1 dateTime|Period|Timing Desired time or schedule for use --></occurrence[x]>
 <authoredOn value="[dateTime]"/><!-- 0..1 When recorded -->
 <requester><!-- 0..1 Reference(Device|Practitioner|PractitionerRole|Organization) Who/what is requesting diagnostics
    --></requester>
 <performerType><!-- 0..1 CodeableConcept Filler role --></performerType>
 <performer><!-- 0..1 Reference(Practitioner|PractitionerRole|Organization|
   CareTeam|HealthcareService|Patient|Device|RelatedPerson) Requested Filler --></performer>
 <reasonCode><!-- 0..* CodeableConcept Coded Reason for request --></reasonCode>
```

```
 <reasonReference><!-- 0..* Reference(Condition|Observation|DiagnosticReport|
   DocumentReference) Linked Reason for request --></reasonReference>
 <insurance><!-- 0..* Reference(Coverage|ClaimResponse) Associated insurance coverage --></insurance>
 <supportingInfo><!-- 0..* Reference(Any) Additional clinical information --></supportingInfo>
 <note><!-- 0..* Annotation Notes or comments --></note>
 <relevantHistory><!-- 0..* Reference(Provenance) Request provenance --></relevantHistory>
</DeviceRequest>
```

JSON Template

```
{
  "resourceType" : "DeviceRequest",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // External Request identifier
  "instantiatesCanonical" : [{ canonical(ActivityDefinition|PlanDefinition) }], // Instantiates FHIR protocol or
    definition
  "instantiatesUri" : ["<uri>"], // Instantiates external protocol or definition
  "basedOn" : [{ Reference(Any) }], // What request fulfills
  "priorRequest" : [{ Reference(Any) }], // What request replaces
  "groupIdentifier" : { Identifier }, // Identifier of composite request
  "status" : "<code>", // draft | active | on-hold | revoked | completed | entered-in-error | unknown
  "intent" : "<code>", // R!  proposal | plan | directive | order | original-order | reflex-order | filler-order |
    instance-order | option
  "priority" : "<code>", // routine | urgent | asap | stat
  // code[x]: Device requested. One of these 2:
  "codeReference" : { Reference(Device) },
  "codeCodeableConcept" : { CodeableConcept },
  "parameter" : [{ // Device details
    "code" : { CodeableConcept }, // Device detail
    // value[x]: Value of detail. One of these 4:
    "valueCodeableConcept" : { CodeableConcept }
    "valueQuantity" : { Quantity }
    "valueRange" : { Range }
    "valueBoolean" : <boolean>
  }],
  "subject" : { Reference(Patient|Group|Location|Device) }, // R!  Focus of request
  "encounter" : { Reference(Encounter) }, // Encounter motivating request
  // occurrence[x]: Desired time or schedule for use. One of these 3:
  "occurrenceDateTime" : "<dateTime>",
  "occurrencePeriod" : { Period },
  "occurrenceTiming" : { Timing },
  "authoredOn" : "<dateTime>", // When recorded
  "requester" : { Reference(Device|Practitioner|PractitionerRole|Organization) }, // Who/what is requesting diagnostics
  "performerType" : { CodeableConcept }, // Filler role
  "performer" : { Reference(Practitioner|PractitionerRole|Organization|
   CareTeam|HealthcareService|Patient|Device|RelatedPerson) }, // Requested Filler
  "reasonCode" : [{ CodeableConcept }], // Coded Reason for request
  "reasonReference" : [{ Reference(Condition|Observation|DiagnosticReport|
   DocumentReference) }], // Linked Reason for request
  "insurance" : [{ Reference(Coverage|ClaimResponse) }], // Associated insurance coverage
  "supportingInfo" : [{ Reference(Any) }], // Additional clinical information
  "note" : [{ Annotation }], // Notes or comments
  "relevantHistory" : [{ Reference(Provenance) }] // Request provenance
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:DeviceRequest;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:DeviceRequest.identifier [ Identifier ], ... ; # 0..* External Request identifier
  fhir:DeviceRequest.instantiatesCanonical [ canonical(ActivityDefinition|PlanDefinition) ], ... ; # 0..* Instantiates
    FHIR protocol or definition
  fhir:DeviceRequest.instantiatesUri [ uri ], ... ; # 0..* Instantiates external protocol or definition
  fhir:DeviceRequest.basedOn [ Reference(Any) ], ... ; # 0..* What request fulfills
  fhir:DeviceRequest.priorRequest [ Reference(Any) ], ... ; # 0..* What request replaces
  fhir:DeviceRequest.groupIdentifier [ Identifier ]; # 0..1 Identifier of composite request
  fhir:DeviceRequest.status [ code ]; # 0..1 draft | active | on-hold | revoked | completed | entered-in-error | unknown
  fhir:DeviceRequest.intent [ code ]; # 1..1 proposal | plan | directive | order | original-order | reflex-order |
    filler-order | instance-order | option
  fhir:DeviceRequest.priority [ code ]; # 0..1 routine | urgent | asap | stat
  # DeviceRequest.code[x] : 1..1 Device requested. One of these 2
    fhir:DeviceRequest.codeReference [ Reference(Device) ]
    fhir:DeviceRequest.codeCodeableConcept [ CodeableConcept ]
  fhir:DeviceRequest.parameter [ # 0..* Device details
    fhir:DeviceRequest.parameter.code [ CodeableConcept ]; # 0..1 Device detail
    # DeviceRequest.parameter.value[x] : 0..1 Value of detail. One of these 4
      fhir:DeviceRequest.parameter.valueCodeableConcept [ CodeableConcept ]
      fhir:DeviceRequest.parameter.valueQuantity [ Quantity ]
      fhir:DeviceRequest.parameter.valueRange [ Range ]
      fhir:DeviceRequest.parameter.valueBoolean [ boolean ]
  ], ...;
  fhir:DeviceRequest.subject [ Reference(Patient|Group|Location|Device) ]; # 1..1 Focus of request
  fhir:DeviceRequest.encounter [ Reference(Encounter) ]; # 0..1 Encounter motivating request
  # DeviceRequest.occurrence[x] : 0..1 Desired time or schedule for use. One of these 3
```

```
    fhir:DeviceRequest.occurrenceDateTime [ dateTime ]
    fhir:DeviceRequest.occurrencePeriod [ Period ]
    fhir:DeviceRequest.occurrenceTiming [ Timing ]
  fhir:DeviceRequest.authoredOn [ dateTime ]; # 0..1 When recorded
  fhir:DeviceRequest.requester [ Reference(Device|Practitioner|PractitionerRole|Organization) ]; # 0..1 Who/what is
    requesting diagnostics
  fhir:DeviceRequest.performerType [ CodeableConcept ]; # 0..1 Filler role
  fhir:DeviceRequest.performer [
    Reference(Practitioner|PractitionerRole|Organization|CareTeam|HealthcareService|Patient|
  Device|RelatedPerson) ]; # 0..1 Requested Filler
  fhir:DeviceRequest.reasonCode [ CodeableConcept ], ... ; # 0..* Coded Reason for request
  fhir:DeviceRequest.reasonReference [ Reference(Condition|Observation|DiagnosticReport|DocumentReference) ], ... ; #
    0..* Linked Reason for request
  fhir:DeviceRequest.insurance [ Reference(Coverage|ClaimResponse) ], ... ; # 0..* Associated insurance coverage
  fhir:DeviceRequest.supportingInfo [ Reference(Any) ], ... ; # 0..* Additional clinical information
  fhir:DeviceRequest.note [ Annotation ], ... ; # 0..* Notes or comments
  fhir:DeviceRequest.relevantHistory [ Reference(Provenance) ], ... ; # 0..* Request provenance
]
```

Changes since Release 3

DeviceRequest
DeviceRequest.instantiatesCanonical | Added Element
DeviceRequest.instantiatesUri | Added Element
DeviceRequest.status | Change value set from http://hl7.org/fhir/ValueSet/request-status to http://hl7.org/fhir/ValueSet/request-status|4.0.1
DeviceRequest.intent | Type changed from CodeableConcept to code Change value set from http://build.fhir.org/valueset-request-intent.html to http://hl7.org/fhir/ValueSet/request-intent|4.0.1
DeviceRequest.priority | Change value set from http://hl7.org/fhir/ValueSet/request-priority to http://hl7.org/fhir/ValueSet/request-priority|4.0.1 Default Value "If missing, normal priority" removed
DeviceRequest.parameter | Added Element
DeviceRequest.parameter.code | Added Element
DeviceRequest.parameter.value[x] | Added Element
DeviceRequest.encounter | Added Element
DeviceRequest.requester | Type changed from BackboneElement to Reference(Device | Practitioner | PractitionerRole | Organization)
DeviceRequest.performer | Type Reference: Added Target Types PractitionerRole, CareTeam
DeviceRequest.reasonReference | Type Reference: Added Target Types Condition, Observation, DiagnosticReport, DocumentReference Type Reference: Removed Target Type Resource
DeviceRequest.insurance | Added Element
DeviceRequest.definition | deleted
DeviceRequest.context | deleted
DeviceRequest.requester.agent | deleted
DeviceRequest.requester.onBehalfOf | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 2 tests that all execute ok. 2 fail round-trip testing and 1 r3 resources are invalid (0 errors).)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

12.19.3.1 Terminology Bindings

Path | Definition | Type | Reference
DeviceRequest.status | Codes representing the status of the request. | Required | RequestStatus
DeviceRequest.intent | The kind of diagnostic request. | Required | RequestIntent
DeviceRequest.priority | Identifies the level of importance to be assigned to actioning the request. | Required | RequestPriority
DeviceRequest.code[x] | Codes for devices that can be requested. | Example | FHIRDeviceTypes
DeviceRequest.parameter.code | A code that identifies the device detail. | Unknown | No details provided yet
DeviceRequest.performerType | Indicates specific responsibility of an individual within the care team, such as "Primary physician", "Team coordinator", "Caregiver", etc. | Example | ParticipantRoles
DeviceRequest.reasonCode | Diagnosis or problem codes justifying the reason for requesting the device. | Example | Condition/Problem/DiagnosisCodes

Notes to reviewers:

At this time, the code bindings are placeholders to be fleshed out upon further review by the community.

12.19.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
authored-on | date | When the request transitioned to being actionable | DeviceRequest.authoredOn
based-on | reference | Plan/proposal/order fulfilled by this request | DeviceRequest.basedOn (Any)
code | token | Code for what is being requested/ordered | (DeviceRequest.code as CodeableConcept) | 13 Resources
device | reference | Reference to resource that is being requested/ordered | (DeviceRequest.code as Reference) (Device)
encounter | reference | Encounter during which request was created | DeviceRequest.encounter (Encounter) | 12 Resources

event-date | date | When service should occur | (DeviceRequest.occurrence as dateTime) | (DeviceRequest.occurrence as Period)
group-identifier | token | Composite request this is part of | DeviceRequest.groupIdentifier
identifier | token | Business identifier for request/order | DeviceRequest.identifier | 30 Resources
instantiates-canonical | reference | Instantiates FHIR protocol or definition | DeviceRequest.instantiatesCanonical (PlanDefinition, ActivityDefinition)
instantiates-uri | uri | Instantiates external protocol or definition | DeviceRequest.instantiatesUri
insurance | reference | Associated insurance coverage | DeviceRequest.insurance (ClaimResponse, Coverage)
intent | token | proposal | plan | original-order |reflex-order | DeviceRequest.intent
patient | reference | Individual the service is ordered for | DeviceRequest.subject.where(resolve() is Patient) (Patient) | 33 Resources
performer | reference | Desired performer for service | DeviceRequest.performer (Practitioner, Organization, CareTeam, Device, Patient, HealthcareService, PractitionerRole, RelatedPerson)
prior-request | reference | Request takes the place of referenced completed or terminated requests | DeviceRequest.priorRequest (Any)
requester | reference | Who/what is requesting service | DeviceRequest.requester (Practitioner, Organization, Device, PractitionerRole)
status | token | entered-in-error | draft | active |suspended | completed | DeviceRequest.status
subject | reference | Individual the service is ordered for | DeviceRequest.subject (Group, Device, Patient, Location)

12.19.5 Resource DeviceRequest - Examples

URL: https://hl7.org/fhir/R4/devicerequest-examples.html

12.19.5 Resource DeviceRequest - Examples

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Device, Encounter, Patient, Practitioner

An introduction to this page as well as proper examples are to be submitted shortly.

Example Name | id | Format
Wheelchair assignment | example | XML | JSON | Turtle
Insulin Pump request | insulinpump | XML | JSON | Turtle
Right Lens prescription using the parameters element | right-lens | XML | JSON | Turtle
Left Lens prescription using the parameters element | left-lens | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

12.19.6 Resource DeviceRequest - Detailed Descriptions

URL: https://hl7.org/fhir/R4/devicerequest-definitions.html

12.19.6 Resource DeviceRequest - Detailed Descriptions

Orders and Observations Work Group | Maturity Level: 1 | Trial Use | Security Category: Patient | Compartments: Device, Encounter, Patient, Practitioner

Detailed Descriptions for the elements in the DeviceRequest resource.

DeviceRequest
Element Id | DeviceRequest
Definition | Represents a request for a patient to employ a medical device. The device may be an implantable device, or an external assistive device, such as a walker.
Cardinality | 0..*
Type | DomainResource
DeviceRequest.identifier
Element Id | DeviceRequest.identifier
Definition | Identifiers assigned to this order by the orderer or by the receiver.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Summary | true
DeviceRequest.instantiatesCanonical
Element Id | DeviceRequest.instantiatesCanonical
Definition | The URL pointing to a FHIR-defined protocol, guideline, orderset or other definition that is adhered to in whole or in part by this DeviceRequest.
Cardinality | 0..*
Type | canonical(ActivityDefinition | PlanDefinition)
Summary | true
Comments | Note: This is a business identifier, not a resource identifier (see discussion). It is best practice for the identifier to only appear on a single resource instance, however business practices may occasionally dictate that multiple resource instances with the same identifier can exist - possibly even with different resource types. For example, multiple Patient and a Person resource instance might share the same social insurance number.
DeviceRequest.instantiatesUri
Element Id | DeviceRequest.instantiatesUri

Definition | The URL pointing to an externally maintained protocol, guideline, orderset or other definition that is adhered to in whole or in part by this DeviceRequest.
Cardinality | 0..*
Type | uri
Summary | true
Comments | This might be an HTML page, PDF, etc. or could just be a non-resolvable URI identifier.
DeviceRequest.basedOn
Element Id | DeviceRequest.basedOn
Definition | Plan/proposal/order fulfilled by this request.
Cardinality | 0..*
Type | Reference(Any)
Summary | true
DeviceRequest.priorRequest
Element Id | DeviceRequest.priorRequest
Definition | The request takes the place of the referenced completed or terminated request(s).
Cardinality | 0..*
Type | Reference(Any)
Summary | true
DeviceRequest.groupIdentifier
Element Id | DeviceRequest.groupIdentifier
Definition | Composite request this is part of.
Cardinality | 0..1
Type | Identifier
Summary | true
DeviceRequest.status
Element Id | DeviceRequest.status
Definition | The status of the request.
Cardinality | 0..1
Terminology Binding | RequestStatus (Required)
Type | code
Is Modifier | true (Reason: This element is labeled as a modifier because it is a status element that contains status entered-in-error which means that the resource should not be treated as valid)
Summary | true
Comments | This element is labeled as a modifier because the status contains the codes cancelled and entered-in-error that mark the request as not currently valid.
DeviceRequest.intent
Element Id | DeviceRequest.intent
Definition | Whether the request is a proposal, plan, an original order or a reflex order.
Cardinality | 1..1
Terminology Binding | RequestIntent (Required)
Type | code
Is Modifier | true (Reason: This element changes the interpretation of all descriptive attributes. For example "the time the request is recommended to occur" vs. "the time the request is authorized to occur" or "who is recommended to perform the request" vs. "who is authorized to perform the request)
Summary | true
DeviceRequest.priority
Element Id | DeviceRequest.priority
Definition | Indicates how quickly the {{title}} should be addressed with respect to other requests.
Cardinality | 0..1
Terminology Binding | Request priority (Required)
Type | code
Meaning if Missing | If missing, normal priority
Summary | true
DeviceRequest.code[x]
Element Id | DeviceRequest.code[x]
Definition | The details of the device to be used.
Cardinality | 1..1
Terminology Binding | FHIR Device Types (Example)
Type | Reference(Device)|CodeableConcept
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true
DeviceRequest.parameter
Element Id | DeviceRequest.parameter
Definition | Specific parameters for the ordered item. For example, the prism value for lenses.
Cardinality | 0..*
DeviceRequest.parameter.code
Element Id | DeviceRequest.parameter.code
Definition | A code or string that identifies the device detail being asserted.
Cardinality | 0..1
Terminology Binding | ParameterCode:
Type | CodeableConcept
DeviceRequest.parameter.value[x]
Element Id | DeviceRequest.parameter.value[x]

Definition | The value of the device detail.
Cardinality | 0..1
Type | CodeableConcept|Quantity|Range|boolean
[x] Note | See Choice of Data Types for further information about how to use [x]
Comments | Range means device should have a value that falls somewhere within the specified range.
DeviceRequest.subject
Element Id | DeviceRequest.subject
Definition | The patient who will use the device.
Cardinality | 1..1
Type | Reference(Patient | Group | Location | Device)
Summary | true
DeviceRequest.encounter
Element Id | DeviceRequest.encounter
Definition | An encounter that provides additional context in which this request is made.
Cardinality | 0..1
Type | Reference(Encounter)
Summary | true
DeviceRequest.occurrence[x]
Element Id | DeviceRequest.occurrence[x]
Definition | The timing schedule for the use of the device. The Schedule data type allows many different expressions, for example. "Every 8 hours"; "Three times a day"; "1/2 an hour before breakfast for 10 days from 23-Dec 2011:"; "15 Oct 2013, 17 Oct 2013 and 1 Nov 2013".
Cardinality | 0..1
Type | dateTime|Period|Timing
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true
DeviceRequest.authoredOn
Element Id | DeviceRequest.authoredOn
Definition | When the request transitioned to being actionable.
Cardinality | 0..1
Type | dateTime
Summary | true
DeviceRequest.requester
Element Id | DeviceRequest.requester
Definition | The individual who initiated the request and has responsibility for its activation.
Cardinality | 0..1
Type | Reference(Device | Practitioner | PractitionerRole | Organization)
Summary | true
DeviceRequest.performerType
Element Id | DeviceRequest.performerType
Definition | Desired type of performer for doing the diagnostic testing.
Cardinality | 0..1
Terminology Binding | Participant Roles (Example)
Type | CodeableConcept
Summary | true
DeviceRequest.performer
Element Id | DeviceRequest.performer
Definition | The desired performer for doing the diagnostic testing.
Cardinality | 0..1
Type | Reference(Practitioner | PractitionerRole | Organization | CareTeam | HealthcareService | Patient | Device | RelatedPerson)
Summary | true
DeviceRequest.reasonCode
Element Id | DeviceRequest.reasonCode
Definition | Reason or justification for the use of this device.
Cardinality | 0..*
Terminology Binding | Condition/Problem/Diagnosis Codes (Example)
Type | CodeableConcept
Summary | true
DeviceRequest.reasonReference
Element Id | DeviceRequest.reasonReference
Definition | Reason or justification for the use of this device.
Cardinality | 0..*
Type | Reference(Condition | Observation | DiagnosticReport | DocumentReference)
Summary | true
DeviceRequest.insurance
Element Id | DeviceRequest.insurance
Definition | Insurance plans, coverage extensions, pre-authorizations and/or pre-determinations that may be required for delivering the requested service.
Cardinality | 0..*
Type | Reference(Coverage | ClaimResponse)
DeviceRequest.supportingInfo
Element Id | DeviceRequest.supportingInfo

Definition | Additional clinical information about the patient that may influence the request fulfilment. For example, this may include where on the subject's body the device will be used (i.e. the target site).
Cardinality | 0..*
Type | Reference(Any)
Requirements | Knowing where the device is targeted is important for tracking if multiple sites are possible.
DeviceRequest.note
Element Id | DeviceRequest.note
Definition | Details about this request that were not represented at all or sufficiently in one of the attributes provided in a class. These may include for example a comment, an instruction, or a note associated with the statement.
Cardinality | 0..*
Type | Annotation
DeviceRequest.relevantHistory
Element Id | DeviceRequest.relevantHistory
Definition | Key events in the history of the request.
Cardinality | 0..*
Type | Reference(Provenance)
Comments | This might not include provenances for all versions of the request - only those deemed "relevant" or important. This SHALL NOT include the Provenance associated with this current version of the resource. (If that provenance is deemed to be a "relevant" change, it will need to be added as part of a later update. Until then, it can be queried directly as the Provenance that points to this version using _revinclude All Provenances should have some historical version of this Request as their subject.

12.19.7 Resource DeviceRequest - Mappings

URL: https://hl7.org/fhir/R4/devicerequest-mappings.html

12.19.7 Resource DeviceRequest - Mappings

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Device, Encounter, Patient, Practitioner

Mappings:

FiveWs Pattern Mapping

Workflow Pattern

HL7 v2 Mapping

RIM Mapping

Quality Improvement and Clinical Knowledge (QUICK)

Mappings for the devicerequest resource (see Mappings to Other Standards for further information & status).

12.19.7.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

DeviceRequest
identifier | FiveWs.identifier
status | FiveWs.status
intent | FiveWs.class
priority | FiveWs.grade
code[x] | FiveWs.what[x]
parameter | FiveWs.what[x]
code | FiveWs.what[x]
value[x] | FiveWs.what[x]
subject | FiveWs.subject[x]
encounter | FiveWs.context
occurrence[x] | FiveWs.planned
authoredOn | FiveWs.recorded
requester | FiveWs.author
performerType | FiveWs.actor
performer | FiveWs.actor
reasonCode | FiveWs.why[x]
reasonReference | FiveWs.why[x]

12.19.7.2 Workflow Pattern (http://hl7.org/fhir/workflow)

DeviceRequest | Request
identifier | Request.identifier
instantiatesCanonical | Request.instantiatesCanonical
instantiatesUri | Request.instantiatesUri
basedOn | Request.basedOn
priorRequest | Request.replaces
groupIdentifier | Request.groupIdentifier
status | Request.status
intent | Request.intent
priority | Request.priority
code[x] | Request.code
subject | Request.subject

encounter | Request.encounter
occurrence[x] | Request.occurrence[x]
authoredOn | Request.authoredOn
requester | Request.requester
performerType | Request.performerType
performer | Request.performer
reasonCode | Request.reasonCode
reasonReference | Request.reasonReference
insurance | Request.insurance
supportingInfo | Request.supportingInfo
note | Request.note
relevantHistory | Request.relevantHistory

12.19.7.3 HL7 v2 Mapping (http://hl7.org/v2)

DeviceRequest | ORC
identifier | ORC.2, ORC.3
instantiatesCanonical | Varies by domain
instantiatesUri | Varies by domain
basedOn | ORC.8 (plus others)
priorRequest | Handled by message location of ORC (ORC.1=RO or RU)
groupIdentifier | ORC.4
status | ORC.5
intent | N/A
priority | TQ1.9
code[x] | Varies by domain
parameter | Varies by domain
code | Varies by domain
value[x] | Varies by domain
subject | Accompanying PID segment
encounter | Accompanying PV1
occurrence[x] | Accompanying TQ1/TQ2 segments
authoredOn | ORC.9
requester | ORC.12
performerType | PRT
performer | PRT
reasonCode | ORC.16
reasonReference | ORC.16
insurance | IN1/IN2
supportingInfo | Accompanying segments
note | NTE
relevantHistory | N/A

12.19.7.4 RIM Mapping (http://hl7.org/v3)

DeviceRequest | Act[moodCode<=INT]
identifier | .identifier
instantiatesCanonical | .outboundRelationship[typeCode=DEFN].target
instantiatesUri | .outboundRelationship[typeCode=DEFN].target
basedOn | .outboundRelationship[typeCode=FLFS].target
priorRequest | .outboundRelationship[typeCode=RPLC].target
groupIdentifier | .inboundRelationship(typeCode=COMP].source[moodCode=INT].identifier
status | .status
intent | .moodCode (nuances beyond PRP/PLAN/RQO would need to be elsewhere)
priority | .priorityCode
code[x] | .code
parameter | .code
code | .code
value[x] | .code
subject | .participation[typeCode=SBJ].role
encounter | .inboundRelationship(typeCode=COMP].source[classCode<=PCPR, moodCode=EVN]
occurrence[x] | .effectiveTime
authoredOn | .participation[typeCode=AUT].time
requester | .participation[typeCode=AUT].role
performerType | .participation[typeCode=PRF].role[scoper.determinerCode=KIND].code
performer | .participation[typeCode=PRF].role[scoper.determinerCode=INSTANCE]
reasonCode | .reasonCode
reasonReference | .outboundRelationship[typeCode=RSON].target
insurance | .outboundRelationship[typeCode=COVBY].target
supportingInfo | .outboundRelationship[typeCode=PERT].target
note | .inboundRelationship(typeCode=SUBJ].source[classCode=ANNGEN, moodCode=EVN].value[xsi:type=ST]
relevantHistory | .inboundRelationship(typeCode=SUBJ].source[classCode=CACT, moodCode=EVN]

12.19.7.5 Quality Improvement and Clinical Knowledge (QUICK) (http://siframework.org/cqf)

DeviceRequest
identifier | ClinicalStatement.identifier
instantiatesCanonical
instantiatesUri
basedOn | Proposal.prnReason.reason
priorRequest
groupIdentifier
status | Action.currentStatus
intent
priority
code[x] | DeviceUse.device
parameter | DeviceUse.device
code | DeviceUse.device
value[x] | DeviceUse.device
subject | ClinicalStatement.subject
encounter | ClinicalStatement.encounter
occurrence[x] | DeviceUse.applicationSchedule
authoredOn | ClinicalStatement.statementDateTime
requester
performerType
performer
reasonCode | Action.indication.reason
reasonReference
insurance
supportingInfo
note | ClinicalStatement.additionalText
relevantHistory

12.19.8 Resource DeviceRequest - Extensions & Profiles

URL: https://hl7.org/fhir/R4/devicerequest-profiles.html

12.19.8 Resource DeviceRequest - Extensions & Profiles

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Device, Encounter, Patient, Practitioner

This table lists profiles and extensions for the DeviceRequest resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

12.19.8.1 Profiles

No Profiles defined for this resource

12.19.8.2 Extensions

devicerequest-patientInstruction | DeviceRequest | patientInstruction | for DeviceRequest HL7 Extensions
procedure-approachBodyStructure | DeviceRequest | approachBodyStructure | for Procedure HL7 Extensions
request-statusReason | DeviceRequest | statusReason | for Request Pattern HL7 Extensions
workflow-episodeOfCare | DeviceRequest | episodeOfCare | for Workflow Pattern HL7 Extensions
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

12.19.8.3 Search Extensions

No Search Extensions defined for this resource

12.19.9 Resource DeviceRequest - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/devicerequest-version-maps.html

No validation errors - all conversions are clean

12.19.9 Resource DeviceRequest - R3/R4 Conversions

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Device, Encounter, Patient, Practitioner

R3 : R4 Conversion maps for DeviceRequest.

Functional status for this map: 2 tests that all execute ok. 2 fail round-trip testing and 1 r3 resources are invalid (0 errors). (see documentation)

12.19.9.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/DeviceRequest3to4" = "R3 to R4 Conversions for DeviceRequest"

uses "http://hl7.org/fhir/3.0/StructureDefinition/DeviceRequest" alias DeviceRequestR3 as source
uses "http://hl7.org/fhir/StructureDefinition/DeviceRequest" alias DeviceRequest as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group DeviceRequest(source src : DeviceRequestR3, target tgt : DeviceRequest) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.basedOn -> tgt.basedOn;
  src.priorRequest -> tgt.priorRequest;
  src.groupIdentifier -> tgt.groupIdentifier;
  src.status -> tgt.status;
  src.intent as vs then {
    vs.coding as c then {
      c.code -> tgt.intent;
    };
  };
  src.priority -> tgt.priority;
  src.code -> tgt.code;
  src.parameter as s -> tgt.parameter as t then DeviceRequestParameter(s, t);
  src.subject -> tgt.subject;
  src.context -> tgt.encounter;
  src.occurrence -> tgt.occurrence;
  src.authoredOn -> tgt.authoredOn;
  src.requester as vs then {
    vs.agent -> tgt.requester;
  };
  src.performerType -> tgt.performerType;
  src.performer -> tgt.performer;
  src.reasonCode -> tgt.reasonCode;
  src.reasonReference -> tgt.reasonReference;
  src.insurance -> tgt.insurance;
  src.supportingInfo -> tgt.supportingInfo;
  src.note -> tgt.note;
  src.relevantHistory -> tgt.relevantHistory;
}

group DeviceRequestParameter(source src, target tgt) extends BackboneElement {
  src.code -> tgt.code;
  src.value -> tgt.value;
}
```

12.19.9.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/DeviceRequest4to3" = "R4 to R3 Conversion for DeviceRequest"

uses "http://hl7.org/fhir/StructureDefinition/DeviceRequest" alias DeviceRequest as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/DeviceRequest" alias DeviceRequestR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group DeviceRequest(source src : DeviceRequestR3, target tgt : DeviceRequest) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.basedOn -> tgt.basedOn;
  src.priorRequest -> tgt.priorRequest;
  src.groupIdentifier -> tgt.groupIdentifier;
  src.status -> tgt.status;
  src.intent as vs ->  tgt.intent as vt,  vt.coding as c,  c.system = 'http://hl7.org/fhir/request-intent',  c.code =
    vs;
  src.priority -> tgt.priority;
  src.code : CodeableConcept as vs -> tgt.code = create('CodeableConcept') as vt then CodeableConcept(vs, vt);
  src.code : Reference as vs -> tgt.code = create('Reference') as vt then Reference(vs, vt);
  src.parameter as s -> tgt.parameter as t then DeviceRequestParameter(s, t);
  src.subject -> tgt.subject;
  src.encounter -> tgt.context;
  src.occurrence -> tgt.occurrence;
  src.authoredOn -> tgt.authoredOn;
  src.requester as vs ->  tgt.requester as vt,  vt.agent = vs;
  src.performerType -> tgt.performerType;
  src.performer -> tgt.performer;
  src.reasonCode -> tgt.reasonCode;
  src.reasonReference -> tgt.reasonReference;
  src.insurance -> tgt.insurance;
  src.supportingInfo -> tgt.supportingInfo;
  src.note -> tgt.note;
```

```
  src.relevantHistory -> tgt.relevantHistory;
}

group DeviceRequestParameter(source src, target tgt) extends BackboneElement {
  src.code -> tgt.code;
  src.value -> tgt.value;
}
```

12.19.9.3 R4 Validation Errors

12.19.10 DeviceRequest HL7 Extensions

URL: https://hl7.org/fhir/R4/devicerequest-extensions.html

Defines common extensions used with or related to the DeviceRequest resource

12.19.10 DeviceRequest HL7 Extensions

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative

12.19.10.1 Content

Extensions:

devicerequest-patientInstruction | patientInstruction : Simple concise instructions to be read by the patient. For example “twice a day” rather than “BID.”.

12.19.11 JSON Schema for DeviceRequest

URL: https://hl7.org/fhir/R4/devicerequest.schema.json.html

12.19.11 JSON Schema for DeviceRequest

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Device, Encounter, Patient, Practitioner

This is a representation of the json schema for DeviceRequest, which is just a part of the full JSON Schema.

```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/DeviceRequest",
  "$ref": "#/definitions/DeviceRequest",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "DeviceRequest": {
      "description": "Represents a request for a patient to employ a medical device. The device may be an implantable
    device, or an external assistive device, such as a walker.",
      "properties": {
        "resourceType": {
          "description": "This is a DeviceRequest resource",
          "const": "DeviceRequest"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
          "$ref": "#/definitions/code"
        },
        "_language": {
          "description": "Extensions for language",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "text": {
          "description": "A human-readable narrative that contains a summary of the resource and can be used to
    represent the content of the resource to a human. The narrative need not encode all the structured data, but is
    required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
    Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
          "$ref": "Narrative.schema.json#/definitions/Narrative"
        },
        "contained": {
```

```
      "description": "These resources do not have an independent existence apart from the resource that contains
them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
      "items": {
        "$ref": "ResourceList.schema.json#/definitions/ResourceList"
      },
      "type": "array"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
 definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource and that modifies the understanding of the element that contains it and/or the understanding of the
containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
met as part of the definition of the extension. Applications processing a resource are required to check for
modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "identifier": {
      "description": "Identifiers assigned to this order by the orderer or by the receiver.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "instantiatesCanonical": {
      "description": "The URL pointing to a FHIR-defined protocol, guideline, orderset or other definition that is
adhered to in whole or in part by this DeviceRequest.",
      "items": {
        "$ref": "canonical.schema.json#/definitions/canonical"
      },
      "type": "array"
    },
    "instantiatesUri": {
      "description": "The URL pointing to an externally maintained protocol, guideline, orderset or other definition
 that is adhered to in whole or in part by this DeviceRequest.",
      "items": {
        "$ref": "#/definitions/uri"
      },
      "type": "array"
    },
    "_instantiatesUri": {
      "description": "Extensions for instantiatesUri",
      "items": {
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "type": "array"
    },
    "basedOn": {
      "description": "Plan/proposal/order fulfilled by this request.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "priorRequest": {
      "description": "The request takes the place of the referenced completed or terminated request(s).",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "groupIdentifier": {
      "description": "Composite request this is part of.",
      "$ref": "Identifier.schema.json#/definitions/Identifier"
    },
    "status": {
      "description": "The status of the request.",
      "$ref": "#/definitions/code"
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "intent": {
      "description": "Whether the request is a proposal, plan, an original order or a reflex order.",
```

```
      "$ref": "#/definitions/code"
    },
    "_intent": {
      "description": "Extensions for intent",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "priority": {
      "description": "Indicates how quickly the {{title}} should be addressed with respect to other requests.",
      "$ref": "#/definitions/code"
    },
    "_priority": {
      "description": "Extensions for priority",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "codeReference": {
      "description": "The details of the device to be used.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "codeCodeableConcept": {
      "description": "The details of the device to be used.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "parameter": {
      "description": "Specific parameters for the ordered item.  For example, the prism value for lenses.",
      "items": {
        "$ref": "#/definitions/DeviceRequest_Parameter"
      },
      "type": "array"
    },
    "subject": {
      "description": "The patient who will use the device.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "encounter": {
      "description": "An encounter that provides additional context in which this request is made.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "occurrenceDateTime": {
      "description": "The timing schedule for the use of the device. The Schedule data type allows many different
expressions, for example. \"Every 8 hours\"; \"Three times a day\"; \"1/2 an hour before breakfast for 10 days from
23-Dec 2011:\"; \"15 Oct 2013, 17 Oct 2013 and 1 Nov 2013\".",
      "pattern": "^([0-9]([0-9]([0-9][1-9]|[1-9]0)|[1-9]00)|[1-9]000)(-(0[1-9]|1[0-2])(-(0[1-9]|[1-2][0-9]|3[0-
1])(T([01][0-9]|2[0-3]):[0-5][0-9]:([0-5][0-9]|60)(\\.[0-9]+)?(Z|(\\+|-)((0[0-9]|1[0-3]):[0-5][0-9]|14:00)))?)?)?$",
      "type": "string"
    },
    "_occurrenceDateTime": {
      "description": "Extensions for occurrenceDateTime",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "occurrencePeriod": {
      "description": "The timing schedule for the use of the device. The Schedule data type allows many different
expressions, for example. \"Every 8 hours\"; \"Three times a day\"; \"1/2 an hour before breakfast for 10 days from
23-Dec 2011:\"; \"15 Oct 2013, 17 Oct 2013 and 1 Nov 2013\".",
      "$ref": "Period.schema.json#/definitions/Period"
    },
    "occurrenceTiming": {
      "description": "The timing schedule for the use of the device. The Schedule data type allows many different
expressions, for example. \"Every 8 hours\"; \"Three times a day\"; \"1/2 an hour before breakfast for 10 days from
23-Dec 2011:\"; \"15 Oct 2013, 17 Oct 2013 and 1 Nov 2013\".",
      "$ref": "Timing.schema.json#/definitions/Timing"
    },
    "authoredOn": {
      "description": "When the request transitioned to being actionable.",
      "$ref": "#/definitions/dateTime"
    },
    "_authoredOn": {
      "description": "Extensions for authoredOn",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "requester": {
      "description": "The individual who initiated the request and has responsibility for its activation.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "performerType": {
      "description": "Desired type of performer for doing the diagnostic testing.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "performer": {
      "description": "The desired performer for doing the diagnostic testing.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "reasonCode": {
      "description": "Reason or justification for the use of this device.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "reasonReference": {
```

```
      "description": "Reason or justification for the use of this device.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "insurance": {
      "description": "Insurance plans, coverage extensions, pre-authorizations and/or pre-determinations that may be
 required for delivering the requested service.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "supportingInfo": {
      "description": "Additional clinical information about the patient that may influence the request fulfilment.
For example, this may include where on the subject\u0027s body the device will be used (i.e. the target site).",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "note": {
      "description": "Details about this request that were not represented at all or sufficiently in one of the
attributes provided in a class. These may include for example a comment, an instruction, or a note associated with
the statement.",
      "items": {
        "$ref": "Annotation.schema.json#/definitions/Annotation"
      },
      "type": "array"
    },
    "relevantHistory": {
      "description": "Key events in the history of the request.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "subject",
    "resourceType"
  ]
},
"DeviceRequest_Parameter": {
  "description": "Represents a request for a patient to employ a medical device. The device may be an implantable
device, or an external assistive device, such as a walker.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "code": {
      "description": "A code or string that identifies the device detail being asserted.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "valueCodeableConcept": {
      "description": "The value of the device detail.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "valueQuantity": {
      "description": "The value of the device detail.",
      "$ref": "Quantity.schema.json#/definitions/Quantity"
```

```
        },
        "valueRange": {
          "description": "The value of the device detail.",
          "$ref": "Range.schema.json#/definitions/Range"
        },
        "valueBoolean": {
          "description": "The value of the device detail.",
          "pattern": "^true|false$",
          "type": "boolean"
        },
        "_valueBoolean": {
          "description": "Extensions for valueBoolean",
          "$ref": "Element.schema.json#/definitions/Element"
        }
      },
      "additionalProperties": false
    }
  }
}
```

12.19.12 Resource DeviceRequest - Maturity Dependency View

URL: https://hl7.org/fhir/R4/devicerequest-dependencies.html

12.19.12 Resource DeviceRequest - Maturity Dependency View

Orders and Observations Work Group | Maturity Level: 1 | Trial Use | Compartments: Device, Encounter, Patient, Practitioner

Dependency Graph for DeviceRequest FMM level 1

Name | Card. | Type | Dependency Analysis
DeviceRequest | | DomainResource
identifier | 0..* | Identifier
instantiatesCanonical | 0..* | canonical(ActivityDefinition | PlanDefinition)
instantiatesUri | 0..* | uri
basedOn | 0..* | Reference(Any)
priorRequest | 0..* | Reference(Any)
groupIdentifier | 0..1 | Identifier
status | 0..1 | code
intent | 1..1 | code
priority | 0..1 | code
code[x] | 1..1
codeReference | | Reference(Device
codeCodeableConcept | | CodeableConcept
parameter | 0..* | BackboneElement
code | 0..1 | CodeableConcept
value[x] | 0..1
valueCodeableConcept | | CodeableConcept
valueQuantity | | Quantity
valueRange | | Range
valueBoolean | | boolean
subject | 1..1 | Reference(Patient | Group | Location | Device)
encounter | 0..1 | Reference(Encounter)
occurrence[x] | 0..1
occurrenceDateTime | | dateTime
occurrencePeriod | | Period
occurrenceTiming | | Timing
authoredOn | 0..1 | dateTime
requester | 0..1 | Reference(Device | Practitioner | PractitionerRole | Organization)
performerType | 0..1 | CodeableConcept
performer | 0..1 | Reference(Practitioner | PractitionerRole | Organization | CareTeam | HealthcareService | Patient | Device | RelatedPerson)
reasonCode | 0..* | CodeableConcept
reasonReference | 0..* | Reference(Condition | Observation | DiagnosticReport | DocumentReference)
insurance | 0..* | Reference(Coverage | ClaimResponse)
supportingInfo | 0..* | Reference(Any)
note | 0..* | Annotation
relevantHistory | 0..* | Reference(Provenance)

12.20 Resource DeviceUseStatement - Content

URL: https://hl7.org/fhir/R4/deviceusestatement.html
Orders and Observations Work Group | Maturity Level: 0 | Trial Use | Security Category: Patient | Compartments: Device, Patient

A record of a device being used by a patient where the record is the result of a report from the patient or another clinician.

12.20.1 Scope and Usage

These resources have not yet undergone proper review by PC, CQI, CDS, and OO. At this time, they are to be considered only as draft resource proposals for potential submission.

This resource is an event resource from a FHIR workflow perspective - see Workflow. It is the intent of the Orders and Observation Workgroup to align this resource with the workflow pattern for event resources.

This resource records the use of a healthcare-related device by a patient. The record is the result of a report of use by the patient, another provider or a related person. The resource can be used to note the use of an assistive device such as a wheelchair or hearing aid, a contraceptive such an intra-uterine device, or other implanted devices such as a pacemaker.

12.20.2 Boundaries and Relationships

This resource is different from DeviceRequest which records a request to use the device. This also is distinct from the Procedure resource which may describe the implantation or explantation of a device.

12.20.3 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
DeviceUseStatement | TU | | DomainResource | Record of use of a device Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | External identifier for this record
basedOn | Σ | 0..* | Reference(ServiceRequest) | Fulfills plan, proposal or order
status | ?!Σ | 1..1 | code | active | completed | entered-in-error + DeviceUseStatementStatus (Required)
subject | Σ | 1..1 | Reference(Patient | Group) | Patient using device
derivedFrom | Σ | 0..* | Reference(ServiceRequest | Procedure | Claim | Observation | QuestionnaireResponse | DocumentReference) | Supporting information
timing[x] | Σ | 0..1 | | How often the device was used
timingTiming | | | Timing
timingPeriod | | | Period
timingDateTime | | | dateTime
recordedOn | Σ | 0..1 | dateTime | When statement was recorded
source | Σ | 0..1 | Reference(Patient | Practitioner | PractitionerRole | RelatedPerson) | Who made the statement
device | Σ | 1..1 | Reference(Device) | Reference to device used
reasonCode | Σ | 0..* | CodeableConcept | Why device was used
reasonReference | Σ | 0..* | Reference(Condition | Observation | DiagnosticReport | DocumentReference | Media) | Why was DeviceUseStatement performed?
bodySite | Σ | 0..1 | CodeableConcept | Target body site SNOMED CT Body Structures (Example)
note | | 0..* | Annotation | Addition details (comments, instructions)
Documentation for this format

UML Diagram (Legend)

XML Template

```
<DeviceUseStatement xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier External identifier for this record --></identifier>
 <basedOn><!-- 0..* Reference(ServiceRequest) Fulfills plan, proposal or order --></basedOn>
 <status value="[code]"/><!-- 1..1 active | completed | entered-in-error + -->
 <subject><!-- 1..1 Reference(Patient|Group) Patient using device --></subject>
 <derivedFrom><!-- 0..* Reference(ServiceRequest|Procedure|Claim|Observation|
   QuestionnaireResponse|DocumentReference) Supporting information --></derivedFrom>
 <timing[x]><!-- 0..1 Timing|Period|dateTime How often  the device was used --></timing[x]>
 <recordedOn value="[dateTime]"/><!-- 0..1 When statement was recorded -->
 <source><!-- 0..1 Reference(Patient|Practitioner|PractitionerRole|RelatedPerson) Who made the statement --></source>
 <device><!-- 1..1 Reference(Device) Reference to device used --></device>
 <reasonCode><!-- 0..* CodeableConcept Why device was used --></reasonCode>
 <reasonReference><!-- 0..* Reference(Condition|Observation|DiagnosticReport|
   DocumentReference|Media) Why was DeviceUseStatement performed? --></reasonReference>
 <bodySite><!-- 0..1 CodeableConcept Target body site --></bodySite>
 <note><!-- 0..* Annotation Addition details (comments, instructions) --></note>
</DeviceUseStatement>
```

JSON Template

```
{
  "resourceType" : "DeviceUseStatement",
  // from Resource: id, meta, implicitRules, and language
```

```
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // External identifier for this record
  "basedOn" : [{ Reference(ServiceRequest) }], // Fulfills plan, proposal or order
  "status" : "<code>", // R!  active | completed | entered-in-error +
  "subject" : { Reference(Patient|Group) }, // R!  Patient using device
  "derivedFrom" : [{ Reference(ServiceRequest|Procedure|Claim|Observation|
   QuestionnaireResponse|DocumentReference) }], // Supporting information
  // timing[x]: How often  the device was used. One of these 3:
  "timingTiming" : { Timing },
  "timingPeriod" : { Period },
  "timingDateTime" : "<dateTime>",
  "recordedOn" : "<dateTime>", // When statement was recorded
  "source" : { Reference(Patient|Practitioner|PractitionerRole|RelatedPerson) }, // Who made the statement
  "device" : { Reference(Device) }, // R!  Reference to device used
  "reasonCode" : [{ CodeableConcept }], // Why device was used
  "reasonReference" : [{ Reference(Condition|Observation|DiagnosticReport|
   DocumentReference|Media) }], // Why was DeviceUseStatement performed?
  "bodySite" : { CodeableConcept }, // Target body site
  "note" : [{ Annotation }] // Addition details (comments, instructions)
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:DeviceUseStatement;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:DeviceUseStatement.identifier [ Identifier ], ... ; # 0..* External identifier for this record
  fhir:DeviceUseStatement.basedOn [ Reference(ServiceRequest) ], ... ; # 0..* Fulfills plan, proposal or order
  fhir:DeviceUseStatement.status [ code ]; # 1..1 active | completed | entered-in-error +
  fhir:DeviceUseStatement.subject [ Reference(Patient|Group) ]; # 1..1 Patient using device
  fhir:DeviceUseStatement.derivedFrom [ Reference(ServiceRequest|Procedure|Claim|Observation|QuestionnaireResponse|
  DocumentReference) ], ... ; # 0..* Supporting information
  # DeviceUseStatement.timing[x] : 0..1 How often  the device was used. One of these 3
    fhir:DeviceUseStatement.timingTiming [ Timing ]
    fhir:DeviceUseStatement.timingPeriod [ Period ]
    fhir:DeviceUseStatement.timingDateTime [ dateTime ]
  fhir:DeviceUseStatement.recordedOn [ dateTime ]; # 0..1 When statement was recorded
  fhir:DeviceUseStatement.source [ Reference(Patient|Practitioner|PractitionerRole|RelatedPerson) ]; # 0..1 Who made the
     statement
  fhir:DeviceUseStatement.device [ Reference(Device) ]; # 1..1 Reference to device used
  fhir:DeviceUseStatement.reasonCode [ CodeableConcept ], ... ; # 0..* Why device was used
  fhir:DeviceUseStatement.reasonReference [ Reference(Condition|Observation|DiagnosticReport|DocumentReference|Media) ],
     ... ; # 0..* Why was DeviceUseStatement performed?
  fhir:DeviceUseStatement.bodySite [ CodeableConcept ]; # 0..1 Target body site
  fhir:DeviceUseStatement.note [ Annotation ], ... ; # 0..* Addition details (comments, instructions)
]
```

Changes since R3

DeviceUseStatement
DeviceUseStatement.basedOn | Added Element
DeviceUseStatement.status | Change value set from http://hl7.org/fhir/ValueSet/device-statement-status to http://hl7.org/fhir/ValueSet/device-statement-status|4.0.1
DeviceUseStatement.derivedFrom | Added Element
DeviceUseStatement.source | Type Reference: Added Target Type PractitionerRole
DeviceUseStatement.reasonCode | Added Element
DeviceUseStatement.reasonReference | Added Element
DeviceUseStatement.whenUsed | deleted
DeviceUseStatement.indication | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 1 test that all execute ok. All tests pass round-trip testing and all r3 resources are valid.)

Structure

Name | Flags | Card. | Type | Description & Constraints
DeviceUseStatement | TU | | DomainResource | Record of use of a device Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | External identifier for this record
basedOn | Σ | 0..* | Reference(ServiceRequest) | Fulfills plan, proposal or order
status | ?!Σ | 1..1 | code | active | completed | entered-in-error + DeviceUseStatementStatus (Required)
subject | Σ | 1..1 | Reference(Patient | Group) | Patient using device
derivedFrom | Σ | 0..* | Reference(ServiceRequest | Procedure | Claim | Observation | QuestionnaireResponse | DocumentReference) | Supporting information
timing[x] | Σ | 0..1 | | How often the device was used
timingTiming | | | Timing
timingPeriod | | | Period
timingDateTime | | | dateTime

recordedOn | Σ | 0..1 | dateTime | When statement was recorded
source | Σ | 0..1 | Reference(Patient | Practitioner | PractitionerRole | RelatedPerson) | Who made the statement
device | Σ | 1..1 | Reference(Device) | Reference to device used
reasonCode | Σ | 0..* | CodeableConcept | Why device was used
reasonReference | Σ | 0..* | Reference(Condition | Observation | DiagnosticReport | DocumentReference | Media) | Why was DeviceUseStatement performed?
bodySite | Σ | 0..1 | CodeableConcept | Target body site SNOMED CT Body Structures (Example)
note | | 0..* | Annotation | Addition details (comments, instructions)
Documentation for this format

UML Diagram (Legend)

XML Template

```
<DeviceUseStatement xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier External identifier for this record --></identifier>
 <basedOn><!-- 0..* Reference(ServiceRequest) Fulfills plan, proposal or order --></basedOn>
 <status value="[code]"/><!-- 1..1 active | completed | entered-in-error + -->
 <subject><!-- 1..1 Reference(Patient|Group) Patient using device --></subject>
 <derivedFrom><!-- 0..* Reference(ServiceRequest|Procedure|Claim|Observation|
   QuestionnaireResponse|DocumentReference) Supporting information --></derivedFrom>
 <timing[x]><!-- 0..1 Timing|Period|dateTime How often  the device was used --></timing[x]>
 <recordedOn value="[dateTime]"/><!-- 0..1 When statement was recorded -->
 <source><!-- 0..1 Reference(Patient|Practitioner|PractitionerRole|RelatedPerson) Who made the statement --></source>
 <device><!-- 1..1 Reference(Device) Reference to device used --></device>
 <reasonCode><!-- 0..* CodeableConcept Why device was used --></reasonCode>
 <reasonReference><!-- 0..* Reference(Condition|Observation|DiagnosticReport|
   DocumentReference|Media) Why was DeviceUseStatement performed? --></reasonReference>
 <bodySite><!-- 0..1 CodeableConcept Target body site --></bodySite>
 <note><!-- 0..* Annotation Addition details (comments, instructions) --></note>
</DeviceUseStatement>
```

JSON Template

```
{
  "resourceType" : "DeviceUseStatement",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // External identifier for this record
  "basedOn" : [{ Reference(ServiceRequest) }], // Fulfills plan, proposal or order
  "status" : "<code>", // R!  active | completed | entered-in-error +
  "subject" : { Reference(Patient|Group) }, // R!  Patient using device
  "derivedFrom" : [{ Reference(ServiceRequest|Procedure|Claim|Observation|
   QuestionnaireResponse|DocumentReference) }], // Supporting information
  // timing[x]: How often  the device was used. One of these 3:
  "timingTiming" : { Timing },
  "timingPeriod" : { Period },
  "timingDateTime" : "<dateTime>",
  "recordedOn" : "<dateTime>", // When statement was recorded
  "source" : { Reference(Patient|Practitioner|PractitionerRole|RelatedPerson) }, // Who made the statement
  "device" : { Reference(Device) }, // R!  Reference to device used
  "reasonCode" : [{ CodeableConcept }], // Why device was used
  "reasonReference" : [{ Reference(Condition|Observation|DiagnosticReport|
   DocumentReference|Media) }], // Why was DeviceUseStatement performed?
  "bodySite" : { CodeableConcept }, // Target body site
  "note" : [{ Annotation }] // Addition details (comments, instructions)
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:DeviceUseStatement;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:DeviceUseStatement.identifier [ Identifier ], ... ; # 0..* External identifier for this record
  fhir:DeviceUseStatement.basedOn [ Reference(ServiceRequest) ], ... ; # 0..* Fulfills plan, proposal or order
  fhir:DeviceUseStatement.status [ code ]; # 1..1 active | completed | entered-in-error +
  fhir:DeviceUseStatement.subject [ Reference(Patient|Group) ]; # 1..1 Patient using device
  fhir:DeviceUseStatement.derivedFrom [ Reference(ServiceRequest|Procedure|Claim|Observation|QuestionnaireResponse|
  DocumentReference) ], ... ; # 0..* Supporting information
  # DeviceUseStatement.timing[x] : 0..1 How often  the device was used. One of these 3
    fhir:DeviceUseStatement.timingTiming [ Timing ]
    fhir:DeviceUseStatement.timingPeriod [ Period ]
    fhir:DeviceUseStatement.timingDateTime [ dateTime ]
  fhir:DeviceUseStatement.recordedOn [ dateTime ]; # 0..1 When statement was recorded
  fhir:DeviceUseStatement.source [ Reference(Patient|Practitioner|PractitionerRole|RelatedPerson) ]; # 0..1 Who made the
     statement
  fhir:DeviceUseStatement.device [ Reference(Device) ]; # 1..1 Reference to device used
  fhir:DeviceUseStatement.reasonCode [ CodeableConcept ], ... ; # 0..* Why device was used
  fhir:DeviceUseStatement.reasonReference [ Reference(Condition|Observation|DiagnosticReport|DocumentReference|Media) ],
     ... ; # 0..* Why was DeviceUseStatement performed?
  fhir:DeviceUseStatement.bodySite [ CodeableConcept ]; # 0..1 Target body site
  fhir:DeviceUseStatement.note [ Annotation ], ... ; # 0..* Addition details (comments, instructions)
]
```

Changes since Release 3

DeviceUseStatement
DeviceUseStatement.basedOn | Added Element
DeviceUseStatement.status | Change value set from http://hl7.org/fhir/ValueSet/device-statement-status to http://hl7.org/fhir/ValueSet/device-statement-status|4.0.1
DeviceUseStatement.derivedFrom | Added Element
DeviceUseStatement.source | Type Reference: Added Target Type PractitionerRole
DeviceUseStatement.reasonCode | Added Element
DeviceUseStatement.reasonReference | Added Element
DeviceUseStatement.whenUsed | deleted
DeviceUseStatement.indication | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 1 test that all execute ok. All tests pass round-trip testing and all r3 resources are valid.)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

12.20.3.1 Terminology Bindings

Path | Definition | Type | Reference
DeviceUseStatement.status | A coded concept indicating the current status of the Device Usage. | Required | DeviceUseStatementStatus
DeviceUseStatement.bodySite | Codes describing anatomical locations. May include laterality. | Example | SNOMEDCTBodyStructures

Notes to reviewers:

At this time, the code bindings are placeholders to be fleshed out upon further review by the community.

12.20.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
device | reference | Search by device | DeviceUseStatement.device (Device)
identifier | token | Search by identifier | DeviceUseStatement.identifier
patient | reference | Search by subject - a patient | DeviceUseStatement.subject (Group, Patient) | 33 Resources
subject | reference | Search by subject | DeviceUseStatement.subject (Group, Patient)

12.20.5 Resource DeviceUseStatement - Examples

URL: https://hl7.org/fhir/R4/deviceusestatement-examples.html

12.20.5 Resource DeviceUseStatement - Examples

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Device, Patient

An introduction to this page as well as proper examples are to be submitted shortly.

Example Name | id | Format
The provision of a wheelchair to a patient | example | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

12.20.6 Resource DeviceUseStatement - Detailed Descriptions

URL: https://hl7.org/fhir/R4/deviceusestatement-definitions.html

12.20.6 Resource DeviceUseStatement - Detailed Descriptions

Orders and Observations Work Group | Maturity Level: 0 | Trial Use | Security Category: Patient | Compartments: Device, Patient

Detailed Descriptions for the elements in the DeviceUseStatement resource.

DeviceUseStatement
Element Id | DeviceUseStatement
Definition | A record of a device being used by a patient where the record is the result of a report from the patient or another clinician.
Cardinality | 0..*
Type | DomainResource
DeviceUseStatement.identifier
Element Id | DeviceUseStatement.identifier
Definition | An external identifier for this statement such as an IRI.
Note | This is a business identifier, not a resource identifier (see discussion)

Cardinality | 0..*
Type | Identifier
Summary | true
DeviceUseStatement.basedOn
Element Id | DeviceUseStatement.basedOn
Definition | A plan, proposal or order that is fulfilled in whole or in part by this DeviceUseStatement.
Cardinality | 0..*
Type | Reference(ServiceRequest)
Requirements | Allows tracing of authorization for the DeviceUseStatement and tracking whether proposals/recommendations were acted upon.
Alternate Names | fulfills
Summary | true
DeviceUseStatement.status
Element Id | DeviceUseStatement.status
Definition | A code representing the patient or other source's judgment about the state of the device used that this statement is about. Generally this will be active or completed.
Cardinality | 1..1
Terminology Binding | DeviceUseStatementStatus (Required)
Type | code
Is Modifier | true (Reason: This element is labelled as a modifier because it is a status element that contains status entered-in-error which means that the resource should not be treated as valid)
Summary | true
Comments | DeviceUseStatment is a statement at a point in time. The status is only representative at the point when it was asserted. The value set for contains codes that assert the status of the use by the patient (for example, stopped or on hold) as well as codes that assert the status of the resource itself (for example, entered in error). This element is labeled as a modifier because the status contains the codes that mark the statement as not currently valid.
DeviceUseStatement.subject
Element Id | DeviceUseStatement.subject
Definition | The patient who used the device.
Cardinality | 1..1
Type | Reference(Patient | Group)
Summary | true
DeviceUseStatement.derivedFrom
Element Id | DeviceUseStatement.derivedFrom
Definition | Allows linking the DeviceUseStatement to the underlying Request, or to other information that supports or is used to derive the DeviceUseStatement.
Cardinality | 0..*
Type | Reference(ServiceRequest | Procedure | Claim | Observation | QuestionnaireResponse | DocumentReference)
Summary | true
Comments | The most common use cases for deriving a DeviceUseStatement comes from creating it from a request or from an observation or a claim. it should be noted that the amount of information that is available varies from the type resource that you derive the DeviceUseStatement from.
DeviceUseStatement.timing[x]
Element Id | DeviceUseStatement.timing[x]
Definition | How often the device was used.
Cardinality | 0..1
Type | Timing|Period|dateTime
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true
DeviceUseStatement.recordedOn
Element Id | DeviceUseStatement.recordedOn
Definition | The time at which the statement was made/recorded.
Cardinality | 0..1
Type | dateTime
Summary | true
DeviceUseStatement.source
Element Id | DeviceUseStatement.source
Definition | Who reported the device was being used by the patient.
Cardinality | 0..1
Type | Reference(Patient | Practitioner | PractitionerRole | RelatedPerson)
Summary | true
DeviceUseStatement.device
Element Id | DeviceUseStatement.device
Definition | The details of the device used.
Cardinality | 1..1
Type | Reference(Device)
Summary | true
DeviceUseStatement.reasonCode
Element Id | DeviceUseStatement.reasonCode
Definition | Reason or justification for the use of the device.
Cardinality | 0..*
Type | CodeableConcept
Summary | true

DeviceUseStatement.reasonReference
Element Id | DeviceUseStatement.reasonReference
Definition | Indicates another resource whose existence justifies this DeviceUseStatement.
Cardinality | 0..*
Type | Reference(Condition | Observation | DiagnosticReport | DocumentReference | Media)
Summary | true
DeviceUseStatement.bodySite
Element Id | DeviceUseStatement.bodySite
Definition | Indicates the anotomic location on the subject's body where the device was used (i.e. the target).
Cardinality | 0..1
Terminology Binding | SNOMED CT Body Structures (Example)
Type | CodeableConcept
Requirements | Knowing where the device is targeted is important for tracking if multiple sites are possible. If more information than just a code is required, use the extension http://hl7.org/fhir/StructureDefinition/bodySite.
Summary | true
DeviceUseStatement.note
Element Id | DeviceUseStatement.note
Definition | Details about the device statement that were not represented at all or sufficiently in one of the attributes provided in a class. These may include for example a comment, an instruction, or a note associated with the statement.
Cardinality | 0..*
Type | Annotation

12.20.7 Resource DeviceUseStatement - Mappings

URL: https://hl7.org/fhir/R4/deviceusestatement-mappings.html

12.20.7 Resource DeviceUseStatement - Mappings

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Device, Patient

Mappings:

FiveWs Pattern Mapping

Workflow Pattern

HL7 v2 Mapping

RIM Mapping

Quality Improvement and Clinical Knowledge (QUICK)

Mappings for the deviceusestatement resource (see Mappings to Other Standards for further information & status).

12.20.7.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

DeviceUseStatement
identifier | FiveWs.identifier
status | FiveWs.status
subject | FiveWs.subject[x]
timing[x] | FiveWs.done[x]
recordedOn | FiveWs.recorded
source | FiveWs.actor
device | FiveWs.actor
reasonCode | FiveWs.why[x]
reasonReference | FiveWs.why[x]

12.20.7.2 Workflow Pattern (http://hl7.org/fhir/workflow)

DeviceUseStatement | Event
identifier | Event.identifier
basedOn | Event.basedOn
status | Event.status
subject | Event.subject
timing[x] | Event.occurrence[x]
source | Event.performer.actor
reasonCode | Event.reasonCode
reasonReference | Event.reasonReference
bodySite | Event.note
note | Event.note

12.20.7.3 HL7 v2 Mapping (http://hl7.org/v2)

DeviceUseStatement
identifier
basedOn | ORC in proximity to EVN segment
status

subject
derivedFrom
timing[x]
recordedOn
source
device
reasonCode
reasonReference | EVN.4 or by domain
bodySite
note

12.20.7.4 RIM Mapping (http://hl7.org/v3)

DeviceUseStatement
basedOn | .outboundRelationship[typeCode=FLFS].target
status | .statusCode
reasonReference | .outboundRelationship[typeCode=RSON].target
bodySite | targetSiteCode

12.20.7.5 Quality Improvement and Clinical Knowledge (QUICK) (http://siframework.org/cqf)

DeviceUseStatement | DeviceUsePerformanceOccurrence
identifier | ClinicalStatement.identifier
basedOn
status | …status
subject | ClinicalStatement.subject
derivedFrom
timing[x] | DeviceUse.applicationSchedule
recordedOn | ClinicalStatement.statementDateTime
source
device | DeviceUse.device
reasonCode | Action.indication.reason
reasonReference
bodySite
note | ClinicalStatement.additionalText

12.20.8 Resource DeviceUseStatement - Extensions & Profiles

URL: https://hl7.org/fhir/R4/deviceusestatement-profiles.html

12.20.8 Resource DeviceUseStatement - Extensions & Profiles

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Device, Patient

This table lists profiles and extensions for the DeviceUseStatement resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

12.20.8.1 Profiles

No Profiles defined for this resource

12.20.8.2 Extensions

event-eventHistory | DeviceUseStatement | eventHistory | for Event Pattern HL7 Extensions
event-statusReason | DeviceUseStatement | statusReason | for Event Pattern HL7 Extensions
procedure-approachBodyStructure | DeviceUseStatement | approachBodyStructure | for Procedure HL7 Extensions
workflow-episodeOfCare | DeviceUseStatement | episodeOfCare | for Workflow Pattern HL7 Extensions
workflow-researchStudy | DeviceUseStatement | researchStudy | for Workflow Pattern HL7 Extensions
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

12.20.8.3 Search Extensions

No Search Extensions defined for this resource

12.20.9 Resource DeviceUseStatement - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/deviceusestatement-version-maps.html

No validation errors - all conversions are clean

12.20.9 Resource DeviceUseStatement - R3/R4 Conversions

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Device, Patient

R3 : R4 Conversion maps for DeviceUseStatement.

Functional status for this map: 1 test that all execute ok. All tests pass round-trip testing and all r3 resources are valid. (see documentation)

12.20.9.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/DeviceUseStatement3to4" = "R3 to R4 Conversions for DeviceUseStatement"

uses "http://hl7.org/fhir/3.0/StructureDefinition/DeviceUseStatement" alias DeviceUseStatementR3 as source
uses "http://hl7.org/fhir/StructureDefinition/DeviceUseStatement" alias DeviceUseStatement as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group DeviceUseStatement(source src : DeviceUseStatementR3, target tgt : DeviceUseStatement) extends DomainResource
    <<type+>> {
  src.identifier -> tgt.identifier;
  src.status -> tgt.status;
  src.subject -> tgt.subject;
  src.whenUsed as v ->  tgt.extension as vt,  vt.url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-
    DeviceUseStatement.whenUsed',  vt.value = v;
  src.timing : Timing as vs -> tgt.timing = create('Timing') as vt then Timing(vs, vt);
  src.timing : Period as vs -> tgt.timing = create('Period') as vt then Period(vs, vt);
  src.timing : dateTime as vs -> tgt.timing = create('dateTime') as vt then dateTime(vs, vt);
  src.recordedOn -> tgt.recordedOn;
  src.source -> tgt.source;
  src.device -> tgt.device;
  src.indication -> tgt.reasonCode;
  src.bodySite -> tgt.bodySite;
  src.note -> tgt.note;
}
```

12.20.9.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/DeviceUseStatement4to3" = "R4 to R3 Conversion for DeviceUseStatement"

uses "http://hl7.org/fhir/StructureDefinition/DeviceUseStatement" alias DeviceUseStatement as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/DeviceUseStatement" alias DeviceUseStatementR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group DeviceUseStatement(source src : DeviceUseStatement, target tgt : DeviceUseStatementR3) extends DomainResource
    <<type+>> {
  src.identifier -> tgt.identifier;
  src.status -> tgt.status;
  src.subject -> tgt.subject;
  src.extension as e where url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-DeviceUseStatement.whenUsed'
    then {
    e.value -> tgt.whenUsed;
  };
  src.timing : Timing as vs -> tgt.timing = create('Timing') as vt then Timing(vs, vt);
  src.timing : Period as vs -> tgt.timing = create('Period') as vt then Period(vs, vt);
  src.timing : dateTime as vs -> tgt.timing = create('dateTime') as vt then dateTime(vs, vt);
  src.recordedOn -> tgt.recordedOn;
  src.source -> tgt.source;
  src.device -> tgt.device;
  src.reasonCode -> tgt.indication "indication";
  src.bodySite -> tgt.bodySite;
  src.note -> tgt.note;
}
```

12.20.9.3 R4 Validation Errors

12.20.10 DeviceUseStatement HL7 Extensions

URL: https://hl7.org/fhir/R4/deviceusestatement-extensions.html

Defines common extensions used with or related to the DeviceUseStatement resource

12.20.10 DeviceUseStatement HL7 Extensions

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative

12.20.10.1 Content

12.20.11 JSON Schema for DeviceUseStatement

URL: https://hl7.org/fhir/R4/deviceusestatement.schema.json.html

12.20.11 JSON Schema for DeviceUseStatement

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Device, Patient

This is a representation of the json schema for DeviceUseStatement, which is just a part of the full JSON Schema.


```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/DeviceUseStatement",
  "$ref": "#/definitions/DeviceUseStatement",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "DeviceUseStatement": {
      "description": "A record of a device being used by a patient where the record is the result of a report from the
    patient or another clinician.",
      "properties": {
        "resourceType": {
          "description": "This is a DeviceUseStatement resource",
          "const": "DeviceUseStatement"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
          "$ref": "#/definitions/code"
        },
        "_language": {
          "description": "Extensions for language",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "text": {
          "description": "A human-readable narrative that contains a summary of the resource and can be used to
    represent the content of the resource to a human. The narrative need not encode all the structured data, but is
    required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
    Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
          "$ref": "Narrative.schema.json#/definitions/Narrative"
        },
        "contained": {
          "description": "These resources do not have an independent existence apart from the resource that contains
    them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
          "items": {
            "$ref": "ResourceList.schema.json#/definitions/ResourceList"
          },
          "type": "array"
        },
        "extension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
     definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
    that SHALL be met as part of the definition of the extension.",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "modifierExtension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the resource and that modifies the understanding of the element that contains it and/or the understanding of the
    containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
    of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
    extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
    met as part of the definition of the extension. Applications processing a resource are required to check for
    modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
```

```
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "identifier": {
      "description": "An external identifier for this statement such as an IRI.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "basedOn": {
      "description": "A plan, proposal or order that is fulfilled in whole or in part by this DeviceUseStatement.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "status": {
      "description": "A code representing the patient or other source\u0027s judgment about the state of the device
used that this statement is about.  Generally this will be active or completed.",
      "enum": [
        "active",
        "completed",
        "entered-in-error",
        "intended",
        "stopped",
        "on-hold"
      ]
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "subject": {
      "description": "The patient who used the device.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "derivedFrom": {
      "description": "Allows linking the DeviceUseStatement to the underlying Request, or to other information that
supports or is used to derive the DeviceUseStatement.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"

      },
      "type": "array"
    },
    "timingTiming": {
      "description": "How often the device was used.",
      "$ref": "Timing.schema.json#/definitions/Timing"
    },
    "timingPeriod": {
      "description": "How often the device was used.",
      "$ref": "Period.schema.json#/definitions/Period"
    },
    "timingDateTime": {
      "description": "How often the device was used.",
      "pattern": "^([0-9]([0-9]([0-9][1-9]|[1-9]0)|[1-9]00)|[1-9]000)(-(0[1-9]|1[0-2])(-(0[1-9]|[1-2][0-9]|3[0-
1])(T([01][0-9]|2[0-3]):[0-5][0-9]:([0-5][0-9]|60)(\\.[0-9]+)?(Z|(\\+|-)((0[0-9]|1[0-3]):[0-5][0-9]|14:00)))?)?)?$",
      "type": "string"
    },
    "_timingDateTime": {
      "description": "Extensions for timingDateTime",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "recordedOn": {
      "description": "The time at which the statement was made/recorded.",
      "$ref": "#/definitions/dateTime"
    },
    "_recordedOn": {
      "description": "Extensions for recordedOn",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "source": {
      "description": "Who reported the device was being used by the patient.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "device": {
      "description": "The details of the device used.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "reasonCode": {
      "description": "Reason or justification for the use of the device.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
```

```
        "reasonReference": {
          "description": "Indicates another resource whose existence justifies this DeviceUseStatement.",
          "items": {
            "$ref": "Reference.schema.json#/definitions/Reference"
          },
          "type": "array"
        },
        "bodySite": {
          "description": "Indicates the anotomic location on the subject\u0027s body where the device was used ( i.e.
    the target).",
          "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
        },
        "note": {
          "description": "Details about the device statement that were not represented at all or sufficiently in one of
    the attributes provided in a class. These may include for example a comment, an instruction, or a note associated
    with the statement.",
          "items": {
            "$ref": "Annotation.schema.json#/definitions/Annotation"
          },
          "type": "array"
        }
      },
      "additionalProperties": false,
      "required": [
        "subject",
        "device",
        "resourceType"
      ]
    }
  }
}
```


12.20.12 Resource DeviceUseStatement - Maturity Dependency View

URL: https://hl7.org/fhir/R4/deviceusestatement-dependencies.html

12.20.12 Resource DeviceUseStatement - Maturity Dependency View

Orders and Observations Work Group | Maturity Level: 0 | Trial Use | Compartments: Device, Patient

Dependency Graph for DeviceUseStatement FMM level 0

Name | Card. | Type | Dependency Analysis
DeviceUseStatement | | DomainResource
identifier | 0..* | Identifier
basedOn | 0..* | Reference(ServiceRequest)
status | 1..1 | code
subject | 1..1 | Reference(Patient | Group)
derivedFrom | 0..* | Reference(ServiceRequest | Procedure | Claim | Observation | QuestionnaireResponse | DocumentReference)
timing[x] | 0..1
timingTiming | | Timing
timingPeriod | | Period
timingDateTime | | dateTime
recordedOn | 0..1 | dateTime
source | 0..1 | Reference(Patient | Practitioner | PractitionerRole | RelatedPerson)
device | 1..1 | Reference(Device)
reasonCode | 0..* | CodeableConcept
reasonReference | 0..* | Reference(Condition | Observation | DiagnosticReport | DocumentReference | Media)
bodySite | 0..1 | CodeableConcept
note | 0..* | Annotation

12.20.13 Value Set Sc-valueset-device-statement-status - Status Code Mapping

URL: https://hl7.org/fhir/R4/sc-valueset-device-statement-status.html

12.20.13 Value Set Sc-valueset-device-statement-status - Status Code Mapping

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Device, Patient

This is the narrative for the resource. See also the XML, JSON or Turtle format.

12.20.14 DeviceUseStatementStatusCanonicalMap (http://hl7.org/fhir/ConceptMap/sc-device-statement-status)

Mapping from http://hl7.org/fhir/ValueSet/device-statement-status to http://hl7.org/fhir/ValueSet/resource-status

DRAFT. Published on 01/11/2019 9:29:23 AM by HL7 (FHIR Project) (http://hl7.org/fhir , fhir@lists.hl7.org).

Canonical Mapping for "A coded concept indicating the current status of the Device Usage."

Source Code	Equivalence	Destination Code
entered-in-error	equivalent	error
intended	equivalent	planned
active	equivalent	active
on-hold	equivalent	suspended
stopped	equivalent	failed
completed	equivalent	complete

12.21 Resource SupplyRequest - Content

URL: https://hl7.org/fhir/R4/supplyrequest.html
Orders and Observations Work Group | Maturity Level: 1 | Trial Use | Security Category: Patient | Compartments: Device, Patient, Practitioner, RelatedPerson

A record of a request for a medication, substance or device used in the healthcare setting.

12.21.1 Scope and Usage

Trial-Use Note: In terms of scope and usage, the Orders and Observations workgroups wish to draw the attention of reviewers and implementers to the following issues and changes.

• We are seeking input from the implementation community on whether to combine SupplyRequest and DeviceRequest for R4.

• The parameters element was added to SupplyRequest for representing custom details for ordering devices.

Feedback is welcome here

This resource is a request resource from a FHIR workflow perspective - see Workflow. It is the intent of the Orders and Observation Workgroup to align this resource with the workflow pattern for request resources.

The scope of the SupplyRequest resource is for recording the request of supplies used in the healthcare process. This includes supplies specifically used in the treatment of patients as well as supply movement within an institution (transport a set of supplies from materials management to a service unit (nurse station). This resource does not include the provisioning of transportation services.

The SupplyRequest resource allows requesting only a single item. If a workflow requires requesting multiple items simultaneously, this is done using multiple instances of this resource. These instances can be linked in different ways, depending on the needs of the workflow. For guidance, refer to the Request pattern

12.21.2 Boundaries and Relationships

Note that the SupplyRequest records the fact that a request was made. To actually act on that request, additional workflow beyond simply the existence of a SupplyRequest is required. This can be achieved by using an Task resource, with the SupplyRequest referenced from the Task.focus, or by using the SupplyRequest resource in the context of an messaging or service workflow where the request is explicit or implicit. The SupplyDelivery resource represents the fulfillment as a result of SupplyRequest being acted upon.

The SupplyRequest resource is used for inventory management. When requesting medication, substances and devices when there is a patient focus or instructions regarding their use, DeviceRequest or MedicationRequest should be used instead

This resource is referenced by SupplyDelivery

12.21.3 Resource Content

• Structure

• UML

• XML

• JSON

• Turtle

• R3 Diff

• All

Structure

Name	Flags	Card.	Type	Description & Constraints
SupplyRequest	TU		DomainResource	Request for a medication, substance or device Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier	Σ	0..*	Identifier	Business Identifier for SupplyRequest
status	?!Σ	0..1	code	draft | active | suspended + SupplyRequestStatus (Required)
category	Σ	0..1	CodeableConcept	The kind of supply (central, non-stock, etc.) Supply Type (Example)
priority	Σ	0..1	code	routine | urgent | asap | stat Request priority (Required)

item[x] | Σ | 1..1 | | Medication, Substance, or Device requested to be supplied SNOMED CT Supply Item (Example)
itemCodeableConcept | | | CodeableConcept
itemReference | | | Reference(Medication | Substance | Device)
quantity | Σ | 1..1 | Quantity | The requested amount of the item indicated
parameter | | 0..* | BackboneElement | Ordered item details
code | | 0..1 | CodeableConcept | Item detail
value[x] | | 0..1 | | Value of detail
valueCodeableConcept | | | CodeableConcept
valueQuantity | | | Quantity
valueRange | | | Range
valueBoolean | | | boolean
occurrence[x] | Σ | 0..1 | | When the request should be fulfilled
occurrenceDateTime | | | dateTime
occurrencePeriod | | | Period
occurrenceTiming | | | Timing
authoredOn | Σ | 0..1 | dateTime | When the request was made
requester | Σ | 0..1 | Reference(Practitioner | PractitionerRole | Organization | Patient | RelatedPerson | Device) | Individual making the request
supplier | Σ | 0..* | Reference(Organization | HealthcareService) | Who is intended to fulfill the request
reasonCode | | 0..* | CodeableConcept | The reason why the supply item was requested SupplyRequestReason (Example)
reasonReference | | 0..* | Reference(Condition | Observation | DiagnosticReport | DocumentReference) | The reason why the supply item was requested
deliverFrom | | 0..1 | Reference(Organization | Location) | The origin of the supply
deliverTo | | 0..1 | Reference(Organization | Location | Patient) | The destination of the supply
Documentation for this format

UML Diagram (Legend)

XML Template

```
<SupplyRequest xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Business Identifier for SupplyRequest --></identifier>
 <status value="[code]"/><!-- 0..1 draft | active | suspended + -->
 <category><!-- 0..1 CodeableConcept The kind of supply (central, non-stock, etc.) --></category>
 <priority value="[code]"/><!-- 0..1 routine | urgent | asap | stat -->
 <item[x]><!-- 1..1 CodeableConcept|Reference(Medication|Substance|Device) Medication, Substance, or Device requested to
     be supplied --></item[x]>
 <quantity><!-- 1..1 Quantity The requested amount of the item indicated --></quantity>
 <parameter>  <!-- 0..* Ordered item details -->
  <code><!-- 0..1 CodeableConcept Item detail --></code>
  <value[x]><!-- 0..1 CodeableConcept|Quantity|Range|boolean Value of detail --></value[x]>
 </parameter>
 <occurrence[x]><!-- 0..1 dateTime|Period|Timing When the request should be fulfilled --></occurrence[x]>
 <authoredOn value="[dateTime]"/><!-- 0..1 When the request was made -->
 <requester><!-- 0..1 Reference(Practitioner|PractitionerRole|Organization|
   Patient|RelatedPerson|Device) Individual making the request --></requester>
 <supplier><!-- 0..* Reference(Organization|HealthcareService) Who is intended to fulfill the request --></supplier>
 <reasonCode><!-- 0..* CodeableConcept The reason why the supply item was requested --></reasonCode>
 <reasonReference><!-- 0..* Reference(Condition|Observation|DiagnosticReport|
   DocumentReference) The reason why the supply item was requested --></reasonReference>
 <deliverFrom><!-- 0..1 Reference(Organization|Location) The origin of the supply --></deliverFrom>
 <deliverTo><!-- 0..1 Reference(Organization|Location|Patient) The destination of the supply --></deliverTo>
</SupplyRequest>
```

JSON Template

```
{
  "resourceType" : "SupplyRequest",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Business Identifier for SupplyRequest
  "status" : "<code>", // draft | active | suspended +
  "category" : { CodeableConcept }, // The kind of supply (central, non-stock, etc.)
  "priority" : "<code>", // routine | urgent | asap | stat
  // item[x]: Medication, Substance, or Device requested to be supplied. One of these 2:
  "itemCodeableConcept" : { CodeableConcept },
  "itemReference" : { Reference(Medication|Substance|Device) },
  "quantity" : { Quantity }, // R!  The requested amount of the item indicated
  "parameter" : [{ // Ordered item details
    "code" : { CodeableConcept }, // Item detail
    // value[x]: Value of detail. One of these 4:
    "valueCodeableConcept" : { CodeableConcept }
    "valueQuantity" : { Quantity }
    "valueRange" : { Range }
    "valueBoolean" : <boolean>
  }],
  // occurrence[x]: When the request should be fulfilled. One of these 3:
  "occurrenceDateTime" : "<dateTime>",
  "occurrencePeriod" : { Period },
  "occurrenceTiming" : { Timing },
  "authoredOn" : "<dateTime>", // When the request was made
  "requester" : { Reference(Practitioner|PractitionerRole|Organization|
   Patient|RelatedPerson|Device) }, // Individual making the request
  "supplier" : [{ Reference(Organization|HealthcareService) }], // Who is intended to fulfill the request
```

```
  "reasonCode" : [{ CodeableConcept }], // The reason why the supply item was requested
  "reasonReference" : [{ Reference(Condition|Observation|DiagnosticReport|
   DocumentReference) }], // The reason why the supply item was requested
  "deliverFrom" : { Reference(Organization|Location) }, // The origin of the supply
  "deliverTo" : { Reference(Organization|Location|Patient) } // The destination of the supply
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:SupplyRequest;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:SupplyRequest.identifier [ Identifier ], ... ; # 0..* Business Identifier for SupplyRequest
  fhir:SupplyRequest.status [ code ]; # 0..1 draft | active | suspended +
  fhir:SupplyRequest.category [ CodeableConcept ]; # 0..1 The kind of supply (central, non-stock, etc.)
  fhir:SupplyRequest.priority [ code ]; # 0..1 routine | urgent | asap | stat
  # SupplyRequest.item[x] : 1..1 Medication, Substance, or Device requested to be supplied. One of these 2
    fhir:SupplyRequest.itemCodeableConcept [ CodeableConcept ]
    fhir:SupplyRequest.itemReference [ Reference(Medication|Substance|Device) ]
  fhir:SupplyRequest.quantity [ Quantity ]; # 1..1 The requested amount of the item indicated
  fhir:SupplyRequest.parameter [ # 0..* Ordered item details
    fhir:SupplyRequest.parameter.code [ CodeableConcept ]; # 0..1 Item detail
    # SupplyRequest.parameter.value[x] : 0..1 Value of detail. One of these 4
      fhir:SupplyRequest.parameter.valueCodeableConcept [ CodeableConcept ]
      fhir:SupplyRequest.parameter.valueQuantity [ Quantity ]
      fhir:SupplyRequest.parameter.valueRange [ Range ]
      fhir:SupplyRequest.parameter.valueBoolean [ boolean ]
  ], ...;
  # SupplyRequest.occurrence[x] : 0..1 When the request should be fulfilled. One of these 3
    fhir:SupplyRequest.occurrenceDateTime [ dateTime ]
    fhir:SupplyRequest.occurrencePeriod [ Period ]
    fhir:SupplyRequest.occurrenceTiming [ Timing ]
  fhir:SupplyRequest.authoredOn [ dateTime ]; # 0..1 When the request was made
  fhir:SupplyRequest.requester [ Reference(Practitioner|PractitionerRole|Organization|Patient|RelatedPerson|Device) ]; #
     0..1 Individual making the request
  fhir:SupplyRequest.supplier [ Reference(Organization|HealthcareService) ], ... ; # 0..* Who is intended to fulfill the
     request
  fhir:SupplyRequest.reasonCode [ CodeableConcept ], ... ; # 0..* The reason why the supply item was requested
  fhir:SupplyRequest.reasonReference [ Reference(Condition|Observation|DiagnosticReport|DocumentReference) ], ... ; #
    0..* The reason why the supply item was requested
  fhir:SupplyRequest.deliverFrom [ Reference(Organization|Location) ]; # 0..1 The origin of the supply
  fhir:SupplyRequest.deliverTo [ Reference(Organization|Location|Patient) ]; # 0..1 The destination of the supply
]
```

Changes since R3

SupplyRequest

SupplyRequest.identifier | Max Cardinality changed from 1 to *

SupplyRequest.status | Change value set from http://hl7.org/fhir/ValueSet/supplyrequest-status to http://hl7.org/fhir/ValueSet/supplyrequest-status|4.0.1

SupplyRequest.priority | Change value set from http://hl7.org/fhir/ValueSet/request-priority to http://hl7.org/fhir/ValueSet/request-priority|4.0.1

SupplyRequest.item[x] | Added Mandatory Element

SupplyRequest.quantity | Added Mandatory Element

SupplyRequest.parameter | Added Element

SupplyRequest.parameter.code | Added Element

SupplyRequest.parameter.value[x] | Added Element

SupplyRequest.requester | Type changed from BackboneElement to Reference(Practitioner | PractitionerRole | Organization | Patient | RelatedPerson | Device)

SupplyRequest.supplier | Type Reference: Added Target Type HealthcareService

SupplyRequest.reasonCode | Added Element

SupplyRequest.reasonReference | Added Element

SupplyRequest.orderedItem | deleted

SupplyRequest.requester.agent | deleted

SupplyRequest.requester.onBehalfOf | deleted

SupplyRequest.reason[x] | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 1 test that all execute ok. All tests pass round-trip testing and 1 r3 resources are invalid (0 errors).)

Structure

Name | Flags | Card. | Type | Description & Constraints

SupplyRequest | TU | | DomainResource | Request for a medication, substance or device Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

identifier | Σ | 0..* | Identifier | Business Identifier for SupplyRequest

status | ?!Σ | 0..1 | code | draft | active | suspended + SupplyRequestStatus (Required)

category | Σ | 0..1 | CodeableConcept | The kind of supply (central, non-stock, etc.) Supply Type (Example)

priority | Σ | 0..1 | code | routine | urgent | asap | stat Request priority (Required)
item[x] | Σ | 1..1 | | Medication, Substance, or Device requested to be supplied SNOMED CT Supply Item (Example)
itemCodeableConcept | | | CodeableConcept
itemReference | | | Reference(Medication | Substance | Device)
quantity | Σ | 1..1 | Quantity | The requested amount of the item indicated
parameter | | 0..* | BackboneElement | Ordered item details
code | | 0..1 | CodeableConcept | Item detail
value[x] | | 0..1 | | Value of detail
valueCodeableConcept | | | CodeableConcept
valueQuantity | | | Quantity
valueRange | | | Range
valueBoolean | | | boolean
occurrence[x] | Σ | 0..1 | | When the request should be fulfilled
occurrenceDateTime | | | dateTime
occurrencePeriod | | | Period
occurrenceTiming | | | Timing
authoredOn | Σ | 0..1 | dateTime | When the request was made
requester | Σ | 0..1 | Reference(Practitioner | PractitionerRole | Organization | Patient | RelatedPerson | Device) | Individual making the request
supplier | Σ | 0..* | Reference(Organization | HealthcareService) | Who is intended to fulfill the request
reasonCode | | 0..* | CodeableConcept | The reason why the supply item was requested SupplyRequestReason (Example)
reasonReference | | 0..* | Reference(Condition | Observation | DiagnosticReport | DocumentReference) | The reason why the supply item was requested
deliverFrom | | 0..1 | Reference(Organization | Location) | The origin of the supply
deliverTo | | 0..1 | Reference(Organization | Location | Patient) | The destination of the supply
Documentation for this format

UML Diagram (Legend)

XML Template

```
<SupplyRequest xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Business Identifier for SupplyRequest --></identifier>
 <status value="[code]"/><!-- 0..1 draft | active | suspended + -->
 <category><!-- 0..1 CodeableConcept The kind of supply (central, non-stock, etc.) --></category>
 <priority value="[code]"/><!-- 0..1 routine | urgent | asap | stat -->
 <item[x]><!-- 1..1 CodeableConcept|Reference(Medication|Substance|Device) Medication, Substance, or Device requested to
     be supplied --></item[x]>
 <quantity><!-- 1..1 Quantity The requested amount of the item indicated --></quantity>
 <parameter>  <!-- 0..* Ordered item details -->
  <code><!-- 0..1 CodeableConcept Item detail --></code>
  <value[x]><!-- 0..1 CodeableConcept|Quantity|Range|boolean Value of detail --></value[x]>
 </parameter>
 <occurrence[x]><!-- 0..1 dateTime|Period|Timing When the request should be fulfilled --></occurrence[x]>
 <authoredOn value="[dateTime]"/><!-- 0..1 When the request was made -->
 <requester><!-- 0..1 Reference(Practitioner|PractitionerRole|Organization|
   Patient|RelatedPerson|Device) Individual making the request --></requester>
 <supplier><!-- 0..* Reference(Organization|HealthcareService) Who is intended to fulfill the request --></supplier>
 <reasonCode><!-- 0..* CodeableConcept The reason why the supply item was requested --></reasonCode>
 <reasonReference><!-- 0..* Reference(Condition|Observation|DiagnosticReport|
   DocumentReference) The reason why the supply item was requested --></reasonReference>
 <deliverFrom><!-- 0..1 Reference(Organization|Location) The origin of the supply --></deliverFrom>
 <deliverTo><!-- 0..1 Reference(Organization|Location|Patient) The destination of the supply --></deliverTo>
</SupplyRequest>
```

JSON Template

```
{
  "resourceType" : "SupplyRequest",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Business Identifier for SupplyRequest
  "status" : "<code>", // draft | active | suspended +
  "category" : { CodeableConcept }, // The kind of supply (central, non-stock, etc.)
  "priority" : "<code>", // routine | urgent | asap | stat
  // item[x]: Medication, Substance, or Device requested to be supplied. One of these 2:
  "itemCodeableConcept" : { CodeableConcept },
  "itemReference" : { Reference(Medication|Substance|Device) },
  "quantity" : { Quantity }, // R!  The requested amount of the item indicated
  "parameter" : [{ // Ordered item details
    "code" : { CodeableConcept }, // Item detail
    // value[x]: Value of detail. One of these 4:
    "valueCodeableConcept" : { CodeableConcept }
    "valueQuantity" : { Quantity }
    "valueRange" : { Range }
    "valueBoolean" : <boolean>
  }],
  // occurrence[x]: When the request should be fulfilled. One of these 3:
  "occurrenceDateTime" : "<dateTime>",
  "occurrencePeriod" : { Period },
  "occurrenceTiming" : { Timing },
  "authoredOn" : "<dateTime>", // When the request was made
  "requester" : { Reference(Practitioner|PractitionerRole|Organization|
```

```
   Patient|RelatedPerson|Device) }, // Individual making the request
  "supplier" : [{ Reference(Organization|HealthcareService) }], // Who is intended to fulfill the request
  "reasonCode" : [{ CodeableConcept }], // The reason why the supply item was requested
  "reasonReference" : [{ Reference(Condition|Observation|DiagnosticReport|
   DocumentReference) }], // The reason why the supply item was requested
  "deliverFrom" : { Reference(Organization|Location) }, // The origin of the supply
  "deliverTo" : { Reference(Organization|Location|Patient) } // The destination of the supply
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:SupplyRequest;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:SupplyRequest.identifier [ Identifier ], ... ; # 0..* Business Identifier for SupplyRequest
  fhir:SupplyRequest.status [ code ]; # 0..1 draft | active | suspended +
  fhir:SupplyRequest.category [ CodeableConcept ]; # 0..1 The kind of supply (central, non-stock, etc.)
  fhir:SupplyRequest.priority [ code ]; # 0..1 routine | urgent | asap | stat
  # SupplyRequest.item[x] : 1..1 Medication, Substance, or Device requested to be supplied. One of these 2
    fhir:SupplyRequest.itemCodeableConcept [ CodeableConcept ]
    fhir:SupplyRequest.itemReference [ Reference(Medication|Substance|Device) ]
  fhir:SupplyRequest.quantity [ Quantity ]; # 1..1 The requested amount of the item indicated
  fhir:SupplyRequest.parameter [ # 0..* Ordered item details
    fhir:SupplyRequest.parameter.code [ CodeableConcept ]; # 0..1 Item detail
    # SupplyRequest.parameter.value[x] : 0..1 Value of detail. One of these 4
      fhir:SupplyRequest.parameter.valueCodeableConcept [ CodeableConcept ]
      fhir:SupplyRequest.parameter.valueQuantity [ Quantity ]
      fhir:SupplyRequest.parameter.valueRange [ Range ]
      fhir:SupplyRequest.parameter.valueBoolean [ boolean ]
  ], ...;
  # SupplyRequest.occurrence[x] : 0..1 When the request should be fulfilled. One of these 3
    fhir:SupplyRequest.occurrenceDateTime [ dateTime ]
    fhir:SupplyRequest.occurrencePeriod [ Period ]
    fhir:SupplyRequest.occurrenceTiming [ Timing ]
  fhir:SupplyRequest.authoredOn [ dateTime ]; # 0..1 When the request was made
  fhir:SupplyRequest.requester [ Reference(Practitioner|PractitionerRole|Organization|Patient|RelatedPerson|Device) ]; #
     0..1 Individual making the request
  fhir:SupplyRequest.supplier [ Reference(Organization|HealthcareService) ], ... ; # 0..* Who is intended to fulfill the
     request
  fhir:SupplyRequest.reasonCode [ CodeableConcept ], ... ; # 0..* The reason why the supply item was requested
  fhir:SupplyRequest.reasonReference [ Reference(Condition|Observation|DiagnosticReport|DocumentReference) ], ... ; #
    0..* The reason why the supply item was requested
  fhir:SupplyRequest.deliverFrom [ Reference(Organization|Location) ]; # 0..1 The origin of the supply
  fhir:SupplyRequest.deliverTo [ Reference(Organization|Location|Patient) ]; # 0..1 The destination of the supply
]
```

Changes since Release 3

SupplyRequest
SupplyRequest.identifier | Max Cardinality changed from 1 to *
SupplyRequest.status | Change value set from http://hl7.org/fhir/ValueSet/supplyrequest-status to http://hl7.org/fhir/ValueSet/supplyrequest-status|4.0.1
SupplyRequest.priority | Change value set from http://hl7.org/fhir/ValueSet/request-priority to http://hl7.org/fhir/ValueSet/request-priority|4.0.1
SupplyRequest.item[x] | Added Mandatory Element
SupplyRequest.quantity | Added Mandatory Element
SupplyRequest.parameter | Added Element
SupplyRequest.parameter.code | Added Element
SupplyRequest.parameter.value[x] | Added Element
SupplyRequest.requester | Type changed from BackboneElement to Reference(Practitioner | PractitionerRole | Organization | Patient | RelatedPerson | Device)
SupplyRequest.supplier | Type Reference: Added Target Type HealthcareService
SupplyRequest.reasonCode | Added Element
SupplyRequest.reasonReference | Added Element
SupplyRequest.orderedItem | deleted
SupplyRequest.requester.agent | deleted
SupplyRequest.requester.onBehalfOf | deleted
SupplyRequest.reason[x] | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 1 test that all execute ok. All tests pass round-trip testing and 1 r3 resources are invalid (0 errors).)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

12.21.3.1 Terminology Bindings

Path | Definition | Type | Reference

SupplyRequest.status | Status of the supply request. | Required | SupplyRequestStatus
SupplyRequest.category | Category of supply request. | Example | SupplyType
SupplyRequest.priority | Identifies the level of importance to be assigned to actioning the request. | Required | RequestPriority
SupplyRequest.item[x] | The item that was requested. | Example | SNOMEDCTSupplyItem
SupplyRequest.parameter.code | A code that identifies the device detail. | Unknown | No details provided yet
SupplyRequest.reasonCode | The reason why the supply item was requested. | Example | SupplyRequestReason

12.21.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
category | token | The kind of supply (central, non-stock, etc.) | SupplyRequest.category
date | date | When the request was made | SupplyRequest.authoredOn | 17 Resources
identifier | token | Business Identifier for SupplyRequest | SupplyRequest.identifier | 30 Resources
requester | reference | Individual making the request | SupplyRequest.requester (Practitioner, Organization, Device, Patient, PractitionerRole, RelatedPerson)
status | token | draft | active | suspended + | SupplyRequest.status
subject | reference | The destination of the supply | SupplyRequest.deliverTo (Organization, Patient, Location)
supplier | reference | Who is intended to fulfill the request | SupplyRequest.supplier (Organization, HealthcareService)

12.21.5 Resource SupplyRequest - Examples

URL: https://hl7.org/fhir/R4/supplyrequest-examples.html

12.21.5 Resource SupplyRequest - Examples

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Device, Patient, Practitioner, RelatedPerson

Example Name | id | Format
Simple order for resupply | simpleorder | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

12.21.6 Resource SupplyRequest - Detailed Descriptions

URL: https://hl7.org/fhir/R4/supplyrequest-definitions.html

12.21.6 Resource SupplyRequest - Detailed Descriptions

Orders and Observations Work Group | Maturity Level: 1 | Trial Use | Security Category: Patient | Compartments: Device, Patient, Practitioner, RelatedPerson

Detailed Descriptions for the elements in the SupplyRequest resource.

SupplyRequest
Element Id | SupplyRequest
Definition | A record of a request for a medication, substance or device used in the healthcare setting.
Cardinality | 0..*
Type | DomainResource
SupplyRequest.identifier
Element Id | SupplyRequest.identifier
Definition | Business identifiers assigned to this SupplyRequest by the author and/or other systems. These identifiers remain constant as the resource is updated and propagates from server to server.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Requirements | Allows identification of the request as it is known by various participating systems and in a way that remains consistent across servers.
Summary | true
Comments | The identifier.type element is used to distinguish between the identifiers assigned by the requester/placer and the performer/filler.
SupplyRequest.status
Element Id | SupplyRequest.status
Definition | Status of the supply request.
Cardinality | 0..1
Terminology Binding | SupplyRequestStatus (Required)
Type | code
Is Modifier | true (Reason: This element is labeled as a modifier because it is a status element that contains status entered-in-error which means that the resource should not be treated as valid)
Summary | true
SupplyRequest.category
Element Id | SupplyRequest.category
Definition | Category of supply, e.g. central, non-stock, etc. This is used to support work flows associated with the supply process.

Cardinality | 0..1
Terminology Binding | Supply Type (Example)
Type | CodeableConcept
Alternate Names | kind
Summary | true
SupplyRequest.priority
Element Id | SupplyRequest.priority
Definition | Indicates how quickly this SupplyRequest should be addressed with respect to other requests.
Cardinality | 0..1
Terminology Binding | Request priority (Required)
Type | code
Meaning if Missing | If missing, this task should be performed with normal priority
Summary | true
SupplyRequest.item[x]
Element Id | SupplyRequest.item[x]
Definition | The item that is requested to be supplied. This is either a link to a resource representing the details of the item or a code that identifies the item from a known list.
Cardinality | 1..1
Terminology Binding | SNOMED CT Supply Item (Example)
Type | CodeableConcept|Reference(Medication | Substance | Device)
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true
Comments | Note that there's a difference between a prescription - an instruction to take a medication, along with a (sometimes) implicit supply, and an explicit request to supply, with no explicit instructions.
SupplyRequest.quantity
Element Id | SupplyRequest.quantity
Definition | The amount that is being ordered of the indicated item.
Cardinality | 1..1
Type | Quantity
Summary | true
SupplyRequest.parameter
Element Id | SupplyRequest.parameter
Definition | Specific parameters for the ordered item. For example, the size of the indicated item.
Cardinality | 0..*
SupplyRequest.parameter.code
Element Id | SupplyRequest.parameter.code
Definition | A code or string that identifies the device detail being asserted.
Cardinality | 0..1
Terminology Binding | ParameterCode:
Type | CodeableConcept
SupplyRequest.parameter.value[x]
Element Id | SupplyRequest.parameter.value[x]
Definition | The value of the device detail.
Cardinality | 0..1
Type | CodeableConcept|Quantity|Range|boolean
[x] Note | See Choice of Data Types for further information about how to use [x]
Comments | Range means device should have a value that falls somewhere within the specified range.
SupplyRequest.occurrence[x]
Element Id | SupplyRequest.occurrence[x]
Definition | When the request should be fulfilled.
Cardinality | 0..1
Type | dateTime|Period|Timing
[x] Note | See Choice of Data Types for further information about how to use [x]
Alternate Names | when; timing
Summary | true
SupplyRequest.authoredOn
Element Id | SupplyRequest.authoredOn
Definition | When the request was made.
Cardinality | 0..1
Type | dateTime
Alternate Names | date; createdOn; signedOn
Summary | true
SupplyRequest.requester
Element Id | SupplyRequest.requester
Definition | The device, practitioner, etc. who initiated the request.
Cardinality | 0..1
Type | Reference(Practitioner | PractitionerRole | Organization | Patient | RelatedPerson | Device)
Alternate Names | author; source
Summary | true
SupplyRequest.supplier
Element Id | SupplyRequest.supplier
Definition | Who is intended to fulfill the request.

Cardinality | 0..*
Type | Reference(Organization | HealthcareService)
Alternate Names | performer
Summary | true
SupplyRequest.reasonCode
Element Id | SupplyRequest.reasonCode
Definition | The reason why the supply item was requested.
Cardinality | 0..*
Terminology Binding | SupplyRequestReason (Example)
Type | CodeableConcept
SupplyRequest.reasonReference
Element Id | SupplyRequest.reasonReference
Definition | The reason why the supply item was requested.
Cardinality | 0..*
Type | Reference(Condition | Observation | DiagnosticReport | DocumentReference)
SupplyRequest.deliverFrom
Element Id | SupplyRequest.deliverFrom
Definition | Where the supply is expected to come from.
Cardinality | 0..1
Type | Reference(Organization | Location)
SupplyRequest.deliverTo
Element Id | SupplyRequest.deliverTo
Definition | Where the supply is destined to go.
Cardinality | 0..1
Type | Reference(Organization | Location | Patient)

12.21.7 Resource SupplyRequest - Mappings

URL: https://hl7.org/fhir/R4/supplyrequest-mappings.html

12.21.7 Resource SupplyRequest - Mappings

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Device, Patient, Practitioner, RelatedPerson

Mappings:

FiveWs Pattern Mapping

Workflow Pattern

HL7 v2 Mapping

RIM Mapping

Mappings for the supplyrequest resource (see Mappings to Other Standards for further information & status).

12.21.7.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

SupplyRequest
identifier | FiveWs.identifier
status | FiveWs.status
category | FiveWs.class
priority | FiveWs.grade
parameter | FiveWs.what[x]
code | FiveWs.what[x]
value[x] | FiveWs.what[x]
occurrence[x] | FiveWs.planned
authoredOn | FiveWs.recorded
requester | FiveWs.author
supplier | FiveWs.actor
reasonCode | FiveWs.why[x]
reasonReference | FiveWs.why[x]

12.21.7.2 Workflow Pattern (http://hl7.org/fhir/workflow)

SupplyRequest | Request
identifier | Request.identifier
status | Request.status
priority | Request.priority
item[x] | Request.code
occurrence[x] | Request.occurrence[x]
authoredOn | Request.authoredOn
requester | Request.requester
supplier | Request.performer
reasonCode | Request.reasonCode
reasonReference | Request.reasonReference

12.21.7.3 HL7 v2 Mapping (http://hl7.org/v2)

SupplyRequest | ORC
identifier | ORC.2, ORC.3
status | ORC.5
category | N/A
priority | TQ1.9
item[x]
quantity
parameter | Varies by domain
code | Varies by domain
value[x] | Varies by domain
occurrence[x] | Accompanying TQ1/TQ2 segments
authoredOn | ORC.9
requester | ORC.12
supplier | PRT
reasonCode | ORC.16
reasonReference | ORC.16
deliverFrom | PRT
deliverTo | PRT

12.21.7.4 RIM Mapping (http://hl7.org/v3)

SupplyRequest | Act[moodCode<=INT]
identifier | .identifier
status | .status
category | .code
priority | .priorityCode
item[x] | .code
quantity | .quantity
code | .code
value[x] | .code
occurrence[x] | .effectiveTime
authoredOn | .participation[typeCode=AUT].time
requester | .participation[typeCode=AUT].role
supplier | .participation[typeCode=PRF].role[scoper.determinerCode=INSTANCE]
reasonCode | .reasonCode
reasonReference | .outboundRelationship[typeCode=RSON].target
deliverFrom | .participation
deliverTo | .participation

12.21.8 Resource SupplyRequest - Extensions & Profiles

URL: https://hl7.org/fhir/R4/supplyrequest-profiles.html

12.21.8 Resource SupplyRequest - Extensions & Profiles

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Device, Patient, Practitioner, RelatedPerson

This table lists profiles and extensions for the SupplyRequest resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

12.21.8.1 Profiles

No Profiles defined for this resource

12.21.8.2 Extensions

request-statusReason | SupplyRequest | statusReason | for Request Pattern HL7 Extensions
workflow-instantiatesCanonical | SupplyRequest | instantiatesCanonical | for Workflow Pattern HL7 Extensions
workflow-instantiatesUri | SupplyRequest | instantiatesUri | for Workflow Pattern HL7 Extensions
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions

workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

12.21.8.3 Search Extensions

No Search Extensions defined for this resource

12.21.9 Resource SupplyRequest - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/supplyrequest-version-maps.html

No validation errors - all conversions are clean

12.21.9 Resource SupplyRequest - R3/R4 Conversions

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Device, Patient, Practitioner, RelatedPerson

R3 : R4 Conversion maps for SupplyRequest.

Functional status for this map: 1 test that all execute ok. All tests pass round-trip testing and 1 r3 resources are invalid (0 errors). (see documentation)

12.21.9.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/SupplyRequest3to4" = "R3 to R4 Conversions for SupplyRequest"

uses "http://hl7.org/fhir/3.0/StructureDefinition/SupplyRequest" alias SupplyRequestR3 as source
uses "http://hl7.org/fhir/StructureDefinition/SupplyRequest" alias SupplyRequest as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group SupplyRequest(source src : SupplyRequestR3, target tgt : SupplyRequest) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.status -> tgt.status;
  src.category -> tgt.category;
  src.priority -> tgt.priority;
  src.orderedItem as s then {
    s.item : CodeableConcept as vs -> tgt.item = create('CodeableConcept') as vt then CodeableConcept(vs, vt);
    s.item : Reference as vs -> tgt.item = create('Reference') as vt then Reference(vs, vt);
    s.quantity -> tgt.quantity;
  };
  src.occurrence : dateTime as vs -> tgt.occurrence = create('dateTime') as vt then dateTime(vs, vt);
  src.occurrence : Period as vs -> tgt.occurrence = create('Period') as vt then Period(vs, vt);
  src.occurrence : Timing as vs -> tgt.occurrence = create('Timing') as vt then Timing(vs, vt);
  src.authoredOn -> tgt.authoredOn;
  src.requester as vs0 -> tgt then agent(vs0, tgt);
  src.supplier -> tgt.supplier;
  src.reason : CodeableConcept as vs -> tgt.reasonCode as vt then CodeableConcept(vs, vt);
  src.reason : Reference as vs -> tgt.reasonReference as vt then Reference(vs, vt);
  src.deliverFrom -> tgt.deliverFrom;
  src.deliverTo -> tgt.deliverTo;
}

group agent(source src, target tgt) {
  src.agent -> tgt.requester;
  src.onBehalfOf as v ->  tgt.extension as vt,  vt.url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-
    SupplyRequest.requester.onBehalfOf',  vt.value = v;
}
```

12.21.9.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/SupplyRequest4to3" = "R4 to R3 Conversion for SupplyRequest"

uses "http://hl7.org/fhir/StructureDefinition/SupplyRequest" alias SupplyRequest as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/SupplyRequest" alias SupplyRequestR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group SupplyRequest(source src : SupplyRequestR3, target tgt : SupplyRequest) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.status -> tgt.status;
  src.category -> tgt.category;
  src.priority -> tgt.priority;
  src.item : CodeableConcept as vs ->  tgt.orderedItem as t,  t.item = create('CodeableConcept') as vt then
    CodeableConcept(vs, vt);
  src.item : Reference as vs ->  tgt.orderedItem as t,  t.item = create('Reference') as vt then Reference(vs, vt);
  src.quantity as vs ->  tgt.orderedItem as t,  t.quantity as vt then Quantity(vs, vt);
  src.occurrence : dateTime as vs -> tgt.occurrence = create('dateTime') as vt then dateTime(vs, vt);
  src.occurrence : Period as vs -> tgt.occurrence = create('Period') as vt then Period(vs, vt);
  src.occurrence : Timing as vs -> tgt.occurrence = create('Timing') as vt then Timing(vs, vt);
  src.authoredOn -> tgt.authoredOn;
  src where requester.exists() or extension.where(url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-
    SupplyRequest.requester.onBehalfOf').exists() -> tgt.requester as vt0 then agent(src, vt0) "requester";
  src.supplier -> tgt.supplier;
  src.reasonCode as vs -> tgt.reason = create('CodeableConcept') as vt then CodeableConcept(vs, vt);
  src.reasonReference as vs -> tgt.reason = create('Reference') as vt then Reference(vs, vt);
  src.deliverFrom -> tgt.deliverFrom;
  src.deliverTo -> tgt.deliverTo;
}
```

```
group agent(source src, target tgt) {
  src.requester -> tgt.agent;
  src.extension as e where url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-
    SupplyRequest.requester.onBehalfOf' then {
    e.value -> tgt.onBehalfOf;
  };
}
```

12.21.9.3 R4 Validation Errors

12.21.10 JSON Schema for SupplyRequest

URL: https://hl7.org/fhir/R4/supplyrequest.schema.json.html

12.21.10 JSON Schema for SupplyRequest

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Device, Patient, Practitioner, RelatedPerson

This is a representation of the json schema for SupplyRequest, which is just a part of the full JSON Schema.


```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/SupplyRequest",
  "$ref": "#/definitions/SupplyRequest",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "SupplyRequest": {
      "description": "A record of a request for a medication, substance or device used in the healthcare setting.",
      "properties": {
        "resourceType": {
          "description": "This is a SupplyRequest resource",
          "const": "SupplyRequest"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
          "$ref": "#/definitions/code"
        },
        "_language": {
          "description": "Extensions for language",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "text": {
          "description": "A human-readable narrative that contains a summary of the resource and can be used to
    represent the content of the resource to a human. The narrative need not encode all the structured data, but is
    required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
    Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
          "$ref": "Narrative.schema.json#/definitions/Narrative"
        },
        "contained": {
          "description": "These resources do not have an independent existence apart from the resource that contains
    them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
          "items": {
            "$ref": "ResourceList.schema.json#/definitions/ResourceList"
          },
          "type": "array"
        },
        "extension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
     definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
    that SHALL be met as part of the definition of the extension.",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "modifierExtension": {
```

```
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource and that modifies the understanding of the element that contains it and/or the understanding of the
containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
met as part of the definition of the extension. Applications processing a resource are required to check for
modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "identifier": {
      "description": "Business identifiers assigned to this SupplyRequest by the author and/or other systems. These
identifiers remain constant as the resource is updated and propagates from server to server.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "status": {
      "description": "Status of the supply request.",
      "enum": [
        "draft",
        "active",
        "suspended",
        "cancelled",
        "completed",
        "entered-in-error",
        "unknown"
      ]
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "category": {
      "description": "Category of supply, e.g.  central, non-stock, etc. This is used to support work flows
associated with the supply process.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "priority": {
      "description": "Indicates how quickly this SupplyRequest should be addressed with respect to other requests.",
      "$ref": "#/definitions/code"
    },
    "_priority": {
      "description": "Extensions for priority",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "itemCodeableConcept": {
      "description": "The item that is requested to be supplied. This is either a link to a resource representing
the details of the item or a code that identifies the item from a known list.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "itemReference": {
      "description": "The item that is requested to be supplied. This is either a link to a resource representing
the details of the item or a code that identifies the item from a known list.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "quantity": {
      "description": "The amount that is being ordered of the indicated item.",
      "$ref": "Quantity.schema.json#/definitions/Quantity"
    },
    "parameter": {
      "description": "Specific parameters for the ordered item.  For example, the size of the indicated item.",
      "items": {
        "$ref": "#/definitions/SupplyRequest_Parameter"
      },
      "type": "array"
    },
    "occurrenceDateTime": {
      "description": "When the request should be fulfilled.",
      "pattern": "^([0-9]([0-9]([0-9][1-9]|[1-9]0)|[1-9]00)|[1-9]000)(-(0[1-9]|1[0-2])(-(0[1-9]|[1-2][0-9]|3[0-
1])(T([01][0-9]|2[0-3]):[0-5][0-9]:([0-5][0-9]|60)(\\.[0-9]+)?(Z|(\\+|-)((0[0-9]|1[0-3]):[0-5][0-9]|14:00)))?)?)?$",
      "type": "string"
    },
    "_occurrenceDateTime": {
      "description": "Extensions for occurrenceDateTime",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "occurrencePeriod": {
      "description": "When the request should be fulfilled.",
      "$ref": "Period.schema.json#/definitions/Period"
    },
    "occurrenceTiming": {
      "description": "When the request should be fulfilled.",
      "$ref": "Timing.schema.json#/definitions/Timing"
    },
```

```
    "authoredOn": {
      "description": "When the request was made.",
      "$ref": "#/definitions/dateTime"
    },
    "_authoredOn": {
      "description": "Extensions for authoredOn",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "requester": {
      "description": "The device, practitioner, etc. who initiated the request.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "supplier": {
      "description": "Who is intended to fulfill the request.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "reasonCode": {
      "description": "The reason why the supply item was requested.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "reasonReference": {
      "description": "The reason why the supply item was requested.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "deliverFrom": {
      "description": "Where the supply is expected to come from.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "deliverTo": {
      "description": "Where the supply is destined to go.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    }
  },
  "additionalProperties": false,
  "required": [
    "quantity",
    "resourceType"
  ]
},
"SupplyRequest_Parameter": {
  "description": "A record of a request for a medication, substance or device used in the healthcare setting.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "code": {
      "description": "A code or string that identifies the device detail being asserted.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "valueCodeableConcept": {
      "description": "The value of the device detail.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "valueQuantity": {
```

```
          "description": "The value of the device detail.",
          "$ref": "Quantity.schema.json#/definitions/Quantity"
        },
        "valueRange": {
          "description": "The value of the device detail.",
          "$ref": "Range.schema.json#/definitions/Range"
        },
        "valueBoolean": {
          "description": "The value of the device detail.",
          "pattern": "^true|false$",
          "type": "boolean"
        },
        "_valueBoolean": {
          "description": "Extensions for valueBoolean",
          "$ref": "Element.schema.json#/definitions/Element"
        }
      },
      "additionalProperties": false
    }
  }
}
```


12.21.11 Resource SupplyRequest - Maturity Dependency View

URL: https://hl7.org/fhir/R4/supplyrequest-dependencies.html

12.21.11 Resource SupplyRequest - Maturity Dependency View

Orders and Observations Work Group | Maturity Level: 1 | Trial Use | Compartments: Device, Patient, Practitioner, RelatedPerson

Dependency Graph for SupplyRequest FMM level 1

Name | Card. | Type | Dependency Analysis
SupplyRequest | | DomainResource
identifier | 0..* | Identifier
status | 0..1 | code
category | 0..1 | CodeableConcept
priority | 0..1 | code
item[x] | 1..1
itemCodeableConcept | | CodeableConcept
itemReference | | Reference(Medication | Substance | Device
quantity | 1..1 | Quantity
parameter | 0..* | BackboneElement
code | 0..1 | CodeableConcept
value[x] | 0..1
valueCodeableConcept | | CodeableConcept
valueQuantity | | Quantity
valueRange | | Range
valueBoolean | | boolean
occurrence[x] | 0..1
occurrenceDateTime | | dateTime
occurrencePeriod | | Period
occurrenceTiming | | Timing
authoredOn | 0..1 | dateTime
requester | 0..1 | Reference(Practitioner | PractitionerRole | Organization | Patient | RelatedPerson | Device)
supplier | 0..* | Reference(Organization | HealthcareService)
reasonCode | 0..* | CodeableConcept
reasonReference | 0..* | Reference(Condition | Observation | DiagnosticReport | DocumentReference)
deliverFrom | 0..1 | Reference(Organization | Location)
deliverTo | 0..1 | Reference(Organization | Location | Patient)

12.21.12 Value Set Sc-valueset-supplyrequest-status - Status Code Mapping

URL: https://hl7.org/fhir/R4/sc-valueset-supplyrequest-status.html

12.21.12 Value Set Sc-valueset-supplyrequest-status - Status Code Mapping

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Device, Patient, Practitioner, RelatedPerson

This is the narrative for the resource. See also the XML, JSON or Turtle format.

12.21.13 SupplyRequestStatusCanonicalMap (http://hl7.org/fhir/ConceptMap/sc-supplyrequest-status)

Mapping from http://hl7.org/fhir/ValueSet/supplyrequest-status to http://hl7.org/fhir/ValueSet/resource-status

DRAFT. Published on 01/11/2019 9:29:23 AM by HL7 (FHIR Project) (http://hl7.org/fhir , fhir@lists.hl7.org).

Canonical Mapping for "Status of the supply request."

Source Code | Equivalence | Destination Code
entered-in-error | equivalent | error
draft | equivalent | draft
active | equivalent | active
suspended | equivalent | suspended
completed | equivalent | complete
cancelled | equivalent | abandoned
unknown | equivalent | unknown

12.22 Resource SupplyDelivery - Content

URL: https://hl7.org/fhir/R4/supplydelivery.html
Orders and Observations Work Group | Maturity Level: 1 | Trial Use | Security Category: Patient | Compartments: Patient, Practitioner

Record of delivery of what is supplied.

12.22.1 Scope and Usage

This resource is an event resource from a FHIR workflow perspective - see Workflow. It is the intent of the Orders and Observation Workgroup to align this resource with the workflow pattern for event resources.

The scope of the supply resource is for supplies used in the healthcare process. This includes supplies specifically used in the treatment of patients as well as supply movement within an institution (transport a set of supplies from materials management to a service unit (nurse station). This resource does not include the provisioning of transportation services.

12.22.2 Boundaries and Relationships

This resource overlaps with others such as Device and Medication. The Supply resource may be used to describe medications and devices when handling them generically (as any other supply). For example, when processing bulk orders, etc. However, when the medication, device or other aspects of the resource are important, (e.g. in a MedicationRequest or when identifying a device as a performer), the more detailed resource must be used.

This resource is referenced by ChargeItem and itself

12.22.3 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
SupplyDelivery | TU | | DomainResource | Delivery of bulk Supplies Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | | 0..* | Identifier | External identifier
basedOn | Σ | 0..* | Reference(SupplyRequest) | Fulfills plan, proposal or order
partOf | Σ | 0..* | Reference(SupplyDelivery | Contract) | Part of referenced event
status | ?!Σ | 0..1 | code | in-progress | completed | abandoned | entered-in-error SupplyDeliveryStatus (Required)
patient | | 0..1 | Reference(Patient) | Patient for whom the item is supplied
type | | 0..1 | CodeableConcept | Category of dispense event Supply Item Type (Required)
suppliedItem | | 0..1 | BackboneElement | The item that is delivered or supplied
quantity | | 0..1 | SimpleQuantity | Amount dispensed
item[x] | | 0..1 | | Medication, Substance, or Device supplied SNOMED CT Supply Item (Example)
itemCodeableConcept | | | CodeableConcept
itemReference | | | Reference(Medication | Substance | Device)
occurrence[x] | Σ | 0..1 | | When event occurred
occurrenceDateTime | | | dateTime
occurrencePeriod | | | Period
occurrenceTiming | | | Timing
supplier | | 0..1 | Reference(Practitioner | PractitionerRole | Organization) | Dispenser
destination | | 0..1 | Reference(Location) | Where the Supply was sent
receiver | | 0..* | Reference(Practitioner | PractitionerRole) | Who collected the Supply
Documentation for this format

UML Diagram (Legend)

XML Template

```
<SupplyDelivery xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier External identifier --></identifier>
 <basedOn><!-- 0..* Reference(SupplyRequest) Fulfills plan, proposal or order --></basedOn>
 <partOf><!-- 0..* Reference(SupplyDelivery|Contract) Part of referenced event --></partOf>
 <status value="[code]"/><!-- 0..1 in-progress | completed | abandoned | entered-in-error -->
 <patient><!-- 0..1 Reference(Patient) Patient for whom the item is supplied --></patient>
 <type><!-- 0..1 CodeableConcept Category of dispense event --></type>
 <suppliedItem>  <!-- 0..1 The item that is delivered or supplied -->
  <quantity><!-- 0..1 Quantity(SimpleQuantity) Amount dispensed --></quantity>
  <item[x]><!-- 0..1 CodeableConcept|Reference(Medication|Substance|Device) Medication, Substance, or Device supplied
    --></item[x]>
 </suppliedItem>
 <occurrence[x]><!-- 0..1 dateTime|Period|Timing When event occurred --></occurrence[x]>
 <supplier><!-- 0..1 Reference(Practitioner|PractitionerRole|Organization) Dispenser --></supplier>
 <destination><!-- 0..1 Reference(Location) Where the Supply was sent --></destination>
 <receiver><!-- 0..* Reference(Practitioner|PractitionerRole) Who collected the Supply --></receiver>
</SupplyDelivery>
```

JSON Template

```
{
  "resourceType" : "SupplyDelivery",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // External identifier
  "basedOn" : [{ Reference(SupplyRequest) }], // Fulfills plan, proposal or order
  "partOf" : [{ Reference(SupplyDelivery|Contract) }], // Part of referenced event
  "status" : "<code>", // in-progress | completed | abandoned | entered-in-error
  "patient" : { Reference(Patient) }, // Patient for whom the item is supplied
  "type" : { CodeableConcept }, // Category of dispense event
  "suppliedItem" : { // The item that is delivered or supplied
    "quantity" : { Quantity(SimpleQuantity) }, // Amount dispensed
    // item[x]: Medication, Substance, or Device supplied. One of these 2:
    "itemCodeableConcept" : { CodeableConcept }
    "itemReference" : { Reference(Medication|Substance|Device) }
  },
  // occurrence[x]: When event occurred. One of these 3:
  "occurrenceDateTime" : "<dateTime>",
  "occurrencePeriod" : { Period },
  "occurrenceTiming" : { Timing },
  "supplier" : { Reference(Practitioner|PractitionerRole|Organization) }, // Dispenser
  "destination" : { Reference(Location) }, // Where the Supply was sent
  "receiver" : [{ Reference(Practitioner|PractitionerRole) }] // Who collected the Supply
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:SupplyDelivery;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:SupplyDelivery.identifier [ Identifier ], ... ; # 0..* External identifier
  fhir:SupplyDelivery.basedOn [ Reference(SupplyRequest) ], ... ; # 0..* Fulfills plan, proposal or order
  fhir:SupplyDelivery.partOf [ Reference(SupplyDelivery|Contract) ], ... ; # 0..* Part of referenced event
  fhir:SupplyDelivery.status [ code ]; # 0..1 in-progress | completed | abandoned | entered-in-error
  fhir:SupplyDelivery.patient [ Reference(Patient) ]; # 0..1 Patient for whom the item is supplied
  fhir:SupplyDelivery.type [ CodeableConcept ]; # 0..1 Category of dispense event
  fhir:SupplyDelivery.suppliedItem [ # 0..1 The item that is delivered or supplied
    fhir:SupplyDelivery.suppliedItem.quantity [ Quantity(SimpleQuantity) ]; # 0..1 Amount dispensed
    # SupplyDelivery.suppliedItem.item[x] : 0..1 Medication, Substance, or Device supplied. One of these 2
      fhir:SupplyDelivery.suppliedItem.itemCodeableConcept [ CodeableConcept ]
      fhir:SupplyDelivery.suppliedItem.itemReference [ Reference(Medication|Substance|Device) ]
  ];
  # SupplyDelivery.occurrence[x] : 0..1 When event occurred. One of these 3
    fhir:SupplyDelivery.occurrenceDateTime [ dateTime ]
    fhir:SupplyDelivery.occurrencePeriod [ Period ]
    fhir:SupplyDelivery.occurrenceTiming [ Timing ]
  fhir:SupplyDelivery.supplier [ Reference(Practitioner|PractitionerRole|Organization) ]; # 0..1 Dispenser
  fhir:SupplyDelivery.destination [ Reference(Location) ]; # 0..1 Where the Supply was sent
  fhir:SupplyDelivery.receiver [ Reference(Practitioner|PractitionerRole) ], ... ; # 0..* Who collected the Supply
]
```

Changes since R3

SupplyDelivery

SupplyDelivery.identifier | Max Cardinality changed from 1 to *

SupplyDelivery.status | Change value set from http://hl7.org/fhir/ValueSet/supplydelivery-status to http://hl7.org/fhir/ValueSet/supplydelivery-status|4.0.1

SupplyDelivery.type | Change value set from http://hl7.org/fhir/ValueSet/supplydelivery-type to http://hl7.org/fhir/ValueSet/supplydelivery-type|4.0.1

SupplyDelivery.supplier | Type Reference: Added Target Type PractitionerRole

SupplyDelivery.receiver | Type Reference: Added Target Type PractitionerRole

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 2 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid.)

Structure

Name | Flags | Card. | Type | Description & Constraints
SupplyDelivery | TU | | DomainResource | Delivery of bulk Supplies Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | | 0..* | Identifier | External identifier
basedOn | Σ | 0..* | Reference(SupplyRequest) | Fulfills plan, proposal or order
partOf | Σ | 0..* | Reference(SupplyDelivery | Contract) | Part of referenced event
status | ?!Σ | 0..1 | code | in-progress | completed | abandoned | entered-in-error SupplyDeliveryStatus (Required)
patient | | 0..1 | Reference(Patient) | Patient for whom the item is supplied
type | | 0..1 | CodeableConcept | Category of dispense event Supply Item Type (Required)
suppliedItem | | 0..1 | BackboneElement | The item that is delivered or supplied
quantity | | 0..1 | SimpleQuantity | Amount dispensed
item[x] | | 0..1 | | Medication, Substance, or Device supplied SNOMED CT Supply Item (Example)
itemCodeableConcept | | | CodeableConcept
itemReference | | | Reference(Medication | Substance | Device)
occurrence[x] | Σ | 0..1 | | When event occurred
occurrenceDateTime | | | dateTime
occurrencePeriod | | | Period
occurrenceTiming | | | Timing
supplier | | 0..1 | Reference(Practitioner | PractitionerRole | Organization) | Dispenser
destination | | 0..1 | Reference(Location) | Where the Supply was sent
receiver | | 0..* | Reference(Practitioner | PractitionerRole) | Who collected the Supply
Documentation for this format

UML Diagram (Legend)

XML Template

```
<SupplyDelivery xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier External identifier --></identifier>
 <basedOn><!-- 0..* Reference(SupplyRequest) Fulfills plan, proposal or order --></basedOn>
 <partOf><!-- 0..* Reference(SupplyDelivery|Contract) Part of referenced event --></partOf>
 <status value="[code]"/><!-- 0..1 in-progress | completed | abandoned | entered-in-error -->
 <patient><!-- 0..1 Reference(Patient) Patient for whom the item is supplied --></patient>
 <type><!-- 0..1 CodeableConcept Category of dispense event --></type>
 <suppliedItem>  <!-- 0..1 The item that is delivered or supplied -->
  <quantity><!-- 0..1 Quantity(SimpleQuantity) Amount dispensed --></quantity>
  <item[x]><!-- 0..1 CodeableConcept|Reference(Medication|Substance|Device) Medication, Substance, or Device supplied
    --></item[x]>
 </suppliedItem>
 <occurrence[x]><!-- 0..1 dateTime|Period|Timing When event occurred --></occurrence[x]>
 <supplier><!-- 0..1 Reference(Practitioner|PractitionerRole|Organization) Dispenser --></supplier>
 <destination><!-- 0..1 Reference(Location) Where the Supply was sent --></destination>
 <receiver><!-- 0..* Reference(Practitioner|PractitionerRole) Who collected the Supply --></receiver>
</SupplyDelivery>
```

JSON Template

```
{
  "resourceType" : "SupplyDelivery",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // External identifier
  "basedOn" : [{ Reference(SupplyRequest) }], // Fulfills plan, proposal or order
  "partOf" : [{ Reference(SupplyDelivery|Contract) }], // Part of referenced event
  "status" : "<code>", // in-progress | completed | abandoned | entered-in-error
  "patient" : { Reference(Patient) }, // Patient for whom the item is supplied
  "type" : { CodeableConcept }, // Category of dispense event
  "suppliedItem" : { // The item that is delivered or supplied
    "quantity" : { Quantity(SimpleQuantity) }, // Amount dispensed
    // item[x]: Medication, Substance, or Device supplied. One of these 2:
    "itemCodeableConcept" : { CodeableConcept }
    "itemReference" : { Reference(Medication|Substance|Device) }
  },
  // occurrence[x]: When event occurred. One of these 3:
  "occurrenceDateTime" : "<dateTime>",
  "occurrencePeriod" : { Period },
  "occurrenceTiming" : { Timing },
  "supplier" : { Reference(Practitioner|PractitionerRole|Organization) }, // Dispenser
  "destination" : { Reference(Location) }, // Where the Supply was sent
  "receiver" : [{ Reference(Practitioner|PractitionerRole) }] // Who collected the Supply
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .
```

```
[ a fhir:SupplyDelivery;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:SupplyDelivery.identifier [ Identifier ], ... ; # 0..* External identifier
  fhir:SupplyDelivery.basedOn [ Reference(SupplyRequest) ], ... ; # 0..* Fulfills plan, proposal or order
  fhir:SupplyDelivery.partOf [ Reference(SupplyDelivery|Contract) ], ... ; # 0..* Part of referenced event
  fhir:SupplyDelivery.status [ code ]; # 0..1 in-progress | completed | abandoned | entered-in-error
  fhir:SupplyDelivery.patient [ Reference(Patient) ]; # 0..1 Patient for whom the item is supplied
  fhir:SupplyDelivery.type [ CodeableConcept ]; # 0..1 Category of dispense event
  fhir:SupplyDelivery.suppliedItem [ # 0..1 The item that is delivered or supplied
    fhir:SupplyDelivery.suppliedItem.quantity [ Quantity(SimpleQuantity) ]; # 0..1 Amount dispensed
    # SupplyDelivery.suppliedItem.item[x] : 0..1 Medication, Substance, or Device supplied. One of these 2
      fhir:SupplyDelivery.suppliedItem.itemCodeableConcept [ CodeableConcept ]
      fhir:SupplyDelivery.suppliedItem.itemReference [ Reference(Medication|Substance|Device) ]
  ];
  # SupplyDelivery.occurrence[x] : 0..1 When event occurred. One of these 3
    fhir:SupplyDelivery.occurrenceDateTime [ dateTime ]
    fhir:SupplyDelivery.occurrencePeriod [ Period ]
    fhir:SupplyDelivery.occurrenceTiming [ Timing ]
  fhir:SupplyDelivery.supplier [ Reference(Practitioner|PractitionerRole|Organization) ]; # 0..1 Dispenser
  fhir:SupplyDelivery.destination [ Reference(Location) ]; # 0..1 Where the Supply was sent
  fhir:SupplyDelivery.receiver [ Reference(Practitioner|PractitionerRole) ], ... ; # 0..* Who collected the Supply
]
```

Changes since Release 3

SupplyDelivery
SupplyDelivery.identifier | Max Cardinality changed from 1 to *
SupplyDelivery.status | Change value set from http://hl7.org/fhir/ValueSet/supplydelivery-status to http://hl7.org/fhir/ValueSet/supplydelivery-status|4.0.1
SupplyDelivery.type | Change value set from http://hl7.org/fhir/ValueSet/supplydelivery-type to http://hl7.org/fhir/ValueSet/supplydelivery-type|4.0.1
SupplyDelivery.supplier | Type Reference: Added Target Type PractitionerRole
SupplyDelivery.receiver | Type Reference: Added Target Type PractitionerRole

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 2 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid.)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

12.22.3.1 Terminology Bindings

Path | Definition | Type | Reference
SupplyDelivery.status | Status of the supply delivery. | Required | SupplyDeliveryStatus
SupplyDelivery.type | The type of supply dispense. | Required | SupplyItemType
SupplyDelivery.suppliedItem.item[x] | The item that was delivered. | Example | SNOMEDCTSupplyItem

12.22.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
identifier | token | External identifier | SupplyDelivery.identifier | 30 Resources
patient | reference | Patient for whom the item is supplied | SupplyDelivery.patient (Patient) | 33 Resources
receiver | reference | Who collected the Supply | SupplyDelivery.receiver (Practitioner, PractitionerRole)
status | token | in-progress | completed | abandoned | entered-in-error | SupplyDelivery.status
supplier | reference | Dispenser | SupplyDelivery.supplier (Practitioner, Organization, PractitionerRole)

12.22.5 Resource SupplyDelivery - Examples

URL: https://hl7.org/fhir/R4/supplydelivery-examples.html

12.22.5 Resource SupplyDelivery - Examples

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Patient, Practitioner

Example Name | id | Format
Simple delivery for resupply | simpledelivery | XML | JSON | Turtle
Delivery of an insulin pump | pumpdelivery | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

12.22.6 Resource SupplyDelivery - Detailed Descriptions

URL: https://hl7.org/fhir/R4/supplydelivery-definitions.html

12.22.6 Resource SupplyDelivery - Detailed Descriptions

Orders and Observations Work Group | Maturity Level: 1 | Trial Use | Security Category: Patient | Compartments: Patient, Practitioner

Detailed Descriptions for the elements in the SupplyDelivery resource.

SupplyDelivery
Element Id | SupplyDelivery
Definition | Record of delivery of what is supplied.
Cardinality | 0..*
Type | DomainResource
SupplyDelivery.identifier
Element Id | SupplyDelivery.identifier
Definition | Identifier for the supply delivery event that is used to identify it across multiple disparate systems.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Comments | This identifier is typically assigned by the dispenser, and may be used to reference the delivery when exchanging information about it with other systems.
SupplyDelivery.basedOn
Element Id | SupplyDelivery.basedOn
Definition | A plan, proposal or order that is fulfilled in whole or in part by this event.
Cardinality | 0..*
Type | Reference(SupplyRequest)
Requirements | Allows tracing of authorization for the event and tracking whether proposals/recommendations were acted upon.
Summary | true
SupplyDelivery.partOf
Element Id | SupplyDelivery.partOf
Definition | A larger event of which this particular event is a component or step.
Cardinality | 0..*
Type | Reference(SupplyDelivery | Contract)
Requirements | E.g. Drug administration as part of a procedure, procedure as part of observation, etc.
Summary | true
Comments | Not to be used to link an event to an Encounter - use Event.context for that. [The allowed reference resources may be adjusted as appropriate for the event resource].
SupplyDelivery.status
Element Id | SupplyDelivery.status
Definition | A code specifying the state of the dispense event.
Cardinality | 0..1
Terminology Binding | SupplyDeliveryStatus (Required)
Type | code
Is Modifier | true (Reason: This element is labelled as a modifier because it is a status element that contains status entered-in-error which means that the resource should not be treated as valid)
Summary | true
Comments | This element is labeled as a modifier because the status contains codes that mark the resource as not currently valid.
SupplyDelivery.patient
Element Id | SupplyDelivery.patient
Definition | A link to a resource representing the person whom the delivered item is for.
Cardinality | 0..1
Type | Reference(Patient)
SupplyDelivery.type
Element Id | SupplyDelivery.type
Definition | Indicates the type of dispensing event that is performed. Examples include: Trial Fill, Completion of Trial, Partial Fill, Emergency Fill, Samples, etc.
Cardinality | 0..1
Terminology Binding | Supply Item Type (Required)
Type | CodeableConcept
SupplyDelivery.suppliedItem
Element Id | SupplyDelivery.suppliedItem
Definition | The item that is being delivered or has been supplied.
Cardinality | 0..1
SupplyDelivery.suppliedItem.quantity
Element Id | SupplyDelivery.suppliedItem.quantity
Definition | The amount of supply that has been dispensed. Includes unit of measure.
Cardinality | 0..1
Type | SimpleQuantity
SupplyDelivery.suppliedItem.item[x]
Element Id | SupplyDelivery.suppliedItem.item[x]
Definition | Identifies the medication, substance or device being dispensed. This is either a link to a resource representing the details of the item or a code that identifies the item from a known list.
Cardinality | 0..1
Terminology Binding | SNOMED CT Supply Item (Example)

Type | CodeableConcept|Reference(Medication | Substance | Device)
[x] Note | See Choice of Data Types for further information about how to use [x]
SupplyDelivery.occurrence[x]
Element Id | SupplyDelivery.occurrence[x]
Definition | The date or time(s) the activity occurred.
Cardinality | 0..1
Type | dateTime|Period|Timing
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true
Comments | [The list of types may be constrained as appropriate for the type of event].
SupplyDelivery.supplier
Element Id | SupplyDelivery.supplier
Definition | The individual responsible for dispensing the medication, supplier or device.
Cardinality | 0..1
Type | Reference(Practitioner | PractitionerRole | Organization)
SupplyDelivery.destination
Element Id | SupplyDelivery.destination
Definition | Identification of the facility/location where the Supply was shipped to, as part of the dispense event.
Cardinality | 0..1
Type | Reference(Location)
SupplyDelivery.receiver
Element Id | SupplyDelivery.receiver
Definition | Identifies the person who picked up the Supply.
Cardinality | 0..*
Type | Reference(Practitioner | PractitionerRole)

12.22.7 Resource SupplyDelivery - Mappings

URL: https://hl7.org/fhir/R4/supplydelivery-mappings.html

12.22.7 Resource SupplyDelivery - Mappings

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Patient, Practitioner

Mappings:

FiveWs Pattern Mapping

Workflow Pattern

HL7 v2 Mapping

RIM Mapping

Mappings for the supplydelivery resource (see Mappings to Other Standards for further information & status).

12.22.7.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

SupplyDelivery
occurrence[x] | FiveWs.done[x]

12.22.7.2 Workflow Pattern (http://hl7.org/fhir/workflow)

SupplyDelivery | Event
identifier | Event.identifier
basedOn | Event.basedOn
partOf | Event.partOf
status | Event.status
patient | Event.subject
type | Event.code
occurrence[x] | Event.occurrence[x]
supplier | Event.performer

12.22.7.3 HL7 v2 Mapping (http://hl7.org/v2)

SupplyDelivery
identifier
basedOn | ORC in proximity to EVN segment
partOf | Varies by domain
status
patient
type
suppliedItem
quantity
item[x]
occurrence[x] | Varies by domain
supplier

destination
receiver

12.22.7.4 RIM Mapping (http://hl7.org/v3)

SupplyDelivery | Supply[moodCode=RQO]
identifier | .id
basedOn | .outboundRelationship[typeCode=FLFS].target
partOf | .inboundRelationship[typeCode=COMP].source[moodCode=EVN]
status | .statusCode, also existence of fulfillment events
patient | .participation[typeCode=SBJ].role
type | .code
suppliedItem | .participation[typeCode=PRD].role
quantity | .quantity
item[x] | .participation[typeCode=PRD].role
occurrence[x] | .effectiveTime
supplier | .participation[typeCode=PFM].role
destination | .participation[typeCode=DST].role
receiver | .participation[typeCode=RCV].role

12.22.8 Resource SupplyDelivery - Extensions & Profiles

URL: https://hl7.org/fhir/R4/supplydelivery-profiles.html

12.22.8 Resource SupplyDelivery - Extensions & Profiles

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Patient, Practitioner

This table lists profiles and extensions for the SupplyDelivery resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

12.22.8.1 Profiles

No Profiles defined for this resource

12.22.8.2 Extensions

event-eventHistory | SupplyDelivery | eventHistory | for Event Pattern HL7 Extensions
event-location | SupplyDelivery | location | for Event Pattern HL7 Extensions
event-statusReason | SupplyDelivery | statusReason | for Event Pattern HL7 Extensions
workflow-instantiatesCanonical | SupplyDelivery | instantiatesCanonical | for Workflow Pattern HL7 Extensions
workflow-instantiatesUri | SupplyDelivery | instantiatesUri | for Workflow Pattern HL7 Extensions
workflow-researchStudy | SupplyDelivery | researchStudy | for Workflow Pattern HL7 Extensions
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

12.22.8.3 Search Extensions

No Search Extensions defined for this resource

12.22.9 Resource SupplyDelivery - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/supplydelivery-version-maps.html

No validation errors - all conversions are clean

12.22.9 Resource SupplyDelivery - R3/R4 Conversions

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Patient, Practitioner

R3 : R4 Conversion maps for SupplyDelivery.

Functional status for this map: 2 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid. (see documentation)

12.22.9.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/SupplyDelivery3to4" = "R3 to R4 Conversions for SupplyDelivery"

uses "http://hl7.org/fhir/3.0/StructureDefinition/SupplyDelivery" alias SupplyDeliveryR3 as source
uses "http://hl7.org/fhir/StructureDefinition/SupplyDelivery" alias SupplyDelivery as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group SupplyDelivery(source src : SupplyDeliveryR3, target tgt : SupplyDelivery) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.basedOn -> tgt.basedOn;
  src.partOf -> tgt.partOf;
  src.status -> tgt.status;
  src.patient -> tgt.patient;
  src.type -> tgt.type;
  src.suppliedItem as vs0 -> tgt.suppliedItem as vt0 then suppliedItem(vs0, vt0);
  src.occurrence : dateTime as vs -> tgt.occurrence = create('dateTime') as vt then dateTime(vs, vt);
  src.occurrence : Period as vs -> tgt.occurrence = create('Period') as vt then Period(vs, vt);
  src.occurrence : Timing as vs -> tgt.occurrence = create('Timing') as vt then Timing(vs, vt);
  src.supplier -> tgt.supplier;
  src.destination -> tgt.destination;
  src.receiver -> tgt.receiver;
}

group suppliedItem(source src, target tgt) extends BackboneElement {
  src.quantity -> tgt.quantity;
  src.item : CodeableConcept as vs -> tgt.item = create('CodeableConcept') as vt then CodeableConcept(vs, vt);
  src.item : Reference as vs -> tgt.item = create('Reference') as vt then Reference(vs, vt);
}
```

12.22.9.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/SupplyDelivery4to3" = "R4 to R3 Conversion for SupplyDelivery"

uses "http://hl7.org/fhir/StructureDefinition/SupplyDelivery" alias SupplyDelivery as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/SupplyDelivery" alias SupplyDeliveryR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group SupplyDelivery(source src : SupplyDelivery, target tgt : SupplyDeliveryR3) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.basedOn -> tgt.basedOn;
  src.partOf -> tgt.partOf;
  src.status -> tgt.status;
  src.patient -> tgt.patient;
  src.type -> tgt.type;
  src.suppliedItem as vs0 -> tgt.suppliedItem as vt0 then suppliedItem(vs0, vt0);
  src.occurrence : dateTime as vs -> tgt.occurrence = create('dateTime') as vt then dateTime(vs, vt);
  src.occurrence : Period as vs -> tgt.occurrence = create('Period') as vt then Period(vs, vt);
  src.occurrence : Timing as vs -> tgt.occurrence = create('Timing') as vt then Timing(vs, vt);
  src.supplier -> tgt.supplier;
  src.destination -> tgt.destination;
  src.receiver -> tgt.receiver;
}

group suppliedItem(source src, target tgt) extends BackboneElement {
  src.quantity -> tgt.quantity;
  src.item : CodeableConcept as vs -> tgt.item = create('CodeableConcept') as vt then CodeableConcept(vs, vt);
  src.item : Reference as vs -> tgt.item = create('Reference') as vt then Reference(vs, vt);
}
```

12.22.9.3 R4 Validation Errors

12.22.10 JSON Schema for SupplyDelivery

URL: https://hl7.org/fhir/R4/supplydelivery.schema.json.html

12.22.10 JSON Schema for SupplyDelivery

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Patient, Practitioner

This is a representation of the json schema for SupplyDelivery, which is just a part of the full JSON Schema.

```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/SupplyDelivery",
  "$ref": "#/definitions/SupplyDelivery",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "SupplyDelivery": {
      "description": "Record of delivery of what is supplied.",
      "properties": {
        "resourceType": {
          "description": "This is a SupplyDelivery resource",
          "const": "SupplyDelivery"
```

```
    },
    "id": {
      "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
 never changes.",
      "$ref": "id.schema.json#/definitions/id"
    },
    "meta": {
      "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
Changes to the content might not always be associated with version changes to the resource.",
      "$ref": "Meta.schema.json#/definitions/Meta"
    },
    "implicitRules": {
      "description": "A reference to a set of rules that were followed when the resource was constructed, and which
must be understood when processing the content. Often, this is a reference to an implementation guide that defines
the special rules along with other profiles etc.",
      "$ref": "#/definitions/uri"
    },
    "_implicitRules": {
      "description": "Extensions for implicitRules",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "language": {
      "description": "The base language in which the resource is written.",
      "$ref": "#/definitions/code"
    },
    "_language": {
      "description": "Extensions for language",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "text": {
      "description": "A human-readable narrative that contains a summary of the resource and can be used to
represent the content of the resource to a human. The narrative need not encode all the structured data, but is
required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
      "$ref": "Narrative.schema.json#/definitions/Narrative"
    },
    "contained": {
      "description": "These resources do not have an independent existence apart from the resource that contains
them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
      "items": {
        "$ref": "ResourceList.schema.json#/definitions/ResourceList"
      },
      "type": "array"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
 definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource and that modifies the understanding of the element that contains it and/or the understanding of the
containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
met as part of the definition of the extension. Applications processing a resource are required to check for
modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "identifier": {
      "description": "Identifier for the supply delivery event that is used to identify it across multiple disparate
 systems.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "basedOn": {
      "description": "A plan, proposal or order that is fulfilled in whole or in part by this event.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "partOf": {
      "description": "A larger event of which this particular event is a component or step.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
```

```
    },
    "status": {
      "description": "A code specifying the state of the dispense event.",
      "enum": [
        "in-progress",
        "completed",
        "abandoned",
        "entered-in-error"
      ]
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "patient": {
      "description": "A link to a resource representing the person whom the delivered item is for.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "type": {
      "description": "Indicates the type of dispensing event that is performed. Examples include: Trial Fill,
Completion of Trial, Partial Fill, Emergency Fill, Samples, etc.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "suppliedItem": {
      "description": "The item that is being delivered or has been supplied.",
      "$ref": "#/definitions/SupplyDelivery_SuppliedItem"
    },
    "occurrenceDateTime": {
      "description": "The date or time(s) the activity occurred.",
      "pattern": "^([0-9]([0-9]([0-9][1-9]|[1-9]0)|[1-9]00)|[1-9]000)(-(0[1-9]|1[0-2])(-(0[1-9]|[1-2][0-9]|3[0-
1])(T([01][0-9]|2[0-3]):[0-5][0-9]:([0-5][0-9]|60)(\\.[0-9]+)?(Z|(\\+|-)((0[0-9]|1[0-3]):[0-5][0-9]|14:00)))?)?)?$",
      "type": "string"
    },
    "_occurrenceDateTime": {
      "description": "Extensions for occurrenceDateTime",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "occurrencePeriod": {
      "description": "The date or time(s) the activity occurred.",
      "$ref": "Period.schema.json#/definitions/Period"
    },
    "occurrenceTiming": {
      "description": "The date or time(s) the activity occurred.",
      "$ref": "Timing.schema.json#/definitions/Timing"
    },
    "supplier": {
      "description": "The individual responsible for dispensing the medication, supplier or device.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "destination": {
      "description": "Identification of the facility/location where the Supply was shipped to, as part of the
dispense event.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "receiver": {
      "description": "Identifies the person who picked up the Supply.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "resourceType"
  ]
},
"SupplyDelivery_SuppliedItem": {
  "description": "Record of delivery of what is supplied.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
```

```
    use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
    extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
     of the definition of the extension. Applications processing a resource are required to check for modifier
    extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
    (including cannot change the meaning of modifierExtension itself).",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "quantity": {
          "description": "The amount of supply that has been dispensed. Includes unit of measure.",
          "$ref": "Quantity.schema.json#/definitions/Quantity"
        },
        "itemCodeableConcept": {
          "description": "Identifies the medication, substance or device being dispensed. This is either a link to a
    resource representing the details of the item or a code that identifies the item from a known list.",
          "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
        },
        "itemReference": {
          "description": "Identifies the medication, substance or device being dispensed. This is either a link to a
    resource representing the details of the item or a code that identifies the item from a known list.",
          "$ref": "Reference.schema.json#/definitions/Reference"
        }
      },
      "additionalProperties": false
    }
  }
}
```


12.22.11 Resource SupplyDelivery - Maturity Dependency View

URL: https://hl7.org/fhir/R4/supplydelivery-dependencies.html

12.22.11 Resource SupplyDelivery - Maturity Dependency View

Orders and Observations Work Group | Maturity Level: 1 | Trial Use | Compartments: Patient, Practitioner

Dependency Graph for SupplyDelivery FMM level 1

Name | Card. | Type | Dependency Analysis
SupplyDelivery | | DomainResource
identifier | 0..* | Identifier
basedOn | 0..* | Reference(SupplyRequest)
partOf | 0..* | Reference(SupplyDelivery | Contract)
status | 0..1 | code
patient | 0..1 | Reference(Patient)
type | 0..1 | CodeableConcept
suppliedItem | 0..1 | BackboneElement
quantity | 0..1 | SimpleQuantity
item[x] | 0..1
itemCodeableConcept | | CodeableConcept
itemReference | | Reference(Medication | Substance | Device
occurrence[x] | 0..1
occurrenceDateTime | | dateTime
occurrencePeriod | | Period
occurrenceTiming | | Timing
supplier | 0..1 | Reference(Practitioner | PractitionerRole | Organization)
destination | 0..1 | Reference(Location)
receiver | 0..* | Reference(Practitioner | PractitionerRole)

12.22.12 Value Set Sc-valueset-supplydelivery-status - Status Code Mapping

URL: https://hl7.org/fhir/R4/sc-valueset-supplydelivery-status.html

12.22.12 Value Set Sc-valueset-supplydelivery-status - Status Code Mapping

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Patient, Practitioner

This is the narrative for the resource. See also the XML, JSON or Turtle format.

12.22.13 SupplyDeliveryStatusCanonicalMap (http://hl7.org/fhir/ConceptMap/sc-supplydelivery-status)

Mapping from http://hl7.org/fhir/ValueSet/supplydelivery-status to http://hl7.org/fhir/ValueSet/resource-status

DRAFT. Published on 01/11/2019 9:29:23 AM by HL7 (FHIR Project) (http://hl7.org/fhir , fhir@lists.hl7.org).

Canonical Mapping for "Status of the supply delivery."

Source Code | Equivalence | Destination Code

entered-in-error | equivalent | error
in-progress | equivalent | active
completed (Delivered) | equivalent | complete
abandoned | equivalent | abandoned

12.23 Resource ExampleScenario - Content

URL: https://hl7.org/fhir/R4/examplescenario.html
FHIR Infrastructure Work Group | Maturity Level: 0 | Trial Use | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Example of workflow instance.

12.23.1 Scope and Usage

Todo

This resource is referenced by itself

12.23.2 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
ExampleScenario | ITU | | DomainResource | Example of workflow instance + Warning: Name should be usable as an identifier for the module by machine processing applications such as code generation Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
url | Σ | 0..1 | uri | Canonical identifier for this example scenario, represented as a URI (globally unique)
identifier | Σ | 0..* | Identifier | Additional identifier for the example scenario
version | Σ | 0..1 | string | Business version of the example scenario
name | ΣI | 0..1 | string | Name for this example scenario (computer friendly)
status | ?!Σ | 1..1 | code | draft | active | retired | unknown PublicationStatus (Required)
experimental | Σ | 0..1 | boolean | For testing purposes, not real usage
date | Σ | 0..1 | dateTime | Date last changed
publisher | Σ | 0..1 | string | Name of the publisher (organization or individual)
contact | Σ | 0..* | ContactDetail | Contact details for the publisher
useContext | Σ | 0..* | UsageContext | The context that the content is intended to support
jurisdiction | Σ | 0..* | CodeableConcept | Intended jurisdiction for example scenario (if applicable) Jurisdiction (Extensible)
copyright | | 0..1 | markdown | Use and/or publishing restrictions
purpose | | 0..1 | markdown | The purpose of the example, e.g. to illustrate a scenario
actor | | 0..* | BackboneElement | Actor participating in the resource
actorId | | 1..1 | string | ID or acronym of the actor
type | | 1..1 | code | person | entity ExampleScenarioActorType (Required)
name | | 0..1 | string | The name of the actor as shown in the page
description | | 0..1 | markdown | The description of the actor
instance | | 0..* | BackboneElement | Each resource and each version that is present in the workflow
resourceId | | 1..1 | string | The id of the resource for referencing
resourceType | | 1..1 | code | The type of the resource ResourceType (Required)
name | | 0..1 | string | A short name for the resource instance
description | | 0..1 | markdown | Human-friendly description of the resource instance
version | | 0..* | BackboneElement | A specific version of the resource
versionId | | 1..1 | string | The identifier of a specific version of a resource
description | | 1..1 | markdown | The description of the resource version
containedInstance | | 0..* | BackboneElement | Resources contained in the instance
resourceId | | 1..1 | string | Each resource contained in the instance
versionId | | 0..1 | string | A specific version of a resource contained in the instance
process | | 0..* | BackboneElement | Each major process - a group of operations
title | Σ | 1..1 | string | The diagram title of the group of operations
description | | 0..1 | markdown | A longer description of the group of operations
preConditions | | 0..1 | markdown | Description of initial status before the process starts
postConditions | | 0..1 | markdown | Description of final status after the process ends
step | | 0..* | BackboneElement | Each step of the process
process | | 0..* | see process | Nested process
pause | | 0..1 | boolean | If there is a pause in the flow
operation | | 0..1 | BackboneElement | Each interaction or action

number | | 1..1 | string | The sequential number of the interaction
type | | 0..1 | string | The type of operation - CRUD
name | | 0..1 | string | The human-friendly name of the interaction
initiator | | 0..1 | string | Who starts the transaction
receiver | | 0..1 | string | Who receives the transaction
description | | 0..1 | markdown | A comment to be inserted in the diagram
initiatorActive | | 0..1 | boolean | Whether the initiator is deactivated right after the transaction
receiverActive | | 0..1 | boolean | Whether the receiver is deactivated right after the transaction
request | | 0..1 | see containedInstance | Each resource instance used by the initiator
response | | 0..1 | see containedInstance | Each resource instance used by the responder
alternative | | 0..* | BackboneElement | Alternate non-typical step action
title | | 1..1 | string | Label for alternative
description | | 0..1 | markdown | A human-readable description of each option
step | | 0..* | see step | What happens in each alternative option
workflow | | 0..* | canonical(ExampleScenario) | Another nested workflow
Documentation for this format

UML Diagram (Legend)

XML Template

```
<ExampleScenario xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <url value="[uri]"/><!-- 0..1 Canonical identifier for this example scenario, represented as a URI (globally unique)
    -->
 <identifier><!-- 0..* Identifier Additional identifier for the example scenario --></identifier>
 <version value="[string]"/><!-- 0..1 Business version of the example scenario -->
 <name value="[string]"/><!--  0..1 Name for this example scenario (computer friendly) -->
 <status value="[code]"/><!-- 1..1 draft | active | retired | unknown -->
 <experimental value="[boolean]"/><!-- 0..1 For testing purposes, not real usage -->
 <date value="[dateTime]"/><!-- 0..1 Date last changed -->
 <publisher value="[string]"/><!-- 0..1 Name of the publisher (organization or individual) -->
 <contact><!-- 0..* ContactDetail Contact details for the publisher --></contact>
 <useContext><!-- 0..* UsageContext The context that the content is intended to support --></useContext>
 <jurisdiction><!-- 0..* CodeableConcept Intended jurisdiction for example scenario (if applicable) --></jurisdiction>
 <copyright value="[markdown]"/><!-- 0..1 Use and/or publishing restrictions -->
 <purpose value="[markdown]"/><!-- 0..1 The purpose of the example, e.g. to illustrate a scenario -->
 <actor>  <!-- 0..* Actor participating in the resource -->
  <actorId value="[string]"/><!-- 1..1 ID or acronym of the actor -->
  <type value="[code]"/><!-- 1..1 person | entity -->
  <name value="[string]"/><!-- 0..1 The name of the actor as shown in the page -->
  <description value="[markdown]"/><!-- 0..1 The description of the actor -->
 </actor>
 <instance>  <!-- 0..* Each resource and each version that is present in the workflow -->
  <resourceId value="[string]"/><!-- 1..1 The id of the resource for referencing -->
  <resourceType value="[code]"/><!-- 1..1 The type of the resource -->
  <name value="[string]"/><!-- 0..1 A short name for the resource instance -->
  <description value="[markdown]"/><!-- 0..1 Human-friendly description of the resource instance -->
  <version>  <!-- 0..* A specific version of the resource -->
   <versionId value="[string]"/><!-- 1..1 The identifier of a specific version of a resource -->
   <description value="[markdown]"/><!-- 1..1 The description of the resource version -->
  </version>
  <containedInstance>  <!-- 0..* Resources contained in the instance -->
   <resourceId value="[string]"/><!-- 1..1 Each resource contained in the instance -->
   <versionId value="[string]"/><!-- 0..1 A specific version of a resource contained in the instance -->
  </containedInstance>
 </instance>
 <process>  <!-- 0..* Each major process - a group of operations -->
  <title value="[string]"/><!-- 1..1 The diagram title of the group of operations -->
  <description value="[markdown]"/><!-- 0..1 A longer description of the group of operations -->
  <preConditions value="[markdown]"/><!-- 0..1 Description of initial status before the process starts -->
  <postConditions value="[markdown]"/><!-- 0..1 Description of final status after the process ends -->
  <step>  <!-- 0..* Each step of the process -->
   <process><!-- 0..* Content as for ExampleScenario.process Nested process --></process>
   <pause value="[boolean]"/><!-- 0..1 If there is a pause in the flow -->
   <operation>  <!-- 0..1 Each interaction or action -->
    <number value="[string]"/><!-- 1..1 The sequential number of the interaction -->
    <type value="[string]"/><!-- 0..1 The type of operation - CRUD -->
    <name value="[string]"/><!-- 0..1 The human-friendly name of the interaction -->
    <initiator value="[string]"/><!-- 0..1 Who starts the transaction -->
    <receiver value="[string]"/><!-- 0..1 Who receives the transaction -->
    <description value="[markdown]"/><!-- 0..1 A comment to be inserted in the diagram -->
    <initiatorActive value="[boolean]"/><!-- 0..1 Whether the initiator is deactivated right after the transaction -->
    <receiverActive value="[boolean]"/><!-- 0..1 Whether the receiver is deactivated right after the transaction -->
    <request><!-- 0..1 Content as for ExampleScenario.instance.containedInstance Each resource instance used by the
    initiator --></request>
    <response><!-- 0..1 Content as for ExampleScenario.instance.containedInstance Each resource instance used by the
    responder --></response>
   </operation>
   <alternative>  <!-- 0..* Alternate non-typical step action -->
    <title value="[string]"/><!-- 1..1 Label for alternative -->
    <description value="[markdown]"/><!-- 0..1 A human-readable description of each option -->
    <step><!-- 0..* Content as for ExampleScenario.process.step What happens in each alternative option --></step>
   </alternative>
  </step>
 </process>
```

```
 <workflow><!-- 0..* canonical(ExampleScenario) Another nested workflow --></workflow>
</ExampleScenario>
```

JSON Template

```
{
  "resourceType" : "ExampleScenario",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "url" : "<uri>", // Canonical identifier for this example scenario, represented as a URI (globally unique)
  "identifier" : [{ Identifier }], // Additional identifier for the example scenario
  "version" : "<string>", // Business version of the example scenario
  "name" : "<string>", // C? Name for this example scenario (computer friendly)
  "status" : "<code>", // R!  draft | active | retired | unknown
  "experimental" : <boolean>, // For testing purposes, not real usage
  "date" : "<dateTime>", // Date last changed
  "publisher" : "<string>", // Name of the publisher (organization or individual)
  "contact" : [{ ContactDetail }], // Contact details for the publisher
  "useContext" : [{ UsageContext }], // The context that the content is intended to support
  "jurisdiction" : [{ CodeableConcept }], // Intended jurisdiction for example scenario (if applicable)
  "copyright" : "<markdown>", // Use and/or publishing restrictions
  "purpose" : "<markdown>", // The purpose of the example, e.g. to illustrate a scenario
  "actor" : [{ // Actor participating in the resource
    "actorId" : "<string>", // R!  ID or acronym of the actor
    "type" : "<code>", // R!  person | entity
    "name" : "<string>", // The name of the actor as shown in the page
    "description" : "<markdown>" // The description of the actor
  }],
  "instance" : [{ // Each resource and each version that is present in the workflow
    "resourceId" : "<string>", // R!  The id of the resource for referencing
    "resourceType" : "<code>", // R!  The type of the resource
    "name" : "<string>", // A short name for the resource instance
    "description" : "<markdown>", // Human-friendly description of the resource instance
    "version" : [{ // A specific version of the resource
      "versionId" : "<string>", // R!  The identifier of a specific version of a resource
      "description" : "<markdown>" // R!  The description of the resource version
    }],
    "containedInstance" : [{ // Resources contained in the instance
      "resourceId" : "<string>", // R!  Each resource contained in the instance
      "versionId" : "<string>" // A specific version of a resource contained in the instance
    }]
  }],
  "process" : [{ // Each major process - a group of operations
    "title" : "<string>", // R!  The diagram title of the group of operations
    "description" : "<markdown>", // A longer description of the group of operations
    "preConditions" : "<markdown>", // Description of initial status before the process starts
    "postConditions" : "<markdown>", // Description of final status after the process ends
    "step" : [{ // Each step of the process
      "process" : [{ Content as for ExampleScenario.process }], // Nested process
      "pause" : <boolean>, // If there is a pause in the flow
      "operation" : { // Each interaction or action
        "number" : "<string>", // R!  The sequential number of the interaction
        "type" : "<string>", // The type of operation - CRUD
        "name" : "<string>", // The human-friendly name of the interaction
        "initiator" : "<string>", // Who starts the transaction
        "receiver" : "<string>", // Who receives the transaction
        "description" : "<markdown>", // A comment to be inserted in the diagram
        "initiatorActive" : <boolean>, // Whether the initiator is deactivated right after the transaction
        "receiverActive" : <boolean>, // Whether the receiver is deactivated right after the transaction
        "request" : { Content as for ExampleScenario.instance.containedInstance }, // Each resource instance used by the
    initiator
        "response" : { Content as for ExampleScenario.instance.containedInstance } // Each resource instance used by the
    responder
      },
      "alternative" : [{ // Alternate non-typical step action
        "title" : "<string>", // R!  Label for alternative
        "description" : "<markdown>", // A human-readable description of each option
        "step" : [{ Content as for ExampleScenario.process.step }] // What happens in each alternative option
      }]
    }]
  }],
  "workflow" : [{ canonical(ExampleScenario) }] // Another nested workflow
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:ExampleScenario;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:ExampleScenario.url [ uri ]; # 0..1 Canonical identifier for this example scenario, represented as a URI
    (globally unique)
  fhir:ExampleScenario.identifier [ Identifier ], ... ; # 0..* Additional identifier for the example scenario
  fhir:ExampleScenario.version [ string ]; # 0..1 Business version of the example scenario
  fhir:ExampleScenario.name [ string ]; # 0..1 Name for this example scenario (computer friendly)
  fhir:ExampleScenario.status [ code ]; # 1..1 draft | active | retired | unknown
  fhir:ExampleScenario.experimental [ boolean ]; # 0..1 For testing purposes, not real usage
```

```
  fhir:ExampleScenario.date [ dateTime ]; # 0..1 Date last changed
  fhir:ExampleScenario.publisher [ string ]; # 0..1 Name of the publisher (organization or individual)
  fhir:ExampleScenario.contact [ ContactDetail ], ... ; # 0..* Contact details for the publisher
  fhir:ExampleScenario.useContext [ UsageContext ], ... ; # 0..* The context that the content is intended to support
  fhir:ExampleScenario.jurisdiction [ CodeableConcept ], ... ; # 0..* Intended jurisdiction for example scenario (if
    applicable)
  fhir:ExampleScenario.copyright [ markdown ]; # 0..1 Use and/or publishing restrictions
  fhir:ExampleScenario.purpose [ markdown ]; # 0..1 The purpose of the example, e.g. to illustrate a scenario
  fhir:ExampleScenario.actor [ # 0..* Actor participating in the resource
    fhir:ExampleScenario.actor.actorId [ string ]; # 1..1 ID or acronym of the actor
    fhir:ExampleScenario.actor.type [ code ]; # 1..1 person | entity
    fhir:ExampleScenario.actor.name [ string ]; # 0..1 The name of the actor as shown in the page
    fhir:ExampleScenario.actor.description [ markdown ]; # 0..1 The description of the actor
  ], ...;
  fhir:ExampleScenario.instance [ # 0..* Each resource and each version that is present in the workflow
    fhir:ExampleScenario.instance.resourceId [ string ]; # 1..1 The id of the resource for referencing
    fhir:ExampleScenario.instance.resourceType [ code ]; # 1..1 The type of the resource
    fhir:ExampleScenario.instance.name [ string ]; # 0..1 A short name for the resource instance
    fhir:ExampleScenario.instance.description [ markdown ]; # 0..1 Human-friendly description of the resource instance
    fhir:ExampleScenario.instance.version [ # 0..* A specific version of the resource
      fhir:ExampleScenario.instance.version.versionId [ string ]; # 1..1 The identifier of a specific version of a
    resource
      fhir:ExampleScenario.instance.version.description [ markdown ]; # 1..1 The description of the resource version
    ], ...;
    fhir:ExampleScenario.instance.containedInstance [ # 0..* Resources contained in the instance
      fhir:ExampleScenario.instance.containedInstance.resourceId [ string ]; # 1..1 Each resource contained in the
    instance
      fhir:ExampleScenario.instance.containedInstance.versionId [ string ]; # 0..1 A specific version of a resource
    contained in the instance
    ], ...;
  ], ...;
  fhir:ExampleScenario.process [ # 0..* Each major process - a group of operations
    fhir:ExampleScenario.process.title [ string ]; # 1..1 The diagram title of the group of operations
    fhir:ExampleScenario.process.description [ markdown ]; # 0..1 A longer description of the group of operations
    fhir:ExampleScenario.process.preConditions [ markdown ]; # 0..1 Description of initial status before the process
    starts
    fhir:ExampleScenario.process.postConditions [ markdown ]; # 0..1 Description of final status after the process ends
    fhir:ExampleScenario.process.step [ # 0..* Each step of the process
      fhir:ExampleScenario.process.step.process [ See ExampleScenario.process ], ... ; # 0..* Nested process
      fhir:ExampleScenario.process.step.pause [ boolean ]; # 0..1 If there is a pause in the flow
      fhir:ExampleScenario.process.step.operation [ # 0..1 Each interaction or action
        fhir:ExampleScenario.process.step.operation.number [ string ]; # 1..1 The sequential number of the interaction
        fhir:ExampleScenario.process.step.operation.type [ string ]; # 0..1 The type of operation - CRUD
        fhir:ExampleScenario.process.step.operation.name [ string ]; # 0..1 The human-friendly name of the interaction
        fhir:ExampleScenario.process.step.operation.initiator [ string ]; # 0..1 Who starts the transaction
        fhir:ExampleScenario.process.step.operation.receiver [ string ]; # 0..1 Who receives the transaction
        fhir:ExampleScenario.process.step.operation.description [ markdown ]; # 0..1 A comment to be inserted in the
    diagram
        fhir:ExampleScenario.process.step.operation.initiatorActive [ boolean ]; # 0..1 Whether the initiator is
    deactivated right after the transaction
        fhir:ExampleScenario.process.step.operation.receiverActive [ boolean ]; # 0..1 Whether the receiver is
    deactivated right after the transaction
        fhir:ExampleScenario.process.step.operation.request [ See ExampleScenario.instance.containedInstance ]; # 0..1
    Each resource instance used by the initiator
        fhir:ExampleScenario.process.step.operation.response [ See ExampleScenario.instance.containedInstance ]; # 0..1
    Each resource instance used by the responder
      ];
      fhir:ExampleScenario.process.step.alternative [ # 0..* Alternate non-typical step action
        fhir:ExampleScenario.process.step.alternative.title [ string ]; # 1..1 Label for alternative
        fhir:ExampleScenario.process.step.alternative.description [ markdown ]; # 0..1 A human-readable description of
    each option
        fhir:ExampleScenario.process.step.alternative.step [ See ExampleScenario.process.step ], ... ; # 0..* What
    happens in each alternative option
      ], ...;
    ], ...;
  ], ...;
  fhir:ExampleScenario.workflow [ canonical(ExampleScenario) ], ... ; # 0..* Another nested workflow
]
```

Changes since R3

This resource did not exist in Release 2

This analysis is available as XML or JSON.

Structure

Name | Flags | Card. | Type | Description & Constraints

ExampleScenario | ITU | | DomainResource | Example of workflow instance + Warning: Name should be usable as an identifier for the module by machine processing applications such as code generation Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

url | Σ | 0..1 | uri | Canonical identifier for this example scenario, represented as a URI (globally unique)

identifier | Σ | 0..* | Identifier | Additional identifier for the example scenario

version | Σ | 0..1 | string | Business version of the example scenario

name | ΣI | 0..1 | string | Name for this example scenario (computer friendly)

status | ?!Σ | 1..1 | code | draft | active | retired | unknown PublicationStatus (Required)

experimental | Σ | 0..1 | boolean | For testing purposes, not real usage

date | Σ | 0..1 | dateTime | Date last changed

publisher | Σ | 0..1 | string | Name of the publisher (organization or individual)
contact | Σ | 0..* | ContactDetail | Contact details for the publisher
useContext | Σ | 0..* | UsageContext | The context that the content is intended to support
jurisdiction | Σ | 0..* | CodeableConcept | Intended jurisdiction for example scenario (if applicable) Jurisdiction (Extensible)
copyright | | 0..1 | markdown | Use and/or publishing restrictions
purpose | | 0..1 | markdown | The purpose of the example, e.g. to illustrate a scenario
actor | | 0..* | BackboneElement | Actor participating in the resource
actorId | | 1..1 | string | ID or acronym of the actor
type | | 1..1 | code | person | entity ExampleScenarioActorType (Required)
name | | 0..1 | string | The name of the actor as shown in the page
description | | 0..1 | markdown | The description of the actor
instance | | 0..* | BackboneElement | Each resource and each version that is present in the workflow
resourceId | | 1..1 | string | The id of the resource for referencing
resourceType | | 1..1 | code | The type of the resource ResourceType (Required)
name | | 0..1 | string | A short name for the resource instance
description | | 0..1 | markdown | Human-friendly description of the resource instance
version | | 0..* | BackboneElement | A specific version of the resource
versionId | | 1..1 | string | The identifier of a specific version of a resource
description | | 1..1 | markdown | The description of the resource version
containedInstance | | 0..* | BackboneElement | Resources contained in the instance
resourceId | | 1..1 | string | Each resource contained in the instance
versionId | | 0..1 | string | A specific version of a resource contained in the instance
process | | 0..* | BackboneElement | Each major process - a group of operations
title | Σ | 1..1 | string | The diagram title of the group of operations
description | | 0..1 | markdown | A longer description of the group of operations
preConditions | | 0..1 | markdown | Description of initial status before the process starts
postConditions | | 0..1 | markdown | Description of final status after the process ends
step | | 0..* | BackboneElement | Each step of the process
process | | 0..* | see process | Nested process
pause | | 0..1 | boolean | If there is a pause in the flow
operation | | 0..1 | BackboneElement | Each interaction or action
number | | 1..1 | string | The sequential number of the interaction
type | | 0..1 | string | The type of operation - CRUD
name | | 0..1 | string | The human-friendly name of the interaction
initiator | | 0..1 | string | Who starts the transaction
receiver | | 0..1 | string | Who receives the transaction
description | | 0..1 | markdown | A comment to be inserted in the diagram
initiatorActive | | 0..1 | boolean | Whether the initiator is deactivated right after the transaction
receiverActive | | 0..1 | boolean | Whether the receiver is deactivated right after the transaction
request | | 0..1 | see containedInstance | Each resource instance used by the initiator
response | | 0..1 | see containedInstance | Each resource instance used by the responder
alternative | | 0..* | BackboneElement | Alternate non-typical step action
title | | 1..1 | string | Label for alternative
description | | 0..1 | markdown | A human-readable description of each option
step | | 0..* | see step | What happens in each alternative option
workflow | | 0..* | canonical(ExampleScenario) | Another nested workflow
Documentation for this format

UML Diagram (Legend)

XML Template

```
<ExampleScenario xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <url value="[uri]"/><!-- 0..1 Canonical identifier for this example scenario, represented as a URI (globally unique)
    -->
 <identifier><!-- 0..* Identifier Additional identifier for the example scenario --></identifier>
 <version value="[string]"/><!-- 0..1 Business version of the example scenario -->
 <name value="[string]"/><!--  0..1 Name for this example scenario (computer friendly) -->
 <status value="[code]"/><!-- 1..1 draft | active | retired | unknown -->
 <experimental value="[boolean]"/><!-- 0..1 For testing purposes, not real usage -->
 <date value="[dateTime]"/><!-- 0..1 Date last changed -->
 <publisher value="[string]"/><!-- 0..1 Name of the publisher (organization or individual) -->
 <contact><!-- 0..* ContactDetail Contact details for the publisher --></contact>
 <useContext><!-- 0..* UsageContext The context that the content is intended to support --></useContext>
 <jurisdiction><!-- 0..* CodeableConcept Intended jurisdiction for example scenario (if applicable) --></jurisdiction>
 <copyright value="[markdown]"/><!-- 0..1 Use and/or publishing restrictions -->
 <purpose value="[markdown]"/><!-- 0..1 The purpose of the example, e.g. to illustrate a scenario -->
 <actor>  <!-- 0..* Actor participating in the resource -->
  <actorId value="[string]"/><!-- 1..1 ID or acronym of the actor -->
  <type value="[code]"/><!-- 1..1 person | entity -->
  <name value="[string]"/><!-- 0..1 The name of the actor as shown in the page -->
  <description value="[markdown]"/><!-- 0..1 The description of the actor -->
 </actor>
 <instance>  <!-- 0..* Each resource and each version that is present in the workflow -->
  <resourceId value="[string]"/><!-- 1..1 The id of the resource for referencing -->
  <resourceType value="[code]"/><!-- 1..1 The type of the resource -->
  <name value="[string]"/><!-- 0..1 A short name for the resource instance -->
```

```
  <description value="[markdown]"/><!-- 0..1 Human-friendly description of the resource instance -->
  <version>  <!-- 0..* A specific version of the resource -->
   <versionId value="[string]"/><!-- 1..1 The identifier of a specific version of a resource -->
   <description value="[markdown]"/><!-- 1..1 The description of the resource version -->
  </version>
  <containedInstance>  <!-- 0..* Resources contained in the instance -->
   <resourceId value="[string]"/><!-- 1..1 Each resource contained in the instance -->
   <versionId value="[string]"/><!-- 0..1 A specific version of a resource contained in the instance -->
  </containedInstance>
 </instance>
 <process>  <!-- 0..* Each major process - a group of operations -->
  <title value="[string]"/><!-- 1..1 The diagram title of the group of operations -->
  <description value="[markdown]"/><!-- 0..1 A longer description of the group of operations -->
  <preConditions value="[markdown]"/><!-- 0..1 Description of initial status before the process starts -->
  <postConditions value="[markdown]"/><!-- 0..1 Description of final status after the process ends -->
  <step>  <!-- 0..* Each step of the process -->
   <process><!-- 0..* Content as for ExampleScenario.process Nested process --></process>
   <pause value="[boolean]"/><!-- 0..1 If there is a pause in the flow -->
   <operation>  <!-- 0..1 Each interaction or action -->
    <number value="[string]"/><!-- 1..1 The sequential number of the interaction -->
    <type value="[string]"/><!-- 0..1 The type of operation - CRUD -->
    <name value="[string]"/><!-- 0..1 The human-friendly name of the interaction -->
    <initiator value="[string]"/><!-- 0..1 Who starts the transaction -->
    <receiver value="[string]"/><!-- 0..1 Who receives the transaction -->
    <description value="[markdown]"/><!-- 0..1 A comment to be inserted in the diagram -->
    <initiatorActive value="[boolean]"/><!-- 0..1 Whether the initiator is deactivated right after the transaction -->
    <receiverActive value="[boolean]"/><!-- 0..1 Whether the receiver is deactivated right after the transaction -->
    <request><!-- 0..1 Content as for ExampleScenario.instance.containedInstance Each resource instance used by the
    initiator --></request>
    <response><!-- 0..1 Content as for ExampleScenario.instance.containedInstance Each resource instance used by the
    responder --></response>
   </operation>
   <alternative>  <!-- 0..* Alternate non-typical step action -->
    <title value="[string]"/><!-- 1..1 Label for alternative -->
    <description value="[markdown]"/><!-- 0..1 A human-readable description of each option -->
    <step><!-- 0..* Content as for ExampleScenario.process.step What happens in each alternative option --></step>
   </alternative>
  </step>
 </process>
 <workflow><!-- 0..* canonical(ExampleScenario) Another nested workflow --></workflow>
</ExampleScenario>
```

JSON Template

```
{
  "resourceType" : "ExampleScenario",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "url" : "<uri>", // Canonical identifier for this example scenario, represented as a URI (globally unique)
  "identifier" : [{ Identifier }], // Additional identifier for the example scenario
  "version" : "<string>", // Business version of the example scenario
  "name" : "<string>", // C? Name for this example scenario (computer friendly)
  "status" : "<code>", // R!  draft | active | retired | unknown
  "experimental" : <boolean>, // For testing purposes, not real usage
  "date" : "<dateTime>", // Date last changed
  "publisher" : "<string>", // Name of the publisher (organization or individual)
  "contact" : [{ ContactDetail }], // Contact details for the publisher
  "useContext" : [{ UsageContext }], // The context that the content is intended to support
  "jurisdiction" : [{ CodeableConcept }], // Intended jurisdiction for example scenario (if applicable)
  "copyright" : "<markdown>", // Use and/or publishing restrictions
  "purpose" : "<markdown>", // The purpose of the example, e.g. to illustrate a scenario
  "actor" : [{ // Actor participating in the resource
    "actorId" : "<string>", // R!  ID or acronym of the actor
    "type" : "<code>", // R!  person | entity
    "name" : "<string>", // The name of the actor as shown in the page
    "description" : "<markdown>" // The description of the actor
  }],
  "instance" : [{ // Each resource and each version that is present in the workflow
    "resourceId" : "<string>", // R!  The id of the resource for referencing
    "resourceType" : "<code>", // R!  The type of the resource
    "name" : "<string>", // A short name for the resource instance
    "description" : "<markdown>", // Human-friendly description of the resource instance
    "version" : [{ // A specific version of the resource
      "versionId" : "<string>", // R!  The identifier of a specific version of a resource
      "description" : "<markdown>" // R!  The description of the resource version
    }],
    "containedInstance" : [{ // Resources contained in the instance
      "resourceId" : "<string>", // R!  Each resource contained in the instance
      "versionId" : "<string>" // A specific version of a resource contained in the instance
    }]
  }],
  "process" : [{ // Each major process - a group of operations
    "title" : "<string>", // R!  The diagram title of the group of operations
    "description" : "<markdown>", // A longer description of the group of operations
    "preConditions" : "<markdown>", // Description of initial status before the process starts
    "postConditions" : "<markdown>", // Description of final status after the process ends
    "step" : [{ // Each step of the process
      "process" : [{ Content as for ExampleScenario.process }], // Nested process
      "pause" : <boolean>, // If there is a pause in the flow
      "operation" : { // Each interaction or action
```

```
      "number" : "<string>", // R!  The sequential number of the interaction
      "type" : "<string>", // The type of operation - CRUD
      "name" : "<string>", // The human-friendly name of the interaction
      "initiator" : "<string>", // Who starts the transaction
      "receiver" : "<string>", // Who receives the transaction
      "description" : "<markdown>", // A comment to be inserted in the diagram
      "initiatorActive" : <boolean>, // Whether the initiator is deactivated right after the transaction
      "receiverActive" : <boolean>, // Whether the receiver is deactivated right after the transaction
      "request" : { Content as for ExampleScenario.instance.containedInstance }, // Each resource instance used by the
    initiator
      "response" : { Content as for ExampleScenario.instance.containedInstance } // Each resource instance used by the
    responder
     },
     "alternative" : [{ // Alternate non-typical step action
      "title" : "<string>", // R!  Label for alternative
      "description" : "<markdown>", // A human-readable description of each option
      "step" : [{ Content as for ExampleScenario.process.step }] // What happens in each alternative option
     }]
    }]
  }],
  "workflow" : [{ canonical(ExampleScenario) }] // Another nested workflow
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:ExampleScenario;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:ExampleScenario.url [ uri ]; # 0..1 Canonical identifier for this example scenario, represented as a URI
    (globally unique)
  fhir:ExampleScenario.identifier [ Identifier ], ... ; # 0..* Additional identifier for the example scenario
  fhir:ExampleScenario.version [ string ]; # 0..1 Business version of the example scenario
  fhir:ExampleScenario.name [ string ]; # 0..1 Name for this example scenario (computer friendly)
  fhir:ExampleScenario.status [ code ]; # 1..1 draft | active | retired | unknown
  fhir:ExampleScenario.experimental [ boolean ]; # 0..1 For testing purposes, not real usage
  fhir:ExampleScenario.date [ dateTime ]; # 0..1 Date last changed
  fhir:ExampleScenario.publisher [ string ]; # 0..1 Name of the publisher (organization or individual)
  fhir:ExampleScenario.contact [ ContactDetail ], ... ; # 0..* Contact details for the publisher
  fhir:ExampleScenario.useContext [ UsageContext ], ... ; # 0..* The context that the content is intended to support
  fhir:ExampleScenario.jurisdiction [ CodeableConcept ], ... ; # 0..* Intended jurisdiction for example scenario (if
    applicable)
  fhir:ExampleScenario.copyright [ markdown ]; # 0..1 Use and/or publishing restrictions
  fhir:ExampleScenario.purpose [ markdown ]; # 0..1 The purpose of the example, e.g. to illustrate a scenario
  fhir:ExampleScenario.actor [ # 0..* Actor participating in the resource
    fhir:ExampleScenario.actor.actorId [ string ]; # 1..1 ID or acronym of the actor
    fhir:ExampleScenario.actor.type [ code ]; # 1..1 person | entity
    fhir:ExampleScenario.actor.name [ string ]; # 0..1 The name of the actor as shown in the page
    fhir:ExampleScenario.actor.description [ markdown ]; # 0..1 The description of the actor
  ], ...;
  fhir:ExampleScenario.instance [ # 0..* Each resource and each version that is present in the workflow
    fhir:ExampleScenario.instance.resourceId [ string ]; # 1..1 The id of the resource for referencing
    fhir:ExampleScenario.instance.resourceType [ code ]; # 1..1 The type of the resource
    fhir:ExampleScenario.instance.name [ string ]; # 0..1 A short name for the resource instance
    fhir:ExampleScenario.instance.description [ markdown ]; # 0..1 Human-friendly description of the resource instance
    fhir:ExampleScenario.instance.version [ # 0..* A specific version of the resource
      fhir:ExampleScenario.instance.version.versionId [ string ]; # 1..1 The identifier of a specific version of a
    resource
      fhir:ExampleScenario.instance.version.description [ markdown ]; # 1..1 The description of the resource version
    ], ...;
    fhir:ExampleScenario.instance.containedInstance [ # 0..* Resources contained in the instance
      fhir:ExampleScenario.instance.containedInstance.resourceId [ string ]; # 1..1 Each resource contained in the
    instance
      fhir:ExampleScenario.instance.containedInstance.versionId [ string ]; # 0..1 A specific version of a resource
    contained in the instance
    ], ...;
  ], ...;
  fhir:ExampleScenario.process [ # 0..* Each major process - a group of operations
    fhir:ExampleScenario.process.title [ string ]; # 1..1 The diagram title of the group of operations
    fhir:ExampleScenario.process.description [ markdown ]; # 0..1 A longer description of the group of operations
    fhir:ExampleScenario.process.preConditions [ markdown ]; # 0..1 Description of initial status before the process
    starts
    fhir:ExampleScenario.process.postConditions [ markdown ]; # 0..1 Description of final status after the process ends
    fhir:ExampleScenario.process.step [ # 0..* Each step of the process
      fhir:ExampleScenario.process.step.process [ See ExampleScenario.process ], ... ; # 0..* Nested process
      fhir:ExampleScenario.process.step.pause [ boolean ]; # 0..1 If there is a pause in the flow
      fhir:ExampleScenario.process.step.operation [ # 0..1 Each interaction or action
        fhir:ExampleScenario.process.step.operation.number [ string ]; # 1..1 The sequential number of the interaction
        fhir:ExampleScenario.process.step.operation.type [ string ]; # 0..1 The type of operation - CRUD
        fhir:ExampleScenario.process.step.operation.name [ string ]; # 0..1 The human-friendly name of the interaction
        fhir:ExampleScenario.process.step.operation.initiator [ string ]; # 0..1 Who starts the transaction
        fhir:ExampleScenario.process.step.operation.receiver [ string ]; # 0..1 Who receives the transaction
        fhir:ExampleScenario.process.step.operation.description [ markdown ]; # 0..1 A comment to be inserted in the
    diagram
        fhir:ExampleScenario.process.step.operation.initiatorActive [ boolean ]; # 0..1 Whether the initiator is
    deactivated right after the transaction
```

```
        fhir:ExampleScenario.process.step.operation.receiverActive [ boolean ]; # 0..1 Whether the receiver is
    deactivated right after the transaction
        fhir:ExampleScenario.process.step.operation.request [ See ExampleScenario.instance.containedInstance ]; # 0..1
    Each resource instance used by the initiator
        fhir:ExampleScenario.process.step.operation.response [ See ExampleScenario.instance.containedInstance ]; # 0..1
    Each resource instance used by the responder
      ];
      fhir:ExampleScenario.process.step.alternative [ # 0..* Alternate non-typical step action
        fhir:ExampleScenario.process.step.alternative.title [ string ]; # 1..1 Label for alternative
        fhir:ExampleScenario.process.step.alternative.description [ markdown ]; # 0..1 A human-readable description of
    each option
        fhir:ExampleScenario.process.step.alternative.step [ See ExampleScenario.process.step ], ... ; # 0..* What
    happens in each alternative option
      ], ...;
    ], ...;
  ], ...;
  fhir:ExampleScenario.workflow [ canonical(ExampleScenario) ], ... ; # 0..* Another nested workflow
]
```

Changes since Release 3

This resource did not exist in Release 2

This analysis is available as XML or JSON.

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

12.23.2.1 Terminology Bindings

Path | Definition | Type | Reference
ExampleScenario.status | The lifecycle status of an artifact. | Required | PublicationStatus
ExampleScenario.jurisdiction | Countries and regions within which this artifact is targeted for use. | Extensible | Jurisdiction ValueSet
ExampleScenario.actor.type | The type of actor - system or human. | Required | ExampleScenarioActorType
ExampleScenario.instance.resourceType | The type of resource. | Required | ResourceType

12.23.2.2 Constraints

id | Level | Location | Description | Expression
esc-0 | Warning | (base) | Name should be usable as an identifier for the module by machine processing applications such as code generation | name.matches('[A-Z]([A-Za-z0-9_]){0,254}')

12.23.3 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
context | token | A use context assigned to the example scenario | (ExampleScenario.useContext.value as CodeableConcept)
context-quantity | quantity | A quantity- or range-valued use context assigned to the example scenario | (ExampleScenario.useContext.value as Quantity) | (ExampleScenario.useContext.value as Range)
context-type | token | A type of use context assigned to the example scenario | ExampleScenario.useContext.code
context-type-quantity | composite | A use context type and quantity- or range-based value assigned to the example scenario | On ExampleScenario.useContext: context-type: code context-quantity: value.as(Quantity) | value.as(Range)
context-type-value | composite | A use context type and value assigned to the example scenario | On ExampleScenario.useContext: context-type: code context: value.as(CodeableConcept)
date | date | The example scenario publication date | ExampleScenario.date
identifier | token | External identifier for the example scenario | ExampleScenario.identifier
jurisdiction | token | Intended jurisdiction for the example scenario | ExampleScenario.jurisdiction
name | string | Computationally friendly name of the example scenario | ExampleScenario.name
publisher | string | Name of the publisher of the example scenario | ExampleScenario.publisher
status | token | The current status of the example scenario | ExampleScenario.status
url | uri | The uri that identifies the example scenario | ExampleScenario.url
version | token | The business version of the example scenario | ExampleScenario.version

12.23.4 Resource ExampleScenario - Examples

URL: https://hl7.org/fhir/R4/examplescenario-examples.html

12.23.4 Resource ExampleScenario - Examples

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Example Name | id | Format
Example of examplescenario | example | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

12.23.5 Resource ExampleScenario - Detailed Descriptions

URL: https://hl7.org/fhir/R4/examplescenario-definitions.html

12.23.5 Resource ExampleScenario - Detailed Descriptions

FHIR Infrastructure Work Group | Maturity Level: 0 | Trial Use | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the ExampleScenario resource.

ExampleScenario
Element Id | ExampleScenario
Definition | Example of workflow instance.
Cardinality | 0..*
Type | DomainResource
Invariants | Defined on this element esc-0 Warning Name should be usable as an identifier for the module by machine processing applications such as code generation name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
Defined on this element
esc-0 | Warning | Name should be usable as an identifier for the module by machine processing applications such as code generation | name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
ExampleScenario.url
Element Id | ExampleScenario.url
Definition | An absolute URI that is used to identify this example scenario when it is referenced in a specification, model, design or an instance; also called its canonical identifier. This SHOULD be globally unique and SHOULD be a literal address at which at which an authoritative instance of this example scenario is (or will be) published. This URL can be the target of a canonical reference. It SHALL remain the same when the example scenario is stored on different servers.
Cardinality | 0..1
Type | uri
Requirements | Allows the example scenario to be referenced by a single globally unique identifier.
Summary | true
Comments | Can be a urn:uuid: or a urn:oid: but real http: addresses are preferred. Multiple instances may share the same URL if they have a distinct version. The determination of when to create a new version of a resource (same url, new version) vs. defining a new artifact is up to the author. Considerations for making this decision are found in Technical and Business Versions. In some cases, the resource can no longer be found at the stated url, but the url itself cannot change. Implementations can use the meta.source element to indicate where the current master source of the resource can be found.
ExampleScenario.identifier
Element Id | ExampleScenario.identifier
Definition | A formal identifier that is used to identify this example scenario when it is represented in other formats, or referenced in a specification, model, design or an instance.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Requirements | Allows externally provided and/or usable business identifiers to be easily associated with the module.
Summary | true
Comments | Typically, this is used for identifiers that can go in an HL7 V3 II (instance identifier) data type, and can then identify this example scenario outside of FHIR, where it is not possible to use the logical URI.
ExampleScenario.version
Element Id | ExampleScenario.version
Definition | The identifier that is used to identify this version of the example scenario when it is referenced in a specification, model, design or instance. This is an arbitrary value managed by the example scenario author and is not expected to be globally unique. For example, it might be a timestamp (e.g. yyyymmdd) if a managed version is not available. There is also no expectation that versions can be placed in a lexicographical sequence.
Note | This is a business versionId, not a resource version id (see discussion)
Cardinality | 0..1
Type | string
Summary | true
Comments | There may be different example scenario instances that have the same identifier but different versions. The version can be appended to the url in a reference to allow a reference to a particular business version of the example scenario with the format [url]|[version].
ExampleScenario.name
Element Id | ExampleScenario.name
Definition | A natural language name identifying the example scenario. This name should be usable as an identifier for the module by machine processing applications such as code generation.
Cardinality | 0..1
Type | string
Requirements | Support human navigation and code generation.
Summary | true
Comments | The name is not expected to be globally unique. The name should be a simple alphanumeric type name to ensure that it is machine-processing friendly.
Invariants | Affect this element inv-0 Warning Name should be usable as an identifier for the module by machine processing applications such as code generation name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
Affect this element
inv-0 | Warning | Name should be usable as an identifier for the module by machine processing applications such as code generation | name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
To Do | & A name should be provided unless the value set is a contained resource (e.g. an anonymous value set in a profile). Most registries will require a name.
ExampleScenario.status

Element Id | ExampleScenario.status
Definition | The status of this example scenario. Enables tracking the life-cycle of the content.
Cardinality | 1..1
Terminology Binding | PublicationStatus (Required)
Type | code
Is Modifier | true (Reason: This is labeled as "Is Modifier" because applications should not use a retired {{title}} without due consideration)
Summary | true
Comments | Allows filtering of example scenarios that are appropriate for use versus not.
ExampleScenario.experimental
Element Id | ExampleScenario.experimental
Definition | A Boolean value to indicate that this example scenario is authored for testing purposes (or education/evaluation/marketing) and is not intended to be used for genuine usage.
Cardinality | 0..1
Type | boolean
Requirements | Enables experimental content to be developed following the same lifecycle that would be used for a production-level example scenario.
Summary | true
Comments | Allows filtering of example scenarios that are appropriate for use versus not.
ExampleScenario.date
Element Id | ExampleScenario.date
Definition | The date (and optionally time) when the example scenario was published. The date must change when the business version changes and it must change if the status code changes. In addition, it should change when the substantive content of the example scenario changes. (e.g. the 'content logical definition').
Cardinality | 0..1
Type | dateTime
Alternate Names | Revision Date
Summary | true
Comments | Note that this is not the same as the resource last-modified-date, since the resource may be a secondary representation of the example scenario. Additional specific dates may be added as extensions or be found by consulting Provenances associated with past versions of the resource.
ExampleScenario.publisher
Element Id | ExampleScenario.publisher
Definition | The name of the organization or individual that published the example scenario.
Cardinality | 0..1
Type | string
Requirements | Helps establish the "authority/credibility" of the example scenario. May also allow for contact.
Summary | true
Comments | Usually an organization but may be an individual. The publisher (or steward) of the example scenario is the organization or individual primarily responsible for the maintenance and upkeep of the example scenario. This is not necessarily the same individual or organization that developed and initially authored the content. The publisher is the primary point of contact for questions or issues with the example scenario. This item SHOULD be populated unless the information is available from context.
ExampleScenario.contact
Element Id | ExampleScenario.contact
Definition | Contact details to assist a user in finding and communicating with the publisher.
Cardinality | 0..*
Type | ContactDetail
Summary | true
Comments | May be a web site, an email address, a telephone number, etc.
ExampleScenario.useContext
Element Id | ExampleScenario.useContext
Definition | The content was developed with a focus and intent of supporting the contexts that are listed. These contexts may be general categories (gender, age, ...) or may be references to specific programs (insurance plans, studies, ...) and may be used to assist with indexing and searching for appropriate example scenario instances.
Cardinality | 0..*
Type | UsageContext
Requirements | Assist in searching for appropriate content.
Summary | true
Comments | When multiple useContexts are specified, there is no expectation that all or any of the contexts apply.
ExampleScenario.jurisdiction
Element Id | ExampleScenario.jurisdiction
Definition | A legal or geographic region in which the example scenario is intended to be used.
Cardinality | 0..*
Terminology Binding | Jurisdiction (Extensible)
Type | CodeableConcept
Summary | true
Comments | It may be possible for the example scenario to be used in jurisdictions other than those for which it was originally designed or intended.
ExampleScenario.copyright
Element Id | ExampleScenario.copyright
Definition | A copyright statement relating to the example scenario and/or its contents. Copyright statements are generally legal restrictions on the use and publishing of the example scenario.
Cardinality | 0..1

Type | markdown
Requirements | Consumers must be able to determine any legal restrictions on the use of the example scenario and/or its content.
Alternate Names | License; Restrictions
Comments | nullFrequently, the copyright differs between the value set and the codes that are included. The copyright statement should clearly differentiate between these when required.
ExampleScenario.purpose
Element Id | ExampleScenario.purpose
Definition | What the example scenario resource is created for. This should not be used to show the business purpose of the scenario itself, but the purpose of documenting a scenario.
Cardinality | 0..1
Type | markdown
Comments | This element does not describe the usage of the example scenario. Instead, it provides traceability of ''why'' the resource is either needed or ''why'' it is defined as it is. This may be used to point to source materials or specifications that drove the structure of this example scenario.
ExampleScenario.actor
Element Id | ExampleScenario.actor
Definition | Actor participating in the resource.
Cardinality | 0..*
ExampleScenario.actor.actorId
Element Id | ExampleScenario.actor.actorId
Definition | ID or acronym of actor.
Cardinality | 1..1
Type | string
Comments | should this be called ID or acronym?
ExampleScenario.actor.type
Element Id | ExampleScenario.actor.type
Definition | The type of actor - person or system.
Cardinality | 1..1
Terminology Binding | ExampleScenarioActorType (Required)
Type | code
ExampleScenario.actor.name
Element Id | ExampleScenario.actor.name
Definition | The name of the actor as shown in the page.
Cardinality | 0..1
Type | string
Comments | Cardinality: is name and description 1..1?
ExampleScenario.actor.description
Element Id | ExampleScenario.actor.description
Definition | The description of the actor.
Cardinality | 0..1
Type | markdown
Comments | Cardinality: is name and description 1..1?
ExampleScenario.instance
Element Id | ExampleScenario.instance
Definition | Each resource and each version that is present in the workflow.
Cardinality | 0..*
ExampleScenario.instance.resourceId
Element Id | ExampleScenario.instance.resourceId
Definition | The id of the resource for referencing.
Cardinality | 1..1
Type | string
ExampleScenario.instance.resourceType
Element Id | ExampleScenario.instance.resourceType
Definition | The type of the resource.
Cardinality | 1..1
Terminology Binding | ResourceType (Required)
Type | code
ExampleScenario.instance.name
Element Id | ExampleScenario.instance.name
Definition | A short name for the resource instance.
Cardinality | 0..1
Type | string
To Do | Should name and description be 1..1?
ExampleScenario.instance.description
Element Id | ExampleScenario.instance.description
Definition | Human-friendly description of the resource instance.
Cardinality | 0..1
Type | markdown
To Do | Should name and description be 1..1?
ExampleScenario.instance.version
Element Id | ExampleScenario.instance.version
Definition | A specific version of the resource.
Note | This is a business versionId, not a resource version id (see discussion)

Cardinality | 0..*
ExampleScenario.instance.version.versionId
Element Id | ExampleScenario.instance.version.versionId
Definition | The identifier of a specific version of a resource.
Cardinality | 1..1
Type | string
ExampleScenario.instance.version.description
Element Id | ExampleScenario.instance.version.description
Definition | The description of the resource version.
Cardinality | 1..1
Type | markdown
ExampleScenario.instance.containedInstance
Element Id | ExampleScenario.instance.containedInstance
Definition | Resources contained in the instance (e.g. the observations contained in a bundle).
Cardinality | 0..*
ExampleScenario.instance.containedInstance.resourceId
Element Id | ExampleScenario.instance.containedInstance.resourceId
Definition | Each resource contained in the instance.
Cardinality | 1..1
Type | string
ExampleScenario.instance.containedInstance.versionId
Element Id | ExampleScenario.instance.containedInstance.versionId
Definition | A specific version of a resource contained in the instance.
Cardinality | 0..1
Type | string
ExampleScenario.process
Element Id | ExampleScenario.process
Definition | Each major process - a group of operations.
Cardinality | 0..*
ExampleScenario.process.title
Element Id | ExampleScenario.process.title
Definition | The diagram title of the group of operations.
Cardinality | 1..1
Type | string
Summary | true
ExampleScenario.process.description
Element Id | ExampleScenario.process.description
Definition | A longer description of the group of operations.
Cardinality | 0..1
Type | markdown
ExampleScenario.process.preConditions
Element Id | ExampleScenario.process.preConditions
Definition | Description of initial status before the process starts.
Cardinality | 0..1
Type | markdown
ExampleScenario.process.postConditions
Element Id | ExampleScenario.process.postConditions
Definition | Description of final status after the process ends.
Cardinality | 0..1
Type | markdown
ExampleScenario.process.step
Element Id | ExampleScenario.process.step
Definition | Each step of the process.
Cardinality | 0..*
ExampleScenario.process.step.process
Element Id | ExampleScenario.process.step.process
Definition | Nested process.
Cardinality | 0..*
Type | See ExampleScenario.process
ExampleScenario.process.step.pause
Element Id | ExampleScenario.process.step.pause
Definition | If there is a pause in the flow.
Cardinality | 0..1
Type | boolean
ExampleScenario.process.step.operation
Element Id | ExampleScenario.process.step.operation
Definition | Each interaction or action.
Cardinality | 0..1
ExampleScenario.process.step.operation.number
Element Id | ExampleScenario.process.step.operation.number
Definition | The sequential number of the interaction, e.g. 1.2.5.
Cardinality | 1..1

Type | string
ExampleScenario.process.step.operation.type
Element Id | ExampleScenario.process.step.operation.type
Definition | The type of operation - CRUD.
Cardinality | 0..1
Type | string
ExampleScenario.process.step.operation.name
Element Id | ExampleScenario.process.step.operation.name
Definition | The human-friendly name of the interaction.
Cardinality | 0..1
Type | string
ExampleScenario.process.step.operation.initiator
Element Id | ExampleScenario.process.step.operation.initiator
Definition | Who starts the transaction.
Cardinality | 0..1
Type | string
ExampleScenario.process.step.operation.receiver
Element Id | ExampleScenario.process.step.operation.receiver
Definition | Who receives the transaction.
Cardinality | 0..1
Type | string
ExampleScenario.process.step.operation.description
Element Id | ExampleScenario.process.step.operation.description
Definition | A comment to be inserted in the diagram.
Cardinality | 0..1
Type | markdown
ExampleScenario.process.step.operation.initiatorActive
Element Id | ExampleScenario.process.step.operation.initiatorActive
Definition | Whether the initiator is deactivated right after the transaction.
Cardinality | 0..1
Type | boolean
ExampleScenario.process.step.operation.receiverActive
Element Id | ExampleScenario.process.step.operation.receiverActive
Definition | Whether the receiver is deactivated right after the transaction.
Cardinality | 0..1
Type | boolean
ExampleScenario.process.step.operation.request
Element Id | ExampleScenario.process.step.operation.request
Definition | Each resource instance used by the initiator.
Cardinality | 0..1
Type | See ExampleScenario.instance.containedInstance
ExampleScenario.process.step.operation.response
Element Id | ExampleScenario.process.step.operation.response
Definition | Each resource instance used by the responder.
Cardinality | 0..1
Type | See ExampleScenario.instance.containedInstance
ExampleScenario.process.step.alternative
Element Id | ExampleScenario.process.step.alternative
Definition | Indicates an alternative step that can be taken instead of the operations on the base step in exceptional/atypical circumstances.
Cardinality | 0..*
ExampleScenario.process.step.alternative.title
Element Id | ExampleScenario.process.step.alternative.title
Definition | The label to display for the alternative that gives a sense of the circumstance in which the alternative should be invoked.
Cardinality | 1..1
Type | string
ExampleScenario.process.step.alternative.description
Element Id | ExampleScenario.process.step.alternative.description
Definition | A human-readable description of the alternative explaining when the alternative should occur rather than the base step.
Cardinality | 0..1
Type | markdown
ExampleScenario.process.step.alternative.step
Element Id | ExampleScenario.process.step.alternative.step
Definition | What happens in each alternative option.
Cardinality | 0..*
Type | See ExampleScenario.process.step
ExampleScenario.workflow
Element Id | ExampleScenario.workflow
Definition | Another nested workflow.
Cardinality | 0..*

Type | canonical(ExampleScenario)

12.23.6 Resource ExampleScenario - Mappings

URL: https://hl7.org/fhir/R4/examplescenario-mappings.html

12.23.6 Resource ExampleScenario - Mappings

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Mappings:

FiveWs Pattern Mapping

Workflow Pattern

Mappings for the examplescenario resource (see Mappings to Other Standards for further information & status).

12.23.6.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

ExampleScenario
url | FiveWs.identifier
identifier | FiveWs.identifier
version | FiveWs.version
status | FiveWs.status
experimental | FiveWs.class
date | FiveWs.recorded
publisher | FiveWs.witness
purpose | FiveWs.why[x]

12.23.6.2 Workflow Pattern (http://hl7.org/fhir/workflow)

ExampleScenario
url | Definition.url
identifier | Definition.identifier
version | Definition.version
status | Definition.status
experimental | Definition.experimental
date | Definition.date
publisher | Definition.publisher
contact | Definition.contact
useContext | Definition.useContext
jurisdiction | Definition.jurisdiction
copyright | Definition.copyright
purpose | Definition.purpose

12.23.7 Resource ExampleScenario - Extensions & Profiles

URL: https://hl7.org/fhir/R4/examplescenario-profiles.html

12.23.7 Resource ExampleScenario - Extensions & Profiles

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

This table lists profiles and extensions for the ExampleScenario resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

12.23.7.1 Profiles

No Profiles defined for this resource

12.23.7.2 Extensions

replaces | ExampleScenario | replaces | for General Extensions for use by FHIR Implementers
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers

replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

12.23.7.3 Search Extensions

No Search Extensions defined for this resource

12.23.8 JSON Schema for ExampleScenario

URL: https://hl7.org/fhir/R4/examplescenario.schema.json.html

12.23.8 JSON Schema for ExampleScenario

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Not linked to any defined compartments

This is a representation of the json schema for ExampleScenario, which is just a part of the full JSON Schema.


```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/ExampleScenario",
  "$ref": "#/definitions/ExampleScenario",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "ExampleScenario": {
      "description": "Example of workflow instance.",
      "properties": {
        "resourceType": {
          "description": "This is a ExampleScenario resource",
          "const": "ExampleScenario"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
          "$ref": "#/definitions/code"
        },
        "_language": {
          "description": "Extensions for language",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "text": {
          "description": "A human-readable narrative that contains a summary of the resource and can be used to
    represent the content of the resource to a human. The narrative need not encode all the structured data, but is
    required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
    Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
          "$ref": "Narrative.schema.json#/definitions/Narrative"
        },
        "contained": {
          "description": "These resources do not have an independent existence apart from the resource that contains
    them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
          "items": {
            "$ref": "ResourceList.schema.json#/definitions/ResourceList"
          },
          "type": "array"
        },
        "extension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
     definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
    that SHALL be met as part of the definition of the extension.",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "modifierExtension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
```

```
the resource and that modifies the understanding of the element that contains it and/or the understanding of the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension. Applications processing a resource are required to check for modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "url": {
      "description": "An absolute URI that is used to identify this example scenario when it is referenced in a specification, model, design or an instance; also called its canonical identifier. This SHOULD be globally unique and SHOULD be a literal address at which at which an authoritative instance of this example scenario is (or will be) published. This URL can be the target of a canonical reference. It SHALL remain the same when the example scenario is stored on different servers.",
      "$ref": "#/definitions/uri"
    },
    "_url": {
      "description": "Extensions for url",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "identifier": {
      "description": "A formal identifier that is used to identify this example scenario when it is represented in other formats, or referenced in a specification, model, design or an instance.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "version": {
      "description": "The identifier that is used to identify this version of the example scenario when it is referenced in a specification, model, design or instance. This is an arbitrary value managed by the example scenario author and is not expected to be globally unique. For example, it might be a timestamp (e.g. yyyymmdd) if a managed version is not available. There is also no expectation that versions can be placed in a lexicographical sequence.",
      "$ref": "#/definitions/string"
    },
    "_version": {
      "description": "Extensions for version",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "name": {
      "description": "A natural language name identifying the example scenario. This name should be usable as an identifier for the module by machine processing applications such as code generation.",
      "$ref": "#/definitions/string"
    },
    "_name": {
      "description": "Extensions for name",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "status": {
      "description": "The status of this example scenario. Enables tracking the life-cycle of the content.",
      "enum": [
        "draft",
        "active",
        "retired",
        "unknown"
      ]
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "experimental": {
      "description": "A Boolean value to indicate that this example scenario is authored for testing purposes (or education/evaluation/marketing) and is not intended to be used for genuine usage.",
      "$ref": "#/definitions/boolean"
    },
    "_experimental": {
      "description": "Extensions for experimental",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "date": {
      "description": "The date  (and optionally time) when the example scenario was published. The date must change when the business version changes and it must change if the status code changes. In addition, it should change when the substantive content of the example scenario changes. (e.g. the \u0027content logical definition\u0027).",
      "$ref": "#/definitions/dateTime"
    },
    "_date": {
      "description": "Extensions for date",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "publisher": {
      "description": "The name of the organization or individual that published the example scenario.",
      "$ref": "#/definitions/string"
    },
    "_publisher": {
```

```
      "description": "Extensions for publisher",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "contact": {
      "description": "Contact details to assist a user in finding and communicating with the publisher.",
      "items": {
        "$ref": "ContactDetail.schema.json#/definitions/ContactDetail"
      },
      "type": "array"
    },
    "useContext": {
      "description": "The content was developed with a focus and intent of supporting the contexts that are listed. These contexts may be general categories (gender, age, ...) or may be references to specific programs (insurance plans, studies, ...) and may be used to assist with indexing and searching for appropriate example scenario instances.",
      "items": {
        "$ref": "UsageContext.schema.json#/definitions/UsageContext"
      },
      "type": "array"
    },
    "jurisdiction": {
      "description": "A legal or geographic region in which the example scenario is intended to be used.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "copyright": {
      "description": "A copyright statement relating to the example scenario and/or its contents. Copyright statements are generally legal restrictions on the use and publishing of the example scenario.",
      "$ref": "#/definitions/markdown"
    },
    "_copyright": {
      "description": "Extensions for copyright",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "purpose": {
      "description": "What the example scenario resource is created for. This should not be used to show the business purpose of the scenario itself, but the purpose of documenting a scenario.",
      "$ref": "#/definitions/markdown"
    },
    "_purpose": {
      "description": "Extensions for purpose",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "actor": {
      "description": "Actor participating in the resource.",
      "items": {
        "$ref": "#/definitions/ExampleScenario_Actor"
      },
      "type": "array"
    },
    "instance": {
      "description": "Each resource and each version that is present in the workflow.",
      "items": {
        "$ref": "#/definitions/ExampleScenario_Instance"
      },
      "type": "array"
    },
    "process": {
      "description": "Each major process - a group of operations.",
      "items": {
        "$ref": "#/definitions/ExampleScenario_Process"
      },
      "type": "array"
    },
    "workflow": {
      "description": "Another nested workflow.",
      "items": {
        "$ref": "canonical.schema.json#/definitions/canonical"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "resourceType"
  ]
},
"ExampleScenario_Actor": {
  "description": "Example of workflow instance.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
```

```
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the element and that modifies the understanding of the element in which it is contained and/or the understanding of the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension. Applications processing a resource are required to check for modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "actorId": {
      "description": "ID or acronym of actor.",
      "$ref": "#/definitions/string"
    },
    "_actorId": {
      "description": "Extensions for actorId",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "type": {
      "description": "The type of actor - person or system.",
      "enum": [
        "person",
        "entity"
      ]
    },
    "_type": {
      "description": "Extensions for type",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "name": {
      "description": "The name of the actor as shown in the page.",
      "$ref": "#/definitions/string"
    },
    "_name": {
      "description": "Extensions for name",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "description": {
      "description": "The description of the actor.",
      "$ref": "#/definitions/markdown"
    },
    "_description": {
      "description": "Extensions for description",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
  "additionalProperties": false
},
"ExampleScenario_Instance": {
  "description": "Example of workflow instance.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the element and that modifies the understanding of the element in which it is contained and/or the understanding of the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension. Applications processing a resource are required to check for modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
```

```
      },
      "type": "array"
    },
    "resourceId": {
      "description": "The id of the resource for referencing.",
      "$ref": "#/definitions/string"
    },
    "_resourceId": {
      "description": "Extensions for resourceId",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "resourceType": {
      "description": "The type of the resource.",
      "$ref": "#/definitions/code"
    },
    "_resourceType": {
      "description": "Extensions for resourceType",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "name": {
      "description": "A short name for the resource instance.",
      "$ref": "#/definitions/string"
    },
    "_name": {
      "description": "Extensions for name",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "description": {
      "description": "Human-friendly description of the resource instance.",
      "$ref": "#/definitions/markdown"
    },
    "_description": {
      "description": "Extensions for description",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "version": {
      "description": "A specific version of the resource.",
      "items": {
        "$ref": "#/definitions/ExampleScenario_Version"
      },
      "type": "array"
    },
    "containedInstance": {
      "description": "Resources contained in the instance (e.g. the observations contained in a bundle).",
      "items": {
        "$ref": "#/definitions/ExampleScenario_ContainedInstance"
      },
      "type": "array"
    }
  },
  "additionalProperties": false
},
"ExampleScenario_Version": {
  "description": "Example of workflow instance.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "versionId": {
      "description": "The identifier of a specific version of a resource.",
      "$ref": "#/definitions/string"
    },
    "_versionId": {
```

```
      "description": "Extensions for versionId",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "description": {
      "description": "The description of the resource version.",
      "$ref": "#/definitions/markdown"
    },
    "_description": {
      "description": "Extensions for description",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
  "additionalProperties": false
},
"ExampleScenario_ContainedInstance": {
  "description": "Example of workflow instance.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "resourceId": {
      "description": "Each resource contained in the instance.",
      "$ref": "#/definitions/string"
    },
    "_resourceId": {
      "description": "Extensions for resourceId",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "versionId": {
      "description": "A specific version of a resource contained in the instance.",
      "$ref": "#/definitions/string"
    },
    "_versionId": {
      "description": "Extensions for versionId",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
  "additionalProperties": false
},
"ExampleScenario_Process": {
  "description": "Example of workflow instance.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
```

```
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "title": {
      "description": "The diagram title of the group of operations.",
      "$ref": "#/definitions/string"
    },
    "_title": {
      "description": "Extensions for title",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "description": {
      "description": "A longer description of the group of operations.",
      "$ref": "#/definitions/markdown"
    },
    "_description": {
      "description": "Extensions for description",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "preConditions": {
      "description": "Description of initial status before the process starts.",
      "$ref": "#/definitions/markdown"
    },
    "_preConditions": {
      "description": "Extensions for preConditions",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "postConditions": {
      "description": "Description of final status after the process ends.",
      "$ref": "#/definitions/markdown"
    },
    "_postConditions": {
      "description": "Extensions for postConditions",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "step": {
      "description": "Each step of the process.",
      "items": {
        "$ref": "#/definitions/ExampleScenario_Step"
      },
      "type": "array"
    }
  },
  "additionalProperties": false
},
"ExampleScenario_Step": {
  "description": "Example of workflow instance.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "process": {
      "description": "Nested process.",
      "items": {
        "$ref": "#/definitions/ExampleScenario_Process"
      },
      "type": "array"
    },
```

```
    "pause": {
      "description": "If there is a pause in the flow.",
      "$ref": "#/definitions/boolean"
    },
    "_pause": {
      "description": "Extensions for pause",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "operation": {
      "description": "Each interaction or action.",
      "$ref": "#/definitions/ExampleScenario_Operation"
    },
    "alternative": {
      "description": "Indicates an alternative step that can be taken instead of the operations on the base step in
exceptional/atypical circumstances.",
      "items": {
        "$ref": "#/definitions/ExampleScenario_Alternative"
      },
      "type": "array"
    }
  },
  "additionalProperties": false
},
"ExampleScenario_Operation": {
  "description": "Example of workflow instance.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "number": {
      "description": "The sequential number of the interaction, e.g. 1.2.5.",
      "$ref": "#/definitions/string"
    },
    "_number": {
      "description": "Extensions for number",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "type": {
      "description": "The type of operation - CRUD.",
      "$ref": "#/definitions/string"
    },
    "_type": {
      "description": "Extensions for type",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "name": {
      "description": "The human-friendly name of the interaction.",
      "$ref": "#/definitions/string"
    },
    "_name": {
      "description": "Extensions for name",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "initiator": {
      "description": "Who starts the transaction.",
      "$ref": "#/definitions/string"
    },
    "_initiator": {
      "description": "Extensions for initiator",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "receiver": {
      "description": "Who receives the transaction.",
```

```
      "$ref": "#/definitions/string"
    },
    "_receiver": {
      "description": "Extensions for receiver",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "description": {
      "description": "A comment to be inserted in the diagram.",
      "$ref": "#/definitions/markdown"
    },
    "_description": {
      "description": "Extensions for description",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "initiatorActive": {
      "description": "Whether the initiator is deactivated right after the transaction.",
      "$ref": "#/definitions/boolean"
    },
    "_initiatorActive": {
      "description": "Extensions for initiatorActive",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "receiverActive": {
      "description": "Whether the receiver is deactivated right after the transaction.",
      "$ref": "#/definitions/boolean"
    },
    "_receiverActive": {
      "description": "Extensions for receiverActive",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "request": {
      "description": "Each resource instance used by the initiator.",
      "$ref": "#/definitions/ExampleScenario_ContainedInstance"
    },
    "response": {
      "description": "Each resource instance used by the responder.",
      "$ref": "#/definitions/ExampleScenario_ContainedInstance"
    }
  },
  "additionalProperties": false
},
"ExampleScenario_Alternative": {
  "description": "Example of workflow instance.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "title": {
      "description": "The label to display for the alternative that gives a sense of the circumstance in which the
alternative should be invoked.",
      "$ref": "#/definitions/string"
    },
    "_title": {
      "description": "Extensions for title",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "description": {
      "description": "A human-readable description of the alternative explaining when the alternative should occur
rather than the base step.",
      "$ref": "#/definitions/markdown"
    },
    "_description": {
      "description": "Extensions for description",
```

```
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "step": {
          "description": "What happens in each alternative option.",
          "items": {
            "$ref": "#/definitions/ExampleScenario_Step"
          },
          "type": "array"
        }
      },
      "additionalProperties": false
    }
  }
}
```


12.23.9 Resource ExampleScenario - Maturity Dependency View

URL: https://hl7.org/fhir/R4/examplescenario-dependencies.html

12.23.9 Resource ExampleScenario - Maturity Dependency View

FHIR Infrastructure Work Group | Maturity Level: 0 | Trial Use | Compartments: Not linked to any defined compartments

Dependency Graph for ExampleScenario FMM level 0

Name | Card. | Type | Dependency Analysis
ExampleScenario | | DomainResource
url | 0..1 | uri
identifier | 0..* | Identifier
version | 0..1 | string
name | 0..1 | string
status | 1..1 | code
experimental | 0..1 | boolean
date | 0..1 | dateTime
publisher | 0..1 | string
contact | 0..* | ContactDetail
useContext | 0..* | UsageContext
jurisdiction | 0..* | CodeableConcept
copyright | 0..1 | markdown
purpose | 0..1 | markdown
actor | 0..* | BackboneElement
actorId | 1..1 | string
type | 1..1 | code
name | 0..1 | string
description | 0..1 | markdown
instance | 0..* | BackboneElement
resourceId | 1..1 | string
resourceType | 1..1 | code
name | 0..1 | string
description | 0..1 | markdown
version | 0..* | BackboneElement
versionId | 1..1 | string
description | 1..1 | markdown
containedInstance | 0..* | BackboneElement
resourceId | 1..1 | string
versionId | 0..1 | string
process | 0..* | BackboneElement
title | 1..1 | string
description | 0..1 | markdown
preConditions | 0..1 | markdown
postConditions | 0..1 | markdown
step | 0..* | BackboneElement
process | 0..* | see process
pause | 0..1 | boolean
operation | 0..1 | BackboneElement
number | 1..1 | string
type | 0..1 | string
name | 0..1 | string
initiator | 0..1 | string
receiver | 0..1 | string
description | 0..1 | markdown
initiatorActive | 0..1 | boolean
receiverActive | 0..1 | boolean
request | 0..1 | see containedInstance
response | 0..1 | see containedInstance

alternative | 0..* | BackboneElement
title | 1..1 | string
description | 0..1 | markdown
step | 0..* | see step
workflow | 0..* | canonical(ExampleScenario)